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Technical Specification

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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

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where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

In the present document, modal verbs have the following meanings:

shall indicates a mandatory requirement to do something

shall not indicates an interdiction (prohibition) to do something

The constructions "shall" and "shall not" are confined to the context of normative provisions, and do not appear in Technical Reports.

The constructions "must" and "must not" are not used as substitutes for "shall" and "shall not". Their use is avoided insofar as possible, and they are not used in a normative context except in a direct citation from an external, referenced, non-3GPP document, or so as to maintain continuity of style when extending or modifying the provisions of such a referenced document.

should indicates a recommendation to do something

should not indicates a recommendation not to do something

may indicates permission to do something

need not indicates permission not to do something

The construction "may not" is ambiguous and is not used in normative elements. The unambiguous constructions "might not" or "shall not" are used instead, depending upon the meaning intended.

can indicates that something is possible

cannot indicates that something is impossible

The constructions "can" and "cannot" are not substitutes for "may" and "need not".

will indicates that something is certain or expected to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

will not indicates that something is certain or expected not to happen as a result of action taken by an agency the behaviour of which is outside the scope of the present document

might indicates a likelihood that something will happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

might not indicates a likelihood that something will not happen as a result of action taken by some agency the behaviour of which is outside the scope of the present document

In addition:

is (or any other verb in the indicative mood) indicates a statement of fact

is not (or any other negative verb in the indicative mood) indicates a statement of fact

The constructions "is" and "is not" do not indicate requirements.

1 Scope

The present document specifies the coding, multiplexing and mapping to physical channels for 5G NR.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document*.

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] void.

[3] void.

[4] 3GPP TS 38.211: "NR; Physical channels and modulation".

[5] 3GPP TS 38.213: "NR; Physical layer procedures for control".

[6] 3GPP TS 38.214: "NR; Physical layer procedures for data".

[7] void.

[8] 3GPP TS 38.321: "NR; Medium Access Control (MAC) protocol specification".

[9] 3GPP TS 38.331: "NR; Radio Resource Control (RRC) protocol specification".

[10] 3GPP TS 38.473: "NG-RAN; F1 Application Protocol (F1AP)".

[11] 3GPP TS 36.212: "Evolved Universal Terrestrial Radio Access (E-UTRA); Multiplexing and channel coding".

[12] 3GPP TS 23.287: "Architecture enhancements for 5G System (5GS) to support Vehicle-to-Everything (V2X) services".

[13] 3GPP TS 38.101-1: "NR; User Equipment (UE) radio transmission and reception; Part 1: Range 1 Standalone".

[14] 3GPP TS 37.213: "Physical layer procedures for shared spectrum channel access".

[15] 3GPP TS 38.101-5: "NR; User Equipment (UE) radio transmission and reception; Part 5: Satellite access Radio Frequency (RF) and performance requirements"

3 Definitions of terms, symbols and abbreviations

3.1 Terms

For the purposes of the present document, the terms given in TR 21.905 [1] and the following apply. A term defined in the present document takes precedence over the definition of the same term, if any, in TR 21.905 [1].

3.2 Symbols

Void.

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

BCH	Broadcast Channel
CAPC	Channel Access Priority Class
CBG	Code Block Group
CBGTI	Code Block Group Transmission Information
CG	Configured Grant
CG-DFI	CG - Downlink Feedback Information
CG-UCI	CG - Uplink Control Information
CLI-RSSI	Cross Link Interference – Received Signal Strength Indicator
CORESET	Control Resource Ret
COT	Channel Occupancy Time
CP	Cyclic Prefix
CQI	Channel Quality Indicator
CRC	Cyclic Redundancy Check
CRI	CSI-RS Resource Indicator
CSI	Channel State Information
CSI-PAI	CSI Prediction Accuracy Indicator
CSI-RS	CSI - Reference Signal
DAI	Downlink Assignment Index
DCI	Downlink Control Information
DL	Downlink
DL-SCH	Downlink - Shared Channel
DMRS	Demodulation Reference Signal
HARQ	Hybrid Automatic repeat Request
HARQ-ACK	Hybrid Automatic repeat Request - Acknowledgement
LDPC	Low Density Parity Check
LI	Layer Indicator
MBS	Multicast Broadcast Services
MCS	Modulation and Coding Scheme
MRI	Measurement Resource Index
NCR	Network-controlled repeater
OFDM	Orthogonal Frequency Division Multiplex
PBCH	Physical Broadcast Channel
PCH	Paging Channel
PDCCH	Physical Downlink Control Channel
PDSCH	Physical Downlink Shared Channel
PMI	Precoding Matrix Indicator
PRB	Physical Resource Block
PRACH	Physical Random Access Channel
PSBCH	Physical Sidelink Broadcast Channel
PSCCH	Physical Sidelink Control Channel
PSFCH	Physical Sidelink Feedback Channel
PSSCH	Physical Sidelink Shared Channel
PTRS	Phase-Tracking Reference Signal
PUCCH	Physical Uplink Control Channel
PUSCH	Physical Uplink Shared Channel
RACH	Random Access Channel
RI	Rank Indicator
RSRP	Reference Signal Received Power
RS-PAI	Reference Signal Prediction Accuracy Indicator
SBFD	Sub-Band Full Duplex

SCI	Sidelink Control Information
SFCI	Sidelink Feedback Control Information
SFN	System Frame Number
SGCS	Squared Generalized Cosine Similarity
SL	Sidelink
SL-BCH	Sidelink - Broadcast Channel
SL PRS	Sidelink Positioning Reference Signal
SL-SCH	Sidelink - Shared Channel
SR	Scheduling Request
SRS	Sounding Reference Signal
SRS-RSRP	SRS - Reference Signal Received Power
SS	Synchronisation Signal
SUL	Supplementary Uplink
TCI	Transmission Configuration Indicator
TPC	Transmit Power Control
TrCH	Transport Channel
TRS	Tracking Reference Signal
UCI	Uplink Control Information
UE	User Equipment
UEIRI	UE Initiated Report Indicator
UL	Uplink
UL-SCH	Uplink Shared Channel
UTO-UCI	Unused Transmission Occasion - Uplink Control Information
VRB	Virtual Resource Block
ZP CSI-RS	Zero power CSI-RS

4 Mapping to physical channels

4.1 Uplink

Table 4.1-1 specifies the mapping of the uplink transport channels to their corresponding physical channels. Table 4.1-2 specifies the mapping of the uplink control channel information to its corresponding physical channel.

Table 4.1-1

TrCH	Physical Channel
UL-SCH	PUSCH
RACH	PRACH

Table 4.1-2

Control information	Physical Channel
UCI	PUCCH, PUSCH

4.2 Downlink

Table 4.2-1 specifies the mapping of the downlink transport channels to their corresponding physical channels.
Table 4.2-2 specifies the mapping of the downlink control channel information to its corresponding physical channel.

Table 4.2-1

TrCH	Physical Channel
DL-SCH	PDSCH
BCH	PBCH
PCH	PDSCH

Table 4.2-2

Control information	Physical Channel
DCI	PDCCH

4.3 Sidelink

Table 4.3-1 specifies the mapping of the sidelink transport channels to their corresponding physical channels. Table 4.3-2 specifies the mapping of the sidelink control information and sidelink feedback control information to their corresponding physical channels.

Table 4.3-1

TrCH	Physical Channel
SL-SCH	PSSCH
SL-BCH	PSBCH

Table 4.3-2

Control information	Physical Channel
1st-stage SCI	PSCCH
2nd-stage SCI	PSSCH
SFCI	PSFCH

5 General procedures

Data and control streams from/to MAC layer are encoded /decoded to offer transport and control services over the radio transmission link. Channel coding scheme is a combination of error detection, error correcting, rate matching, interleaving and transport channel or control information mapping onto/splitting from physical channels.

5.1 CRC calculation

Denote the input bits to the CRC computation by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, and the parity bits by $p_0, p_1, p_2, p_3, \ldots, p_{L-1}$, where A is the size of the input sequence and L is the number of parity bits. The parity bits are generated by one of the following cyclic generator polynomials:

- $g_{\text{CRC24A}}(D) = [D^{24} + D^{23} + D^{18} + D^{17} + D^{14} + D^{11} + D^{10} + D^7 + D^6 + D^5 + D^4 + D^3 + D + 1]$ for a CRC length $L = 24$;
- $g_{\text{CRC24B}}(D) = [D^{24} + D^{23} + D^6 + D^5 + D + 1]$ for a CRC length $L = 24$;
- $g_{\text{CRC24C}}(D) = [D^{24} + D^{23} + D^{21} + D^{20} + D^{17} + D^{15} + D^{13} + D^{12} + D^8 + D^4 + D^2 + D + 1]$ for a CRC length $L = 24$;
- $g_{\text{CRC16}}(D) = [D^{16} + D^{12} + D^5 + 1]$ for a CRC length $L = 16$;
- $g_{\text{CRC11}}(D) = [D^{11} + D^{10} + D^9 + D^5 + 1]$ for a CRC length $L = 11$;
- $g_{\text{CRC6}}(D) = [D^6 + D^5 + 1]$ for a CRC length $L = 6$.

The encoding is performed in a systematic form, which means that in GF(2), the polynomial:

$$a_0 D^{A+L-1} + a_1 D^{A+L-2} + \ldots + a_{A-1} D^L + p_0 D^{L-1} + p_1 D^{L-2} + \ldots + p_{L-2} D^1 + p_{L-1}$$

yields a remainder equal to 0 when divided by the corresponding CRC generator polynomial.

The bits after CRC attachment are denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B = A + L$. The relation between a_k and b_k is:

$b_k = a_k$ for $k = 0,1,2,\ldots,A-1$

$b_k = p_{k-A}$ for $k = A, A+1, A+2, \ldots, A+L-1$.

5.2 Code block segmentation and code block CRC attachment

5.2.1 Polar coding

The input bit sequence to the code block segmentation is denoted by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, where $A > 0$.

if $I_{seg} = 1$

Number of code blocks: $C = 2$;

else

Number of code blocks: $C = 1$

end if

$A' = \lceil A/C \rceil \cdot C$;

for $i = 0$ to $A' - A - 1$

$a'_i = 0$;

end for

for $i = A' - A$ to $A' - 1$

$a'_i = a_{i-(A'-A)}$;

end for

$s = 0$;

for $r = 0$ to $C - 1$

for $k = 0$ to $A'/C - 1$

$c_{rk} = a'_s$;

$s = s + 1$;

end for

The sequence $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(A'/C-1)}$ is used to calculate the CRC parity bits $p_{r0}, p_{r1}, p_{r2}, \ldots, p_{r(L-1)}$ according to Clause 5.1 with a generator polynomial of length L.

for $k = A'/C$ to $A'/C + L - 1$

$c_{rk} = p_{r(k-A'/C)}$;

end for

end for

The value of A is no larger than 1706.

5.2.2 Low density parity check coding

The input bit sequence to the code block segmentation is denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B > 0$. If B is larger than the maximum code block size K_{cb}, segmentation of the input bit sequence is performed and an additional CRC sequence of $L = 24$ bits is attached to each code block.

For LDPC base graph 1, the maximum code block size is:

- $K_{cb} = 8448$.

For LDPC base graph 2, the maximum code block size is:

- $K_{cb} = 3840$.

Total number of code blocks C is determined by:

if $B \le K_{cb}$

$L = 0$

Number of code blocks: $C = 1$

$B' = B$

else

$L = 24$

Number of code blocks: $C = \lceil B / (K_{cb} - L) \rceil$.

$B' = B + C \cdot L$

end if

The bits output from code block segmentation are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r - 1)}$, where $0 \le r < C$ is the code block number, and $K_r = K$ is the number of bits for the code block number r.

The number of bits K in each code block is calculated as:

$K' = B'/C$;

For LDPC base graph 1,

$K_b = 22$.

For LDPC base graph 2,

if $B > 640$

$K_b = 10$;

elseif $B > 560$

$K_b = 9$;

elseif $B > 192$

$K_b = 8$;

else

$K_b = 6$;
end if
find the minimum value of Z in all sets of lifting sizes in Table 5.3.2-1, denoted as Z_c, such that $K_b \cdot Z_c \geq K'$, and set $K = 22 Z_c$ for LDPC base graph 1 and $K = 10 Z_c$ for LDPC base graph 2;

The bit sequence c_{rk} is calculated as:

$s = 0$;
for $r = 0$ to $C-1$
  for $k = 0$ to $K'-L-1$
    $c_{rk} = b_s$;
    $s = s+1$;
  end for
  if $C > 1$
    The sequence $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K'-L-1)}$ is used to calculate the CRC parity bits $p_{r0}, p_{r1}, p_{r2}, \ldots, p_{r(L-1)}$ according to Clause 5.1 with the generator polynomial $g_{\text{CRC24B}}(D)$.
    for $k = K'-L$ to $K'-1$
      $c_{rk} = p_{r(k+L-K')}$;
    end for
  end if
  for $k = K'$ to $K-1$ -- Insertion of filler bits
    $c_{rk} = <NULL>$;
  end for
end for

5.3 Channel coding

Usage of coding scheme for the different types of TrCH is shown in table 5.3-1. Usage of coding scheme for the different control information types is shown in table 5.3-2.

Table 5.3-1: Usage of channel coding scheme for TrCHs

TrCH	Coding scheme
UL-SCH	LDPC
DL-SCH	
PCH	
BCH	Polar code

Table 5.3-2: Usage of channel coding scheme for control information

Control Information	Coding scheme
DCI	Polar code
UCI	Block code
	Polar code

5.3.1 Polar coding

The bit sequence input for a given code block to channel coding is denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits to encode. After encoding the bits are denoted by $d_0, d_1, d_2, \ldots, d_{N-1}$, where $N = 2^n$ and the value of n is determined by the following:

Denote by E the rate matching output sequence length as given in Clause 5.4.1;

If $E \le (9/8) \cdot 2^{(\lceil \log_2 E \rceil - 1)}$ and $K/E < 9/16$

$n_1 = \lceil \log_2 E \rceil - 1$;

else

$n_1 = \lceil \log_2 E \rceil$;

end if

$R_{\min} = 1/8$;

$n_2 = \lceil \log_2 (K / R_{\min}) \rceil$;

$n = \max\{\min\{n_1, n_2, n_{\max}\}, n_{\min}\}$

where $n_{\min} = 5$.

UE is not expected to be configured with $K + n_{PC} > E$, where n_{PC} is the number of parity check bits defined in Clause 5.3.1.2.

5.3.1.1 Interleaving

The bit sequence $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$ is interleaved into bit sequence $c'_0, c'_1, c'_2, c'_3, \ldots, c'_{K-1}$ as follows:

$c'_k = c_{\Pi(k)}$, $k = 0, 1, \ldots, K-1$

where the interleaving pattern $\Pi(k)$ is given by the following:

if $I_{IL} = 0$

$\Pi(k) = k$, $k = 0, 1, \ldots, K-1$

else

$k = 0$;

for $m = 0$ to $K_{IL}^{\max} - 1$

if $\Pi_{IL}^{\max}(m) \ge K_{IL}^{\max} - K$

$\Pi(k) = \Pi_{IL}^{\max}(m) - (K_{IL}^{\max} - K)$;

$k = k + 1$;

end if

end for

end if

where $\Pi_{IL}^{\max}(m)$ is given by Table 5.3.1.1-1 and $K_{IL}^{\max} = 164$.

Table 5.3.1.1-1: Interleaving pattern $\Pi_{IL}^{\max}(m)$

m	$\Pi_{IL}^{\max}(m)$	m	$\Pi_{IL}^{\max}(m)$	m	$\Pi_{IL}^{\max}(m)$	m	$\Pi_{IL}^{\max}(m)$	m	$\Pi_{IL}^{\max}(m)$	m	$\Pi_{IL}^{\max}(m)$
0	0	28	67	56	122	84	68	112	33	140	38
1	2	29	69	57	123	85	73	113	36	141	144
2	4	30	70	58	126	86	78	114	44	142	39
3	7	31	71	59	127	87	84	115	47	143	145
4	9	32	72	60	129	88	90	116	64	144	40
5	14	33	76	61	132	89	92	117	74	145	146
6	19	34	77	62	134	90	94	118	79	146	41
7	20	35	81	63	138	91	96	119	85	147	147
8	24	36	82	64	139	92	99	120	97	148	148
9	25	37	83	65	140	93	102	121	100	149	149
10	26	38	87	66	1	94	105	122	103	150	150
11	28	39	88	67	3	95	107	123	117	151	151
12	31	40	89	68	5	96	109	124	125	152	152
13	34	41	91	69	8	97	112	125	131	153	153
14	42	42	93	70	10	98	114	126	136	154	154
15	45	43	95	71	15	99	116	127	142	155	155
16	49	44	98	72	21	100	121	128	12	156	156
17	50	45	101	73	27	101	124	129	17	157	157
18	51	46	104	74	29	102	128	130	23	158	158
19	53	47	106	75	32	103	130	131	37	159	159
20	54	48	108	76	35	104	133	132	48	160	160
21	56	49	110	77	43	105	135	133	75	161	161
22	58	50	111	78	46	106	141	134	80	162	162
23	59	51	113	79	52	107	6	135	86	163	163
24	61	52	115	80	55	108	11	136	137		
25	62	53	118	81	57	109	16	137	143		
26	65	54	119	82	60	110	22	138	13		
27	66	55	120	83	63	111	30	139	18		

5.3.1.2 Polar encoding

The Polar sequence $Q_0^{N_{\max}-1}=\left\{Q_0^{N_{\max}},Q_1^{N_{\max}},\ldots,Q_{N_{\max}-1}^{N_{\max}}\right\}$ is given by Table 5.3.1.2-1, where $0\le Q_i^{N_{\max}}\le N_{\max}-1$ denotes a bit index before Polar encoding for $i=0,1,\ldots,N_{\max}-1$ and $N_{\max}=1024$. The Polar sequence $Q_0^{N_{\max}-1}$ is in ascending order of reliability $W\left(Q_0^{N_{\max}}\right)<W\left(Q_1^{N_{\max}}\right)<\ldots<W\left(Q_{N_{\max}-1}^{N_{\max}}\right)$, where $W\left(Q_i^{N_{\max}}\right)$ denotes the reliability of bit index $Q_i^{N_{\max}}$.

For any code block encoded to N bits, a same Polar sequence $Q_0^{N-1}=\left\{Q_0^N,Q_1^N,Q_2^N,\ldots,Q_{N-1}^N\right\}$ is used. The Polar sequence Q_0^{N-1} is a subset of Polar sequence $Q_0^{N_{\max}-1}$ with all elements $Q_i^{N_{\max}}$ of values less than N, ordered in ascending order of reliability $W\left(Q_0^N\right)<W\left(Q_1^N\right)<W\left(Q_2^N\right)<\ldots<W\left(Q_{N-1}^N\right)$.

Denote $\bar{Q}_I^N$ as a set of bit indices in Polar sequence Q_0^{N-1}, and $\bar{Q}_F^N$ as the set of other bit indices in Polar sequence Q_0^{N-1}, where $\bar{Q}_I^N$ and $\bar{Q}_F^N$ are given in Clause 5.4.1.1, $|\bar{Q}_I^N|=K+n_{PC}$, $|\bar{Q}_F^N|=N-|\bar{Q}_I^N|$, and n_{PC} is the number of parity check bits.

Denote $G_N=\left(G_2\right)^{\otimes n}$ as the n-th Kronecker power of matrix G_2, where $G_2=\begin{bmatrix}1 & 0\\ 1 & 1\end{bmatrix}$.

For a bit index j with $j=0,1,\ldots,N-1$, denote g_j as the j-th row of G_N and $w\left(g_j\right)$ as the row weight of g_j, where $w\left(g_j\right)$ is the number of ones in g_j. Denote the set of bit indices for parity check bits as Q_{PC}^N, where $|Q_{PC}^N|=n_{PC}$. A number of $\left(n_{PC}-n_{PC}^{wm}\right)$ parity check bits are placed in the $\left(n_{PC}-n_{PC}^{wm}\right)$ least reliable bit indices in $\bar{Q}_I^N$. A number of n_{PC}^{wm} other parity check bits are placed in the bit indices of minimum row weight in $\widetilde{Q}_I^N$, where $\widetilde{Q}_I^N$ denotes the $\left(|\bar{Q}_I^N|-n_{PC}\right)$ most reliable bit indices in $\bar{Q}_I^N$; if there

are more than n_{PC}^{wm} bit indices of the same minimum row weight in $\widetilde{Q}_I^N$, the n_{PC}^{wm} other parity check bits are placed in the n_{PC}^{wm} bit indices of the highest reliability and the minimum row weight in $\widetilde{Q}_I^N$.

Generate $u=[u_0 u_1 u_2 \ldots u_{N-1}]$ according to the following:

$k=0;$

if $n_{PC}>0$

$y_0=0;\ y_1=0;\ y_2=0;\ y_3=0;\ y_4=0;$

for $n=0$ to $N-1$

$y_t=y_0;\ y_0=y_1;\ y_1=y_2;\ y_2=y_3;\ y_3=y_4;\ y_4=y_t;$

if $n \in \bar{Q}_I^N$

if $n \in Q_{PC}^N$

$u_n=y_0;$

else

$u_n=c_k^{'};$

$k=k+1;$

$y_0=y_0 \oplus u_n;$

end if

else

$u_n=0;$

end if

end for

else

for $n=0$ to $N-1$

if $n \in \bar{Q}_I^N$

$u_n=c_k^{'};$

$k=k+1;$

else

$u_n=0;$

end if

end for

end if

The output after encoding $d=[d_0 d_1 d_2 \ldots d_{N-1}]$ is obtained by $d=\mathbf{u}\mathbf{G}_N$. The encoding is performed in GF(2).

Table 5.3.1.2-1: Polar sequence $Q_0^{N_{\max}-1}$ and its corresponding reliability $W\left(Q_i^{N_{\max}}\right)$

$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$
0	0	128	518	256	94	384	214	512	364	640	414	768	819	896	966
1	1	129	54	257	204	385	309	513	654	641	223	769	814	897	755
2	2	130	83	258	298	386	188	514	659	642	663	770	439	898	859
3	4	131	57	259	400	387	449	515	335	643	692	771	929	899	940
4	8	132	521	260	608	388	217	516	480	644	835	772	490	900	830
5	16	133	112	261	352	389	408	517	315	645	619	773	623	901	911
6	32	134	135	262	325	390	609	518	221	646	472	774	671	902	871
7	3	135	78	263	533	391	596	519	370	647	455	775	739	903	639
8	5	136	289	264	155	392	551	520	613	648	796	776	916	904	888
9	64	137	194	265	210	393	650	521	422	649	809	777	463	905	479
10	9	138	85	266	305	394	229	522	425	650	714	778	843	906	946
11	6	139	276	267	547	395	159	523	451	651	721	779	381	907	750
12	17	140	522	268	300	396	420	524	614	652	837	780	497	908	969
13	10	141	58	269	109	397	310	525	543	653	716	781	930	909	508

$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$
14	18	142	168	270	184	398	541	526	235	654	864	782	821	910	861
15	128	143	139	271	534	399	773	527	412	655	810	783	726	911	757
16	12	144	99	272	537	400	610	528	343	656	606	784	961	912	970
17	33	145	86	273	115	401	657	529	372	657	912	785	872	913	919
18	65	146	60	274	167	402	333	530	775	658	722	786	492	914	875
19	20	147	280	275	225	403	119	531	317	659	696	787	631	915	862
20	256	148	89	276	326	404	600	532	222	660	377	788	729	916	758
21	34	149	290	277	306	405	339	533	426	661	435	789	700	917	948
22	24	150	529	278	772	406	218	534	453	662	817	790	443	918	977
23	36	151	524	279	157	407	368	535	237	663	319	791	741	919	923
24	7	152	196	280	656	408	652	536	559	664	621	792	845	920	972
25	129	153	141	281	329	409	230	537	833	665	812	793	920	921	761
26	66	154	101	282	110	410	391	538	804	666	484	794	382	922	877
27	512	155	147	283	117	411	313	539	712	667	430	795	822	923	952
28	11	156	176	284	212	412	450	540	834	668	838	796	851	924	495
29	40	157	142	285	171	413	542	541	661	669	667	797	730	925	703
30	68	158	530	286	776	414	334	542	808	670	488	798	498	926	935
31	130	159	321	287	330	415	233	543	779	671	239	799	880	927	978
32	19	160	31	288	226	416	555	544	617	672	378	800	742	928	883
33	13	161	200	289	549	417	774	545	604	673	459	801	445	929	762
34	48	162	90	290	538	418	175	546	433	674	622	802	471	930	503
35	14	163	545	291	387	419	123	547	720	675	627	803	635	931	925
36	72	164	292	292	308	420	658	548	816	676	437	804	932	932	878
37	257	165	322	293	216	421	612	549	836	677	380	805	687	933	735
38	21	166	532	294	416	422	341	550	347	678	818	806	903	934	993
39	132	167	263	295	271	423	777	551	897	679	461	807	825	935	885
40	35	168	149	296	279	424	220	552	243	680	496	808	500	936	939
41	258	169	102	297	158	425	314	553	662	681	669	809	846	937	994
42	26	170	105	298	337	426	424	554	454	682	679	810	745	938	980
43	513	171	304	299	550	427	395	555	318	683	724	811	826	939	926
44	80	172	296	300	672	428	673	556	675	684	841	812	732	940	764
45	37	173	163	301	118	429	583	557	618	685	629	813	446	941	941
46	25	174	92	302	332	430	355	558	898	686	351	814	962	942	967
47	22	175	47	303	579	431	287	559	781	687	467	815	936	943	886
48	136	176	267	304	540	432	183	560	376	688	438	816	475	944	831
49	260	177	385	305	389	433	234	561	428	689	737	817	853	945	947
50	264	178	546	306	173	434	125	562	665	690	251	818	867	946	507
51	38	179	324	307	121	435	557	563	736	691	462	819	637	947	889
52	514	180	208	308	553	436	660	564	567	692	442	820	907	948	984
53	96	181	386	309	199	437	616	565	840	693	441	821	487	949	751
54	67	182	150	310	784	438	342	566	625	694	469	822	695	950	942
55	41	183	153	311	179	439	316	567	238	695	247	823	746	951	996
56	144	184	165	312	228	440	241	568	359	696	683	824	828	952	971
57	28	185	106	313	338	441	778	569	457	697	842	825	753	953	890
58	69	186	55	314	312	442	563	570	399	698	738	826	854	954	509
59	42	187	328	315	704	443	345	571	787	699	899	827	857	955	949
60	516	188	536	316	390	444	452	572	591	700	670	828	504	956	973
61	49	189	577	317	174	445	397	573	678	701	783	829	799	957	1000
62	74	190	548	318	554	446	403	574	434	702	849	830	255	958	892
63	272	191	113	319	581	447	207	575	677	703	820	831	964	959	950
64	160	192	154	320	393	448	674	576	349	704	728	832	909	960	863
65	520	193	79	321	283	449	558	577	245	705	928	833	719	961	759
66	288	194	269	322	122	450	785	578	458	706	791	834	477	962	1008
67	528	195	108	323	448	451	432	579	666	707	367	835	915	963	510
68	192	196	578	324	353	452	357	580	620	708	901	836	638	964	979
69	544	197	224	325	561	453	187	581	363	709	630	837	748	965	953
70	70	198	166	326	203	454	236	582	127	710	685	838	944	966	763
71	44	199	519	327	63	455	664	583	191	711	844	839	869	967	974
72	131	200	552	328	340	456	624	584	782	712	633	840	491	968	954
73	81	201	195	329	394	457	587	585	407	713	711	841	699	969	879
74	50	202	270	330	527	458	780	586	436	714	253	842	754	970	981
75	73	203	641	331	582	459	705	587	626	715	691	843	858	971	982
76	15	204	523	332	556	460	126	588	571	716	824	844	478	972	927
77	320	205	275	333	181	461	242	589	465	717	902	845	968	973	995
78	133	206	580	334	295	462	565	590	681	718	686	846	383	974	765
79	52	207	291	335	285	463	398	591	246	719	740	847	910	975	956
80	23	208	59	336	232	464	346	592	707	720	850	848	815	976	887
81	134	209	169	337	124	465	456	593	350	721	375	849	976	977	985
82	384	210	560	338	205	466	358	594	599	722	444	850	870	978	997
83	76	211	114	339	182	467	405	595	668	723	470	851	917	979	986
84	137	212	277	340	643	468	303	596	790	724	483	852	727	980	943
85	82	213	156	341	562	469	569	597	460	725	415	853	493	981	891

$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$	$W\left(Q_i^{N_{\max}}\right)$	$Q_i^{N_{\max}}$
86	56	214	87	342	286	470	244	598	249	726	485	854	873	982	998
87	27	215	197	343	585	471	595	599	682	727	905	855	701	983	766
88	97	216	116	344	299	472	189	600	573	728	795	856	931	984	511
89	39	217	170	345	354	473	566	601	411	729	473	857	756	985	988
90	259	218	61	346	211	474	676	602	803	730	634	858	860	986	1001
91	84	219	531	347	401	475	361	603	789	731	744	859	499	987	951
92	138	220	525	348	185	476	706	604	709	732	852	860	731	988	1002
93	145	221	642	349	396	477	589	605	365	733	960	861	823	989	893
94	261	222	281	350	344	478	215	606	440	734	865	862	922	990	975
95	29	223	278	351	586	479	786	607	628	735	693	863	874	991	894
96	43	224	526	352	645	480	647	608	689	736	797	864	918	992	1009
97	98	225	177	353	593	481	348	609	374	737	906	865	502	993	955
98	515	226	293	354	535	482	419	610	423	738	715	866	933	994	1004
99	88	227	388	355	240	483	406	611	466	739	807	867	743	995	1010
100	140	228	91	356	206	484	464	612	793	740	474	868	760	996	957
101	30	229	584	357	95	485	680	613	250	741	636	869	881	997	983
102	146	230	769	358	327	486	801	614	371	742	694	870	494	998	958
103	71	231	198	359	564	487	362	615	481	743	254	871	702	999	987
104	262	232	172	360	800	488	590	616	574	744	717	872	921	1000	1012
105	265	233	120	361	402	489	409	617	413	745	575	873	501	1001	999
106	161	234	201	362	356	490	570	618	603	746	913	874	876	1002	1016
107	576	235	336	363	307	491	788	619	366	747	798	875	847	1003	767
108	45	236	62	364	301	492	597	620	468	748	811	876	992	1004	989
109	100	237	282	365	417	493	572	621	655	749	379	877	447	1005	1003
110	640	238	143	366	213	494	219	622	900	750	697	878	733	1006	990
111	51	239	103	367	568	495	311	623	805	751	431	879	827	1007	1005
112	148	240	178	368	832	496	708	624	615	752	607	880	934	1008	959
113	46	241	294	369	588	497	598	625	684	753	489	881	882	1009	1011
114	75	242	93	370	186	498	601	626	710	754	866	882	937	1010	1013
115	266	243	644	371	646	499	651	627	429	755	723	883	963	1011	895
116	273	244	202	372	404	500	421	628	794	756	486	884	747	1012	1006
117	517	245	592	373	227	501	792	629	252	757	908	885	505	1013	1014
118	104	246	323	374	896	502	802	630	373	758	718	886	855	1014	1017
119	162	247	392	375	594	503	611	631	605	759	813	887	924	1015	1018
120	53	248	297	376	418	504	602	632	848	760	476	888	734	1016	991
121	193	249	770	377	302	505	410	633	690	761	856	889	829	1017	1020
122	152	250	107	378	649	506	231	634	713	762	839	890	965	1018	1007
123	77	251	180	379	771	507	688	635	632	763	725	891	938	1019	1015
124	164	252	151	380	360	508	653	636	482	764	698	892	884	1020	1019
125	768	253	209	381	539	509	248	637	806	765	914	893	506	1021	1021
126	268	254	284	382	111	510	369	638	427	766	752	894	749	1022	1022
127	274	255	648	383	331	511	190	639	904	767	868	895	945	1023	1023

5.3.2 Low density parity check coding

The bit sequence input for a given code block to channel coding is denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits to encode as defined in Clause 5.2.2. After encoding the bits are denoted by $d_0, d_1, d_2, \ldots, d_{N-1}$, where $N = 66 Z_c$ for LDPC base graph 1 and $N = 50 Z_c$ for LDPC base graph 2, and the value of Z_c is given in Clause 5.2.2.

For a code block encoded by LDPC, the following encoding procedure applies:

1) Find the set with index i_{LS} in Table 5.3.2-1 which contains Z_c.
2) for $k = 2Z_c$ to $K-1$

   if $c_k \neq$ <NULL>
      $d_{k-2Z_c} = c_k$;
   else
      $c_k = 0$;
      $d_{k-2Z_c} =$ <NULL>;
   end if

end for

3) Generate $N+2Z_c-K$ parity bits $w=\left[w_0,w_1,w_2,\ldots,w_{N+2Z_c-K-1}\right]^T$ such that $H\times\begin{bmatrix}c\\w\end{bmatrix}=0$, where $c=\left[c_0,c_1,c_2,\ldots,c_{K-1}\right]^T$; $\mathbf{O}$ is a column vector of all elements equal to 0. The encoding is performed in GF(2).

For LDPC base graph 1, a matrix of H_{BG} has 46 rows with row indices $i=0,1,2,\ldots,45$ and 68 columns with column indices $j=0,1,2,\ldots,67$. For LDPC base graph 2, a matrix of H_{BG} has 42 rows with row indices $i=0,1,2,\ldots,41$ and 52 columns with column indices $j=0,1,2,\ldots,51$. The elements in H_{BG} with row and column indices given in Table 5.3.2-2 (for LDPC base graph 1) and Table 5.3.2-3 (for LDPC base graph 2) are of value 1, and all other elements in H_{BG} are of value 0.

The matrix H is obtained by replacing each element of H_{BG} with a $Z_c\times Z_c$ matrix, according to the following:

- Each element of value 0 in H_{BG} is replaced by an all zero matrix $\mathbf{O}$ of size $Z_c\times Z_c$;
- Each element of value 1 in H_{BG} is replaced by a circular permutation matrix $I(P_{i,j})$ of size $Z_c\times Z_c$, where i and j are the row and column indices of the element, and $I(P_{i,j})$ is obtained by circularly shifting the identity matrix $\boldsymbol{I}$ of size $Z_c\times Z_c$ to the right $P_{i,j}$ times. The value of $P_{i,j}$ is given by $P_{i,j}=\mathrm{mod}(V_{i,j},Z_c)$. The value of $V_{i,j}$ is given by Tables 5.3.2-2 and 5.3.2-3 according to the set index i_{LS} and LDPC base graph.

4) for $k=K$ to $N+2Z_c-1$

$d_{k-2Z_c}=w_{k-K}$;

end for

Table 5.3.2-1: Sets of LDPC lifting size Z

Set index (i_{LS})	*Set of lifting sizes (Z)*
0	{2, 4, 8, 16, 32, 64, 128, 256}
1	{3, 6, 12, 24, 48, 96, 192, 384}
2	{5, 10, 20, 40, 80, 160, 320}
3	{7, 14, 28, 56, 112, 224}
4	{9, 18, 36, 72, 144, 288}
5	{11, 22, 44, 88, 176, 352}
6	{13, 26, 52, 104, 208}
7	{15, 30, 60, 120, 240}

Table 5.3.2-2: LDPC base graph 1 (H_{BG}) and its parity check matrices ($V_{i,j}$)

H_{BG}		$V_{i,j}$								H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}								Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7	i	j	0	1	2	3	4	5	6	7
0	0	250	307	73	223	211	294	0	135	15	1	96	2	290	120	0	348	6	138
	1	69	19	15	16	198	118	0	227		10	65	210	60	131	183	15	81	220
	2	226	50	103	94	188	167	0	126		13	63	318	130	209	108	81	182	173
	3	159	369	49	91	186	330	0	134		18	75	55	184	209	68	176	53	142
	5	100	181	240	74	219	207	0	84		25	179	269	51	81	64	113	46	49
	6	10	216	39	10	4	165	0	83		37	0	0	0	0	0	0	0	0
	9	59	317	15	0	29	243	0	53	16	1	64	13	69	154	270	190	88	78
	10	229	288	162	205	144	250	0	225		3	49	338	140	164	13	293	198	152
	11	110	109	215	216	116	1	0	205		11	49	57	45	43	99	332	160	84

H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7
	12	191	17	164	21	216	339	0	128
	13	9	357	133	215	115	201	0	75
	15	195	215	298	14	233	53	0	135
	16	23	106	110	70	144	347	0	217
	18	190	242	113	141	95	304	0	220
	19	35	180	16	198	216	167	0	90
	20	239	330	189	104	73	47	0	105
	21	31	346	32	81	261	188	0	137
	22	1	1	1	1	1	1	0	1
	23	0	0	0	0	0	0	0	0
1	0	2	76	303	141	179	77	22	96
	2	239	76	294	45	162	225	11	236
	3	117	73	27	151	223	96	124	136
	4	124	288	261	46	256	338	0	221
	5	71	144	161	119	160	268	10	128
	7	222	331	133	157	76	112	0	92
	8	104	331	4	133	202	302	0	172
	9	173	178	80	87	117	50	2	56
	11	220	295	129	206	109	167	16	11
	12	102	342	300	93	15	253	60	189
	14	109	217	76	79	72	334	0	95
	15	132	99	266	9	152	242	6	85
	16	142	354	72	118	158	257	30	153
	17	155	114	83	194	147	133	0	87
	19	255	331	260	31	156	9	168	163
	21	28	112	301	187	119	302	31	216
	22	0	0	0	0	0	0	105	0
	23	0	0	0	0	0	0	0	0
	24	0	0	0	0	0	0	0	0
2	0	106	205	68	207	258	226	132	189
	1	111	250	7	203	167	35	37	4
	2	185	328	80	31	220	213	21	225
	4	63	332	280	176	133	302	180	151
	5	117	256	38	180	243	111	4	236
	6	93	161	227	186	202	265	149	117
	7	229	267	202	95	218	128	48	179
	8	177	160	200	153	63	237	38	92
	9	95	63	71	177	0	294	122	24
	10	39	129	106	70	3	127	195	68
	13	142	200	295	77	74	110	155	6
	14	225	88	283	214	229	286	28	101
	15	225	53	301	77	0	125	85	33
	17	245	131	184	198	216	131	47	96
	18	205	240	246	117	269	163	179	125
	19	251	205	230	223	200	210	42	67
	20	117	13	276	90	234	7	66	230
	24	0	0	0	0	0	0	0	0
	25	0	0	0	0	0	0	0	0
3	0	121	276	220	201	187	97	4	128
	1	89	87	208	18	145	94	6	23
	3	84	0	30	165	166	49	33	162
	4	20	275	197	5	108	279	113	220
	6	150	199	61	45	82	139	49	43
	7	131	153	175	142	132	166	21	186
	8	243	56	79	16	197	91	6	96
	10	136	132	281	34	41	106	151	1
	11	86	305	303	155	162	246	83	216
	12	246	231	253	213	57	345	154	22
	13	219	341	164	147	36	269	87	24
	14	211	212	53	69	115	185	5	167
	16	240	304	44	96	242	249	92	200
	17	76	300	28	74	165	215	173	32
	18	244	271	77	99	0	143	120	235
	20	144	39	319	30	113	121	2	172
	21	12	357	68	158	108	121	142	219
	22	1	1	1	1	1	1	0	1
	25	0	0	0	0	0	0	0	0
4	0	157	332	233	170	246	42	24	64
	1	102	181	205	10	235	256	204	211
	26	0	0	0	0	0	0	0	0
5	0	205	195	83	164	261	219	185	2
	1	236	14	292	59	181	130	100	171
	3	194	115	50	86	72	251	24	47
	12	231	166	318	80	283	322	65	143
	16	28	241	201	182	254	295	207	210
	21	123	51	267	130	79	258	161	180
	22	115	157	279	153	144	283	72	180
	27	0	0	0	0	0	0	0	0
6	0	183	278	289	158	80	294	6	199
	6	22	257	21	119	144	73	27	22
	10	28	1	293	113	169	330	163	23
	11	67	351	13	21	90	99	50	100
	13	244	92	232	63	59	172	48	92
	17	11	253	302	51	177	150	24	207

H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7
	20	51	289	115	189	54	331	122	5
	22	154	57	300	101	0	114	182	205
	38	0	0	0	0	0	0	0	0
17	0	7	260	257	56	153	110	91	183
	14	164	303	147	110	137	228	184	112
	16	59	81	128	200	0	247	30	106
	17	1	358	51	63	0	116	3	219
	21	144	375	228	4	162	190	155	129
	39	0	0	0	0	0	0	0	0
18	1	42	130	260	199	161	47	1	183
	12	233	163	294	110	151	286	41	215
	13	8	280	291	200	0	246	167	180
	18	155	132	141	143	241	181	68	143
	19	147	4	295	186	144	73	148	14
	40	0	0	0	0	0	0	0	0
19	0	60	145	64	8	0	87	12	179
	1	73	213	181	6	0	110	6	108
	7	72	344	101	103	118	147	166	159
	8	127	242	270	198	144	258	184	138
	10	224	197	41	8	0	204	191	196
	41	0	0	0	0	0	0	0	0
20	0	151	187	301	105	265	89	6	77
	3	186	206	162	210	81	65	12	187
	9	217	264	40	121	90	155	15	203
	11	47	341	130	214	144	244	5	167
	22	160	59	10	183	228	30	30	130
	42	0	0	0	0	0	0	0	0
21	1	249	205	79	192	64	162	6	197
	5	121	102	175	131	46	264	86	122
	16	109	328	132	220	266	346	96	215
	20	131	213	283	50	9	143	42	65
	21	171	97	103	106	18	109	199	216
	43	0	0	0	0	0	0	0	0
22	0	64	30	177	53	72	280	44	25
	12	142	11	20	0	189	157	58	47
	13	188	233	55	3	72	236	130	126
	17	158	22	316	148	257	113	131	178
	44	0	0	0	0	0	0	0	0
23	1	156	24	249	88	180	18	45	185
	2	147	89	50	203	0	6	18	127
	10	170	61	133	168	0	181	132	117
	18	152	27	105	122	165	304	100	199
	45	0	0	0	0	0	0	0	0
24	0	112	298	289	49	236	38	9	32
	3	86	158	280	157	199	170	125	178
	4	236	235	110	64	0	249	191	2
	11	116	339	187	193	266	288	28	156
	22	222	234	281	124	0	194	6	58
	46	0	0	0	0	0	0	0	0
25	1	23	72	172	1	205	279	4	27
	6	136	17	295	166	0	255	74	141
	7	116	383	96	65	0	111	16	11
	14	182	312	46	81	183	54	28	181
	47	0	0	0	0	0	0	0	0
26	0	195	71	270	107	0	325	21	163
	2	243	81	110	176	0	326	142	131
	4	215	76	318	212	0	226	192	169
	15	61	136	67	127	277	99	197	98
	48	0	0	0	0	0	0	0	0
27	1	25	194	210	208	45	91	98	165
	6	104	194	29	141	36	326	140	232
	8	194	101	304	174	72	268	22	9
	49	0	0	0	0	0	0	0	0
28	0	128	222	11	146	275	102	4	32
	4	165	19	293	153	0	1	1	43
	19	181	244	50	217	155	40	40	200
	21	63	274	234	114	62	167	93	205
	50	0	0	0	0	0	0	0	0
29	1	86	252	27	150	0	273	92	232
	14	236	5	308	11	180	104	136	32
	18	84	147	117	53	0	243	106	118
	25	6	78	29	68	42	107	6	103
	51	0	0	0	0	0	0	0	0
30	0	216	159	91	34	0	171	2	170
	10	73	229	23	130	90	16	88	199
	13	120	260	105	210	252	95	112	26
	24	9	90	135	123	173	212	20	105
	52	0	0	0	0	0	0	0	0
31	1	95	100	222	175	144	101	4	73
	7	177	215	308	49	144	297	49	149
	22	172	258	66	177	166	279	125	175
	25	61	256	162	128	19	222	194	108
	53	0	0	0	0	0	0	0	0
32	0	221	102	210	192	0	351	6	103

H_{BG}		$V_{i,j}$								H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}								Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7	i	j	0	1	2	3	4	5	6	7
	18	157	18	138	136	151	284	38	52		12	112	201	22	209	211	265	126	110
	20	211	225	235	116	108	305	91	13		14	199	175	271	58	36	338	63	151
	28	0	0	0	0	0	0	0	0		24	121	287	217	30	162	83	20	211
	0	220	9	12	17	169	3	145	77		54	0	0	0	0	0	0	0	0
	1	44	62	88	76	189	103	88	146		1	2	323	170	114	0	56	10	199
	4	159	316	207	104	154	224	112	209		2	187	8	20	49	0	304	30	132
7	7	31	333	50	100	184	297	153	32	33	11	41	361	140	161	76	141	6	172
	8	167	290	25	150	104	215	159	166		21	211	105	33	137	18	101	92	65
	14	104	114	76	158	164	39	76	18		55	0	0	0	0	0	0	0	0
	29	0	0	0	0	0	0	0	0		0	127	230	187	82	197	60	4	161
	0	112	307	295	33	54	348	172	181		7	167	148	296	186	0	320	153	237
	1	4	179	133	95	0	75	2	105	34	15	164	202	5	68	108	112	197	142
	3	7	165	130	4	252	22	131	141		17	159	312	44	150	0	54	155	180
	12	211	18	231	217	41	312	141	223		56	0	0	0	0	0	0	0	0
8	16	102	39	296	204	98	224	96	177		1	161	320	207	192	199	100	4	231
	19	164	224	110	39	46	17	99	145		6	197	335	158	173	278	210	45	174
	21	109	368	269	58	15	59	101	199	35	12	207	2	55	26	0	195	168	145
	22	241	67	245	44	230	314	35	153		22	103	266	285	187	205	268	185	100
	24	90	170	154	201	54	244	116	38		57	0	0	0	0	0	0	0	0
	30	0	0	0	0	0	0	0	0		0	37	210	259	222	216	135	6	11
	0	103	366	189	9	162	156	6	169		14	105	313	179	157	16	15	200	207
	1	182	232	244	37	159	88	10	12	36	15	51	297	178	0	0	35	177	42
	10	109	321	36	213	93	293	145	206		18	120	21	160	6	0	188	43	100
	11	21	133	286	105	134	111	53	221		58	0	0	0	0	0	0	0	0
9	13	142	57	151	89	45	92	201	17		1	198	269	298	81	72	319	82	59
	17	14	303	267	185	132	152	4	212	37	13	220	82	15	195	144	236	2	204
	18	61	63	135	109	76	23	164	92		23	122	115	115	138	0	85	135	161
	20	216	82	209	218	209	337	173	205		59	0	0	0	0	0	0	0	0
	31	0	0	0	0	0	0	0	0		0	167	185	151	123	190	164	91	121
	1	98	101	14	82	178	175	126	116		9	151	177	179	90	0	196	64	90
	2	149	339	80	165	1	253	77	151	38	10	157	289	64	73	0	209	198	26
	4	167	274	211	174	28	27	156	70		12	163	214	181	10	0	246	100	140
10	7	160	111	75	19	267	231	16	230		60	0	0	0	0	0	0	0	0
	8	49	383	161	194	234	49	12	115		1	173	258	102	12	153	236	4	115
	14	58	354	311	103	201	267	70	84		3	139	93	77	77	0	264	28	188
	32	0	0	0	0	0	0	0	0	39	7	149	346	192	49	165	37	109	168
	0	77	48	16	52	55	25	184	45		19	0	297	208	114	117	272	188	52
	1	41	102	147	11	23	322	194	115		61	0	0	0	0	0	0	0	0
	12	83	8	290	2	274	200	123	134		0	157	175	32	67	216	304	10	4
11	16	182	47	289	35	181	351	16	1	40	8	137	37	80	45	144	237	84	103
	21	78	188	177	32	273	166	104	152		17	149	312	197	96	2	135	12	30
	22	252	334	43	84	39	338	109	165		62	0	0	0	0	0	0	0	0
	23	22	115	280	201	26	192	124	107		1	167	52	154	23	0	123	2	53
	33	0	0	0	0	0	0	0	0		3	173	314	47	215	0	77	75	189
	0	160	77	229	142	225	123	6	186	41	9	139	139	124	60	0	25	142	215
	1	42	186	235	175	162	217	20	215		18	151	288	207	167	183	272	128	24
	10	21	174	169	136	244	142	203	124		63	0	0	0	0	0	0	0	0
12	11	32	232	48	3	151	110	153	180		0	149	113	226	114	27	288	163	222
	13	234	50	105	28	238	176	104	98	42	4	157	14	65	91	0	83	10	170
	18	7	74	52	182	243	76	207	80		24	137	218	126	78	35	17	162	71
	34	0	0	0	0	0	0	0	0		64	0	0	0	0	0	0	0	0
	0	177	313	39	81	231	311	52	220		1	151	113	228	206	52	210	1	22
	3	248	177	302	56	0	251	147	185		16	163	132	69	22	243	3	163	127
13	7	151	266	303	72	216	265	1	154	43	18	173	114	176	134	0	53	99	49
	20	185	115	160	217	47	94	16	178		25	139	168	102	161	270	167	98	125
	23	62	370	37	78	36	81	46	150		65	0	0	0	0	0	0	0	0
	35	0	0	0	0	0	0	0	0		0	139	80	234	84	18	79	4	191
	0	206	142	78	14	0	22	1	124		7	157	78	227	4	0	244	6	211
	12	55	248	299	175	186	322	202	144	44	9	163	163	259	9	0	293	142	187
	15	206	137	54	211	253	277	118	182		22	173	274	260	12	57	272	3	148
14	16	127	89	61	191	16	156	130	95		66	0	0	0	0	0	0	0	0
	17	16	347	179	51	0	66	1	72		1	149	135	101	184	168	82	181	177
	21	229	12	258	43	79	78	2	76	45	6	151	149	228	121	0	67	45	114
	36	0	0	0	0	0	0	0	0		10	167	15	126	29	144	235	153	93
15	0	40	241	229	90	170	176	173	39		67	0	0	0	0	0	0	0	0

Table 5.3.2-3: LDPC base graph 2 (H_{BG}) and its parity check matrices ($V_{i,j}$)

H_{BG}		$V_{i,j}$								H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}								Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7	i	j	0	1	2	3	4	5	6	7
	0	9	174	0	72	3	156	143	145	16	26	0	0	0	0	0	0	0	0
	1	117	97	0	110	26	143	19	131		1	254	158	0	48	120	134	57	196
	2	204	166	0	23	53	14	176	71		5	124	23	24	132	43	23	201	173
0	3	26	66	0	181	35	3	165	21	17	11	114	9	109	206	65	62	142	195
	6	189	71	0	95	115	40	196	23		12	64	6	18	2	42	163	35	218
	9	205	172	0	8	127	123	13	112		27	0	0	0	0	0	0	0	0
	10	0	0	0	1	0	0	0	1	18	0	220	186	0	68	17	173	129	128

H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7
	11	0	0	0	0	0	0	0	0
1	0	167	27	137	53	19	17	18	142
	3	166	36	124	156	94	65	27	174
	4	253	48	0	115	104	63	3	183
	5	125	92	0	156	66	1	102	27
	6	226	31	88	115	84	55	185	96
	7	156	187	0	200	98	37	17	23
	8	224	185	0	29	69	171	14	9
	9	252	3	55	31	50	133	180	167
	11	0	0	0	0	0	0	0	0
	12	0	0	0	0	0	0	0	0
2	0	81	25	20	152	95	98	126	74
	1	114	114	94	131	106	168	163	31
	3	44	117	99	46	92	107	47	3
	4	52	110	9	191	110	82	183	53
	8	240	114	108	91	111	142	132	155
	10	1	1	1	0	1	1	1	0
	12	0	0	0	0	0	0	0	0
	13	0	0	0	0	0	0	0	0
3	1	8	136	38	185	120	53	36	239
	2	58	175	15	6	121	174	48	171
	4	158	113	102	36	22	174	18	95
	5	104	72	146	124	4	127	111	110
	6	209	123	12	124	73	17	203	159
	7	54	118	57	110	49	89	3	199
	8	18	28	53	156	128	17	191	43
	9	128	186	46	133	79	105	160	75
	10	0	0	0	1	0	0	0	1
	13	0	0	0	0	0	0	0	0
4	0	179	72	0	200	42	86	43	29
	1	214	74	136	16	24	67	27	140
	11	71	29	157	101	51	83	117	180
	14	0	0	0	0	0	0	0	0
5	0	231	10	0	185	40	79	136	121
	1	41	44	131	138	140	84	49	41
	5	194	121	142	170	84	35	36	169
	7	159	80	141	219	137	103	132	88
	11	103	48	64	193	71	60	62	207
	15	0	0	0	0	0	0	0	0
6	0	155	129	0	123	109	47	7	137
	5	228	92	124	55	87	154	34	72
	7	45	100	99	31	107	10	198	172
	9	28	49	45	222	133	155	168	124
	11	158	184	148	209	139	29	12	56
	16	0	0	0	0	0	0	0	0
7	1	129	80	0	103	97	48	163	86
	5	147	186	45	13	135	125	78	186
	7	140	16	148	105	35	24	143	87
	11	3	102	96	150	108	47	107	172
	13	116	143	78	181	65	55	58	154
	17	0	0	0	0	0	0	0	0
8	0	142	118	0	147	70	53	101	176
	1	94	70	65	43	69	31	177	169
	12	230	152	87	152	88	161	22	225
	18	0	0	0	0	0	0	0	0
9	1	203	28	0	2	97	104	186	167
	8	205	132	97	30	40	142	27	238
	10	61	185	51	184	24	99	205	48
	11	247	178	85	83	49	64	81	68
	19	0	0	0	0	0	0	0	0
10	0	11	59	0	174	46	111	125	38
	1	185	104	17	150	41	25	60	217
	6	0	22	156	8	101	174	177	208
	7	117	52	20	56	96	23	51	232
	20	0	0	0	0	0	0	0	0
11	0	11	32	0	99	28	91	39	178
	7	236	92	7	138	30	175	29	214
	9	210	174	4	110	116	24	35	168
	13	56	154	2	99	64	141	8	51
	21	0	0	0	0	0	0	0	0
12	1	63	39	0	46	33	122	18	124
	3	111	93	113	217	122	11	155	122
	11	14	11	48	109	131	4	49	72
	22	0	0	0	0	0	0	0	0
13	0	83	49	0	37	76	29	32	48
	1	2	125	112	113	37	91	53	57
	8	38	35	102	143	62	27	95	167
	13	222	166	26	140	47	127	186	219
	23	0	0	0	0	0	0	0	0
14	1	115	19	0	36	143	11	91	82
	6	145	118	138	95	51	145	20	232
	11	3	21	57	40	130	8	52	204
	13	232	163	27	116	97	166	109	162
	24	0	0	0	0	0	0	0	0

H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7
	6	194	6	18	16	106	31	203	211
	7	50	46	86	156	142	22	140	210
	28	0	0	0	0	0	0	0	0
19	0	87	58	0	35	79	13	110	39
	1	20	42	158	138	28	135	124	84
	10	185	156	154	86	41	145	52	88
	29	0	0	0	0	0	0	0	0
20	1	26	76	0	6	2	128	196	117
	4	105	61	148	20	103	52	35	227
	11	29	153	104	141	78	173	114	6
	30	0	0	0	0	0	0	0	0
21	0	76	157	0	80	91	156	10	238
	8	42	175	17	43	75	166	122	13
	13	210	67	33	81	81	40	23	11
	31	0	0	0	0	0	0	0	0
22	1	222	20	0	49	54	18	202	195
	2	63	52	4	1	132	163	126	44
	32	0	0	0	0	0	0	0	0
23	0	23	106	0	156	68	110	52	5
	3	235	86	75	54	115	132	170	94
	5	238	95	158	134	56	150	13	111
	33	0	0	0	0	0	0	0	0
24	1	46	182	0	153	30	113	113	81
	2	139	153	69	88	42	108	161	19
	9	8	64	87	63	101	61	88	130
	34	0	0	0	0	0	0	0	0
25	0	228	45	0	211	128	72	197	66
	5	156	21	65	94	63	136	194	95
	35	0	0	0	0	0	0	0	0
26	2	29	67	0	90	142	36	164	146
	7	143	137	100	6	28	38	172	66
	12	160	55	13	221	100	53	49	190
	13	122	85	7	6	133	145	161	86
	36	0	0	0	0	0	0	0	0
27	0	8	103	0	27	13	42	168	64
	6	151	50	32	118	10	104	193	181
	37	0	0	0	0	0	0	0	0
28	1	98	70	0	216	106	64	14	7
	2	101	111	126	212	77	24	186	144
	5	135	168	110	193	43	149	46	16
	38	0	0	0	0	0	0	0	0
29	0	18	110	0	108	133	139	50	25
	4	28	17	154	61	25	161	27	57
	39	0	0	0	0	0	0	0	0
30	2	71	120	0	106	87	84	70	37
	5	240	154	35	44	56	173	17	139
	7	9	52	51	185	104	93	50	221
	9	84	56	134	176	70	29	6	17
	40	0	0	0	0	0	0	0	0
31	1	106	3	0	147	80	117	115	201
	13	1	170	20	182	139	148	189	46
	41	0	0	0	0	0	0	0	0
32	0	242	84	0	108	32	116	110	179
	5	44	8	20	21	89	73	0	14
	12	166	17	122	110	71	142	163	116
	42	0	0	0	0	0	0	0	0
33	2	132	165	0	71	135	105	163	46
	7	164	179	88	12	6	137	173	2
	10	235	124	13	109	2	29	179	106
	43	0	0	0	0	0	0	0	0
34	0	147	173	0	29	37	11	197	184
	12	85	177	19	201	25	41	191	135
	13	36	12	78	69	114	162	193	141
	44	0	0	0	0	0	0	0	0
35	1	57	77	0	91	60	126	157	85
	5	40	184	157	165	137	152	167	225
	11	63	18	6	55	93	172	181	175
	45	0	0	0	0	0	0	0	0
36	0	140	25	0	1	121	73	197	178
	2	38	151	63	175	129	154	167	112
	7	154	170	82	83	26	129	179	106
	46	0	0	0	0	0	0	0	0
37	10	219	37	0	40	97	167	181	154
	13	151	31	144	12	56	38	193	114
	47	0	0	0	0	0	0	0	0
38	1	31	84	0	37	1	112	157	42
	5	66	151	93	97	70	7	173	41
	11	38	190	19	46	1	19	191	105
	48	0	0	0	0	0	0	0	0
39	0	239	93	0	106	119	109	181	167
	7	172	132	24	181	32	6	157	45
	12	34	57	138	154	142	105	173	189
	49	0	0	0	0	0	0	0	0
40	2	0	103	0	98	6	160	193	78

H_{BG}		$V_{i,j}$								H_{BG}		$V_{i,j}$							
Row index	Column index	Set index i_{LS}								Row index	Column index	Set index i_{LS}							
i	j	0	1	2	3	4	5	6	7	i	j	0	1	2	3	4	5	6	7
15	0	51	68	0	116	139	137	174	38		10	75	107	36	35	73	156	163	67
	10	175	63	73	200	96	103	108	217		13	120	163	143	36	102	82	179	180
	11	213	81	99	110	128	40	102	157		50	0	0	0	0	0	0	0	0
	25	0	0	0	0	0	0	0	0	41	1	129	147	0	120	48	132	191	53
16	1	203	87	0	75	48	78	125	170		5	229	7	2	101	47	6	197	215
	9	142	177	79	158	9	158	31	23		11	118	60	55	81	19	8	167	230
	11	8	135	111	134	28	17	54	175		51	0	0	0	0	0	0	0	0
	12	242	64	143	97	8	165	176	202										

5.3.3 Channel coding of small block lengths

The bit sequence input for a given code block to channel coding is denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits to encode. After encoding the bits are denoted by $d_0, d_1, d_2, \ldots, d_{N-1}$.

5.3.3.1 Encoding of 1-bit information

For $K=1$, the code block is encoded according to Table 5.3.3.1-1, where $N=Q_m$ and Q_m is the modulation order for the code block.

Table 5.3.3.1-1: Encoding of 1-bit information

Q_m	Encoded bits $d_0, d_1, d_2, \ldots, d_{N-1}$
1	$[c_0]$
2	$[c_0$ y$]$
4	$[c_0$ y x x$]$
6	$[c_0$ y x x x x $]$
8	$[c_0$ y *x x x x x* x $]$

The "x" and "y" in Table 5.3.3.1-1 are placeholders for Clauses 6.3.1.1, 6.3.2.5.1, 6.3.2.6.1 of [4, TS 38.211] to scramble the information bits in a way that maximizes the Euclidean distance of the modulation symbols carrying the information bits.

5.3.3.2 Encoding of 2-bit information

For $K=2$, the code block is encoded according to Table 5.3.3.2-1, where $c_2=(c_0+c_1) \bmod 2$, $N=3Q_m$, and Q_m is the modulation order for the code block.

Table 5.3.3.2-1: Encoding of 2-bit information

Q_m	Encoded bits $d_0, d_1, d_2, \ldots, d_{N-1}$
1	$[c_0 c_1 c_2]$
2	$[c_0 c_1 c_2 c_0 c_1 c_2]$
4	$[c_0 c_1$ x x $c_2 c_0$ x x $c_1 c_2$ x x$]$
6	$[c_0 c_1$ x x x x $c_2 c_0$ x x x x $c_1 c_2$ x x x x$]$
8	$[c_0 c_1$ x *x x x x* x $c_2 c_0$ x *x x x x* x $c_1 c_2$ x *x x x x* x$]$

The "x" in Table 5.3.3.2-1 are placeholders for Clause 6.3.1.1 of [4, TS 38.211] to scramble the information bits in a way that maximizes the Euclidean distance of the modulation symbols carrying the information bits.

5.3.3.3 Encoding of other small block lengths

For $3 \le K \le 11$, the code block is encoded by $d_i = \left(\sum_{k=0}^{K-1} c_k \cdot M_{i,k} \right) \bmod 2$, where $i = 0, 1, \cdots, N-1$, $N = 32$, and $M_{i,k}$ represents the basis sequences as defined in Table 5.3.3.3-1.

Table 5.3.3.3-1: Basis sequences for (32, K) code

i	$M_{i,0}$	$M_{i,1}$	$M_{i,2}$	$M_{i,3}$	$M_{i,4}$	$M_{i,5}$	$M_{i,6}$	$M_{i,7}$	$M_{i,8}$	$M_{i,9}$	$M_{i,10}$
0	1	1	0	0	0	0	0	0	0	0	1
1	1	1	1	0	0	0	0	0	0	1	1
2	1	0	0	1	0	0	1	0	1	1	1
3	1	0	1	1	0	0	0	0	1	0	1
4	1	1	1	1	0	0	0	1	0	0	1
5	1	1	0	0	1	0	1	1	1	0	1
6	1	0	1	0	1	0	1	0	1	1	1
7	1	0	0	1	1	0	0	1	1	0	1
8	1	1	0	1	1	0	0	1	0	1	1
9	1	0	1	1	1	0	1	0	0	1	1
10	1	0	1	0	0	1	1	1	0	1	1
11	1	1	1	0	0	1	1	0	1	0	1
12	1	0	0	1	0	1	0	1	1	1	1
13	1	1	0	1	0	1	0	1	0	1	1
14	1	0	0	0	1	1	0	1	0	0	1
15	1	1	0	0	1	1	1	1	0	1	1
16	1	1	1	0	1	1	1	0	0	1	0
17	1	0	0	1	1	1	0	0	1	0	0
18	1	1	0	1	1	1	1	1	0	0	0
19	1	0	0	0	0	1	1	0	0	0	0
20	1	0	1	0	0	0	1	0	0	0	1
21	1	1	0	1	0	0	0	0	0	1	1
22	1	0	0	0	1	0	0	1	1	0	1
23	1	1	1	0	1	0	0	0	1	1	1
24	1	1	1	1	1	0	1	1	1	1	0
25	1	1	0	0	0	1	1	1	0	0	1
26	1	0	1	1	0	1	0	0	1	1	0
27	1	1	1	1	0	1	0	1	1	1	0
28	1	0	1	0	1	1	1	0	1	0	0
29	1	0	1	1	1	1	1	1	1	0	0
30	1	1	1	1	1	1	1	1	1	1	1
31	1	0	0	0	0	0	0	0	0	0	0

5.4 Rate matching

5.4.1 Rate matching for Polar code

The rate matching for Polar code is defined per coded block and consists of sub-block interleaving, bit collection, and bit interleaving. The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$. The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

5.4.1.1 Sub-block interleaving

The bits input to the sub-block interleaver are the coded bits $d_0, d_1, d_2, \ldots, d_{N-1}$. The coded bits $d_0, d_1, d_2, \ldots, d_{N-1}$ are divided into 32 sub-blocks. The bits output from the sub-block interleaver are denoted as $y_0, y_1, y_2, \ldots, y_{N-1}$, generated as follows:

for $n = 0$ to $N-1$

$i = \lfloor 32n/N \rfloor$;
$J(n) = P(i) \times (N/32) + \mathrm{mod}(n, N/32)$;
$y_n = d_{J(n)}$;
end for

where the sub-block interleaver pattern $P(i)$ is given by Table 5.4.1.1-1.

Table 5.4.1.1-1: Sub-block interleaver pattern $P(i)$

i	$P(i)$	i	$P(i)$	i	$P(i)$	i	$P(i)$	i	$P(i)$	i	$P(i)$	i	$P(i)$	i	$P(i)$
0	0	4	3	8	8	12	10	16	12	20	14	24	24	28	27
1	1	5	5	9	16	13	18	17	20	21	22	25	25	29	29
2	2	6	6	10	9	14	11	18	13	22	15	26	26	30	30
3	4	7	7	11	17	15	19	19	21	23	23	27	28	31	31

The sets of bit indices $\bar{Q}_I^N$ and $\bar{Q}_F^N$ are determined as follows, where K, n_{PC}, and Q_0^{N-1} are defined in Clause 5.3.1

$\bar{Q}_{F,tmp}^N = \emptyset$
if $E < N$
  if $K/E \le 7/16$ -- puncturing
    for $n = 0$ to $N-E-1$
      $\bar{Q}_{F,tmp}^N = \bar{Q}_{F,tmp}^N \cup \{J(n)\}$;
    end for
    if $E \ge 3N/4$
      $\bar{Q}_{F,tmp}^N = \bar{Q}_{F,tmp}^N \cup \{0,1,\ldots,\lceil 3N/4 - E/2 \rceil - 1\}$;
    else
      $\bar{Q}_{F,tmp}^N = \bar{Q}_{F,tmp}^N \cup \{0,1,\ldots,\lceil 9N/16 - E/4 \rceil - 1\}$;
    end if
  else -- shortening
    for $n = E$ to $N-1$
      $\bar{Q}_{F,tmp}^N = \bar{Q}_{F,tmp}^N \cup \{J(n)\}$;
    end for
  end if
end if
$\bar{Q}_{I,tmp}^N = Q_0^{N-1} \setminus \bar{Q}_{F,tmp}^N$;
$\bar{Q}_I^N$ comprises $(K + n_{PC})$ most reliable bit indices in $\bar{Q}_{I,tmp}^N$;
$\bar{Q}_F^N = Q_0^{N-1} \setminus \bar{Q}_I^N$;

5.4.1.2 Bit selection

The bit sequence after the sub-block interleaver $y_0, y_1, y_2, \ldots, y_{N-1}$ from Clause 5.4.1.1 is written into a circular buffer of length N.

Denoting by E the rate matching output sequence length, the bit selection output bit sequence e_k, $k = 0,1,2,\ldots,E-1$, is generated as follows:

if $E \ge N$ -- repetition
  for $k = 0$ to $E-1$

$e_k = y_{\mathrm{mod}(k,N)}$;
end for
else
if $K/E \le 7/16$ -- puncturing
for $k = 0$ to $E-1$
$e_k = y_{k+N-E}$;
end for
else -- shortening
for $k = 0$ to $E-1$
$e_k = y_k$;
end for
end if
end if

5.4.1.3 Interleaving of coded bits

The bit sequence $e_0, e_1, e_2, \ldots, e_{E-1}$ is interleaved into bit sequence $f_0, f_1, f_2, \ldots, f_{E-1}$, as follows:

If $I_{BIL} = 1$

Denote T as the smallest integer such that $T(T+1)/2 \ge E$;

$k = 0$;

for $i = 0$ to $T-1$

for $j = 0$ to $T-1-i$
if $k < E$
$v_{i,j} = e_k$;
else
$v_{i,j} = <NULL>$¿¿;
end if
$k = k+1$;
end for
end for

$k = 0$;

for $j = 0$ to $T-1$

for $i = 0$ to $T-1-j$
if $v_{i,j} \neq$ ¿NULL>¿¿
$f_k = v_{i,j}$;
$k = k+1$
end if
end for
end for

else

for $i = 0$ to $E-1$

$f_i = e_i$;
end for

end if

The value of E is no larger than 8192.

5.4.2 Rate matching for LDPC code

The rate matching for LDPC code is defined per coded block and consists of bit selection and bit interleaving. The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$. The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

5.4.2.1 Bit selection

The bit sequence after encoding $d_0, d_1, d_2, \ldots, d_{N-1}$ from Clause 5.3.2 is written into a circular buffer of length N_{cb} for the r-th coded block, where N is defined in Clause 5.3.2.

For the r-th code block, let $N_{cb} = N$ if $I_{LBRM} = 0$ and $N_{cb} = \min(N, N_{ref})$ otherwise, where $N_{ref} = \left\lfloor \frac{TBS_{LBRM}}{C \cdot R_{\text{LBRM}}} \right\rfloor$, $R_{\text{LBRM}} = 2/3$, TBS_{LBRM} is determined according to Clause 6.1.4.2 in [6, TS 38.214] for UL-SCH and Clause 5.1.3.2 in [6, TS 38.214] for DL-SCH/PCH, assuming the following:

For one TB for DL-SCH with PDSCH scheduled by DCI format 4_0/4_1/4_2:

- if the PDSCH is scheduled by DCI format 4_1/4_2:
  - maximum number of layers is given by X, where:
    - if the higher layer parameter *maxMIMO-Layers* of *pdsch-ConfigMulticast* is configured, X is given by that parameter;
    - otherwise, X equals to 1;
  - if the higher layer parameter *mcs-Table* given by a *pdsch-ConfigMulticast* or by *pdsch-ConfigMTCH* for at least one common frequency resource (CFR) is set to 'qam256', maximum modulation order $Q_m = 8$ is assumed for DL-SCH; otherwise a maximum modulation order $Q_m = 6$ is assumed for DL-SCH;
- if the PDSCH is scheduled by DCI format 4_0:
  - maximum number of layers is 1;
  - if the higher layer parameter *mcs-Table* given by a *pdsch-ConfigMCCH* is set to 'qam256', maximum modulation order $Q_m = 8$ is assumed for DL-SCH; otherwise a maximum modulation order $Q_m = 6$ is assumed for DL-SCH;
  - if the higher layer parameter *mcs-Table* given by a *pdsch-ConfigMTCH* is set to 'qam256', maximum modulation order $Q_m = 8$ is assumed for DL-SCH; otherwise a maximum modulation order $Q_m = 6$ is assumed for DL-SCH;
- $n_{PRB} = n_{PRB,LBRM}$ is given by Table 5.4.2.1-1, where the value of $n_{PRB,LBRM}$ for DL-SCH is determined according to the size of the associated CFR if configured to the UE;
- maximum coding rate of 948/1024;
- $N_{\text{RE}} = 156 \cdot n_{PRB}$;
- C is the number of code blocks of the transport block determined according to Clause 5.2.2.

For one TB for UL-SCH, or for one TB for DL-SCH/PCH except for DL-SCH with PDSCH scheduled by DCI format 4_0/4_1/4_2:

- maximum number of layers for one TB for UL-SCH is given by the minimum of X and 4, where:
  - if the higher layer parameter *maxMIMO-Layers* of *PUSCH-ServingCellConfig* of the serving cell is configured and if neither the higher layer parameter *multipanelSchemeSFN* nor the higher layer parameter *multipanelSchemeSDM* is configured, X is given by that parameter;

  - elseif the higher layer parameter *maxMIMO-Layers* of *PUSCH-ServingCellConfig* of the serving cell is configured and if the higher layer parameter *multipanelSchemeSFN* is configured, X is given by max{*maxMIMO-Layers*, *maxMIMO-LayersforSFN*};
  - elseif the higher layer parameter *maxMIMO-Layers* of *PUSCH-ServingCellConfig* of the serving cell is configured and if the higher layer parameter *multipanelSchemeSDM* is configured, X is given by max{*maxMIMO-Layers*, 2**maxMIMO-LayersforSDM*};
  - elseif the higher layer parameter *maxRank* of *pusch-Config* of the serving cell is configured and if neither the higher layer parameter *multipanelSchemeSFN* nor the higher layer parameter *multipanelSchemeSDM* is configured, X is given by the maximum value of *maxRank* across all BWPs of the serving cell;
  - elseif the higher layer parameter *maxRank* of *pusch-Config* of the serving cell is configured and if the higher layer parameter *multipanelSchemeSFN* is configured, X is given by max{*maxRank*, *maxRankSFN*} across all BWPs of the serving cell;
  - elseif the higher layer parameter *maxRank* of *pusch-Config* of the serving cell is configured and if the higher layer parameter *multipanelSchemeSDM* is configured, X is given by max{*maxRank*, 2**maxRankSDM*} across all BWPs of the serving cell;
  - otherwise, X is given by the maximum number of layers for PUSCH supported by the UE for the serving cell;
- maximum number of layers for one TB for DL-SCH/PCH is given by the minimum of X and 4, where:
  - if the higher layer parameter *maxMIMO-Layers* of *PDSCH-ServingCellConfig* of the serving cell is configured, X is given by that parameter;
  - otherwise, X is given by the maximum number of layers for PDSCH supported by the UE for the serving cell;
- if the higher layer parameter *mcs-Table-r17* or *mcs-TableDCI-1-2-r17* given by a *pdsch-Config* for at least one DL BWP of the serving cell is set to 'qam1024', maximum modulation order $Q_m = 10$ is assumed for DL-SCH, except if the UE indicated support for *pdsch-1024QAM-2MIMO-FR1-r17* and X > 2, maximum modulation order $Q_m = 8$ is assumed for DL-SCH, else if the higher layer parameter *mcs-Table* or *mcs-TableDCI-1-2* given by a *pdsch-Config* for at least one DL BWP of the serving cell is set to 'qam256', maximum modulation order $Q_m = 8$ is assumed for DL-SCH; otherwise a maximum modulation order $Q_m = 6$ is assumed for DL-SCH;
- if the higher layer parameter *mcs-Table* or *mcs-TableTransformPrecoder* or *mcs-TableDCI-0-2* or *mcs-TableTransformPrecoderDCI-0-2* given by a *pusch-Config* or the higher layer parameter *mcs-Table* or *mcs-TableTransformPrecoder* given by *configuredGrantConfig* for at least one UL BWP of the serving cell is set to 'qam256', maximum modulation order $Q_m = 8$ is assumed for UL-SCH; otherwise a maximum modulation order $Q_m = 6$ is assumed for UL-SCH;
- maximum coding rate of 948/1024;
- $n_{PRB} = n_{PRB,LBRM}$ is given by Table 5.4.2.1-1, where the value of $n_{PRB,LBRM}$ for DL-SCH is determined according to the initial downlink bandwidth part if there is no other downlink bandwidth part configured to the UE;
- $N_{RE} = 156 \cdot n_{PRB}$;
- C is the number of code blocks of the transport block determined according to Clause 5.2.2.

Table 5.4.2.1-1: Value of $\eta_{PRB,LBRM}$

Maximum number of PRBs across all configured DL BWPs and UL BWPs of a carrier for DL-SCH and UL-SCH, respectively, or Maximum number of PRBs across all CFRs of a carrier for DL-SCH with PDSCH scheduled by DCI format 4_0/4_1/4_2	$\eta_{PRB,LBRM}$
Less than 33	32
33 to 66	66
67 to 107	107
108 to 135	135
136 to 162	162
163 to 217	217
Larger than 217	273

Denoting by E_r the rate matching output sequence length for the r-th coded block, where the value of E_r is determined as follows:

Set $j=0$

for $r=0$ to $C-1$

if the r-th coded block is not scheduled for transmission as indicated by CBGTI according to Clause 5.1.7.2 for DL-SCH and 6.1.5.2 for UL-SCH in [6, TS 38.214]:

$E_r=0$;

else

if $j\le C'-\mathrm{mod}\left(G/\left(N_L\cdot Q_m\right),C'\right)-1$

$$E_r=N_L\cdot Q_m\cdot\left\lfloor\frac{G}{N_L\cdot Q_m\cdot C'}\right\rfloor;$$

else

$$E_r=N_L\cdot Q_m\cdot\left\lceil\frac{G}{N_L\cdot Q_m\cdot C'}\right\rceil;$$

end if

$j=j+1$;

end if

end for

where:

- N_L is the number of transmission layers that the transport block is mapped onto;
- Q_m is the modulation order;
- G is the total number of coded bits available for transmission of the transport block;
- $C'=C$ if CBGTI is not present in the DCI scheduling the transport block and C' is the number of scheduled code blocks of the transport block if CBGTI is present in the DCI scheduling the transport block.

Denote by rv_{id} the redundancy version number for this transmission (rv_{id} = 0, 1, 2 or 3), the rate matching output bit sequence e_k, $k=0,1,2,\ldots,E-1$, is generated as follows, where k_0 is given by Table 5.4.2.1-2 according to the value of rv_{id} and LDPC base graph:

$k=0$;

$j=0$;

while $k<E$

if $d_{(k_0+j) \bmod N_{cb}} \neq$ ¿ NULL> ¿ ¿
$e_k = d_{(k_0+j) \bmod N_{cb}}$;
$k = k+1$;
end if
$j = j+1$;
end while

Table 5.4.2.1-2: Starting position of different redundancy versions, k_0

rv_{id}	k_0	
	LDPC base graph 1	LDPC base graph 2
0	0	0
1	$\left\lfloor \frac{17 N_{cb}}{66 Z_c} \right\rfloor Z_c$	$\left\lfloor \frac{13 N_{cb}}{50 Z_c} \right\rfloor Z_c$
2	$\left\lfloor \frac{33 N_{cb}}{66 Z_c} \right\rfloor Z_c$	$\left\lfloor \frac{25 N_{cb}}{50 Z_c} \right\rfloor Z_c$
3	$\left\lfloor \frac{56 N_{cb}}{66 Z_c} \right\rfloor Z_c$	$\left\lfloor \frac{43 N_{cb}}{50 Z_c} \right\rfloor Z_c$

5.4.2.2 Bit interleaving

The bit sequence $e_0, e_1, e_2, \ldots, e_{E-1}$ is interleaved to bit sequence $f_0, f_1, f_2, \ldots, f_{E-1}$, according to the following, where the value of Q_m is the modulation order:

for $j = 0$ to $E/Q_m - 1$
for $i = 0$ to $Q_m - 1$
$f_{i+j \cdot Q_m} = e_{i \cdot E/Q_m + j}$;
end for
end for

5.4.3 Rate matching for channel coding of small block lengths

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$. The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$, where E is the rate matching output sequence length. The bit sequence $f_0, f_1, f_2, \ldots, f_{E-1}$ is obtained by the following:

for $k = 0$ to $E-1$

$f_k = d_{k \bmod N}$;

end for

5.5 Code block concatenation

The input bit sequence for the code block concatenation block are the sequences f_{rk}, for $r = 0, \ldots, C-1$ and $k = 0, \ldots, E_r - 1$, where E_r is the number of rate matched bits for the r-th code block. The output bit sequence from the code block concatenation block is the sequence g_k for $k = 0, \ldots, G-1$.

The code block concatenation consists of sequentially concatenating the rate matching outputs for the different code blocks. Therefore:

Set $k=0$ and $r=0$

while $r<C$

Set $j=0$

while $j<E_r$

$g_k=f_{rj}$
$k=k+1$
$j=j+1$
end while

$r=r+1$

end while

6 Uplink transport channels and control information

6.1 Random access channel

The sequence index for the random access channel is received from higher layers and is processed according to [4, TS 38.211].

6.2 Uplink shared channel

6.2.1 Transport block CRC attachment

Error detection is provided on each UL-SCH transport block through a Cyclic Redundancy Check (CRC).

The entire transport block is used to calculate the CRC parity bits. Denote the bits in a transport block delivered to layer 1 by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, and the parity bits by $p_0, p_1, p_2, p_3, \ldots, p_{L-1}$, where A is the payload size and L is the number of parity bits. The lowest order information bit a_0 is mapped to the most significant bit of the transport block as defined in Clause 6.1.1 of [TS38.321].

The parity bits are computed and attached to the UL-SCH transport block according to Clause 5.1, by setting L to 24 bits and using the generator polynomial $g_{\text{CRC24A}}(D)$ if $A>3824$; and by setting L to 16 bits and using the generator polynomial $g_{\text{CRC16}}(D)$ otherwise.

The bits after CRC attachment are denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B=A+L$.

6.2.2 LDPC base graph selection

For initial transmission of a transport block with coding rate R indicated by the MCS index according to Clause 6.1.4.1 in [6, TS 38.214] and subsequent re-transmission of the same transport block, each code block of the transport block is encoded with either LDPC base graph 1 or 2 according to the following:

- if $A\leq 292$, or if $A\leq 3824$ and $R\leq 0.67$, or if $R\leq 0.25$, LDPC base graph 2 is used;
- otherwise, LDPC base graph 1 is used,

where A is the payload size as described in Clause 6.2.1.

6.2.3 Code block segmentation and code block CRC attachment

The bits input to the code block segmentation are denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$ where B is the number of bits in the transport block (including CRC).

Code block segmentation and code block CRC attachment are performed according to Clause 5.2.2.

The bits after code block segmentation are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$, where r is the code block number and K_r is the number of bits for code block number r according to Clause 5.2.2.

When the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is larger than 1, the value of B is no larger than 3840 if $R \leq 0.25$ and no larger than 8448 otherwise, where coding rate R is indicated by the MCS index according to Clause 6.1.4.1 in [6, TS 38.214].

6.2.4 Channel coding of UL-SCH

Code blocks are delivered to the channel coding block. The bits in a code block are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$, where r is the code block number, and K_r is the number of bits in code block number r. The total number of code blocks is denoted by C and each code block is individually LDPC encoded according to Clause 5.3.2.

After encoding the bits are denoted by $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$, where the values of N_r is given in Clause 5.3.2.

6.2.5 Rate matching

Coded bits for each code block, denoted as $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$, are delivered to the rate match block, where r is the code block number, and N_r is the number of encoded bits in code block number r. The total number of code blocks is denoted by C and each code block is individually rate matched according to Clause 5.4.2 by setting $I_{LBRM}=1$ if higher layer parameter *rateMatching* is set to *limitedBufferRM* and by setting $I_{LBRM}=0$ otherwise, if *numberOfSlotsTBoMS* is not present in the resource allocation table, or if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is equal to 1. When the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is larger than 1, each code block is individually rate matched per slot according to Clause 5.4.2 by setting:

- $I_{LBRM}=1$ if higher layer parameter *rateMatching* is set to *limitedBufferRM* and by setting $I_{LBRM}=0$ otherwise;
- G as the total number of coded bits available for transmission of the transport block in the slot;
- k_0 as given by Table 5.4.2.1-2 according to the value of rv_{id} and LDPC base graph if the slot is the first slot within the N_s slots allocated for the transmission of TB processing over multiple slots, and setting $k_0=\left(k_0^{'}+H+\tau\right) mod\, N_{cb}$ if the slot is a slot except for the first one within the N_s slots, where N_s is the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI, $k_0^{'}$ denotes the index of starting coded bit in the previous slot within the N_s slots, H is the total number of coded bits available for transmission of the transport block in the previous slot within the N_s slots assuming no UCI multiplexing, and τ denotes the number of skipped filler bits if any in the previous slot within the N_s slots according to Clause 5.4.2.1 by assuming no UCI multiplexing.

After rate matching, the bits are denoted by $f_{r0}, f_{r1}, f_{r2}, f_{r3}, \ldots, f_{r(E_r-1)}$, where E_r is the number of rate matched bits for code block number r.

6.2.6 Code block concatenation

The input bit sequence for the code block concatenation block are the sequences $f_{r0}, f_{r1}, f_{r2}, f_{r3}, \ldots, f_{r(E_r-1)}$, for $r=0,\ldots,C-1$ and where E_r is the number of rate matched bits for the r-th code block.

Code block concatenation is performed according to Clause 5.5.

The bits after code block concatenation are denoted by $g_0, g_1, g_2, g_3, \ldots, g_{G-1}$, where G is the total number of coded bits for transmission.

6.2.7 Data and control multiplexing

In case where there are more than one UL-SCH transport blocks for the PUSCH transmission, the UCI information is multiplexed only on the UL-SCH transport block with highest I_{MCS} value for the initial PUSCH, where I_{MCS} is as defined in Clause 6.1.4.1 in [6, TS 38.214]. In case the two transport blocks have the same I_{MCS} value for the initial PUSCH, the UCI information is multiplexed with data only on the first transport block. The PUSCH for UCI multiplexing in this Clause refers to the UL-SCH transport block for UCI multiplexing.

If the higher layer parameter *nrofBitsInUTO-UCI* is configured, the procedure in this clause 6.2.7 applies by replacing CG-UCI with UTO-UCI in all the notations and texts, and replacing "when higher layer parameter *cg-UCI-Multiplexing* is configured" with "when UTO-UCI and HARQ-ACK are transmitted on a PUSCH".

Denote the coded bits for UL-SCH as $g_0^{\text{UL-SCH}}, g_1^{\text{UL-SCH}}, g_2^{\text{UL-SCH}}, g_3^{\text{UL-SCH}}, \ldots, g_{G^{\text{UL-SCH}}-1}^{\text{UL-SCH}}$.

Denote the coded bits for HARQ-ACK or jointly coded bits for HARQ-ACK and CG-UCI when the high layer parameter *cg-UCI-Multiplexing* is configured, if any, as $g_0^{\text{ACK}}, g_1^{\text{ACK}}, g_2^{\text{ACK}}, g_3^{\text{ACK}}, \ldots, g_{G^{\text{ACK}}-1}^{\text{ACK}}$.

Denote the coded bits for CSI part 1, if any, as $g_0^{\text{CSI-part1}}, g_1^{\text{CSI-part1}}, g_2^{\text{CSI-part1}}, g_3^{\text{CSI-part1}}, \ldots, g_{G^{\text{CSI-part1}}-1}^{\text{CSI-part1}}$.

Denote the coded bits for CSI part 2, if any, as $g_0^{\text{CSI-part2}}, g_1^{\text{CSI-part2}}, g_2^{\text{CSI-part2}}, g_3^{\text{CSI-part2}}, \ldots, g_{G^{\text{CSI-part2}}-1}^{\text{CSI-part2}}$.

Denote the coded bits for CG-UCI without HARQ-ACK, if any, as $g_0^{CG\text{-}UCI}$, $g_1^{CG\text{-}UCI}$, $g_2^{CG\text{-}UCI}$, $g_3^{CG\text{-}UCI}$, …, $g_{G^{CG\text{-}UCI}-1}^{CG\text{-}UCI}$.

Denote the multiplexed data and control coded bit sequence as $g_0, g_1, g_2, g_3, \ldots, g_{G-1}$.

Denote l as the OFDM symbol index of the PUSCH transmission, starting from 0 to $N_{\text{symb,all}}^{\text{PUSCH}}-1$, where $N_{\text{symb,all}}^{\text{PUSCH}}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS.

Denote k as the subcarrier index of the PUSCH transmission, starting from 0 to $M_{\text{sc}}^{\text{PUSCH}}-1$, where $M_{\text{sc}}^{\text{PUSCH}}$ is expressed as a number of subcarriers.

Denote $\Phi_l^{\text{UL-SCH}}$ as the set of resource elements, in ascending order of indices k, available for transmission of data in OFDM symbol l, for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$.

Denote $M_{\text{sc}}^{\text{UL-SCH}}(l)=\left|\Phi_l^{\text{UL-SCH}}\right|$ as the number of elements in set $\Phi_l^{\text{UL-SCH}}$. Denote $\Phi_l^{\text{UL-SCH}}(j)$ as the j-th element in $\Phi_l^{\text{UL-SCH}}$.

Denote Φ_l^{UCI} as the set of resource elements, in ascending order of indices k, available for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$. Denote $M_{\text{sc}}^{\text{UCI}}(l)=\left|\Phi_l^{\text{UCI}}\right|$ as the number of elements in set Φ_l^{UCI}. Denote $\Phi_l^{\text{UCI}}(j)$ as the j-th element in Φ_l^{UCI}. For any OFDM symbol that carries DMRS of the PUSCH, $\Phi_l^{\text{UCI}}=\varnothing$. For any OFDM symbol that does not carry DMRS of the PUSCH, $\Phi_l^{\text{UCI}}=\Phi_l^{\text{UL-SCH}}$.

If frequency hopping is configured for the PUSCH,

- denote $l^{(1)}$ as the OFDM symbol index of the first OFDM symbol after the first set of consecutive OFDM symbol(s) carrying DMRS in the first hop;
- denote $l^{(2)}$ as the OFDM symbol index of the first OFDM symbol after the first set of consecutive OFDM symbol(s) carrying DMRS in the second hop;
- denote $l_{\text{CSI}}^{(1)}$ as the OFDM symbol index of the first OFDM symbol that does not carry DMRS in the first hop;
- denote $l_{\text{CSI}}^{(2)}$ as the OFDM symbol index of the first OFDM symbol that does not carry DMRS in the second hop;
- if HARQ-ACK is present for transmission on the PUSCH with UL-SCH or if both HARQ-ACK and CG-UCI are present on the same PUSCH with UL-SCH, let:
  - $G^{\text{ACK}}(1)=N_L\cdot Q_m\cdot\left\lfloor G^{\text{ACK}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor$ and $G^{\text{ACK}}(2)=N_L\cdot Q_m\cdot\left\lceil G^{\text{ACK}}/\left(2\cdot N_L\cdot Q_m\right)\right\rceil$;
- if CSI is present for transmission on the PUSCH with UL-SCH, let:
  - $G^{\text{CSI-part1}}(1)=N_L\cdot Q_m\cdot\left\lfloor G^{\text{CSI-part1}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor$;
  - $G^{\text{CSI-part1}}(2)=N_L\cdot Q_m\cdot\left\lceil G^{\text{CSI-part1}}/\left(2\cdot N_L\cdot Q_m\right)\right\rceil$;
  - $G^{\text{CSI-part2}}(1)=N_L\cdot Q_m\cdot\left\lfloor G^{\text{CSI-part2}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor$; and
  - $G^{\text{CSI-part2}}(2)=N_L\cdot Q_m\cdot\left\lceil G^{\text{CSI-part2}}/\left(2\cdot N_L\cdot Q_m\right)\right\rceil$;
- if CG-UCI is present for transmission on the PUSCH with UL-SCH and without HARQ-ACK, let:
  - $G^{CG\text{-}UCI}(1)=N_L\cdot Q_m\cdot\left\lfloor G^{CG\text{-}UCI}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor$ and $G^{CG\text{-}UCI}(2)=N_L\cdot Q_m\cdot\left\lceil G^{CG\text{-}UCI}/\left(2\cdot N_L\cdot Q_m\right)\right\rceil$
- if only HARQ-ACK and CSI part 1 are present for transmission on the PUSCH without UL-SCH, let:
  - $G^{\text{ACK}}(1)=\min\left(N_L\cdot Q_m\cdot\left\lfloor G^{\text{ACK}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor,\ M_3\cdot N_L\cdot Q_m\right)$;
  - $G^{\text{ACK}}(2)=G^{\text{ACK}}-G^{\text{ACK}}(1)$;
  - $G^{\text{CSI-part1}}(1)=M_1\cdot N_L\cdot Q_m-G^{\text{ACK}}(1)$; and
  - $G^{\text{CSI-part1}}(2)=G^{\text{CSI-part1}}-G^{\text{CSI-part1}}(1)$;
- if HARQ-ACK, CSI part 1 and CSI part 2 are present for transmission on the PUSCH without UL-SCH, let:
  - $G^{\text{ACK}}(1)=\min\left(N_L\cdot Q_m\cdot\left\lfloor G^{\text{ACK}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor,\ M_3\cdot N_L\cdot Q_m\right)$;
  - $G^{\text{ACK}}(2)=G^{\text{ACK}}-G^{\text{ACK}}(1)$;
  - if the number of HARQ-ACK information bits is more than 2,
    $G^{\text{CSI-part1}}(1)=\min\left(N_L\cdot Q_m\cdot\left\lfloor G^{\text{CSI-part1}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor,M_1\cdot N_L\cdot Q_m-G^{\text{ACK}}(1)\right)$; otherwise,
    $G^{\text{CSI-part1}}(1)=\min\left(N_L\cdot Q_m\cdot\left\lfloor G^{\text{CSI-part1}}/\left(2\cdot N_L\cdot Q_m\right)\right\rfloor,\ M_1\cdot N_L\cdot Q_m-G_{rvd}^{\text{ACK}}(1)\right)$
  - $G^{\text{CSI-part1}}(2)=G^{\text{CSI-part1}}-G^{\text{CSI-part1}}(1)$;

- $G^{\text{CSI-part2}}(1)=M_1\cdot N_L\cdot Q_m-G^{\text{CSI-part1}}(1)$ if the number of HARQ-ACK information bits is no more than 2, and $G^{\text{CSI-part2}}(1)=M_1\cdot N_L\cdot Q_m-G^{\text{ACK}}(1)-G^{\text{CSI-part1}}(1)$ otherwise; and
- $G^{\text{CSI-part2}}(2)=M_2\cdot N_L\cdot Q_m-G^{\text{CSI-part1}}(2)$ if the number of HARQ-ACK information bits is no more than 2, and $G^{\text{CSI-part2}}(2)=M_2\cdot N_L\cdot Q_m-G^{\text{ACK}}(2)-G^{\text{CSI-part1}}(2)$ otherwise;

- if only CSI part 1 and CSI part 2 are present for transmission on the PUSCH without UL-SCH, let:
  - $G^{\text{CSI-part1}}(1)=\min\left(N_L\cdot Q_m\cdot\left\lfloor G^{\text{CSI-part1}}/(2\cdot N_L\cdot Q_m)\right\rfloor,\ M_1\cdot N_L\cdot Q_m-G_{rvd}^{\text{ACK}}(1)\right)$;
  - $G^{\text{CSI-part1}}(2)=G^{\text{CSI-part1}}-G^{\text{CSI-part1}}(1)$;
  - $G^{\text{CSI-part2}}(1)=M_1\cdot N_L\cdot Q_m-G^{\text{CSI-part1}}(1)$; and
  - $G^{\text{CSI-part2}}(2)=M_2\cdot N_L\cdot Q_m-G^{\text{CSI-part1}}(2)$;
- let $N_{\text{hop}}^{\text{PUSCH}}=2$, and denote $N_{\text{symb,hop}}^{\text{PUSCH}}(1)$, $N_{\text{symb,hop}}^{\text{PUSCH}}(2)$ as the number of OFDM symbols of the PUSCH in the first and second hop, respectively;
- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH;
- $M_1=\sum_{l=0}^{N_{\text{symb,hop}}^{\text{PUSCH}}(1)-1}M_{\text{SC}}^{\text{UCI}}(l)$;
- $M_2=\sum_{l=N_{\text{symb,hop}}^{\text{PUSCH}}(1)}^{N_{\text{symb,hop}}^{\text{PUSCH}}(1)+N_{\text{symb,hop}}^{\text{PUSCH}}(2)-1}M_{\text{SC}}^{\text{UCI}}(l)$;
- $M_3=\sum_{l=l^{(1)}}^{N_{\text{symb,hop}}^{\text{PUSCH}}(1)-1}M_{\text{SC}}^{\text{UCI}}(l)$.

If frequency hopping is not configured for the PUSCH,

- denote $l^{(1)}$ as the OFDM symbol index of the first OFDM symbol after the first set of consecutive OFDM symbol(s) carrying DMRS;
- denote $l_{\text{CSI}}^{(1)}$ as the OFDM symbol index of the first OFDM symbol that does not carry DMRS;
- if HARQ-ACK is present for transmission on the PUSCH or if both HARQ-ACK and CG-UCI are present on the same PUSCH with UL-SCH, let $G^{\text{ACK}}(1)=G^{\text{ACK}}$;
- if CSI is present for transmission on the PUSCH, let $G^{\text{CSI-part1}}(1)=G^{\text{CSI-part1}}$ and $G^{\text{CSI-part2}}(1)=G^{\text{CSI-part2}}$;
- if CG-UCI is present for transmission on the PUSCH without HARQ-ACK, let $G^{CG\text{-}UCI}(1)=G^{CG\text{-}UCI}$;
- let $N_{\text{hop}}^{\text{PUSCH}}=1$ and $N_{\text{symb,hop}}^{\text{PUSCH}}(1)=N_{\text{symb,all}}^{\text{PUSCH}}$.

The multiplexed data and control coded bit sequence $g_0,g_1,g_2,g_3,\ldots,g_{G-1}$ is obtained according to the following:

Step 1:

Set $\bar{\Phi}_l^{\text{UL-SCH}}=\Phi_l^{\text{UL-SCH}}$ for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$;

Set $\bar{M}_{\text{sc}}^{\text{UL-SCH}}(l)=\left|\bar{\Phi}_l^{\text{UL-SCH}}\right|$ for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$;

Set $\bar{\Phi}_l^{\text{UCI}}=\Phi_l^{\text{UCI}}$ for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$;

Set $\bar{M}_{sc}^{UCI}(l) = \left|\bar{\Phi}_l^{UCI}\right|$ for $l = 0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$;

if the number of HARQ-ACK information bits to be transmitted on PUSCH is 0, 1 or 2 bits and without CG-UCI:

the number of reserved resource elements for potential HARQ-ACK transmission is calculated according to Clause 6.3.2.4.2.1, by setting $O_{ACK} = 2$;

denote G_{rvd}^{ACK} as the number of coded bits for potential HARQ-ACK transmission using the reserved resource elements;

if frequency hopping is configured for the PUSCH, let $G_{rvd}^{ACK}(1) = N_L \cdot Q_m \cdot \left\lfloor G_{rvd}^{ACK} / (2 \cdot N_L \cdot Q_m) \right\rfloor$ and $G_{rvd}^{ACK}(2) = N_L \cdot Q_m \cdot \left\lceil G_{rvd}^{ACK} / (2 \cdot N_L \cdot Q_m) \right\rceil$;

if frequency hopping is not configured for the PUSCH, let $G_{rvd}^{ACK}(1) = G_{rvd}^{ACK}$;

denote $\bar{\Phi}_l^{rvd}$ as the set of reserved resource elements for potential HARQ-ACK transmission, in OFDM symbol l, for $l = 0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$;

Set $m_{count}^{ACK}(1) = 0$;

Set $m_{count}^{ACK}(2) = 0$;

$\bar{\Phi}_l^{rvd} = \varnothing$ for $l = 0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$;

for $i = 1$ to N_{hop}^{PUSCH}

$l = l^{(i)}$;

while $m_{count}^{ACK}(i) < G_{rvd}^{ACK}(i)$

if $\bar{M}_{sc}^{UCI}(l) > 0$

if $G_{rvd}^{ACK}(i) - m_{count}^{ACK}(i) \geq \bar{M}_{sc}^{UCI}(l) \cdot N_L \cdot Q_m$

$d = 1$;

$m_{count}^{RE} = \bar{M}_{sc}^{UL\text{-}SCH}(l)$;

end if

if $G_{rvd}^{ACK}(i) - m_{count}^{ACK}(i) < \bar{M}_{sc}^{UCI}(l) \cdot N_L \cdot Q_m$

$d = \left\lfloor \bar{M}_{sc}^{UCI}(l) \cdot N_L \cdot Q_m / \left(G_{rvd}^{ACK}(i) - m_{count}^{ACK}(i)\right) \right\rfloor$;

$m_{count}^{RE} = \left\lceil \left(G_{rvd}^{ACK}(i) - m_{count}^{ACK}(i)\right) / (N_L \cdot Q_m) \right\rceil$;

end if

for $j = 0$ to $m_{count}^{RE} - 1$

$\bar{\Phi}_l^{rvd} = \bar{\Phi}_l^{rvd} \cup \left\{ \bar{\Phi}_l^{UL\text{-}SCH}(j \cdot d) \right\}$

$m_{count}^{ACK}(i) = m_{count}^{ACK}(i) + N_L \cdot Q_m$;

end for

end if

$l = l + 1$;

end while

end for

else

$\bar{\Phi}_l^{\text{rvd}} = \varnothing$ for $l = 0, 1, 2, \ldots, N_{\text{symb,all}}^{\text{PUSCH}} - 1$;

end if

Denote $\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l) = |\bar{\Phi}_l^{\text{rvd}}|$ as the number of elements in $\bar{\Phi}_l^{\text{rvd}}$.

Step 2:

if HARQ-ACK is present for transmission on the PUSCH and the number of HARQ-ACK information bits is more than 2 or if both HARQ-ACK and CG-UCI are present on the same PUSCH with UL-SCH:

Set $m_{\text{count}}^{\text{ACK}}(1) = 0$;

Set $m_{\text{count}}^{\text{ACK}}(2) = 0$;

Set $m_{\text{count,all}}^{\text{ACK}} = 0$;

for $i = 1$ to $N_{\text{hop}}^{\text{PUSCH}}$

$l = l^{(i)}$;

while $m_{\text{count}}^{\text{ACK}}(i) < G^{\text{ACK}}(i)$

if $\bar{M}_{\text{sc}}^{\text{UCI}}(l) > 0$

if $G^{\text{ACK}}(i) - m_{\text{count}}^{\text{ACK}}(i) \geq \bar{M}_{\text{sc}}^{\text{UCI}}(l) \cdot N_L \cdot Q_m$

$d = 1$;

$m_{\text{count}}^{\text{RE}} = \bar{M}_{\text{sc}}^{\text{UCI}}(l)$;

end if

if $G^{\text{ACK}}(i) - m_{\text{count}}^{\text{ACK}}(i) < \bar{M}_{\text{sc}}^{\text{UCI}}(l) \cdot N_L \cdot Q_m$

$d = \left\lfloor \bar{M}_{\text{sc}}^{\text{UCI}}(l) \cdot N_L \cdot Q_m \big/ \left(G^{\text{ACK}}(i) - m_{\text{count}}^{\text{ACK}}(i)\right) \right\rfloor$;

$m_{\text{count}}^{\text{RE}} = \left\lceil \left(G^{\text{ACK}}(i) - m_{\text{count}}^{\text{ACK}}(i)\right) / \left(N_L \cdot Q_m\right) \right\rceil$;

end if

for $j = 0$ to $m_{\text{count}}^{\text{RE}} - 1$

$k = \bar{\Phi}_l^{\text{UCI}}(j \cdot d)$;

for $v = 0$ to $N_L \cdot Q_m - 1$

$\bar{g}_{l,k,v} = g_{m_{\text{count, all}}^{\text{ACK}}}^{\text{ACK}}$;

$m_{\text{count,all}}^{\text{ACK}} = m_{\text{count,all}}^{\text{ACK}} + 1$;

$m_{\text{count}}^{\text{ACK}}(i) = m_{\text{count}}^{\text{ACK}}(i) + 1$;

end for

end for

$\bar{\Phi}_{l,tmp}^{\text{UCI}} = \varnothing$;

for $j = 0$ to $m_{\text{count}}^{\text{RE}} - 1$

$\bar{\Phi}_{l,tmp}^{\text{UCI}} = \bar{\Phi}_{l,tmp}^{\text{UCI}} \cup \bar{\Phi}_l^{\text{UCI}}(j \cdot d)$;

end for

$\bar{\Phi}_l^{\text{UCI}} = \bar{\Phi}_l^{\text{UCI}} \setminus \bar{\Phi}_{l,tmp}^{\text{UCI}}$;

$\bar{\Phi}_l^{\text{UL-SCH}} = \bar{\Phi}_l^{\text{UL-SCH}} \setminus \bar{\Phi}_{l,tmp}^{\text{UCI}}$;

$\bar{M}_{sc}^{\text{UCI}}(l) = \left|\bar{\Phi}_l^{\text{UCI}}\right|$;
$\bar{M}_{sc}^{\text{UL-SCH}}(l) = \left|\bar{\Phi}_l^{\text{UL-SCH}}\right|$;
end if
$l = l + 1$;
end while
end for

end if

Step 2A:

If CG-UCI is present for transmission on the PUSCH without HARQ-ACK:

Set $m_{count}^{CG\text{-}UCI}(1) = 0$;
Set $m_{count}^{CG\text{-}UCI}(2) = 0$;
Set $m_{count,all}^{CG\text{-}UCI} = 0$;
for $i = 1$ to N_{hop}^{PUSCH}
$l = l^{(i)}$;
while $m_{count}^{CG\text{-}UCI}(i) < G^{CG\text{-}UCI}(i)$
if $\overline{M}_{sc}^{UCI}(l) > 0$
if $G^{CG\text{-}UCI}(i) - m_{count}^{CG\text{-}UCI}(1) \geq \overline{M}_{sc}^{UCI}(l).N_L.Q_m$
$d = 1$;
$m_{count}^{\text{RE}} = \overline{M}_{sc}^{UCI}(l)$;
end if
if $G^{CG\text{-}UCI}(i) - m_{count}^{CG\text{-}UCI}(1) < \overline{M}_{sc}^{UCI}(l).N_L.Q_m$
$d = \lfloor \overline{M}_{sc}^{UCI}(l).N_L.Q_m / \left(G^{CG\text{-}UCI}(i) - m_{count}^{CG\text{-}UCI}(i)\right) \rfloor$;
$m_{count}^{\text{RE}} = \lceil \left(G^{CG\text{-}UCI}(i) - m_{count}^{CG\text{-}UCI}(i)\right) / \left(N_L.Q_m\right) \rceil$;
end if
for $j = 0$ to $m_{count}^{\text{RE}} - 1$
$k = \overline{\Phi}_l^{UCI}(j.d)$;
for $v = 0$ to $N_L.Q_m - 1$
$\overline{g}_{l,k,v} = g_{m_{count,all}^{CG\text{-}UCI}}^{CG\text{-}UCI}$;
$m_{count,all}^{CG\text{-}UCI} = m_{count,all}^{CG\text{-}UCI} + 1$;
$m_{count}^{CG\text{-}UCI}(i) = m_{count}^{CG\text{-}UCI}(i) + 1$;
end for
end for
$\overline{\Phi}_{l,tmp}^{UCI} = \varnothing$;
for $j = 0$ to $m_{count}^{\text{RE}} - 1$
$\overline{\Phi}_{l,tmp}^{UCI} = \overline{\Phi}_{l,tmp}^{UCI} \cup \overline{\Phi}_l^{UCI}(j.d)$;
end for
$\overline{\Phi}_l^{UCI} = \overline{\Phi}_l^{UCI} \{ \overline{\Phi} ¿_{l,tmp}^{UCI}$;
$\overline{\Phi}_l^{UL\text{-}SCH} = \overline{\Phi}_l^{UL\text{-}SCH} \{ \overline{\Phi} ¿_{l,tmp}^{UCI}$;
$\overline{M}_{sc}^{UCI}(l) = \left|\overline{\Phi}_l^{UCI}\right|$;
$\overline{M}_{sc}^{UL\text{-}SCH}(l) = \left|\overline{\Phi}_l^{UL\text{-}SCH}\right|$;
end if
$l = l + 1$;

end while
end for

end if

Step 3:

if CSI is present for transmission on the PUSCH:

Set $m_{\text{count}}^{\text{CSI-part1}}(1)=0$;

Set $m_{\text{count}}^{\text{CSI-part1}}(2)=0$;

Set $m_{\text{count,all}}^{\text{CSI-part1}}=0$;

for $i=1$ to $N_{\text{hop}}^{\text{PUSCH}}$

$l=l_{\text{CSI}}^{(i)}$;

while $\bar{M}_{\text{sc}}^{\text{UCI}}(l)-\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l)\le 0$

$l=l+1$;

end while

while $m_{\text{count}}^{\text{CSI-part1}}(i)<G^{\text{CSI-part1}}(i)$

if $\bar{M}_{\text{sc}}^{\text{UCI}}(l)-\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l)>0$

if $G^{\text{CSI-part1}}(i)-m_{\text{count}}^{\text{CSI-part1}}(i)\ge\left(\bar{M}_{\text{sc}}^{\text{UCI}}(l)-\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l)\right)\cdot N_L\cdot Q_m$

$d=1$;

$m_{\text{count}}^{\text{RE}}=\bar{M}_{\text{sc}}^{\text{UCI}}(l)-\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l)$;

end if

if $G^{\text{CSI-part1}}(i)-m_{\text{count}}^{\text{CSI-part1}}(i)<\left(\bar{M}_{\text{sc}}^{\text{UCI}}(l)-\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l)\right)\cdot N_L\cdot Q_m$

$d=\left\lfloor\left(\bar{M}_{\text{sc}}^{\text{UCI}}(l)-M_{\text{sc, rvd}}^{\bar{\Phi}}(l)\right)\cdot N_L\cdot Q_m\Big/\left(G^{\text{CSI-part1}}(i)-m_{\text{count}}^{\text{CSI-part1}}(i)\right)\right\rfloor$;

$m_{\text{count}}^{\text{RE}}=\left\lceil\left(G^{\text{CSI-part1}}(i)-m_{\text{count}}^{\text{CSI-part1}}(i)\right)/\left(N_L\cdot Q_m\right)\right\rceil$;

end if

$\bar{\Phi}_l^{\text{temp}}=\bar{\Phi}_l^{\text{UCI}}\setminus\bar{\Phi}_l^{\text{rvd}}$;

for $j=0$ to $m_{\text{count}}^{\text{RE}}-1$

$k=\bar{\Phi}_l^{\text{temp}}(j\cdot d)$;

for $v=0$ to $N_L\cdot Q_m-1$

$\bar{g}_{l,k,v}=g_{m_{\text{count, all}}^{\text{CSI-part1}}}^{\text{CSI-part1}}$;

$m_{\text{count,all}}^{\text{CSI-part1}}=m_{\text{count,all}}^{\text{CSI-part1}}+1$;

$m_{\text{count}}^{\text{CSI-part1}}(i)=m_{\text{count}}^{\text{CSI-part1}}(i)+1$;

end for

end for

$\bar{\Phi}_{l,tmp}^{\text{UCI}}=\varnothing$;

for $j=0$ to $m_{\text{count}}^{\text{RE}}-1$

$\bar{\Phi}_{l,tmp}^{\text{UCI}}=\bar{\Phi}_{l,tmp}^{\text{UCI}}\cup\bar{\Phi}_l^{\text{temp}}(j\cdot d)$;

end for

$\bar{\Phi}_l^{\text{UCI}} = \bar{\Phi}_l^{\text{UCI}} \setminus \bar{\Phi}_{l,tmp}^{\text{UCI}}$;
$\bar{\Phi}_l^{\text{UL-SCH}} = \bar{\Phi}_l^{\text{UL-SCH}} \setminus \bar{\Phi}_{l,tmp}^{\text{UCI}}$;
$\bar{M}_{\text{sc}}^{\text{UCI}}(l) = \left|\bar{\Phi}_l^{\text{UCI}}\right|$;
$\bar{M}_{\text{sc}}^{\text{UL-SCH}}(l) = \left|\bar{\Phi}_l^{\text{UL-SCH}}\right|$;
end if
$l = l + 1$;
end while
end for

Set $m_{\text{count}}^{\text{CSI-part2}}(1) = 0$;

Set $m_{\text{count}}^{\text{CSI-part2}}(2) = 0$;

Set $m_{\text{count,all}}^{\text{CSI-part2}} = 0$;

for $i = 1$ to $N_{\text{hop}}^{\text{PUSCH}}$
$l = l_{\text{CSI}}^{(i)}$;
while $\bar{M}_{\text{sc}}^{\text{UCI}}(l) \le 0$
$l = l + 1$;
end while
while $m_{\text{count}}^{\text{CSI-part2}}(i) < G^{\text{CSI-part2}}(i)$
if $\bar{M}_{\text{sc}}^{\text{UCI}}(l) > 0$
if $G^{\text{CSI-part2}}(i) - m_{\text{count}}^{\text{CSI-part2}}(i) \ge \bar{M}_{\text{sc}}^{\text{UCI}}(l) \cdot N_L \cdot Q_m$
$d = 1$;
$m_{\text{count}}^{\text{RE}} = \bar{M}_{\text{sc}}^{\text{UCI}}(l)$;
end if
if $G^{\text{CSI-part2}}(i) - m_{\text{count}}^{\text{CSI-part2}}(i) < \bar{M}_{\text{sc}}^{\text{UCI}}(l) \cdot N_L \cdot Q_m$
$d = \left\lfloor \bar{M}_{\text{sc}}^{\text{UCI}}(l) \cdot N_L \cdot Q_m \Big/ \left(G^{\text{CSI-part2}}(i) - m_{\text{count}}^{\text{CSI-part2}}(i)\right) \right\rfloor$;
$m_{\text{count}}^{\text{RE}} = \left\lceil \left(G^{\text{CSI-part2}}(i) - m_{\text{count}}^{\text{CSI-part2}}(i)\right) / \left(N_L \cdot Q_m\right) \right\rceil$;
end if
for $j = 0$ to $m_{\text{count}}^{\text{RE}} - 1$
$k = \bar{\Phi}_l^{\text{UCI}}(j \cdot d)$;
for $v = 0$ to $N_L \cdot Q_m - 1$
$\bar{g}_{l,k,v} = g_{m_{\text{count, all}}^{\text{CSI-part2}}}$;
$m_{\text{count,all}}^{\text{CSI-part2}} = m_{\text{count,all}}^{\text{CSI-part2}} + 1$;
$m_{\text{count}}^{\text{CSI-part2}}(i) = m_{\text{count}}^{\text{CSI-part2}}(i) + 1$;
end for
end for
$\bar{\Phi}_{l,tmp}^{\text{UCI}} = \varnothing$;
for $j = 0$ to $m_{\text{count}}^{\text{RE}} - 1$

$\bar{\Phi}_{l,tmp}^{\text{UCI}} = \bar{\Phi}_{l,tmp}^{\text{UCI}} \cup \bar{\Phi}_{l}^{\text{UCI}}(j \cdot d)$;
end for
$\bar{\Phi}_{l}^{\text{UCI}} = \bar{\Phi}_{l}^{\text{UCI}} \setminus \bar{\Phi}_{l,tmp}^{\text{UCI}}$;
$\bar{\Phi}_{l}^{\text{UL-SCH}} = \bar{\Phi}_{l}^{\text{UL-SCH}} \setminus \bar{\Phi}_{l,tmp}^{\text{UCI}}$;
$\bar{M}_{\text{sc}}^{\text{UCI}}(l) = \left|\bar{\Phi}_{l}^{\text{UCI}}\right|$;
$\bar{M}_{\text{sc}}^{\text{UL-SCH}}(l) = \left|\bar{\Phi}_{l}^{\text{UL-SCH}}\right|$;
end if
$l = l + 1$;
end while
end for

end if

Step 4:

if UL-SCH is present for transmission on the PUSCH:

Set $m_{\text{count}}^{\text{UL-SCH}} = 0$;
for $l = 0$ to $N_{\text{symb,all}}^{\text{PUSCH}} - 1$
if $\bar{M}_{\text{sc}}^{\text{UL-SCH}}(l) > 0$
for $j = 0$ to $\bar{M}_{\text{sc}}^{\text{UL-SCH}}(l) - 1$
$k = \bar{\Phi}_{l}^{\text{UL-SCH}}(j)$;
for $v = 0$ to $N_L \cdot Q_m - 1$
$\bar{g}_{l,k,v} = g_{m_{\text{count}}^{\text{UL-SCH}}}$;
$m_{\text{count}}^{\text{UL-SCH}} = m_{\text{count}}^{\text{UL-SCH}} + 1$;
end for
end for
end if
end for

end if

Step 5:

if HARQ-ACK is present for transmission on the PUSCH without CG-UCI and the number of HARQ-ACK information bits is no more than 2:

Set $m_{\text{count}}^{\text{ACK}}(1) = 0$;
Set $m_{\text{count}}^{\text{ACK}}(2) = 0$;
Set $m_{\text{count,all}}^{\text{ACK}} = 0$;
for $i = 1$ to $N_{\text{hop}}^{\text{PUSCH}}$
$l = l^{(i)}$;
while $m_{\text{count}}^{\text{ACK}}(i) < G^{\text{ACK}}(i)$
if $\bar{M}_{\text{sc, rvd}}^{\bar{\Phi}}(l) > 0$

if $G^{\mathrm{ACK}}(i) - m_{\mathrm{count}}^{\mathrm{ACK}}(i) \geq \bar{M}_{\mathrm{sc,\,rvd}}^{\bar{\Phi}}(l) \cdot N_L \cdot Q_m$

$d = 1$;

$m_{\mathrm{count}}^{\mathrm{RE}} = \bar{M}_{\mathrm{sc,\,rvd}}^{\bar{\Phi}}(l)$;

end if

if $G^{\mathrm{ACK}}(i) - m_{\mathrm{count}}^{\mathrm{ACK}}(i) < \bar{M}_{\mathrm{sc,\,rvd}}^{\bar{\Phi}}(l) \cdot N_L \cdot Q_m$

$d = \left\lfloor \bar{M}_{\mathrm{sc,\,rvd}}^{\bar{\Phi}}(l) \cdot N_L \cdot Q_m \Big/ \left(G^{\mathrm{ACK}}(i) - m_{\mathrm{count}}^{\mathrm{ACK}}(i)\right) \right\rfloor$;

$m_{\mathrm{count}}^{\mathrm{RE}} = \left\lceil \left(G^{\mathrm{ACK}}(i) - m_{\mathrm{count}}^{\mathrm{ACK}}(i)\right) / \left(N_L \cdot Q_m\right) \right\rceil$;

end if

for $j = 0$ to $m_{\mathrm{count}}^{\mathrm{RE}} - 1$

$k = \bar{\Phi}_l^{\mathrm{rvd}}(j \cdot d)$;

for $v = 0$ to $N_L \cdot Q_m - 1$

$\bar{g}_{l,k,v} = g_{m_{\mathrm{count,\,all}}^{\mathrm{ACK}}}$;

$m_{\mathrm{count,all}}^{\mathrm{ACK}} = m_{\mathrm{count,all}}^{\mathrm{ACK}} + 1$;

$m_{\mathrm{count}}^{\mathrm{ACK}}(i) = m_{\mathrm{count}}^{\mathrm{ACK}}(i) + 1$;

end for

end for

end if

$l = l + 1$;

end while

end for

end if

Step 6:

Set $t = 0$;

for $l = 0$ to $N_{\mathrm{symb,all}}^{\mathrm{PUSCH}} - 1$

for $j = 0$ to $M_{\mathrm{sc}}^{\mathrm{UL\text{-}SCH}}(l) - 1$

$k = \Phi_l^{\mathrm{UL\text{-}SCH}}(j)$;

for $v = 0$ to $N_L \cdot Q_m - 1$

$g_t = \bar{g}_{l,k,v}$;

$t = t + 1$;

end for

end for

end for

6.3 Uplink control information

6.3.1 Uplink control information on PUCCH

The procedure in this clause applies to PUCCH formats 2/3/4.

The following clauses 6.3.1.2, 6.3.1.3 and 6.3.1.5 apply regardless of whether the higher layer parameter *uci-MuxWithDiffPrio* is configured or not. The following clauses 6.3.1.1, 6.3.1.4 and 6.3.1.6 apply by assuming *uci-*

MuxWithDiffPrio is not configured, or *uci-MuxWithDiffPrio* is configured and the UCIs for transmission on a PUCCH are of the same priority index, unless stated otherwise.

If the UE is configured with a PUCCH-SCell, *uci-MuxWithDiffPrio* is replaced by *uci-MuxWithDiffPrioSecondaryPUCCHgroup* for the secondary PUCCH group in this clause.

If UEIRI is transmitted on a PUCCH, the procedure in this clause 6.3.1 applies by replacing SR with UEIRI in all the notations and texts, when applicable.

If joint indication information for SR and UEIRI as given by Clause 9.2.5.1 of [5, TS38.213] is transmitted on a PUCCH, the procedure in this clause 6.3.1 applies assuming SR representing the joint indication information for SR and UEIRI in all the notations and texts, when applicable.

6.3.1.1 UCI bit sequence generation

6.3.1.1.1 HARQ-ACK/SR only

If only HARQ-ACK bits are transmitted on a PUCCH, the UCI bit sequence $a_0, a_1, a_2, a_3, ..., a_{A-1}$ is determined by setting $a_i = \tilde{o}_i^{ACK}$ for $i = 0, 1, ..., O^{\text{ACK}} - 1$ and $A = O^{\text{ACK}}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK}, \tilde{o}_1^{ACK}, ..., \tilde{o}_{O^{ACK}-1}^{ACK}$ is given by Clause 9.1 of [5, TS38.213].

If only HARQ-ACK and SR bits are transmitted on a PUCCH, the UCI bit sequence $a_0, a_1, a_2, a_3, ..., a_{A-1}$ is determined by setting $a_i = \tilde{o}_i^{ACK}$ for $i = 0, 1, ..., O^{\text{ACK}} - 1$, $a_i = \tilde{o}_{i-O^{ACK}}^{SR}$ for $i = O^{\text{ACK}}, O^{\text{ACK}} + 1, ..., O^{\text{ACK}} + O^{\text{SR}} - 1$, and $A = O^{\text{ACK}} + O^{\text{SR}}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK}, \tilde{o}_1^{ACK}, ..., \tilde{o}_{O^{ACK}-1}^{ACK}$ is given by Clause 9.1 of [5, TS 38.213], and the SR bit sequence $\tilde{o}_0^{SR}, \tilde{o}_1^{SR}, ..., \tilde{o}_{O^{\text{SR}}-1}^{SR}$ is given by Clause 9.2.5.1 of [5, TS 38.213].

6.3.1.1.2 CSI only

If *cqi-BitsPerSubband* is configured, this Clause 6.3.1.1.2 applies by taking Subband CQI as Subband differential CQI and replacing the corresponding number of bits 2 by 4.

If *CSI-ReportSubConfig* is configured, for a corresponding CSI sub-report, the bitwidth of a CSI field of the CSI sub-report is determined following the procedure in this clause 6.3.1.1.2 by taking configurations in *CSI-ReportSubConfig* when applicable. If *CSI-ReportSubConfig* configures a list of CSI-RS resource IDs, for the determination of the bitwidth of a CRI field, the value of K_s^{CSI-RS} is the number of CSI-RS resources configured in the corresponding *CSI-ReportSubConfig*.

The bitwidth for PMI of *codebookType=typeI-SinglePanel* with 2 CSI-RS ports is 2 for Rank=1 and 1 for Rank=2, according to Clause 5.2.2.2.1 in [6, TS 38.214].

The bitwidth for PMI of *codebookType=typeI-SinglePanel* with more than 2 CSI-RS ports is provided in Table 6.3.1.1.2-1, where the values of (N_1, N_2) and (O_1, O_2) are given by Clause 5.2.2.2.1 in [6, TS 38.214].

Table 6.3.1.1.2-1: PMI of *codebookType=typeI-SinglePanel*

	Information field X_1 for wideband PMI			Information field X_2 for wideband PMI or per subband PMI	
	$(i_{1,1}, i_{1,2})$		$i_{1,3}$	i_2	
	***codebookMode*=1**	***codebookMode*=2**		***codebookMode*=1**	***codebookMode*=2**
Rank = 1 with >2 CSI-RS ports, $N_2>1$	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$	$\left(\left\lceil \log_2 \frac{N_1O_1}{2} \right\rceil, \left\lceil \log_2 \frac{N_2O_2}{2} \right\rceil\right)$	N/A	2	4
Rank = 1 with >2 CSI-RS ports, $N_2=1$	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$	$\left(\left\lceil \log_2\left(\frac{N_1O_1}{2}\right) \right\rceil, 0\right)$	N/A	2	4
Rank=2 with 4 CSI-RS ports, $N_2=1$	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$	$\left(\left\lceil \log_2\left(\frac{N_1O_1}{2}\right) \right\rceil, 0\right)$	1	1	3
Rank=2 with >4 CSI-RS ports, $N_2>1$	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$	$\left(\left\lceil \log_2 \frac{N_1O_1}{2} \right\rceil, \left\lceil \log_2 \frac{N_2O_2}{2} \right\rceil\right)$	2	1	3
Rank=2 with >4 CSI-RS ports, $N_2=1$	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$	$\left(\left\lceil \log_2\left(\frac{N_1O_1}{2}\right) \right\rceil, 0\right)$	2	1	3
Rank=3 or 4, with 4 CSI-RS ports	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$		0	1	
Rank=3 or 4, with 8 or 12 CSI-RS ports	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$		2	1	
Rank=3 or 4 , with >=16 CSI-RS ports	$\left(\left\lceil \log_2 \frac{N_1O_1}{2} \right\rceil, \lceil \log_2 N_2O_2 \rceil\right)$		2	1	
Rank=5 or 6	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$		N/A	1	
Rank=7 or 8, $N_1=4, N_2=1$	$\left(\left\lceil \log_2 \frac{N_1O_1}{2} \right\rceil, \lceil \log_2 N_2O_2 \rceil\right)$		N/A	1	
Rank=7 or 8, $N_1>2, N_2=2$	$\left(\lceil \log_2 N_1O_1 \rceil, \left\lceil \log_2 \frac{N_2O_2}{2} \right\rceil\right)$		N/A	1	
Rank=7 or 8, with $N_1>4, N_2=1$ or $N_1=2, N_2=2$ or $N_1>2, N_2>2$	$\left(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil\right)$		N/A	1	

The bitwidth for PMI of *codebookType=typeI-SinglePanel-r19* is provided in Table 6.3.1.1.2-1A/1B for *typeI-codebookMode=ModeA* and *typeI-codebookMode=ModeB* respectively, where the values of (N_1, N_2) and (O_1, O_2) are given by Clause 5.2.2.2.1a in [6, TS 38.214].

Table 6.3.1.1.2-1A: PMI of codebookType=typeI-SinglePanel-r19 and typeI-codebookMode=ModeA

	Information field X_1 for wideband PMI					Information field X_2 for wideband PMI or per subband PMI
	$(i_{1,1}, i_{1,2})$	$i_{1,3}$	$(i_{1,1,2}, i_{1,2,2})$	$(i_{1,1,3}, i_{1,2,3})$	$(i_{1,1,4}, i_{1,2,4})$	i_2
Rank = 1	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$	N/A	N/A	N/A	N/A	2
Rank=2, 3 or 4	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$	2	N/A	N/A	N/A	1
Rank=5 or 6,	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$	1	If $i_{1,3}=0$, $(\lceil \log_2 N_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$; If $i_{1,3}=1$, $(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 \rceil)$.	If $i_{1,3}=0$, $(\lceil \log_2 N_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$; If $i_{1,3}=1$, $(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 \rceil)$.	N/A	1
Rank=7 or 8	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$	1	If $i_{1,3}=0$, $(\lceil \log_2 N_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$; If $i_{1,3}=1$, $(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 \rceil)$.	If $i_{1,3}=0$, $(\lceil \log_2 N_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$; If $i_{1,3}=1$, $(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 \rceil)$.	If $i_{1,3}=0$, $(\lceil \log_2 N_1 \rceil, \lceil \log_2 N_2 O_2 \rceil)$; If $i_{1,3}=1$, $(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 \rceil)$.	1

Table 6.3.1.1.2-1B: PMI of codebookType=typeI-SinglePanel-r19 and typeI-codebookMode=ModeB

	Information field X_1 for wideband PMI						Information field X_2 for wideband PMI or per subband PMI			
	$i_{1,1}$	$i_{1,2,1}$	$i_{1,2,2}$	$i_{1,2,3}$	$i_{1,2,4}$	$i_{1,2}$	$i_{2,1}$	$i_{2,2}$	$i_{2,3}$	$i_{2,4}$

Rank=1	$\lceil \log_2(O_1$	$\lceil \log_2(N_1$	N/A	N/A	N/A	N/A	2	N/A	N/A	N/A
Rank=2	$\lceil \log_2(O_1$	$\lceil \log_2(N_1$	$\lceil \log_2(N_1$	N/A	N/A	N/A	2	2	N/A	N/A
Rank=3	$\lceil \log_2(O_1$	$\lceil \log_2(N_1$	$\lceil \log_2(N_1$	$\lceil \log_2(N_1$	N/A	N/A	2	2	2	N/A
Rank= 4	$\lceil \log_2(O_1$	$\lceil \log_2(N_1$	$\lceil \log_2(N_1$	$\lceil \log_2(N_1$	$\lceil \log_2(N_1$	N/A	2	2	2	2
Rank=5	$\lceil \log_2(O_1$	N/A	N/A	N/A	N/A	$\lceil \log_2\left(\frac{N_1}{3}\right.$	1	1	2	N/A
Rank=6	$\lceil \log_2(O_1$	N/A	N/A	N/A	N/A	$\lceil \log_2\left(\frac{N_1}{3}\right.$	1	1	1	N/A
Rank=7	$\lceil \log_2(O_1$	N/A	N/A	N/A	N/A	$\lceil \log_2\left(\frac{N_1}{4}\right.$	1	2	1	1
Rank=8	$\lceil \log_2(O_1$	N/A	N/A	N/A	N/A	$\lceil \log_2\left(\frac{N_1}{4}\right.$	1	1	1	1

The bitwidth for PMI of *codebookType= typeI-MultiPanel* is provided in Table 6.3.1.1.2-2, where the values of (N_g, N_1, N_2) and (O_1, O_2) are given by Clause 5.2.2.2.2 in [6, TS 38.214].

Table 6.3.1.1.2-2: PMI of *codebookType= typeI-MultiPanel*

	Information fields X_1 for wideband					Information fields X_2 for wideband or per subband			
	$(i_{1,1}, i_{1,2})$	$i_{1,3}$	$i_{1,4,1}$	$i_{1,4,2}$	$i_{1,4,3}$	i_2	$i_{2,0}$	$i_{2,1}$	$i_{2,2}$
Rank=1 with $N_g=2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	N/A	2	N/A	N/A	2	N/A	N/A	N/A
Rank=1 with $N_g=4$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	N/A	2	2	2	2	N/A	N/A	N/A
Rank=2 with $N_g=2$, $N_1N_2=2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	1	2	N/A	N/A	1	N/A	N/A	N/A
Rank=3 or 4 with $N_g=2$, $N_1N_2=2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	0	2	N/A	N/A	1	N/A	N/A	N/A
Rank=2 or 3 or 4 with $N_g=2$, $N_1N_2>2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	2	2	N/A	N/A	1	N/A	N/A	N/A
Rank=2 with $N_g=4$, $N_1N_2=2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	1	2	2	2	1	N/A	N/A	N/A
Rank=3 or 4 with $N_g=4$, $N_1N_2=2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	0	2	2	2	1	N/A	N/A	N/A
Rank=2 or 3 or 4 with $N_g=4$, $N_1N_2>2$ *codebookMode=1*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	2	2	2	2	1	N/A	N/A	N/A
Rank=1 with $N_g=2$ *codebookMode=2*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	N/A	2	2	N/A	N/A	2	1	1
Rank=2 with $N_g=2$, $N_1N_2=2$ *codebookMode=2*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	1	2	2	N/A	N/A	1	1	1
Rank=3 or 4 with $N_g=2$, $N_1N_2=2$ *codebookMode=2*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	0	2	2	N/A	N/A	1	1	1
Rank=2 or 3 or 4 with $N_g=2$, $N_1N_2>2$ *codebookMode=2*	$(\lceil \log_2 N_1O_1 \rceil, \lceil \log_2 N_2O_2 \rceil)$	2	2	2	N/A	N/A	1	1	1

The bitwidth for PMI of *codebookType= typeI-MultiPanel-r19* is provided in Table 6.3.1.1.2-2A, where the values of $(N_g,\ N_1,\ N_2)$ and $(O_1,\ O_2)$ are given by Clause 5.2.2.2.2a in [6, TS 38.214].

Table 6.3.1.1.2-2A: PMI of codebookType= typeI-MultiPanel-r19

	Information fields X_1 for wideband											Information fields X_2 for wideband or per subband			
	$(i_{1,1,1},\ i_{1,2,1})$	$i_{1,3,1}$	$(i_{1,1,2},\ i_{1,2,2})$	$i_{1,3,2}$	$(i_{1,1,3},\ i_{1,2,3})$	$i_{1,3,3}$	$(i_{1,1,4},\ i_{1,2,4})$	$i_{1,3,4}$	$i_{1,4,1}$	$i_{1,4,2}$	$i_{1,4,3}$	$i_{2,1}$	$i_{2,2}$	$i_{2,3}$	$i_{2,4}$
Rank=1 with $N_g=2$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	N/A	N/A	N/A	N/A	2	N/A	N/A	2	2	N/A	N/A
Rank=1 with $N_g=4$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	2	2	2	2	2	2	2
Rank=2 with $N_g=2$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	N/A	N/A	N/A	N/A	2	N/A	N/A	1	1	N/A	N/A
Rank=2 with $N_g=4$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	2	2	2	1	1	1	1
Rank=3 with $N_g=2$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	N/A	N/A	N/A	N/A	2	N/A	N/A	1	1	N/A	N/A
Rank=3 with $N_g=4$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	2	2	2	1	1	1	1
Rank=4 with $N_g=2$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	N/A	N/A	N/A	N/A	2	N/A	N/A	1	1	N/A	N/A
Rank=4 with $N_g=4$	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	$(\lceil \log_2 N_1 O_1 \rceil, \lceil \log_2 N_2 O \rceil)$	2	2	2	2	1	1	1	1

The bitwidth for PMI with 1 CSI-RS port is 0.

The bitwidth for RI/LI/CQI/CRI of *codebookType=typeI-SinglePanel* or *reportQuantity* set to 'cri-RI-CQI' or 1 CSI-RS port is provided in Table 6.3.1.1.2-3.

Table 6.3.1.1.2-3: RI, LI, CQI, and CRI of *codebookType=typeI-SinglePanel*, or *reportQuantity* set to 'cri-RI-CQI', or 1 CSI-RS port

Field	Bitwidth				
	1 antenna port	2 antenna ports	4 antenna ports	>4 antenna ports	
				Rank1~4	Rank5~8
Rank Indicator when *codebookType=typeI-SinglePanel*	0	$\min\left(1,\lceil \log_2 n_{\mathrm{RI}} \rceil\right)$	$\min\left(2,\lceil \log_2 n_{\mathrm{RI}} \rceil\right)$	$\lceil \log_2 n_{\mathrm{RI}} \rceil$	$\lceil \log_2 n_{\mathrm{RI}} \rceil$
Rank Indicator when *reportQuantity* set to 'cri-RI-CQI'	0	1	2	3	3
Layer Indicator	0	$\lceil \log_2 \upsilon \rceil$	$\min\left(2,\lceil \log_2 \upsilon \rceil\right)$	$\min\left(2,\lceil \log_2 \upsilon \rceil\right)$	$\min\left(2,\lceil \log_2 \upsilon \rceil\right)$
Wide-band CQI for the first TB	4	4	4	4	4
Wideband CQI for the second TB	0	0	0	0	4
Subband differential CQI for the first TB	2	2	2	2	2
Subband differential CQI for the second TB	0	0	0	0	2
CRI	$\lceil \log_2\left(K_s^{\mathrm{CSI-RS}}\right) \rceil$	$\lceil \log_2\left(K_s^{\mathrm{CSI-RS}}\right) \rceil$	$\lceil \log_2\left(K_s^{\mathrm{CSI-RS}}\right) \rceil$	$\lceil \log_2\left(K_s^{\mathrm{CSI-RS}}\right) \rceil$	$\lceil \log_2\left(K_s^{\mathrm{CSI-RS}}\right) \rceil$

n_{RI} in Table 6.3.1.1.2-3 is the number of allowed rank indicator values according to Clause 5.2.2.2.1 [6, TS 38.214]. υ is the value of the rank. The value of $K_s^{\mathrm{CSI-RS}}$ is the number of CSI-RS resources in the corresponding resource set. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. For higher layer parameter *reportQuantity* set to 'cri-RI-CQI', the values of the rank indicator field are mapped to rank indicator values with increasing order, where '0' is mapped to rank-1.

Table 6.3.1.1.2-3A: RI, LI, CQI, and CRI associated with one CSI-RS resource pair and *csi-ReportMode= Mode 1* or *Mode 2*

Field	Bitwidth	
	1 antenna port per Resource	> 1 antenna ports per Resource
Rank Combination Indicator	0	$min\left(2,\lceil \log_2 n_{RI,NC} \right.$
The first Layer Indicator	0	$\lceil \log_2\left(v_1\right) \rceil$
The second Layer Indicator	0	$\lceil \log_2\left(v_2\right) \rceil$
Wide-band CQI for the first TB	4	4
Subband differential CQI for the first TB	2	2
CRI if *csi-ReportMode= Mode 1*	$\lceil \log_2 N \rceil$	$\lceil \log_2 N \rceil$
CRI if *csi-ReportMode= Mode 2*	$\lceil \log_2\left(M_1+M_2+N\right) \rceil$	$\lceil \log_2\left(M_1+M_2+N\right) \rceil$

Table 6.3.1.1.2-3B: RI, LI, CQI, and CRI associated with one CSI-RS resource and *csi-ReportMode= Mode 1 or Mode 2*

Field	Bitwidth				
	1 antenna port	2 antenna ports	4 antenna ports	>4 antenna ports	
				Rank1~4	Rank5~8
Rank Indicator	0	$min(1, \lceil \log_2 n$	$min(2, \lceil \log_2 n$	$\lceil \log_2 n_{RI,sTRP} \rceil$	$\lceil \log_2 n_{RI,sTRP} \rceil$
Layer Indicator	0	$\lceil \log_2(v) \rceil$	$min(2, \lceil \log_2($	$min(2, \lceil \log_2($	$min(2, \lceil \log_2($
Wide-band CQI for the first TB	4	4	4	4	4
Wideband CQI for the second TB	0	0	0	0	4
Subband differential CQI for the first TB	2	2	2	2	2
Subband differential CQI for the second TB	0	0	0	0	2
CRI if *csi-ReportMode= Mode 1* and *numberOfSingleTRP-CSI-Mode1 = 1*	$\lceil \log_2(M_1+M_2) \rceil$	$\lceil \log_2(M_1+M_2) \rceil$	$\lceil \log_2(M_1+M_2) \rceil$	$\lceil \log_2(M_1+M_2) \rceil$	$\lceil \log_2(M_1+M_2) \rceil$
CRI if *csi-ReportMode= Mode 1* and *numberOfSingleTRP-CSI-Mode1 = 2*	$\lceil \log_2(M_1) \rceil$ for the first CRI; $\lceil \log_2(M_2) \rceil$ for the second CRI	$\lceil \log_2(M_1) \rceil$ for the first CRI; $\lceil \log_2(M_2) \rceil$ for the second CRI	$\lceil \log_2(M_1) \rceil$ for the first CRI; $\lceil \log_2(M_2) \rceil$ for the second CRI	$\lceil \log_2(M_1) \rceil$ for the first CRI; $\lceil \log_2(M_2) \rceil$ for the second CRI	$\lceil \log_2(M_1) \rceil$ for the first CRI; $\lceil \log_2(M_2) \rceil$ for the second CRI
CRI if *csi-ReportMode= Mode 2*	$\lceil \log_2(M_1+M_2+$	$\lceil \log_2(M_1+M_2+$	$\lceil \log_2(M_1+M_2+$	$\lceil \log_2(M_1+M_2+$	$\lceil \log_2(M_1+M_2+$

$n_{RI,NCJT}$ in Table 6.3.1.1.2-3A is the number of allowed rank combination indicator values associated with one CSI-RS resource pair according to Clause 5.2.1.4.2 [6, TS 38.214]. The values of the rank combination indicator field are mapped to allowed rank combinations in the following order: {1,1}, {1,2}, {2,1},{2,2}, where '0' is mapped to the first allowed rank combination. v_1 and v_2 are the values of the first and the second rank associated with two CSI-RS resources of the CSI-RS resource pair respectively.

$n_{RI,sTRP}$ in Table 6.3.1.1.2-3B is the number of allowed rank indicator values associated with one CSI-RS resource according to Clause 5.2.1.4.2 [6, TS 38.214]. v is the value of the rank associated with the CSI-RS resource. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value.

The value of N in Table 6.3.1.1.2-3A and Table 6.3.1.1.2-3B is the number of CSI-RS resource pairs configured within a CSI-RS resource set. The values of M_1 and M_2 in Table 6.3.1.1.2-3A and Table 6.3.1.1.2-3B are given by

- If *sharedCMR* = "Enabled", $M_1 = K_1$ and $M_2 = K_2$
- If *sharedCMR* is absent and $N = 1$, $M_1 = K_1 - 1$ and $M_2 = K_2 - 1$
- If *sharedCMR* is absent and $N = 2$,
  - $M_1 = K_1 - 2$ and $M_2 = K_2 - 2$, if the two resource pairs do not share any CSI-RS resource
  - $M_1 = K_1 - 1$ and $M_2 = K_2 - 2$, if the two resource pairs share the same CSI-RS resource from the first CSI-RS resource group
  - $M_1 = K_1 - 2$ and $M_2 = K_2 - 1$, if the two resource pairs share the same CSI-RS resource from the second CSI-RS resource group

where the values of K_1 and K_2 are the numbers of CSI-RS resources in the first and second CSI-RS resource groups within the CSI-RS resource set respectively.

The bitwidth for RI/LI/CQI/CRI of *codebookType=typeI-SinglePanel-r19* is provided in Table 6.3.1.1.2-3C.

Table 6.3.1.1.2-3C: RI, LI, CQI, and CRI of *codebookType=typeI-SinglePanel-r19*

Field	Bitwidth	
	Rank1~4	Rank5~8
Rank Indicator	$\lceil \log_2 n_{RI} \rceil$	$\lceil \log_2 n_{RI} \rceil$
Layer Indicator	$min(2, \lceil \log_2 \upsilon$	$min(2, \lceil \log_2 \upsilon \rceil)$
Wide-band CQI for the first TB	4	4
Wideband CQI for the second TB	0	4
Subband differential CQI for the first TB	2	2
Subband differential CQI for the second TB	0	2
CRI	0	0
NOTE: n_{RI} is the number of allowed rank indicator values according to Clause 5.2.2.2.1a in [6, TS 38.214]. υ is the value of the rank. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value.		

The bitwidth for RI/LI/CQI/CRI of *codebookType= typeI-MultiPanel* or *codebookType=typeI-MultiPanel-r19* is provided in Table 6.3.1.1.2-4.

Table 6.3.1.1.2-4: RI, LI, CQI, and CRI of codebookType=typeI-MultiPanel or codebookType=typeI-MultiPanel-r19

Field	Bitwidth
Rank Indicator	$\min(2, \lceil \log_2 n_{\mathrm{RI}} \rceil)$
Layer Indicator	$\min(2, \lceil \log_2 \upsilon \rceil)$
Wide-band CQI	4
Subband differential CQI	2
CRI if *codebookType=typeI-MultiPanel*	$\lceil \log_2 \left(K_s^{\mathrm{CSI-RS}} \right) \rceil$
CRI if *codebookType=typeI-MultiPanel-r19*	0
NOTE: n_{RI} is the number of allowed rank indicator values according to Clause 5.2.2.2.2 and 5.2.2.2.2a in [6, TS 38.214], υ is the value of the rank, and K_s^{CSI-RS} is the number of CSI-RS resources in the corresponding resource set. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value.	

The bitwidth for RI/LI/CQI of *codebookType= typeII* or *codebookType=typeII-PortSelection* is provided in Table 6.3.1.1.2-5.

Table 6.3.1.1.2-5: RI, LI, and CQI of *codebookType=typeII or typeII-PortSelection*

Field	Bitwidth
Rank Indicator	$\min\left(1,\lceil \log_2 n_{\mathrm{RI}} \rceil\right)$
Layer Indicator	$\min\left(2,\lceil \log_2 \upsilon \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
Indicator of the number of non-zero wideband amplitude coefficients M_l for layer l	$\lceil \log_2(2L-1) \rceil$
NOTE: n_{RI} is the number of allowed rank indicator values according to Clauses 5.2.2.2.3 and 5.2.2.2.4 [6, TS 38.214] and υ is the value of the rank. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value.	

The bitwidth for CRI, SSBRI, RSRP, differential RSRP, and CapabilityIndex are provided in Table 6.3.1.1.2-6.

Table 6.3.1.1.2-6: CRI, SSBRI, RSRP, and CapabilityIndex

Field	Bitwidth
CRI	$\lceil \log_2\left(K_s^{\mathrm{CSI-RS}}\right) \rceil$
SSBRI	$\lceil \log_2\left(K_s^{\mathrm{SSB}}\right) \rceil$
RSRP	7
Differential RSRP	4
CapabilityIndex	2
NOTE: K_s^{CSI-RS} is the number of CSI-RS resources in the corresponding resource set, K_s^{SSB} is the configured number of SS/PBCH blocks in the corresponding resource set for reporting 'ssb-Index-RSRP' or 'ssb-Index-RSRP-Index'.	

The bitwidth for CRI, SSBRI, SINR, differential SINR, and CapabilityIndex are provided in Table 6.3.1.1.2-6A.

Table 6.3.1.1.2-6A: CRI, SSBRI, SINR, and CapabilityIndex

Field	Bitwidth
CRI	$\lceil \log_2\left(K_s^{CSI-RS}\right) \rceil$
SSBRI	$\lceil \log_2\left(K_s^{SSB}\right) \rceil$
SINR	7
Differential SINR	4
CapabilityIndex	2
NOTE: K_s^{CSI-RS} is the number of CSI-RS resources in the corresponding resource set, and K_s^{SSB} is the configured number of SS/PBCH blocks in the corresponding resource set for reporting 'ssb-Index-SINR' or 'ssb-Index-SINR-Index'.	

The bitwidth for MRI and CLI-RSSI are provided in Table 6.3.1.1.2-6B.

Table 6.3.1.1.2-6B: Bitwidth for MRI and CLI-RSSI

Field	Bitwidth
MRI	$\lceil \log_2\left(K_s^{CLI-RSSI}\right) \rceil$
CLI-RSSI	7
Differential CLI-RSSI	4
NOTE: $K_s^{CLI-RSSI}$ is the number of CLI-RSSI measurement resources in the corresponding resource set.	

The bitwidth for predicted CRI, predicted SSBRI, predicted RSRP, differential predicted RSRP are provided in Table 6.3.1.1.2-6C.

Table 6.3.1.1.2-6C: Predicted CRI, Predicted SSBRI, Predicted RSRP and Differential predicted RSRP

Field	Bitwidth
Predicted CRI	$\lceil \log_2\left(K_s^{CSI-RS}\right) \rceil$
Predicted SSBRI	$\lceil \log_2\left(K_s^{SSB}\right) \rceil$
Predicted RSRP	7
Differential predicted RSRP	4
Time instance indicator	$\lceil \log_2\left(N\right) \rceil$
NOTE: K_s^{CSI-RS} is the number of CSI-RS resources in the resource set configured by *resourcesForSetA*, K_s^{SSB} is the number of SS/PBCH blocks in the resource set configured by *resourcesForSetA,* and N is the number of predicted time instances configured by *nroftimeinstance.*	

The bitwidth for RS-PAI is provided in Table 6.3.1.1.2-6D.

Table 6.3.1.1.2-6D: RS-PAI

Field	Bitwidth
RS-PAI	$\lceil \log_2\left(N^{Mon}+1\right) \rceil$
NOTE: N^{Mon} is the number of transmission occasion(s) of monitoring resources configured by *nroftransmissionOccasion.*	

If *CSI-ReportSubConfig* is configured, for a corresponding CSI sub-report, the mapping order of CSI fields of one CSI sub-report is determined following the procedure in this clause 6.3.1.1.2, by replacing CSI report #n in the following Tables 6.3.1.1.2-7, 6.3.1.1.2-9 and 6.3.1.1.2-10 with CSI sub-report #n, and taking only Tables 6.3.1.1.2-1/1A/1B/2/2A/3/4 for the determination of the bitwidth of a CSI field.

Table 6.3.1.1.2-7: Mapping order of CSI fields of one CSI report, *pmi-FormatIndicator=widebandPMI* and *cqi-FormatIndicator=widebandCQI* or *reportQuantity* set to 'cri-RI-CQI' and *cqi-FormatIndicator=widebandCQI*

CSI report number	CSI fields
CSI report #n	CRI as in Tables 6.3.1.1.2-3/4, if reported
	Rank Indicator as in Tables 6.3.1.1.2-3/3C/4, if reported
	Layer Indicator as in Tables 6.3.1.1.2-3/3C/4, if reported
	Zero padding bits O_P, if needed
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if reported
	Wideband CQI for the first TB as in Tables 6.3.1.1.2-3/3C/4, if reported
	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3/3C/4, if reported

The number of zero padding bits O_P in Table 6.3.1.1.2-7 is 0 for 1 CSI-RS port and $O_P = N_{\max} - N_{\text{reported}}$ for more than 1 CSI-RS port, where:

- $N_{\max} = \max_{r \in S_{\text{Rank}}} B(r)$ and S_{Rank} is the set of rank values r that are allowed to be reported;

- $N_{\text{reported}} = B(R)$, where R is the reported rank;
- For 2 CSI-RS ports, $B(r) = N_{\text{PMI}}(r) + N_{\text{CQI}}(r) + N_{\text{LI}}(r)$;
- For more than 2 CSI-RS ports, $B(r) = N_{\text{PMI,i1}}(r) + N_{\text{PMI,i2}}(r) + N_{\text{CQI}}(r) + N_{\text{LI}}(r)$;
- if PMI is reported, $N_{\text{PMI}}(1) = 2$ and $N_{\text{PMI}}(2) = 1$; otherwise, $N_{\text{PMI}}(r) = 0$;
- if PMI $i1$ is reported, $N_{\text{PMI,i1}}(r)$ is obtained according to Tables 6.3.1.1.2-1/1A/1B/2/2A; otherwise, $N_{\text{PMI,i1}}(r) = 0$;
- if PMI $i2$ is reported, $N_{\text{PMI,i2}}(r)$ is obtained according to Tables 6.3.1.1.2-1/1A/1B/2/2A; otherwise, $N_{\text{PMI,i2}}(r) = 0$;
- if CQI is reported, $N_{\text{CQI}}(r)$ is obtained according to Tables 6.3.1.1.2-3/3C/4; otherwise, $N_{\text{CQI}}(r) = 0$;
- if LI is reported, $N_{\text{LI}}(r)$ is obtained according to Tables 6.3.1.1.2-3/3C/4; otherwise, $N_{\text{LI}}(r) = 0$.

Table 6.3.1.1.2-7A: Mapping order of CSI fields of one CSI report, *pmi-FormatIndicator=widebandPMI*, *cqi-FormatIndicator=widebandCQI, csi-ReportMode= Mode 1* and *numberOfSingleTRP-CSI-Mode1=0*

CSI report number	CSI fields
CSI report #n	CRI as in Tables 6.3.1.1.2-3A, if reported
	Rank Combination Indicator as in Tables 6.3.1.1.2-3A, if reported
	Two Layer Indicators as in Table 6.3.1.1.2-3A, where the first Layer Indicator and the second Layer Indicator are associated with the first resource and the second resource within the resource pair respectively and if reported;
	Zero padding bits O_P, if needed
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the first resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the first CSI-RS resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the second resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the second CSI-RS resource within the CSI-RS resource pair, if reported
	Wideband CQI for the first TB as in Tables 6.3.1.1.2-3A, if reported

The number of zero padding bits O_P in Table 6.3.1.1.2-7A is 0 for 1 CSI-RS port and $O_P = N_{max} - N_{reported}$ for more than 1 CSI-RS port, where:

- $N_{max} = \underset{r \in S_{Rank}}{max}\, B(r)$ and S_{Rank} is the set of rank combination values of $r = \{r_1, r_2\}$ that are allowed to be reported;
- $N_{reported} = B(R)$ where R is the reported rank combination;
- For 2 CSI-RS ports, $B(r) = N_{PMI}(r_1) + N_{PMI}(r_2) + N_{CQI}(r) + N_{LI}(r_1) + N_{LI}(r_2)$;
- For more than 2 CSI-RS ports, $B(r) = N_{PMI,i_1}(r_1) + N_{PMI,i_1}(r_2) + N_{PMI,i_2}(r_1) + N_{PMI,i_2}(r_2) + N_{CQI}(r) + N_{LI}(r_1) + N_{LI}(r_2)$;
- if PMI is reported, $N_{PMI}(1) = 2$ and $N_{PMI}(2) = 1$; otherwise, $N_{PMI} = 0$;

- if PMI i_1 is reported, $N_{PMI,i_1}(r_1)$ and $N_{PMI,i_1}(r_2)$ are obtained according to Tables 6.3.1.1.2-1; otherwise, $N_{PMI,i_1}=0$;
- if PMI i_2 is reported, $N_{PMI,i_2}(r_1)$ and $N_{PMI,i_2}(r_2)$ are obtained according to Tables 6.3.1.1.2-1; otherwise, $N_{PMI,i_2}=0$;
- if CQI is reported, $N_{CQI}(r)$ is obtained according to Tables 6.3.1.1.2-3A; otherwise, $N_{CQI}(r)=0$;
- if LI is reported, $N_{LI}(r_1)$ and $N_{LI}(r_2)$ are obtained according to Tables 6.3.1.1.2-3A; otherwise , $N_{LI}=0$.

Table 6.3.1.1.2-7B: Mapping order of CSI fields of one CSI report configured with *valueOfM*, *pmi-FormatIndicator=widebandPMI* and *cqi-FormatIndicator=widebandCQI*

CSI report number	CSI fields
CSI report #n	The first reported CRI as in Tables 6.3.1.1.2-3, if reported
	Rank Indicator for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	Layer Indicator for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	Zero padding bits $O_{P,1}$ for the first reported CRI, if needed
	PMI wideband information fields X_1 for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the first reported CRI according to Clause 5.2.2.2.1 in [6, TS38.214], if reported
	Wideband CQI for the first TB for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	Wideband CQI for the second TB for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	…
	The *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	Rank Indicator for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	Layer Indicator for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	Zero padding bits $O_{P,M}$ for the *M*-th reported CRI, if needed
	PMI wideband information fields X_1 for the *M*-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the *M*-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the *M*-th reported CRI according to Clause 5.2.2.2.1 in [6, TS38.214], if reported
	Wideband CQI for the first TB for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	Wideband CQI for the second TB for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported

The number of zero padding bits $O_{P,k}$ for *k*-th reported CRI in Table 6.3.1.1.2-7B is 0 for 1 CSI-RS port and $O_{P,k}=N_{max}-N_{reported,k}$ for more than 1 CSI-RS port, where:

- $N_{max}=\max_{j\in Q}(N_{max,j})$, where Q is the set of CRIs corresponding to the K_s resources and $N_{max,j}$ is the maximum payload size of associated CSI fields for the *j*-th CRI;
- $N_{max,j}=N_{RI}(j)+\max_{r_j\in S_{rank,j}} B(r_j)$, where $N_{RI}(j)$ is the payload size of RI field for the *j*-th CRI, and $S_{rank,j}$ is the set of rank values r_j that are allowed to be reported for the *j*-th CRI obtained according to Table 6.3.1.1.2-3;
- $N_{reported,k}=N_{RI}(k)+B(R_k)$, where R_k is the reported rank for the *k*-th CRI;
- For 2 CSI-RS ports, $B(r)=N_{\mathrm{PMI}}(r)+N_{\mathrm{CQI}}(r)+N_{\mathrm{LI}}(r)$;

- For more than 2 CSI-RS ports, $B(r)=N_{\mathrm{PMI},i_1}(r)+N_{\mathrm{PMI},i_2}(r)+N_{\mathrm{CQI}}(r)+N_{\mathrm{LI}}(r)$;
- if PMI is reported, $N_{\mathrm{PMI}}(1)=2$ and $N_{\mathrm{PMI}}(2)=1$; otherwise, $N_{\mathrm{PMI}}(r)=0$;
- if PMI $i1$ is reported, $N_{\mathrm{PMI},i_1}(r)$ is obtained according to Table 6.3.1.1.2-1; otherwise, $N_{\mathrm{PMI},i_1}(r)=0$;
- if PMI $i2$ is reported, $N_{\mathrm{PMI},i_2}(r)$ is obtained according to Table 6.3.1.1.2-1; otherwise, $N_{\mathrm{PMI},i_2}(r)=0$;
- if CQI is reported, $N_{\mathrm{CQI}}(r)$ is obtained according to Table 6.3.1.1.2-3; otherwise, $N_{\mathrm{CQI}}(r)=0$;
- if LI is reported, $N_{\mathrm{LI}}(r)$ is obtained according to Table 6.3.1.1.2-3; otherwise, $N_{\mathrm{LI}}(r)=0$.

Table 6.3.1.1.2-8: Mapping order of CSI fields of one report for CRI/RSRP or SSBRI/RSRP or CRI/RSRP/CapabilityIndex or SSBRI/RSRP/CapabilityIndex reporting, or mapping order of CSI fields of one report for inter-cell SSBRI/RSRP reporting

CSI report number	CSI fields
CSI report #n	CRI or SSBRI #1 as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #2 as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #3 as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #4 as in Table 6.3.1.1.2-6, if reported
	RSRP #1 as in Table 6.3.1.1.2-6, if reported
	Differential RSRP #2 as in Table 6.3.1.1.2-6, if reported
	Differential RSRP #3 as in Table 6.3.1.1.2-6, if reported
	Differential RSRP #4 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #1 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #2 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #3 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #4 as in Table 6.3.1.1.2-6, if reported

Table 6.3.1.1.2-8A: Mapping order of CSI fields of one report for CRI/SINR or SSBRI/SINR or CRI/SINR/CapabilityIndex or SSBRI/SINR/CapabilityIndex reporting

CSI report number	CSI fields
CSI report #n	CRI or SSBRI #1 as in Table 6.3.1.1.2-6A, if reported
	CRI or SSBRI #2 as in Table 6.3.1.1.2-6A, if reported
	CRI or SSBRI #3 as in Table 6.3.1.1.2-6A, if reported
	CRI or SSBRI #4 as in Table 6.3.1.1.2-6A, if reported
	SINR #1 as in Table 6.3.1.1.2-6A, if reported
	Differential SINR #2 as in Table 6.3.1.1.2-6A, if reported
	Differential SINR #3 as in Table 6.3.1.1.2-6A, if reported
	Differential SINR #4 as in Table 6.3.1.1.2-6A, if reported
	CapabilityIndex #1 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #2 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #3 as in Table 6.3.1.1.2-6, if reported
	CapabilityIndex #4 as in Table 6.3.1.1.2-6, if reported

Table 6.3.1.1.2-8B: Mapping order of CSI fields of one report for group-based CRI/RSRP or SSBRI/RSRP reporting

CSI report number	CSI fields
CSI report #n	Resource set indicator
	CRI or SSBRI #1 of 1st resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #2 of 1st resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #1 of 2nd resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #2 of 2nd resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #1 of 3rd resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #2 of 3rd resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #1 of 4th resource group as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #2 of 4th resource group as in Table 6.3.1.1.2-6, if reported
	RSRP of CRI or SSBRI #1 of 1st resource group as in Table 6.3.1.1.2-6
	Differential RSRP of CRI or SSBRI #2 of 1st resource group as in Table 6.3.1.1.2-6
	Differential RSRP of CRI or SSBRI #1 of 2nd resource group as in Table 6.3.1.1.2-6, if reported
	Differential RSRP of CRI or SSBRI #2 of 2nd resource group as in Table 6.3.1.1.2-6, if reported
	Differential RSRP of CRI or SSBRI #1 of 3rd resource group as in Table 6.3.1.1.2-6, if reported
	Differential RSRP of CRI or SSBRI #2 of 3rd resource group as in Table 6.3.1.1.2-6, if reported
	Differential RSRP of CRI or SSBRI #1 of 4th resource group as in Table 6.3.1.1.2-6, if reported
	Differential RSRP of CRI or SSBRI #2 of 4th resource group as in Table 6.3.1.1.2-6, if reported

where the 1-bit resource set indicator, with value of 0 or 1, indicates the 1st or the 2nd channel measurement resource set respectively, from which CRI or SSBRI #1 of 1st resource group is reported from; and all remaining resource groups, if reported, follow the same mapping order as the 1st resource group where CRI or SSBRI #1 of all remaining resource groups is reported from the indicated channel measurement resource set. For all reported resource groups, CRI or SSBRI #1 and CRI or SSBRI #2 are reported from different channel measurement resource sets.

Table 6.3.1.1.2-8C: Mapping order of CSI fields of one report for SSBRI/RSRP or CRI/RSRP reporting for L1/L2-triggered mobility

CSI report number	CSI fields
CSI report #n	CRI or SSBRI #1 as in Table 6.3.1.1.2-6, if reported
	CRI or SSBRI #2 as in Table 6.3.1.1.2-6, if reported
	…
	CRI or SSBRI # $L \times M$ as in Table 6.3.1.1.2-6, if reported
	RSRP #1 as in Table 6.3.1.1.2-6, if reported
	Differential RSRP #2 as in Table 6.3.1.1.2-6, if reported
	…
	Differential RSRP # $L \times M$ as in Table 6.3.1.1.2-6, if reported
NOTE: *L* is the number of reported cells provided by higher layer parameter *nrOfReportedCells* and *M* is the number of reported SSBRI/RSRP or CRI/RSRP pairs per cell and equal to the value provided by higher layer parameter *nrofReportedRS-PerCell.*	

Table 6.3.1.1.2-8D: Mapping order of CSI fields of one report for MRI/CLI-RSSI reporting

CSI report number	CSI fields
CSI report #n	MRI #1 as in Table 6.3.1.1.2-6B
	MRI #2 as in Table 6.3.1.1.2-6B, if reported
	MRI #3 as in Table 6.3.1.1.2-6B, if reported
	MRI #4 as in Table 6.3.1.1.2-6B, if reported
	CLI-RSSI #1 as in Table 6.3.1.1.2-6B
	Differential CLI-RSSI #2 as in Table 6.3.1.1.2-6B, if reported
	Differential CLI-RSSI #3 as in Table 6.3.1.1.2-6B, if reported
	Differential CLI-RSSI #4 as in Table 6.3.1.1.2-6B, if reported

Table 6.3.1.1.2-8E: Mapping order of CSI fields of one report for Predicted CRI/RSRP or Predicted SSBRI/RSRP reporting

CSI report number	CSI fields
CSI report #n	Predicted CRI or SSBRI #1 as in Table 6.3.1.1.2-6C
	Predicted CRI or SSBRI #2 as in Table 6.3.1.1.2-6C, if reported
	…
	Predicted CRI or SSBRI #K as in Table 6.3.1.1.2-6C, if reported
	Predicted RSRP #1 as in Table 6.3.1.1.2-6C, if reported
	Differential predicted RSRP #2 as in Table 6.3.1.1.2-6C, if reported
	…
	Differential predicted RSRP #K as in Table 6.3.1.1.2-6C, if reported
NOTE: K is the number of predicted CRI or predicted SSBRI to be reported configured by *nrofreportedpredictedrs.*	

Table 6.3.1.1.2-8F: Mapping order of CSI fields of one report for Time instance indicator/Predicted CRI/Predicted RSRP or Time instance indicator/Predicted SSBRI/Predicted RSRP reporting

CSI report number	CSI fields
CSI report #n	Time instance indicator, if reported
	Predicted CRI or SSBRI #1 as in Table 6.3.1.1.2-6C of time instance #1
	Predicted CRI or SSBRI #2 as in Table 6.3.1.1.2-6C of time instance #1, if reported
	…
	Predicted CRI or SSBRI #K as in Table 6.3.1.1.2-6C of time instance #1, if reported
	Predicted CRI or SSBRI #1 as in Table 6.3.1.1.2-6C of time instance #2, if reported
	Predicted CRI or SSBRI #2 as in Table 6.3.1.1.2-6C of time instance #2, if reported
	…
	Predicted CRI or SSBRI #K as in Table 6.3.1.1.2-6C of time instance #2, if reported
	…
	Predicted CRI or SSBRI #1 as in Table 6.3.1.1.2-6C of time instance #N, if reported
	Predicted CRI or SSBRI #2 as in Table 6.3.1.1.2-6C of time instance #N, if reported
	…
	Predicted CRI or SSBRI #K as in Table 6.3.1.1.2-6C of time instance #N, if reported
	Predicted RSRP #1 as in Table 6.3.1.1.2-6C of time instance #1, if reported
	Differential predicted RSRP #2 as in Table 6.3.1.1.2-6C of time instance #1, if reported
	…
	Differential predicted RSRP #K as in Table 6.3.1.1.2-6C of time instance #1, if reported
	Differential predicted RSRP #1 as in Table 6.3.1.1.2-6C of time instance #2, if reported
	Differential predicted RSRP #2 as in Table 6.3.1.1.2-6C of time instance #2, if reported
	…
	Differential predicted RSRP #K as in Table 6.3.1.1.2-6C of time instance #2, if reported
	…
	Differential predicted RSRP #1 as in Table 6.3.1.1.2-6C of time instance #N, if reported
	Differential predicted RSRP #2 as in Table 6.3.1.1.2-6C of time instance #N, if reported
	…
	Differential predicted RSRP #K as in Table 6.3.1.1.2-6C of time instance #N, if reported

NOTE:
K is the number of predicted CRI or predicted SSBRI to be reported for a time instance configured by the higher layer parameter *nrofreportedpredictedrs*.

N is the number of time instances for prediction and the value of N is configured by the higher layer parameter *nroftimeinstance*. Time instance indicator is reported only when N is larger than 1. The N time instances for prediction are ordered in ascending order in time domain. The time instance indicator with value n, $n = 0,1,\ldots,N-1$, indicates the (n+1)-th time instance among the N time instances. Time instance #1 corresponds to the time instance indicated by the time instance indicator among the N time instances if N is larger than 1, and corresponds to the time instance for prediction otherwise. Time instance # i, $i = 2,3,\ldots,N$, corresponds to the (i-1)-th time instance among the remaining $N-1$ time instances.

Table 6.3.1.1.2-8G: Mapping order of CSI fields of one report for CRI/RSRP or SSBRI/RSRP reporting, if *nrofReportedRS* is configured

CSI report number	CSI fields
CSI report #n	CRI or SSBRI #1 as in Table 6.3.1.1.2-6
	CRI or SSBRI #2 as in Table 6.3.1.1.2-6, if reported
	…
	CRI or SSBRI # M as in Table 6.3.1.1.2-6, if reported
	RSRP #1 as in Table 6.3.1.1.2-6
	Differential RSRP #2 as in Table 6.3.1.1.2-6, if reported
	…
	Differential RSRP # M as in Table 6.3.1.1.2-6, if reported
NOTE: The value of M is configured by the higher layer parameter *nrofReportedRS*. If M is equal to the number of SSB/CSI-RS resources in the corresponding resource set for channel measurement, Differential RSRP #m, $m = 2,\ldots,M$, corresponds to the ($m-1$)-th SSB/CSI-RS resources among the remaining $M-1$ SSB/CSI-RS resources other than the resource corresponding to CRI/SSBRI#1 in the resource set.	

Table 6.3.1.1.2-8H: Mapping order of CSI fields of one report for RS-PAI reporting

CSI report number	CSI fields
CSI report #n	RS-PAI as in Table 6.3.1.1.2-6D

Table 6.3.1.1.2-9: Mapping order of CSI fields of one CSI report, CSI part 1, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report number	CSI fields
CSI report #n CSI part 1	CRI as in Tables 6.3.1.1.2-3/4, if reported
	Rank Indicator as in Tables 6.3.1.1.2-3/3C/4/5, if reported
	Wideband CQI for the first TB as in Tables 6.3.1.1.2-3/3C/4/5, if reported
	Subband differential CQI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3/3C/4/5, if reported
	Indicator of the number of non-zero wideband amplitude coefficients M_0 for layer 0 as in Table 6.3.1.1.2-5, if reported
	Indicator of the number of non-zero wideband amplitude coefficients M_1 for layer 1 as in Table 6.3.1.1.2-5 (if the rank according to the reported RI is equal to one, this field is set to all zeros), if 2-layer PMI reporting is allowed according to the rank restriction in Clauses 5.2.2.2.3 and 5.2.2.2.4 [6, TS 38.214] and if reported
NOTE: Subbands for given CSI report n indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.	

Table 6.3.1.1.2-9A: Mapping order of CSI fields of one CSI report, CSI part 1, *csi-ReportMode= Mode 1*

CSI report number	CSI fields
CSI report #n CSI part 1	CRI as in Tables 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported
	Rank Combination Indicator as in Tables 6.3.1.1.2-3A, if reported
	Wideband CQI for the first TB as in Tables 6.3.1.1.2-3A, if reported
	Subband differential CQI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3A, if reported
	CRI as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource, *numberOfSingleTRP-CSI-Mode1 = 1* and if reported; First CRI as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource, *numberOfSingleTRP-CSI-Mode1 = 2* and if reported
	Rank Indicator associated with CRI as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Rank Indicator associated with the first CRI as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Wideband CQI associated with CRI for the first TB as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Wideband CQI associated with the first CRI for the first TB as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1 = 2* and if reported
	Subband differential CQI associated with CRI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 1 if reported; Subband differential CQI associated with the first CRI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Second CRI as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Rank Indicator associated with the second CRI as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Wideband CQI associated with the second CRI for the first TB as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Subband differential CQI associated with the second CRI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.	

Table 6.3.1.1.2-9B: Mapping order of CSI fields of one CSI report, CSI part 1, *csi-ReportMode= Mode 2*

CSI report number	CSI fields
CSI report #n CSI part 1	CRI as in Tables 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; CRI as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported
	Rank Combination Indicator as in Tables 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; Rank Indicator as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported; Zero padding bits O_P, if needed
	Wideband CQI for the first TB as in Tables 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; Wideband CQI for the first TB as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported
	Subband differential CQI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; Subband differential CQI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.	

The number of zero padding bits O_P in Table 6.3.1.1.2-9B is 0 for 1 CSI-RS port and $O_P = N_{max} - N_{reported}(R)$ for more than 1 CSI-RS port, where:

- $N_{max} = \max_{r \in S_{Rank}} N(r)$. S_{Rank} is the set of rank and rank combination values r that are allowed to be reported. $N(r)$ is obtained according to Tables 6.3.1.1.2-3A/3B for rank combination indicator and rank indicator respectively.
- $N_{reported}(R)$ is obtained according to Tables 6.3.1.1.2-3A for rank combination indicator and R is the reported rank combination.
- $N_{reported}(R)$ is obtained according to Tables 6.3.1.1.2-3B for rank indicator and R is the reported rank.

Table 6.3.1.1.2-9C: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 1, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report number	CSI fields
CSI report #n CSI part 1	CRI for the first reported CRI as in Tables 6.3.1.1.2-3
	Rank Indicator for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	Zero padding bits $O_{P,1}$ for the first reported CRI, if needed
	Wideband CQI for the first TB for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	Subband differential CQI for the first TB with increasing order of subband number for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	…
	CRI for the *M*-th reported CRI as in Tables 6.3.1.1.2-3
	Rank Indicator for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	Zero padding bits $O_{P,M}$ for the *M*-th reported CRI, if needed
	Wideband CQI for the first TB for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	Subband differential CQI for the first TB with increasing order of subband number for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.	

The number of zero padding bits $O_{P,k}$ in Table 6.3.1.1.2-9C is 0 for 1 CSI-RS port and $O_{P,k} = N_{max} - N_{reported,k}$ for more than 1 CSI-RS port, where:

- $N_{max} = \max_{i \in S}(N_i)$, where S is the set of CRIs corresponding to the K_s resources and N_i is the payload size of RI field for the i-th CRI obtained according to Table 6.3.1.1.2-3;
- $N_{reported,k}$ is the payload size of RI field for the k-th CRI obtained according to Table 6.3.1.1.2-3.

Table 6.3.1.1.2-10: Mapping order of CSI fields of one CSI report, CSI part 2 wideband, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3/3C/4/5, if present and reported
	Layer Indicator as in Tables 6.3.1.1.2-3/3C/4/5, if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported

Table 6.3.1.1.2-10A: Mapping order of CSI fields of one CSI report, CSI part 2 wideband, *csi-ReportMode= Mode 1*

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Two Layer Indicators as in Table 6.3.1.1.2-3A, where the first Layer Indicator and the second Layer Indicator are associated with the first resource and the second resource within the resource pair respectively and if reported;
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the first resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the first CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the second resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the second CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and if reported
	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if associated with CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if associated with the first CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Layer Indicator as in Table 6.3.1.1.2-3B, if associated with CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Layer Indicator as in Table 6.3.1.1.2-3B, if associated with the first CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if associated with CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if associated with the first CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if associated with CRI in CSI part 1, *pmi-FormatIndicator= widebandPMI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if associated with the first CRI in CSI part 1, *pmi-FormatIndicator= widebandPMI, numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if associated with the second CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Layer Indicator as in Table 6.3.1.1.2-3B, if associated with the second CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if associated with the second CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if associated with the second CRI in CSI part 1, *pmi-FormatIndicator= widebandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported

Table 6.3.1.1.2-10B: Mapping order of CSI fields of one CSI report, CSI part 2 wideband, *csi-ReportMode= Mode 2*

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if reported part 1 is associated with one CSI-RS resource and if reported
	Two Layer Indicators as in Table 6.3.1.1.2-3A, if reported part 1 is associated with one CSI-RS resource pair, where the first Layer Indicator and the second Layer Indicator are associated with the first resource and the second resource within the resource pair respectively and if reported; Layer Indicator as in Table 6.3.1.1.2-3B, if reported part 1 is associated with one CSI-RS resource and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the first resource within the CSI-RS resource pair, if reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the first CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the second CSI-RS resource within the CSI-RS resource pair, if reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the second CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if reported part 1 is associated with one CSI-RS resource and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and reported part 1 is associated with one CSI-RS resource and if reported

Table 6.3.1.1.2-10C: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 2 wideband, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Wideband CQI for the second TB for the first reported CRI as in Tables 6.3.1.1.2-3, if present and reported
	Layer Indicator for the first reported CRI as in Tables 6.3.1.1.2-3, if reported
	PMI wideband information fields X_1 for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the first reported CRI according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported
	…
	Wideband CQI for the second TB for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if present and reported
	Layer Indicator for the *M*-th reported CRI as in Tables 6.3.1.1.2-3, if reported
	PMI wideband information fields X_1 for the *M*-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the *M*-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the first reported CRI according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported

Table 6.3.1.1.2-11: Mapping order of CSI fields of one CSI report, CSI part 2 subband, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report #n Part 2 subband	Subband differential CQI for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4/5, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4/5, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.	

Table 6.3.1.1.2-11A: Mapping order of CSI fields of one CSI report, CSI part 2 subband, *csi-ReportMode= Mode 1*

CSI report #n Part 2 subband	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported

	Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with the first CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with the second CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with the first CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with the second CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported

Table 6.3.1.1.2-11B: Mapping order of CSI fields of one CSI report, CSI part 2 subband, *csi-ReportMode= Mode 2*

CSI report #n Part 2 subband	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with one CSI-RS resource, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI* and reported part 1 is associated with one CSI-RS resource and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with one CSI-RS resource according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with one CSI-RS resource, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI* and reported part 1 is associated with one CSI-RS resource and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with one CSI-RS resource according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource and if reported

Table 6.3.1.1.2-11C: Mapping order of CSI fields of one CSI report containing N_n^{sub} CSI sub-report(s), CSI part 2 subband, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report #n Part 2 subband	Subband differential CQI of CSI sub-report #1 for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #1 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI of CSI sub-report #2 for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #2 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI of CSI sub-report #N_n^{sub} for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #N_n^{sub} of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #N_n^{sub} according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI of CSI sub-report #1 for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #1 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI of CSI sub-report #2 for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index of for 2 antenna ports CSI sub-report #2 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI of CSI sub-report #N_n^{sub} for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #N_n^{sub} of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #N_n^{sub} according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported

NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0. CSI sub-report #1, CSI sub-report #2, ⋯, CSI sub-report #N_n^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.

Table 6.3.1.1.2-11D: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 2 subband, *pmi-FormatIndicator= subbandPMI* or *cqi-FormatIndicator=subbandCQI*

CSI report #n Part 2 subband	Subband differential CQI for the second TB of all even subbands with increasing order of subband number for the first reported CRI, as in Table 6.3.1.1.2-3, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number for the first reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number for the first reported CRI, as in Table 6.3.1.1.2-3, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number for the first reported CRI, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number for the first reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number for the *M*-th reported CRI, as in Table 6.3.1.1.2-3, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number for the *M*-th reported CRI, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number for the *M*-th reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number for the *M*-th reported CRI, as in Table 6.3.1.1.2-3, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number for the *M*-th reported CRI, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number for the *M*-th reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.	

If none of the CSI reports for transmission on a PUCCH is of two parts, the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.1.1.2-12, are mapped to the UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ starting with a_0. The most significant bit of each field is mapped to the lowest order information bit for that field, e.g. the most significant bit of the first field is mapped to a_0.

Table 6.3.1.1.2-12: Mapping order of CSI reports to UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, without two-part CSI report(s)

UCI bit sequence	CSI report number
a_0	CSI report #1 as in Table 6.3.1.1.2-7/7A/7B/8/8A/8B/8C/8D/8E/8F/8G/8H
a_1	CSI report #2 as in Table 6.3.1.1.2-7/7A/7B/8/8A/8B/8C/8D/8E/8F/8G/8H
a_2	…
a_3	CSI report #n as in Table 6.3.1.1.2-7/7A/7B/8/8A/8B/8C/8D/8E/8F/8G/8H
$\vdots$	
a_{A-1}	
NOTE: For a CSI report #i containing N_i^{sub} CSI sub-reports, where $i \in \{1,2,\ldots,n\}$, all CSI sub-reports within the CSI report #i are mapped to the corresponding segment of the UCI bit sequence of CSI report #i, from upper part to lower part of the segment, in increasing order of CSI sub-report number. CSI sub-report #1, CSI sub-report #2, …, CSI sub-report #N_i^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.	

If at least one of the CSI reports for transmission on a PUCCH is of two parts, two UCI bit sequences are generated, $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ and $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$. The CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.1.1.2-13, are mapped to the UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ starting with $a_0^{(1)}$. The most significant bit of each field is mapped to the lowest order information bit for that field, e.g. the most significant bit of the first field is mapped to $a_0^{(1)}$. The CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.1.1.2-14, are mapped to the UCI bit sequence $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ starting with $a_0^{(2)}$. The most significant bit of each field is mapped to the lowest order information bit for that field, e.g. the most significant bit of the first field is mapped to $a_0^{(2)}$. If the length of UCI bit sequence $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ is less than 3 bits, zeros shall be appended to the UCI bit sequence until its length equals 3.

Table 6.3.1.1.2-13: Mapping order of CSI reports to UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$, with two-part CSI report(s)

UCI bit sequence	CSI report number
$a_0^{(1)}$	CSI report #1 if CSI report #1 is not of two parts, or CSI report #1, CSI part 1, if CSI report #1 is of two parts, as in Table 6.3.1.1.2-7/7A/7B/8/8A/8B/8C/8D/8E/8F/8G/8H/9/9A/9B/9C
$a_1^{(1)}$	CSI report #2 if CSI report #2 is not of two parts, or CSI report #2, CSI part 1, if CSI report #2 is of two parts, as in Table 6.3.1.1.2-7/7A/7B/8/8A/8B/8C/8D/8E/8F/8G/8H/9/9A/9B/9C
$a_2^{(1)}$	…
$a_3^{(1)}$	CSI report #n if CSI report #n is not of two parts, or CSI report #n, CSI part 1, if CSI report #n is of two parts, as in Table 6.3.1.1.2-7/7A/7B/8/8A/8B/8C/8D/8E/8F/8G/8H/9/9A/9B/9C
$\vdots$	
$a_{A^{(1)}-1}^{(1)}$	
NOTE: For a CSI report #i containing N_i^{sub} CSI sub-reports, where $i \in \{1,2,\ldots,n\}$, either all CSI sub-reports not of two parts or CSI part 1 of all CSI sub-reports of two parts, are mapped to the corresponding segment of the UCI bit sequence of CSI report #i, from upper part to lower part of the segment, in increasing order of CSI sub-report number. CSI sub-report #1, CSI sub-report #2, …, CSI sub-report #N_i^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.	

where CSI report #1, CSI report #2, ⋯, CSI report #n in Table 6.3.1.1.2-13 correspond to the CSI reports in increasing order of CSI report priority values according to Clause 5.2.5 of [6, TS38.214].

Table 6.3.1.1.2-14: Mapping order of CSI reports to UCI bit sequence $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$, with two-part CSI report(s)

UCI bit sequence	CSI report number
$a_0^{(2)}$ $a_1^{(2)}$ $a_2^{(2)}$ $a_3^{(2)}$ ⋮ $a_{A^{(2)}-1}^{(2)}$	CSI report #1, CSI part 2 wideband, as in Table 6.3.1.1.2-10/10A/10B/10C if CSI part 2 exists for CSI report #1
	CSI report #2, CSI part 2 wideband, as in Table 6.3.1.1.2-10/10A/10B/10C if CSI part 2 exists for CSI report #2
	…
	CSI report #n, CSI part 2 wideband, as in Table 6.3.1.1.2-10/10A/10B/10C if CSI part 2 exists for CSI report #n
	CSI report #1, CSI part 2 subband, as in Table 6.3.1.1.2-11/11A/11B/11C/11D if CSI part 2 exists for CSI report #1
	CSI report #2, CSI part 2 subband, as in Table 6.3.1.1.2-11/11A/11B/11C/11D if CSI part 2 exists for CSI report #2
	…
	CSI report #n, CSI part 2 subband, as in Table 6.3.1.1.2-11/11A/11B/11C/11D if CSI part 2 exists for CSI report #n
NOTE: For a CSI report #i containing N_i^{sub} CSI sub-reports, where $i \in \{1,2,\ldots,n\}$: - CSI part 2 widebands of all CSI sub-reports are mapped to the corresponding segment of the UCI bit sequence of CSI report #i, from upper part to lower part of the segment, in increasing order of CSI sub-report number; - CSI sub-report #1, CSI sub-report #2, ⋯, CSI sub-report #N_i^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.	

where CSI report #1, CSI report #2, ⋯, CSI report #n in Table 6.3.1.1.2-14 correspond to the CSI reports in increasing order of CSI report priority values according to Clause 5.2.5 of [6, TS38.214].

6.3.1.1.3 HARQ-ACK/SR and CSI

If none of the CSI reports for transmission on a PUCCH is of two parts, the UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ is generated according to the following, where $A = O^{\text{ACK}} + O^{\text{SR}} + O^{\text{CSI}}$:

- if there is HARQ-ACK for transmission on the PUCCH, the HARQ-ACK bits are mapped to the UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{O^{\text{ACK}}-1}$, where $a_i = \tilde{o}_i^{ACK}$ for $i = 0,1,\ldots,O^{\text{ACK}}-1$, the HARQ-ACK bit sequence $\tilde{o}_0^{ACK}, \{\tilde{o}_1^{ACK}, \ldots, \tilde{o}_{O^{ACK}-1}^{ACK}¿$ is given by Clause 9.1 of [5, TS38.213], and O^{ACK} is number of HARQ-ACK bits; if there is no HARQ-ACK for transmission on the PUCCH, set $O^{\text{ACK}} = 0$;
- if there is SR for transmission on the PUCCH, set $a_i = \tilde{o}_{i-O^{ACK}}^{SR}$ for $i = O^{\text{ACK}}, O^{\text{ACK}}+1, \ldots, O^{\text{ACK}}+O^{\text{SR}}-1$, where the SR bit sequence $\tilde{o}_0^{SR}, \{\tilde{o}_1^{SR}, \ldots, \tilde{o}_{O^{SR}-1}^{SR}¿$ is given by Clause 9.2.5.1 of [5, TS 38.213]; if there is no SR for transmission on the PUCCH, set $O^{\text{SR}} = 0$;
- the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.1.1.2-12, are mapped to the UCI bit sequence $a_{O^{\text{ACK}}+O^{\text{SR}}}, a_{O^{\text{ACK}}+O^{\text{SR}}+1}, \ldots, a_{O^{\text{ACK}}+O^{\text{SR}}+O^{\text{CSI}}-1}$ starting with $a_{O^{\text{ACK}}+O^{\text{SR}}}$, where O^{CSI} is the number of CSI bits.

If at least one of the CSI reports for transmission on a PUCCH is of two parts, two UCI bit sequences are generated, $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ and $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$, according to the following, where $A^{(1)} = O^{\text{ACK}} + O^{\text{SR}} + O^{\text{CSI-part1}}$ and $A^{(2)} = O^{\text{CSI-part2}}$:

- if there is HARQ-ACK for transmission on the PUCCH, the HARQ-ACK bits are mapped to the UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{O^{\text{ACK}}-1}^{(1)}$, where $a_i^{(1)} = \tilde{o}_i^{ACK}$ for $i = 0,1,\ldots,O^{\text{ACK}}-1$, the HARQ-ACK bit sequence $\tilde{o}_0^{ACK}, \{\tilde{o}_1^{ACK}, \ldots, \tilde{o}_{O^{ACK}-1}^{ACK}¿$ is given by Clause 9.1 of [5, TS38.213], and O^{ACK} is number of HARQ-ACK bits; if there is no HARQ-ACK for transmission on the PUCCH, set $O^{\text{ACK}} = 0$;

- if there is SR for transmission on the PUCCH, set $a_i = \tilde{o}^{SR}_{i-O^{ACK}}$ for $i = O^{\text{ACK}}, O^{\text{ACK}}+1, \ldots, O^{\text{ACK}}+O^{\text{SR}}-1$, where the SR bit sequence $\tilde{o}^{SR}_0, \{\tilde{o}^{SR}_1, \ldots, \tilde{o}^{SR}_{O^{\text{SR}}-1}$ is given by Clause 9.2.5.1 of [5, TS 38.213]; if there is no SR for transmission on the PUCCH, set $O^{\text{SR}} = 0$;
- the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.1.1.2-13, are mapped to the UCI bit sequence $a^{(1)}_{O^{\text{ACK}}+O^{\text{SR}}}, a^{(1)}_{O^{\text{ACK}}+O^{\text{SR}}+1}, \ldots, a^{(1)}_{O^{\text{ACK}}+O^{\text{SR}}+O^{\text{CSI-part1}}-1}$ starting with $a^{(1)}_{O^{\text{ACK}}+O^{\text{SR}}}$, where $O^{\text{CSI-part1}}$ is the number of CSI bits in CSI part 1 of all CSI reports;
- the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.1.1.2-14, are mapped to the UCI bit sequence $a^{(2)}_0, a^{(2)}_1, a^{(2)}_2, a^{(2)}_3, \ldots, a^{(2)}_{A^{(2)}-1}$ starting with $a^{(2)}_0$, where $O^{\text{CSI-part2}}$ is the number of CSI bits in CSI part 2 of all CSI reports. If the length of UCI bit sequence $a^{(2)}_0, a^{(2)}_1, a^{(2)}_2, a^{(2)}_3, \ldots, a^{(2)}_{A^{(2)}-1}$ is less than 3 bits, zeros shall be appended to the UCI bit sequence until its length equals 3.

6.3.1.1.4 UCI with different priority indexes

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1, and SR associated with priority index 1 if any are transmitted on a PUCCH, two UCI bit sequences are generated, $a^{(1)}_0, a^{(1)}_1, a^{(1)}_2, a^{(1)}_3, \ldots, a^{(1)}_{A^{(1)}-1}$ and $a^{(2)}_0, a^{(2)}_1, a^{(2)}_2, a^{(2)}_3, \ldots, a^{(2)}_{A^{(2)}-1}$, according to the following, where $A^{(1)} = O^{ACK-HP} + O^{SR-HP}$ and $A^{(2)} = O^{ACK-LP}$:

- the HARQ-ACK bits associated with priority index 1 are mapped to the UCI bit sequence $a^{(1)}_0, a^{(1)}_1, a^{(1)}_2, a^{(1)}_3, \ldots, a^{(1)}_{O^{ACK-HP}-1}$, where $a^{(1)}_i = \tilde{o}^{ACK-HP}_i$ for $i = 0, 1, \ldots, O^{ACK-HP}-1$, the HARQ-ACK bit sequence $\tilde{o}^{ACK-HP}_0, \tilde{o}^{ACK-HP}_1, \ldots, \tilde{o}^{ACK-HP}_{O^{ACK-HP}-1}$ is given by Clause 9.1 of [5, TS 38.213], and O^{ACK-HP} is the number of HARQ-ACK bits associated with priority index 1;
- if there is SR associated with priority index 1 for transmission on the PUCCH, set $a^{(1)}_i = \tilde{o}^{SR-HP}_{i-O^{ACK-HP}}$ for $i = O^{ACK-HP}, O^{ACK-HP}+1, \ldots, O^{ACK-HP}+O^{SR-HP}-1$, where the SR bit sequence $\tilde{o}^{SR-HP}_0, \tilde{o}^{SR-HP}_1, \ldots, \tilde{o}^{SR-HP}_{O^{SR-HP}-1}$ is given by Clause 9.2.5.1 of [5, TS 38.213]; if there is no SR associated with priority index 1 for transmission on the PUCCH, set $O^{SR-HP} = 0$;
- the HARQ-ACK bits associated with priority index 0 are mapped to the UCI bit sequence $a^{(2)}_0, a^{(2)}_1, a^{(2)}_2, a^{(2)}_3, \ldots, a^{(2)}_{O^{ACK-LP}-1}$, where $a^{(2)}_i = \tilde{o}^{ACK-LP}_i$ for $i = 0, 1, \ldots, O^{ACK-LP}-1$, the HARQ-ACK bit sequence $\tilde{o}^{ACK-LP}_0, \tilde{o}^{ACK-LP}_1, \ldots, \tilde{o}^{ACK-LP}_{O^{ACK-LP}-1}$ is given by Clause 9.1 of [5, TS 38.213], and O^{ACK-LP} is the number of HARQ-ACK bits associated with priority index 0.

6.3.1.2 Code block segmentation and CRC attachment

The UCI bit sequence from clause 6.3.1.1 is denoted by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, where A is the payload size. The procedure in Clause 6.3.1.2.1 applies for $A \geq 12$ and the procedure in Clause 6.3.1.2.2 applies for $A \leq 11$.

6.3.1.2.1 UCI encoded by Polar code

If the payload size $A \geq 12$, code block segmentation and CRC attachment is performed according to Clause 5.2.1. If ($A \geq 360$ and $E \geq 1088$) or if $A \geq 1013$, $I_{seg} = 1$; otherwise $I_{seg} = 0$, where E is the rate matching output sequence length as given in Clauses 6.3.1.4.1 and 6.3.1.4.3.

If $12 \le A \le 19$, the parity bits $p_{r0}, p_{r1}, p_{r2}, \ldots, p_{r(L-1)}$ in Clause 5.2.1 are computed by setting L to 6 bits and using the generator polynomial $g_{\text{CRC6}}(D)$ in Clause 5.1, resulting in the sequence $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$ where r is the code block number and K_r is the number of bits for code block number r.

If $A \ge 20$, the parity bits $p_{r0}, p_{r1}, p_{r2}, \ldots, p_{r(L-1)}$ in Clause 5.2.1 are computed by setting L to 11 bits and using the generator polynomial $g_{\text{CRC11}}(D)$ in Clause 5.1, resulting in the sequence $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$ where r is the code block number and K_r is the number of bits for code block number r.

6.3.1.2.2 UCI encoded by channel coding of small block lengths

If the payload size $A \le 11$, CRC bits are not attached.

The output bit sequence is denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where $c_i = a_i$ for $i = 0, 1, \ldots, A-1$ and $K = A$.

6.3.1.3 Channel coding of UCI

6.3.1.3.1 UCI encoded by Polar code

Information bits are delivered to the channel coding block. They are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$, where r is the code block number, and K_r is the number of bits in code block number r. The total number of code blocks is denoted by C and each code block is individually encoded by the following:

If $18 \le K_r \le 25$, the information bits are encoded via Polar coding according to Clause 5.3.1, by setting $n_{\max} = 10$, $I_{IL} = 0$, $n_{PC} = 3$, $n_{PC}^{wm} = 1$ if $E_r - K_r + 3 > 192$ and $n_{PC}^{wm} = 0$ if $E_r - K_r + 3 \le 192$, where E_r is the rate matching output sequence length as given in Clauses 6.3.1.4.1 and 6.3.1.4.3.

If $K_r > 30$, the information bits are encoded via Polar coding according to Clause 5.3.1, by setting $n_{\max} = 10$, $I_{IL} = 0$, $n_{PC} = 0$, and $n_{PC}^{wm} = 0$.

After encoding the bits are denoted by $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$, where N_r is the number of coded bits in code block number r.

6.3.1.3.2 UCI encoded by channel coding of small block lengths

Information bits are delivered to the channel coding block. They are denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits.

The information bits are encoded according to Clause 5.3.3.

After encoding the bits are denoted by $d_0, d_1, d_2, d_3, \ldots, d_{N-1}$, where N is the number of coded bits.

6.3.1.4 Rate matching

For PUCCH formats 2/3/4, the total rate matching output sequence length E_{tot} is given by Table 6.3.1.4-1, where $N_{\text{symb, UCI}}^{\text{PUCCH, 2}}$, $N_{\text{symb, UCI}}^{\text{PUCCH, 3}}$, and $N_{\text{symb, UCI}}^{\text{PUCCH, 4}}$ are the number of symbols carrying UCI for PUCCH formats 2/3/4 respectively; $N_{\text{PRB}}^{\text{PUCCH, 2}}$, $N_{\text{PRB}}^{\text{PUCCH, 3}}$ and $N_{PRB}^{PUCCH,4}$ are the number of PRBs that are determined by the UE for PUCCH formats 2/3/4 transmission respectively according to Clause 9.2 of [5, TS38.213]; and $N_{SF}^{PUCCH,2}$, $N_{SF}^{PUCCH,3}$, and $N_{\text{SF}}^{\text{PUCCH, 4}}$ are the spreading factors for PUCCH format 2, PUCCH format 3, and PUCCH format 4, respectively.

Table 6.3.1.4-1: Total rate matching output sequence length E_{tot}

PUCCH format	*Modulation order*	
	QPSK	π/2-BPSK
PUCCH format 2	$16 \cdot N_{symb,UCI}^{PUCCH,2} \cdot N_{PRB}^{PUCCH,2} / N_{SF}^{PUCCH,2}$	N/A
PUCCH format 3	$24 \cdot N_{symb,UCI}^{PUCCH,3} \cdot N_{PRB}^{PUCCH,3} / N_{SF}^{PUCCH,3}$	$12 \cdot N_{symb,UCI}^{PUCCH,3} \cdot N_{PRB}^{PUCCH,3} / N_{SF}^{PUCCH,}$
PUCCH format 4	$24 \cdot N_{symb,UCI}^{PUCCH,4} \cdot N_{PRB}^{PUCCH,4} / N_{SF}^{PUCCH,4}$	$12 \cdot N_{symb,UCI}^{PUCCH,4} \cdot N_{PRB}^{PUCCH,4} / N_{SF}^{PUCCH,}$

6.3.1.4.1 UCI encoded by Polar code

The input bit sequence to rate matching is $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Table 6.3.1.4.1-1: Rate matching output sequence length E_{UCI}

UCI(s) for transmission on a PUCCH	UCI for encoding	Value of E_{UCI}
HARQ-ACK	HARQ-ACK	$E_{\text{UCI}} = E_{\text{tot}}$
HARQ-ACK, SR	HARQ-ACK, SR	$E_{\text{UCI}} = E_{\text{tot}}$
CSI (CSI not of two parts)	CSI	$E_{\text{UCI}} = E_{\text{tot}}$
HARQ-ACK, CSI (CSI not of two parts)	HARQ-ACK, CSI	$E_{\text{UCI}} = E_{\text{tot}}$
HARQ-ACK, SR, CSI (CSI not of two parts)	HARQ-ACK, SR, CSI	$E_{\text{UCI}} = E_{\text{tot}}$
CSI (CSI of two parts)	CSI part 1	$E_{\text{UCI}} = \min\left(E_{\text{tot}}, \lceil (O^{\text{CSI-part1}} + L)/R_{\text{UCI}}^{\max}/Q_m \rceil \cdot Q_m\right)$
	CSI part 2	$E_{\text{UCI}} = E_{\text{tot}} - \min\left(E_{\text{tot}}, \lceil (O^{\text{CSI-part1}} + L)/R_{\text{UCI}}^{\max}/Q_m \rceil \cdot Q_m\right)$
HARQ-ACK, CSI (CSI of two parts)	HARQ-ACK, CSI part 1	$E_{\text{UCI}} = \min\left(E_{\text{tot}}, \lceil (O^{\text{ACK}} + O^{\text{CSI-part1}} + L)/R_{\text{UCI}}^{\max}/Q_m \rceil \cdot Q_m\right)$
	CSI part 2	$E_{\text{UCI}} = E_{\text{tot}} - \min\left(E_{\text{tot}}, \lceil (O^{\text{ACK}} + O^{\text{CSI-part1}} + L)/R_{\text{UCI}}^{\max}/Q_m \rceil \cdot Q_m\right)$
HARQ-ACK, SR, CSI (CSI of two parts)	HARQ-ACK, SR, CSI part 1	$E_{\text{UCI}} = \min\left(E_{\text{tot}}, \lceil (O^{\text{ACK}} + O^{\text{SR}} + O^{\text{CSI-part1}} + L)/R_{\text{UCI}}^{\max}/Q_m \rceil \cdot Q_m\right)$
	CSI part 2	$E_{\text{UCI}} = E_{\text{tot}} - \min\left(E_{\text{tot}}, \lceil (O^{\text{ACK}} + O^{\text{SR}} + O^{\text{CSI-part1}} + L)/R_{\text{UCI}}^{\max}/Q_m \rceil \cdot Q_m\right)$

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL} = 1$ and the rate matching output sequence length to $E_r = \lfloor E_{\text{UCI}}/C_{\text{UCI}} \rfloor$, where C_{UCI} is the number of code blocks for UCI determined according to Clause 6.3.1.2.1 and the value of E_{UCI} is given by Table 6.3.1.4.1-1:

- O^{ACK} is the number of bits for HARQ-ACK for transmission on the current PUCCH;
- O^{SR} is the number of bits for SR for transmission on the current PUCCH;
- $O^{\text{CSI-part1}}$ is the number of bits for CSI part 1 for transmission on the current PUCCH;
- $O^{\text{CSI-part2}}$ is the number of bits for CSI part 2 for transmission on the current PUCCH;
- if $A \geq 360$, $L = 11$; otherwise, L is the number of CRC bits determined according to Clause 6.3.1.2.1, where A equals $O^{\text{CSI-part1}}$ for "CSI (CSI of two parts)", equals $O^{\text{ACK}} + O^{\text{CSI-part1}}$ for "HARQ-ACK, CSI (CSI of two parts)", and equals $O^{\text{ACK}} + O^{\text{SR}} + O^{\text{CSI-part1}}$ for "HARQ-ACK, SR, CSI (CSI of two parts)" respectively in Table 6.3.1.4.1-1;
- $R_{\text{UCI}}^{\max}$ is the configured maximum PUCCH coding rate;

- E_{tot} is given by Table 6.3.1.4-1.

The output bit sequence after rate matching is denoted as $f_{r0}, f_{r1}, f_{r2}, \ldots, f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.1.4.2 UCI encoded by channel coding of small block lengths

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$.

The value of E_{UCI} is determined according to Table 6.3.1.4.1-1 by setting $L=0$.

Rate matching is performed according to Clause 5.4.3 by setting the rate matching output sequence length $E=E_{\text{UCI}}$.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

6.3.1.4.3 UCI with different priority indexes encoded by Polar code

The following procedure in this clause 6.3.1.4.3 applies if *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1 and SR associated with priority index 1 if any are transmitted on a PUCCH.

The input bit sequence to rate matching is $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Table 6.3.1.4.3-1: Rate matching output sequence length E_{UCI} for UCIs with different priority indexes

UCIs for transmission on a PUCCH	UCI for encoding	Value of E_{UCI}
HARQ-ACK of priority index 1, HARQ-ACK of priority index 0	HARQ-ACK of priority index 1	$E_{UCI}=min\left(E_{tot}, \lceil \left(O^{ACK-HP}+L\right)/R_{UCI}^{max-HP}/Q_m \rceil \cdot Q_m\right)$
	HARQ-ACK of priority index 0	$E_{UCI}=E_{tot}-min\left(E_{tot}, \lceil \left(O^{ACK-HP}+L\right)/R_{UCI}^{max-HP}/Q_m \rceil \cdot Q_m\right)$
HARQ-ACK of priority index 1, SR of priority index 1, HARQ-ACK of priority index 0	HARQ-ACK of priority index 1, SR of priority index 1	$E_{UCI}=min\left(E_{tot}, \lceil \left(O^{ACK-HP}+O^{SR-HP}+L\right)/R_{UCI}^{max-HP}/Q_m \rceil \cdot$
	HARQ-ACK of priority index 0	$E_{UCI}=E_{tot}-min\left(E_{tot}, \lceil \left(O^{ACK-HP}+O^{SR-HP}+L\right)/R_{UCI}^{max-HP}/Q$

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL}=1$ and the rate matching output sequence length to $E_r=\lfloor E_{UCI}/C_{UCI} \rfloor$, where C_{UCI} is the number of code blocks for UCI determined according to Clause 6.3.1.2.1 and the value of E_{UCI} is given by Table 6.3.1.4.3-1:

- O^{ACK-HP} is the number of bits for HARQ-ACK associated with priority index 1 for transmission on the current PUCCH;
- O^{SR-HP} is the number of bits for SR associated with priority index 1 for transmission on the current PUCCH;
- if $A \geq 360$, L=11; otherwise, L is the number of CRC bits determined according to clause 6.3.1.2.1, where A equals O^{ACK-HP} for the case of "HARQ-ACK of priority index 1, HARQ-ACK of priority index 0", and equals $O^{ACK-HP}+O^{SR-HP}$ for the case of "HARQ-ACK of priority index 1, SR of priority index 1, HARQ-ACK of priority index 0" respectively in Table 6.3.1.4.3-1;
- R_{UCI}^{max-HP} is the configured maximum PUCCH coding rate of priority index 1;

- E_{tot} is given by Table 6.3.1.4-1.

The output bit sequence after rate matching is denoted as $f_{r0}, f_{r1}, f_{r2}, \ldots, f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.1.4.4 UCI with different priority indexes encoded by channel coding of small block lengths

The following procedure in this clause 6.3.1.4.4 applies if *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1 and SR associated with priority index 1 if any are transmitted on a PUCCH.

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$.

The value of E_{UCI} is determined according to Table 6.3.1.4.3-1 by setting L=0.

Rate matching is performed according to Clause 5.4.3 by setting the rate matching output sequence length $E = E_{UCI}$.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

6.3.1.5 Code block concatenation

The input bit sequence for the code block concatenation block are the sequences $f_{r0}, f_{r1}, f_{r2}, \ldots, f_{r(E_r-1)}$, for $r = 0, \ldots, C-1$ and where E_r is the number of rate matched bits for the r-th code block.

Code block concatenation is performed according to Clause 5.5.

The bits after code block concatenation are denoted by $g_0, g_1, g_2, g_3, \ldots, g_{G'-1}$, where $G' = \lfloor E_{\mathrm{UCI}} / C_{\mathrm{UCI}} \rfloor \cdot C_{\mathrm{UCI}}$ with the values of E_{UCI} and C_{UCI} given in Clauses 6.3.1.4.1 and 6.3.1.4.3. Let G be the total number of coded bits for transmission and $G = G' + \mathrm{mod}(E_{\mathrm{UCI}}, C_{\mathrm{UCI}})$. Set $g_i = 0$ for $i = G', G'+1, \ldots, G-1$.

6.3.1.6 Multiplexing of coded UCI bits to PUCCH

If CSI of two parts or UCIs with different priority indexes are transmitted on a PUCCH, the coded bits corresponding to UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ is denoted by $g_0^{(1)}, g_1^{(1)}, g_2^{(1)}, g_3^{(1)}, \ldots, g_{G^{(1)}-1}^{(1)}$ and the coded bits corresponding to UCI bit sequence $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ is denoted by $g_0^{(2)}, g_1^{(2)}, g_2^{(2)}, g_3^{(2)}, \ldots, g_{G^{(2)}-1}^{(2)}$.

For PUCCH format 2 when *uci-MuxWithDiffPrio* is configured, the coded bit sequence $g_0, g_1, g_2, g_3, \ldots, g_{G-1}$ is generated for UCIs with different priority indexes by setting $g_i = g_i^{(1)}$ for $i = 0, 1, \ldots, G^{(1)} - 1$, and setting $g_i = g_{i-G^{(1)}}^{(2)}$ for $i = G^{(1)}, G^{(1)}+1, \ldots, G^{(1)}+G^{(2)}-1$.

For PUCCH format 3/4, the coded bit sequence $g_0, g_1, g_2, g_3, \ldots, g_{G-1}$, where $G = G^{(1)} + G^{(2)}$, is generated according to the following.

Table 6.3.1.6-1: PUCCH DMRS and UCI symbols

PUCCH duration (symbols)	PUCCH DMRS symbol indices	Number of UCI symbol indices sets $N_{\text{UCI}}^{\text{set}}$	1st UCI symbol indices set $S_{\text{UCI}}^{(1)}$	2nd UCI symbol indices set $S_{\text{UCI}}^{(2)}$	3rd UCI symbol indices set $S_{\text{UCI}}^{(3)}$
4	{1}	2	{0,2}	{3}	-
4	{0,2}	1	{1,3}	-	-
5	{0, 3}	1	{1, 2, 4}	-	-
6	{1, 4}	1	{0, 2, 3, 5}	-	-
7	{1, 4}	2	{0, 2, 3, 5}	{6}	-
8	{1, 5}	2	{0, 2, 4, 6}	{3, 7}	-
9	{1, 6}	2	{0, 2, 5, 7}	{3, 4, 8}	-
10	{2, 7}	2	{1, 3, 6, 8}	{0, 4, 5, 9}	-
10	{1, 3, 6, 8}	1	{0,2,4,5,7,9}	-	-
11	{2, 7}	3	{1,3,6,8}	{0,4,5,9}	{10}
11	{1,3,6,9}	1	{0,2,4,5,7,8,10}	-	-
12	{2, 8}	3	{1,3,7,9}	{0,4,6,10}	{5, 11}
12	{1,4,7,10}	1	{0,2,3,5,6,8,9,11}	-	-
13	{2, 9}	3	{1,3,8,10}	{0,4,7,11}	{5,6,12}
13	{1,4,7,11}	2	{0,2,3,5,6,8,10,12}	{9}	-
14	{3, 10}	3	{2,4,9,11}	{1,5,8,12}	{0,6,7,13}
14	{1,5,8,12}	2	{0,2,4,6,7,9,11,13}	{3, 10}	-

Denote s_l as UCI OFDM symbol index. Denote $N_{\text{UCI}}^{(i)}$ as the number of elements in UCI symbol indices set $S_{\text{UCI}}^{(i)}$ for $i=1,\ldots,N_{\text{UCI}}^{\text{set}}$, where $S_{\text{UCI}}^{(i)}$ and $N_{\text{UCI}}^{\text{set}}$ are given by Table 6.3.1.6-1 according to the PUCCH duration and the PUCCH DMRS configuration. Denote $N_{\text{symb, UCI}}^{\text{PUCCH,}}=\sum_{i=1}^{N_{\text{UCI}}^{\text{set}}} N_{\text{UCI}}^{(i)}$ as the number of OFDM symbols carrying UCI in the PUCCH. Denote Q_m as the modulation order of the PUCCH.

For PUCCH formats 3/4, set $N_{\text{UCI}}^{\text{symbol}}=12\cdot N_{\text{PRB}}^{\text{PUCCH,s}}/N_{\text{SF}}^{\text{PUCCH,}s}$, where $N_{\text{PRB}}^{\text{PUCCH,s}}$ is the number of PRBs that is determined by the UE for the corresponding PUCCH format transmission according to Clause 9.2 of [5, TS 38.213], and $N_{\text{SF}}^{\text{PUCCH,}s}$ is the spreading factor for the corresponding PUCCH format [4, TS 38.211], where $s\in\{3,4\}$.

Find the smallest $j>0$ such that $\left(\sum_{i=1}^{j} N_{\text{UCI}}^{(i)}\right)\cdot N_{\text{UCI}}^{\text{symbol}}\cdot Q_m\geq G^{(1)}$.

Set $n_1=0$;

Set $n_2=0$;

Set $\bar{N}_{\text{UCI}}^{\text{symbol}}=\left\lfloor\left(G^{(1)}-\left(\sum_{i=1}^{j-1} N_{\text{UCI}}^{(i)}\right)\cdot N_{\text{UCI}}^{\text{symbol}}\cdot Q_m\right)/\left(N_{\text{UCI}}^{(j)}\cdot Q_m\right)\right\rfloor$;

Set $M=\text{mod}\left(\left(G^{(1)}-\left(\sum_{i=1}^{j-1} N_{\text{UCI}}^{(i)}\right)\cdot N_{\text{UCI}}^{\text{symbol}}\cdot Q_m\right)/Q_m, N_{\text{UCI}}^{(j)}\right)$;

for $l=0$ to $N_{\text{symb, UCI}}^{\text{PUCCH,}}-1$

if $s_l\in\bigcup_{i=1}^{j-1} S_{\text{UCI}}^{(i)}$

for $k=0$ to $N_{\text{UCI}}^{\text{symbol}}-1$
for $v=0$ to Q_m-1
$\bar{g}_{l,k,v}=g_{n_1}^{(1)}$;
$n_1=n_1+1$;
end for
end for
elseif $s_l \in S_{\text{UCI}}^{(j)}$
if $M>0$
$\gamma=1$;
else
$\gamma=0$;
end if
$M=M-1$;
for $k=0$ to $\bar{N}_{\text{UCI}}^{\text{symbol}}+\gamma-1$
for $v=0$ to Q_m-1
$\bar{g}_{l,k,v}=g_{n_1}^{(1)}$;
$n_1=n_1+1$;
end for
end for
for $k=\bar{N}_{\text{UCI}}^{\text{symbol}}+\gamma$ to $N_{\text{UCI}}^{\text{symbol}}-1$
for $v=0$ to Q_m-1
$\bar{g}_{l,k,v}=g_{n_2}^{(2)}$;
$n_2=n_2+1$;
end for
end for
else
for $k=0$ to $N_{\text{UCI}}^{\text{symbol}}-1$
for $v=0$ to Q_m-1
$\bar{g}_{l,k,v}=g_{n_2}^{(2)}$;
$n_2=n_2+1$;
end for
end for
end if
end for

Set $n=0$

for $l=0$ to $N_{\text{symb, UCI}}^{\text{PUCCH,}}-1$
for $k=0$ to $N_{\text{UCI}}^{\text{symbol}}-1$
for $v=0$ to Q_m-1
$g_n=\bar{g}_{l,k,v}$;
$n=n+1$;
end for
end for

end for

6.3.2 Uplink control information on PUSCH

The following clauses 6.3.2.2, 6.3.2.3, and 6.3.2.5 apply regardless of whether the higher layer parameter *uci-MuxWithDiffPrio* is configured or not. The following clauses 6.3.2.1, 6.3.2.4, and 6.3.2.6 apply by assuming *uci-MuxWithDiffPrio* is not configured, or *uci-MuxWithDiffPrio* is configured and the UCIs for transmission on a PUSCH are of the same priority index, unless stated otherwise.

If the UE is configured with a PUCCH-SCell, *uci-MuxWithDiffPrio* is replaced by *uci-MuxWithDiffPrioSecondaryPUCCHgroup* for the secondary PUCCH group in this clause.

6.3.2.1 UCI bit sequence generation

6.3.2.1.1 HARQ-ACK

If HARQ-ACK bits are transmitted on a PUSCH, the UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ is determined as follows:

- If UCI is transmitted on PUSCH without UL-SCH and the UCI includes CSI part 1 without CSI part 2,
  - if there is no HARQ-ACK bit given by Clause 9.1 of [5, TS 38.213], set $a_0 = 0$, $a_1 = 0$, and $A = 2$;
  - if there is only one HARQ-ACK bit $\tilde{o}_0^{ACK}$ given by Clause 9.1 of [5, TS 38.213], set $a_0 = \tilde{o}_0^{ACK}$, $a_1 = 0$, and $A = 2$;
- otherwise, set $a_i = \tilde{o}_i^{ACK}$ for $i = 0, 1, \ldots, O^{\text{ACK}} - 1$ and $A = O^{\text{ACK}}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK}, \{\tilde{o}_1^{ACK}, \ldots, \tilde{o}_{O^{ACK}-1}^{ACK}\}$ is given by Clause 9.1 of [5, TS 38.213].

6.3.2.1.2 CSI

If *cqi-BitsPerSubband* is configured, this Clause 6.3.2.1.2 applies by taking Subband CQI as Subband differential CQI and replacing the corresponding number of bits 2 by 4.

If *CSI-ReportSubConfig* is configured, for a corresponding CSI sub-report, the bitwidth of a CSI field of the CSI sub-report is determined following the procedure in this clause 6.3.2.1.2 by taking configurations in *CSI-ReportSubConfig* when applicable. If *CSI-ReportSubConfig* configures a list of CSI-RS resource IDs, for the determination of the bitwidth of a CRI field, the value of $K_s^{CSI\text{-}RS}$ is the number of CSI-RS resources configured in the corresponding *CSI-ReportSubConfig*.

The bitwidth for PMI of *codebookType=typeI-SinglePanel*, *codebookType=typeI-SinglePanel-r19*, *codebookType=typeI-MultiPanel* and *codebookType=typeI-MultiPanel-r19* is specified in Clause 6.3.1.1.2.

The bitwidth for RI/LI/CQI/CRI of *codebookType=typeI-SinglePanel*, *codebookType=typeI-SinglePanel-r19*, *codebookType=typeI-MultiPanel* and *codebookType=typeI-MultiPanel-r19* is specified in Clause 6.3.1.1.2.

The bitwidth for PMI/RI/LI/CQI/CRI with 1 CSI-RS port is specified in Clause 6.3.1.1.2.

The bitwidth for PMI of *codebookType=typeII* is provided in Tables 6.3.2.1.2-1, where the values of (N_1, N_2), (O_1, O_2), L, N_{PSK}, M_1, M_2, and $K^{(2)}$ are given by Clause 5.2.2.2.3 in [6, TS 38.214].

Table 6.3.2.1.2-1: PMI of *codebookType= typeII*

	Information fields X_1 for wideband PMI						Information fields X_2 for wideband PMI or per subband PMI			
	$i_{1,1}$	$i_{1,2}$	$i_{1,3,1}$	$i_{1,4,1}$	$i_{1,3,2}$	$i_{1,4,2}$	$i_{2,1,1}$	$i_{2,1,2}$	$i_{2,2,1}$	$i_{2,2,2}$
Rank=1 SBAmp off	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2\binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	N/A	N/A	$(M_1-1)\cdot \log_2 N_{\text{PSK}}$	N/A	N/A	N/A
Rank=2 SBAmp off	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2\binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$(M_1-1)\cdot \log_2 N_{\text{PSK}}$	$(M_2-1)\cdot \log_2 N_{\text{PSK}}$	N/A	N/A
Rank=1 SBAmp on	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2\binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	N/A	N/A	$\min(M_1, K^{(2)})\cdot \log_2 N_{\text{PSK}} - \log_2 N_{\text{PSK}} + 2\cdot(M_1 - \min(M_1, K^{(2)}))$	N/A	$\min(M_1, K^{(2)})-1$	N/A
Rank=2 SBAmp on	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2\binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$\min(M_1, K^{(2)})\cdot \log_2 N_{\text{PSK}} - \log_2 N_{\text{PSK}} + 2\cdot(M_1 - \min(M_1, K^{(2)}))$	$\min(M_2, K^{(2)})\cdot \log_2 N_{\text{PSK}} - \log_2 N_{\text{PSK}} + 2\cdot(M_2 - \min(M_2, K^{(2)}))$	$\min(M_1, K^{(2)})-1$	$\min(M_2, K^{(2)})-1$

The bitwidth for PMI of *codebookType=typeII-r16* and *codebookType=etypeII-r19* is provided in Tables 6.3.2.1.2-1A, where the values of (N_1, N_2), (O_1, O_2), L, K^{NZ}, N_3, and $\{M_l\}_{l=1,\ldots,\upsilon}$ are given by Clause 5.2.2.2.5 and 5.2.2.2.5a in [6, TS 38.214].

Table 6.3.2.1.2-1A: PMI of *codebookType= typeII-r16* or *codebookType=etypeII-r19*

	Information fields X_1					
	$i_{1,1}$	$i_{1,2}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2 K^{NZ} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	N/A	N/A
Rank=3 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	N/A
Rank=4 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$
Rank=1 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2 K^{NZ} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	N/A	N/A
Rank=3 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	N/A
Rank=4 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$	$\lceil \log_2(2L) \rceil$

	Information fields X_2											
	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$i_{1,5}$	$i_{1,6,1}$	$i_{1,6,2}$	$i_{1,6,3}$	$i_{1,6,4}$	$\{i_{2,4,l}\}_{l=}$	$\{i_{2,5,l}\}_{l=1}$	$\{i_{1,7,l}\}_{l=}$
Rank =1 $N_3 \le 1$	4	N/A	N/A	N/A	N/A	$\lceil \log_2 \binom{N_3}{M_1} \rceil$	N/A	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$2LM_1$
Rank =2 $N_3 \le 1$	4	4	N/A	N/A	N/A	$\lceil \log_2 \binom{N_3}{M_2} \rceil$	$\lceil \log_2 \binom{N_3}{M_2} \rceil$	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$4LM_2$
Rank =3 $N_3 \le 1$	4	4	4	N/A	N/A	$\lceil \log_2 \binom{N_3}{M_3} \rceil$	$\lceil \log_2 \binom{N_3}{M_3} \rceil$	$\lceil \log_2 \binom{N_3}{M_3} \rceil$	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$6LM_3$
Rank =4 $N_3 \le 1$	4	4	4	4	N/A	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$3(K^{NZ}$	$4(K^{NZ}$	$8LM_4$
Rank =1 $N_3 > 1$	4	N/A	N/A	N/A	$\lceil \log_2(2$	$\lceil \log_2 \binom{2M}{M_1} \rceil$	N/A	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$2LM_1$
Rank =2 $N_3 > 1$	4	4	N/A	N/A	$\lceil \log_2(2$	$\lceil \log_2 \binom{2M}{M_2} \rceil$	$\lceil \log_2 \binom{2M}{M} \rceil$	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$4LM_2$
Rank =3 $N_3 > 1$	4	4	4	N/A	$\lceil \log_2(2$	$\lceil \log_2 \binom{2M}{M_3} \rceil$	$\lceil \log_2 \binom{2M}{M} \rceil$	$\lceil \log_2 \binom{2M}{M_3} \rceil$	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$6LM_3$
Rank =4 $N_3 > 1$	4	4	4	4	$\lceil \log_2(2$	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$\lceil \log_2 \binom{2M}{M} \rceil$	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$3(K^{NZ}$	$4(K^{NZ}$	$8LM_4$

Note: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-1A is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $2LM_\upsilon$, $3\left(K_l^{NZ}-1\right)$, and $4\left(K_l^{NZ}-1\right)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f}^{(3)}$, $k_{l,i,f}^{(2)}$, and $c_{l,i,f}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.5 in TS 38.214

[6] is the number of nonzero coefficients for layer l such that $K^{NZ}=\sum_{l=1}^{\upsilon} K_l^{NZ}$.

The bitwidth for PMI of *codebookType=typeII-CJT* is provided in Tables 6.3.2.1.2-1B, where the values of $(N_1,\ N_2)$, $(O_1,\ O_2)$, O_3, K^{NZ}, N_3, N_0, L_n, $\sigma(n)$ and $\{M_l\}_{l=1,\ldots,\upsilon}$ are given by Clause 5.2.2.2.8 in [6, TS 38.214].

Table 6.3.2.1.2-1B: PMI of *codebookType= typeII-CJT*

	Information fields X_1					
	$i_{1,1}$	$i_{1,2}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1 $N_3\le 19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2 K^{NZ}\rceil$	N/A	N/A	N/A
Rank=2 $N_3\le 19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n}$	N/A	N/A
Rank=3 $N_3\le 19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma}$	N/A
Rank=4 $N_3\le 19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n)})\rceil$
Rank=1 $N_3>19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2 K^{NZ}\rceil$	N/A	N/A	N/A
Rank=2 $N_3>19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n}$	N/A	N/A
Rank=3 $N_3>19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma}$	N/A
Rank=4 $N_3>19$	$N_0\lceil\log_2(C$	$\sum_{n=1}^{N_0}\lceil\log_2\binom{N_1N_2}{L_{\sigma(n)}}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma}$	$\lceil\log_2(\sum_{n=1}^{N_0}2L_{\sigma(n)})\rceil$

	Information fields X_2						
	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$i_{1,5}$	$i_{1,6,1}$	$i_{1,6,2}$
Rank=1 $N_3\le 19$	4	N/A	N/A	N/A	N/A	$\lceil\log_2\binom{N_3-1}{M_1-1}$	N/A
Rank=2 $N_3\le 19$	4	4	N/A	N/A	N/A	$\lceil\log_2\binom{N_3-1}{M_2-1}$	$\lceil\log_2\binom{N_3-1}{M_2-1}\rceil$
Rank=3 $N_3\le 19$	4	4	4	N/A	N/A	$\lceil\log_2\binom{N_3-1}{M_3-1}$	$\lceil\log_2\binom{N_3-1}{M_3-1}\rceil$
Rank=4 $N_3\le 19$	4	4	4	4	N/A	$\lceil\log_2\binom{N_3-1}{M_4-1}$	$\lceil\log_2\binom{N_3-1}{M_4-1}\rceil$
Rank=1 $N_3>19$	4	N/A	N/A	N/A	$\lceil\log_2(2M_1)\rceil$	$\lceil\log_2\binom{2M_1-}{M_1-}$	N/A
Rank=2 $N_3>19$	4	4	N/A	N/A	$\lceil\log_2(2M_2)\rceil$	$\lceil\log_2\binom{2M_2-}{M_2-}$	$\lceil\log_2\binom{2M_2-1}{M_2-1}\rceil$

Rank=3 $N_3>19$	4	4	4	N/A	$\lceil \log_2(2M_3) \rceil$	$\lceil \log_2 \binom{2M_3}{M_3} \rceil$	$\lceil \log_2 \binom{2M_3-1}{M_3-1} \rceil$
Rank=4 $N_3>19$	4	4	4	4	$\lceil \log_2(2M_4) \rceil$	$\lceil \log_2 \binom{2M_4}{M_4} \rceil$	$\lceil \log_2 \binom{2M_4-1}{M_4-1} \rceil$
	Information fields X_2						
	$i_{1,6,3}$	$i_{1,6,4}$	$i_{1,9}$	$\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$	
Rank=1 $N_3\le 19$	N/A	N/A	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	$2\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_1$	
Rank=2 $N_3\le 19$	N/A	N/A	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	$4\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_2$	
Rank=3 $N_3\le 19$	$\lceil \log_2 \binom{N_3-1}{M_3-1}$	N/A	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$6\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_3$	
Rank=4 $N_3\le 19$	$\lceil \log_2 \binom{N_3-1}{M_4-1}$	$\lceil \log_2 \binom{N_3-1}{M_4-1}$	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$8\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_4$	
Rank=1 $N_3>19$	N/A	N/A	$(N_0-1)\lceil \log_2 O_3 N$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	$2\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_1$	
Rank=2 $N_3>19$	N/A	N/A	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	$4\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_2$	
Rank=3 $N_3>19$	$\lceil \log_2 \binom{2M_3-1}{M_3-1}$	N/A	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$6\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_3$	
Rank=4 $N_3>19$	$\lceil \log_2 \binom{2M_4-1}{M_4-1}$	$\lceil \log_2 \binom{2M_4}{M_4}$	$(N_0-1)\lceil \log_2 O_3$ if *Mode1* is configured, NA otherwise	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$8\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_4$	

NOTE: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-1B is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $2\sum_{n=1}^{N_0} L_{\sigma(n)} M_\upsilon$, $3\left(K_l^{NZ}-1\right)$, and $4\left(K_l^{NZ}-1\right)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f,j}^{(3)}$, $k_{l,i,f,j}^{(2)}$, and $c_{l,i,f,j}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.8 in [6, TS 38.214] is the number of nonzero coefficients for layer l such that $K^{NZ}=\sum_{l=1}^{\upsilon} K_l^{NZ}$.

The bitwidth for PMI of *codebookType=typeII-Doppler* and *codebookType=typeII-Doppler-r19* is provided in Tables 6.3.2.1.2-1C, where the values of $(N_1,\ N_2)$, $(O_1,\ O_2)$, L, K^{NZ}, N_3, N_4, Q and $\{M_l\}_{l=1,\ldots,\upsilon}$ are given by Clause 5.2.2.2.10 and 5.2.2.2.11a in [6, TS 38.214].

Table 6.3.2.1.2-1C: PMI of *codebookType=typeII-Doppler* or *codebookType=typeII-Doppler-r19*

	Information fields X_1

	$i_{1,1}$	$i_{1,2}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2 K^{NZ} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ) \rceil$	N/A	N/A
Rank=3 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ) \rceil$	$\lceil \log_2(2LQ$	N/A
Rank=4 $N_3 \le 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ) \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ$
Rank=1 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2 K^{NZ} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ) \rceil$	N/A	N/A
Rank=3 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ) \rceil$	$\lceil \log_2(2LQ$	N/A
Rank=4 $N_3 > 19$	$\lceil \log_2(O_1 O_2) \rceil$	$\lceil \log_2 \binom{N_1 N_2}{L} \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ) \rceil$	$\lceil \log_2(2LQ$	$\lceil \log_2(2LQ$

	Information fields X_2								
	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$i_{1,5}$	$i_{1,6,1}$	$i_{1,6,2}$	$i_{1,6,3}$	$i_{1,6,4}$
Rank=1 $N_3 \le 19$	4	N/A	N/A	N/A	N/A	$\lceil \log_2 \binom{N_3-1}{M_1-1} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 \le 19$	4	4	N/A	N/A	N/A	$\lceil \log_2 \binom{N_3-1}{M_2-1} \rceil$	$\lceil \log_2 \binom{N_3-1}{M_2-1}$	N/A	N/A
Rank=3 $N_3 \le 19$	4	4	4	N/A	N/A	$\lceil \log_2 \binom{N_3-1}{M_3-1} \rceil$	$\lceil \log_2 \binom{N_3-1}{M_3-1}$	$\lceil \log_2 \binom{N_3-1}{M_3-1}$	N/A
Rank=4 $N_3 \le 19$	4	4	4	4	N/A	$\lceil \log_2 \binom{N_3-1}{M_4-1} \rceil$	$\lceil \log_2 \binom{N_3-1}{M_4-1}$	$\lceil \log_2 \binom{N_3-1}{M_4-1}$	$\lceil \log_2 \binom{N_3-1}{M_4-1}$
Rank=1 $N_3 > 19$	4	N/A	N/A	N/A	$\lceil \log_2(2M$	$\lceil \log_2 \binom{2M_1-}{M_1-1}$	N/A	N/A	N/A
Rank=2 $N_3 > 19$	4	4	N/A	N/A	$\lceil \log_2(2M$	$\lceil \log_2 \binom{2M_2-}{M_2-1}$	$\lceil \log_2 \binom{2M_2-1}{M_2-1}$	N/A	N/A
Rank=3 $N_3 > 19$	4	4	4	N/A	$\lceil \log_2(2M$	$\lceil \log_2 \binom{2M_3-}{M_3-1}$	$\lceil \log_2 \binom{2M_3-1}{M_3-1}$	$\lceil \log_2 \binom{2M_3}{M_3-}$	N/A
Rank=4 $N_3 > 19$	4	4	4	4	$\lceil \log_2(2M$	$\lceil \log_2 \binom{2M_4-}{M_4-1}$	$\lceil \log_2 \binom{2M_4-1}{M_4-1}$	$\lceil \log_2 \binom{2M_4}{M_4-}$	$\lceil \log_2 \binom{2M_4-}{M_4-}$

	Information fields X_2						
	$i_{1,10,1}$	$i_{1,10,2}$	$i_{1,10,3}$	$i_{1,10,4}$	$\{i_{2,4,l}\}_{l=1,\ldots,}$	$\{i_{2,5,l}\}_{l=1,\ldots,}$	$\{i_{1,7,l}\}_{l=1,\ldots,}$
Rank=1 $N_3 \le 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4 > 2$ and Q=2, NA otherwise	NA	NA	NA	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	$2LM_1Q$
Rank=2 $N_3 \le 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4 > 2$ and	$\lceil \log_2(N_4-1) \rceil$ if $N_4 > 2$ and	NA	NA	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	$4LM_2Q$

	Q=2, NA otherwise	Q=2, NA otherwise					
Rank=3 $N_3 \le 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	NA	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$6LM_3Q$
Rank=4 $N_3 \le 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$8LM_4Q$
Rank=1 $N_3 > 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	NA	NA	NA	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	$2LM_1Q$
Rank=2 $N_3 > 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	NA	NA	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	$4LM_2Q$
Rank=3 $N_3 > 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	NA	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$6LM_3Q$
Rank=4 $N_3 > 19$	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$\lceil \log_2(N_4-1) \rceil$ if $N_4>2$ and Q=2, NA otherwise	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$8LM_4Q$
NOTE: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-1C is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $2LM_\upsilon Q$, $3(K_l^{NZ}-1)$, and $4(K_l^{NZ}-1)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f,\tau}^{(3)}$, $k_{l,i,f,\tau}^{(2)}$, and $c_{l,i,f,\tau}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.10 in [6, TS 38.214] is the number of nonzero coefficients for layer l such that $K^{NZ}=\sum_{l=1}^{\upsilon} K_l^{NZ}$.							

The bitwidth for PMI of *codebookType*= *typeII-PortSelection* is provided in Tables 6.3.2.1.2-2, where the values of p_{CSI-RS}, $\boldsymbol{d}$, L, N_{PSK}, M_1, M_2, and $K^{(2)}$ are given by Clause 5.2.2.2.4 in [6, TS 38.214].

Table 6.3.2.1.2-2: PMI of *codebookType= typeII-PortSelection*

	Information fields X_1 for wideband PMI					Information fields X_2 for wideband PMI or per subband PMI			
	$i_{1,1}$	$i_{1,3,1}$	$i_{1,4,1}$	$i_{1,3,2}$	$i_{1,4,2}$	$i_{2,1,1}$	$i_{2,1,2}$	$i_{2,2,1}$	$i_{2,2,2}$
Rank =1 SBAmp off	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	N/A	N/A	$(M_1-1)\cdot \log_2 N_{\text{PSK}}$	N/A	N/A	N/A
Rank =2 SBAmp off	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$(M_1-1)\cdot \log_2 N_{\text{PSK}}$	$(M_2-1)\cdot \log_2 N_{\text{PSK}}$	N/A	N/A
Rank =1 SBAmp on	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	N/A	N/A	$\min(M_1, K^{(2)})\cdot \log_2 N_{\text{PSK}} - \log_2 N_{\text{PSK}} + 2\cdot(M_1 - \min(M_1, K^{(2)}))$	N/A	$\min(M_1, K^{(2)})-1$	N/A
Rank =2 SBAmp on	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$\lceil \log_2(2L) \rceil$	$3(2L-1)$	$\min(M_1, K^{(2)})\cdot \log_2 N_{\text{PSK}} - \log_2 N_{\text{PSK}} + 2\cdot(M_1 - \min(M_1, K^{(2)}))$	$\min(M_2, K^{(2)})\cdot \log_2 N_{\text{PSK}} - \log_2 N_{\text{PSK}} + 2\cdot(M_2 - \min(M_2, K^{(2)}))$	$\min(M_1, K^{(2)})-1$	$\min(M_2, K^{(2)})-1$

The bitwidth for PMI of *codebookType=typeII-PortSelection-r16* is provided in Tables 6.3.2.1.2-2A, where the values of $P_{CSI\text{-}RS}$, d, L, K^{NZ}, N_3, and $\{M_l\}_{l=1,\ldots,\upsilon}$ are given by Clause 5.2.2.2.6 in [6, TS 38.214].

Table 6.3.2.1.2-2A: PMI of *codebookType= typeII-PortSelection-r16*

	Information fields X_1				
	$i_{1,1}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1 $N_3 \le 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 K^{NZ} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 \le 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	N/A	N/A
Rank=3 $N_3 \le 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	N/A
Rank=4 $N_3 \le 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$
Rank=1 $N_3 > 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 K^{NZ} \rceil$	N/A	N/A	N/A
Rank=2 $N_3 > 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	N/A	N/A
Rank=3 $N_3 > 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	N/A
Rank=4 $N_3 > 19$	$\lceil \log_2 \lceil \frac{P_{CSI-RS}}{2d} \rceil \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$	$\lceil \log_2 (2L) \rceil$
	Information fields X_2				

	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$i_{1,5}$	$i_{1,6,1}$	$i_{1,6,2}$	$i_{1,6,3}$	$i_{1,6,4}$	$\{i_{2,4,l}\}_{l=1}$	$\{i_{2,5,l}\}_{l=1}$	$\{i_{1,7,l}\}_{l=1}$
Rank=1 $N_3 \le 1$	4	N/A	N/A	N/A	N/A	$\lceil \log_2 \binom{N_3}{M_1} \rceil$	N/A	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$2LM_1$
Rank=2 $N_3 \le 1$	4	4	N/A	N/A	N/A	$\lceil \log_2 \binom{N_3}{M_2} \rceil$	$\lceil \log_2 \binom{N_3}{M_2} \rceil$	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$4LM_2$
Rank=3 $N_3 \le 1$	4	4	4	N/A	N/A	$\lceil \log_2 \binom{N_3}{M_3} \rceil$	$\lceil \log_2 \binom{N_3}{M_3} \rceil$	$\lceil \log_2 \binom{N_3}{M_3} \rceil$	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$6LM_3$
Rank=4 $N_3 \le 1$	4	4	4	4	N/A	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$\lceil \log_2 \binom{N_3}{M_4} \rceil$	$3(K^{NZ}$	$4(K^{NZ}$	$8LM_4$
Rank=1 $N_3 > 1$	4	N/A	N/A	N/A	$\lceil \log_2($2	$\lceil \log_2 \binom{2M}{M_1} \rceil$	N/A	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$2LM_1$
Rank=2 $N_3 > 1$	4	4	N/A	N/A	$\lceil \log_2($2	$\lceil \log_2 \binom{2M}{M_2} \rceil$	$\lceil \log_2 \binom{2M}{M_2} \rceil$	N/A	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$4LM_2$
Rank=3 $N_3 > 1$	4	4	4	N/A	$\lceil \log_2($2	$\lceil \log_2 \binom{2M}{M_3} \rceil$	$\lceil \log_2 \binom{2M}{M_3} \rceil$	$\lceil \log_2 \binom{2M}{M_3} \rceil$	N/A	$3(K^{NZ}$	$4(K^{NZ}$	$6LM_3$
Rank=4 $N_3 > 1$	4	4	4	4	$\lceil \log_2($2	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$\lceil \log_2 \binom{2M}{M_4} \rceil$	$3(K^{NZ}$	$4(K^{NZ}$	$8LM_4$

NOTE: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-2A is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $2LM_\upsilon$, $3\left(K_l^{NZ}-1\right)$, and $4\left(K_l^{NZ}-1\right)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f}^{(3)}$, $k_{l,i,f}^{(2)}$, and $c_{l,i,f}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.5 in TS 38.214 [6] is the number of nonzero coefficients for layer l such that $K^{NZ}=\sum_{l=1}^{\upsilon} K_l^{NZ}$.

The bitwidth for PMI of *codebookType=typeII-PortSelection-r17* and *codebookType=typeII-FePortSelection-r19* is provided in Tables 6.3.2.1.2-2B, where the values of P_{CSI-RS}, K_1, K^{NZ}, N_3, N and M are given by Clause 5.2.2.2.7 and 5.2.2.2.9a in [6, TS 38.214].

Table 6.3.2.1.2-2B: PMI of *codebookType= typeII-PortSelection-r17* and *typeII-FePortSelection-r19*

	Information fields X_1					
	$i_{1,2}$	$i_{1,6}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1	$\lceil \log_2\left(\frac{P_{CSI-RS}}{K_1/2}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	N/A	N/A	N/A
Rank=2	$\lceil \log_2\left(\frac{P_{CSI-RS}}{K_1/2}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	N/A	N/A
Rank=3	$\lceil \log_2\left(\frac{P_{CSI-RS}}{K_1/2}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	N/A
Rank=4	$\lceil \log_2\left(\frac{P_{CSI-RS}}{K_1/2}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$

	Information fields X_2						
	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$
Rank=1	4	N/A	N/A	N/A	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	N/A if $K^{NZ}=K_1 M$; $K_1 M$ otherwise
Rank=2	4	4	N/A	N/A	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	N/A if $K^{NZ}=2K_1 M$; $2K_1 M$ otherwise
Rank=3	4	4	4	N/A	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$3K_1 M$
Rank=4	4	4	4	4	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$4K_1 M$

NOTE: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-2B is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $K_1 M$, $3\left(K_l^{NZ}-1\right)$, and $4\left(K_l^{NZ}-1\right)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f}^{(3)}$, $k_{l,i,f}^{(2)}$, and $c_{l,i,f}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.7 in TS 38.214 [6] is the number of nonzero coefficients for layer l such that $K^{NZ}=\sum_{l=1}^{\upsilon} K_l^{NZ}$.

If *CSI-ReportSubConfig* is configured, for a corresponding CSI sub-report, the mapping order of CSI fields of one CSI sub-report is determined following the procedure in this clause 6.3.2.1.2, by replacing CSI report #n in the following Tables 6.3.2.1.2-3 and 6.3.2.1.2-4 with CSI sub-report #n, and taking only Tables 6.3.1.1.2-1/1A/1B/2/2A/3/4 for the determination of the bitwidth of a CSI field.

The bitwidth for PMI of *codebookType=typeII-CJT-PortSelection* is provided in Tables 6.3.2.1.2-2C, where the values of P_{CSI-RS}, $\left\{K_{1,n}\right\}$, K^{NZ}, N_3, N, M and $\sigma(n)$ are given by Clause 5.2.2.2.9 in [6, TS 38.214].

Table 6.3.2.1.2-2C: PMI of *codebookType= typeII-CJT-PortSelection*

	Information fields X_1					
	$i_{1,2}$	$i_{1,6}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1	$\sum_{n=1}^{N_0} \lceil \log_2 \binom{P_{CSI}}{K_{1,\sigma}}$	$\lceil \log_2(N-1)$ if $N > M=2$, N/A otherwise	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	N/A	N/A	N/A
Rank=2	$\sum_{n=1}^{N_0} \lceil \log_2 \binom{P_{CSI}}{K_{1,\sigma}}$	$\lceil \log_2(N-1)$ if $N > M=2$, N/A otherwise	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	N/A	N/A
Rank=3	$\sum_{n=1}^{N_0} \lceil \log_2 \binom{P_{CSI}}{K_{1,\sigma}}$	$\lceil \log_2(N-1)$ if $N > M=2$, N/A otherwise	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	N/A
Rank=4	$\sum_{n=1}^{N_0} \lceil \log_2 \binom{P_{CSI}}{K_{1,\sigma}}$	$\lceil \log_2(N-1)$ if $N > M=2$, N/A otherwise	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma(n}$	$\lceil \log_2 \left(\sum_{n=1}^{N_0} K_{1,\sigma}$

	Information fields X_2							
	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$	$i_{1,9}$	$\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$
Rank=1	4	N/A	N/A	N/A	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	$(N_0-1)\lceil \log_2 O \rceil$ if *Mode1* is configured, NA otherwise	N/A if $K^{NZ} = \sum_{n=1}^{N_0} K_{1,\sigma(n)} M$; $\sum_{n=1}^{N_0} K_{1,\sigma(n)} M$ otherwise
Rank=2	4	4	N/A	N/A	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	$(N_0-1)\lceil \log_2 O \rceil$ if *Mode1* is configured, NA otherwise	N/A if $K^{NZ} = 2\sum_{n=1}^{N_0} K_{1,\sigma(n)} M$; $2\sum_{n=1}^{N_0} K_{1,\sigma(n)} M$ otherwise
Rank=3	4	4	4	N/A	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$(N_0-1)\lceil \log_2 O \rceil$ if *Mode1* is configured, NA otherwise	$3 \cdot \sum_{n=1}^{N_0} K_{1,\sigma(n)} M$
Rank=4	4	4	4	4	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$(N_0-1)\lceil \log_2 O \rceil$ if *Mode1* is configured, NA otherwise	$4 \cdot \sum_{n=1}^{N_0} K_{1,\sigma(n)} M$

NOTE: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-2C is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $\sum_{n=1}^{N_0} K_{1,\sigma(n)} M$, $3(K_l^{NZ}-1)$, and $4(K_l^{NZ}-1)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f,j}^{(3)}$, $k_{l,i,f,j}^{(2)}$, and $c_{l,i,f,j}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.9 in [6, TS 38.214] is the number of nonzero coefficients for layer l such that $K^{NZ} = \sum_{l=1}^{\upsilon} K_l^{NZ}$.

The bitwidth for PMI of *codebookType=typeII-Doppler-PortSelection* is provided in Tables 6.3.2.1.2-2D, where the values of P_{CSI-RS}, K_1, K^{NZ}, N_3, N and M are given by Clause 5.2.2.2.11 in [6, TS 38.214].

Table 6.3.2.1.2-2D: PMI of *codebookType= typeII-Doppler-PortSelection*

	Information fields X_1					
	$i_{1,2}$	$i_{1,6}$	$i_{1,8,1}$	$i_{1,8,2}$	$i_{1,8,3}$	$i_{1,8,4}$
Rank=1	$\lceil \log_2\left(\begin{matrix}P_{CSI-RS}/2\\K_1/2\end{matrix}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	N/A	N/A	N/A
Rank=2	$\lceil \log_2\left(\begin{matrix}P_{CSI-RS}/2\\K_1/2\end{matrix}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	N/A	N/A
Rank=3	$\lceil \log_2\left(\begin{matrix}P_{CSI-RS}/2\\K_1/2\end{matrix}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	N/A
Rank=4	$\lceil \log_2\left(\begin{matrix}P_{CSI-RS}/2\\K_1/2\end{matrix}\right) \rceil$	$\lceil \log_2(N-1) \rceil$ if $N > M=2$, N/A otherwise	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$	$\lceil \log_2(K_1 M) \rceil$

	Information fields X_2						
	$i_{2,3,1}$	$i_{2,3,2}$	$i_{2,3,3}$	$i_{2,3,4}$	$\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$	$\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$
Rank=1	4	N/A	N/A	N/A	$3(K^{NZ}-1)$	$4(K^{NZ}-1)$	N/A if $K^{NZ}=K_1 M$; $K_1 M$ otherwise
Rank=2	4	4	N/A	N/A	$3(K^{NZ}-2)$	$4(K^{NZ}-2)$	N/A if $K^{NZ}=2K_1 M$; $2K_1 M$ otherwise
Rank=3	4	4	4	N/A	$3(K^{NZ}-3)$	$4(K^{NZ}-3)$	$3K_1 M$
Rank=4	4	4	4	4	$3(K^{NZ}-4)$	$4(K^{NZ}-4)$	$4K_1 M$

NOTE: the bitwidth for $\{i_{1,7,l}\}_{l=1,\ldots,\upsilon}$, $\{i_{2,4,l}\}_{l=1,\ldots,\upsilon}$ and $\{i_{2,5,l}\}_{l=1,\ldots,\upsilon}$ shown in Table 6.3.2.1.2-2D is the total bitwidth of $\{i_{1,7,l}\}$, $\{i_{2,4,l}\}$ and $\{i_{2,5,l}\}$ up to Rank = υ, respectively, and the corresponding per layer bitwidths are $K_1 M$, $3(K_l^{NZ}-1)$, and $4(K_l^{NZ}-1)$, (i.e., 1, 3, and 4 bits for each respective indicator elements $k_{l,i,f}^{(3)}$, $k_{l,i,f}^{(2)}$, and $c_{l,i,f}$, respectively), where K_l^{NZ} as defined in Clause 5.2.2.2.11 in [6, TS 38.214] is the number of nonzero coefficients for layer l such that $K^{NZ}=\sum_{l=1}^{\upsilon} K_l^{NZ}$.

The bitwidth for MRI and SRS-RSRP are provided in Table 6.3.2.1.2-2E.

Table 6.3.2.1.2-2E: Bitwidth for MRI and SRS-RSRP

Field	Bitwidth
MRI	$\lceil \log_2(K_s^{SRS-RSRP}) \rceil$
SRS-RSRP	7
Differential RSRP	4
NOTE: $K_s^{SRS-RSRP}$ is the number of SRS-RSRP measurement resources in the corresponding resource set.	

For CSI on PUSCH, two UCI bit sequences are generated, $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ and $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$. The CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-6, are mapped to the UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ starting with $a_0^{(1)}$. The CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-7, are mapped to the UCI bit sequence $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ starting with $a_0^{(2)}$.

The mapping order of CSI fields of one report for CSI reporting for L1/L2-triggered mobility and handover as defined in Clause 5.2.4a of [6, TS 38.214] is provided in Table 6.3.1.1.2-7 by taking only Tables 6.3.1.1.2-1/3 for the determination of the bitwidth of a CSI field. The mapping order of CSI fields of one report for CRI/RSRP or SSBRI/RSRP or CRI/RSRP/CapabilityIndex or SSBRI/RSRP/CapabilityIndex reporting is provided in Table 6.3.1.1.2-8. The mapping order of CSI fields of one report for inter-cell SSBRI/RSRP reporting is provided in Table 6.3.1.1.2-8. The mapping order of CSI fields of one report for CRI/SINR or SSBRI/SINR or CRI/SINR/CapabilityIndex or SSBRI/SINR/CapabilityIndex reporting is provided in Table 6.3.1.1.2-8A. The mapping order of CSI fields of one report for group-based CRI/RSRP or SSBRI/RSRP reporting is provided in Table 6.3.1.1.2-8B. The mapping order of CSI fields of one report for SSBRI/RSRP or CRI/RSRP reporting for L1/L2-triggered mobility is provided in Table 6.3.1.1.2-8C. The mapping order of CSI fields of one report for MRI/CLI-RSSI is provided in Table 6.3.1.1.2-8D. The mapping order of CSI fields of one report for predicted CRI/RSRP or predicted SSBRI/RSRP reporting is provided in Table 6.3.1.1.2-8E. The mapping order of CSI fields of one report for time instance indicator/predicted CRI/predicted RSRP or time instance indicator/predicted SSBRI/predicted RSRP reporting is provided in Table 6.3.1.1.2-8F. The mapping order of CSI fields of one report for CRI/RSRP or SSBRI/RSRP if *nrofReportedRS* is configured is provided in Table 6.3.1.1.2-8G. The mapping order of CSI fields of one report for RS-PAI is provided in Table 6.3.1.1.2-8H. The procedure in clause 6.3.2 described for CSI part 1 is also applicable for one report for CRI/RSRP, SSBRI/RSRP, predicted CRI/RSRP, predicted SSBRI/RSRP, time instance indicator/predicted CRI/predicted RSRP, time instance indicator/predicted SSBRI/predicted RSRP, RS-PAI, CSI-PAI, CRI/SINR, SSBRI/SINR, MRI/SRS-RSRP, MRI/CLI-RSSI reporting, CSI reporting for L1/L2-triggered mobility, CSI reporting for handover, TDCP reporting, delay offset reporting, frequency offset reporting, both delay offset and frequency offset reporting, and phase offset reporting.

Table 6.3.2.1.2-3: Mapping order of CSI fields of one CSI report, CSI part 1

CSI report number	CSI fields
CSI report #n CSI part 1	CRI as in Tables 6.3.1.1.2-3/4/6, if reported
	Rank Indicator as in Tables 6.3.1.1.2-3//3C 4/5 or 6.3.2.1.2-8/8A/8B/9/9A, if reported
	Wideband CQI for the first TB as in Tables 6.3.1.1.2-3//3C 4/5 or 6.3.2.1.2-8/8A/8B/9/9A, if reported
	Subband differential CQI for the first TB with increasing order of subband number as in Tables 6.3.1.1.2-3//3C 4/5 or 6.3.2.1.2-8/8A/8B/9/9A, if reported
	Indicator of the number of non-zero wideband amplitude coefficients M_0 for layer 0 as in Table 6.3.1.1.2-5, if reported
	Indicator of the number of non-zero wideband amplitude coefficients M_1 for layer 1 as in Table 6.3.1.1.2-5 (if the rank according to the reported RI is equal to one, this field is set to all zeros), if 2-layer PMI reporting is allowed according to the rank restriction in Clauses 5.2.2.2.3 and 5.2.2.2.4 [6, TS 38.214] and if reported
	Indicator of the N_0 selected CSI-RS resources by a bitmap with N_{TRP} bits, this field is present only if $N_{TRP}>1$ and *restrictedCMR-Selection* is configured to *OFF*
	Indicator of selected L_n value combination or α_n value combination with bitwidth of $\lceil \log_2(N_L) \rceil$, this field is present only if $N_L>1$
	Indicator of the total number of non-zero coefficients summed across all layers K^{NZ} as in Tables 6.3.2.1.2-8/8A/8B/9/9A, if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* as subband 0.	

Table 6.3.2.1.2-3A: Mapping order of CSI fields of one CSI report, CSI part 1, *csi-ReportMode= Mode 1*

CSI report number	CSI fields
CSI report #n CSI part 1	CRI as in Table 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported
	Rank Combination Indicator as in Table 6.3.1.1.2-3A, if reported
	Wideband CQI for the first TB as in Table 6.3.1.1.2-3A, if reported
	Subband differential CQI for the first TB with increasing order of subband number as in Table 6.3.1.1.2-3A, if reported
	CRI as in Tables 6.3.1.1.2-3B, if associated with one CSI-RS resource, *numberOfSingleTRP-CSI-Mode1 = 1* and if reported; First CRI as in Table 6.3.1.1.2-3B, if associated with one CSI-RS resource, *numberOfSingleTRP-CSI-Mode1 = 2* and if reported
	Rank Indicator associated with CRI as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Rank Indicator associated with the first CRI as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Wideband CQI associated with CRI for the first TB as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Wideband CQI associated with the first CRI for the first TB as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1 = 2* and if reported
	Subband differential CQI associated with CRI for the first TB with increasing order of subband number as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 1 if reported; Subband differential CQI associated with the first CRI for the first TB with increasing order of subband number as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Second CRI as in Table 6.3.1.1.2-3B, if associated with one CSI-RS resource, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Rank Indicator associated with the second CRI as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Wideband CQI associated with the second CRI for the first TB as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Subband differential CQI associated with the second CRI for the first TB with increasing order of subband number as in Table 6.3.1.1.2-3B, if *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* as subband 0.	

Table 6.3.2.1.2-3B: Mapping order of CSI fields of one CSI report, CSI part 1, *csi-ReportMode= Mode 2*

CSI report number	CSI fields
CSI report #n CSI part 1	CRI as in Table 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; CRI as in Table 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported
	Rank Combination Indicator as in Table 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; Rank Indicator as in Table 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported; Zero padding bits O_P, if needed
	Wideband CQI for the first TB as in Table 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; Wideband CQI for the first TB as in Table 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported
	Subband differential CQI for the first TB with increasing order of subband number as in Table 6.3.1.1.2-3A, if associated with one CSI-RS resource pair and if reported; Subband differential CQI for the first TB with increasing order of subband number as in Table 6.3.1.1.2-3B, if associated with one CSI-RS resource and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* as subband 0.	

The number of zero padding bits O_P in Table 6.3.2.1.2-3B is 0 for 1 CSI-RS port and $O_P = N_{max} - N_{reported}(R)$ for more than 1 CSI-RS port, where:

- $N_{max} = \max_{r \in S_{Rank}} N(r)$. S_{Rank} is the set of rank and rank combination values r that are allowed to be reported. $N(r)$ is obtained according to Tables 6.3.1.1.2-3A/3B for rank combination indicator and rank indicator respectively.
- $N_{reported}(R)$ is obtained according to Tables 6.3.1.1.2-3A for rank combination indicator and R is the reported rank combination
- $N_{reported}(R)$ is obtained according to Tables 6.3.1.1.2-3B for rank indicator and R is the reported rank

Table 6.3.2.1.2-3C: Mapping order of CSI fields of one CSI report for *reportQuantity=tdcp*

CSI report number	CSI fields
CSI report #n	Amplitude value for the configured delay values as in Table 6.3.2.1.2-10 based on the order from the first configured delay to the last configured delay
	Phase value for the configured delay values as in Table 6.3.2.1.2-10 based on the order from the first configured delay to the last configured delay, if reported

Table 6.3.2.1.2-3D: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 1

CSI report number	CSI fields
CSI report #n CSI part 1	Rank Indicator for the first CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Wideband CQI for the first TB for the first CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Subband differential CQI for the first TB with increasing order of subband number for the first CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Indicator of the total number of non-zero coefficients summed across all layers K^{NZ} for the first CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8D, if reported
	…
	Rank Indicator for the M_R-th CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Wideband CQI for the first TB for the M_R-th CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Subband differential CQI for the first TB with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Indicator of the total number of non-zero coefficients summed across all layers K^{NZ} for the M_R-th CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8D, if reported
	…
	First reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Rank Indicator for the first reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Zero padding bits $O_{P,1}$ for the first reported CRI, if needed
	Wideband CQI for the first TB for the first reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Subband differential CQI for the first TB with increasing order of subband number for the first reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Indicator of the total number of non-zero coefficients summed across all layers K^{NZ} for the first reported CRI as in Tables 6.3.2.1.2-8D, if reported
	…
	The (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Rank Indicator for the (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Zero padding bits $O_{P,M-M_R}$ for the (M-M_R)-th reported CRI, if needed
	Wideband CQI for the first TB for the (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Subband differential CQI for the first TB with increasing order of subband number for the (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8C or 6.3.2.1.2-8D, if reported
	Indicator of the total number of non-zero coefficients summed across all layers K^{NZ} for the (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8D, if reported
NOTE1: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* as subband 0.	
NOTE2: M_R is set to 0 if higher layer parameter *mrSelectedResources* is not configured for the CSI report.	

The number of zero padding bits $O_{P,k}$ in Table 6.3.1.1.2-3D is 0 for 1 CSI-RS port and $O_{P,k} = N_{max} - N_{reported,k}$ for more than 1 CSI-RS port, where:

- $N_{max} = \max_{i \in S}\left(N_i\right)$, where S is the set of CRIs corresponding to the K_s-M_R resources and N_i is the payload size of RI field for the i-th CRI obtained according to Table 6.3.2.1.2-8C or 6.3.2.1.2-8D;
- $N_{reported,k}$ is the payload size of RI field for the k-th CRI obtained according to Table 6.3.2.1.2-8C or 6.3.2.1.2-8D.

Table 6.3.2.1.2-3E: Mapping order of CSI fields of one CSI report for *reportQuantity=cjtc-Dd*

CSI report number	CSI fields
CSI report #n	The reference TRS resource set index as in Table 6.3.2.1.2-11.
	The delay offset for each TRS resource set configured for the CSI report as in Table 6.3.2.1.2-11, except that indicated by reference TRS resource set, ordered from the lowest to highest CSI-RS resource set ID.
	The inside/outside CP indicator for each TRS resource set configured for the CSI report as in Table 6.3.2.1.2-11, except that indicated by reference TRS resource set, ordered from the lowest to highest CSI-RS resource set ID.

Table 6.3.2.1.2-3F: Mapping order of CSI fields of one CSI report for *reportQuantity=cjtc-F*

CSI report number	CSI fields
CSI report #n	The reference TRS resource set index as in Table 6.3.2.1.2-12.
	The frequency offset for each TRS resource set configured for the CSI report as in Table 6.3.2.1.2-12, except that indicated by reference TRS resource set, ordered from the lowest to highest CSI-RS resource set ID.

Table 6.3.2.1.2-3G: Mapping order of CSI fields of one CSI report for *reportQuantity=cjtc-Dd-F*

CSI report number	CSI fields
CSI report #n	The reference TRS resource set index for delay offset as in Table 6.3.2.1.2-11.
	The reference TRS resource set index for frequency offset as in Table 6.3.2.1.2-12.
	The delay offset for each TRS resource set configured for the CSI report as in Table 6.3.2.1.2-11, except that indicated by reference TRS resource set, ordered from the lowest to highest CSI-RS resource set ID.
	The inside/outside CP indicator for each TRS resource set configured for the CSI report as in Table 6.3.2.1.2-11, except that indicated by reference TRS resource set, ordered from the lowest to highest CSI-RS resource set ID.
	The frequency offset for each TRS resource set configured for the CSI report as in Table 6.3.2.1.2-12, except that indicated by reference TRS resource set, ordered from the lowest to highest CSI-RS resource set ID.

Table 6.3.2.1.2-3H: Mapping order of CSI fields of one CSI report for *reportQuantity=cjtc-P*

CSI report number	CSI fields
CSI report #n	The reference CSI-RS resource index as in Table 6.3.2.1.2-13.
	Subband phase offsets for CSI-RS resource #0 as in Table 6.3.2.1.2-13 if subband reporting is configured, with increasing order of subband number; otherwise, wideband phase offset for CSI-RS resource #0 as in Table 6.3.2.1.2-13.
	Subband phase offsets for CSI-RS resource #1 as in Table 6.3.2.1.2-13 if subband reporting is configured, with increasing order of subband number; otherwise, wideband phase offset for CSI-RS resource #1 as in Table 6.3.2.1.2-13.
	…
	Subband phase offsets for CSI-RS resource #$\left(N_{TRP} - 1 \right)$ as in Table 6.3.2.1.2-13 if subband reporting is configured, with increasing order of subband number; otherwise, wideband phase offset for CSI-RS resource #$\left(N_{TRP} - 1 \right)$ as in Table 6.3.2.1.2-13.
NOTE: The CSI-RS resources are ordered from the lowest to highest CSI-RS resource ID, except that indicated by reference CSI-RS resource index. N_{TRP} is the number of CSI-RS resources configured for the CSI report.	

Table 6.3.2.1.2-3I: Mapping order of CSI fields of one report for CSI report configuration indicator/CRI/RSRP/Condition met indicator, or CSI report configuration indicator/SSBRI/RSRP/Condition met indicator

CSI report number	CSI fields
CSI report #n	CSI report configuration indicator as in Table 6.3.2.1.2-14, if reported
	CRI or SSBRI #1 as in Table 6.3.2.1.2-14
	CRI or SSBRI #2 as in Table 6.3.2.1.2-14, if reported
	CRI or SSBRI #3 as in Table 6.3.2.1.2-14, if reported
	CRI or SSBRI #4 as in Table 6.3.2.1.2-14, if reported
	RSRP #1 as in Table 6.3.2.1.2-14
	Differential RSRP #2 as in Table 6.3.2.1.2-14, if reported
	Differential RSRP #3 as in Table 6.3.2.1.2-14, if reported
	Differential RSRP #4 as in Table 6.3.2.1.2-14, if reported
	Condition met indicator #1 in Table 6.3.2.1.2-14, if reported
	Condition met indicator #2 in Table 6.3.2.1.2-14, if reported
	Condition met indicator #3 in Table 6.3.2.1.2-14, if reported
	Condition met indicator #4 in Table 6.3.2.1.2-14, if reported
	RSRP as in Table 6.3.2.1.2-14 for the reference signal in the indicated TCI state or the SS/PBCH block which is QCLed with the reference signal in the indicated TCI state if *eventType* is set to 'event1' and *enabledCurrentBeamReport* is configured; Differentatial RSRP as in Table 6.3.2.1.2-14 for the reference signal in the indicated TCI state or the SS/PBCH block which is QCLed with the reference signal in the indicated TCI state if *eventType* is set to 'event2' and if *enabledCurrentBeamReport* is configured; Differentatial RSRP as in Table 6.3.2.1.2-14 for the reference signal with the *valueOfQ* highest L1-RSRP out of reference signals among the activated TCI states or the SS/PBCH block which is QCLed with the reference signal with the *valueOfQ* highest L1-RSRP out of refence signals among the activated TCI states if *eventType* is set to 'event7' and if *enabledCurrentBeamReport* is configured;
	Zero padding bits O_P, if needed

NOTE:
The CSI report configuration indicator with value i indicates the (i+1)-th CSI report configuration among the CSI report configurations associated with a same PUCCH resource configured by *firstPUCCHResourceConfig-UEIBR*, which are ordered in ascending order of *CSI-ReportConfigID*.

The condition met indicators are reported only when the UE is configured with higher layer parameter *PresenceOfConditionMetIndicator*. If the reference signal associated with CRI or SSBRI *#j* satisfies the condition for transmitting UEIRI as defined in Clause 5.2.1.5.4.1 in [6, TS 38.214] when *eventDetectionTimeWindowLength* is configured, the value of the condition met indicator *#j* is set to 1; otherwise set to 0.

The number of zero padding bits O_P in Table 6.3.2.1.2-3I is 0 if there is only one CSI report configuration associated with the PUCCH carrying UEIRI; otherwise $O_P = N_{max} - N_{reported}$, where:

- $N_{max} = \max_{i \in S}\left(N_i\right)$, where S is the set of CSI report configurations associated with a same PUCCH resource configured by *firstPUCCHResourceConfig-UEIBR*, and N_i is the payload size of the report for the *i*-th CSI report configuration, obtained according to Table 6.3.2.1.2-3I prior to padding;
- $N_{reported}$ is the payload size of the current report, obtained according to Table 6.3.2.1.2-3I prior to padding.

Table 6.3.2.1.2-3J: Mapping order of CSI fields of one CSI report for *reportQuantity=cli-SRS-RSRP*

CSI report number	CSI fields
CSI report #n	MRI #1 as in Table 6.3.2.1.2-2E
	MRI #2 as in Table 6.3.2.1.2-2E, if reported
	MRI #3 as in Table 6.3.2.1.2-2E, if reported
	MRI #4 as in Table 6.3.2.1.2-2E, if reported
	SRS-RSRP #1 as in Table 6.3.2.1.2-2E
	Differential SRS-RSRP #2 as in Table 6.3.2.1.2-2E, if reported
	Differential SRS-RSRP #3 as in Table 6.3.2.1.2-2E, if reported
	Differential SRS-RSRP #4 as in Table 6.3.2.1.2-2E, if reported

Table 6.3.2.1.2-3K: Mapping order of CSI fields of one CSI report for CSI-PAI

CSI report number	CSI fields
CSI report #n CSI part 1	$SGCS_1^1$ as in Table 6.3.2.1.2-15
	$SGCS_2^1$ as in Table 6.3.2.1.2-15, if reported
	…
	$SGCS_{\upsilon_{Max}}^1$ as in Table 6.3.2.1.2-15, if reported
	$SGCS_1^2$ as in Table 6.3.2.1.2-15
	$SGCS_2^2$ as in Table 6.3.2.1.2-15, if reported
	…
	$SGCS_{\upsilon_{Max}}^2$ as in Table 6.3.2.1.2-15, if reported
NOTE: υ_{Max} is the number of max allowed rank according to the rank restriction for the CSI report. For layer $l=1,\ldots,\upsilon$, where υ is the value of the reported Rank Indicator for the associated CSI report of predicted PMI configured with the higher layer parameter *XYZ*, the values of $SGCS_l^1$ and $SGCS_l^2$ are determined according to Clause 5.2.1.4.6 TS 38.214 [6]. For layer $l=\upsilon+1,\ldots,\upsilon_{Max}$, if any, the values of $SGCS_l^1$ and $SGCS_l^2$ are set to '1111'.	

Table 6.3.2.1.2-4: Mapping order of CSI fields of one CSI report, CSI part 2 wideband

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3/3C/4/5, if present and reported
	Layer Indicator as in Tables 6.3.1.1.2-3/3C/4/5, if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A or 6.3.2.1.2-1/2, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A or 6.3.2.1.2-1/2, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported

Table 6.3.2.1.2-4A: Mapping order of CSI fields of one CSI report, CSI part 2 wideband, *csi-ReportMode= Mode 1*

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Two Layer Indicators as in Table 6.3.1.1.2-3A, where the first Layer Indicator and the second Layer Indicator are associated with the first resource and the second resource within the resource pair respectively and if reported;
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the first resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the first CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the second resource within the CSI-RS resource pair, if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the second CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and if reported
	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if associated with CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if associated with the first CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Layer Indicator as in Table 6.3.1.1.2-3B, if associated with CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Layer Indicator as in Table 6.3.1.1.2-3B, if associated with the first CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if associated with CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if associated with the first CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if associated with CRI in CSI part 1, *pmi-FormatIndicator= widebandPMI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if associated with the first CRI in CSI part 1, *pmi-FormatIndicator= widebandPMI, numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if associated with the second CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Layer Indicator as in Table 6.3.1.1.2-3B, if associated with the second CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if associated with the second CRI in CSI part 1, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if associated with the second CRI in CSI part 1, *pmi-FormatIndicator= widebandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported

Table 6.3.2.1.2-4B: Mapping order of CSI fields of one CSI report, CSI part 2 wideband, *csi-ReportMode= Mode 2*

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Wideband CQI for the second TB as in Tables 6.3.1.1.2-3B, if reported part 1 is associated with one CSI-RS resource and if reported
	Two Layer Indicators as in Table 6.3.1.1.2-3A, if reported part 1 is associated with one CSI-RS resource pair, where the first Layer Indicator and the second Layer Indicator are associated with the first resource and the second resource within the resource pair respectively and if reported; Layer Indicator as in Table 6.3.1.1.2-3B, if reported part 1 is associated with one CSI-RS resource and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the first resource within the CSI-RS resource pair, if reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the first CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1 associated with the second CSI-RS resource within the CSI-RS resource pair, if reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] associated with the second CSI-RS resource within the CSI-RS resource pair, if *pmi-FormatIndicator= widebandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI wideband information fields X_1, from left to right as in Tables 6.3.1.1.2-1, if reported part 1 is associated with one CSI-RS resource and if reported
	PMI wideband information fields X_2, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and reported part 1 is associated with one CSI-RS resource and if reported

Table 6.3.2.1.2-4C: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 2 wideband

CSI report number	CSI fields
CSI report #n CSI part 2 wideband	Wideband CQI for the second TB for the first CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C, if present and reported
	Layer Indicator for the first CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C, if reported
	PMI wideband information fields X_1 for the first CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the first CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the first CRI among the non-reported M_R CRIs according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported
	…
	Wideband CQI for the second TB for the M_R-th CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C, if present and reported
	Layer Indicator for the M_R-th CRI among the non-reported M_R CRIs as in Tables 6.3.2.1.2-8C, if reported
	PMI wideband information fields X_1 for the M_R-th CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the M_R-th CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the M_R-th CRI among the non-reported M_R CRIs according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported
	…
	Wideband CQI for the second TB for the first reported CRI as in Tables 6.3.2.1.2-8C, if present and reported
	Layer Indicator for the first reported CRI as in Tables 6.3.2.1.2-8C, if reported
	PMI wideband information fields X_1 for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the first reported CRI according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported
	…
	Wideband CQI for the second TB for the (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8C, if present and reported
	Layer Indicator for the (M-M_R)-th reported CRI as in Tables 6.3.2.1.2-8C, if reported
	PMI wideband information fields X_1 for the (M-M_R)-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, if reported
	PMI wideband information fields X_2 for the (M-M_R)-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports for the (M-M_R)-th reported CRI according to Clause 5.2.2.2.1 in [6, TS38.214], if *pmi-FormatIndicator= widebandPMI* and if reported
NOTE: M_R is set to 0 if higher layer parameter [mrSelectedResources] is not configured for the CSI report.	

Table 6.3.2.1.2-5: Mapping order of CSI fields of one CSI report, CSI part 2 subband

CSI report #n Part 2 subband	Subband differential CQI for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4/5, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A or 6.3.2.1.2-1/2, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4/5, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A or 6.3.2.1.2-1/2, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
NOTE: Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* as subband 0.	

Table 6.3.2.1.2-5A: Mapping order of CSI fields of one CSI report, CSI part 2 of *codebookType=typeII-r16 or typeII-PortSelection-r16*

CSI report number	CSI fields
CSI report #n CSI part 2, group 0	PMI fields X_1, from left to right as in Tables 6.3.2.1.2-1A/2A, if reported
CSI report #n CSI part 2, group 1	The following PMI fields X_2, from left to right, as in Tables 6.3.2.1.2-1A/2A: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$, $i_{1,5}$, $\{i_{1,6,l}: l=1,\ldots,\upsilon\}$ and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $v*2LM_\upsilon - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2, from left to right, as in Tables 6.3.2.1.2-1A/2A : $min\left(K^{NZ} - v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $min\left(K^{NZ} - v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported

Table 6.3.2.1.2-5B: Mapping order of CSI fields of one CSI report, CSI part 2 of *codebookType=typeII-PortSelection-r17* or *typeII-Doppler-PortSelection*

CSI report number	CSI fields
CSI report #n CSI part 2, group 0	PMI fields X_1, from left to right as in Table 6.3.2.1.2-2B/2D, if reported
CSI report #n CSI part 2, group 1	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-2B/2D: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$ $(max\,(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon)) \times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}, (max\,(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon)) \times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $v * K_1 M - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-2B/2D : $(min(K^{NZ} - v,\ \lfloor \frac{K^{NZ}}{2} \rfloor)) \times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}, (min(K^{NZ} - v,\ \lfloor \frac{K^{NZ}}{2} \rfloor)) \times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported

Table 6.3.2.1.2-5C: Mapping order of CSI fields of one CSI report, CSI part 2 subband, *ReportMode= Mode 1*

CSI report #n Part 2 subband	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with the first CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Subband differential CQI for the second TB of all even subbands with increasing order of subband

	number associated with the second CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with the first CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 1 and if reported; PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with the second CRI in CSI part 1, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second CRI in CSI part 1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI*, *numberOfSingleTRP-CSI-Mode1* = 2 and if reported

Table 6.3.2.1.2-5D: Mapping order of CSI fields of one CSI report, CSI part 2 subband, *ReportMode= Mode 2*

CSI report #n Part 2 subband	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the first resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with the second resource within the CSI-RS resource pair, according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource pair and if reported
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number associated with one CSI-RS resource, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI* and reported part 1 is associated with one CSI-RS resource and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports associated with one CSI-RS resource according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number associated with one CSI-RS resource, as in Tables 6.3.1.1.2-3B, if *cqi-FormatIndicator=subbandCQI* and reported part 1 is associated with one CSI-RS resource and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/2, or codebook index for 2 antenna ports associated with one CSI-RS resource according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and reported part 1 is associated with one CSI-RS resource and if reported

Table 6.3.2.1.2-5E: Mapping order of CSI fields of one CSI report, CSI part 2 of *codebookType=typeII-CJT*

CSI report number	CSI fields
CSI report #n CSI part 2, group 0	PMI fields X_1, from left to right as in Table 6.3.2.1.2-1B, if reported;
CSI report #n CSI part 2, group 1	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-1B: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$, $i_{1,5}$, $\{i_{1,6,l}: l=1,\ldots,\upsilon\}$, $i_{1,9}$, and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\nu * 2\cdot\sum_{n=1}^{N_0} L_{\sigma(n)} M_\upsilon - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l, i_j, f, j)$ defined in clause 5.2.3 of TS 38.214 [6] where j is equivalent to n, if present and if reported;
CSI report #n CSI part 2, group 2	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-1B : $min\left(K^{NZ}-\nu, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $min\left(K^{NZ}-\nu, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l, i_j, f, j)$ defined in clause 5.2.3 of TS 38.214 [6] where j is equivalent to n, if reported;

Table 6.3.2.1.2-5F: Mapping order of CSI fields of one CSI report, CSI part 2 of *codebookType= typeII-Doppler*

CSI report number	CSI fields
CSI report #n CSI part 2, group 0	PMI fields X_1, from left to right as in Table 6.3.2.1.2-1C, if reported; The second time-domain wideband CQI as in Table 6.3.2.1.2-8B, if present and reported
CSI report #n CSI part 2, group 1	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-1C: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$, $i_{1,5}$, $\{i_{1,6,l}: l=1,\ldots,\upsilon\}$, $\{i_{1,10,l}: l=1,\ldots,\upsilon\}$, and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right) \times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right) \times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $v*2LM_{\upsilon}Q - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ for $N_4=1$ or $Pri(l,i,f,j)$ for $N_4>1$ defined in clause 5.2.3 of TS 38.214 [6], if present and if reported; The second time-domain subband differential CQI of all even subbands with increasing order of subband number, as in Table 6.3.2.1.2-8B, if present and if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-1C : $min\left(K^{NZ} - v, \lfloor \frac{K^{NZ}}{2} \rfloor\right) \times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $min\left(K^{NZ} - v, \lfloor \frac{K^{NZ}}{2} \rfloor\right) \times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ for $N_4=1$ or $Pri(l,i,f,j)$ for $N_4>1$ defined in clause 5.2.3 of TS 38.214 [6], if reported; The second time-domain subband differential CQI of all odd subbands with increasing order of subband number, as in Table 6.3.2.1.2-8B, if present and if reported

Table 6.3.2.1.2-5G: Mapping order of CSI fields of one CSI report, CSI part 2 of *codebookType=typeII-CJT-PortSelection*

CSI report number	CSI fields
CSI report #n CSI part 2, group 0	PMI fields X_1, from left to right as in Table 6.3.2.1.2-2C, if reported
CSI report #n CSI part 2, group 1	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-2C: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$ $(max\,(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon)) \times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}, (max\,(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon)) \times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$, $i_{1,9}$ and $v * \sum_{n=1}^{N_0} K_{1,\sigma(n)} M - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,\ i_j, f, j)$ defined in clause 5.2.3 of TS 38.214 [6] where j is equivalent to n, if present and if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2, from left to right, as in Table 6.3.2.1.2-2C : $(min\left(K^{NZ} - v,\ \lfloor \frac{K^{NZ}}{2} \rfloor\right)) \times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}, (min\left(K^{NZ} - v,\ \lfloor \frac{K^{NZ}}{2} \rfloor\right)) \times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,\ i_j, f, j)$ defined in clause 5.2.3 of TS 38.214 [6] where j is equivalent to n, if reported

Table 6.3.2.1.2-5H: Mapping order of CSI fields of one CSI report containing N_n^{sub} CSI sub-report(s), CSI part 2 subband

CSI report #n Part 2 subband	Subband differential CQI of CSI sub-report #1 for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #1 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI of CSI sub-report #2 for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #2 of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #2 according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI of CSI sub-report #N_n^{sub} for the second TB of all even subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #N_n^{sub} of all even subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #N_n^{sub} according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI of CSI sub-report #1 for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #1 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #1 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI of CSI sub-report #2 for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #2 of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #2 according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI of CSI sub-report #N_n^{sub} for the second TB of all odd subbands with increasing order of subband number, as in Tables 6.3.1.1.2-3/3C/4, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of CSI sub-report #N_n^{sub} of all odd subbands with increasing order of subband number, from left to right as in Tables 6.3.1.1.2-1/1A/1B/2/2A, or codebook index for 2 antenna ports of CSI sub-report #N_n^{sub} according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number, if *pmi-FormatIndicator= subbandPMI* and if reported

Note:
Subbands for given CSI report *n* indicated by the higher layer parameter *csi-ReportingBand* with value set to '1' are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* with value set to '1' as subband 0.

CSI sub-report #1, CSI sub-report #2, ⋯, CSI sub-report #N_n^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.

Table 6.3.2.1.2-5l: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 2 subband

CSI report #n Part 2 subband	Subband differential CQI for the second TB of all even subbands with increasing order of subband number for the first CRI among the non-reported M_R CRIs, as in Tables 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number for the first CRI among the non-reported M_R CRIs, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number for the first CRI among the non-reported M_R CRIs, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number for the first CRI among the non-reported M_R CRIs, as in Table 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number for the first CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number for the first CRI among the non-reported M_R CRIs, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs, as in Tables 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs, as in Table 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number for the M_R-th CRI among the non-reported M_R CRIs, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number for the first reported CRI, as in Tables 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number for the first reported CRI, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number for the first reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number for the first reported CRI, as in Table 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number for the first reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number for the first reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
	…
	Subband differential CQI for the second TB of all even subbands with increasing order of subband number for the (M-M_R)-th reported CRI, as in Tables 6.3.2.1.2-8C, if *cqi-FormatIndicator=subbandCQI* and if reported
	PMI subband information fields X_2 of all even subbands with increasing order of subband number for the (M-M_R)-th reported CRI, from left to right as in Table 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all even subbands with increasing order of subband number for the (M-M_R)-th reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
	Subband differential CQI for the second TB of all odd subbands with increasing order of subband number for the (M-M_R)-th reported CRI, as in Table 6.3.2.1.2-8C, if *cqi-*

	FormatIndicator=subbandCQI and if reported
	PMI subband information fields X_2 of all odd subbands with increasing order of subband number for the (M-M_R)-th reported CRI, from left to right as in Tables 6.3.1.1.2-1, or codebook index for 2 antenna ports according to Clause 5.2.2.2.1 in [6, TS38.214] of all odd subbands with increasing order of subband number for the (M-M_R)-th reported CRI, if *pmi-FormatIndicator= subbandPMI* and if reported
NOTE1: Subbands for given CSI report n indicated by the higher layer parameter *csi-ReportingBand* are numbered continuously in the increasing order with the lowest subband of *csi-ReportingBand* as subband 0.	
NOTE2: M_R is set to 0 if higher layer parameter [mrSelectedResources] is not configured for the CSI report.	

Table 6.3.2.1.2-5J: Mapping order of CSI fields of one CSI report configured with *valueOfM*, CSI part 2 of *codebookType=typeII-r16*

CSI report number	CSI fields
CSI report #n CSI part 2, group 0	PMI fields X_1 for the first CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.2.1.2-1A, if reported … PMI fields X_1 for the M_R-th CRI among the non-reported M_R CRIs, from left to right as in Tables 6.3.2.1.2-1A, if reported PMI fields X_1 for the first reported CRI, from left to right as in Tables 6.3.2.1.2-1A, if reported … PMI fields X_1 for the (M-M_R)-th reported CRI, from left to right as in Tables 6.3.2.1.2-1A, if reported
CSI report #n CSI part 2, group 1	The following PMI fields X_2 for the first CRI among the non-reported M_R CRIs, from left to right, as in Tables 6.3.2.1.2-1A: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$, $i_{1,5}$, $\{i_{1,6,l}: l=1,\ldots,\upsilon\}$ and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $v*2LM_\upsilon - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2 for the first CRI among the non-reported M_R CRIs, from left to right, as in Tables 6.3.2.1.2-1A: $min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
	…
CSI report #n CSI part 2, group 1	The following PMI fields X_2 for the M_R-th CRI among the non-reported M_R CRIs, from left to right, as in Tables 6.3.2.1.2-1A: $\{i_{2,3,l}: l=1,\ldots,\upsilon\}$, $i_{1,5}$, $\{i_{1,6,l}: l=1,\ldots,\upsilon\}$ and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 3$ highest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 4$ highest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $v*2LM_\upsilon - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2 for the M_R-th CRI among the non-reported M_R CRIs, from left to right, as in Tables 6.3.2.1.2-1A: $min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 3$ lowest priority bits of $\{i_{2,4,l}: l=1,\ldots,\upsilon\}$, $min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 4$ lowest priority bits of $\{i_{2,5,l}: l=1,\ldots,\upsilon\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\{i_{1,7,l}: l=1,\ldots,\upsilon\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
	…
CSI report #n	The following PMI fields X_2 for the first reported CRI, from left to right, as in Tables 6.3.2.1.2-

CSI part 2, group 1	1A: $\left\{i_{2,3,l}: l=1,\ldots,\upsilon\right\}$, $i_{1,5}$, $\left\{i_{1,6,l}: l=1,\ldots,\upsilon\right\}$ and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 3$ highest priority bits of $\left\{i_{2,4,l}: l=1,\ldots,\upsilon\right\}, max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 4$ highest priority bits of $\left\{i_{2,5,l}: l=1,\ldots,\upsilon\right\}$ and $v*2LM_{\upsilon} - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\left\{i_{1,7,l}: l=1,\ldots,\upsilon\right\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2 for the first reported CRI, from left to right, as in Tables 6.3.2.1.2-1A: $min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 3$ lowest priority bits of $\left\{i_{2,4,l}: l=1,\ldots,\upsilon\right\}, min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 4$ lowest priority bits of $\left\{i_{2,5,l}: l=1,\ldots,\upsilon\right\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\left\{i_{1,7,l}: l=1,\ldots,\upsilon\right\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
	...
CSI report #n CSI part 2, group 1	The following PMI fields X_2 for the (M-M_R)-th reported CRI, from left to right, as in Tables 6.3.2.1.2-1A: $\left\{i_{2,3,l}: l=1,\ldots,\upsilon\right\}$, $i_{1,5}$, $\left\{i_{1,6,l}: l=1,\ldots,\upsilon\right\}$ and $max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 3$ highest priority bits of $\left\{i_{2,4,l}: l=1,\ldots,\upsilon\right\}, max\left(0, \lceil \frac{K^{NZ}}{2} \rceil - \upsilon\right)\times 4$ highest priority bits of $\left\{i_{2,5,l}: l=1,\ldots,\upsilon\right\}$ and $v*2LM_{\upsilon} - \lfloor K^{NZ}/2 \rfloor$ highest priority bits of $\left\{i_{1,7,l}: l=1,\ldots,\upsilon\right\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
CSI report #n CSI part 2, group 2	The following PMI fields X_2 for the (M-M_R)-th reported CRI, from left to right, as in Tables 6.3.2.1.2-1A: $min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 3$ lowest priority bits of $\left\{i_{2,4,l}: l=1,\ldots,\upsilon\right\}, min\left(K^{NZ}-v, \lfloor \frac{K^{NZ}}{2} \rfloor\right)\times 4$ lowest priority bits of $\left\{i_{2,5,l}: l=1,\ldots,\upsilon\right\}$ and $\lfloor K^{NZ}/2 \rfloor$ lowest priority bits of $\left\{i_{1,7,l}: l=1,\ldots,\upsilon\right\}$, in decreasing order of priority based on the corresponding function $Pri(l,i,f)$ defined in clause 5.2.3 of TS 38.214 [6], if reported
NOTE: M_R is set to 0 if higher layer parameter [mrSelectedResources] is not configured for the CSI report.	

Table 6.3.2.1.2-6: Mapping order of CSI reports to UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$, with two-part CSI report(s)

UCI bit sequence	CSI report number
$a_0^{(1)}$ $a_1^{(1)}$	CSI part 1 of CSI report #1 as in Table 6.3.2.1.2-3/3A/3B/3C/3D/3E/3F/3G/3H/3I/3J/3K or Table 6.3.1.1.2-7/8/8A/8B/8C/8D/8E/8F/8G/8H
$a_2^{(1)}$ $a_3^{(1)}$	CSI part 1 of CSI report #2 as in Table 6.3.2.1.2-3/3A/3B/3C/3D/3E/3F/3G/3H/3I/3J/3K or Table 6.3.1.1.2-8/8A/8B/8C/8D/8E/8F/8G/8H
	…
$\vdots$ $a_{A^{(1)}-1}^{(1)}$	CSI part 1 of CSI report #n as in Table 6.3.2.1.2-3/3A/3B/3C/3D/3E/3F/3G/3H/3I/3J/3K or Table 6.3.1.1.2-8/8A/8B/8C/8D/8E/8F/8G/8H
NOTE: For a CSI report #i containing N_i^{sub} CSI sub-reports, where $i \in \{1,2,\ldots,n\}$, CSI part 1 of all CSI sub-reports are mapped to the corresponding segment of the UCI bit sequence of CSI report #i, from upper part to lower part of the segment, in increasing order of CSI sub-report number. CSI sub-report #1, CSI sub-report #2, …, CSI sub-report #N_i^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.	

where CSI report #1, CSI report #2, ⋯, CSI report #n in Table 6.3.2.1.2-6 correspond to the CSI reports in increasing order of CSI report priority values according to Clause 5.2.5 of [6, TS38.214].

Table 6.3.2.1.2-7: Mapping order of CSI reports to UCI bit sequence $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$, with two-part CSI report(s)

UCI bit sequence	CSI report number
$a_0^{(2)}$ $a_1^{(2)}$ $a_2^{(2)}$ $a_3^{(2)}$ $\vdots$ $a_{A^{(2)}-1}^{(2)}$	CSI report #1, CSI part 2 wideband, as in Table 6.3.2.1.2-4/4A/4B/4C, or CSI part 2 with group 0, as in Table 6.3.2.1.2-5A/5B/5E/5F/5G/5J, if CSI part 2 exists for CSI report #1
	CSI report #2, CSI part 2 wideband, as in Table 6.3.2.1.2-4/4A/4B/4C, or CSI part 2 with group 0, as in Table 6.3.2.1.2-5A/5B/5E/5F/5G/5J, if CSI part 2 exists for CSI report #2
	…
	CSI report #n, CSI part 2 wideband, as in Table 6.3.2.1.2-4/4A/4B/4C, or CSI part 2 with group 0, as in Table 6.3.2.1.2-5A/5B/5E/5F/5G/5J, if CSI part 2 exists for CSI report #n
	CSI report #1, CSI part 2 subband, as in Table 6.3.2.1.2-5/5C/5D/5H/5I, or CSI part 2 with group 1 and 2, as in Table 6.3.2.1.2-5A/5B/5E/5F/5G/5J, if CSI part 2 exists for CSI report #1
	CSI report #2, CSI part 2 subband, as in Table 6.3.2.1.2-5/5C/5D/5H/5I, or CSI part 2 with group 1 and 2, as in Table 6.3.2.1.2-5A/5B/5E/5F/5G/5J, if CSI part 2 exists for CSI report #2
	…
	CSI report #n, CSI part 2 subband, as in Table 6.3.2.1.2-5/5C/5D/5H/5I, or CSI part 2 with group 1 and 2, as in Table 6.3.2.1.2-5A/5B/5E/5F/5G/5J, if CSI part 2 exists for CSI report #n

NOTE: For a CSI report #i containing N_i^{sub} CSI sub-reports, where $i \in \{1,2,\ldots,n\}$,

- CSI part 2 wideband of all CSI sub-reports are mapped to the corresponding segment of the UCI bit sequence of CSI report #i, from upper part to lower part of the segment, in increasing order of CSI sub-report number;
- CSI sub-report #1, CSI sub-report #2, ⋯, CSI sub-report #N_i^{sub} correspond to the CSI sub-reports in increasing order of *reportSubConfigId*.

where CSI report #1, CSI report #2, ⋯, CSI report #n in Table 6.3.2.1.2-7 correspond to the CSI reports in increasing order of CSI report priority values according to Clause 5.2.5 of [6, TS38.214].

The bitwidth for RI/CQI of *codebookType= typeII-r16*, *codebookType=typeII-PortSelection-r16* or *codebookType= eTypeII-r19* is provided in Table 6.3.2.1.2-8.

Table 6.3.2.1.2-8: RI and CQI of codebookType=typeII-r16, typeII-PortSelection-r16 or codebookType= eTypeII-r19

Field	Bitwidth
Rank Indicator	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
Indicator of the total number of non-zero coefficients summed across all layers K^{NZ}	$\lceil \log_2(K_0) \rceil$ if max allowed rank is 1; $\lceil \log_2(2K_0) \rceil$ otherwise
NOTE: n_{RI} is the number of allowed rank indicator values according to Clauses 5.2.2.2.5, 5.2.2.2.5a and 5.2.2.2.6 [6, TS 38.214], $K_0 = \lceil 2L \lceil p_1 \times \frac{N_3}{R} \rceil \beta \rceil$, where p_1, N_3, R, and β are given by Clause 5.2.2.2.5, 5.2.2.2.5a and 5.2.2.2.6 in [6, TS 38.214]. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. The values of the K^{NZ} indicator field are mapped to the allowed values of K^{NZ}, according to Clauses 5.2.2.2.5, 5.2.2.2.5a and 5.2.2.2.6 [6, TS 38.214], with increasing order, where '0' is mapped to $K^{NZ}=1$.	

The bitwidth for RI/CQI of *codebookType= typeII-CJT* is provided in Table 6.3.2.1.2-8A.

Table 6.3.2.1.2-8A: RI and CQI of *codebookType= typeII-CJT*

Field	Bitwidth
Rank Indicator	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
Indicator of the total number of non-zero coefficients summed across all layers, and all CSI-RS resources if configured, K^{NZ}	$\lceil \log_2 \lceil 2\beta M_1 L_{max} \rceil \rceil$ if max allowed rank is 1; $\lceil \log_2 2 \lceil 2\beta M_1 L_{max} \rceil \rceil$ otherwise
NOTE: n_{RI} is the number of allowed rank indicator values according to Clause 5.2.2.2.8 TS 38.214 [6], L_{max} is the maximum of $\sum_{n=1}^{N_{TRP}} L_n$ for all N_L configured L_n combinations, where $\left(L_n\right)_{n=1,\ldots,N_{TRP}}$, M_1, and β are given by Clause 5.2.2.2.8 in TS 38.214 [6]. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. The values of the K^{NZ} indicator field are mapped to the allowed values of K^{NZ}, according to Clause 5.2.2.2.8 TS 38.214 [6], with increasing order, where '0' is mapped to $K^{NZ}=1$.	

The bitwidth for RI/CQI of *codebookType= typeII-Doppler* and *codebookType= typeII-Doppler-r19* is provided in Table 6.3.2.1.2-8B.

Table 6.3.2.1.2-8B: RI and CQI of *codebookType= typeII-Doppler or codebookType= typeII-Doppler-r19*

Field	Bitwidth
Rank Indicator	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
The second time-domain wide-band CQI	4
The second time-domain subband differential CQI	2
Indicator of the total number of non-zero coefficients summed across all layers K^{NZ}	$\lceil \log_2(K_0) \rceil$ if max allowed rank is 1; $\lceil \log_2(2K_0) \rceil$ otherwise
NOTE: n_{RI} is the number of allowed rank indicator values according to Clause 5.2.2.2.10 and 5.2.2.2.11a in [6, TS 38.214], $K_0 = \lceil 2L \lceil p_1 \times \frac{N_3}{R} \rceil \beta Q \rceil$, where p_1, N_3, R, β and Q are given by Clause 5.2.2.2.10 and 5.2.2.2.11a in [6, TS 38.214]. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. The values of the K^{NZ} indicator field are mapped to the allowed values of K^{NZ}, according to Clause 5.2.2.2.10 and 5.2.2.2.11a in [6, TS 38.214], with increasing order, where '0' is mapped to $K^{NZ}=1$.	

The bitwidth for RI/LI/CQI/CRI of a CSI report configured with *valueOfM* and *codebookType=typeI-SinglePanel* is provided in Table 6.3.2.1.2-8C.

Table 6.3.2.1.2-8C: RI, LI, CQI, and CRI of a CSI report configured with *valueOfM* and co*codebookType=typeI-SinglePanel*

Field	Bitwidth			
	2 antenna ports	4 antenna ports	>4 antenna ports	
			Rank1~4	Rank5~8
Rank Indicator	$min\left(1, \lceil \log_2 n_{RI} \rceil\right)$	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$	$\lceil \log_2 n_{RI} \rceil$	$\lceil \log_2 n_{RI} \rceil$
Layer Indicator	$\lceil \log_2 \upsilon \rceil$	$min\left(2, \lceil \log_2 \upsilon \rceil\right)$	$min\left(2, \lceil \log_2 \upsilon \rceil\right)$	$min\left(2, \lceil \log_2 \upsilon \rceil\right)$
Wide-band CQI for the first TB	4	4	4	4
Wideband CQI for the second TB	0	0	0	4
Subband differential CQI for the first TB	2	2	2	2
Subband differential CQI for the second TB	0	0	0	2
CRI	$\lceil \log_2\left(K_S^{CSI-RS} - M_R\right) \rceil$	$\lceil \log_2\left(K_S^{CSI-RS} - M_R\right) \rceil$	$\lceil \log_2\left(K_S^{CSI-RS} - M_R\right) \rceil$	$\lceil \log_2\left(K_S^{CSI-RS} - M_R\right) \rceil$
NOTE: n_{RI} is the number of allowed rank indicator values according to Clause 5.2.2.2.1 [6, TS 38.214]. υ is the value of the rank. The value of K_S^{CSI-RS} is the number of CSI-RS resources in the corresponding resource set, M_R is the number of CSI-RS resources that are configured to report their corresponding CSI (except CRI), and is set to 0 if higher layer parameter *mrSelectedResources* is not configured for the CSI report. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value.				

The bitwidth for RI/CQI/CRI of a CSI report configured with *valueOfM* and *codebookType= typeII-r16* is provided in Table 6.3.2.1.2-8D.

Table 6.3.2.1.2-8D: RI, CQI and CRI of a CSI report configured with *valueOfM* and *codebookType=typeII-r16*

Field	Bitwidth
Rank Indicator	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
CRI	$\lceil \log_2\left(K_S^{CSI-RS} - M_R\right) \rceil$
Indicator of the total number of non-zero coefficients summed across all layers K^{NZ}	$\lceil \log_2(K_0) \rceil$ if max allowed rank is 1; $\lceil \log_2(2K_0) \rceil$ otherwise

NOTE: n_{RI} is the number of allowed rank indicator values according to Clauses 5.2.2.2.5 and 5.2.2.2.6 [6, TS 38.214], K_S^{CSI-RS} is the number of CSI-RS resources in the corresponding resource set, M_R is the number of CSI-RS resources that are configured to report their corresponding CSI (except CRI), and is set to 0 if higher layer parameter *mrSelectedResources* is not configured for the CSI report, $K_0 = \lceil 2L \lceil p_1 \times \frac{N_3}{R} \rceil \beta \rceil$, where p_1, N_3, R, and β are given by Clause 5.2.2.2.5 and 5.2.2.2.6 in [6, TS 38.214]. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. The values of the K^{NZ} indicator field are mapped to the allowed values of K^{NZ}, according to Clauses 5.2.2.2.5 and 5.2.2.2.6 [6, TS 38.214], with increasing order, where '0' is mapped to $K^{NZ} = 1$.

The bitwidth for RI/CQI of *codebookType=typeII-PortSelection-r17*, *typeII-Doppler-PortSelection* or *codebookType= typeII-FePortSelection-r19* is provided in Table 6.3.2.1.2-9.

Table 6.3.2.1.2-9: RI and CQI of codebookType=typeII-PortSelection-r17, typeII-Doppler-PortSelection, or codebookType= typeII-FePortSelection-r19

Field	Bitwidth
Rank Indicator	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
Indicator of the total number of non-zero coefficients summed across all layers K^{NZ}	$\lceil \log_2(K_0) \rceil$ if max allowed rank is 1; $\lceil \log_2(2K_0) \rceil$ otherwise

NOTE: n_{RI} is the number of allowed rank indicator values according to Clauses 5.2.2.2.7, 5.2.2.2.9a and Clauses 5.2.2.2.11 [6, TS 38.214], $K_0 = \lceil K_1 M \beta \rceil$, where K_1, M, and β are given by Clause 5.2.2.2.7, 5.2.2.2.9a and Clauses 5.2.2.2.11 in [6, TS 38.214]. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. The values of the K^{NZ} indicator field are mapped to the allowed values of K^{NZ}, according to Clauses 5.2.2.2.7, 5.2.2.2.9a and Clauses 5.2.2.2.11 [6, TS 38.214], with increasing order, where '0' is mapped to $K^{NZ} = 1$.

The bitwidth for RI/CQI of *codebookType=typeII-CJT-PortSelection* is provided in Table 6.3.2.1.2-9A.

Table 6.3.2.1.2-9A: RI and CQI of codebookType= typeII-CJT-PortSelection

Field	Bitwidth
Rank Indicator	$min\left(2, \lceil \log_2 n_{RI} \rceil\right)$
Wide-band CQI	4
Subband differential CQI	2
Indicator of the total number of non-zero coefficients summed across all layers, and all CSI-RS resources if configured, K^{NZ}	$\lceil \log_2 \lceil \beta M K_{1,max} \rceil \rceil$ if max allowed rank is 1; $\lceil \log_2 2 \lceil \beta M K_{1,max} \rceil \rceil$ otherwise
NOTE: n_{RI} is the number of allowed rank indicator values according to Clause 5.2.2.2.9 TS 38.214 [6], $K_{1,max}$ is the maximum of $\sum_{n=1}^{N_{TRP}} K_{1,n}$ for all N_L configured α_n combinations, where $\left(K_{1.n}\right)_{n=1,\ldots,N_{TRP}}$, M, and β are given by Clause 5.2.2.2.9 in TS 38.214 [6], and. The values of the rank indicator field are mapped to allowed rank indicator values with increasing order, where '0' is mapped to the smallest allowed rank indicator value. The values of the K^{NZ} indicator field are mapped to the allowed values of K^{NZ}, according to Clause 5.2.2.2.9 TS 38.214 [6], with increasing order, where '0' is mapped to $K^{NZ}=1$.	

The bitwidth for *reportQuantity=tdcp* is provided in Table 6.3.2.1.2-10.

Table 6.3.2.1.2-10: Amplitude and phase values for *reportQuantity=tdcp*

Field	Bitwidth
Amplitude value	4
Phase value	4

The bitwidth for *reportQuantity=cjtc-Dd* is provided in Table 6.3.2.1.2-11.

Table 6.3.2.1.2-11: Reference TRS resource set index, delay offset and inside/outside indicator for *reportQuantity=cjtc-Dd*

Field	Bitwidth
Reference TRS resource set index	$\lceil \log_2 N_{TRP} \rceil$, where N_{TRP} is the number of TRS resource sets configured for the CSI report.
Delay offset	$\lceil \log_2 M_D \rceil$, where M_D is configured by *valueOfMD*.
Inside/outside CP indicator	1

The bitwidth for *reportQuantity=cjtc-F* is provided in Table 6.3.2.1.2-12.

Table 6.3.2.1.2-12: Reference TRS resource set index, and frequency offset for *reportQuantity=cjtc-F*

Field	Bitwidth
Reference TRS resource set index	$\lceil \log_2 N_{TRP} \rceil$, where N_{TRP} is the number of TRS resource sets configured for the CSI report.
Frequency offset	$\lceil \log_2 M_{FO} \rceil$, where M_{FO} is configured by *valueofMFO*.

The bitwidth for *reportQuantity=cjtc-P* is provided in Table 6.3.2.1.2-13.

Table 6.3.2.1.2-13: Reference CSI-RS resource index, and phase offset for *reportQuantity=cjtc-P*

Field	Bitwidth
Reference CSI-RS resource index	$\lceil \log_2 N_{TRP} \rceil$, where N_{TRP} is the number of CSI-RS resources configured for the CSI report.
Wideband Phase offset	$\lceil \log_2 M_{\Phi} \rceil$, where M_{Φ} is configured by *valueOfMPhi*.
Subband Phase offset	$\lceil \log_2 M_{\Phi} \rceil$, where M_{Φ} is configured by *valueOfMPhi*.

The bitwidth for CRI, SSBRI, RSRP, differential RSRP, CSI report configuration indicator and Condition met indicator are provided in Table 6.3.2.1.2-14.

Table 6.3.2.1.2-14: CRI, SSBRI, RSRP, CapabilityIndex, CSI report configuration indicator and Condition met indicator

Field	Bitwidth
CRI	$\lceil \log_2(K_s^{CSI\text{-}RS}) \rceil$
SSBRI	$\lceil \log_2(K_s^{SSB}) \rceil$
RSRP	7
Differential RSRP	4
CSI report configuration indicator	$\lceil \log_2(K_s^{CSI_config}) \rceil$
Condition met indicator	1
NOTE: $K_s^{CSI\text{-}RS}$ is the number of CSI-RS resources in the corresponding resource set, K_s^{SSB} is the configured number of SS/PBCH blocks in the corresponding resource set for reporting 'ssb-Index-RSRP' or 'ssb-Index-RSRP-Index', $K_s^{CSI_config}$ is the number of CSI report configurations associated with a same PUCCH resource configured by *firstPUCCHResourceConfig-UEIBR*.	

The bitwidth for CSI-PAI is provided in Table 6.3.2.1.2-15.

Table 6.3.2.1.2-15: CSI-PAI

Field	Bitwidth
$SGCS_l^1$	4
$SGCS_l^2$	4
NOTE: $SGCS_l^1$ and $SGCS_l^2$ for the l-th layer are the SGCS values as defined by Clause 5.2.1.4.6 in [6, TS 38.214].	

6.3.2.1.3 CG-UCI

For CG-UCI bits transmitted on a CG PUSCH when the higher layer parameter *cg-RetransmissionTimer* is configured, the CG-UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ is determined as follows:

- set $a_i = \tilde{o}_i^{CG\text{-}UCI}$ for $i = 0,1, \ldots, O^{CG\text{-}UCI}-1$ and $A = O^{CG\text{-}UCI}$, where the CG-UCI bit sequence $\tilde{o}_0^{CG\text{-}UCI}, \tilde{o}_1^{CG\text{-}UCI}, \ldots, \tilde{o}_{O^{CG\text{-}UCI}-1}^{CG\text{-}UCI}$ is given by Table 6.3.2.1.3-1, mapped in the order from upper part to lower part.

Table 6.3.2.1.3-1: Mapping order of CG-UCI fields

Field	Bitwidth
HARQ process number	5 if *nrofHARQ-Processes-v1700* in *ConfiguredGrantConfig* is configured; 4 otherwise.
Redundancy version	2
New data indicator	1
Channel Occupancy Time (COT) sharing information	$\lceil \log_2 C \rceil$ if both higher layer parameter *ul-toDL-COT-SharingED-Threshold* and higher layer parameter *cg-COT-SharingList* are configured, or if both higher layer parameter *semiStaticChannelAccessConfigUE* and higher layer parameter *cg-COT-SharingList* are configured, or if higher layer parameter *cg-COT-SharingList* is configured in frequency range 2-2, where C is the number of combinations configured in *cg-COT-SharingList;* 1 if higher layer parameter *ul-toDL-COT-SharingED-Threshold* is not configured, and if higher layer parameter *semiStaticChannelAccessConfigUE* is not configured, and if higher layer parameter *cg-COT-SharingOffset* is configured; 0 otherwise. If a UE indicates COT sharing other than "no sharing" in a CG PUSCH within the UE's initiated COT, the UE should provide consistent COT sharing information in all the subsequent CG PUSCHs, if any, occurring within the same UE's initiated COT such that the same DL starting point and duration are maintained.

6.3.2.1.3A UTO-UCI

For UTO-UCI bits transmitted on a CG PUSCH when the higher layer parameter *nrofBitsInUTO-UCI* is configured, the UTO-UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ is determined as follows:

- set $a_i = \tilde{o}_i^{UTO-UCI}$ for $i = 0,1, \ldots, O^{UTO-UCI} - 1$ and $A = O^{UTO-UCI}$, where $O^{UTO-UCI}$ is provided by *nrofBitsInUTO-UCI*, and the UTO-UCI bit sequence $\tilde{o}_0^{UTO-UCI}, \tilde{o}_1^{UTO-UCI}, \ldots, \tilde{o}_{O^{UTO-UCI}-1}^{UTO-UCI}$ is given by clause 9.3.1 of [5, TS 38.213].

6.3.2.1.3B UEIRI

If UEIRI bits are transmitted on a PUSCH, the UEIRI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ is determined as follows:

- set $a_i = \tilde{o}_i^{UEIRI}$ for $i = 0,1, \ldots, O^{UEIRI} - 1$ and $A = O^{UEIRI}$, where O^{UEIRI} is the number of PUCCH resources for UEIRI given by Clause 9 of [5, TS38.213], and the UEIRI bit sequence $\tilde{o}_0^{UEIRI}, \tilde{o}_1^{UEIRI}, \ldots, \tilde{o}_{O^{UEIRI}-1}^{UEIRI}$ is mapped to the PUCCH resources according to an ascending order of *pucch-ResourceId*. If the associated UEIRI of $\tilde{o}_i^{UEIRI}$ is positive, $\tilde{o}_i^{UEIRI} = 1$; otherwise, $\tilde{o}_i^{UEIRI} = 0$.

6.3.2.1.4 HARQ-ACK and CG-UCI/UTO-UCI

If the higher layer parameter *nrofBitsInUTO-UCI* is configured, the procedure in this clause 6.3.2.1.4 applies by replacing CG-UCI with UTO-UCI in all the notations and texts, replacing "When higher layer parameter *cg-UCI-Multiplexing* is configured" with "When UTO-UCI and HARQ-ACK have the same priority index and are jointly encoded and transmitted on a PUSCH" and replacing "is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part " with "is given by Clause 9.3.1 of [5, TS 38.213]".

When higher layer parameter *cg-UCI-Multiplexing* is configured, the UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$ is determined as follows, where $A = O^{CG-UCI} + O^{ACK}$.

- The CG-UCI bits are mapped to the UCI bit sequence $a_0, a_1, a_2, a_3, \ldots, a_{O^{CG-UCI}-1}$, where $a_i = \tilde{o}_i^{CG-UCI}$ for $i = 0,1, \ldots, O^{CG-UCI}-1$. The CG-UCI bit sequence $\tilde{o}_0^{CG-UCI}, \tilde{o}_1^{CG-UCI}, \ldots, \tilde{o}_{O^{CG-UCI}-1}^{CG-UCI}$ is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part, and O^{CG-UCI} is number of CG-UCI bits;
- The HARQ-ACK bits are mapped to the UCI bit sequence $a_{O^{CG-UCI}}, a_{O^{CG-UCI}+1}, \ldots, a_{O^{CG-UCI}+O^{ACK}-1}$, where $a_{i+O^{CG-UCI}} = \tilde{o}_i^{ACK}$ for $i = 0,1, \ldots, O^{ACK}-1$. The HARQ-ACK bit sequence $\tilde{o}_0^{ACK}, \tilde{o}_1^{ACK}, \ldots, \tilde{o}_{O^{ACK}-1}^{ACK}$ is given by Clause 9.1 of [5, TS38.213], and O^{ACK} is number of HARQ-ACK bits.

6.3.2.1.5 UCI with different priority indexes

If the higher layer parameter *nrofBitsInUTO-UCI* is configured, the procedure in this clause 6.3.2.1.5 applies by replacing CG-UCI with UTO-UCI in all the notations and texts, and replacing "is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part" with "is given by clause 9.3.1 of [5, TS 38.213]".

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 1, the following UCI bit sequences are generated, $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$, and $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ if any, according to the following:

- If CSI part 1 is also transmitted on the PUSCH,
  - Set $a_i^{(1)}$ for $i = 0,1,\ldots,A^{(1)}-1$ as the bit sequence of CSI part 1, where the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-6, are mapped to the UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ starting with $a_0^{(1)}$.
  - Set $a_i^{(2)} = \tilde{o}_i^{ACK-LP}$ for $i = 0,1,\ldots,O^{ACK-LP}-1$ and $A^{(2)} = O^{ACK-LP}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-LP}, \tilde{o}_1^{ACK-LP}, \ldots, \tilde{o}_{O^{ACK-LP}-1}^{ACK-LP}$ associated with priority index 0 is given by Clause 9.1 of [5, TS 38.213].
- Otherwise, set $a_i^{(1)} = \tilde{o}_i^{ACK-LP}$ for $i = 0,1,\ldots,O^{ACK-LP}-1$ and $A^{(1)} = O^{ACK-LP}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-LP}, \tilde{o}_1^{ACK-LP}, \ldots, \tilde{o}_{O^{ACK-LP}-1}^{ACK-LP}$ associated with priority index 0 is given by Clause 9.1 of [5, TS 38.213].

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 1, and CSI if any are transmitted on a PUSCH associated with priority index 0, the following UCI bit sequences are generated, $a_0, a_1, a_2, a_3, \ldots, a_{A-1}, a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ if any, and $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ if any, according to the following:

- If HARQ-ACK bits associated with priority index 1 and CSI are transmitted on the PUSCH without UL-SCH and the CSI includes CSI part 1 without CSI part 2, and there is only one HARQ-ACK bit associated with priority index 1 given by Clause 9.1 of [5, TS 38.213], set $a_0 = \tilde{o}_0^{ACK-HP}$, $a_1 = 0$, and $A = 2$; otherwise, set $a_i = \tilde{o}_i^{ACK-HP}$ for $i = 0,1,\ldots,O^{ACK-HP}-1$ and $A = O^{ACK-HP}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-HP}, \tilde{o}_1^{ACK-HP}, \ldots, \tilde{o}_{O^{ACK-HP}-1}^{ACK-HP}$ associated with priority index 1 is given by Clause 9.1 of [5, TS 38.213];
- Set $a_i^{(1)}$ for $i = 0,1,\ldots,A^{(1)}-1$ as the bit sequence of CSI part 1, if CSI part 1 is also transmitted on the PUSCH, where the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-6, are mapped to the UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ starting with $a_0^{(1)}$;

- Set $a_i^{(2)}$ for $i=0,1,\ldots,A^{(2)}-1$ as the bit sequence of CSI part 2, if CSI part 2 is also transmitted on the PUSCH, where the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-7, are mapped to the UCI bit sequence $a_0^{(2)}$, $a_1^{(2)}$, $a_2^{(2)}$, $a_3^{(2)}$, …, $a_{A^{(1)}-1}^{(2)}$ starting with $a_0^{(2)}$.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1 and/or CG-UCI associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH, the following UCI bit sequences are generated, a_0, a_1, a_2, a_3, …, a_{A-1}, $a_0^{(1)}$, $a_1^{(1)}$, $a_2^{(1)}$, $a_3^{(1)}$, …, $a_{A^{(1)}-1}^{(1)}$, and $a_0^{(2)}$, $a_1^{(2)}$, $a_2^{(2)}$, $a_3^{(2)}$, …, $a_{A^{(2)}-1}^{(2)}$ if any, according to the following:

- Set $a_i=\tilde{o}_i^{ACK-HP}$ for $i=0,1,\ldots,O^{ACK-HP}-1$ and $A=O^{ACK-HP}$ if HARQ-ACK bits associated with priority index 1 are transmitted without CG-UCI associated with priority index 1, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-HP}$, $\tilde{o}_1^{ACK-HP}$, …, $\tilde{o}_{O^{ACK-HP}-1}^{ACK-HP}$ associated with priority index 1 is given by Clause 9.1 of [5, TS 38.213];
- Set $a_i=\tilde{o}_i^{CG-UCI}$ for $i=0,1,\ldots,O^{CG-UCI}-1$ and $A=O^{CG-UCI}$ if CG-UCI associated with priority index 1 is transmitted without HARQ-ACK bits associated with priority index 1, where the CG-UCI bit sequence $\tilde{o}_0^{CG-UCI}$, $\tilde{o}_1^{CG-UCI}$, …, $\tilde{o}_{O^{CG-UCI}-1}^{CG-UCI}$ associated with priority index 1 is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part;
- Set a_0, a_1, a_2, a_3, …, a_{A-1} as follows, if both CG-UCI associated with priority index 1 and HARQ-ACK bits associated with priority index 1 are transmitted, where $A=O^{CG-UCI}+O^{ACK-HP}$
  - The CG-UCI bits are mapped to the UCI bit sequence a_0, a_1, a_2, a_3, …, $a_{O^{CG-UCI}-1}$, where $a_i=\tilde{o}_i^{CG-UCI}$ for $i=0,1,\ldots,O^{CG-UCI}-1$. The CG-UCI bit sequence $\tilde{o}_0^{CG-UCI}$, $\tilde{o}_1^{CG-UCI}$, …, $\tilde{o}_{O^{CG-UCI}-1}^{CG-UCI}$ is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part, and O^{CG-UCI} is number of CG-UCI bits
  - The HARQ-ACK bits are mapped to the UCI bit sequence $a_{O^{CG-UCI}}$, $a_{O^{CG-UCI}+1}$, …, $a_{O^{CG-UCI}+O^{ACK-HP}-1}$, where $a_{i+O^{CG-UCI}}=\tilde{o}_i^{ACK-HP}$ for $i=0,1,\ldots,O^{ACK-HP}-1$. The HARQ-ACK bit sequence $\tilde{o}_0^{ACK-HP}$, $\tilde{o}_1^{ACK-HP}$, …, $\tilde{o}_{O^{ACK-HP}-1}^{ACK-HP}$ associated with priority index 1 is given by Clause 9.1 of [5, TS 38.213].
- If CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 1,
  - Set $a_i^{(1)}$ for $i=0,1,\ldots,A^{(1)}-1$ as the bit sequence of CSI part 1, where the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-6, are mapped to the UCI bit sequence $a_0^{(1)}$, $a_1^{(1)}$, $a_2^{(1)}$, $a_3^{(1)}$, …, $a_{A^{(1)}-1}^{(1)}$ starting with $a_0^{(1)}$.
  - Set $a_i^{(2)}=\tilde{o}_i^{ACK-LP}$ for $i=0,1,\ldots,O^{ACK-LP}-1$ and $A^{(2)}=O^{ACK-LP}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-LP}$, $\tilde{o}_1^{ACK-LP}$, …, $\tilde{o}_{O^{ACK-LP}-1}^{ACK-LP}$ associated with priority index 0 is given by Clause 9.1 of [5, TS 38.213].
- Otherwise,
  - Set $a_i^{(1)}=\tilde{o}_i^{ACK-LP}$ for $i=0,1,\ldots,O^{ACK-LP}-1$ and $A^{(1)}=O^{ACK-LP}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-LP}$, $\tilde{o}_1^{ACK-LP}$, …, $\tilde{o}_{O^{ACK-LP}-1}^{ACK-LP}$ associated with priority index 0 is given by Clause 9.1 of [5, TS 38.213].
  - Set $a_i^{(2)}=\tilde{a}_i^{(1)}$ for $i=0,1,\ldots,\tilde{A}^{(1)}-1$ and $A^{(2)}=\tilde{A}^{(1)}$, if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0, where the CSI part 1 sequence $\tilde{a}_0^{(1)}$, $\tilde{a}_1^{(1)}$, $\tilde{a}_2^{(1)}$, $\tilde{a}_3^{(1)}$, …, $\tilde{a}_{\tilde{A}^{(1)}-1}^{(1)}$ is given by Table 6.3.2.1.2-6 by replacing $a_0^{(1)}$, $a_1^{(1)}$, $a_2^{(1)}$, $a_3^{(1)}$, …, $a_{A^{(1)}-1}^{(1)}$, and the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-6, are mapped to the CSI part 1 sequence $\tilde{a}_0^{(1)}$, $\tilde{a}_1^{(1)}$, $\tilde{a}_2^{(1)}$, $\tilde{a}_3^{(1)}$, …, $\tilde{a}_{\tilde{A}^{(1)}-1}^{(1)}$ starting with $\tilde{a}_0^{(1)}$.

If *uci-MuxWithDiffPrio* is configured, and CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 if any, HARQ-ACK bits associated with priority index 1, and CSI part 1 if any are transmitted

on a PUSCH associated with priority index 0, the following UCI bit sequences are generated, $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$, and $a_0^{(2)}, a_1^{(2)}, a_2^{(2)}, a_3^{(2)}, \ldots, a_{A^{(2)}-1}^{(2)}$ if any, according to the following:

- Set $a_i = \tilde{o}_i^{ACK-HP}$ for $i = 0,1,\ldots,O^{ACK-HP}-1$ and $A = O^{ACK-HP}$, where the HARQ-ACK bit sequence $\tilde{o}_0^{ACK-HP}, \tilde{o}_1^{ACK-HP}, \ldots, \tilde{o}_{O^{ACK-HP}-1}^{ACK-HP}$ associated with priority index 1 is given by Clause 9.1 of [5, TS 38.213];
- Set $a_i^{(1)} = \tilde{o}_i^{CG-UCI}$ for $i = 0,1, \ldots, O^{CG-UCI}-1$ and $A^{(1)} = O^{CG-UCI}$ if CG-UCI associated with priority index 0 is transmitted without HARQ-ACK bits associated with priority index 0, where the CG-UCI bit sequence $\tilde{o}_0^{CG-UCI}, \tilde{o}_1^{CG-UCI}, \ldots, \tilde{o}_{O^{CG-UCI}-1}^{CG-UCI}$ associated with priority index 0 is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part;
- Set $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$ as follows if both CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 are transmitted, where $A^{(1)} = O^{CG-UCI} + O^{ACK-LP}$
  - The CG-UCI bits are mapped to the UCI bit sequence $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{O^{CG-UCI}-1}^{(1)}$, where $a_i^{(1)} = \tilde{o}_i^{CG-UCI}$ for $i = 0,1, \ldots, O^{CG-UCI}-1$. The CG-UCI bit sequence $\tilde{o}_0^{CG-UCI}, \tilde{o}_1^{CG-UCI}, \ldots, \tilde{o}_{O^{CG-UCI}-1}^{CG-UCI}$ is given by Table 6.3.2.1.3-1 mapped in the order from upper part to lower part, and O^{CG-UCI} is number of CG-UCI bits
  - The HARQ-ACK bits are mapped to the UCI bit sequence $a_{O^{CG-UCI}}^{(1)}, a_{O^{CG-UCI}+1}^{(1)}, \ldots, a_{O^{CG-UCI}+O^{ACK-LP}-1}^{(1)}$, where $a_{i+O^{CG-UCI}}^{(1)} = \tilde{o}_i^{ACK-LP}$ for $i = 0,1, \ldots, O^{ACK-LP}-1$. The HARQ-ACK bit sequence $\tilde{o}_0^{ACK-LP}, \tilde{o}_1^{ACK-LP}, \ldots, \tilde{o}_{O^{ACK-LP}-1}^{ACK-LP}$ associated with priority index 0 is given by Clause 9.1 of [5, TS 38.213].
- Set $a_i^{(2)} = \tilde{a}_i^{(1)}$ for $i = 0,1,\ldots,\tilde{A}^{(1)}-1$ and $A^{(2)} = \tilde{A}^{(1)}$, if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0, where the CSI part 1 sequence $\tilde{a}_0^{(1)}, \tilde{a}_1^{(1)}, \tilde{a}_2^{(1)}, \tilde{a}_3^{(1)}, \ldots, \tilde{a}_{\tilde{A}^{(1)}-1}^{(1)}$ is given by Table 6.3.2.1.2-6 by replacing $a_0^{(1)}, a_1^{(1)}, a_2^{(1)}, a_3^{(1)}, \ldots, a_{A^{(1)}-1}^{(1)}$, and the CSI fields of all CSI reports, in the order from upper part to lower part in Table 6.3.2.1.2-6, are mapped to the CSI part 1 sequence $\tilde{a}_0^{(1)}, \tilde{a}_1^{(1)}, \tilde{a}_2^{(1)}, \tilde{a}_3^{(1)}, \ldots, \tilde{a}_{\tilde{A}^{(1)}-1}^{(1)}$ starting with $\tilde{a}_0^{(1)}$.

6.3.2.2 Code block segmentation and CRC attachment

Denote the bits of the payload by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, where A is the payload size. The procedure in 6.3.2.2.1 applies for $A \geq 12$ and the procedure in Clause 6.3.2.2.2 applies for $A \leq 11$.

6.3.2.2.1 UCI encoded by Polar code

Code block segmentation and CRC attachment is performed according to Clause 6.3.1.2.1.

6.3.2.2.2 UCI encoded by channel coding of small block lengths

The procedure in Clause 6.3.1.2.2 applies.

6.3.2.3 Channel coding of UCI

6.3.2.3.1 UCI encoded by Polar code

Channel coding is performed according to Clause 6.3.1.3.1, except that the rate matching output sequence length E_r is given in Clause 6.3.2.4.1.

6.3.2.3.2 UCI encoded by channel coding of small block lengths

Information bits are delivered to the channel coding block. They are denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits.

The information bits are encoded according to Clause 5.3.3.

After encoding the bits are denoted by $d_0, d_1, d_2, d_3, \ldots, d_{N-1}$, where N is the number of coded bits.

6.3.2.4 Rate matching

In case where there are more than one UL-SCH transport blocks for the PUSCH transmission, the UCI information is multiplexed only on the UL-SCH transport block with highest I_{MCS} value for the initial PUSCH, where I_{MCS} is as defined in Clause 6.1.4.1 in [6, TS 38.214]. In case the two transport blocks have the same I_{MCS} value for the initial PUSCH, the UCI information is multiplexed with data only on the first transport block. The PUSCH for UCI multiplexing in this Clause refers to the UL-SCH transport block for UCI multiplexing.

6.3.2.4.1 UCI encoded by Polar code

If the higher layer parameter *nrofBitsInUTO-UCI* is configured, the procedures in this clause and the clauses it refers to apply by replacing CG-UCI with UTO-UCI in all the notations and texts, when applicable.

6.3.2.4.1.1 HARQ-ACK

For HARQ-ACK transmission on PUSCH not using repetition type B with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table, or if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is equal to 1, the number of coded modulation symbols per layer for HARQ-ACK transmission, denoted as Q'_{ACK}, is determined as follows:

$$Q'_{\text{ACK}} = \min\left\{ \left\lceil \frac{\left(O_{\text{ACK}} + L_{\text{ACK}}\right) \cdot \beta_{\text{offset}}^{\text{PUSCH}} \cdot \sum_{l=0}^{N_{\text{symb,all}}^{\text{PUSCH}}-1} M_{\text{sc}}^{\text{UCI}}(l)}{\sum_{r=0}^{C_{\text{UL-SCH}}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=l_0}^{N_{\text{symb,all}}^{\text{PUSCH}}-1} M_{\text{sc}}^{\text{UCI}}(l) \right\rceil \right\}$$

where

- O_{ACK} is the number of HARQ-ACK bits;
- if $O_{\text{ACK}} \geq 360$, $L_{\text{ACK}} = 11$; otherwise L_{ACK} is the number of CRC bits for HARQ-ACK determined according to Clause 6.3.1.2.1;
- $\beta_{\text{offset}}^{\text{PUSCH}} = \beta_{\text{offset}}^{\text{HARQ-ACK}}$;
- $C_{\text{UL-SCH}}$ is the number of code blocks for UL-SCH of the PUSCH transmission;
- if the DCI format scheduling the PUSCH transmission includes a CBGTI field indicating that the UE shall not transmit the r-th code block, $K_r = 0$; otherwise, K_r is the r-th code block size for UL-SCH of the PUSCH transmission;
- $M_{\text{sc}}^{\text{PUSCH}}$ is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];

- $M_{sc}^{PT\text{-}RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission;
- $M_{sc}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{sc}^{UCI}(l)=0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{sc}^{UCI}(l)=M_{sc}^{PUSCH}-M_{sc}^{PT\text{-}RS}(l)-M_{sc}^{Muted}(l)$;
- α is configured by higher layer parameter *scaling*;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission.

For HARQ-ACK transmission on PUSCH not using repetition type B with UL-SCH, and if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is larger than 1, the number of coded modulation symbols per layer for HARQ-ACK transmission, denoted as Q'_{ACK}, is determined as follows:

$$Q'_{ACK}=min\left\{\left\lceil\frac{(O_{ACK}+L_{ACK})\cdot\beta_{offset}^{PUSCH}\cdot\sum_{l=0}^{N_{symb,all}^{PUSCH}-1}M_{sc}^{UCI}(l)}{\frac{1}{N_s}\sum_{r=0}^{C_{UL\text{-}SCH}-1}K_r}\right\rceil,\left\lceil\alpha\cdot\sum_{l=l_0}^{N_{symb,all}^{PUSCH}-1}M_{sc}^{UCI}(l)\right\rceil\right\}$$

where

- N_s is the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI;
- $M_{sc}^{PT\text{-}RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK transmission;
- $M_{sc}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH in the slot, including all OFDM symbols used for DMRS;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK transmission;
- and all the other notations in the formula are defined the same as for PUSCH not using repetition type B and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

For HARQ-ACK transmission on an actual repetition of a PUSCH with repetition Type B with UL-SCH, the number of coded modulation symbols per layer for HARQ-ACK transmission, denoted as Q'_{ACK}, is determined as follows:

$$Q'_{ACK} = min\left\{ \left\lceil \frac{\left(O_{ACK} + L_{ACK}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,nominal}^{PUSCH}-1} M_{sc,nominal}^{UCI}(l)}{\sum_{r=0}^{C_{UL-SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=0}^{N_{symb,nominal}^{PUSCH}-1} M_{sc,nominal}^{UCI}(l) \right\rceil, \sum_{l=0}^{N_{symb,actual}^{PUSCH}-1} M_{sc,actual}^{UCI}(l) \right\}$$

where

- $M_{sc,nominal}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0, 1, 2, \cdots, N_{symb,nominal}^{PUSCH} - 1$, in the PUSCH transmission assuming a nominal repetition without segmentation, and $N_{symb,nominal}^{PUSCH}$ is the total number of OFDM symbols in a nominal repetition of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH assuming a nominal repetition without segmentation, $M_{sc,nominal}^{UCI}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH assuming a nominal repetition without segmentation, $M_{sc,nominal}^{UCI}(l) = M_{sc}^{PUSCH} - M_{sc,nominal}^{PT\text{-}RS}(l) - M_{sc,nominal}^{Muted}(l)$ where $M_{sc,nominal}^{PT\text{-}RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission assuming a nominal repetition without segmentation, $M_{sc,nominal}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission, assuming a nominal repetition without segmentation;
- $M_{sc,actual}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0, 1, 2, \cdots, N_{symb,actual}^{PUSCH} - 1$, in the actual repetition of the PUSCH transmission, and $N_{symb,actual}^{PUSCH}$ is the total number of OFDM symbols in the actual repetition of the PUSCH transmission, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the actual repetition of the PUSCH transmission, $M_{sc,actual}^{UCI}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the actual repetition of the PUSCH transmission, $M_{sc,actual}^{UCI}(l) = M_{sc}^{PUSCH} - M_{sc,actual}^{PT\text{-}RS}(l) - M_{sc,actual}^{Muted}(l)$ where $M_{sc,actual}^{PT\text{-}RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the actual repetition of the PUSCH transmission, $M_{sc,actual}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the actual repetition of the PUSCH transmission;
- and all the other notations in the formula are defined the same as for PUSCH not using repetition type B and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

For HARQ-ACK transmission on PUSCH without UL-SCH, the number of coded modulation symbols per layer for HARQ-ACK transmission, denoted as Q'_{ACK}, is determined as follows:

$$Q'_{ACK} = \min\left\{ \left\lceil \frac{\left(O_{ACK} + L_{ACK}\right) \cdot \beta_{offset}^{PUSCH}}{R \cdot Q_m} \right\rceil, \left\lceil \alpha \cdot \sum_{l=l_0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil \right\}$$

where

- O_{ACK} is the number of HARQ-ACK bits;
- if $O_{ACK} \geq 360$, $L_{ACK} = 11$; otherwise L_{ACK} is the number of CRC bits for HARQ-ACK defined according to Clause 6.3.1.2.1;;

- $\beta_{\text{offset}}^{\text{PUSCH}} = \beta_{\text{offset}}^{\text{HARQ- ACK}}$;
- $M_{\text{sc}}^{\text{PUSCH}}$ is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{\text{sc}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;
- $M_{\text{sc}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$, in the PUSCH transmission and $N_{\text{symb,all}}^{\text{PUSCH}}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=M_{\text{sc}}^{\text{PUSCH}}-M_{\text{sc}}^{\text{PT-RS}}(l)-M_{\text{sc}}^{Muted}(l)$;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission;
- R is the code rate of the PUSCH, determined according to Clause 6.1.4.1 of [6, TS38.214];
- Q_m is the modulation order of the PUSCH;
- α is configured by higher layer parameter *scaling*.

The input bit sequence to rate matching is $d_{r0},d_{r1},d_{r2},d_{r3},\ldots,d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL}=1$ and the rate matching output sequence length to $E_r=\lfloor E_{\text{UCI}}/C_{\text{UCI}} \rfloor$, where

- C_{UCI} is the number of code blocks for UCI determined according to Clause 5.2.1;
- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH;
- $E_{\text{UCI}}=N_L \cdot Q'_{\text{ACK}} \cdot Q_m$.

The output bit sequence after rate matching is denoted as $f_{r0},f_{r1},f_{r2},\ldots,f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.2.4.1.2 CSI part 1

For CSI part 1 transmission on PUSCH not using repetition type B with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table, or if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is equal to 1, the number of coded modulation symbols per layer for CSI part 1 transmission, denoted as $Q'_{\text{CSI-part1}}$, is determined as follows:

$$Q'_{CSI-1} = min\left(\left\lceil \frac{\left(O_{CSI-1} + L_{CSI-1}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\sum_{r=0}^{C_{UL-SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil - Q'_{ACK/CG-UCI} \right)$$

where

- $O_{\text{CSI-1}}$ is the number of bits for CSI part 1;
- if $O_{\text{CSI-1}} \geq 360$, $L_{\text{CSI-1}} = 11$; otherwise $L_{\text{CSI-1}}$ is the number of CRC bits for CSI part 1 determined according to Clause 6.3.1.2.1;
- $\beta_{\text{offset}}^{\text{PUSCH}} = \beta_{\text{offset}}^{\text{CSI-part1}}$;
- $C_{\text{UL-SCH}}$ is the number of code blocks for UL-SCH of the PUSCH transmission;
- if the DCI format scheduling the PUSCH transmission includes a CBGTI field indicating that the UE shall not transmit the r-th code block, $K_r = 0$; otherwise, K_r is the r-th code block size for UL-SCH of the PUSCH transmission;
- $M_{\text{sc}}^{\text{PUSCH}}$ is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{\text{sc}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;
- $Q'_{ACK/CG-UCI} = Q'_{ACK}$ if HARQ-ACK is present for transmission on the same PUSCH with UL-SCH and without CG-UCI, where Q'_{ACK} is the number of coded modulation symbols per layer for HARQ-ACK transmitted on the PUSCH as defined in clause 6.3.2.4.1.1 if number of HARQ-ACK information bits is more than 2, and $Q'_{\text{ACK}} = \sum_{l=0}^{N_{\text{symb,all}}^{\text{PUSCH}}-1} \bar{M}_{\text{sc, rvd}}^{\text{ACK}}(l)$ if the number of HARQ-ACK information bits is no more than 2 bits, where $\bar{M}_{\text{sc, rvd}}^{\text{ACK}}(l)$ is the number of reserved resource elements for potential HARQ-ACK transmission in OFDM symbol l, for $l = 0, 1, 2, \ldots, N_{\text{symb,all}}^{\text{PUSCH}} - 1$, in the PUSCH transmission, defined in Clause 6.2.7; or
- $Q'_{ACK/CG-UCI} = Q'_{ACK}$ if both HARQ-ACK and CG-UCI are present on the same PUSCH with UL-SCH, where Q'_{ACK} is the number of coded modulation symbols per layer for HARQ-ACK and CG-UCI transmitted on the PUSCH as defined in clause 6.3.2.4.1.5; or
- $Q'_{ACK/CG-UCI} = Q'_{CG-UCI}$ if CG-UCI is present on the same PUSCH with UL-SCH and without HARQ-ACK, where Q'_{CG-UCI} is the number of coded modulation symbols per layer for CG-UCI transmitted on the PUSCH as defined in clause 6.3.2.4.1.4;
- $M_{\text{sc}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0, 1, 2, \ldots, N_{\text{symb,all}}^{\text{PUSCH}} - 1$, in the PUSCH transmission and $N_{\text{symb,all}}^{\text{PUSCH}}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;

- for any OFDM symbol that carries DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=0$;
- for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=M_{\text{sc}}^{\text{PUSCH}}-M_{\text{sc}}^{\text{PT-RS}}(l)-M_{\text{sc}}^{Muted}(l)$;

- α is configured by higher layer parameter *scaling*.

For CSI part 1 transmission on PUSCH not using repetition type B with UL-SCH, and if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is larger than 1, the number of coded modulation symbols per layer for CSI part 1 transmission, denoted as $Q'_{CSI-part\,1}$, is determined as follows:

$$Q'_{CSI-1}=min\left\{\left\lceil\frac{\left(O_{CSI-1}+L_{CSI-1}\right)\cdot\beta_{offset}^{PUSCH}\cdot\sum_{l=0}^{N_{symb,all}^{PUSCH}-1}M_{sc}^{UCI}(l)}{\frac{1}{N_s}\sum_{r=0}^{C_{UL-SCH}-1}K_r}\right\rceil,\left\lceil\alpha\cdot\sum_{l=0}^{N_{symb,all}^{PUSCH}-1}M_{sc}^{UCI}(l)\right\rceil-Q'_{ACK/CG-UCI}\right\}$$

where

- N_s is the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI;
- $M_{sc}^{PT-RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission of TB processing over multiple slots in the slot with the CSI part 1 transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission of TB processing over multiple slots in the slot with the CSI part 1 transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission of TB processing over multiple slots in the slot with the CSI part 1 transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH in the slot, including all OFDM symbols used for DMRS;
- and all the other notations in the formula are defined the same as for PUSCH not using repetition type B and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

For CSI part 1 transmission on an actual repetition of a PUSCH with repetition Type B with UL-SCH, the number of coded modulation symbols per layer for CSI part 1 transmission, denoted as $Q'_{\text{CSI-part 1}}$, is determined as follows:

$$Q'_{\text{CSI-1}}=min\left\{\left\lceil\frac{\left(O_{\text{CSI-1}}+L_{\text{CSI-1}}\right)\cdot\beta_{\text{offset}}^{\text{PUSCH}}\cdot\sum_{l=0}^{N_{\text{symb,nominal}}^{\text{PUSCH}}-1}M_{\text{sc,nominal}}^{\text{UCI}}(l)}{\sum_{r=0}^{C_{\text{UL-SCH}}-1}K_r}\right\rceil,\quad\left\lceil\alpha\cdot\sum_{l=0}^{N_{\text{symb,nominal}}^{\text{PUSCH}}-1}M_{\text{sc,nominal}}^{\text{UCI}}(l)\right\rceil-Q'_{ACK/CG-UCI},\quad\sum_{l=0}^{N_{\text{symb,actual}}^{\text{PUSCH}}}\right.$$

where

- $M_{\text{sc,nominal}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,\ 1,\ 2,\ \cdots,\ N_{\text{symb,nominal}}^{\text{PUSCH}}-1$, in the PUSCH transmission assuming a nominal repetition without segmentation, and $N_{\text{symb,nominal}}^{\text{PUSCH}}$ is the total number of OFDM symbols in a nominal repetition of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH assuming a nominal repetition without segmentation, $M_{\text{sc,nominal}}^{\text{UCI}}(l)=0$;

  - for any OFDM symbol that does not carry DMRS of the PUSCH assuming a nominal repetition without segmentation, $M_{\text{sc,nominal}}^{\text{UCI}}(l) = M_{\text{sc}}^{\text{PUSCH}} - M_{\text{sc,nominal}}^{\text{PT-RS}}(l) - M_{\text{sc,nominal}}^{Muted}(l)$ where $M_{\text{sc,nominal}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission assuming a nominal repetition without segmentation, $M_{\text{sc,nominal}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission, assuming a nominal repetition without segmentation;
- $M_{\text{sc,actual}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0, 1, 2, \cdots, N_{\text{symb,actual}}^{\text{PUSCH}} - 1$, in the actual repetition of the PUSCH transmission, and $N_{\text{symb,actual}}^{\text{PUSCH}}$ is the total number of OFDM symbols in the actual repetition of the PUSCH transmission, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the actual repetition of the PUSCH transmission, $M_{\text{sc,actual}}^{\text{UCI}}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the actual repetition of the PUSCH transmission, $M_{\text{sc,actual}}^{\text{UCI}}(l) = M_{\text{sc}}^{\text{PUSCH}} - M_{\text{sc,actual}}^{\text{PT-RS}}(l) - M_{\text{sc,actual}}^{Muted}(l)$ where $M_{\text{sc,actual}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the actual repetition of the PUSCH transmission, $M_{\text{sc,actual}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the actual repetition of the PUSCH transmission;
- and all the other notations in the formula are defined the same as for PUSCH not using repetition type B and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

For CSI part 1 transmission on PUSCH without UL-SCH, the number of coded modulation symbols per layer for CSI part 1 transmission, denoted as $Q'_{\text{CSI-part1}}$, is determined as follows:

if there is CSI part 2 to be transmitted on the PUSCH,

$$Q'_{\text{CSI-1}} = \min\left\{\left\lceil \frac{(O_{\text{CSI-1}} + L_{\text{CSI-1}}) \cdot \beta_{\text{offset}}^{\text{PUSCH}}}{R \cdot Q_m} \right\rceil, \sum_{l=0}^{N_{\text{symb,all}}^{\text{PUSCH}} - 1} M_{\text{sc}}^{\text{UCI}}(l) - Q'_{\text{ACK}}\right\}$$

else

$$Q'_{\text{CSI-1}} = \sum_{l=0}^{N_{\text{symb,all}}^{\text{PUSCH}} - 1} M_{\text{sc}}^{\text{UCI}}(l) - Q'_{\text{ACK}}$$

end if

where

- $O_{\text{CSI-1}}$ is the number of bits for CSI part 1;
- if $O_{\text{CSI-1}} \geq 360$, $L_{\text{CSI-1}} = 11$; otherwise $L_{\text{CSI-1}}$ is the number of CRC bits for CSI part 1 determined according to Clause 6.3.1.2.1;
- $\beta_{\text{offset}}^{\text{PUSCH}} = \beta_{\text{offset}}^{\text{CSI-part1}}$;
- $M_{\text{sc}}^{\text{PUSCH}}$ is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{\text{sc}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;

- Q'_{ACK} is the number of coded modulation symbols per layer for HARQ-ACK transmitted on the PUSCH if number of HARQ-ACK information bits is more than 2, and $Q'_{\text{ACK}} = \sum_{l=0}^{N_{\text{symb,all}}^{\text{PUSCH}}-1} \bar{M}_{\text{sc, rvd}}^{\text{ACK}}(l)$ if the number of HARQ-ACK information bits is no more than 2 bits, where $\bar{M}_{\text{sc, rvd}}^{\text{ACK}}(l)$ is the number of reserved resource elements for potential HARQ-ACK transmission in OFDM symbol l, for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$, in the PUSCH transmission, defined in Clause 6.2.7;
- $M_{\text{sc}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$, in the PUSCH transmission and $N_{\text{symb,all}}^{\text{PUSCH}}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=M_{\text{sc}}^{\text{PUSCH}}-M_{\text{sc}}^{\text{PT-RS}}(l)-M_{\text{sc}}^{Muted}(l)$;
- R is the code rate of the PUSCH, determined according to Clause 6.1.4.1 of [6, TS38.214];
- Q_m is the modulation order of the PUSCH.

The input bit sequence to rate matching is $d_{r0},d_{r1},d_{r2},d_{r3},\ldots,d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL}=1$ and the rate matching output sequence length to $E_r=\lfloor E_{\text{UCI}}/C_{\text{UCI}} \rfloor$, where

- C_{UCI} is the number of code blocks for UCI determined according to Clause 5.2.1;
- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH;
- $E_{\text{UCI}}=N_L \cdot Q'_{\text{CSI,1}} \cdot Q_m$.

The output bit sequence after rate matching is denoted as $f_{r0},f_{r1},f_{r2},\ldots,f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.2.4.1.3 CSI part 2

For CSI part 2 transmission on PUSCH not using repetition type B with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table, or if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is equal to 1, the number of coded modulation symbols per layer for CSI part 2 transmission, denoted as $Q'_{\text{CSI-part2}}$, is determined as follows:

$$Q'_{CSI-2}=min\left(\left\lceil \frac{\left(O_{CSI-2}+L_{CSI-2}\right)\cdot \beta_{offset}^{PUSCH}\cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\sum_{r=0}^{C_{UL-SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil - Q'_{ACK/CG-UCI} - Q'_{CSI-1}\right)$$

where

- $O_{\text{CSI-2}}$ is the number of bits for CSI part 2;
- if $O_{\text{CSI-2}} \geq 360$, $L_{\text{CSI-2}} = 11$; otherwise $L_{\text{CSI-2}}$ is the number of CRC bits for CSI part 2 determined according to Clause 6.3.1.2.1;
- $\beta_{\text{offset}}^{\text{PUSCH}} = \beta_{\text{offset}}^{\text{CSI-part2}}$;
- $C_{\text{UL-SCH}}$ is the number of code blocks for UL-SCH of the PUSCH transmission;
- if the DCI format scheduling the PUSCH transmission includes a CBGTI field indicating that the UE shall not transmit the r-th code block, $K_r = 0$; otherwise, K_r is the r-th code block size for UL-SCH of the PUSCH transmission;
- $M_{\text{sc}}^{\text{PUSCH}}$ is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{\text{sc}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;
- $Q'_{ACK/CG\text{-}UCI} = Q'_{ACK}$ if HARQ-ACK is present for transmission on the same PUSCH with UL-SCH and without CG-UCI, where Q'_{ACK} is the number of coded modulation symbols per layer for HARQ-ACK transmitted on the PUSCH as defined in clause 6.3.2.4.1.1 if number of HARQ-ACK information bits is more than 2, and $Q'_{\text{ACK}} = 0$ if the number of HARQ-ACK information bits is 1 or 2 bits; or
- $Q'_{ACK/CG\text{-}UCI} = Q'_{ACK}$ if both HARQ-ACK and CG-UCI are present on the same PUSCH with UL-SCH, where Q'_{ACK} is the number of coded modulation symbols per layer for HARQ-ACK and CG-UCI transmitted on the PUSCH as defined in clause 6.3.2.4.1.5; or
- $Q'_{ACK/CG\text{-}UCI} = Q'_{CG\text{-}UCI}$ if CG-UCI is present on the same PUSCH with UL-SCH and without HARQ-ACK, where $Q'_{CG\text{-}UCI}$ is the number of coded modulation symbols per layer for CG-UCI transmitted on the PUSCH as defined in clause 6.3.2.4.1.4;
- $Q'_{\text{CSI-1}}$ is the number of coded modulation symbols per layer for CSI part 1 transmitted on the PUSCH;
- $M_{\text{sc}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}} - 1$, in the PUSCH transmission and $N_{\text{symb,all}}^{\text{PUSCH}}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l) = M_{\text{sc}}^{\text{PUSCH}} - M_{\text{sc}}^{\text{PT-RS}}(l) - M_{\text{sc}}^{Muted}(l)$.
- α is configured by higher layer parameter *scaling*.

For CSI part 2 transmission on PUSCH not using repetition type B with UL-SCH, and if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is larger than 1, the number of coded modulation symbols per layer for CSI part 2 transmission, denoted as $Q'_{CSI\text{-}part2}$, is determined as follows:

$$Q'_{CSI-2} = min\left\{ \left\lceil \frac{\left(O_{CSI-2} + L_{CSI-2}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\frac{1}{N_s} \sum_{r=0}^{C_{UL-SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil - Q'_{ACK/CG-UCI} - Q'_{CSI-1} \right\}$$

where

- N_s is the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI;
- $M_{sc}^{PT-RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission of TB processing over multiple slots in the slot with the CSI part 2 transmission;
- $M_{sc}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission of TB processing over multiple slots in the slot with the CSI part 2 transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission of TB processing over multiple slots in the slot with the CSI part 2 transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH in the slot, including all OFDM symbols used for DMRS;
- and all the other notations in the formula are defined the same as for PUSCH not using repetition type B and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

For CSI part 2 transmission on an actual repetition of a PUSCH with repetition Type B with UL-SCH, the number of coded modulation symbols per layer for CSI part 2 transmission, denoted as $Q'_{\text{CSI-part 2}}$, is determined as follows:

$$Q'_{\text{CSI-2}} = min\left\{ \left\lceil \frac{\left(O_{\text{CSI-2}} + L_{\text{CSI-2}}\right) \cdot \beta_{\text{offset}}^{\text{PUSCH}} \cdot \sum_{l=0}^{N_{\text{symb,nominal}}^{\text{PUSCH}}-1} M_{\text{sc,nominal}}^{\text{UCI}}(l)}{\sum_{r=0}^{C_{\text{UL-SCH}}-1} K_r} \right\rceil, \quad \left\lceil \alpha \cdot \sum_{l=0}^{N_{\text{symb,nominal}}^{\text{PUSCH}}-1} M_{\text{sc,nominal}}^{\text{UCI}}(l) \right\rceil - Q'_{ACK/CG-UCI} - Q'_{\text{CSI-1}} \right.,$$

where

- $M_{\text{sc,nominal}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0,\ 1,\ 2,\ \cdots,\ N_{\text{symb,nominal}}^{\text{PUSCH}}-1$, in the PUSCH transmission assuming a nominal repetition without segmentation, and $N_{\text{symb,nominal}}^{\text{PUSCH}}$ is the total number of OFDM symbols in a nominal repetition of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH assuming a nominal repetition without segmentation, $M_{\text{sc,nominal}}^{\text{UCI}}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH assuming a nominal repetition without segmentation, $M_{\text{sc,nominal}}^{\text{UCI}}(l) = M_{\text{sc}}^{\text{PUSCH}} - M_{\text{sc,nominal}}^{\text{PT-RS}}(l) - M_{\text{sc,nominal}}^{Muted}(l)$ where $M_{\text{sc,nominal}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission assuming a nominal repetition without segmentation, $M_{\text{sc,nominal}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission, assuming a nominal repetition without segmentation;
- $M_{\text{sc,actual}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0,\ 1,\ 2,\ \cdots,\ N_{\text{symb,actual}}^{\text{PUSCH}}-1$, in the actual repetition of the PUSCH transmission, and $N_{\text{symb,actual}}^{\text{PUSCH}}$ is the total number of OFDM symbols in the actual repetition of the PUSCH transmission, including all OFDM symbols used for DMRS;

- for any OFDM symbol that carries DMRS of the actual repetition of the PUSCH transmission, $M_{\text{sc,actual}}^{\text{UCI}}(l)=0$;
- for any OFDM symbol that does not carry DMRS of the actual repetition of the PUSCH transmission, $M_{\text{sc,actual}}^{\text{UCI}}(l)=M_{\text{sc}}^{\text{PUSCH}}-M_{\text{sc,actual}}^{\text{PT-RS}}(l)-M_{\text{sc,actual}}^{Muted}(l)$ where $M_{\text{sc,actual}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the actual repetition of the PUSCH transmission, $M_{\text{sc,actual}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the actual repetition of the PUSCH transmission;

- and all the other notations in the formula are defined the same as for PUSCH not using repetition type B and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

For CSI part 2 transmission on PUSCH without UL-SCH, the number of coded modulation symbols per layer for CSI part 2 transmission, denoted as $Q'_{\text{CSI-part2}}$, is determined as follows:

$$Q'_{\text{CSI-2}}=\sum_{l=0}^{N_{\text{symb,all}}^{\text{PUSCH}}-1}M_{\text{sc}}^{\text{UCI}}(l)-Q'_{\text{ACK}}-Q'_{\text{CSI-1}}$$

where

- $M_{\text{sc}}^{\text{PUSCH}}$ is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{\text{sc}}^{\text{PT-RS}}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;
- Q'_{ACK} is the number of coded modulation symbols per layer for HARQ-ACK transmitted on the PUSCH if number of HARQ-ACK information bits is more than 2, and $Q'_{\text{ACK}}=0$ if the number of HARQ-ACK information bits is 1 or 2 bits;
- $Q'_{\text{CSI-1}}$ is the number of coded modulation symbols per layer for CSI part 1 transmitted on the PUSCH;
- $M_{\text{sc}}^{\text{UCI}}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l=0,1,2,\ldots,N_{\text{symb,all}}^{\text{PUSCH}}-1$, in the PUSCH transmission and $N_{\text{symb,all}}^{\text{PUSCH}}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{\text{sc}}^{\text{UCI}}(l)=M_{\text{sc}}^{\text{PUSCH}}-M_{\text{sc}}^{\text{PT-RS}}(l)-M_{\text{sc}}^{Muted}(l)$.

The input bit sequence to rate matching is $d_{r0},d_{r1},d_{r2},d_{r3},\ldots,d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL}=1$ and the rate matching output sequence length to $E_r=\lfloor E_{\text{UCI}}/C_{\text{UCI}}\rfloor$, where

- C_{UCI} is the number of code blocks for UCI determined according to Clause 5.2.1;
- N_L is the number of transmission layers of the PUSCH;

- Q_m is the modulation order of the PUSCH;
- $E_{\text{UCI}} = N_L \cdot Q'_{\text{CSI,2}} \cdot Q_m$.

The output bit sequence after rate matching is denoted as $f_{r0}, f_{r1}, f_{r2}, \ldots, f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.2.4.1.4 CG-UCI

For CG-UCI transmission on PUSCH with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table, or if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is equal to 1, the number of coded modulation symbols per layer for CG-UCI transmission, denoted as Q'_{CG-UCI}, is determined as follows:

$$Q'_{CG-UCI} = min\left(\left\lceil \frac{\left(O_{CG-UCI} + L_{CG-UCI}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\sum_{r=0}^{C_{UL-SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=l_0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil \right)$$

where

- O_{CG-UCI} is the number of CG-UCI bits;
- L_{CG-UCI} is the number of CRC bits for CG-UCI determined according to Clause 6.3.1.2.1;
- $\beta_{offset}^{PUSCH} = \beta_{offset}^{CG-UCI}$;
- C_{UL-SCH} is the number of code blocks for UL-SCH of the PUSCH transmission;
- K_r is the *r*-th code block size for UL-SCH of the PUSCH transmission;
- M_{sc}^{PUSCH} is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, $M_{\text{sc}}^{\text{PUSCH}}$ only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{sc}^{PT-RS}(l)$ is the number of subcarriers in OFDM symbol *l* that carries PTRS, in the PUSCH transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol *l*, for l=0,1,2,…, $N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{sc}^{UCI}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{sc}^{UCI}(l) = M_{sc}^{PUSCH} - M_{sc}^{PT-RS}(l) - M_{\text{sc}}^{Muted}(l)$;
- α is configured by higher layer parameter *scaling*;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission.

For CG-UCI transmission on PUSCH with UL-SCH, and if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment

field in DCI is larger than 1, the number of coded modulation symbols per layer for CG-UCI transmission, denoted as Q'_{CG-UCI}, is determined as follows:

$$Q'_{CG-UCI} = min\left\{ \left\lceil \frac{\left(O_{CG-UCI} + L_{CG-UCI}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\frac{1}{N_s} \sum_{r=0}^{C_{UL-SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=l_0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil \right\}$$

where

- N_s is the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI;
- $M_{sc}^{PT-RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission of TB processing over multiple slots in the slot with the CG-UCI transmission;
- $M_{sc}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission of TB processing over multiple slots in the slot with the CG-UCI transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission of TB processing over multiple slots in the slot with the CG-UCI transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH in the slot, including all OFDM symbols used for DMRS;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission of TB processing over multiple slots in the slot with the CG-UCI transmission;
- and all the other notations in the formula are defined the same as for PUSCH with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

The input bit sequence to rate matching is $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL} = 1$ and the rate matching output sequence length to $E_r = \lfloor E_{UCI}/C_{UCI} \rfloor$, where

- C_{UCI} is the number of code blocks for UCI determined according to Clause 5.2.1;
- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH;
- $E_{UCI} = N_L \cdot Q'_{CG-UCI} \cdot Q_m$.

The output bit sequence after rate matching is denoted as $f_{r0}, f_{r1}, f_{r2}, \ldots, f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.2.4.1.5 HARQ-ACK and CG-UCI

For HARQ-ACK and CG-UCI transmission on PUSCH with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table, or if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is equal to 1, the number of coded modulation symbols per layer for HARQ-ACK and CG-UCI transmission, denoted as Q'_{ACK}, is determined as follows:

$$Q'_{ACK} = min\left\{ \left\lceil \frac{\left(O_{ACK} + O_{CG\text{-}UCI} + L_{ACK}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\sum_{r=0}^{C_{UL\text{-}SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=l_0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil \right\}$$

where

- O_{ACK} is the number of HARQ-ACK bits;
- $O_{CG\text{-}UCI}$ is the number of CG-UCI bits;
- if $O_{ACK} + O_{CG\text{-}UCI} \geq 360$, $L_{ACK} = 11$; otherwise L_{ACK} is the number of CRC bits for HARQ-ACK and CG-UCI determined according to Clause 6.3.1.2.1;
- $\beta_{offset}^{PUSCH} = \beta_{offset}^{HARQ\text{-}ACK}$;
- $C_{UL\text{-}SCH}$ is the number of code blocks for UL-SCH of the PUSCH transmission;
- K_r is the *r*-th code block size for UL-SCH of the PUSCH transmission;
- M_{sc}^{PUSCH} is the scheduled bandwidth of the PUSCH transmission, expressed as a number of subcarriers;
  - If the PUSCH transmission is in SBFD symbols, M_{sc}^{PUSCH} only includes subcarriers of the scheduled bandwidth that are both in the active UL BWP and in the UL sub-band, as described in Clause 6.1.2.2.1 in [6, TS 38.214];
- $M_{sc}^{PT\text{-}RS}(l)$ is the number of subcarriers in OFDM symbol *l* that carries PTRS, in the PUSCH transmission;
- $M_{sc}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l, in the PUSCH transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol *l*, for l=0,1,2,…, $N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH, including all OFDM symbols used for DMRS;
  - for any OFDM symbol that carries DMRS of the PUSCH, $M_{sc}^{UCI}(l) = 0$;
  - for any OFDM symbol that does not carry DMRS of the PUSCH, $M_{sc}^{UCI}(l) = M_{sc}^{PUSCH} - M_{sc}^{PT\text{-}RS}(l) - M_{sc}^{Muted}(l)$;
- α is configured by higher layer parameter *scaling*;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission.

For HARQ-ACK and CG-UCI transmission on PUSCH with UL-SCH, and if *numberOfSlotsTBoMS* is present in the resource allocation table and the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI is larger than 1, the number of coded modulation symbols per layer for HARQ-ACK and CG-UCI transmission, denoted as Q'_{ACK}, is determined as follows:

$$Q'_{ACK} = min\left\{ \left\lceil \frac{\left(O_{ACK} + O_{CG\text{-}UCI} + L_{ACK}\right) \cdot \beta_{offset}^{PUSCH} \cdot \sum_{l=0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l)}{\frac{1}{N_s}\sum_{r=0}^{C_{UL\text{-}SCH}-1} K_r} \right\rceil, \left\lceil \alpha \cdot \sum_{l=l_0}^{N_{symb,all}^{PUSCH}-1} M_{sc}^{UCI}(l) \right\rceil \right\}$$

where

- N_s is the value of *numberOfSlotsTBoMS* in the row indicated by the Time domain resource assignment field in DCI;
- $M_{sc}^{PT\text{-}RS}(l)$ is the number of subcarriers in OFDM symbol l that carries PTRS, in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK and CG-UCI transmission;
- $M_{\text{sc}}^{Muted}(l)$ is the number of muted subcarriers in OFDM symbol l in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK and CG-UCI transmission;
- $M_{sc}^{UCI}(l)$ is the number of resource elements that can be used for transmission of UCI in OFDM symbol l, for $l = 0,1,2,\ldots,N_{symb,all}^{PUSCH}-1$, in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK and CG-UCI transmission and $N_{symb,all}^{PUSCH}$ is the total number of OFDM symbols of the PUSCH in the slot, including all OFDM symbols used for DMRS;
- l_0 is the symbol index of the first OFDM symbol that does not carry DMRS of the PUSCH, after the first DMRS symbol(s), in the PUSCH transmission of TB processing over multiple slots in the slot with the HARQ-ACK and CG-UCI transmission;
- and all the other notations in the formula are defined the same as for PUSCH with UL-SCH and if *numberOfSlotsTBoMS* is not present in the resource allocation table.

The input bit sequence to rate matching is $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$ where r is the code block number, and N_r is the number of coded bits in code block number r.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL}=1$ and the rate matching output sequence length to $E_r = \lfloor E_{UCI}/C_{UCI} \rfloor$, where

- C_{UCI} is the number of code blocks for UCI determined according to Clause 5.2.1;
- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH;
- $E_{UCI} = N_L \cdot Q'_{ACK} \cdot Q_m$.

The output bit sequence after rate matching is denoted as $f_{r0}, f_{r1}, f_{r2}, \ldots, f_{r(E_r-1)}$ where E_r is the length of rate matching output sequence in code block number r.

6.3.2.4.1.6 UCI with different priority indexes

In this clause, $\beta_{offset}^{HARQ\text{-}ACK\text{-}LP}$ is equal to $\beta_{offset}^{HARQ\text{-}ACK,0}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 1, and equal to $\beta_{offset}^{HARQ\text{-}ACK}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 0. $\beta_{offset}^{HARQ\text{-}ACK\text{-}HP}$ is equal to $\beta_{offset}^{HARQ\text{-}ACK,1}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 0, and equal to $\beta_{offset}^{HARQ\text{-}ACK}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 1.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 1:

- If CSI part 1 is also transmitted on the PUSCH,
  - Perform rate matching for CSI part 1 according to clause 6.3.2.4.1.2, by assuming the number of HARQ-ACK information bits to be transmitted on PUSCH in clause 6.3.2.4.1.2 is 0 bit.
  - Perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.1.3, by taking HARQ-ACK with priority index 0 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ\text{-}ACK\text{-}LP}$, and assuming the number of HARQ-ACK information bits to be transmitted on PUSCH in clause 6.3.2.4.1.3 is 0 bit.
- Otherwise, perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.1.2, by taking HARQ-ACK with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ\text{-}ACK\text{-}LP}$, and

assuming the number of HARQ-ACK information bits to be transmitted on PUSCH in clause 6.3.2.4.1.2 is 0 bit.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 1, and CSI if any are transmitted on a PUSCH associated with priority index 0:

- Perform rate matching for HARQ-ACK with priority index 1 according to clause 6.3.2.4.1.1, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$.
- Perform rate matching for CSI part 1 according to clause 6.3.2.4.1.2, by taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 1 is also transmitted on the PUSCH.
- Perform rate matching for CSI part 2 according to clause 6.3.2.4.1.3, by taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 2 is also transmitted on the PUSCH.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1 and/or CG-UCI associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH:

- Perform rate matching for HARQ-ACK with priority index 1 according to clause 6.3.2.4.1.1, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$, if HARQ-ACK bits associated with priority index 1 are transmitted without CG-UCI associated with priority index 1.
- Perform rate matching for CG-UCI with priority index 1 according to clause 6.3.2.4.1.4, if CG-UCI associated with priority index 1 is transmitted without HARQ-ACK bits associated with priority index 1.
- Perform rate matching for CG-UCI with priority index 1 and HARQ-ACK with priority index 1 according to clause 6.3.2.4.1.5, if both CG-UCI associated with priority index 1 and HARQ-ACK bits associated with priority index 1 are transmitted, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$.
- If CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 1,
  - Perform rate matching for CSI part 1 according to clause 6.3.2.4.1.2, by taking HARQ-ACK with priority index 1 if any as HARQ-ACK, and taking CG-UCI associated with priority index 1 if any as CG-UCI.
  - Perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.1.3, by taking HARQ-ACK with priority index 0 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and taking HARQ-ACK with priority index 1 if any as HARQ-ACK, and taking CG-UCI associated with priority index 1 if any as CG-UCI.
- Otherwise,
  - Perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.1.2, by taking HARQ-ACK with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and taking HARQ-ACK with priority index 1 if any as HARQ-ACK, and taking CG-UCI associated with priority index 1 if any as CG-UCI.
  - Perform rate matching for CSI part 1 according to clause 6.3.2.4.1.3, by taking CSI part 1 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{CSI-part1}$, taking HARQ-ACK with priority index 0 as CSI-part 1 and taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0.

If *uci-MuxWithDiffPrio* is configured, and CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 if any, HARQ-ACK bits associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 0:

- Perform rate matching for HARQ-ACK with priority index 1 according to clause 6.3.2.4.1.1, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$.
- Perform rate matching for CG-UCI associated with priority index 0 according to clause 6.3.2.4.1.2, if CG-UCI associated with priority index 0 is transmitted without HARQ-ACK bits associated with priority index 0, by taking CG-UCI associated with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by β_{offset}^{CG-UCI}, and taking HARQ-ACK with priority index 1 as HARQ-ACK.
- Perform rate matching for CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 according to clause 6.3.2.4.1.2, if both CG-UCI associated with priority index 0 and HARQ-

ACK bits associated with priority index 0 are transmitted, by taking CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and taking HARQ-ACK with priority index 1 as HARQ-ACK.

- Perform rate matching for CSI part 1 according to clause 6.3.2.4.1.3, by taking CSI part 1 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{CSI-part1}$, taking CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 if any as CSI-part 1 and taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0.

6.3.2.4.2 UCI encoded by channel coding of small block lengths

If the higher layer parameter *nrofBitsInUTO-UCI* is configured, the procedures in this clause and the clauses it refers to apply by replacing CG-UCI with UTO-UCI in all the notations and texts.

6.3.2.4.2.1 HARQ-ACK

For HARQ-ACK transmission on PUSCH, the number of coded modulation symbols per layer for HARQ-ACK transmission, denoted as Q'_{ACK}, is determined according to Clause 6.3.2.4.1.1, by setting the number of CRC bits $L=0$.

The input bit sequence to rate matching is $d_0,d_1,d_2,\ldots,d_{N-1}$.

Rate matching is performed according to Clause 5.4.3, by setting the rate matching output sequence length $E=N_L\cdot Q'_{\text{ACK}}\cdot Q_m$, where

- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH.

The output bit sequence after rate matching is denoted as $f_0,f_1,f_2,\ldots,f_{E-1}$.

6.3.2.4.2.2 CSI part 1

For CSI part 1 transmission on PUSCH, the number of coded modulation symbols per layer for CSI part 1 transmission, denoted as $Q'_{\text{CSI,1}}$, is determined according to Clause 6.3.2.4.1.2, by setting the number of CRC bits $L=0$.

Rate matching is performed according to Clause 5.4.3, by setting the rate matching output sequence length $E=N_L\cdot Q'_{\text{CSI,1}}\cdot Q_m$, where

- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH.

The output bit sequence after rate matching is denoted as $f_0,f_1,f_2,\ldots,f_{E-1}$.

6.3.2.4.2.3 CSI part 2

For CSI part 2 transmission on PUSCH, the number of coded modulation symbols per layer for CSI part 2 transmission, denoted as $Q'_{\text{CSI,2}}$, is determined according to Clause 6.3.2.4.1.3, by setting the number of CRC bits $L=0$.

Rate matching is performed according to Clause 5.4.3, by setting the rate matching output sequence length $E=N_L\cdot Q'_{\text{CSI,2}}\cdot Q_m$, where

- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

6.3.2.4.2.4 CG-UCI

For CG-UCI transmission on PUSCH, the number of coded modulation symbols per layer for CG-UCI transmission, denoted as $Q'_{CG\text{-}UCI}$, is determined according to Clause 6.3.2.4.1.4, by setting the number of CRC bits $L_{CG\text{-}UCI} = 0$.

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$.

Rate matching is performed according to Clause 5.4.3, by setting the rate matching output sequence length $E = N_L \cdot Q'_{CG\text{-}UCI} \cdot Q_m$, where

- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

6.3.2.4.2.5 HARQ-ACK and CG-UCI

For HARQ-ACK and CG-UCI transmission on PUSCH, the number of coded modulation symbols per layer for HARQ-ACK and CG-UCI transmission, denoted as Q'_{ACK}, is determined according to Clause 6.3.2.4.1.5, by setting the number of CRC bits $L_{ACK} = 0$.

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$.

Rate matching is performed according to Clause 5.4.3, by setting the rate matching output sequence length $E = N_L \cdot Q'_{ACK} \cdot Q_m$, where

- N_L is the number of transmission layers of the PUSCH;
- Q_m is the modulation order of the PUSCH.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

6.3.2.4.2.6 UCI with different priority indexes

In this clause, $\beta_{offset}^{HARQ\text{-}ACK\text{-}LP}$ is equal to $\beta_{offset}^{HARQ\text{-}ACK,0}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 1, and equal to $\beta_{offset}^{HARQ\text{-}ACK}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 0. $\beta_{offset}^{HARQ\text{-}ACK\text{-}HP}$ is equal to $\beta_{offset}^{HARQ\text{-}ACK,1}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 0, and equal to $\beta_{offset}^{HARQ\text{-}ACK}$ defined in [5, TS38.213] in case of PUSCH associated with priority index 1.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 1:

- If CSI part 1 is also transmitted on the PUSCH,
  - Perform rate matching for CSI part 1 according to clause 6.3.2.4.2.2, by assuming the number of HARQ-ACK information bits to be transmitted on PUSCH in clause 6.3.2.4.2.2 is 0 bit.

  - Perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.2.3, by taking HARQ-ACK with priority index 0 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and assuming the number of HARQ-ACK information bits to be transmitted on PUSCH in clause 6.3.2.4.2.3 is 0 bit.
- Otherwise, perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.2.2, by taking HARQ-ACK with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and assuming the number of HARQ-ACK information bits to be transmitted on PUSCH in clause 6.3.2.4.2.2 is 0 bit.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 1, and CSI if any are transmitted on a PUSCH associated with priority index 0:

- Perform rate matching for HARQ-ACK with priority index 1 according to clause 6.3.2.4.2.1, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$.
- Perform rate matching for CSI part 1 according to clause 6.3.2.4.2.2, by taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 1 is also transmitted on the PUSCH.
- Perform rate matching for CSI part 2 according to clause 6.3.2.4.2.3, by taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 2 is also transmitted on the PUSCH.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1 and/or CG-UCI associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH:

- Perform rate matching for HARQ-ACK with priority index 1 according to clause 6.3.2.4.2.1, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$, if HARQ-ACK bits associated with priority index 1 are transmitted without CG-UCI associated with priority index 1.
- Perform rate matching for CG-UCI with priority index 1 according to clause 6.3.2.4.2.4, if CG-UCI associated with priority index 1 is transmitted without HARQ-ACK bits associated with priority index 1.
- Perform rate matching for CG-UCI with priority index 1 and HARQ-ACK with priority index 1 according to clause 6.3.2.4.2.5, if both CG-UCI associated with priority index 1 and HARQ-ACK bits associated with priority index 1 are transmitted, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$.
- If CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 1,
  - Perform rate matching for CSI part 1 according to clause 6.3.2.4.2.2, by taking HARQ-ACK with priority index 1 if any as HARQ-ACK, and taking CG-UCI associated with priority index 1 if any as CG-UCI.
  - Perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.2.3, by taking HARQ-ACK with priority index 0 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and taking HARQ-ACK with priority index 1 if any as HARQ-ACK, and taking CG-UCI associated with priority index 1 if any as CG-UCI.
- Otherwise,
  - Perform rate matching for HARQ-ACK with priority index 0 according to clause 6.3.2.4.2.2, by taking HARQ-ACK with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and taking HARQ-ACK with priority index 1 if any as HARQ-ACK, and taking CG-UCI associated with priority index 1 if any as CG-UCI.
  - Perform rate matching for CSI part 1 according to clause 6.3.2.4.2.3, by taking CSI part 1 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{CSI-part\,1}$, taking HARQ-ACK with priority index 0 as CSI-part 1 and taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0.

If *uci-MuxWithDiffPrio* is configured, and CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 if any, HARQ-ACK bits associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 0:

- Perform rate matching for HARQ-ACK with priority index 1 according to clause 6.3.2.4.2.1, by taking HARQ-ACK with priority index 1 as HARQ-ACK and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-HP}$.

- Perform rate matching for CG-UCI associated with priority index 0 according to clause 6.3.2.4.2.2, if CG-UCI associated with priority index 0 is transmitted without HARQ-ACK bits associated with priority index 0, by taking CG-UCI associated with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by β_{offset}^{CG-UCI}, and taking HARQ-ACK with priority index 1 as HARQ-ACK.
- Perform rate matching for CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 according to clause 6.3.2.4.2.2, if both CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 are transmitted, by taking CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 as CSI-part 1 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{HARQ-ACK-LP}$, and taking HARQ-ACK with priority index 1 as HARQ-ACK.
- Perform rate matching for CSI part 1 according to clause 6.3.2.4.2.3, by taking CSI part 1 as CSI part 2 and replacing β_{offset}^{PUSCH} by $\beta_{offset}^{CSI-part1}$, taking CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 if any as CSI-part 1 and taking HARQ-ACK with priority index 1 as HARQ-ACK, if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0.

6.3.2.5 Code block concatenation

Code block concatenation is performed according to Clause 6.3.1.5, except that the values of E_{UCI} and C_{UCI} given in Clause 6.3.2.4.1.

6.3.2.6 Multiplexing of coded UCI bits to PUSCH

The coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7.

6.3.2.7 Multiplexing of coded UCI bits with different priority indexes to PUSCH

If the higher layer parameter *nrofBitsInUTO-UCI* is configured, the procedure in this clause 6.3.2.7 applies by replacing CG-UCI with UTO-UCI in all the notations and texts, when applicable.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 1,

- If CSI part 1 is also transmitted on the PUSCH, the coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7 by taking HARQ-ACK with priority index 0 as CSI part 2, and assuming the number of HARQ-ACK information in Clause 6.2.7 is 0 bit;
- Otherwise, the coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7 by taking HARQ-ACK with priority index 0 as CSI-part 1, and assuming the number of HARQ-ACK information in Clause 6.2.7 is 0 bit.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 1, and CSI if any are transmitted on a PUSCH associated with priority index 0, the coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7 by taking HARQ-ACK with priority index 1 as HARQ-ACK.

If *uci-MuxWithDiffPrio* is configured, and HARQ-ACK bits associated with priority index 0, HARQ-ACK bits associated with priority index 1 and/or CG-UCI associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH,

- if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 1, the coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7 by taking HARQ-ACK with priority index 1 as HARQ-ACK, and taking HARQ-ACK with priority index 0 as CSI part 2;
- otherwise, the coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7 by taking HARQ-ACK with priority index 1 if any as HARQ-ACK, taking CG-UCI associated with priority index 1 if any as CG-UCI, taking HARQ-ACK with priority index 0 as CSI part 1, and taking CSI part 1 as CSI part 2 if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0.

If *uci-MuxWithDiffPrio* is configured, and CG-UCI associated with priority index 0 and HARQ-ACK bits associated with priority index 0 if any, HARQ-ACK bits associated with priority index 1, and CSI part 1 if any are transmitted on a PUSCH associated with priority index 0, the coded UCI bits are multiplexed onto PUSCH according to the procedures in Clause 6.2.7 by taking HARQ-ACK with priority index 1 as HARQ-ACK, taking CG-UCI associated

with priority index 0 and HARQ-ACK bits associated with priority index 0 if any as CSI part 1, and taking CSI part 1 as CSI part 2 if CSI part 1 is also transmitted on the PUSCH and the PUSCH is associated with priority index 0.

7 Downlink transport channels and control information

7.1 Broadcast channel

Data arrives to the coding unit in the form of a maximum of one transport block every 80ms. The following coding steps can be identified:

- Payload generation
- Scrambling
- Transport block CRC attachment
- Channel coding
- Rate matching

7.1.1 PBCH payload generation

Denote the bits in a transport block delivered to layer 1 by $\bar{a}_0, \bar{a}_1, \bar{a}_2, \bar{a}_3, \ldots, \bar{a}_{\bar{A}-1}$, where $\bar{A}$ is the payload size generated by higher layers. The lowest order information bit $\bar{a}_0$ is mapped to the most significant bit of the transport block as defined in Clause 6.1.1 of [8, TS 38.321].

Generate the following additional timing related PBCH payload bits $\bar{a}_{\bar{A}}, \bar{a}_{\bar{A}+1}, \bar{a}_{\bar{A}+2}, \bar{a}_{\bar{A}+3}, \ldots, \bar{a}_{\bar{A}+7}$, where:

- $\bar{a}_{\bar{A}}, \bar{a}_{\bar{A}+1}, \bar{a}_{\bar{A}+2}, \bar{a}_{\bar{A}+3}$ are the 4th, 3rd, 2nd, and 1st LSB of SFN, respectively;
- $\bar{a}_{\bar{A}+4}$ is the half frame bit $\bar{a}_{\text{HRF}}$;
- if $\bar{L}_{max} = 10$ as defined in Clause 4.1 of [5, TS38.213],

  $\bar{a}_{\bar{A}+5}$ is the MSB of k_{SSB} as defined in Clause 7.4.3.1 of [4, TS 38.211].
  $\bar{a}_{\bar{A}+6}$ is reserved.
  $\bar{a}_{\bar{A}+7}$ is the MSB of candidate SS/PBCH block index.
- else if $\bar{L}_{max} = 20$ as defined in Clause 4.1 of [5, TS38.213],

  $\bar{a}_{\bar{A}+5}$ is the MSB of k_{SSB} as defined in Clause 7.4.3.1 of [4, TS 38.211].
  $\bar{a}_{\bar{A}+6}$, $\bar{a}_{\bar{A}+7}$ are the 5th and 4th bits of the candidate SS/PBCH block index, respectively.
- else if $\bar{L}_{max} = 64$ as defined in Clause 4.1 of [5, TS38.213],

  $\bar{a}_{\bar{A}+5}$, $\bar{a}_{\bar{A}+6}$, $\bar{a}_{\bar{A}+7}$ are the 6th, 5th, and 4th bits of the candidate SS/PBCH block index, respectively.
- else

  $\bar{a}_{\bar{A}+5}$ is the MSB of k_{SSB} as defined in Clause 7.4.3.1 of [4, TS 38.211].
  $\bar{a}_{\bar{A}+6}$ is reserved.
  $\bar{a}_{\bar{A}+7}$ is the enabling/disabling bit for PDCCH repetition of Type0-PDCCH CSS set of *searchSpaceZero* as defined by Clause 13 of [5, TS38.213].
- end if

Let $A = \bar{A} + 8$; $j_{\text{SFN}} = 0$; $j_{\text{HRF}} = 10$; $j_{\text{SSB}} = 11$; $j_{\text{other}} = 14$;

for $i=0$ to $A-1$

if $\bar{a}_i$ is an SFN bit

$a_{G(j_{\text{SFN}})}=\bar{a}_i$;

$j_{\text{SFN}}=j_{\text{SFN}}+1$;

elseif $\bar{a}_i$ is the half radio frame bit

$a_{G(j_{\text{HRF}})}=\bar{a}_i$

elseif $\bar{A}+5\le i\le\bar{A}+7$

$a_{G(j_{\text{SSB}})}=\bar{a}_i$;

$j_{\text{SSB}}=j_{\text{SSB}}+1$;

else

$a_{G(j_{\text{Other}})}=\bar{a}_i$;

$j_{\text{Other}}=j_{\text{Other}}+1$;

end if

end for

where $\overline{L}_{max}$ is the number of candidate SS/PBCH blocks in a half frame according to Clause 4.1 of [5, TS38.213], and the value of $G(j)$ is given by Table 7.1.1-1.

Table 7.1.1-1: Value of PBCH payload interleaver pattern $G(j)$

j	$G(j)$	j	$G(j)$	j	$G(j)$	j	$G(j)$	j	$G(j)$	j	$G(j)$	j	$G(j)$	j	$G(j)$
0	16	4	8	8	24	12	3	16	9	20	14	24	21	28	27
1	23	5	30	9	7	13	2	17	11	21	15	25	22	29	28
2	18	6	10	10	0	14	1	18	12	22	19	26	25	30	29
3	17	7	6	11	5	15	4	19	13	23	20	27	26	31	31

7.1.2 Scrambling

For PBCH transmission in a frame, the bit sequence $a_0,a_1,a_2,a_3,\ldots,a_{A-1}$ is scrambled into a bit sequence $a'_0,a'_1,a'_2,a'_3,\ldots,a'_{A-1}$, where $a'_i=(a_i+s_i)\bmod 2$ for $i=0,1,\ldots,A-1$ and $s_0,s_1,s_2,s_3,\ldots,s_{A-1}$ is generated according to the following:

$i=0$;

$j=0$;

while $i<A$

if a_i corresponds to any one of the bits belonging to the candidate SS/PBCH block index, the half frame index, and 2nd and 3rd least significant bits of the system frame number

$s_i=0$;

else

$s_i=c(j+vM)$;

$j = j+1$;
end if
$i = i+1$;
end while

The scrambling sequence $c(i)$ is given by Clause 5.2.1of [4, TS38.211] and initialized with $c_{\text{init}} = N_{ID}^{cell}$ at the start of each SFN satisfying $\text{mod}(SFN, 8) = 0$; $M = A-3$ for $\overline{L}_{max} = 4$ or $\overline{L}_{max} = 8$, $M = A-4$ for $\overline{L}_{max} = 10$, $M = A-5$ for $\overline{L}_{max} = 20$, and $M = A-6$ for $\overline{L}_{max} = 64$, where $\overline{L}_{max}$ is the number of candidate SS/PBCH blocks in a half frame according to Clause 4.1 of [5, TS38.213]; and v is determined according to Table 7.1.2-1 using the 3rd and 2nd LSB of the SFN in which the PBCH is transmitted.

Table 7.1.2-1: Value of v for PBCH scrambling

(3rd LSB of SFN, 2nd LSB of SFN)	*Value of v*
(0, 0)	0
(0, 1)	1
(1, 0)	2
(1, 1)	3

7.1.3 Transport block CRC attachment

Error detection is provided on BCH transport blocks through a Cyclic Redundancy Check (CRC).

The entire transport block is used to calculate the CRC parity bits. The input bit sequence is denoted by $a'_0, a'_1, a'_2, a'_3, \ldots, a'_{A-1}$, and the parity bits by $p_0, p_1, p_2, p_3, \ldots, p_{L-1}$, where A is the payload size and L is the number of parity bits.

The parity bits are computed and attached to the BCH transport block according to Clause 5.1 by setting L to 24 bits and using the generator polynomial $g_{\text{CRC24C}}(D)$, resulting in the sequence $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B = A+L$.

The bit sequence $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$ is the input bit sequence $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$ to the channel encoder, where $c_i = b_i$ for $i = 0, 1, \ldots, B-1$ and $K = B$.

7.1.4 Channel coding

Information bits are delivered to the channel coding block. They are denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits, and they are encoded via Polar coding according to Clause 5.3.1, by setting $n_{\max} = 9$, $I_{IL} = 1$, $n_{PC} = 0$, and $n_{PC}^{wm} = 0$.

After encoding the bits are denoted by $d_0, d_1, d_2, d_3, \ldots, d_{N-1}$, where N is the number of coded bits.

7.1.5 Rate matching

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$.

The rate matching output sequence length $E = 864$.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL} = 0$.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

7.2 Downlink shared channel and paging channel

7.2.1 Transport block CRC attachment

Error detection is provided on each transport block through a Cyclic Redundancy Check (CRC).

The entire transport block is used to calculate the CRC parity bits. Denote the bits in a transport block delivered to layer 1 by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, and the parity bits by $p_0, p_1, p_2, p_3, \ldots, p_{L-1}$, where A is the payload size and L is the number of parity bits. The lowest order information bit a_0 is mapped to the most significant bit of the transport block as defined in Clause 6.1.1 of [TS38.321].

The parity bits are computed and attached to the DL-SCH transport block according to Clause 5.1, by setting L to 24 bits and using the generator polynomial $g_{\text{CRC24A}}(D)$ if $A > 3824$; and by setting L to 16 bits and using the generator polynomial $g_{\text{CRC16}}(D)$ otherwise.

The bits after CRC attachment are denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$, where $B = A + L$.

7.2.2 LDPC base graph selection

For initial transmission of a transport block with coding rate R indicated by the MCS index according to Clause 5.1.3.1 in [6, TS 38.214] and subsequent re-transmission of the same transport block, each code block of the transport block is encoded with either LDPC base graph 1 or 2 according to the following:

- if $A \leq 292$, or if $A \leq 3824$ and $R \leq 0.67$, or if $R \leq 0.25$, LDPC base graph 2 is used;
- otherwise, LDPC base graph 1 is used,

where A is the payload size in Clause 7.2.1.

7.2.3 Code block segmentation and code block CRC attachment

The bits input to the code block segmentation are denoted by $b_0, b_1, b_2, b_3, \ldots, b_{B-1}$ where B is the number of bits in the transport block (including CRC).

Code block segmentation and code block CRC attachment are performed according to Clause 5.2.2.

The bits after code block segmentation are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$, where r is the code block number and K_r is the number of bits for code block number r according to Clause 5.2.2.

7.2.4 Channel coding

Code blocks are delivered to the channel coding block. The bits in a code block are denoted by $c_{r0}, c_{r1}, c_{r2}, c_{r3}, \ldots, c_{r(K_r-1)}$, where r is the code block number, and K_r is the number of bits in code block number r. The total number of code blocks is denoted by C and each code block is individually LDPC encoded according to Clause 5.3.2.

After encoding the bits are denoted by $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$, where the values of N_r is given in Clause 5.3.2.

7.2.5 Rate matching

Coded bits for each code block, denoted as $d_{r0}, d_{r1}, d_{r2}, d_{r3}, \ldots, d_{r(N_r-1)}$, are delivered to the rate match block, where r is the code block number, and N_r is the number of encoded bits in code block number r. The total

number of code blocks is denoted by C and each code block is individually rate matched according to Clause 5.4.2 by setting $I_{LBRM}=1$.

After rate matching, the bits are denoted by $f_{r0},f_{r1},f_{r2},f_{r3},\ldots,f_{r(E_r-1)}$, where E_r is the number of rate matched bits for code block number r.

7.2.6 Code block concatenation

The input bit sequence for the code block concatenation block are the sequences $f_{r0},f_{r1},f_{r2},f_{r3},\ldots,f_{r(E_r-1)}$, for $r=0,\ldots,C-1$ and where E_r is the number of rate matched bits for the r-th code block.

Code block concatenation is performed according to Clause 5.5.

The bits after code block concatenation are denoted by $g_0,g_1,g_2,g_3,\ldots,g_{G-1}$, where G is the total number of coded bits for transmission.

7.3 Downlink control information

A DCI transports downlink control information for one or more cells with one RNTI.

The following coding steps can be identified:

- Information element multiplexing
- CRC attachment
- Channel coding
- Rate matching

7.3.1 DCI formats

The DCI formats defined in table 7.3.1-1 are supported.

Table 7.3.1-1: DCI formats

DCI format	Usage
0_0	Scheduling of PUSCH in one cell
0_1	Scheduling of one or multiple PUSCH in one cell, or indicating downlink feedback information for configured grant PUSCH (CG-DFI)
0_2	Scheduling of PUSCH in one cell
0_3	Scheduling of one or multiple PUSCHs in one cell, or multiple PUSCHs in multiple cells with one or multiple PUSCHs per cell
1_0	Scheduling of PDSCH in one cell
1_1	Scheduling of one or multiple PDSCH in one cell, and/or triggering one shot HARQ-ACK codebook feedback
1_2	Scheduling of PDSCH in one cell
1_3	Scheduling of one or multiple PDSCHs in one cell, or multiple PDSCHs in multiple cells with one or multiple PDSCHs per cell
2_0	Notifying a group of UEs of the slot format, available RB sets, COT duration and search space set group switching
2_1	Notifying a group of UEs of the PRB(s) and OFDM symbol(s) where UE may assume no transmission is intended for the UE
2_2	Transmission of TPC commands for PUCCH and PUSCH
2_3	Transmission of a group of TPC commands for SRS transmissions by one or more UEs
2_4	Notifying a group of UEs of the PRB(s) and OFDM symbol(s) where UE cancels the corresponding UL transmission from the UE
2_5	Notifying the availability of soft resources as defined in Clause 9.3.1 of [10, TS 38.473]
2_6	Notifying the power saving information outside DRX Active Time for one or more UEs
2_7	Notifying paging early indication and TRS availability indication for one or more UEs
2_8	Notifying the aperiodic beam indication and associated time resources

DCI format	Usage
2_9	Activating or de-activating the cell DTX and/or DRX configuration of one or multiple serving cells for one or more UEs, and/or for providing NES-mode indication of the primary cell for one or more UEs, or for adapting SSB periodicity of one or multiple serving cells for one or more UEs
3_0	Scheduling of NR sidelink in one cell
3_1	Scheduling of LTE sidelink in one cell
3_2	Scheduling of NR SL PRS in one cell
4_0	Scheduling of PDSCH with CRC scrambled by MCCH-RNTI/G-RNTI for broadcast or by Multicast MCCH-RNTI for multicast in RRC_INACITIVE state
4_1	Scheduling of PDSCH with CRC scrambled by G-RNTI/G-CS-RNTI for multicast in RRC_CONNECTED state or by G-RNTI for multicast in RRC_INACITIVE state
4_2	Scheduling of PDSCH with CRC scrambled by G-RNTI/G-CS-RNTI for multicast in RRC_CONNECTED state

The fields defined in the DCI formats below are mapped to the information bits a_0 to a_{A-1} as follows.

Each field is mapped in the order in which it appears in the description, including the zero-padding bit(s), if any, with the first field mapped to the lowest order information bit a_0 and each successive field mapped to higher order information bits. The most significant bit of each field is mapped to the lowest order information bit for that field, e.g. the most significant bit of the first field is mapped to a_0.

If the number of information bits in a DCI format is less than 12 bits, zeros shall be appended to the DCI format until the payload size equals 12.

The size of each DCI format except for DCI format 0_3/1_3 is determined by the configuration of the corresponding active bandwidth part of the scheduled cell and shall be adjusted as described in clause 7.3.1.0 if necessary.

For a cell set configured by higher layer parameter *mc-DCI-SetofCellsToAddModList*, the size of DCI format 0_3/1_3 is determined as follows and shall be adjusted as described in Clause 7.3.1.0 if necessary:

- If *scheduledCellComboListDCI-0-3* for the cell set is configured, the size of DCI format 0_3 is determined by the configuration of the corresponding active bandwidth part(s) of the scheduled cells in the entry which results in the largest size among the entries in the higher layer parameter *scheduledCellComboListDCI-0-3*; Otherwise, the size of DCI format 0_3 is determined by the configuration of the corresponding active bandwidth part(s) of the cells configured by higher layer parameter *scheduledCellListDCI-0-3* for the cell set.
- If *scheduledCellComboListDCI-1-3* for the cell set is configured, the size of DCI format 1_3 is determined by the configuration of the corresponding active bandwidth part(s) of the scheduled cells in the entry which results in the largest size among the entries in the higher layer parameter *scheduledCellComboListDCI-1-3*; Otherwise, the size of DCI format 1_3 is determined by the configuration of the corresponding active bandwidth part(s) of the cells configured by higher layer parameter *scheduledCellListDCI-1-3* for the cell set.

If a UE is configured with *pdsch-HARQ-ACK-CodebookList-r16*, *pdsch-HARQ-ACK-Codebook* is replaced by the relevant entry in *pdsch-HARQ-ACK-CodebookList-r16* in this clause.

If a UE is configured with *pdsch-HARQ-ACK-CodebookListMulticast-r17*, *pdsch-HARQ-ACK-Codebook* is replaced by the relevant entry in *pdsch-HARQ-ACK-CodebookListMulticast-r17* in this clause.

For a cell detected in cell search procedure with synchronization raster defined in Table 5.4.3.1-2 or Table 5.4.3.1-3 of [13, TS 38.101-1] or Table 5.4.3.1-2 of [15, TS 38.101-5], the size of CORESET 0 for the cell in this clause refers to the size of punctured CORESET 0 as defined in clause 7.3.2.2 of [4, TS 38.211] if any.

7.3.1.0 DCI size alignment

If necessary, padding or truncation shall be applied to the DCI formats according to the following steps executed in the order below:

Step 0:

- Determine DCI format 0_0 monitored in a common search space according to clause 7.3.1.1.1 where $N_{\mathrm{RB}}^{\mathrm{UL,BWP}}$ is the size of the initial UL bandwidth part.

- Determine DCI format 1_0 monitored in a common search space according to clause 7.3.1.2.1 where $N_{RB}^{DL,BWP}$ is given by
  - the size of CORESET 0 if CORESET 0 is configured for the cell; and
  - the size of initial DL bandwidth part if CORESET 0 is not configured for the cell.
- If DCI format 0_0 is monitored in common search space and if the number of information bits in the DCI format 0_0 prior to padding is less than the payload size of the DCI format 1_0 monitored in common search space for scheduling the same serving cell, a number of zero padding bits are generated for the DCI format 0_0 until the payload size equals that of the DCI format 1_0.
- If DCI format 0_0 is monitored in common search space and if the number of information bits in the DCI format 0_0 prior to truncation is larger than the payload size of the DCI format 1_0 monitored in common search space for scheduling the same serving cell, the bitwidth of the frequency domain resource assignment field in the DCI format 0_0 is reduced by truncating the first few most significant bits such that the size of DCI format 0_0 equals the size of the DCI format 1_0.

Step 1:

- Determine DCI format 0_0 monitored in a UE-specific search space according to clause 7.3.1.1.1 where $N_{RB}^{UL,BWP}$ is the size of the active UL bandwidth part.
- Determine DCI format 1_0 monitored in a UE-specific search space according to clause 7.3.1.2.1 where $N_{RB}^{DL,BWP}$ is the size of the active DL bandwidth part.
- For a UE configured with *supplementaryUplink* in *ServingCellConfig* in a cell, if PUSCH is configured to be transmitted on both the SUL and the non-SUL of the cell and if the number of information bits in DCI format 0_0 in UE-specific search space for the SUL is not equal to the number of information bits in DCI format 0_0 in UE-specific search space for the non-SUL, a number of zero padding bits are generated for the smaller DCI format 0_0 until the payload size equals that of the larger DCI format 0_0.
- If DCI format 0_0 is monitored in UE-specific search space and if the number of information bits in the DCI format 0_0 prior to padding is less than the payload size of the DCI format 1_0 monitored in UE-specific search space for scheduling the same serving cell, a number of zero padding bits are generated for the DCI format 0_0 until the payload size equals that of the DCI format 1_0.
- If DCI format 1_0 is monitored in UE-specific search space and if the number of information bits in the DCI format 1_0 prior to padding is less than the payload size of the DCI format 0_0 monitored in UE-specific search space for scheduling the same serving cell, zeros shall be appended to the DCI format 1_0 until the payload size equals that of the DCI format 0_0

Step 2:

- Determine DCI format 0_1 monitored in a UE-specific search space according to clause 7.3.1.1.2.
- Determine DCI format 1_1 monitored in a UE-specific search space according to clause 7.3.1.2.2.
- For a UE configured with *supplementaryUplink* in *ServingCellConfig* in a cell, if PUSCH is configured to be transmitted on both the SUL and the non-SUL of the cell and if the number of information bits in format 0_1 for the SUL is not equal to the number of information bits in format 0_1 for the non-SUL, zeros shall be appended to smaller format 0_1 until the payload size equals that of the larger format 0_1.
- If the size of DCI format 0_1 monitored in a UE-specific search space equals that of a DCI format 0_0/1_0 monitored in another UE-specific search space, one bit of zero padding shall be appended to DCI format 0_1.
- If the size of DCI format 1_1 monitored in a UE-specific search space equals that of a DCI format 0_0/1_0 monitored in another UE-specific search space, one bit of zero padding shall be appended to DCI format 1_1.

Step 2A:

- Determine DCI format 0_2 monitored in a UE-specific search space according to clause 7.3.1.1.3.
- Determine DCI format 1_2 monitored in a UE-specific search space according to clause 7.3.1.2.3.
- For a UE configured with *supplementaryUplink* in *ServingCellConfig* in a cell, if PUSCH is configured to be transmitted on both the SUL and the non-SUL of the cell and if the number of information bits in

format 0_2 for the SUL is not equal to the number of information bits in format 0_2 for the non-SUL, zeros shall be appended to smaller format 0_2 until the payload size equals that of the larger format 0_2.

Step 2B:

- If the cell is the serving cell for counting the size of one or both DCI format 0_3 and DCI format 1_3 as defined in Clause 10.1 of [5, TS38.213],
  - Determine DCI format 0_3 monitored in a UE-specific search space according to clause 7.3.1.1.4.
  - Determine DCI format 1_3 monitored in a UE-specific search space according to clause 7.3.1.2.4.

Step 3:

- If both of the following conditions are fulfilled the size alignment procedure is complete
  - the total number of different DCI sizes configured to monitor is no more than 4 for the cell
  - the total number of different DCI sizes with C-RNTI configured to monitor is no more than 3 for the cell

Step 4:

- Otherwise

  Step 4A:
  - Remove the padding bit (if any) introduced in step 2 above.
  - Determine DCI format 1_0 monitored in a UE-specific search space according to clause 7.3.1.2.1 where $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is given by
    - the size of CORESET 0 if CORESET 0 is configured for the cell; and
    - the size of initial DL bandwidth part if CORESET 0 is not configured for the cell.
  - Determine DCI format 0_0 monitored in a UE-specific search space according to clause 7.3.1.1.1 where $N_{\mathrm{RB}}^{\mathrm{UL,BWP}}$ is the size of the initial UL bandwidth part.
  - If the number of information bits in the DCI format 0_0 monitored in a UE-specific search space prior to padding is less than the payload size of the DCI format 1_0 monitored in UE-specific search space for scheduling the same serving cell, a number of zero padding bits are generated for the DCI format 0_0 monitored in a UE-specific search space until the payload size equals that of the DCI format 1_0 monitored in a UE-specific search space.
  - If the number of information bits in the DCI format 0_0 monitored in a UE-specific search space prior to truncation is larger than the payload size of the DCI format 1_0 monitored in UE-specific search space for scheduling the same serving cell, the bitwidth of the frequency domain resource assignment field in the DCI format 0_0 is reduced by truncating the first few most significant bits such that the size of DCI format 0_0 monitored in a UE-specific search space equals the size of the DCI format 1_0 monitored in a UE-specific search space.

  Step 4B:
  - If the total number of different DCI sizes configured to monitor is more than 4 for the cell after applying the above steps, or if the total number of different DCI sizes with C-RNTI configured to monitor is more than 3 for the cell after applying the above steps
    - If the number of information bits in the DCI format 0_2 prior to padding is less than the payload size of the DCI format 1_2 for scheduling the same serving cell, a number of zero padding bits are generated for the DCI format 0_2 until the payload size equals that of the DCI format 1_2.
    - If the number of information bits in the DCI format 1_2 prior to padding is less than the payload size of the DCI format 0_2 for scheduling the same serving cell, zeros shall be appended to the DCI format 1_2 until the payload size equals that of the DCI format 0_2.

  Step 4C:
  - If the total number of different DCI sizes configured to monitor is more than 4 for the cell after applying the above steps, or if the total number of different DCI sizes with C-RNTI configured to monitor is more than 3 for the cell after applying the above steps
    - If the number of information bits in the DCI format 0_1 prior to padding is less than the payload size of the DCI format 1_1 for scheduling the same serving cell, a number of zero padding bits are generated for the DCI format 0_1 until the payload size equals that of the DCI format 1_1.
    - If the number of information bits in the DCI format 1_1 prior to padding is less than the payload size of the DCI format 0_1 for scheduling the same serving cell, zeros shall be appended to the DCI format 1_1 until the payload size equals that of the DCI format 0_1.

  Step 4D:
  - If the total number of different DCI sizes configured to monitor is more than 4 for the cell after applying the above steps and the cell is the serving cell for counting the size of one or both DCI format 0_3 and DCI format 1_3 as defined in Clause 10.1 of [5, TS38.213], or if the total number of different DCI sizes with C-

RNTI configured to monitor is more than 3 for the cell after applying the above steps and the cell is the serving cell for counting the size of one or both DCI format 0_3 and DCI format 1_3 as defined in Clause 10.1 of [5, TS38.213]

- If the number of information bits in the DCI format 0_3 prior to padding is less than the payload size of the DCI format 1_3 for scheduling the same cell set, a number of zero padding bits are generated for the DCI format 0_3 until the payload size equals that of the DCI format 1_3.
- If the number of information bits in the DCI format 1_3 prior to padding is less than the payload size of the DCI format 0_3 for scheduling the same cell set, zeros shall be appended to the DCI format 1_3 until the payload size equals that of the DCI format 0_3.

The UE is not expected to handle a configuration that, after applying the above steps, results in

- the total number of different DCI sizes configured to monitor is more than 4 for the cell; or
- the total number of different DCI sizes with C-RNTI configured to monitor is more than 3 for the cell; or
- the size of DCI format 0_0 in a UE-specific search space is equal to DCI format 0_1 in another UE-specific search space; or
- the size of DCI format 1_0 in a UE-specific search space is equal to DCI format 1_1 in another UE-specific search space; or
- the size of DCI format 0_0 in a UE-specific search space is equal to DCI format 0_2 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 0_0 and 0_2 are mapped to the same resource; or
- the size of DCI format 1_0 in a UE-specific search space is equal to DCI format 1_2 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 1_0 and 1_2 are mapped to the same resource; or
- the size of DCI format 0_1 in a UE-specific search space is equal to DCI format 0_2 in the same or another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 0_1 and 0_2 are mapped to the same resource; or
- the size of DCI format 1_1 in a UE-specific search space is equal to DCI format 1_2 in the same or another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 1_1 and 1_2 are mapped to the same resource; or
- the size of DCI format 0_0 in a UE-specific search space is equal to DCI format 0_3 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 0_0 and 0_3 are mapped to the same resource; or
- the size of DCI format 1_0 in a UE-specific search space is equal to DCI format 1_3 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 1_0 and 1_3 are mapped to the same resource; or
- the size of DCI format 0_1 in a UE-specific search space is equal to DCI format 0_3 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 0_1 and 0_3 are mapped to the same resource; or
- the size of DCI format 1_1 in a UE-specific search space is equal to DCI format 1_3 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 1_1 and 1_3 are mapped to the same resource.
- the size of DCI format 0_2 in a UE-specific search space is equal to DCI format 0_3 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 0_2 and 0_3 are mapped to the same resource; or
- the size of DCI format 1_2 in a UE-specific search space is equal to DCI format 1_3 in another UE-specific search space when at least one pair of the corresponding PDCCH candidates of DCI formats 1_2 and 1_3 are mapped to the same resource.

7.3.1.0.1 DCI size alignment for DCI formats for scheduling of sidelink

If DCI format 3_0, and/or DCI format 3_1, and/or DCI format 3_2 is monitored on a cell, DCI size alignment for DCI format 3_0, DCI format 3_1, and DCI format 3_2 is performed as described in this clause after performing the DCI size alignment described in Clause 7.3.1.0. The size(s) of the DCI formats configured to monitor for a cell in this clause refers to that after performing the DCI size alignment described in Clause 7.3.1.0.

If DCI format 3_0, and/or DCI format 3_1, and/or DCI format 3_2 is monitored on a cell and the total number of DCI sizes of the DCI formats configured to monitor for the cell and DCI format 3_0, and/or DCI format 3_1, and/or DCI format 3_2 is more than 4, zeros shall be appended to DCI format 3_0 if configured, to DCI format 3_1 if configured, and to DCI format 3_2 if configured, until the payload size of DCI format 3_0, DCI format 3_1, and DCI format 3_2 equals that of the smallest DCI format configured to monitor for the cell that is larger than DCI format 3_0, DCI format 3_1, and DCI format 3_2.

The UE is not expected to handle a configuration that results in:

- the total number of different DCI sizes configured to monitor for the cell and DCI format 3_0, and/or DCI format 3_1, and/or DCI format 3_2 is more than 4; and
- the payload size of DCI format 3_0, and/or DCI format 3_1, and/or DCI format 3_2 is larger than the payload size of all other DCI formats configured to monitor for the cell.

7.3.1.1 DCI formats for scheduling of PUSCH

7.3.1.1.1 Format 0_0

DCI format 0_0 is used for the scheduling of PUSCH in one cell.

The following information is transmitted by means of the DCI format 0_0 with CRC scrambled by C-RNTI or CS-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bit
  - The value of this bit field is always set to 0, indicating an UL DCI format
- Frequency domain resource assignment - number of bits determined by the following:
  - $\lceil \log_2(N_{RB}^{UL,BWP}(N_{RB}^{UL,BWP}+1)/2) \rceil$ bits if neither of the higher layer parameters *useInterlacePUCCH-PUSCH* in *BWP-UplinkCommon* and *useInterlacePUCCH-PUSCH* in *BWP-UplinkDedicated* is configured, where $N_{RB}^{UL,BWP}$ is defined in clause 7.3.1.0
    - For PUSCH hopping with resource allocation type 1:
      - N_{UL_hop} MSB bits are used to indicate the frequency offset according to Clause 6.3 of [6, TS 38.214], where $N_{UL_hop}=1$ if the higher layer parameter *frequencyHoppingOffsetLists* contains two offset values and $N_{UL_hop}=2$ if the higher layer parameter *frequencyHoppingOffsetLists* contains four offset values
      - $\lceil \log_2(N_{RB}^{UL,BWP}(N_{RB}^{UL,BWP}+1)/2) \rceil - N_{UL_hop}$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
    - For non-PUSCH hopping with resource allocation type 1:
      - $\lceil \log_2(N_{RB}^{UL,BWP}(N_{RB}^{UL,BWP}+1)/2) \rceil$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
  - If any of the higher layer parameters *useInterlacePUCCH-PUSCH* in *BWP-UplinkCommon* and *useInterlacePUCCH-PUSCH* in *BWP-UplinkDedicated* is configured
    - 5+Y bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 30 kHz.
    - 6+Y bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 15 kHz.

    If the DCI format 0_0 is monitored in a UE-specific search space, the value of Y is determined by $\lceil \log_2\left(\frac{N_{RB\text{-}set,UL}^{BWP}\left(N_{RB\text{-}set,UL}^{BWP}+1\right)}{2}\right) \rceil$ where $N_{RB\text{-}set,UL}^{BWP}$ is the number of RB sets contained in the active UL BWP as defined in clause 7 of [6, TS38.214]. If the DCI 0_0 is monitored in a common search space Y = 0.
- Time domain resource assignment - 4 bits as defined in Clause 6.1.2.1 of [6, TS 38.214]
- Frequency hopping flag - 1 bit according to Table 7.3.1.1.1-3, as defined in Clause 6.3 of [6, TS 38.214]

- Modulation and coding scheme - 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- HARQ process number - 4 bits
- TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS 38.213]
- ChannelAccess-CPext - 2 bits indicating combinations of channel access type and CP extension as defined in Table 7.3.1.1.1-4, or Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1; 2 bits indicating channel access type as defined in Table 7.3.1.1.1-4B if *ChannelAccessMode2-r17* is provided for operation in a cell in frequency range 2-2; 0 bit otherwise.
- Padding bits, if required.
- UL/SUL indicator - 1 bit for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell as defined in Table 7.3.1.1.1-1 and the number of bits for DCI format 1_0 before padding is larger than the number of bits for DCI format 0_0 before padding; 0 bit otherwise. The UL/SUL indicator, if present, locates in the last bit position of DCI format 0_0, after the padding bit(s).
  - If the UL/SUL indicator is present in DCI format 0_0 and the higher layer parameter *pusch-Config* is not configured on both UL and SUL the UE ignores the UL/SUL indicator field in DCI format 0_0, and the corresponding PUSCH scheduled by the DCI format 0_0 is for the UL or SUL for which high layer parameter *pucch-Config* is configured;
  - If the UL/SUL indicator is not present in DCI format 0_0 and *pucch-Config* is configured, the corresponding PUSCH scheduled by the DCI format 0_0 is for the UL or SUL for which high layer parameter *pucch-Config* is configured.
  - If the UL/SUL indicator is not present in DCI format 0_0 and *pucch-Config* is not configured, the corresponding PUSCH scheduled by the DCI format 0_0 is for the uplink on which the latest PRACH is transmitted.

The following information is transmitted by means of the DCI format 0_0 with CRC scrambled by TC-RNTI:

- Identifier for DCI formats - 1 bit
  - The value of this bit field is always set to 0, indicating an UL DCI format
- Frequency domain resource assignment - number of bits determined by the following:
  - $\lceil \log_2(N_{RB}^{UL,BWP}(N_{RB}^{UL,BWP}+1)/2) \rceil$ bits if the higher layer parameter *useInterlacePUCCH-PUSCH* in *BWP-UplinkCommon* is not configured, where
    - $N_{RB}^{UL,BWP}$ is the size of the initial UL bandwidth part.
    - For PUSCH hopping with resource allocation type 1:
      - N_{UL_hop} MSB bits are used to indicate the frequency offset according to Table 8.3-1 in Clause 8.3 of [5, TS 38.213], where $N_{UL_hop}=1$ if $N_{RB}^{UL,BWP}<50$ and $N_{UL_hop}=2$ otherwise
      - $\lceil \log_2(N_{RB}^{UL,BWP}(N_{RB}^{UL,BWP}+1)/2) \rceil - N_{UL_hop}$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
    - For non-PUSCH hopping with resource allocation type 1:
      - $\lceil \log_2(N_{RB}^{UL,BWP}(N_{RB}^{UL,BWP}+1)/2) \rceil$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
  - If the higher layer parameter *useInterlacePUCCH-PUSCH* in *BWP-UplinkCommon* is configured
    - 5 bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 30 kHz
    - 6 bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 15 kHz
- Time domain resource assignment - 4 bits as defined in Clause 6.1.2.1 of [6, TS 38.214]
- Frequency hopping flag - 1 bit according to Table 7.3.1.1.1-3, as defined in Clause 6.3 of [6, TS 38.214]
- Modulation and coding scheme - 5 bits

  - If the UE requests repetition of PUSCH scheduled by RAR UL grant [8, TS 38.321], 5 bits as defined in Clause 6.1.2.1 and Clause 6.1.4.1 of [6, TS 38.214];
  - otherwise 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214].
- New data indicator - 1 bit, reserved
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- HARQ process number - 4 bits, reserved
- TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS 38.213]
- ChannelAccess-CPext - 2 bits indicating combinations of channel access type and CP extension as defined in Table 7.3.1.1.1-4, or Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1; 2 bits indicating channel access type as defined in Table 7.3.1.1.1-4B if *ChannelAccessMode2-r17* is provided for operation in a cell in frequency range 2-2; 0 bit otherwise
- Padding bits, if required.
- UL/SUL indicator - 1 bit if the cell has two ULs and the number of bits for DCI format 1_0 before padding is larger than the number of bits for DCI format 0_0 before padding; 0 bit otherwise. The UL/SUL indicator, if present, locates in the last bit position of DCI format 0_0, after the padding bit(s).
  - If 1 bit, reserved, and the corresponding PUSCH is always on the same UL carrier as the previous transmission of the same TB

Table 7.3.1.1.1-1: UL/SUL indicator

Value of UL/SUL indicator	Uplink
0	The non-supplementary uplink
1	The supplementary uplink

Table 7.3.1.1.1-2: Redundancy version

Value of the Redundancy version field	Value of rv_{id} to be applied
00	0
01	1
10	2
11	3

Table 7.3.1.1.1-3: Frequency hopping indication

Bit field mapped to index	PUSCH frequency hopping
0	Disabled
1	Enabled

Table 7.3.1.1.1-4: Channel access type & CP extension for DCI format 0_0 and DCI format 1_0 for frequency range 1

Bit field mapped to index	Channel Access Type	The CP extension T_"ext" index defined in Clause 5.3.1 of [4, TS 38.211]
0	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	2
1	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	3
2	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	1
3	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	0

Table 7.3.1.1.1-4A: Channel access type & CP extension if *channelAccessMode-r16* = "*semiStatic*" is provided

Bit field mapped to index	Channel Access Type	The CP extension T_"ext" index defined in Clause 5.3.1 of [4, TS 38.211]	Initiator of the channel occupancy associated with the UL transmission as described in Clause 4.3.1 in TS 37.213
0	No sensing as defined in Clause 4.3 in TS 37.213 [14]	0	gNB
1	No sensing as defined in Clause 4.3 in TS 37.213 [14]	2	gNB
2	Sensing within a 25us interval as defined in Clause 4.3 in TS 37.213 [14]	0	gNB
3	Sensing as defined in Clause 4.3.1.2 in TS 37.213 [14]	0	UE
NOTE: Row index 3 is only applicable if *semiStaticChannelAccessConfigUE* is provided. Otherwise, the row is reserved.			

Table 7.3.1.1.1-4B: Channel access type for DCI format 0_0 and DCI format 1_0 for frequency range 2-2

Bit field mapped to index	Channel Access Type
0	Type 1 channel access defined in clause 4.4.1 of TS 37.213 [14]
1	Type 2 channel access defined in clause 4.4.2 of TS 37.213 [14]
2	Type 3 channel access defined in clause 4.4.3 of TS 37.213 [14]
3	Reserved

7.3.1.1.2 Format 0_1

DCI format 0_1 is used for the scheduling of one or multiple PUSCH in one cell, or indicating CG downlink feedback information (CG-DFI) to a UE.

The following information is transmitted by means of the DCI format 0_1 with CRC scrambled by C-RNTI or CS-RNTI or SP-CSI-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bit
  - The value of this bit field is always set to 0, indicating an UL DCI format
- Carrier indicator - 0 or 3 bits, as defined in Clause 10.1 of [5, TS38.213]. This field is reserved when this format is carried by PDCCH on the primary cell and the UE is configured for scheduling on the primary cell from an SCell, with the same number of bits as that in this format carried by PDCCH on the SCell for scheduling on the primary cell.
- DFI flag - 0 or 1 bit
  - 1 bit if the UE is configured to monitor DCI format 0_1 with CRC scrambled by CS-RNTI and for operation in a cell with shared spectrum channel access when the higher layer parameter *cg-RetransmissionTimer* is configured. For a DCI format 0_1 with CRC scrambled by CS-RNTI, the bit value of 0 indicates activating or releasing type 2 CG transmission and the bit value of 1 indicates CG-DFI. For a DCI format 0_1 with CRC scrambled by C-RNTI/SP-CSI-RNTI/MCS-C-RNTI and for operation in a cell with shared spectrum channel access, the bit is reserved.
  - 0 bit otherwise;

If DCI format 0_1 is used for indicating CG-DFI, all the remaining fields are set as follows:

- HARQ-ACK bitmap - 16 bits if *nrofHARQ-Processes-v1700* in *ConfiguredGrantConfig* is not configured or 32 bits if *nrofHARQ-Processes-v1700* in *ConfiguredGrantConfig* is configured, where the order of the bitmap to HARQ process index mapping is such that HARQ process indices are mapped in ascending order from MSB to LSB of the bitmap. For each bit of the bitmap, value 1 indicates ACK, and value 0 indicates NACK.
- TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS38.213]
- All the remaining bits in format 0_1 are set to zero.

Otherwise, all the remaining fields are set as follows:

- UL/SUL indicator - 0 bit for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell or UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell but only one carrier in the cell is configured for PUSCH transmission; otherwise, 1 bit as defined in Table 7.3.1.1.1-1.
- Bandwidth part indicator - 0, 1 or 2 bits as determined by the number of UL BWPs $n_{\text{BWP,RRC}}$ configured by higher layers, excluding the initial UL bandwidth part. The bitwidth for this field is determined as $\lceil \log_2(n_{\text{BWP}}) \rceil$ bits, where
  - $n_{\text{BWP}} = n_{\text{BWP,RRC}} + 1$ if $n_{\text{BWP,RRC}} \leq 3$, in which case the bandwidth part indicator is equivalent to the ascending order of the higher layer parameter *BWP-Id*;
  - otherwise $n_{\text{BWP}} = n_{\text{BWP,RRC}}$, in which case the bandwidth part indicator is defined in Table 7.3.1.1.2-1;

  If a UE does not support active BWP change via DCI, the UE ignores this bit field.
- Frequency domain resource assignment - number of bits determined by the following, where $N_{\text{RB}}^{\text{UL,BWP}}$ is the size of the active UL bandwidth part:
  - If higher layer parameter *useInterlacePUCCH-PUSCH* in *BWP-UplinkDedicated* is not configured
    - N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 6.1.2.2.1 of [6, TS 38.214],
    - $\lceil \log_2(N_{\text{RB}}^{\text{UL,BWP}}(N_{\text{RB}}^{\text{UL,BWP}}+1)/2) \rceil$ bits if only resource allocation type 1 is configured, or $\max\left(\lceil \log_2(N_{\text{RB}}^{\text{UL,BWP}}(N_{\text{RB}}^{\text{UL,BWP}}+1)/2) \rceil, N_{\text{RBG}} \right)+1$ bits if *resourceAllocation* is configured as '*dynamicSwitch*'.
    - If *resourceAllocation* is configured as '*dynamicSwitch*', the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
    - For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 6.1.2.2.1 of [6, TS 38.214].
    - For resource allocation type 1, the $\lceil \log_2(N_{\text{RB}}^{\text{UL,BWP}}(N_{\text{RB}}^{\text{UL,BWP}}+1)/2) \rceil$ LSBs provide the resource allocation as follows:
      - For PUSCH hopping with resource allocation type 1:
        - $N_{\text{UL_hop}}$ MSB bits are used to indicate the frequency offset according to Clause 6.3 of [6, TS 38.214], where $N_{\text{UL_hop}} = 1$ if the higher layer parameter *frequencyHoppingOffsetLists* contains two offset values and $N_{\text{UL_hop}} = 2$ if the higher layer parameter *frequencyHoppingOffsetLists* contains four offset values
        - $\lceil \log_2(N_{\text{RB}}^{\text{UL,BWP}}(N_{\text{RB}}^{\text{UL,BWP}}+1)/2) \rceil - N_{\text{UL_hop}}$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
      - For non-PUSCH hopping with resource allocation type 1:
        - $\lceil \log_2(N_{\text{RB}}^{\text{UL,BWP}}(N_{\text{RB}}^{\text{UL,BWP}}+1)/2) \rceil$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]

    If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and if *resourceAllocation* is configured as '*dynamicSwitch*' for the indicated bandwidth part, the UE assumes resource allocation type 0 for the indicated bandwidth part if the bitwidth of the "Frequency domain resource assignment" field of the active bandwidth part is smaller than the bitwidth of the "Frequency domain resource assignment" field of the indicated bandwidth part.

- If the higher layer parameter *useInterlacePUCCH-PUSCH* in *BWP-UplinkDedicated* is configured
  - 5 + Y bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 30 kHz. The 5 MSBs provide the interlace allocation and the Y LSBs provide the RB set allocation.
  - 6 + Y bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 15 kHz. The 6 MSBs provide the interlace allocation and the Y LSBs provide the RB set allocation.

  The value of Y is determined by $\lceil \log_2\left(\frac{N_{RB\text{-}set,UL}^{BWP}\left(N_{RB\text{-}set,UL}^{BWP}+1\right)}{2}\right)\rceil$ where $N_{\text{RB-set,UL}}^{\text{BWP}}$ is the number of RB sets contained in the active UL BWP as defined in clause 7 of [6, TS38.214].
- Time domain resource assignment - 0, 1, 2, 3, 4, 5, or 6 bits
  - If the higher layer parameter *pusch-TimeDomainAllocationListDCI-0-1* is not configured and if the higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH* is not configured and if the higher layer parameter *pusch-TimeDomainAllocationList* is configured, 0, 1, 2, 3, or 4 bits as defined in Clause 6.1.2.1 of [6, TS38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *pusch-TimeDomainAllocationList*;
  - If the higher layer parameter *pusch-TimeDomainAllocationListDCI-0-1* is configured or if the higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH* is configured, 0, 1, 2, 3, 4, 5 or 6 bits as defined in Clause 6.1.2.1 of [6, TS38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *pusch-TimeDomainAllocationListDCI-0-1* or *pusch-TimeDomainAllocationListForMultiPUSCH*;
  - otherwise the bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the default table.
- Frequency hopping flag - 0 or 1 bit:
  - 0 bit if only resource allocation type 0 is configured, or if the higher layer parameter *frequencyHopping* is not configured and the higher layer parameter *pusch-RepTypeIndicatorDCI-0-1* is not configured to *pusch-RepTypeB*, or if the higher layer parameter *frequencyHoppingDCI-0-1* is not configured and *pusch-RepTypeIndicatorDCI-0-1* is configured to *pusch-RepTypeB*, or if only resource allocation type 2 is configured;
  - 1 bit according to Table 7.3.1.1.1-3 otherwise, only applicable to resource allocation type 1, as defined in Clause 6.3 of [6, TS 38.214].

For transport block 1:

- Modulation and coding scheme - 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214]
- New data indicator - 1 bit if the number of scheduled PUSCH indicated by the Time domain resource assignment field is 1; otherwise 2, 3, 4, 5, 6, 7 or 8 bits determined based on the maximum number of schedulable PUSCH among all entries in the higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH*, where each bit corresponds to one scheduled PUSCH as defined in clause 6.1.4 in [6, TS 38.214].
- Redundancy version - - number of bits determined by the following:
  - 2 bits as defined in Table 7.3.1.1.1-2 if the number of scheduled PUSCH indicated by the Time domain resource assignment field is 1;
  - otherwise 2, 3, 4, 5, 6, 7 or 8 bits determined by the maximum number of schedulable PUSCHs among all entries in the higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH*, where each bit corresponds to one scheduled PUSCH as defined in clause 6.1.4 in [6, TS 38.214] and redundancy version is determined according to Table 7.3.1.1.2-34.

For transport block 2 (only present if *maxRank* or *maxMIMO-Layers* is larger than 4):

- Modulation and coding scheme - 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part, *maxRank* is larger than 4 or the value of *maxMIMO-Layers* for the indicated bandwidth part is larger than 4 and the value of *maxRank* or *maxMIMO-Layers* for the active bandwidth part is no more than 4, the UE assumes zeros are padded when interpreting the "Modulation and coding scheme", "New data indicator", and "Redundancy version" fields for transport block 2 according to Clause 12 of [5, TS38.213], and the UE ignores the "Modulation and coding scheme", "New data indicator", and "Redundancy version" fields of transport block 2 for the indicated bandwidth part.

- Transform precoder indicator - 0 or 1 bit
  - 1 bit if the higher layer parameter *dynamicTransformPrecoderFieldPresenceDCI-0-1* is configured to 'enabled ' and if the UE is configured to monitor DCI format 0_1 with CRC scrambled by C-RNTI or CS-RNTI or MCS-C-RNTI, where the bit value of 0 indicates that transform precoder is enabled and the bit value of 1 indicates that transform precoder is disabled. For a DCI format 0_1 with CRC scrambled by CS-RNTI and the value indicated by new data indicator field is 0, or for a DCI format 0_1 with CRC scrambled by SP-CSI-RNTI, the bit is reserved.
  - 0 bit otherwise.
- HARQ process number - 5 bits if higher layer parameter *harq-ProcessNumberSizeDCI-0-1* or *harq-ProcessNumberSizeDCI-0-1-Ext* is configured; otherwise 4 bits
- 1st downlink assignment index - 1, 2 or 4 bits:
  - 1 bit for semi-static HARQ-ACK codebook for unicast and multicast if *pdsch-HARQ-ACK-Codebook = semiStatic* is configured for both unicast and multicast and the higher layer parameter *fdmed-ReceptionMulticast* is not configured; otherwise for semi-static HARQ-ACK codebook for unicast;
  - 2 bits for dynamic HARQ-ACK codebook for unicast, or for enhanced dynamic HARQ-ACK codebook without *UL-TotalDAI-Included* configured;
  - 4 bits for enhanced dynamic HARQ-ACK codebook and with *UL-TotalDAI-Included = true*.
    When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookList* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-1* is configured, if the bit width of the 1st downlink assignment index in DCI format 0_1 for one HARQ-ACK codebook is not equal to that of the 1st downlink assignment index in DCI format 0_1 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 1st downlink assignment index until the bit width of the 1st downlink assignment index in DCI format 0_1 for the two HARQ-ACK codebooks are the same.
- 2nd downlink assignment index - 0, 2 or 4 bits:
  - 2 bits for dynamic HARQ-ACK codebook with two HARQ-ACK sub-codebooks for unicast, or for enhanced dynamic HARQ-ACK codebook with two HARQ-ACK sub-codebooks and without *UL-TotalDAI-Included* configured;
  - 4 bits for enhanced dynamic HARQ-ACK codebook with two HARQ-ACK sub-codebooks and with *UL-TotalDAI-Included = true*;
  - 0 bit otherwise.
    When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookList* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-1* is configured, if the bit width of the 2nd downlink assignment index in DCI format 0_1 for one HARQ-ACK codebook is not equal to that of the 2nd downlink assignment index in DCI format 0_1 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 2nd downlink assignment index until the bit width of the 2nd downlink assignment index in DCI format 0_1 for the two HARQ-ACK codebooks are the same.
- 3rd downlink assignment index - 0, 1 or 2 bits:
  - 1 bit for semi-static HARQ-ACK codebook for multicast if the higher layer parameter *fdmed-ReceptionMulticast* is configured;
  - 2 bits for the dynamic HARQ-ACK codebook for multicast;
  - 0 bit otherwise.
    When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookListMulticast* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-1* is configured, if the bit width of the 3rd downlink assignment index in DCI format 0_1 for one HARQ-ACK codebook is not equal to that of the 3rd downlink assignment index in DCI format 0_1 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 3rd downlink assignment index until the bit width of the 3rd downlink assignment index in DCI format 0_1 for the two HARQ-ACK codebooks are the same.
- TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS38.213]
- Second TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS38.213] if higher layer parameter *SecondTPCFieldDCI-0-1* is configured; 0 bit otherwise.
- SRS resource set indicator - 0 or 2 bits
  - 2 bits according to Table 7.3.1.1.2-36 if
    - *txConfig = nonCodeBook*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModList* and associated with the *usage* of value '*nonCodeBook*', and is not

configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided, or

- *txConfig=codebook*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModList* and associated with *usage* of value '*codebook*', and is not configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided;

- 0 bit otherwise.

- SRS resource indicator -number of bits determined by the following:
  - $\lceil \log_2 \left(\sum_{k=1}^{\min\{L_{\max}, N_{\text{SRS}}\}} \binom{N_{\text{SRS}}}{k} \right) \rceil$ bits according to Tables 7.3.1.1.2-28/28A/29/29B/30/30B/31/31B/31C/31D/31E/31F if the higher layer parameter *txConfig* = *nonCodebook*, where
    - N_{SRS} is the number of configured SRS resources in the SRS resource set indicated by SRS resource set indicator field if present,
    - N_{SRS} is the number of configured SRS resources in the SRS resource set associated with the *coresetPoolIndex* value for the CORESET used for the PDCCH carrying the DCI format 0_1, if the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*,
    - otherwise N_{SRS} is the number of configured SRS resources in the SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModList* and associated with the higher layer parameter *usage* of value '*nonCodeBook*',

    and
    - if UE supports operation with *maxMIMO-Layers* and the higher layer parameter *maxMIMO-Layers* of *PUSCH-ServingCellConfig* of the serving cell is configured,
      - L_{max} is given by max{*maxMIMO-Layers*, *maxMIMO-LayersforSDM*} if *maxMIMO-LayersforSDM* is configured
      - L_{max} is given by max{*maxMIMO-Layers*, *maxMIMO-LayersforSFN*} if *maxMIMO-LayersforSFN* is configured
      - L_{max} is given by *maxMIMO-Layers* otherwise
    - otherwise, L_{max} is given by the maximum number of layers for PUSCH supported by the UE for the serving cell for non-codebook based operation.
  - $\lceil \log_2 (N_{\text{SRS}}) \rceil$ bits according to Tables 7.3.1.1.2-32, 7.3.1.1.2-32A and 7.3.1.1.2-32B if the higher layer parameter *txConfig* = *codebook*, where
    - N_{SRS} is the number of configured SRS resources in the SRS resource set indicated by SRS resource set indicator field if present,
    - N_{SRS} is the number of configured SRS resources in the SRS resource set associated with the *coresetPoolIndex* value for the CORESET used for the PDCCH carrying the DCI format 0_1, if the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*,
    - otherwise N_{SRS} is the number of configured SRS resources in the SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModList* and associated with the higher layer parameter *usage* of value '*codeBook*'.

    When the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModList* and associated with *usage* of value '*codebook*' or '*nonCodeBook*', the first SRS resource set is associated with *coresetPoolIndex* value 0 and the second SRS resource set is associated with *coresetPoolIndex* value 1, where the first and the second SRS resource sets are respectively the ones with lower and higher *srs-ResourceSetId* of the two SRS resources sets.

- Second SRS resource indicator - number of bits determined by the following:
  - $\lceil \log_2 (\max_{k \in \{1,2,\ldots,\min\{L_{max}, N_{SRS}\}\}} \binom{N_{SRS}}{k}) \rceil$ bits according to Tables 7.3.1.1.2-28/29A/30A/31A with the same number of layers indicated by SRS resource indicator field if the higher layer parameter *txConfig* =

nonCodebook, the higher layer paramtere *maxMIMO-LayersforSDM* is not configured, and SRS resource set indicator field is present, where N_{SRS} is the number of configured SRS resources in the second SRS resource set, and
  - if UE supports operation with *maxMIMO-Layers* and the higher layer parameter *maxMIMO-Layers* of *PUSCH-ServingCellConfig* of the serving cell is configured,
    - L_{max} is given by *maxMIMO-LayersforSFN* if *maxMIMO-LayersforSFN* is configured
    - L_{max} is given by *maxMIMO-Layers* otherwise
  - otherwise, L_{max} is given by the maximum number of layers for PUSCH supported by the UE for the serving cell for non-codebook based operation.
- $\lceil \log_2 \left(\sum_{k=1}^{\min\{L_{\max}, N_{\text{SRS}}\}} \binom{N_{\text{SRS}}}{k} \right) \rceil$ bits according to Tables 7.3.1.1.2-28/29 if the higher layer parameter *txConfig = nonCodebook*, the higher layer paramtere *maxMIMO-LayersforSDM* is configured and SRS resource set indicator field is present, where N_{SRS} is the number of configured SRS resources in the second SRS resource set, and L_{max} is given by *maxMIMO-LayersforSDM*.
- $\lceil \log_2(N_{SRS}) \rceil$ bits according to Tables 7.3.1.1.2-32, 7.3.1.1.2-32A and 7.3.1.1.2-32B if the higher layer parameter *txConfig = codebook* and SRS resource set indicator field is present, where N_{SRS} is the number of configured SRS resources in the second SRS resource set.
- 0 bit otherwise.

- Precoding information and number of layers - number of bits determined by the following:
  - 0 bits if the higher layer parameter *txConfig = nonCodeBook*;
  - 0 bits for 1 antenna port and if the higher layer parameter *txConfig = codebook*;
  - 4, 5, or 6 bits according to Table 7.3.1.1.2-2 for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRank* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSFN*} if *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSDM*} if *multipanelSchemeSDM* is configured, and *codebookSubset*;
  - 4 or 5 bits according to Table 7.3.1.1.2-2A for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRank=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSFN*} = 2 if *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSDM*} = 2 if *multipanelSchemeSDM* is configured, transform precoder is disabled, and according to the values of higher layer parameter *codebookSubset*;
  - 4 or 6 bits according to Table 7.3.1.1.2-2B for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRank=3 or 4*, transform precoder is disabled, and according to the values of higher layer parameter *codebookSubset*;
  - 2, 4, or 5 bits according to Table 7.3.1.1.2-3 for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and *maxRank=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSFN*} = 1 if *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSDM*} = 1 if *multipanelSchemeSDM* is configured, and *codebookSubset*;
  - 3 or 4 bits according to Table 7.3.1.1.2-3A for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1*, and according to whether transform precoder is enabled, or disabled and *maxRank*=1 if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSFN*} = 1 if *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSDM*} = 1 if *multipanelSchemeSDM* is configured, and the values of higher layer parameter *codebookSubset*;
  - 2 or 4 bits according to Table7.3.1.1.2-4 for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRank* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSFN*} if *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSDM*} if *multipanelSchemeSDM* is configured, and *codebookSubset*;
  - 2 bits according to Table 7.3.1.1.2-4A for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1*, transform precoder is disabled, *maxRank=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSFN*} = 2 if *multipanelSchemeSFN* is configured or max{*maxRank*, *maxRankSDM*} = 2 if *multipanelSchemeSDM* is configured, and *codebookSubset=nonCoherent*;

- 1 or 3 bits according to Table7.3.1.1.2-5 for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and *maxRank=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank, maxRankSFN*}*=1* if *multipanelSchemeSFN* is configured or max{*maxRank, maxRankSDM*}*=1* if *multipanelSchemeSDM* is configured, and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-5A for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1,* and according to whether transform precoder is enabled, or disabled and *maxRank*=1 if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRank, maxRankSFN*} = 1 if *multipanelSchemeSFN* is configured or max{*maxRank, maxRankSDM*} = 1 if *multipanelSchemeSDM* is configured, and the values of higher layer parameter *codebookSubset*;
- 7 bits according to Table 7.3.1.1.2-5B for 8 antenna ports, if *CodebookTypeUL= codebook1*, transform precoder is disabled, *maxRank* = 8, and according to *codebook1*;
- 7 bits according to Table 7.3.1.1.2-5C for 8 antenna ports, if *CodebookTypeUL= codebook1*, transform precoder is disabled, *maxRank* = 7, and according to *codebook1*;
- 7 bits according to Table 7.3.1.1.2-5D for 8 antenna ports, if *CodebookTypeUL= codebook1*, transform precoder is disabled, *maxRank* = 4, 5 or 6, and according to *maxRank*;
- 4, 6 or 7 bits according to Table 7.3.1.1.2-5E for 8 antenna ports, if *CodebookTypeUL= codebook1*, transform precoder is enabled or *maxRank* = 1, 2 or 3 if transform precoder is disabled, and according to transform precoder and *maxRank*;
- 8 bits according to Table 7.3.1.1.2-5F for 8 antenna ports, if *CodebookTypeUL= codebook4*, transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to *maxRank*;
- 6 or 7 or 8 bits according to Table 7.3.1.1.2-5G for 8 antenna ports, if *CodebookTypeUL= codebook4*, transform precoder is disabled, *maxRank* = 2, 3 or 4, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to *maxRank*;
- 3 bits according to Table 7.3.1.1.2-5H for 8 antenna ports, if *CodebookTypeUL= codebook4*, transform precoder is enabled or *maxRank* = 1 if transform precoder is disabled, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*.
- 10 bits according to Table 7.3.1.1.2-5I for 8 antenna ports, if *CodebookTypeUL=codebook2,* transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to *maxRank*;
- 5, 9 or 10 bits according to Table 7.3.1.1.2-5J for 8 antenna ports, if *CodebookTypeUL=codebook2,* transform precoder is enabled or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to transform precoder and *maxRank*;
- 10 bits according to Table 7.3.1.1.2-5K for 8 antenna ports, if *CodebookTypeUL=codebook3*, transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to *maxRank*;
- 4, 7, 9 or 10 bits according to Table 7.3.1.1.2-5L for 8 antenna ports, if *CodebookTypeUL=codebook3*, transform precoder is enabled or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to transform precoder and *maxRank*;
- 6 or 7 or 8 bits according to Table 7.3.1.1.2-5M for 8 antenna ports, if *CodebookTypeUL=codebook4*, transform precoder is disabled, *maxRank* = 2, 3 or 4, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to *maxRank*;
- 4 bits according to Table 7.3.1.1.2-5N for 8 antenna ports, if *CodebookTypeUL=codebook4*, transform precoder is enabled or *maxRank* = 1 if transform precoder is disabled, *ul-FullPowerTransmission* is configured to *fullpowerMode1*.
- 6, 9 or 10 bits according to Table 7.3.1.1.2-5O for 8 antenna ports, if *CodebookTypeUL=codebook2,* transform precoder is enabled or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to transform precoder and *maxRank*;
- 5, 7, 9 or 10 bits according to Table 7.3.1.1.2-5P for 8 antenna ports, if *CodebookTypeUL=codebook3,* transform precoder is enabled or *maxRank* = 1, 2, 3, or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is configured to *fullpowerMode1,* and according to transform precoder and *maxRank*;
- 8 or 9 bits according to Table 7.3.1.1.2-5Q for 8 antenna ports, if *CodebookTypeUL=codebook4*, transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to *maxRank*;

- 10 bits according to Table 7.3.1.1.2-5R for 8 antenna ports, if *CodebookTypeUL=codebook2*, transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to *maxRank*;
- 10 bits according to Table 7.3.1.1.2-5S for 8 antenna ports, if *CodebookTypeUL=codebook3*, transform precoder is disabled, *maxRank* = 5, 6, 7, or 8, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to *maxRank*;
- 3 bits according to Table 7.3.1.1.2-5T for 3 antenna ports, if *txConfig* = *codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRank*;
- 2 bits according to Table 7.3.1.1.2-5U for 3 antenna ports, if *txConfig* = *codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRank*;

For the higher layer parameter *txConfig=codebook*, if *ul-FullPowerTransmission* is configured to *fullpowerMode2*, maxRank is configured to be larger than 2, and at least one SRS resource with 4 antenna ports or 8 antenna ports is configured in the SRS resource set indicated by SRS resource set indicator field if present, otherwise in an SRS resource set with usage set to 'codebook', and an SRS resource with 2 antenna ports is indicated via SRI in the same SRS resource set, then Table 7.3.1.1.2-4 is used.

For the higher layer parameter *txConfig=codebook*, if *ul-FullPowerTransmission* is configured to *fullpowerMode2*, *maxRank* is configured to be larger than 4, and at least one SRS resource with 8 antenna ports is configured in the SRS resource set with usage set to 'codebook', and an SRS resource with 4 antenna ports is indicated via SRI in the same SRS resource set, then Table 7.3.1.1.2-2 is used.

For the higher layer parameter *txConfig* = *codebook*, if different SRS resources with different number of antenna ports are configured, the bitwidth is determined according to the maximum number of ports in an SRS resource among the configured SRS resources in all SRS resource set(s) with usage set to 'codebook'. If the number of ports for a configured SRS resource in the set is less than the maximum number of ports in an SRS resource among the configured SRS resources, a number of most significant bits with value set to '0' are inserted to the field.

When the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModList* and associated with *usage* of value '*codebook*' or '*nonCodeBook*', the Precoding information and number of layers field is associated with the SRS resource set that is associated with the *coresetPoolIndex* value for the CORESET used for the PDCCH carrying the DCI format 0_1.

For the higher layer parameter *txConfig* = *codebook*, when the Transform precoder indicator field is present, if the bit width of the Precoding information and number of layers field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the Precoding information and number of layers field for the case with smaller bit width until the bit width of the Precoding information and number of layers field for the two cases are the same.

- Second Precoding information - number of bits determined by the following:
  - 0 bits if SRS resource set indicator field is not present;
  - 0 bits if the higher layer parameter *txConfig* = *nonCodeBook*;
  - 0 bits for 1 antenna port and if the higher layer parameter *txConfig* = *codebook*;
  - 3, 4, or 5 bits according to Table 7.3.1.1.2-2C with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports, if SRS resource set indicator field is present, *txConfig* = *codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRank* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN* if *multipanelSchemeSFN* is configured, and *codebookSubset*;
  - 3 or 4 bits according to Table 7.3.1.1.2-2D with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports, if SRS resource set indicator field is present, *txConfig* = *codebook*, *ul-FullPowerTransmission* = *fullpowerMode1*, *maxRank=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN=2* if *multipanelSchemeSFN* is configured, transform precoder is disabled, and according to the values of higher layer parameter *codebookSubset*;
  - 3 or 4 bits according to Table 7.3.1.1.2-2E with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports, if SRS resource set indicator field is

present, *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRank=3 or 4,* transform precoder is disabled, and according to the values of higher layer parameter *codebookSubset*;

- 2, 4, or 5 bits according to Table 7.3.1.1.2-3 with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRank* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN* if *multipanelSchemeSFN* is configured, and *codebookSubset*;
- 3 or 4 bits according to Table 7.3.1.1.2-3A with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRank=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN=1* if *multipanelSchemeSFN* is configured, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameter *codebookSubset*;
- 1 or 3 bits according to Table7.3.1.1.2-4B with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* transform precoder is disabled, and according to the values of higher layer parameters *maxRank* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN* if *multipanelSchemeSFN* is configured, and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-4C with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1*, transform precoder is disabled, *maxRank=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN=2* if *multipanelSchemeSFN* is configured, and *codebookSubset=nonCoherent*;
- 1 or 3 bits according to Table7.3.1.1.2-5 with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRank* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN* if *multipanelSchemeSFN* is configured, and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-5A with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRank=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN=1* if *multipanelSchemeSFN* is configured, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameter *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-5V with the same number of layers indicated by Precoding information and number of layers field for 3 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpower,* transform precoder is disabled, and according to the values of higher layer parameters *maxRank*;
- 2 bits according to Table 7.3.1.1.2-5U with the same number of layers indicated by Precoding information and number of layers field for 3 antenna ports, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRank*;
- 4, 5, or 6 bits according to Table 7.3.1.1.2-2 for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* transform precoder is disabled, and according to the values of higher layer parameters *maxRankSDM* if *multipanelSchemeSDM* is configured, and *codebookSubset*;
- 4 or 5 bits according to Table 7.3.1.1.2-2A for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRankSDM = 2* if *multipanelSchemeSDM* is configured, transform precoder is disabled, and according to the values of higher layer parameter *codebookSubset*;
- 2, 4, or 5 bits according to Table 7.3.1.1.2-3 for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRankSDM* if *multipanelSchemeSDM* is configured, and *codebookSubset*;
- 3 or 4 bits according to Table 7.3.1.1.2-3A for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRankSDM = 1* if *multipanelSchemeSDM* is configured,

and according to whether transform precoder is enabled or disabled, and the values of higher layer parameter *codebookSubset*;
- 2 or 4 bits according to Table7.3.1.1.2-4 for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* transform precoder is disabled, and according to the values of higher layer parameters *maxRankSDM* if *multipanelSchemeSDM* is configured, and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-4A for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1*, transform precoder is disabled, *maxRankSDM = 2* if *multipanelSchemeSDM* is configured, and *codebookSubset=nonCoherent*;
- 1 or 3 bits according to Table7.3.1.1.2-5 for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and *maxRankSDM= 1* if *multipanelSchemeSDM* is configured, and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-5A for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission = fullpowerMode1, maxRankSDM = 1* if *multipanelSchemeSDM* is configured, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameter *codebookSubset*;

For the higher layer parameter *txConfig=codebook*, if *ul-FullPowerTransmission* is configured to *fullpowerMode2*, maxRank is configured to be larger than 2, and at least one SRS resource with 4 antenna ports is configured in the SRS resource set indicated by SRS resource set indicator field, and an SRS resource with 2 antenna ports is indicated via Second SRS resource indicator field in the same SRS resource set, then Table 7.3.1.1.2-4B is used.

For the higher layer parameter *txConfig = codebook*, if different SRS resources with different number of antenna ports are configured, the bitwidth is determined according to the maximum number of ports in an SRS resource among the configured SRS resources in the second SRS resource set with usage set to 'codebook' as defined in Table 7.3.1.1.2-36. If the number of ports for a configured SRS resource in the set is less than the maximum number of ports in an SRS resource among the configured SRS resources, a number of most significant bits with value set to '0' are inserted to the field.

For the higher layer parameter *txConfig = codebook*, when the Transform precoder indicator field is present, if the bit width of the Second Precoding information field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the Second Precoding information field for the case with smaller bit width until the bit width of the Second Precoding information field for the two cases are the same.

- Antenna ports - number of bits determined by the following

  - 2 bits as defined by Tables 7.3.1.1.2-6, if transform precoder is enabled, *dmrs-Type*=1, and *maxLength*=1, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2 BPSK modulation is used;
  - 2 bits as defined by Tables 7.3.1.1.2-6A, if transform precoder is enabled and *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2 BPSK modulation is used, *dmrs-Type*=1, and *maxLength*=1, where n_{SCID} is the scrambling identity for antenna ports defined in Clause 6.4.1.1.1.2, TS 38.211 [4];
  - 4 bits as defined by Tables 7.3.1.1.2-7, if transform precoder is enabled, *dmrs-Type*=1, and *maxLength*=2, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2 BPSK modulation is used;
  - 4 bits as defined by Tables 7.3.1.1.2-7A, if transform precoder is enabled and *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2 BPSK modulation is used, *dmrs-Type*=1, and *maxLength*=2, where n_{SCID} is the scrambling identity for antenna ports defined in Clause 6.4.1.1.1.2, TS 38.211 [4];
  - 3 bits as defined by Tables 7.3.1.1.2-8/9/10/10A/11 according to the value of rank, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, and *maxLength*=1;
  - 4 bits as defined by Tables 7.3.1.1.2-12/13/14/14A/15/15A/15B/15C/15D according to the value of rank, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, and *maxLength*=2;
  - 4 bits as defined by Tables 7.3.1.1.2-16/17/18/18A/19/19A/19B according to the value of rank, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, and *maxLength*=1;
  - 5 bits as defined by Tables 7.3.1.1.2-20/21/22/22A/23/23A/23B/23C/23D according to the value of rank, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, and *maxLength*=2.
  - 4 bits as defined by Tables 7.3.1.1.2-38/39/40/40A/41/42/43/44/45, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, and *maxLength*=1;

- 5 bits as defined by Tables 7.3.1.1.2-46/47/48/48A/49/50/51/52/53, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, and *maxLength*=2;
- 5 bits as defined by Tables 7.3.1.1.2-54/55/56/56A/57/58/59/60/61, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, and *maxLength*=1;
- 6 bits as defined by Tables 7.3.1.1.2-62/63/64/64A/65/66/67/68/69, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, and *maxLength*=2.

where the number of CDM groups without data of values 1, 2, and 3 in Tables 7.3.1.1.2-6 to 7.3.1.1.2-23 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively, and the value of rank is:

- the sum of the value determined according to the SRS resource indicator field and the value determined according to the second SRS resource indicator field, if *txConfig* = *nonCodebook*, *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "10"
- the sum of the value determined according to the Precoding information and number of layers field and the value determined according to the Second Precoding information, if *txConfig* = *codebook*, *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "10"
- determined according to the SRS resource indicator field if the higher layer parameter *txConfig* = *nonCodebook* and *multipanelSchemeSDM* is not configured, or if the higher layer parameter *txConfig* = *nonCodebook*, *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "00" or “01”,
- determined according to the Precoding information and number of layers field if the higher layer parameter *txConfig* = *codebook* and *multipanelSchemeSDM* is not configured, or if the higher layer parameter *txConfig* = *codebook*, *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "00" or "01".

If a UE is configured with both *dmrs-UplinkForPUSCH-MappingTypeA* and *dmrs-UplinkForPUSCH-MappingTypeB*, the bitwidth of this field equals $\max\{x_A, x_B\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-UplinkForPUSCH-MappingTypeA* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-UplinkForPUSCH-MappingTypeB*. A number of $|x_A - x_B|$ zeros are padded in the MSB of this field, if the mapping type of the PUSCH corresponds to the smaller value of x_A and x_B.

When the Transform precoder indicator field is present, if the bit width of the Antenna ports field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the Antenna ports field for the case with smaller bit width until the bit width of the Antenna ports field for the two cases are the same.

- SRS request - 2 bits as defined by Table 7.3.1.1.2-24 for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell; 3 bits for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second and third bits are defined by Table 7.3.1.1.2-24. This bit field may also indicate the associated CSI-RS according to Clause 6.1.1.2 of [6, TS 38.214].

- SRS offset indicator - 0, 1 or 2 bits.
  - 0 bit if higher layer parameter *AvailableSlotOffset* is not configured for any aperiodic SRS resource set in the scheduled cell, or if higher layer parameter *AvailableSlotOffset* is configured for at least one aperiodic SRS resource set in the scheduled cell and the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) is 1;
  - otherwise, $\lceil \log_2(K) \rceil$ bits are used to indicate available slot offset according to Table 7.3.1.1.2-37 and Clause 6.2.1 of [6, TS 38.214], where K is the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) in the scheduled cell;
- CSI request - 0, 1, 2, 3, 4, 5, or 6 bits determined by higher layer parameter *reportTriggerSize*.

- CBG transmission information (CBGTI) - 0 bit if higher layer parameter *codeBlockGroupTransmission* for PUSCH is not configured or if the number of scheduled PUSCH indicated by the Time domain resource assignment field is larger than 1; otherwise, 2, 4, 6, or 8 bits as defined in Clause 6.1.5 of [6, TS38.214], determined by higher layer parameter *maxCodeBlockGroupsPerTransportBlock* and *maxRank* or *maxMIMO-Layers* for PUSCH.

- PTRS-DMRS association - number of bits determined as follows

- 0 bit if *PTRS-UplinkConfi*g is not configured in either *dmrs-UplinkForPUSCH-MappingTypeA* or *dmrs-UplinkForPUSCH-MappingTypeB* and transform precoder is disabled, or if transform precoder is enabled, or if *maxRank=1* and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, or if *maxMIMO-Layers*=1 and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, or if *maxRank=1* and *maxRankSFN=1*, or if *maxMIMO-Layers*=1 and *maxMIMO-LayersforSFN*=1, or if *maxRank=1* and *maxRankSDM=1* when two PTRS ports are configured by *maxNrofPorts-SDM*, or if *maxMIMO-Layers*=1 and *maxMIMO-LayersforSDM*=1 when two PTRS ports are configured by *maxNrofPorts-SDM*;
- 1 or 2 or 4 bits otherwise, where Table 7.3.1.1.2-25/7.3.1.1.2-25A/7.3.1.1.2-25B/7.3.1.1.2-26/7.3.1.1.2-26A/7.3.1.1.2-26B are used to indicate the association between PTRS port(s) and DMRS port(s), and the DMRS ports are indicated by the Antenna ports field.
  - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 2, 4, or 8 antenna ports, SRS resource set indicator field is absent or SRS resource set indicator field is present and equals "00" or "01", and *maxRank*<=4 or *maxMIMO-Layers*<=4, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25 and 7.3.1.1.2-26.
  - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 4 or 8 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRank*=3 or 4 or *maxMIMO-Layers*=3 or 4, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25 and 7.3.1.1.2-26.
  - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 2, 4, or 8 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRank*=2 or *maxMIMO-Layers*=2, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, the MSB of this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator and/or Precoding information and number of layers field, and the LSB of this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to Second SRS resource indicator field and/or Second Precoding information field, according to Table 7.3.1.1.2-25A.
  - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, SRS resource set indicator field is absent or SRS resource set indicator field is present and equals "00" or "01", *maxRank*<=3 or *maxMIMO-Layers*<=3, this field indicates the association between PTRS port and DMRS port corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25.
  - 1 bit when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, SRS resource set indicator field is absent or SRS resource set indicator field is present and equals "00" or "01", *maxRank*<=3 or *maxMIMO-Layers*<=3, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-26B.
  - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRank*=3 or *maxMIMO-Layers*=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port and DMRS port corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25.
  - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRank*=2 or *maxMIMO-Layers*=2, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, the MSB of this field indicates the association between PTRS port and DMRS port corresponding to SRS resource indicator and/or Precoding information and number of layers field, and the LSB of this field indicates the association between PTRS port and DMRS port corresponding to Second SRS resource indicator field and/or Second Precoding information field, according to Table 7.3.1.1.2-25A.
  - 1 bit when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRank*<=3 or *maxMIMO-Layers*<=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-26B.
  - 2 bits when two PTRS ports are configured by *maxNrofPorts-SDM* in *PTRS-UplinkConfig*, the SRS resource set indicator field is present and equals "10" and *multipanelSchemeSDM* is configured, the

MSB of this field indicates the association between PTRS port 0 and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field, and the LSB of this field indicates the association between PTRS port 1 and DMRS port(s) corresponding to Second SRS resource indicator field and/or Second Precoding information field, according to Table 7.3.1.1.2-25A.
- 2 bits when one PTRS port is configured by *maxNrofPorts-SDM* in *PTRS-UplinkConfig*, SRS resource set indicator field is present and equals "10" and *multipanelSchemeSDM* is configured, this field indicates the association between PTRS port and DMRS ports corresponding to SRS resource indicator field and Second SRS resource indicator field and/or Precoding information and number of layers field and Second Precoding information field according to Table 7.3.1.1.2-25.
- 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig*, SRS resource set indicator field is present and equals "10", *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25 and 7.3.1.1.2-26.
- 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig*, the SRS resource set indicator field is absent, *maxRank*>4 or *maxMIMO-Layers*>4, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port and DMRS port(s) corresponding to the selected codeword according to Table 7.3.1.1.2-25B, where the selected codeword is the codeword with higher MCS for the initial PUSCH if the MCS indices of the two codewords are different for the initial PUSCH, or codeword 0 otherwise.
- 4 bits when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig*, the SRS resource set indicator field is absent, *maxRank*>4 or *maxMIMO-Layers*>4, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Table 7.3.1.1.2-26A.

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and the "PTRS-DMRS association" field is present for the indicated bandwidth part but not present for the active bandwidth part, the UE assumes the "PTRS-DMRS association" field is not present for the indicated bandwidth part.

When the Transform precoder indicator field is present, if the bit width of PTRS-DMRS association field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the PTRS-DMRS association field for the case with smaller bit width until the bit width of the PTRS-DMRS association field for the two cases are the same.

- Second PTRS-DMRS association
  - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 4 or 8 antenna ports, PTRS-DMRS association field is present, SRS resource set indicator field is present and equals "10" or "11", *maxRank*>2 or *maxMIMO-Layers*>2, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured;
  - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, PTRS-DMRS association field is present, SRS resource set indicator field is present and equals "10" or "11", *maxRank*=3 or *maxMIMO-Layers*=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured;
  - 1 bit when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, PTRS-DMRS association field is present, SRS resource set indicator field is present and equals "10" or "11", *maxRank*<=3 or *maxMIMO-Layers*<=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured;
  - 0 bit otherwise.

  Tables 7.3.1.1.2-25 and 7.3.1.1.2-26/7.3.1.1.2-26B are used to indicate the association between PTRS port(s) and DMRS port(s) corresponding to Second SRS resource indicator field and/or Second precoding information field when one PT-RS port and two PT-RS ports are configured by maxNrofPorts in PTRS-UplinkConfig respectively, and the DMRS ports are indicated by the Antenna ports field.

- beta_offset indicator - 0 if the higher layer parameter *betaOffsets* = *semiStatic*; otherwise 2 bits as defined by Table 9.3-3 in [5, TS 38.213].

  When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookList* or by *pdsch-HARQ-ACK-CodebookListMulticast* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-1* is configured, if the bit width of the beta_offset indicator in DCI format 0_1 for one HARQ-ACK codebook is not equal to that of the beta_offset indicator in DCI format 0_1 for the other HARQ-ACK codebook, a

number of most significant bits with value set to '0' are inserted to smaller beta_offset indicator until the bit width of the beta_offset indicator in DCI format 0_1 for the two HARQ-ACK codebooks are the same.

- DMRS sequence initialization – 0 bit if transform precoder is enabled by higher layers and the Transform precoder indicator field is not present; 1 bit if transform precoder is disabled by higher layers or if the Transform precoder indicator field is present. If the Transform precoder indicator field is present and set to '0', the bit is reserved.

- UL-SCH indicator - 0 or 1 bit as follows

  - 0 bit if the number of scheduled PUSCH indicated by the Time domain resource assignment field is larger than 1;
  - 1 bit otherwise. A value of "1" indicates UL-SCH shall be transmitted on the PUSCH and a value of "0" indicates UL-SCH shall not be transmitted on the PUSCH. If a UE does not support triggering SRS only in DCI, except for DCI format 0_1 with CRC scrambled by SP-CSI-RNTI, the UE is not expected to receive a DCI format 0_1 with UL-SCH indicator of "0" and CSI request of all zero(s). If a UE supports triggering SRS only in DCI, except for DCI format 0_1 with CRC scrambled by SP-CSI-RNTI, the UE is not expected to receive a DCI format 0_1 with UL-SCH indicator of "0", CSI request of all zero(s) and SRS request of all zero(s). The UE is not expected to receive a DCI format 0_1 with UL-SCH indicator of "0", when the indicated number of layers is larger than 4.
- ChannelAccess-CPext-CAPC - 0, 1, 2, 3, 4, 5 or 6 bits. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *ul-AccessConfigListDCI-0-1* or in Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1, or for operation in frequency range 2-2 if *ChannelAccessMode2-r17* is provided; otherwise 0 bit. One or more entries from Table 7.3.1.1.2-35 or Table 7.3.1.1.2-35A are configured by the higher layer parameter *ul-AccessConfigListDCI-0-1*.

- Open-loop power control parameter set indication - 0 or 1 or 2 bits.

  - 0 bit if the higher layer parameter *p0-PUSCH-SetList* is not configured;
  - 1 or 2 bits otherwise,
    - 1 bit if SRS resource indicator is present in the DCI format 0_1;
    - 1 or 2 bits as determined by higher layer parameter *olpc-ParameterSetDCI-0-1* if SRS resource indicator is not present in the DCI format 0_1.
- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-0-1* is not configured; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213].

- Invalid symbol pattern indicator - 0 bit if higher layer parameter *invalidSymbolPatternIndicatorDCI-0-1* is not configured; otherwise 1 bit as defined in Clause 6.1.2.1 in [6, TS 38.214].

- Minimum applicable scheduling offset indicator - 0 or 1 bit

  - 0 bit if higher layer parameter *minimumSchedulingOffsetK2* is not configured;
  - 1 bit if higher layer parameter *minimumSchedulingOffsetK2* is configured. The 1 bit indication is used to determine the minimum applicable K2 for the active UL BWP and the minimum applicable K0 value for the active DL BWP, if configured respectively, according to Table 7.3.1.1.2-33. If the minimum applicable K0 is indicated, the minimum applicable value of the aperiodic CSI-RS triggering offset for an active DL BWP shall be the same as the minimum applicable K0 value.
- SCell dormancy indication - 0 bit if higher layer parameter *dormancyGroupWithinActiveTime* is not configured; otherwise 1, 2, 3, 4 or 5 bits bitmap determined according to the number of different *DormancyGroupID(s)* provided by higher layer parameter *dormancyGroupWithinActiveTime*, where each bit corresponds to one of the SCell group(s) configured by higher layers parameter *dormancyGroupWithinActiveTime*, with MSB to LSB of the bitmap corresponding to the first to last configured SCell group in ascending order of *DormancyGroupID*. The field is only present when this format is carried by PDCCH on the primary cell within DRX Active Time and the UE is configured with at least two DL BWPs for an SCell.

- Sidelink assignment index - 0, 1 or 2 bits:

  - 1 bit if the UE is configured with *pdsch-HARQ-ACK-Codebook* = *semi-static* and, in addition, the UE is configured with a SL configured grant type 1 or to monitor DCI format 3_0 with CRC scrambled by SL-RNTI or SL-CS-RNTI;
  - 2 bits if the UE is configured with *pdsch-HARQ-ACK-Codebook* = *dynamic* and, in addition, the UE is configured with a SL configured grant type 1 or to monitor DCI format 3_0 with CRC scrambled by SL-RNTI or SL-CS-RNTI;

  - 0 bit otherwise.
- PDCCH monitoring adaptation indication - 0, 1 or 2 bits
  - 1 or 2 bits, if *searchSpaceGroupIdList-r17* is not configured and if *pdcch-SkippingDurationList* is configured
    - 1 bit if the UE is configured with only one duration by *pdcch-SkippingDurationList;*
    - 2 bits if the UE is configured with more than one duration by *pdcch-SkippingDurationList.*
  - 1 or 2 bits, if *pdcch-SkippingDurationList* is not configured and if *searchSpaceGroupIdList-r17* is configured
    - 1 bit if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0 and search space set(s) with group index 1, and if the UE is not configured by *searchSpaceGroupIdList-r17* with any search space set with group index 2;
    - 2 bits if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0, search space set(s) with group index 1 and search space set(s) with group index 2;
  - 2 bits, if *pdcch-SkippingDurationList* is configured and if *searchSpaceGroupIdList-r17* is configured
  - 0 bit, otherwise
- Measurement gap cancellation – 0 bit if higher layer parameter *mg-CancellationDCI-0-1* is not configured; otherwise 1 bit as defined in Clause 10.6 in [5, TS 38.213].

A UE does not expect that the bit width of a field in DCI format 0_1 with CRC scrambled by CS-RNTI is larger than corresponding bit width of same field in DCI format 0_1 with CRC scrambled by C-RNTI for the same serving cell. If the bit width of a field in the DCI format 0_1 with CRC scrambled by CS-RNTI is not equal to that of the corresponding field in the DCI format 0_1 with CRC scrambled by C-RNTI for the same serving cell, a number of most significant bits with value set to '0' are inserted to the field in DCI format 0_1 with CRC scrambled by CS-RNTI until the bit width equals that of the corresponding field in the DCI format 0_1 with CRC scrambled by C-RNTI for the same serving cell.

If the number of information bits in DCI format 0_1 scheduling a single PUSCH prior to padding is not equal to the number of information bits in DCI format 0_1 scheduling multiple PUSCHs for the same serving cell, zeros shall be appended to the DCI format 0_1 with smaller size until the payload size is the same for scheduling a single PUSCH and multiple PUSCHs.

For a UE configured with scheduling on the primary cell from an SCell, if prior to padding the number of information bits in DCI format 0_1 carried by PDCCH on the primary cell is not equal to the number of information bits in DCI format 0_1 carried by PDCCH on the SCell for scheduling on the primary cell, zeros shall be appended to the DCI format 0_1 with smaller size until the payload size is the same.

- If application of step 4C in clause 7.3.1.0 results in additional zero padding for DCI format 0_1 for scheduling on the primary cell, corresponding zeros shall be appended to both DCI format 0_1 monitored on the primary cell and DCI format 0_1 monitored on the SCell for scheduling on the primary cell.
- If the SCell is deactivated and *firstActiveDownlinkBWP-Id* is not set to dormant BWP, the UE determines the number of information bits in DCI format 0_1 carried by PDCCH on the primary cell based on a DL BWP provided by *firstActiveDownlinkBWP-Id* for the SCell. If the active DL BWP of the SCell is a dormant DL BWP, or if the SCell is deactivated and *firstActiveDownlinkBWP-Id* is set to dormant BWP, the UE determines the number of information bits in DCI format 0_1 carried by PDCCH on the primary cell based on a DL BWP provided by *firstWithinActiveTimeBWP-Id* for the SCell if provided; otherwise, based on a DL BWP provided by *firstOutsideActiveTimeBWP-Id* for the SCell.

Table 7.3.1.1.2-1: Bandwidth part indicator

Value of BWP indicator field 2 bits	Bandwidth part
00	Configured BWP with BWP-Id = 1
01	Configured BWP with BWP-Id = 2
10	Configured BWP with BWP-Id = 3
11	Configured BWP with BWP-Id = 4

Table 7.3.1.1.2-2: Precoding information and number of layers or Second Precoding information, for 4 antenna ports, if transform precoder is disabled, *maxRank* = 2 or 3 or 4 or max{*maxRank*, *maxRankSFN*} = 2 or 3 or 4 or max{*maxRank*, *maxRankSDM*} = 2 or 3 or 4 or *maxRankSDM*= 2, and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*codebookSubset* = *fullyAndPartialAndNonCoherent*	Bit field mapped to index	*codebookSubset* = *partialAndNonCoherent*	Bit field mapped to index	*codebookSubset*= *nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…	…	…
3	1 layer: TPMI=3	3	1 layer: TPMI=3	3	1 layer: TPMI=3
4	2 layers: TPMI=0	4	2 layers: TPMI=0	4	2 layers: TPMI=0
…	…	…	…	…	…
9	2 layers: TPMI=5	9	2 layers: TPMI=5	9	2 layers: TPMI=5
10	3 layers: TPMI=0	10	3 layers: TPMI=0	10	3 layers: TPMI=0
11	4 layers: TPMI=0	11	4 layers: TPMI=0	11	4 layers: TPMI=0
12	1 layer: TPMI=4	12	1 layer: TPMI=4	12-15	reserved
…	…	…	…		
19	1 layer: TPMI=11	19	1 layer: TPMI=11		
20	2 layers: TPMI=6	20	2 layers: TPMI=6		
…	…	…	…		
27	2 layers: TPMI=13	27	2 layers: TPMI=13		
28	3 layers: TPMI=1	28	3 layers: TPMI=1		
29	3 layers: TPMI=2	29	3 layers: TPMI=2		
30	4 layers: TPMI=1	30	4 layers: TPMI=1		
31	4 layers: TPMI=2	31	4 layers: TPMI=2		
32	1 layers: TPMI=12				
…	…				
47	1 layers: TPMI=27				
48	2 layers: TPMI=14				
…	…				
55	2 layers: TPMI=21				
56	3 layers: TPMI=3				
…	…				
59	3 layers: TPMI=6				
60	4 layers: TPMI=3				
61	4 layers: TPMI=4				
62-63	reserved				

Table 7.3.1.1.2-2A: Precoding information and number of layers for 4 antenna ports or Second Precoding information,, if transform precoder is disabled, *maxRank* = 2 or max{*maxRank*, *maxRankSFN*} = 2 or max{*maxRank*, *maxRankSDM*} = 2 or *maxRankSDM*= 2, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset* = *partialAndNonCoherent*	Bit field mapped to index	*codebookSubset*= *nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…
3	1 layer: TPMI=3	3	1 layer: TPMI=3
4	2 layers: TPMI=0	4	2 layers: TPMI=0
…	…	…	…
9	2 layers: TPMI=5	9	2 layers: TPMI=5
10	1 layer: TPMI=13	10	1 layer: TPMI=13
11	2 layer: TPMI=6	11	2 layer: TPMI=6
12	1 layer: TPMI=4	12-15	Reserved
…	…		
20	1 layer: TPMI=12		
21	1 layer: TPMI=14		
22	1 layer: TPMI=15		
23	2 layers: TPMI=7		
…	…		
29	2 layers: TPMI=13		
30-31	Reserved		

Table 7.3.1.1.2-2B: Precoding information and number of layers for 4 antenna ports, if transform precoder is disabled, *maxRank* = 3 or 4, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset* = *partialAndNonCoherent*	Bit field mapped to index	*codebookSubset*= *nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…
3	1 layer: TPMI=3	3	1 layer: TPMI=3
4	2 layers: TPMI=0	4	2 layers: TPMI=0
…	…	…	…
9	2 layers: TPMI=5	9	2 layers: TPMI=5
10	3 layers: TPMI=0	10	3 layers: TPMI=0
11	4 layers: TPMI=0	11	4 layers: TPMI=0
12	1 layer: TPMI=13	12	1 layer: TPMI=13
13	2 layer: TPMI=6	13	2 layer: TPMI=6
14	3 layer: TPMI=1	14	3 layer: TPMI=1
15	1 layer: TPMI=4	15	Reserved
…	…		
23	1 layer: TPMI=12		
24	1 layer: TPMI=14		
25	1 layer: TPMI=15		
26	2 layers: TPMI=7		
…	…		
32	2 layers: TPMI=13		
33	3 layers: TPMI=2		
34	4 layers: TPMI=1		
35	4 layers: TPMI=2		
36-63	Reserved		

Table 7.3.1.1.2-2C: Second precoding information, for 4 antenna ports, if transform precoder is disabled, *maxRank* = 2 or 3 or 4 or *maxRankSFN* = 2, and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*codebookSubset = fullyAndPartialAndNonCoherent*	Bit field mapped to index	*codebookSubset = partialAndNonCoherent*	Bit field mapped to index	*codebookSubset = nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…
27	1 layer: TPMI=27	11	1 layer: TPMI=11	3	1 layer: TPMI=3
28-31	1 layer: reserved	12-15	1 layer: reserved	4-7	1 layer: reserved
0	2 layers: TPMI=0	0	2 layers: TPMI=0	0	2 layers: TPMI=0
…	…	…	…	…	…
21	2 layers: TPMI=21	13	2 layers: TPMI=13	5	2 layers: TPMI=5
22-31	2 layers: reserved	14-15	2 layers: reserved	6-7	2 layers: reserved
0	3 layers: TPMI=0	0	3 layers: TPMI=0	0	3 layers: TPMI=0
…	…	…	…	1-7	3 layers: reserved
6	3 layers: TPMI=6	2	3 layers: TPMI=2	0	4 layers: TPMI=0
7-31	3 layers: reserved	3-15	3 layers: reserved	1-7	4 layers: reserved
0	4 layers: TPMI=0	0	4 layers: TPMI=0		
…	…	…	…		
4	4 layers: TPMI=4	2	4 layers: TPMI=2		
5-31	4 layers: reserved	3-15	4 layers: reserved		

Table 7.3.1.1.2-2D: Second precoding information for 4 antenna ports, if transform precoder is disabled, *maxRank* = 2 or *maxRankSFN* = 2, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset = partialAndNonCoherent*	Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…
14	1 layer: TPMI=14	3	1 layer: TPMI=3
15	1 layer: TPMI=15	4	1 layer: TPMI=13
0	2 layers: TPMI=0	5-7	1 layer: reserved
…	…	0	2 layers: TPMI=0
13	2 layers: TPMI=13	…	…
14-15	2 layers: reserved	6	2 layers: TPMI=6
		7	2 layers: reserved

Table 7.3.1.1.2-2E: Second precoding information for 4 antenna ports, if transform precoder is disabled, *maxRank* = 3 or 4, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset = partialAndNonCoherent*	Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…
14	1 layer: TPMI=14	3	1 layer: TPMI=3
15	1 layer: TPMI=15	4	1 layer: TPMI=13
0	2 layers: TPMI=0	5-7	1 layer: reserved
…	…	0	2 layers: TPMI=0
13	2 layers: TPMI=13	…	…
14-15	2 layers: reserved	6	2 layers: TPMI=6
0	3 layers: TPMI=0	7	2 layers: reserved
…	…	0	3 layers: TPMI=0
2	3 layers: TPMI=2	1	3 layer: TPMI=1
3-15	3 layers: reserved	2-7	3 layers: reserved
0	4 layers: TPMI=0	0	4 layers: TPMI=0
…	…	1-7	4 layers: reserved
2	4 layers: TPMI=2		
3-15	4 layers: reserved		

Table 7.3.1.1.2-3: Precoding information and number of layers or Second Precoding information, for 4 antenna ports, if transform precoder is enabled and *ul-FullPowerTransmission* is either not configured or configured to *fullpowerMode2* or configured to *fullpower*, or if transform precoder is disabled, *maxRank* = 1 or max{*maxRank*, *maxRankSFN*} = 1 or max{*maxRank*, *maxRankSDM*} = 1 or *maxRankSDM*= 1 or *maxRankSFN*= 1, and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*codebookSubset = fullyAndPartialAndNonCoherent*	Bit field mapped to index	*codebookSubset= partialAndNonCoherent*	Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…	…	…
3	1 layer: TPMI=3	3	1 layer: TPMI=3	3	1 layer: TPMI=3
4	1 layer: TPMI=4	4	1 layer: TPMI=4		
…	…	…	…		
11	1 layer: TPMI=11	11	1 layer: TPMI=11		
12	1 layers: TPMI=12	12-15	reserved		
…	…				
27	1 layers: TPMI=27				
28-31	reserved				

Table 7.3.1.1.2-3A: Precoding information and number of layers or Second Precoding information, for 4 antenna ports, if transform precoder is enabled and *ul-FullPowerTransmission* = *fullpowerMode1*, or if transform precoder is disabled, *maxRank* = 1 or max{*maxRank*, *maxRankSFN*} = 1 or max{*maxRank*, *maxRankSDM*} = 1 or *maxRankSDM*= 1 or *maxRankSFN*= 1, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset= partialAndNonCoherent*	Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…
3	1 layer: TPMI=3	3	1 layer: TPMI=3
4	1 layer: TPMI=13	4	1 layer: TPMI=13
5	1 layer: TPMI=4	5-7	Reserved
…	…		
13	1 layer: TPMI=12		
14	1 layer: TPMI=14		
15	1 layer: TPMI=15		

Table 7.3.1.1.2-4: Precoding information and number of layers or Second Precoding information, for 2 antenna ports, if transform precoder is disabled, *maxRank* = 2 or max{*maxRank*, *maxRankSFN*} = 2 or max{*maxRank*, *maxRankSDM*} = 2 or *maxRankSDM*= 2, and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*codebookSubset = fullyAndPartialAndNonCoherent*	Bit field mapped to index	*codebookSubset = nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
2	2 layers: TPMI=0	2	2 layers: TPMI=0
3	1 layer: TPMI=2	3	reserved
4	1 layer: TPMI=3		
5	1 layer: TPMI=4		
6	1 layer: TPMI=5		
7	2 layers: TPMI=1		
8	2 layers: TPMI=2		
9-15	reserved		

Table 7.3.1.1.2-4A: Precoding information and number of layers or Second Precoding information, for 2 antenna ports, if transform precoder is disabled, *maxRank* = 2 or max{*maxRank*, *maxRankSFN*} = 2 or max{*maxRank*, *maxRankSDM*} = 2 or *maxRankSDM*= 2, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0
1	1 layer: TPMI=1
2	2 layers: TPMI=0
3	1 layer: TPMI=2

Table 7.3.1.1.2-4B: Second precoding information, for 2 antenna ports, if transform precoder is disabled, *maxRank* = 2 or *maxRankSFN* = 2, and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*codebookSubset* = *fullyAndPartialAndNonCoherent*	Bit field mapped to index	*codebookSubset* = *nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	0	2 layers: TPMI=0
5	1 layer: TPMI=5	1	2 layers: reserved
6-7	1 layer: reserved		
0	2 layers: TPMI=0		
…	…		
2	2 layers: TPMI=2		
3-7	2 layers: reserved		

Table 7.3.1.1.2-4C: Second precoding information, for 2 antenna ports, if transform precoder is disabled, *maxRank* = 2 or *maxRankSFN* = 2, and *ul-FullPowerTransmission* = *fullpowerMode1*

Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0
…	…
2	1 layer: TPMI=2
3	1 layer: reserved
0	2 layers: TPMI=0
1-3	2 layers: reserved

Table 7.3.1.1.2-5: Precoding information and number of layers or Second Precoding information, for 2 antenna ports, if transform precoder is enabled and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, or if transform precoder is disabled, *maxRank* = 1 or max{*maxRank*, *maxRankSFN*} = 1 or max{*maxRank*, *maxRankSDM*} = 1 or *maxRankSDM*= 1 or *maxRankSFN*= 1, and and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*codebookSubset* = *fullyAndPartialAndNonCoherent*	Bit field mapped to index	*codebookSubset* = *nonCoherent*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
2	1 layer: TPMI=2		
3	1 layer: TPMI=3		
4	1 layer: TPMI=4		
5	1 layer: TPMI=5		
6-7	reserved		

Table 7.3.1.1.2-5A: Precoding information and number of layers or Second Precoding information, for 2 antenna ports or Second Precoding information, if transform precoder is enabled and *ul-FullPowerTransmission = fullpowerMode1*, or if transform precoder is disabled, *maxRank* = 1 or max{*maxRank*, *maxRankSFN*} = 1 or max{*maxRank*, *maxRankSDM*} = 1 or *maxRankSDM*= 1 or *maxRankSFN*= 1, and *ul-FullPowerTransmission = fullpowerMode1*

Bit field mapped to index	*codebookSubset= nonCoherent*
0	1 layer: TPMI=0
1	1 layer: TPMI=1
2	1 layer: TPMI=2
3	Reserved

Table 7.3.1.1.2-5B: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 8, and *CodebookTypeUL=codebook1*

Bit field mapped to index	*codebook1 = ng1n4n1*	Bit field mapped to index	*codebook1 = ng1n2n2*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15
16	2 layers: TPMI=0	16	2 layer2: TPMI=0
17	2 layers: TPMI=1	17	2 layer2: TPMI=1
…	…	…	…
47	2 layers: TPMI=31	47	2 layers: TPMI=31
48	3 layers: TPMI=0	48	3 layers: TPMI=0
49	3 layers: TPMI=1	49	3 layers: TPMI=1
…	…	…	…
71	3 layers: TPMI=23	71	3 layers: TPMI=23
72	4 layers: TPMI=0	72	4 layers: TPMI=0
73	4 layers: TPMI=1	73	4 layers: TPMI=1
…	…	…	…
95	4 layers: TPMI=23	95	4 layers: TPMI=23
96	5 layers: TPMI=0	96	5 layers: TPMI=0
97	5 layers: TPMI=1	97	5 layers: TPMI=1
…	…	…	…
103	5 layers: TPMI=7	103	5 layers: TPMI=7
104	6 layers: TPMI=0	104	6 layers: TPMI=0
105	6 layers: TPMI=1	105	6 layers: TPMI=1
…	…	…	…
111	6 layers: TPMI=7	111	6 layers: TPMI=7
112	7 layers: TPMI=0	112	7 layers: TPMI=0
113	7 layers: TPMI=1	113	7 layers: TPMI=1
…	…	…	…
115	7 layers: TPMI=3	119	7 layers: TPMI=7
116	8 layers: TPMI=0	120	8 layers: TPMI=0
117	8 layers: TPMI=1	121	8 layers: TPMI=1
…	…	…	…
119	8 layers: TPMI=3	127	8 layers: TPMI=7
120-127	reserved		

Table 7.3.1.1.2-5C: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 7, and *CodebookTypeUL=codebook1*

Bit field mapped to index	*codebook1= ng1n4n1*	Bit field mapped to index	*codebook1= ng1n2n2*
0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15
16	2 layers: TPMI=0	16	2 layer2: TPMI=0
17	2 layers: TPMI=1	17	2 layer2: TPMI=1
…	…	…	…
47	2 layers: TPMI=31	47	2 layers: TPMI=31
48	3 layers: TPMI=0	48	3 layers: TPMI=0
49	3 layers: TPMI=1	49	3 layers: TPMI=1
…	…	…	…
71	3 layers: TPMI=23	71	3 layers: TPMI=23
72	4 layers: TPMI=0	72	4 layers: TPMI=0
73	4 layers: TPMI=1	73	4 layers: TPMI=1
…	…	…	…
95	4 layers: TPMI=23	95	4 layers: TPMI=23
96	5 layers: TPMI=0	96	5 layers: TPMI=0
97	5 layers: TPMI=1	97	5 layers: TPMI=1
…	…	…	…
103	5 layers: TPMI=7	103	5 layers: TPMI=7
104	6 layers: TPMI=0	104	6 layers: TPMI=0
105	6 layers: TPMI=1	105	6 layers: TPMI=1
…	…	…	…
111	6 layers: TPMI=7	111	6 layers: TPMI=7
112	7 layers: TPMI=0	112	7 layers: TPMI=0
113	7 layers: TPMI=1	113	7 layers: TPMI=1
…	…	…	…
115	7 layers: TPMI=3	119	7 layers: TPMI=7
116-127	reserved	120-127	reserved

Table 7.3.1.1.2-5D: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 4, 5 or 6, *CodebookTypeUL=codebook1*

Bit field mapped to index	*maxRank* = 4	Bit field mapped to index	*maxRank* = 5	Bit field mapped to index	*maxRank = 6*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15
16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layer2: TPMI=0
17	2 layers: TPMI=1	17	2 layers: TPMI=1	17	2 layer2: TPMI=1
…	…	…	…	…	…
47	2 layers: TPMI=31	47	2 layers: TPMI=31	47	2 layers: TPMI=31
48	3 layers: TPMI=0	48	3 layers: TPMI=0	48	3 layers: TPMI=0
49	3 layers: TPMI=1	49	3 layers: TPMI=1	49	3 layers: TPMI=1
…	…	…	…	…	…
71	3 layers: TPMI=23	71	3 layers: TPMI=23	71	3 layers: TPMI=23
72	4 layers: TPMI=0	72	4 layers: TPMI=0	72	4 layers: TPMI=0
73	4 layers: TPMI=1	73	4 layers: TPMI=1	73	4 layers: TPMI=1
…	…	…	…	…	…
95	4 layers: TPMI=23	95	4 layers: TPMI=23	95	4 layers: TPMI=23
96-127	reserved	96	5 layers: TPMI=0	96	5 layers: TPMI=0
		97	5 layers: TPMI=1	97	5 layers: TPMI=1
		…	…	…	…
		103	5 layers: TPMI=7	103	5 layers: TPMI=7
		104-127	reserved	104	6 layers: TPMI=0
				105	6 layers: TPMI=1
				…	…
				111	6 layers: TPMI=7
				112-127	reserved

Table 7.3.1.1.2-5E: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled or *maxRank*=1 or 2 or 3 if transform precoder is disabled, *CodebookTypeUL=codebook1*

Bit field mapped to index	transform precoder is enabled, or *maxRank*=1 if transform precoder is disabled	Bit field mapped to index	transform precoder is disabled, and *maxRank*=2	Bit field mapped to index	transform precoder is disabled, and *maxRank=3*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
1	1 layer: TPMI=1	1	1 layer: TPMI=1	1	1 layer: TPMI=1
…	…	…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15
		16	2 layer2: TPMI=0	16	2 layer2: TPMI=0
		17	2 layer2: TPMI=1	17	2 layer2: TPMI=1
		…	…	…	…
		47	2 layers: TPMI=31	47	2 layers: TPMI=31
		48-63	reserved	48	3 layers: TPMI=0
				49	3 layers: TPMI=1
				…	…
				71	3 layers: TPMI=23
				72-127	reserved

Table 7.3.1.1.2-5F: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, and *CodebookTypeUL=codebook4*

Bit field mapped to index	*maxRank = 5*	Bit field mapped to index	*maxRank = 6*	Bit field mapped to index	*maxRank = 7*	Bit field mapped to index	*maxRank = 8*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
7	1 layer: TPMI=7	7	1 layer: TPMI=7	7	1 layer: TPMI=7	7	1 layer: TPMI=7
8	2 layers: TPMI=8	8	2 layers: TPMI=8	8	2 layers: TPMI=8	8	2 layers: TPMI=8
…	…	…	…	…	…	…	…
35	2 layers: TPMI=35	35	2 layers: TPMI=35	35	2 layers: TPMI=35	35	2 layers: TPMI=35
36	3 layers: TPMI=36	36	3 layers: TPMI=36	36	3 layers: TPMI=36	36	3 layers: TPMI=36
…	…	…	…	…	…	…	…
91	3 layers: TPMI=91	91	3 layers: TPMI=91	91	3 layers: TPMI=91	91	3 layers: TPMI=91
92	4 layers: TPMI=92	92	4 layers: TPMI=92	92	4 layers: TPMI=92	92	4 layers: TPMI=92
…	…	…	…	…	…	…	…
161	4 layers: TPMI=161	161	4 layers: TPMI=161	161	4 layers: TPMI=161	161	4 layers: TPMI=161
162	5 layers: TPMI=162	162	5 layers: TPMI=162	162	5 layers: TPMI=162	162	5 layers: TPMI=162
…	…	…	…	…	…	…	…
217	5 layers: TPMI=217	217	5 layers: TPMI=217	217	5 layers: TPMI=217	217	5 layers: TPMI=217
218-255	reserved	218	6 layers: TPMI=218	218	6 layers: TPMI=218	218	6 layers: TPMI=218
		…	…	…	…	…	…
		245	6 layers: TPMI=245	245	6 layers: TPMI=245	245	6 layers: TPMI=245
		246-255	reserved	246	7 layers: TPMI=246	246	7 layers: TPMI=246
				…	…	…	…
				253	7 layers: TPMI=253	253	7 layers: TPMI=253
				254-255	reserved	254	8 layers: TPMI=254
						255	reserved

Table 7.3.1.1.2-5G: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 2, 3 or 4, *CodebookTypeUL=codebook4,* and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	*maxRank = 2*	Bit field mapped to index	*maxRank = 3*	Bit field mapped to index	*maxRank = 4*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…
7	1 layer: TPMI=7	7	1 layer: TPMI=7	7	1 layer: TPMI=7
8	2 layers: TPMI=8	8	2 layers: TPMI=8	8	2 layers: TPMI=8
…	…	…	…	…	…
35	2 layers: TPMI=35	35	2 layers: TPMI=35	35	2 layers: TPMI=35
36-63	reserved	36	3 layers: TPMI=36	36	3 layers: TPMI=36
		…	…	…	…
		91	3 layers: TPMI=91	91	3 layers: TPMI=91
		92-127	reserved	92	4 layers: TPMI=92
				…	…
				161	4 layers: TPMI=161
				162-255	reserved

Table 7.3.1.1.2-5H: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled or *maxRank*=1 if transform is disabled, *CodebookTypeUL=codebook4,* and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	Precoding information and number of layers
0	1 layer: TPMI=0
…	…
7	1 layer: TPMI=7

Table 7.3.1.1.2-5l: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, and *CodebookTypeUL=codebook2*

Bit field mapped to index	*maxRank = 5*	Bit field mapped to index	*maxRank = 6*	Bit field mapped to index	*maxRank = 7*	Bit field mapped to index	*maxRank = 8*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31
32	2 layers: TPMI=0	32	2 layers: TPMI=0	32	2 layers: TPMI=0	32	2 layers: TPMI=0
…	…	…	…	…	…	…	…
303	2 layers: TPMI=271	303	2 layers: TPMI=271	303	2 layers: TPMI=271	303	2 layers: TPMI=271
304	3 layers: TPMI=0	304	3 layers: TPMI=0	304	3 layers: TPMI=0	304	3 layers: TPMI=0
…	…	…	…	…	…	…	…
567	3 layers: TPMI=263	567	3 layers: TPMI=263	567	3 layers: TPMI=263	567	3 layers: TPMI=263
568	4 layers: TPMI=0	568	4 layers: TPMI=0	568	4 layers: TPMI=0	568	4 layers: TPMI=0
…	…	…	…	…	…	…	…
635	4 layers: TPMI=67	635	4 layers: TPMI=67	635	4 layers: TPMI=67	635	4 layers: TPMI=67
636	5 layers: TPMI=0	636	5 layers: TPMI=0	636	5 layers: TPMI=0	636	5 layers: TPMI=0
…	…	…	…	…	…	…	…
667	5 layers: TPMI=31	667	5 layers: TPMI=31	667	5 layers: TPMI=31	667	5 layers: TPMI=31
698-1023	reserved	668	6 layers: TPMI=0	668	6 layers: TPMI=0	668	6 layers: TPMI=0
		…	…	…	…	…	…
		683	6 layers: TPMI=15	683	6 layers: TPMI=15	683	6 layers: TPMI=15
		684-1023	reserved	684	7 layers: TPMI=0	684	7 layers: TPMI=0
				…	…	…	…
				691	7 layers: TPMI=7	691	7 layers: TPMI=7
				692-1023	reserved	692	8 layers: TPMI=0
						…	…
						695	8 layers: TPMI=3
						696-1023	reserved

Table 7.3.1.1.2-5J: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled, or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *CodebookTypeUL=codebook2,* and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	Transform precoder is enabled, or *maxRank* = 1 if transform precoder is disabled	Bit field mapped to index	transform precoder is disabled *and maxRank = 2*	Bit field mapped to index	transform precoder is disabled *and maxRank = 3*	Bit field mapped to index	transform precoder is disabled *and maxRank = 4*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31
		32	2 layers: TPMI=0	32	2 layers: TPMI=0	32	2 layers: TPMI=0
		…	…	…	…	…	…
		303	2 layers: TPMI=271	303	2 layers: TPMI=271	303	2 layers: TPMI=271
		303-511	reserved	304	3 layers: TPMI=0	304	3 layers: TPMI=0
				…	…	…	…
				567	3 layers: TPMI=263	567	3 layers: TPMI=263
				568-1023	reserved	568	4 layers: TPMI=0
						…	…
						635	4 layers: TPMI=67
						636-1023	reserved

Table 7.3.1.1.2-5K: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 5, 6, 7 or 8, and *CodebookTypeUL=codebook3*

Bit field mapped to index	*maxRank = 5*	Bit field mapped to index	*maxRank = 6*	Bit field mapped to index	*maxRank = 7*	Bit field mapped to index	*maxRank = 8*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15
16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layers: TPMI=0
…	…	…	…	…	…	…	…
119	2 layers: TPMI=103	119	2 layers: TPMI=103	119	2 layers: TPMI=103	119	2 layers: TPMI=103
120	3 layers: TPMI=0	120	3 layers: TPMI=0	120	3 layers: TPMI=0	120	3 layers: TPMI=0
…	…	…	…	…	…	…	…
423	3 layers: TPMI=303	423	3 layers: TPMI=303	423	3 layers: TPMI=303	423	3 layers: TPMI=303
424	4 layers: TPMI=0	424	4 layers: TPMI=0	424	4 layers: TPMI=0	424	4 layers: TPMI=0
…	…	…	…	…	…	…	…
703	4 layers: TPMI=279	703	4 layers: TPMI=279	703	4 layers: TPMI=279	703	4 layers: TPMI=279
704	5 layers: TPMI=0	704	5 layers: TPMI=0	704	5 layers: TPMI=0	704	5 layers: TPMI=0
…	…	…	…	…	…	…	…
863	5 layers: TPMI=159	863	5 layers: TPMI=159	863	5 layers: TPMI=159	863	5 layers: TPMI=159
864-1023	reserved	864	6 layers: TPMI=0	864	6 layers: TPMI=0	864	6 layers: TPMI=0
		…	…	…	…	…	…
		943	6 layers: TPMI=79	943	6 layers: TPMI=79	943	6 layers: TPMI=79
		944-1023	reserved	944	7 layers: TPMI=0	944	7 layers: TPMI=0
				…	…	…	…
				975	7 layers: TPMI=31	975	7 layers: TPMI=31
				976-1023	reserved	976	8 layers: TPMI=0
						…	…
						991	8 layers: TPMI=15
						992-1023	reserved

Table 7.3.1.1.2-5L: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled, or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *CodebookTypeUL=codebook3,* and *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*

Bit field mapped to index	Transform precoder is enabled, or *maxRank* = 1 if transform precoder is disabled	Bit field mapped to index	transform precoder is disabled *and maxRank = 2*	Bit field mapped to index	transform precoder is disabled *and maxRank = 3*	Bit field mapped to index	transform precoder is disabled *and maxRank = 4*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15
		16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layers: TPMI=0
		…	…	…	…	…	…
		119	2 layers: TPMI=103	119	2 layers: TPMI=103	119	2 layers: TPMI=103
		119-127	reserved	120	3 layers: TPMI=0	120	3 layers: TPMI=0
				…	…	…	…
				423	3 layers: TPMI=303	423	3 layers: TPMI=303
				424-511	reserved	424	4 layers: TPMI=0
						…	…
						703	4 layers: TPMI=279
						704-1024	reserved

Table 7.3.1.1.2-5M: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 2, 3 or 4, *CodebookTypeUL=codebook4,* and *ul-FullPowerTransmission* configured to *fullpowerMode1*

Bit field mapped to index	*maxRank = 2*	Bit field mapped to index	*maxRank = 3*	Bit field mapped to index	*maxRank = 4*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	...	…
7	1 layer: TPMI=7	7	1 layer: TPMI=7	7	1 layer: TPMI=7
8	2 layers: TPMI=8	8	2 layers: TPMI=8	8	2 layers: TPMI=8
…	…	…	…	...	…
35	2 layers: TPMI=35	35	2 layers: TPMI=35	35	2 layers: TPMI=35
36	1 layer: TPMI=255	36	3 layers: TPMI=36	36	3 layers: TPMI=36
37	2 layers: TPMI=256	…	…	...	…
38-63	reserved	91	3 layers: TPMI=91	91	3 layers: TPMI=91
		92	1 layer: TPMI=255	92	4 layers: TPMI=92
		93	2 layers: TPMI=256	...	…
		94	3 layers: TPMI=257	161	4 layers: TPMI=161
		95-127	reserved	162	1 layer: TPMI=255
				163	2 layers: TPMI=256
				164	3 layers: TPMI=257
				165	4 layers: 258
				166-255	reserved

Table 7.3.1.1.2-5N: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled or *maxRank*=1 if transform is disabled, *CodebookTypeUL=codebook4,* and *ul-FullPowerTransmission* configured to *fullpowerMode1*

Bit field mapped to index	Precoding information and number of layers
0	1 layer: TPMI=0
…	…
7	1 layer: TPMI=7
8	1 layer: TPMI=255
9-15	reserved

Table 7.3.1.1.2-5O: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled, or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *CodebookTypeUL=codebook2,* and *ul-FullPowerTransmission* configured to *fullpowerMode1*

Bit field mapped to index	Transform precoder is enabled, or maxRank = 1 if transform precoder is disabled	Bit field mapped to index	Transform precoder is disabled *and maxRank = 2*	Bit field mapped to index	Transform precoder is disabled *and maxRank = 3*	Bit field mapped to index	Transform precoder is disabled *and maxRank = 4*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31
32	1 layer: TPMI=32	32	2 layers: TPMI=0	32	2 layers: TPMI=0	32	2 layers: TPMI=0
33-63	reserved	…	…	…	…	…	…
		303	2 layers: TPMI=271	303	2 layers: TPMI=271	303	2 layers: TPMI=271
		304	1 layer: TPMI=32	304	3 layers: TPMI=0	304	3 layers: TPMI=0
		305-511	reserved	…	…	…	…
				567	3 layers: TPMI=263	567	3 layers: TPMI=263
				568	1 layer: TPMI=32	568	4 layers: TPMI=0
				569-1023	reserved	…	…
						635	4 layers: TPMI=67
						636	1 layer: TPMI=32
						637-1023	reserved

Table 7.3.1.1.2-5P: Precoding information and number of layers, for 8 antenna ports, if transform precoder is enabled, or *maxRank* = 1, 2, 3 or 4 if transform precoder is disabled, *CodebookTypeUL=codebook3,* and *ul-FullPowerTransmission* is configured to *fullpowerMode1*

Bit field mapped to index	Transform precoder is enabled, or *maxRank* = 1 if transform precoder is disabled	Bit field mapped to index	transform precoder is disabled *and maxRank = 2*	Bit field mapped to index	transform precoder is disabled *and maxRank = 3*	Bit field mapped to index	transform precoder is disabled *and maxRank = 4*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15
16	1 layer: TPMI=16	16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layers: TPMI=0
17-31	reserved	…	…	…	…	…	…
		119	2 layers: TPMI=103	119	2 layers: TPMI=103	119	2 layers: TPMI=103
		120	1 layer: TPMI=16	120	3 layers: TPMI=0	120	3 layers: TPMI=0
		121	2 layers: TPMI=104	…	…	…	…
		122-127	reserved	423	3 layers: TPMI=303	423	3 layers: TPMI=303
				424	1 layer: TPMI=16	424	4 layers: TPMI=0
				425	2 layers: TPMI=104	…	…
				426	3 layers: 304	703	4 layers: TPMI=279
				427-511	reserved	704	1 layer: TPMI=16
						705	2 layers: TPMI=104
						706	3 layers: TPMI=304
						707-1023	reserved

Table 7.3.1.1.2-5Q: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 5, 6, 7, 8, *CodebookTypeUL=codebook4,* and *ul-FullPowerTransmission* is configured to *fullpowerMode1*

Bit field mapped to index	*maxRank* = 5	Bit field mapped to index	*maxRank* = 6	Bit field mapped to index	*maxRank* = 7	Bit field mapped to index	*maxRank* = 8
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
7	1 layer: TPMI=7	7	1 layer: TPMI=7	7	1 layer: TPMI=7	7	1 layer: TPMI=7
8	2 layers: TPMI=8	8	2 layers: TPMI=8	8	2 layers: TPMI=8	8	2 layers: TPMI=8
…	…	…	…	…	…	…	…
35	2 layers: TPMI=35	35	2 layers: TPMI=35	35	2 layers: TPMI=35	35	2 layers: TPMI=35
36	3 layers: TPMI=36	36	3 layers: TPMI=36	36	3 layers: TPMI=36	36	3 layers: TPMI=36
…	…	…	…	…	…	…	…
91	3 layers: TPMI=91	91	3 layers: TPMI=91	91	3 layers: TPMI=91	91	3 layers: TPMI=91
92	4 layers: TPMI=92	92	4 layers: TPMI=92	92	4 layers: TPMI=92	92	4 layers: TPMI=92
…	…	…	…	…	…	…	…
161	4 layers: TPMI=161	161	4 layers: TPMI=161	161	4 layers: TPMI=161	161	4 layers: TPMI=161
162	5 layers: TPMI=162	162	5 layers: TPMI=162	162	5 layers: TPMI=162	162	5 layers: TPMI=162
…	…	…	…	…	…	…	…
217	5 layers: TPMI=217	217	5 layers: TPMI=217	217	5 layers: TPMI=217	217	5 layers: TPMI=217
218	1 layer: TPMI=255	218	6 layers: TPMI=218	218	6 layers: TPMI=218	218	6 layers: TPMI=218
219	2 layers: TPMI=256	…	…	…	…	…	…
220	3 layers: TPMI=257	245	6 layers: TPMI=245	245	6 layers: TPMI=245	245	6 layers: TPMI=245
221	4 layers: TPMI=258	246	1 layer: TPMI=255	246	7 layers: TPMI=246	246	7 layers: TPMI=246
222-255	reserved	247	2 layers: TPMI=256	…	…	…	…
		248	3 layers: TPMI=257	253	7 layers: TPMI=253	253	7 layers: TPMI=253
		249	4 layers: TPMI=258	254	1 layer: TPMI=255	254	8 layers: TPMI=254
		250-255	reserved	255	2 layers: TPMI=256	255	1 layer: TPMI=255

				256	3 layers: TPMI=257	256	2 layers: TPMI=256
				257	4 layers: TPMI=258	257	3 layers: TPMI=257
				258-511	reserved	258	4 layers: TPMI=258
						259-511	reserved

Table 7.3.1.1.2-5R: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 5, 6, 7, 8, *CodebookTypeUL=codebook2,* and *ul-FullPowerTransmission* is configured to *fullpowerMode1*

Bit field mapped to index	*maxRank = 5*	Bit field mapped to index	*maxRank = 6*	Bit field mapped to index	*maxRank = 7*	Bit field mapped to index	*maxRank = 8*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31	31	1 layer: TPMI=31
32	2 layers: TPMI=0	32	2 layers: TPMI=0	32	2 layers: TPMI=0	32	2 layers: TPMI=0
…	…	…	…	…	…	…	…
303	2 layers: TPMI=271	303	2 layers: TPMI=271	303	2 layers: TPMI=271	303	2 layers: TPMI=271
304	3 layers: TPMI=0	304	3 layers: TPMI=0	304	3 layers: TPMI=0	304	3 layers: TPMI=0
…	…	…	…	…	…	…	…
567	3 layers: TPMI=263	567	3 layers: TPMI=263	567	3 layers: TPMI=263	567	3 layers: TPMI=263
568	4 layers: TPMI=0	568	4 layers: TPMI=0	568	4 layers: TPMI=0	568	4 layers: TPMI=0
…	…	…	…	…	…	…	…
635	4 layers: TPMI=67	635	4 layers: TPMI=67	635	4 layers: TPMI=67	635	4 layers: TPMI=67
636	5 layers: TPMI=0	636	5 layers: TPMI=0	636	5 layers: TPMI=0	636	5 layers: TPMI=0
…	…	…	…	…	…	…	…
667	5 layers: TPMI=31	667	5 layers: TPMI=31	667	5 layers: TPMI=31	667	5 layers: TPMI=31
668	1 layer: TPMI=32	668	6 layers: TPMI=0	668	6 layers: TPMI=0	668	6 layers: TPMI=0
669-1023	reserved	…	…	…	…	…	…
		683	6 layers: TPMI=15	683	6 layers: TPMI=15	683	6 layers: TPMI=15
		684	1 layer: TPMI=32	684	7 layers: TPMI=0	684	7 layers: TPMI=0
		685-1023	reserved	…	…	…	…
				691	7 layers: TPMI=7	691	7 layers: TPMI=7
				692	1 layer: TPMI=32	692	8 layers: TPMI=0
				693-1023	reserved	…	…

						695	8 layers: TPMI=3
						696	1 layer: TPMI=32
						697-1023	reserved

Table 7.3.1.1.2-5S: Precoding information and number of layers, for 8 antenna ports, if transform precoder is disabled, *maxRank* = 5, 6, 7, 8, *CodebookTypeUL=codebook3,* and *ul-FullPowerTransmission* is configured to *fullpowerMode1*

Bit field mapped to index	*maxRank = 5*	Bit field mapped to index	*maxRank = 6*	Bit field mapped to index	*maxRank = 7*	Bit field mapped to index	*maxRank = 8*
0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0	0	1 layer: TPMI=0
…	…	…	…	…	…	…	…
15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15	15	1 layer: TPMI=15
16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layers: TPMI=0	16	2 layers: TPMI=0
…	…	…	…	…	…	…	…
119	2 layers: TPMI=103	119	2 layers: TPMI=103	119	2 layers: TPMI=103	119	2 layers: TPMI=103
120	3 layers: TPMI=0	120	3 layers: TPMI=0	120	3 layers: TPMI=0	120	3 layers: TPMI=0
…	…	…	…	…	…	…	…
423	3 layers: TPMI=303	423	3 layers: TPMI=303	423	3 layers: TPMI=303	423	3 layers: TPMI=303
424	4 layers: TPMI=0	424	4 layers: TPMI=0	424	4 layers: TPMI=0	424	4 layers: TPMI=0
…	…	…	…	…	…	…	…
703	4 layers: TPMI=279	703	4 layers: TPMI=279	703	4 layers: TPMI=279	703	4 layers: TPMI=279
704	5 layers: TPMI=0	704	5 layers: TPMI=0	704	5 layers: TPMI=0	704	5 layers: TPMI=0
…	…	…	…	…	…	…	…
863	5 layers: TPMI=159	863	5 layers: TPMI=159	863	5 layers: TPMI=159	863	5 layers: TPMI=159
864	1 layer: TPMI=16	864	6 layers: TPMI=0	864	6 layers: TPMI=0	864	6 layers: TPMI=0
865	2 layers: TPMI=104	…	…	…	…	…	…
866	3 layers: TPMI=304	943	6 layers: TPMI=79	943	6 layers: TPMI=79	943	6 layers: TPMI=79
867-1023	reserved	944	1 layer: TPMI=16	944	7 layers: TPMI=0	944	7 layers: TPMI=0
		945	2 layers: TPMI=104	…	…	…	…
		946	3 layers: TPMI=304	975	7 layers: TPMI=31	975	7 layers: TPMI=31
		944-1023	reserved	976	1 layer: TPMI=16	976	8 layers: TPMI=0
				977	2 layers: TPMI=104	…	…

				978	3 layers: TPMI=304	991	8 layers: TPMI=15
				979-1023	reserved	992	1 layer: TPMI=16
						993	2 layers: TPMI=104
						994	3 layers: TPMI=304
						995-1023	reserved

Table 7.3.1.1.2-5T: Precoding information and number of layers, for 3 antenna ports, if transform precoder is disabled, *maxRank* = 2 or 3, and *ul-FullPowerTransmission* is not configured or configured to *fullpower*

Bit field mapped to index	***codebookSubset= nonCoherent***
0	1 layer: TPMI=0
1	1 layer: TPMI=1
2	1 layer: TPMI=2
3	2 layers: TPMI=0
4	2 layers: TPMI=1
5	2 layers: TPMI=2
6	3 layers: TPMI=0
7	reserved

Table 7.3.1.1.2-5U: Precoding information and number of layers or Second Precoding information, for 3 antenna ports, if transform precoder is enabled and *ul-FullPowerTransmission* is not configured or configured to *fullpower*, or if transform precoder is disabled, *maxRank* = 1, and *ul-FullPowerTransmission* is not configured or configured to *fullpower*

Bit field mapped to index	***codebookSubset= nonCoherent***
0	1 layer: TPMI=0
1	1 layer: TPMI=1
2	1 layer: TPMI=2
3	reserved

Table 7.3.1.1.2-5V: Second precoding information, for 3 antenna ports, if transform precoder is disabled, *maxRank* = 2 or 3, and *ul-FullPowerTransmission* is not configured or configured to *fullpower*

Bit field mapped to index	***codebookSubset= nonCoherent***
0	1 layer: TPMI=0
1	1 layer: TPMI=1
2	1 layer: TPMI=2
3	reserved
0	2 layers: TPMI=0
1	2 layers: TPMI=1
2	2 layers: TPMI=2
3	reserved
0	3 layers: TPMI=0
1-3	reserved

Table 7.3.1.1.2-6: Antenna port(s), transform precoder is enabled, *dmrs-Type*=1, *maxLength*=1, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2-BPSK modulation is used

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Orthogonal sequence index for OCC length =2	Orthogonal sequence index for OCC length =4
0	2	0	0	0
1	2	1	1	1
2	2	2	0	2
3	2	3	1	3
NOTE: The two columns for orthogonal sequence index in this table are applicable only when OCC length 2 or 4 is indicated by the field of Time domain resource assignment in DCI format 0_1 or DCI format 0_2. The corresponding orthogonal sequence of an orthogonal sequence index is as defined in Table 6.3.2.5A-1 or Table 6.3.2.5A-2 of [4, TS 38.211].				

Table 7.3.1.1.2-6A: Antenna port(s), transform precoder is enabled, *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2-BPSK modulation is used, *dmrs-Type*=1, *maxLength*=1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Orthogonal sequence index for OCC length =2	Orthogonal sequence index for OCC length =4
0	2	0, n_{SCID}= 0	0	0
1	2	0, n_{SCID}= 1	1	1
2	2	2, n_{SCID}= 0	0	2
3	2	2, n_{SCID}= 1	1	3
NOTE: The two columns for orthogonal sequence index in this table are applicable only when OCC length 2 or 4 is indicated by the field of Time domain resource assignment in DCI format 0_1 or DCI format 0_2. The corresponding orthogonal sequence of an orthogonal sequence index is as defined in Table 6.3.2.5A-1 or Table 6.3.2.5A-2 of [4, TS 38.211].				

Table 7.3.1.1.2-7: Antenna port(s), transform precoder is enabled, *dmrs-Type*=1, *maxLength*=2, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2-BPSK modulation is used

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Orthogonal sequence index for OCC length =2	Orthogonal sequence index for OCC length =4
0	2	0	1	0	0
1	2	1	1	1	1
2	2	2	1	0	2
3	2	3	1	1	3
4	2	0	2	0	0
5	2	1	2	1	1
6	2	2	2	0	2
7	2	3	2	1	3
8	2	4	2	0	0
9	2	5	2	1	1
10	2	6	2	0	2
11	2	7	2	1	3
12-15	Reserved	Reserved	Reserved	Reserved	Reserved
NOTE: The two columns for orthogonal sequence index in this table are applicable only when OCC length 2 or 4 is indicated by the field of Time domain resource assignment in DCI format 0_1 or DCI format 0_2. The corresponding orthogonal sequence of an orthogonal sequence index is as defined in Table 6.3.2.5A-1 or Table 6.3.2.5A-2 of [4, TS 38.211].					

Table 7.3.1.1.2-7A: Antenna port(s), transform precoder is enabled, *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2-BPSK modulation is used, *dmrs-Type=1, maxLength=2*

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Orthogonal sequence index for OCC length =2	Orthogonal sequence index for OCC length =4
0	2	0, n_{SCID}= 0	1	0	0
1	2	0, n_{SCID}= 1	1	1	1
2	2	2, n_{SCID}= 0	1	0	2
3	2	2, n_{SCID}= 1	1	1	3
4	2	0, n_{SCID}= 0	2	0	0
5	2	0, n_{SCID}= 1	2	1	1
6	2	2, n_{SCID}= 0	2	0	2
7	2	2, n_{SCID}= 1	2	1	3
8	2	4, n_{SCID}= 0	2	0	0
9	2	4, n_{SCID}= 1	2	1	1
10	2	6, n_{SCID}= 0	2	0	2
11	2	6, n_{SCID}= 1	2	1	3
12-15	Reserved	Reserved	Reserved	Reserved	Reserved
NOTE: The two columns for orthogonal sequence index in this table are applicable only when OCC length 2 or 4 is indicated by the field of Time domain resource assignment in DCI format 0_1 or DCI format 0_2. The corresponding orthogonal sequence of an orthogonal sequence index is as defined in Table 6.3.2.5A-1 or Table 6.3.2.5A-2 of [4, TS 38.211].					

Table 7.3.1.1.2-8: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank = 1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	2	0
3	2	1
4	2	2
5	2	3
6-7	Reserved	Reserved

Table 7.3.1.1.2-9: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank = 2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0,1
1	2	0,1
2	2	2,3
3	2	0,2
4-7	Reserved	Reserved

Table 7.3.1.1.2-10: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is not configured, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1-7	Reserved	Reserved

Table 7.3.1.1.2-10A: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is configured, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	2	0,2,3
2-7	Reserved	Reserved

Table 7.3.1.1.2-11: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank = 4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3
1-7	Reserved	Reserved

Table 7.3.1.1.2-12: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1
1	1	1	1
2	2	0	1
3	2	1	1
4	2	2	1
5	2	3	1
6	2	0	2
7	2	1	2
8	2	2	2
9	2	3	2
10	2	4	2
11	2	5	2
12	2	6	2
13	2	7	2
14-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-13: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0,1	1
1	2	0,1	1
2	2	2,3	1
3	2	0,2	1
4	2	0,1	2
5	2	2,3	2
6	2	4,5	2
7	2	6,7	2
8	2	0,4	2
9	2	2,6	2
10-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-14: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is not configured, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	2	0,1,4	2
2	2	2,3,6	2
3-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-14A: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is configured, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	2	0,1,4	2
2	2	2,3,6	2
3	2	0,2,3	1
4-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-15: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-3	1
1	2	0,1,4,5	2
2	2	2,3,6,7	2
3	2	0,2,4,6	2
4-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-15A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-4	2
1-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-15B Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,4,6	2
1-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-15C: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 7

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,4,5,6	2
1-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-15D: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 8

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,4,5,6,7	2
1-15	Reserved	Reserved	Reserved

Table 7.3.1.1.2-16: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank=1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	2	0
3	2	1
4	2	2
5	2	3
6	3	0
7	3	1
8	3	2
9	3	3
10	3	4
11	3	5
12-15	Reserved	Reserved

Table 7.3.1.1.2-17: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank=2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0,1
1	2	0,1
2	2	2,3
3	3	0,1
4	3	2,3
5	3	4,5
6	2	0,2
7-15	Reserved	Reserved

Table 7.3.1.1.2-18: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is not configured, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank =3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	3	0-2
2	3	3-5
3-15	Reserved	Reserved

Table 7.3.1.1.2-18A: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is configured, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank =3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	3	0-2
2	3	3-5
3	2	0,2,3
4	3	0,2,3
3-15	Reserved	Reserved

Table 7.3.1.1.2-19: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank =4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3
1	3	0-3
2-15	Reserved	Reserved

Table 7.3.1.1.2-19A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	3	0-4
1-15	Reserved	Reserved

Table 7.3.1.1.2-19B: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *enhanced-dmrs-Typedmrs-TypeEnh* is not configured, *maxLength*=1, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	3	0-5
1-15	Reserved	Reserved

Table 7.3.1.1.2-20: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank=1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1
1	1	1	1
2	2	0	1
3	2	1	1
4	2	2	1
5	2	3	1
6	3	0	1
7	3	1	1
8	3	2	1
9	3	3	1
10	3	4	1
11	3	5	1
12	3	0	2
13	3	1	2
14	3	2	2
15	3	3	2
16	3	4	2
17	3	5	2
18	3	6	2
19	3	7	2
20	3	8	2
21	3	9	2
22	3	10	2
23	3	11	2
24	1	0	2
25	1	1	2
26	1	6	2
27	1	7	2
28-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-21: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank=2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0,1	1
1	2	0,1	1
2	2	2,3	1
3	3	0,1	1
4	3	2,3	1
5	3	4,5	1
6	2	0,2	1
7	3	0,1	2
8	3	2,3	2
9	3	4,5	2
10	3	6,7	2
11	3	8,9	2
12	3	10,11	2
13	1	0,1	2
14	1	6,7	2
15	2	0,1	2
16	2	2,3	2
17	2	6,7	2
18	2	8,9	2
19-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-22: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is not configured, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank=3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	3	0-2	1
2	3	3-5	1
3	3	0,1,6	2
4	3	2,3,8	2
5	3	4,5,10	2
6-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-22A: Antenna port(s), transform precoder is disabled, *multipanelSchemeSDM* is configured, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank=3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	3	0-2	1
2	3	3-5	1
3	3	0,1,6	2
4	3	2,3,8	2
5	3	4,5,10	2
6	2	0,2,3	1
7	3	0,2,3	1
8-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-23: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank=4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-3	1
1	3	0-3	1
2	3	0,1,6,7	2
3	3	2,3,8,9	2
4	3	4,5,10,11	2
5-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-23A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	3	0-4	1
1	2	0,1,2,3,6	2
12-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-23B Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	3	0-5	1
1	2	0,1,2,3,6,8	2
2-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-23C: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 7

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,6,7,8	2
1-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-23D: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2, rank = 8

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,6,7,8,9	2
1-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-24: SRS request

Value of SRS request field, or value of 'SRS request' index for each cell in the scheduled cell set indicated by SRS request field in DCI 0_3 or 1_3	Triggered aperiodic SRS resource set(s) for DCI format 0_1, 0_2, 0_3, 1_1, 1_2, 1_3, and 2_3 configured with higher layer parameter *srs-TPC-PDCCH-Group* set to 'typeB'	Triggered aperiodic SRS resource set(s) for DCI format 2_3 configured with higher layer parameter *srs-TPC-PDCCH-Group* set to 'typeA'
00	No aperiodic SRS resource set triggered	No aperiodic SRS resource set triggered
01	SRS resource set(s) configured by *SRS-ResourceSet* with higher layer parameter *aperiodicSRS-ResourceTrigger* set to 1 or an entry in *aperiodicSRS-ResourceTriggerList* set to 1 SRS resource set(s) configured by *SRS-PosResourceSet* with an entry in *aperiodicSRS-ResourceTriggerList* set to 1 when triggered by DCI formats 0_1, 0_2, 0_3, 1_1, 1_2 and 1_3	SRS resource set(s) configured with higher layer parameter *usage* in *SRS-ResourceSet* set to '*antennaSwitching*' and *resourceType* in *SRS-ResourceSet* set to 'aperiodic' for a 1st set of serving cells configured by higher layers
10	SRS resource set(s) configured by *SRS-ResourceSet* with higher layer parameter *aperiodicSRS-ResourceTrigger* set to 2 or an entry in *aperiodicSRS-ResourceTriggerList* set to 2 SRS resource set(s) configured by *SRS-PosResourceSet* with an entry in *aperiodicSRS-ResourceTriggerList* set to 2 when triggered by DCI formats 0_1, 0_2, 0_3, 1_1, 1_2 and 1_3	SRS resource set(s) configured with higher layer parameter *usage* in *SRS-ResourceSet* set to '*antennaSwitching*' and *resourceType* in *SRS-ResourceSet* set to 'aperiodic' for a 2nd set of serving cells configured by higher layers
11	SRS resource set(s) configured by *SRS-ResourceSet* with higher layer parameter *aperiodicSRS-ResourceTrigger* set to 3 or an entry in *aperiodicSRS-ResourceTriggerList* set to 3 SRS resource set(s) configured by *SRS-PosResourceSet* with an entry in *aperiodicSRS-ResourceTriggerList* set to 3 when triggered by DCI formats 0_1, 0_2, 0_3, 1_1, 1_2 and 1_3	SRS resource set(s) configured with higher layer parameter *usage* in *SRS-ResourceSet* set to '*antennaSwitching*' and *resourceType* in *SRS-ResourceSet* set to 'aperiodic' for a 3rd set of serving cells configured by higher layers

Table 7.3.1.1.2-25: PTRS-DMRS association or Second PTRS-DMRS association for UL PTRS port 0

Value	DMRS port
0	1st scheduled DMRS port
1	2nd scheduled DMRS port
2	3rd scheduled DMRS port
3	4th scheduled DMRS port

Table 7.3.1.1.2-25A: PTRS-DMRS association for UL PTRS port 0 or for the actual UL PT-RS port if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, or PTRS-DMRS association for UL PTRS port 0 and 1 if *multipanelSchemeSDM* is configured and *maxNrofPorts-SDM* is set to 2

Value of MSB	DMRS port		Value of LSB	DMRS port
0	1st scheduled DMRS port corresponding to SRS resource indicator field and/or Precoding information and number of layers field		0	1st scheduled DMRS port corresponding to Second SRS resource indicator field and/or Second Precoding information field
1	2nd scheduled DMRS port corresponding to SRS resource indicator field and/or Precoding information and number of layers field		1	2nd scheduled DMRS port corresponding to Second SRS resource indicator field and/or Second Precoding information field

Table 7.3.1.1.2-25B: PTRS-DMRS association for UL PTRS port 0, *maxRank*>4 or *maxMIMO-Layers*>4

Value	DMRS port
0	1st scheduled DMRS port corresponding to the selected Codeword
1	2nd scheduled DMRS port corresponding to the selected Codeword
2	3rd scheduled DMRS port corresponding to the selected Codeword
3	4th scheduled DMRS port corresponding to the selected Codeword

Table 7.3.1.1.2-26: PTRS-DMRS association or Second PTRS-DMRS association for UL PTRS ports 0 and 1

Value of MSB	DMRS port		Value of LSB	DMRS port
0	1st DMRS port which shares PTRS port 0		0	1st DMRS port which shares PTRS port 1
1	2nd DMRS port which shares PTRS port 0		1	2nd DMRS port which shares PTRS port 1

Table 7.3.1.1.2-26A: PTRS-DMRS association for UL PTRS ports 0 and 1, *maxRank*>4 or *maxMIMO-Layers*>4

Value of 2 MSBs	DMRS port	Value of 2 LSBs	DMRS port
0	1st DMRS port which shares PTRS port 0	0	1st DMRS port which shares PTRS port 1
1	2nd DMRS port which shares PTRS port 0	1	2nd DMRS port which shares PTRS port 1
2	3rd DMRS port which shares PTRS port 0	2	3rd DMRS port which shares PTRS port 1
3	4th DMRS port which shares PTRS port 0	3	4th DMRS port which shares PTRS port 1

Table 7.3.1.1.2-26B: PTRS-DMRS association or Second PTRS-DMRS association for the actual UL PTRS port

Value	DMRS port
0	1st DMRS port which shares the actual PTRS port
1	2nd DMRS port which shares the actual PTRS port
Note: - The actual PTRS port is PTRS port 0 if *txConfig = codebook* - The actual PTRS port is the PTRS port associated with more than one SRS resources indidcated by SRS resource indicator, if *txConfig = nonCodeBook*	

Table 7.3.1.1.2-27: void

Table 7.3.1.1.2-28: SRI indication or Second SRI indication, for non-codebook based PUSCH transmission, $L_{\max}=1$

Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=2$	Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=3$	Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=4$
0	0	0	0	0	0
1	1	1	1	1	1
		2	2	2	2
		3	reserved	3	3

Table 7.3.1.1.2-28A: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=1$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5-7	reserved	5	5	5	5	5	5
		6-7	reserved	6	6	6	6
				7	reserved	7	7

Table 7.3.1.1.2-29: SRI indication or Second SRI indication, for non-codebook based PUSCH transmission, $L_{\max}=2$

Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=2$	Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=3$	Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=4$
0	0	0	0	0	0
1	1	1	1	1	1
2	0,1	2	2	2	2
3	reserved	3	0,1	3	3
		4	0,2	4	0,1
		5	1,2	5	0,2
		6-7	reserved	6	0,3
				7	1,2
				8	1,3
				9	2,3
				10-15	reserved

Table 7.3.1.1.2-29A: Second SRI indication for non-codebook based PUSCH transmission, $L_{max}=2$

Bit field mapped to index	SRI(s), $N_{\text{SRS}}=2$	Bit field mapped to index	SRI(s), $N_{\text{SRS}}=3$	Bit field mapped to index	SRI(s), $N_{\text{SRS}}=4$
0	0	0	0	0	0
1	1	1	1	1	1
0	0,1	2	2	2	2
1	2 layers: reserved	3	1 layer: reserved	3	3
		0	0,1	4-7	1 layer: reserved
		1	0,2	0	0,1
		2	1,2	1	0,2
		3	2 layers: reserved	2	0,3
				3	1,2
				4	1,3
				5	2,3
				6-7	2 layers: reserved

Table 7.3.1.1.2-29B: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=2$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	reserved	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
		21-31	reserved	27	s_0^{20}, s_1^{20}	…	…
				28-31	reserved	35	s_0^{27}, s_1^{27}
						36-63	reserved
where SRIs $\left(s_{0,}^{B} s_{1,\ldots,}^{B} s_{\upsilon\text{-}1}^{B}\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)\text{-}1\right\}$ are as given in Table 7.3.1.1.2-29B-1.							

Table 7.3.1.1.2-29B-1: SRI(s) for 2 layers, $N_{SRS}>4$

SRI(s)	SRI(s) for 2 layers, $N_{SRS}=5$	SRI(s) for 2 layers, $N_{SRS}=6$	SRI(s) for 2 layers, $N_{SRS}=7$	SRI(s) for 2 layers, $N_{SRS}=8$
s_0^0, s_1^0	0,1	0,1	0,1	0,1
s_0^1, s_1^1	0,2	0,2	0,2	0,2
s_0^2, s_1^2	0,3	0,3	0,3	0,3
s_0^3, s_1^3	0,4	0,4	0,4	0,4
s_0^4, s_1^4	1,2	0,5	0,5	0,5
s_0^5, s_1^5	1,3	1,2	0,6	0,6
s_0^6, s_1^6	1,4	1,3	1,2	0,7
s_0^7, s_1^7	2,3	1,4	1,3	1,2
s_0^8, s_1^8	2,4	1,5	1,4	1,3
s_0^9, s_1^9	3,4	2,3	1,5	1,4
s_0^{10}, s_1^{10}		2,4	1,6	1,5
s_0^{11}, s_1^{11}		2,5	2,3	1,6
s_0^{12}, s_1^{12}		3,4	2,4	1,7
s_0^{13}, s_1^{13}		3,5	2,5	2,3
s_0^{14}, s_1^{14}		4,5	2,6	2,4
s_0^{15}, s_1^{15}			3,4	2,5
s_0^{16}, s_1^{16}			3,5	2,6
s_0^{17}, s_1^{17}			3,6	2,7
s_0^{18}, s_1^{18}			4,5	3,4
s_0^{19}, s_1^{19}			4,6	3,5
s_0^{20}, s_1^{20}			5,6	3,6
s_0^{21}, s_1^{21}				3,7
s_0^{22}, s_1^{22}				4,5
s_0^{23}, s_1^{23}				4,6
s_0^{24}, s_1^{24}				4,7
s_0^{25}, s_1^{25}				5,6
s_0^{26}, s_1^{26}				5,7
s_0^{27}, s_1^{27}				6,7

Table 7.3.1.1.2-30: SRI indication for non-codebook based PUSCH transmission, $L_{\max}=3$

Bit field mapped to index	SRI(s), $N_{SRS}=2$	Bit field mapped to index	SRI(s), $N_{SRS}=3$	Bit field mapped to index	SRI(s), $N_{SRS}=4$
0	0	0	0	0	0
1	1	1	1	1	1
2	0,1	2	2	2	2
3	reserved	3	0,1	3	3
		4	0,2	4	0,1
		5	1,2	5	0,2
		6	0,1,2	6	0,3
		7	reserved	7	1,2
				8	1,3
				9	2,3
				10	0,1,2
				11	0,1,3
				12	0,2,3
				13	1,2,3
				14-15	reserved

Table 7.3.1.1.2-30A: Second SRI indication for non-codebook based PUSCH transmission, if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, $L_{max}=3$

Bit field mapped to index	SRI(s), $N_{SRS}=2$	Bit field mapped to index	SRI(s), $N_{SRS}=3$	Bit field mapped to index	SRI(s), $N_{SRS}=4$
0	0	0	0	0	0
1	1	1	1	1	1
0	0,1	2	2	2	2
1	2 layers: reserved	3	1 layer: reserved	3	3
		0	0,1	4-7	1 layer: reserved
		1	0,2	0	0,1
		2	1,2	1	0,2
		3	2 layers: reserved	2	0,3
		0	0,1,2	3	1,2
		1-3	3 layers: reserved	4	1,3
				5	2,3
				6-7	2 layers: reserved
				0	0,1,2
				1	0,1,3
				2	0,2,3
				3	1,2,3
				4-7	3 layers: reserved

Table 7.3.1.1.2-30B: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=3$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	s_0^0, s_1^0, s_2^0	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
…	…	21	s_0^0, s_1^0, s_2^0	27	s_0^{20}, s_1^{20}	…	…
24	s_0^9, s_1^9, s_2^9	…	…	28	s_0^0, s_1^0, s_2^0	35	s_0^{27}, s_1^{27}
25-31	reserved	40	$s_0^{19}, s_1^{19}, s_2^{19}$	…	…	36	s_0^0, s_1^0, s_2^0
		41-63	reserved	62	$s_0^{34}, s_1^{34}, s_2^{34}$	…	…
				63-127	reserved	91	$s_0^{55}, s_1^{55}, s_2^{55}$
						92-127	reserved

where SRIs $\left(s_{0,}^{B} s_{1,\ldots,}^{B} s_{\upsilon-1}^{B}\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)-1\right\}$ are as given in Table 7.3.1.1.2-29B-1 and Table 7.3.1.1.2-30B-1.

Table 7.3.1.1.2-30B-1: SRI combinations for 3 layers, $N_{SRS}>4$

	SRI(s) for 3 layers, $N_{SRS}=5$	SRI(s) for 3 layers, $N_{SRS}=6$	SRI(s) for 3 layers, $N_{SRS}=7$	SRI(s) for 3 layers, $N_{SRS}=8$
s_0^0, s_1^0, s_2^0	0,1,2	0,1,2	0,1,2	0,1,2
s_0^1, s_1^1, s_2^1	0,1,3	0,1,3	0,1,3	0,1,3
s_0^2, s_1^2, s_2^2	0,1,4	0,1,4	0,1,4	0,1,4
s_0^3, s_1^3, s_2^3	0,2,3	0,1,5	0,1,5	0,1,5
s_0^4, s_1^4, s_2^4	0,2,4	0,2,3	0,1,6	0,1,6
s_0^5, s_1^5, s_2^5	0,3,4	0,2,4	0,2,3	0,1,7
s_0^6, s_1^6, s_2^6	1,2,3	0,2,5	0,2,4	0,2,3
s_0^7, s_1^7, s_2^7	1,2,4	0,3,4	0,2,5	0,2,4
s_0^8, s_1^8, s_2^8	1,3,4	0,3,5	0,2,6	0,2,5
s_0^9, s_1^9, s_2^9	2,3,4	0,4,5	0,3,4	0,2,6
$s_0^{10}, s_1^{10}, s_2^{10}$		1,2,3	0,3,5	0,2,7
$s_0^{11}, s_1^{11}, s_2^{11}$		1,2,4	0,3,6	0,3,4
$s_0^{12}, s_1^{12}, s_2^{12}$		1,2,5	0,4,5	0,3,5
$s_0^{13}, s_1^{13}, s_2^{13}$		1,3,4	0,4,6	0,3,6
$s_0^{14}, s_1^{14}, s_2^{14}$		1,3,5	0,5,6	0,3,7
$s_0^{15}, s_1^{15}, s_2^{15}$		1,4,5	1,2,3	0,4,5
$s_0^{16}, s_1^{16}, s_2^{16}$		2,3,4	1,2,4	0,4,6
$s_0^{17}, s_1^{17}, s_2^{17}$		2,3,5	1,2,5	0,4,7

	SRI(s) for 3 layers, $N_{SRS}=5$	SRI(s) for 3 layers, $N_{SRS}=6$	SRI(s) for 3 layers, $N_{SRS}=7$	SRI(s) for 3 layers, $N_{SRS}=8$
$s_0^{18}, s_1^{18}, s_2^{18}$		2,4,5	1,2,6	0,5,6
$s_0^{19}, s_1^{19}, s_2^{19}$		3,4,5	1,3,4	0,5,7
$s_0^{20}, s_1^{20}, s_2^{20}$			1,3,5	0,6,7
$s_0^{21}, s_1^{21}, s_2^{21}$			1,3,6	1,2,3
$s_0^{22}, s_1^{22}, s_2^{22}$			1,4,5	1,2,4
$s_0^{23}, s_1^{23}, s_2^{23}$			1,4,6	1,2,5
$s_0^{24}, s_1^{24}, s_2^{24}$			1,5,6	1,2,6
$s_0^{25}, s_1^{25}, s_2^{25}$			2,3,4	1,2,7
$s_0^{26}, s_1^{26}, s_2^{26}$			2,3,5	1,3,4
$s_0^{27}, s_1^{27}, s_2^{27}$			2,3,6	1,3,5
$s_0^{28}, s_1^{28}, s_2^{28}$			2,4,5	1,3,6
$s_0^{29}, s_1^{29}, s_2^{29}$			2,4,6	1,3,7
$s_0^{30}, s_1^{30}, s_2^{30}$			2,5,6	1,4,5
$s_0^{31}, s_1^{31}, s_2^{31}$			3,4,5	1,4,6
$s_0^{32}, s_1^{32}, s_2^{32}$			3,4,6	1,4,7
$s_0^{33}, s_1^{33}, s_2^{33}$			3,5,6	1,5,6
$s_0^{34}, s_1^{34}, s_2^{34}$			4,5,6	1,5,7
$s_0^{35}, s_1^{35}, s_2^{35}$				1,6,7
$s_0^{36}, s_1^{36}, s_2^{36}$				2,3,4
$s_0^{37}, s_1^{37}, s_2^{37}$				2,3,5
$s_0^{38}, s_1^{38}, s_2^{38}$				2,3,6
$s_0^{39}, s_1^{39}, s_2^{39}$				2,3,7
$s_0^{40}, s_1^{40}, s_2^{40}$				2,4,5
$s_0^{41}, s_1^{41}, s_2^{41}$				2,4,6
$s_0^{42}, s_1^{42}, s_2^{42}$				2,4,7
$s_0^{43}, s_1^{43}, s_2^{43}$				2,5,6
$s_0^{44}, s_1^{44}, s_2^{44}$				2,5,7
$s_0^{45}, s_1^{45}, s_2^{45}$				2,6,7
$s_0^{46}, s_1^{46}, s_2^{46}$				3,4,5
$s_0^{47}, s_1^{47}, s_2^{47}$				3,4,6
$s_0^{48}, s_1^{48}, s_2^{48}$				3,4,7
$s_0^{49}, s_1^{49}, s_2^{49}$				3,5,6
$s_0^{50}, s_1^{50}, s_2^{50}$				3,5,7
$s_0^{51}, s_1^{51}, s_2^{51}$				3,6,7
$s_0^{52}, s_1^{52}, s_2^{52}$				4,5,6
$s_0^{53}, s_1^{53}, s_2^{53}$				4,5,7
$s_0^{54}, s_1^{54}, s_2^{54}$				4,6,7
$s_0^{55}, s_1^{55}, s_2^{55}$				5,6,7

Table 7.3.1.1.2-31: SRI indication for non-codebook based PUSCH transmission, $L_{\max}=4$

Bit field mapped to index	SRI(s), $N_{SRS}=2$	Bit field mapped to index	SRI(s), $N_{SRS}=3$	Bit field mapped to index	SRI(s), $N_{SRS}=4$
0	0	0	0	0	0
1	1	1	1	1	1
2	0,1	2	2	2	2
3	reserved	3	0,1	3	3
		4	0,2	4	0,1
		5	1,2	5	0,2
		6	0,1,2	6	0,3
		7	reserved	7	1,2
				8	1,3
				9	2,3
				10	0,1,2
				11	0,1,3
				12	0,2,3
				13	1,2,3
				14	0,1,2,3
				15	reserved

Table 7.3.1.1.2-31A: Second SRI indication for non-codebook based PUSCH transmission, if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, $L_{max}=4$

Bit field mapped to index	SRI(s), $N_{SRS}=2$	Bit field mapped to index	SRI(s), $N_{SRS}=3$	Bit field mapped to index	SRI(s), $N_{SRS}=4$
0	0	0	0	0	0
1	1	1	1	1	1
0	0,1	2	2	2	2
1	2 layers: reserved	3	1 layer: reserved	3	3
		0	0,1	4-7	1 layer: reserved
		1	0,2	0	0,1
		2	1,2	1	0,2
		3	2 layers: reserved	2	0,3
		0	0,1,2	3	1,2
		1-3	3 layers: reserved	4	1,3
				5	2,3
				6-7	2 layers: reserved
				0	0,1,2
				1	0,1,3
				2	0,2,3
				3	1,2,3
				4-7	3 layer: reserved
				0	0,1,2,3
				1-7	4 layers: reserved

Table 7.3.1.1.2-31B: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=4$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	s_0^0, s_1^0, s_2^0	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
…	…	21	s_0^0, s_1^0, s_2^0	27	s_0^{20}, s_1^{20}	…	…
24	s_0^9, s_1^9, s_2^9	…	…	28	s_0^0, s_1^0, s_2^0	35	s_0^{27}, s_1^{27}
25	$s_0^0, s_1^0, s_2^0, s_3^0$	40	$s_0^{19}, s_1^{19}, s_2^{19}$	…	…	36	s_0^0, s_1^0, s_2^0
…	…	41	$s_0^0, s_1^0, s_2^0, s_3^0$	62	$s_0^{34}, s_1^{34}, s_2^{34}$	…	…
29	$s_0^4, s_1^4, s_2^4, s_3^4$	…	…	63	$s_0^0, s_1^0, s_2^0, s_3^0$	91	$s_0^{55}, s_1^{55}, s_2^{55}$
30-31	reserved	55	$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}$	…	…	92	$s_0^0, s_1^0, s_2^0, s_3^0$
		56-63	reserved	97	$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}$	…	…
				98-127	reserved	161	$s_0^{69}, s_1^{69}, s_2^{69}, s_3^6$
						162-255	reserved
where SRIs $\left(s_{0,}^B s_{1,\ldots,}^B s_{\upsilon-1}^B\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)-1\right\}$ are as given in Table 7.3.1.1.2-29B-1, Table 7.3.1.1.2-30B-1 and Table 7.3.1.1.2-31B-1.							

Table 7.3.1.1.2-31B-1: SRI combinations for 4 layers, $N_{SRS}>4$

	SRI(s) for 4 layers, $N_{SRS}=5$	SRI(s) for 4 layers, $N_{SRS}=6$	SRI(s) for 4 layers, $N_{SRS}=7$	SRI(s) for 4 layers, $N_{SRS}=8$
$s_0^0, s_1^0, s_2^0, s_3^0$	0,1,2,3	0,1,2,3	0,1,2,3	0,1,2,3
$s_0^1, s_1^1, s_2^1, s_3^1$	0,1,2,4	0,1,2,4	0,1,2,4	0,1,2,4
$s_0^2, s_1^2, s_2^2, s_3^2$	0,1,3,4	0,1,2,5	0,1,2,5	0,1,2,5
$s_0^3, s_1^3, s_2^3, s_3^3$	0,2,3,4	0,1,3,4	0,1,2,6	0,1,2,6
$s_0^4, s_1^4, s_2^4, s_3^4$	1,2,3,4	0,1,3,5	0,1,3,4	0,1,2,7
$s_0^5, s_1^5, s_2^5, s_3^5$		0,1,4,5	0,1,3,5	0,1,3,4
$s_0^6, s_1^6, s_2^6, s_3^6$		0,2,3,4	0,1,3,6	0,1,3,5
$s_0^7, s_1^7, s_2^7, s_3^7$		0,2,3,5	0,1,4,5	0,1,3,6
$s_0^8, s_1^8, s_2^8, s_3^8$		0,2,4,5	0,1,4,6	0,1,3,7
$s_0^9, s_1^9, s_2^9, s_3^9$		0,3,4,5	0,1,5,6	0,1,4,5
$s_0^{10}, s_1^{10}, s_2^{10}, s_3^{10}$		1,2,3,4	0,2,3,4	0,1,4,6
$s_0^{11}, s_1^{11}, s_2^{11}, s_3^{11}$		1,2,3,5	0,2,3,5	0,1,4,7
$s_0^{12}, s_1^{12}, s_2^{12}, s_3^{12}$		1,2,4,5	0,2,3,6	0,1,5,6
$s_0^{13}, s_1^{13}, s_2^{13}, s_3^{13}$		1,3,4,5	0,2,4,5	0,1,5,7
$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}$		2,3,4,5	0,2,4,6	0,1,6,7

	SRI(s) for 4 layers, $N_{SRS}=5$	SRI(s) for 4 layers, $N_{SRS}=6$	SRI(s) for 4 layers, $N_{SRS}=7$	SRI(s) for 4 layers, $N_{SRS}=8$
$s_0^{15}, s_1^{15}, s_2^{15}, s_3^{15}$			0,2,5,6	0,2,3,4
$s_0^{16}, s_1^{16}, s_2^{16}, s_3^{16}$			0,3,4,5	0,2,3,5
$s_0^{17}, s_1^{17}, s_2^{17}, s_3^{17}$			0,3,4,6	0,2,3,6
$s_0^{18}, s_1^{18}, s_2^{18}, s_3^{18}$			0,3,5,6	0,2,3,7
$s_0^{19}, s_1^{19}, s_2^{19}, s_3^{19}$			0,4,5,6	0,2,4,5
$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}$			1,2,3,4	0,2,4,6
$s_0^{21}, s_1^{21}, s_2^{21}, s_3^{21}$			1,2,3,5	0,2,4,7
$s_0^{22}, s_1^{22}, s_2^{22}, s_3^{22}$			1,2,3,6	0,2,5,6
$s_0^{23}, s_1^{23}, s_2^{23}, s_3^{23}$			1,2,4,5	0,2,5,7
$s_0^{24}, s_1^{24}, s_2^{24}, s_3^{24}$			1,2,4,6	0,2,6,7
$s_0^{25}, s_1^{25}, s_2^{25}, s_3^{25}$			1,2,5,6	0,3,4,5
$s_0^{26}, s_1^{26}, s_2^{26}, s_3^{26}$			1,3,4,5	0,3,4,6
$s_0^{27}, s_1^{27}, s_2^{27}, s_3^{27}$			1,3,4,6	0,3,4,7
$s_0^{28}, s_1^{28}, s_2^{28}, s_3^{28}$			1,3,5,6	0,3,5,6
$s_0^{29}, s_1^{29}, s_2^{29}, s_3^{29}$			1,4,5,6	0,3,5,7
$s_0^{30}, s_1^{30}, s_2^{30}, s_3^{30}$			2,3,4,5	0,3,6,7
$s_0^{31}, s_1^{31}, s_2^{31}, s_3^{31}$			2,3,4,6	0,4,5,6
$s_0^{32}, s_1^{32}, s_2^{32}, s_3^{32}$			2,3,5,6	0,4,5,7
$s_0^{33}, s_1^{33}, s_2^{33}, s_3^{33}$			2,4,5,6	0,4,6,7
$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}$			3,4,5,6	0,5,6,7
$s_0^{35}, s_1^{35}, s_2^{35}, s_3^{35}$				1,2,3,4
$s_0^{36}, s_1^{36}, s_2^{36}, s_3^{36}$				1,2,3,5
$s_0^{37}, s_1^{37}, s_2^{37}, s_3^{37}$				1,2,3,6
$s_0^{38}, s_1^{38}, s_2^{38}, s_3^{38}$				1,2,3,7
$s_0^{39}, s_1^{39}, s_2^{39}, s_3^{39}$				1,2,4,5
$s_0^{40}, s_1^{40}, s_2^{40}, s_3^{40}$				1,2,4,6
$s_0^{41}, s_1^{41}, s_2^{41}, s_3^{41}$				1,2,4,7
$s_0^{42}, s_1^{42}, s_2^{42}, s_3^{42}$				1,2,5,6
$s_0^{43}, s_1^{43}, s_2^{43}, s_3^{43}$				1,2,5,7
$s_0^{44}, s_1^{44}, s_2^{44}, s_3^{44}$				1,2,6,7
$s_0^{45}, s_1^{45}, s_2^{45}, s_3^{45}$				1,3,4,5
$s_0^{46}, s_1^{46}, s_2^{46}, s_3^{46}$				1,3,4,6
$s_0^{47}, s_1^{47}, s_2^{47}, s_3^{47}$				1,3,4,7
$s_0^{48}, s_1^{48}, s_2^{48}, s_3^{48}$				1,3,5,6
$s_0^{49}, s_1^{49}, s_2^{49}, s_3^{49}$				1,3,5,7
$s_0^{50}, s_1^{50}, s_2^{50}, s_3^{50}$				1,3,6,7
$s_0^{51}, s_1^{51}, s_2^{51}, s_3^{51}$				1,4,5,6
$s_0^{52}, s_1^{52}, s_2^{52}, s_3^{52}$				1,4,5,7
$s_0^{53}, s_1^{53}, s_2^{53}, s_3^{53}$				1,4,6,7

	SRI(s) for 4 layers, $N_{SRS}=5$	SRI(s) for 4 layers, $N_{SRS}=6$	SRI(s) for 4 layers, $N_{SRS}=7$	SRI(s) for 4 layers, $N_{SRS}=8$
$s_0^{54}, s_1^{54}, s_2^{54}, s_3^{54}$				1,5,6,7
$s_0^{55}, s_1^{55}, s_2^{55}, s_3^{55}$				2,3,4,5
$s_0^{56}, s_1^{56}, s_2^{56}, s_3^{56}$				2,3,4,6
$s_0^{57}, s_1^{57}, s_2^{57}, s_3^{57}$				2,3,4,7
$s_0^{58}, s_1^{58}, s_2^{58}, s_3^{58}$				2,3,5,6
$s_0^{59}, s_1^{59}, s_2^{59}, s_3^{59}$				2,3,5,7
$s_0^{60}, s_1^{60}, s_2^{60}, s_3^{60}$				2,3,6,7
$s_0^{61}, s_1^{61}, s_2^{61}, s_3^{61}$				2,4,5,6
$s_0^{62}, s_1^{62}, s_2^{62}, s_3^{62}$				2,4,5,7
$s_0^{63}, s_1^{63}, s_2^{63}, s_3^{63}$				2,4,6,7
$s_0^{64}, s_1^{64}, s_2^{64}, s_3^{64}$				2,5,6,7
$s_0^{65}, s_1^{65}, s_2^{65}, s_3^{65}$				3,4,5,6
$s_0^{66}, s_1^{66}, s_2^{66}, s_3^{66}$				3,4,5,7
$s_0^{67}, s_1^{67}, s_2^{67}, s_3^{67}$				3,4,6,7
$s_0^{68}, s_1^{68}, s_2^{68}, s_3^{68}$				3,5,6,7
$s_0^{69}, s_1^{69}, s_2^{69}, s_3^{69}$				4,5,6,7

Table 7.3.1.1.2-31C: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=5$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	s_0^0, s_1^0, s_2^0	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
…	…	21	s_0^0, s_1^0, s_2^0	27	s_0^{20}, s_1^{20}	…	…
24	s_0^9, s_1^9, s_2^9	…	…	28	s_0^0, s_1^0, s_2^0	35	s_0^{27}, s_1^{27}
25	$s_0^0, s_1^0, s_2^0, s_3^0$	40	$s_0^{19}, s_1^{19}, s_2^{19}$	…	…	36	s_0^0, s_1^0, s_2^0
…	…	41	$s_0^0, s_1^0, s_2^0, s_3^0$	62	$s_0^{34}, s_1^{34}, s_2^{34}$	…	…
29	$s_0^4, s_1^4, s_2^4, s_3^4$	…	…	63	$s_0^0, s_1^0, s_2^0, s_3^0$	91	$s_0^{55}, s_1^{55}, s_2^{55}$
30	0,1,2,3,4	55	$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}$	…	…	92	$s_0^0, s_1^0, s_2^0, s_3^0$
31	reserved	56	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	97	$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}$	…	…
		…	…	98	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	161	$s_0^{69}, s_1^{69}, s_2^{69}, s$
		61	$s_0^5, s_1^5, s_2^5, s_3^5, s_4^5$	…	…	162	$s_0^0, s_1^0, s_2^0, s_3^0, s$
		62-63	reserved	118	$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}, s_4^{20}$	…	…
				119-127	reserved	217	$s_0^{55}, s_1^{55}, s_2^{55}, s$
						218-255	reserved

where SRIs $\left(s_{0,}^{B} s_{1,\ldots,}^{B} s_{\upsilon-1}^{B}\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)-1\right\}$ are as given in Table 7.3.1.1.2-29B-1, Table 7.3.1.1.2-30B-1, Table 7.3.1.1.2-31B-1 and Table 7.3.1.1.2-31C-1.

Table 7.3.1.1.2-31C-1: SRI combinations for 5 layers, $N_{SRS}>4$

	SRI(s) for 5 layers, $N_{SRS}=6$	SRI(s) for 5 layers, $N_{SRS}=7$	SRI(s) for 5 layers, $N_{SRS}=8$
$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	0,1,2,3,4	0,1,2,3,4	0,1,2,3,4
$s_0^1, s_1^1, s_2^1, s_3^1, s_4^1$	0,1,2,3,5	0,1,2,3,5	0,1,2,3,5
$s_0^2, s_1^2, s_2^2, s_3^2, s_4^2$	0,1,2,4,5	0,1,2,3,6	0,1,2,3,6
$s_0^3, s_1^3, s_2^3, s_3^3, s_4^3$	0,1,3,4,5	0,1,2,4,5	0,1,2,3,7
$s_0^4, s_1^4, s_2^4, s_3^4, s_4^4$	0,2,3,4,5	0,1,2,4,6	0,1,2,4,5
$s_0^5, s_1^5, s_2^5, s_3^5, s_4^5$	1,2,3,4,5	0,1,2,5,6	0,1,2,4,6
$s_0^6, s_1^6, s_2^6, s_3^6, s_4^6$		0,1,3,4,5	0,1,2,4,7
$s_0^7, s_1^7, s_2^7, s_3^7, s_4^7$		0,1,3,4,6	0,1,2,5,6
$s_0^8, s_1^8, s_2^8, s_3^8, s_4^8$		0,1,3,5,6	0,1,2,5,7
$s_0^9, s_1^9, s_2^9, s_3^9, s_4^9$		0,1,4,5,6	0,1,2,6,7
$s_0^{10}, s_1^{10}, s_2^{10}, s_3^{10}, s_4^1$		0,2,3,4,5	0,1,3,4,5
$s_0^{11}, s_1^{11}, s_2^{11}, s_3^{11}, s_4^{11}$		0,2,3,4,6	0,1,3,4,6

	SRI(s) for 5 layers, $N_{SRS}=6$	SRI(s) for 5 layers, $N_{SRS}=7$	SRI(s) for 5 layers, $N_{SRS}=8$
$s_0^{12}, s_1^{12}, s_2^{12}, s_3^{12}, s_4^{1}$		0,2,3,5,6	0,1,3,4,7
$s_0^{13}, s_1^{13}, s_2^{13}, s_3^{13}, s_4^{1}$		0,2,4,5,6	0,1,3,5,6
$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}, s_4^{1}$		0,3,4,5,6	0,1,3,5,7
$s_0^{15}, s_1^{15}, s_2^{15}, s_3^{15}, s_4^{1}$		1,2,3,4,5	0,1,3,6,7
$s_0^{16}, s_1^{16}, s_2^{16}, s_3^{16}, s_4^{1}$		1,2,3,4,6	0,1,4,5,6
$s_0^{17}, s_1^{17}, s_2^{17}, s_3^{17}, s_4^{1}$		1,2,3,5,6	0,1,4,5,7
$s_0^{18}, s_1^{18}, s_2^{18}, s_3^{18}, s_4^{1}$		1,2,4,5,6	0,1,4,6,7
$s_0^{19}, s_1^{19}, s_2^{19}, s_3^{19}, s_4^{1}$		1,3,4,5,6	0,1,5,6,7
$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}, s_4^{2}$		2,3,4,5,6	0,2,3,4,5
$s_0^{21}, s_1^{21}, s_2^{21}, s_3^{21}, s_4^{2}$			0,2,3,4,6
$s_0^{22}, s_1^{22}, s_2^{22}, s_3^{22}, s_4^{2}$			0,2,3,4,7
$s_0^{23}, s_1^{23}, s_2^{23}, s_3^{23}, s_4^{2}$			0,2,3,5,6
$s_0^{24}, s_1^{24}, s_2^{24}, s_3^{24}, s_4^{2}$			0,2,3,5,7
$s_0^{25}, s_1^{25}, s_2^{25}, s_3^{25}, s_4^{2}$			0,2,3,6,7
$s_0^{26}, s_1^{26}, s_2^{26}, s_3^{26}, s_4^{2}$			0,2,4,5,6
$s_0^{27}, s_1^{27}, s_2^{27}, s_3^{27}, s_4^{2}$			0,2,4,5,7
$s_0^{28}, s_1^{28}, s_2^{28}, s_3^{28}, s_4^{2}$			0,2,4,6,7
$s_0^{29}, s_1^{29}, s_2^{29}, s_3^{29}, s_4^{2}$			0,2,5,6,7
$s_0^{30}, s_1^{30}, s_2^{30}, s_3^{30}, s_4^{3}$			0,3,4,5,6
$s_0^{31}, s_1^{31}, s_2^{31}, s_3^{31}, s_4^{3}$			0,3,4,5,7
$s_0^{32}, s_1^{32}, s_2^{32}, s_3^{32}, s_4^{3}$			0,3,4,6,7
$s_0^{33}, s_1^{33}, s_2^{33}, s_3^{33}, s_4^{3}$			0,3,5,6,7
$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}, s_4^{3}$			0,4,5,6,7
$s_0^{35}, s_1^{35}, s_2^{35}, s_3^{35}, s_4^{3}$			1,2,3,4,5
$s_0^{36}, s_1^{36}, s_2^{36}, s_3^{36}, s_4^{3}$			1,2,3,4,6
$s_0^{37}, s_1^{37}, s_2^{37}, s_3^{37}, s_4^{3}$			1,2,3,4,7
$s_0^{38}, s_1^{38}, s_2^{38}, s_3^{38}, s_4^{3}$			1,2,3,5,6
$s_0^{39}, s_1^{39}, s_2^{39}, s_3^{39}, s_4^{3}$			1,2,3,5,7
$s_0^{40}, s_1^{40}, s_2^{40}, s_3^{40}, s_4^{4}$			1,2,3,6,7
$s_0^{41}, s_1^{41}, s_2^{41}, s_3^{41}, s_4^{4}$			1,2,4,5,6
$s_0^{42}, s_1^{42}, s_2^{42}, s_3^{42}, s_4^{4}$			1,2,4,5,7
$s_0^{43}, s_1^{43}, s_2^{43}, s_3^{43}, s_4^{4}$			1,2,4,6,7
$s_0^{44}, s_1^{44}, s_2^{44}, s_3^{44}, s_4^{4}$			1,2,5,6,7
$s_0^{45}, s_1^{45}, s_2^{45}, s_3^{45}, s_4^{4}$			1,3,4,5,6
$s_0^{46}, s_1^{46}, s_2^{46}, s_3^{46}, s_4^{4}$			1,3,4,5,7
$s_0^{47}, s_1^{47}, s_2^{47}, s_3^{47}, s_4^{4}$			1,3,4,6,7
$s_0^{48}, s_1^{48}, s_2^{48}, s_3^{48}, s_4^{4}$			1,3,5,6,7
$s_0^{49}, s_1^{49}, s_2^{49}, s_3^{49}, s_4^{4}$			1,4,5,6,7
$s_0^{50}, s_1^{50}, s_2^{50}, s_3^{50}, s_4^{5}$			2,3,4,5,6

	SRI(s) for 5 layers, $N_{SRS}=6$	SRI(s) for 5 layers, $N_{SRS}=7$	SRI(s) for 5 layers, $N_{SRS}=8$
$s_0^{51}, s_1^{51}, s_2^{51}, s_3^{51}, s_4^{5}$			2,3,4,5,7
$s_0^{52}, s_1^{52}, s_2^{52}, s_3^{52}, s_4^{5}$			2,3,4,6,7
$s_0^{53}, s_1^{53}, s_2^{53}, s_3^{53}, s_4^{5}$			2,3,5,6,7
$s_0^{54}, s_1^{54}, s_2^{54}, s_3^{54}, s_4^{5}$			2,4,5,6,7
$s_0^{55}, s_1^{55}, s_2^{55}, s_3^{55}, s_4^{5}$			3,4,5,6,7

Table 7.3.1.1.2-31D: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=6$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	s_0^0, s_1^0, s_2^0	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
…	…	21	s_0^0, s_1^0, s_2^0	27	s_0^{20}, s_1^{20}	…	…
24	s_0^9, s_1^9, s_2^9	…	…	28	s_0^0, s_1^0, s_2^0	35	s_0^{27}, s_1^{27}
25	$s_0^0, s_1^0, s_2^0, s_3^0$	40	$s_0^{19}, s_1^{19}, s_2^{19}$	…	…	36	s_0^0, s_1^0, s_2^0
…	…	41	$s_0^0, s_1^0, s_2^0, s_3^0$	62	$s_0^{34}, s_1^{34}, s_2^{34}$	…	…
29	$s_0^4, s_1^4, s_2^4, s_3^4$	…	…	63	$s_0^0, s_1^0, s_2^0, s_3^0$	91	$s_0^{55}, s_1^{55}, s_2^{55}$
30	0,1,2,3,4	55	$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}$	…	…	92	$s_0^0, s_1^0, s_2^0, s_3^0$
31	reserved	56	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	97	$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}$	…	…
		…	…	98	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	161	$s_0^{69}, s_1^{69}, s_2^{69}, s_3^{69}$
		61	$s_0^5, s_1^5, s_2^5, s_3^5, s_4^5$	…	…	162	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$
		62	0,1,2,3,4,5	118	$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}, s_4^{20}$	…	…
		63	reserved	119	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s_5^0$	217	$s_0^{55}, s_1^{55}, s_2^{55}, s_3^{55}, s$
				…	…	218	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s$
				125	$s_0^6, s_1^6, s_2^6, s_3^6, s_4^6, s_5^6$	…	…
				126-127	reserved	245	$s_0^{27}, s_1^{27}, s_2^{27}, s_3^{27}, s$
						246-255	reserved

where SRIs $\left(s_{0,}^{B} s_{1,\ldots,}^{B} s_{\upsilon-1}^{B}\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)-1\right\}$ are as given in Table 7.3.1.1.2-29B-1, Table 7.3.1.1.2-30B-1, Table 7.3.1.1.2-31B-1, Table 7.3.1.1.2-31C-1 and Table 7.3.1.1.2-31D-1.

Table 7.3.1.1.2-31D-1: SRI combinations for 6 layers, $N_{SRS}>6$

	SRI(s) for 6 layers, $N_{SRS}=7$	SRI(s) for 6 layers, $N_{SRS}=8$
$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s_5^0$	0,1,2,3,4,5	0,1,2,3,4,5

	SRI(s) for 6 layers, $N_{SRS}=7$	SRI(s) for 6 layers, $N_{SRS}=8$
$s_0^1, s_1^1, s_2^1, s_3^1, s_4^1, s_5^1$	0,1,2,3,4,6	0,1,2,3,4,6
$s_0^2, s_1^2, s_2^2, s_3^2, s_4^2, s_5^2$	0,1,2,3,5,6	0,1,2,3,4,7
$s_0^3, s_1^3, s_2^3, s_3^3, s_4^3, s_5^3$	0,1,2,4,5,6	0,1,2,3,5,6
$s_0^4, s_1^4, s_2^4, s_3^4, s_4^4, s_5^4$	0,1,3,4,5,6	0,1,2,3,5,7
$s_0^5, s_1^5, s_2^5, s_3^5, s_4^5, s_5^5$	0,2,3,4,5,6	0,1,2,3,6,7
$s_0^6, s_1^6, s_2^6, s_3^6, s_4^6, s_5^6$	1,2,3,4,5,6	0,1,2,4,5,6
$s_0^7, s_1^7, s_2^7, s_3^7, s_4^7, s_5^7$		0,1,2,4,5,7
$s_0^8, s_1^8, s_2^8, s_3^8, s_4^8, s_5^8$		0,1,2,4,6,7
$s_0^9, s_1^9, s_2^9, s_3^9, s_4^9, s_5^9$		0,1,2,5,6,7
$s_0^{10}, s_1^{10}, s_2^{10}, s_3^{10}, s_4^1$		0,1,3,4,5,6
$s_0^{11}, s_1^{11}, s_2^{11}, s_3^{11}, s_4^{11}$		0,1,3,4,5,7
$s_0^{12}, s_1^{12}, s_2^{12}, s_3^{12}, s_4^1$		0,1,3,4,6,7
$s_0^{13}, s_1^{13}, s_2^{13}, s_3^{13}, s_4^1$		0,1,3,5,6,7
$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}, s_4^1$		0,1,4,5,6,7
$s_0^{15}, s_1^{15}, s_2^{15}, s_3^{15}, s_4^1$		0,2,3,4,5,6
$s_0^{16}, s_1^{16}, s_2^{16}, s_3^{16}, s_4^1$		0,2,3,4,5,7
$s_0^{17}, s_1^{17}, s_2^{17}, s_3^{17}, s_4^1$		0,2,3,4,6,7
$s_0^{18}, s_1^{18}, s_2^{18}, s_3^{18}, s_4^1$		0,2,3,5,6,7
$s_0^{19}, s_1^{19}, s_2^{19}, s_3^{19}, s_4^1$		0,2,4,5,6,7
$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}, s_4^2$		0,3,4,5,6,7
$s_0^{21}, s_1^{21}, s_2^{21}, s_3^{21}, s_4^2$		1,2,3,4,5,6
$s_0^{22}, s_1^{22}, s_2^{22}, s_3^{22}, s_4^2$		1,2,3,4,5,7
$s_0^{23}, s_1^{23}, s_2^{23}, s_3^{23}, s_4^2$		1,2,3,4,6,7
$s_0^{24}, s_1^{24}, s_2^{24}, s_3^{24}, s_4^2$		1,2,3,5,6,7
$s_0^{25}, s_1^{25}, s_2^{25}, s_3^{25}, s_4^2$		1,2,4,5,6,7
$s_0^{26}, s_1^{26}, s_2^{26}, s_3^{26}, s_4^2$		1,3,4,5,6,7
$s_0^{27}, s_1^{27}, s_2^{27}, s_3^{27}, s_4^2$		2,3,4,5,6,7

Table 7.3.1.1.2-31E: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=7$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	s_0^0, s_1^0, s_2^0	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
…	…	21	s_0^0, s_1^0, s_2^0	27	s_0^{20}, s_1^{20}	…	…
24	s_0^9, s_1^9, s_2^9	…	…	28	s_0^0, s_1^0, s_2^0	35	s_0^{27}, s_1^{27}
25	$s_0^0, s_1^0, s_2^0, s_3^0$	40	$s_0^{19}, s_1^{19}, s_2^{19}$	…	…	36	s_0^0, s_1^0, s_2^0
…	…	41	$s_0^0, s_1^0, s_2^0, s_3^0$	62	$s_0^{34}, s_1^{34}, s_2^{34}$	…	…
29	$s_0^4, s_1^4, s_2^4, s_3^4$	…	…	63	$s_0^0, s_1^0, s_2^0, s_3^0$	91	$s_0^{55}, s_1^{55}, s_2^{55}$
30	0,1,2,3,4	55	$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}$	…	…	92	$s_0^0, s_1^0, s_2^0, s_3^0$
31	reserved	56	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	97	$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}$	…	…
		…	…	98	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	161	$s_0^{69}, s_1^{69}, s_2^{69}, s_3^{69}$
		61	$s_0^5, s_1^5, s_2^5, s_3^5, s_4^5$	…	…	162	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$
		62	0,1,2,3,4,5	118	$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}, s_4^2$	…	…
		63	reserved	119	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s_5^0$	217	$s_0^{55}, s_1^{55}, s_2^{55}, s_3^{55}, s_4^{55}$
				…	…	218	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s_5^0$
				125	$s_0^6, s_1^6, s_2^6, s_3^6, s_4^6, s_5^6$	…	…
				126	0,1,2,3,4,5,6	245	$s_0^{27}, s_1^{27}, s_2^{27}, s_3^{27}, s_4^{27}, s_5^{27}$
				127	reserved	246	0,1,2,3,4,5,6
						247	0,1,2,3,4,5,7
						248	0,1,2,3,4,6,7
						249	0,1,2,3,5,6,7
						250	0,1,2,4,5,6,7
						251	0,1,3,4,5,6,7
						252	0,2,3,4,5,6,7
						253	1,2,3,4,5,6,7
						254-255	reserved
where SRIs $\left(s_{0,}^B s_{1,\ldots,}^B s_{\upsilon\text{-}1}^B\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)\text{-}1\right\}$ are as given in Table 7.3.1.1.2-29B-1, Table 7.3.1.1.2-30B-1, Table 7.3.1.1.2-31B-1, Table 7.3.1.1.2-31C-1 and Table 7.3.1.1.2-31D-1.							

Table 7.3.1.1.2-31F: SRI indication, for non-codebook based PUSCH transmission, $N_{SRS}>4$, $L_{max}=8$

Bit field mapped to index	SRI(s), $N_{SRS}=5$	Bit field mapped to index	SRI(s), $N_{SRS}=6$	Bit field mapped to index	SRI(s), $N_{SRS}=7$	Bit field mapped to index	SRI(s), $N_{SRS}=8$
0	0	0	0	0	0	0	0
1	1	1	1	1	1	1	1
2	2	2	2	2	2	2	2
3	3	3	3	3	3	3	3
4	4	4	4	4	4	4	4
5	s_0^0, s_1^0	5	5	5	5	5	5
…	…	6	s_0^0, s_1^0	6	6	6	6
14	s_0^9, s_1^9	…	…	7	s_0^0, s_1^0	7	7
15	s_0^0, s_1^0, s_2^0	20	s_0^{14}, s_1^{14}	…	…	8	s_0^0, s_1^0
…	…	21	s_0^0, s_1^0, s_2^0	27	s_0^{20}, s_1^{20}	…	…
24	s_0^9, s_1^9, s_2^9	…	…	28	s_0^0, s_1^0, s_2^0	35	s_0^{27}, s_1^{27}
25	$s_0^0, s_1^0, s_2^0, s_3^0$	40	$s_0^{19}, s_1^{19}, s_2^{19}$	…	…	36	s_0^0, s_1^0, s_2^0
…	…	41	$s_0^0, s_1^0, s_2^0, s_3^0$	62	$s_0^{34}, s_1^{34}, s_2^{34}$	…	…
29	$s_0^4, s_1^4, s_2^4, s_3^4$	…	…	63	$s_0^0, s_1^0, s_2^0, s_3^0$	91	$s_0^{55}, s_1^{55}, s_2^{55}$
30	0,1,2,3,4	55	$s_0^{14}, s_1^{14}, s_2^{14}, s_3^{14}$	…	…	92	$s_0^0, s_1^0, s_2^0, s_3^0$
31	reserved	56	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	97	$s_0^{34}, s_1^{34}, s_2^{34}, s_3^{34}$	…	…
		…	…	98	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$	161	$s_0^{69}, s_1^{69}, s_2^{69}, s_3^{69}$
		61	$s_0^5, s_1^5, s_2^5, s_3^5, s_4^5$	…	…	162	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0$
		62	0,1,2,3,4,5	118	$s_0^{20}, s_1^{20}, s_2^{20}, s_3^{20}, s_4^{20}$	…	…
		63	reserved	119	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s_5^0$	217	$s_0^{55}, s_1^{55}, s_2^{55}, s_3^{55}, s_4^{55}$
				…	…	218	$s_0^0, s_1^0, s_2^0, s_3^0, s_4^0, s_5^0$
				125	$s_0^6, s_1^6, s_2^6, s_3^6, s_4^6, s_5^6$	…	…
				126	0,1,2,3,4,5,6	245	$s_0^{27}, s_1^{27}, s_2^{27}, s_3^{27}, s_4^{27}, s_5^{2}$
				127	reserved	246	0,1,2,3,4,5,6
						247	0,1,2,3,4,5,7
						248	0,1,2,3,4,6,7
						249	0,1,2,3,5,6,7
						250	0,1,2,4,5,6,7
						251	0,1,3,4,5,6,7
						252	0,2,3,4,5,6,7
						253	1,2,3,4,5,6,7
						254	0,1,2,3,4,5,6,7
						255	reserved
where SRIs $\left(s_{0,}^{B} s_{1,\ldots,}^{B} s_{\upsilon\text{-}1}^{B}\right)$, $B \in \left\{0,1,\ldots,C\left(N_{SRS,}\upsilon\right)\text{-}1\right\}$ are as given in Table 7.3.1.1.2-29B-1, Table 7.3.1.1.2-30B-1, Table 7.3.1.1.2-31B-1, Table 7.3.1.1.2-31C-1 and Table 7.3.1.1.2-31D-1.							

Table 7.3.1.1.2-32: SRI indication or Second SRI indication, for codebook based PUSCH transmission, if *ul-FullPowerTransmission* is not configured, or *ul-FullPowerTransmission = fullpowerMode1, or ul-FullPowerTransmission = fullpowerMode2, or ul-FullPowerTransmission = fullpower* and $N_{SRS}=2$

Bit field mapped to index	SRI(s), $N_{\mathrm{SRS}}=2$
0	0
1	1

Table 7.3.1.1.2-32A: SRI indication or Second SRI indication, for codebook based PUSCH transmission, if *ul-FullPowerTransmission* = *fullpowerMode2* and $N_{SRS}=3$

Bit field mapped to index	SRI(s), $N_{SRS}=3$
0	0
1	1
2	2
3	Reserved

Table 7.3.1.1.2-32B: SRI indication or Second SRI indication, for codebook based PUSCH transmission, if *ul-FullPowerTransmission* = *fullpowerMode2* and $N_{SRS}=4$

Bit field mapped to index	SRI(s), $N_{SRS}=4$
0	0
1	1
2	2
3	3

Table 7.3.1.1.2-33: Joint indication of minimum applicable scheduling offset K0/K2

Bit field mapped to index	Minimum applicable K0 for the active DL BWP, if *minimumSchedulingOffsetK0* is configured for the DL BWP	Minimum applicable K2 for the active UL BWP, if *minimumSchedulingOffsetK2* is configured for the UL BWP
0	The first value configured by *minimumSchedulingOffsetK0* for the active DL BWP	The first value configured by *minimumSchedulingOffsetK2* for the active UL BWP
1	The second value configured by *minimumSchedulingOffsetK0* for the active DL BWP if the second value is configured; 0 otherwise	The second value configured by *minimumSchedulingOffsetK2* for the active UL BWP if the second value is configured; 0 otherwise

Table 7.3.1.1.2-34: Redundancy version

Value of the Redundancy version field	Value of rv_{id} to be applied
0	0
1	2

Table 7.3.1.1.2-35: Allowed entries for DCI format 0_1/0_3 and DCI format 0_2, configured by higher layer parameter *ul-AccessConfigListDCI-0-1* and *ul-AccessConfigListDCI-0-2*, respectively, in frequency range 1

Entry index	Channel Access Type	The CP extension T_"ext" index defined in Clause 5.3.1 of [4, 38.211]	CAPC
0	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	0	1
1	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	0	2
2	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	0	3
3	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	0	4
4	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	2	1
5	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	2	2
6	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in	2	3

Entry index	Channel Access Type	The CP extension T_"ext" index defined in Clause 5.3.1 of [4, 38.211]	CAPC
	TS 37.213 [14]		
7	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	2	4
8	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	0	1
9	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	0	2
10	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	0	3
11	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	0	4
12	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	2	1
13	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	2	2
14	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	2	3
15	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	2	4
16	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	0	1
17	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	0	2
18	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	0	3
19	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	0	4
20	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	1	1
21	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	1	2
22	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	1	3
23	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	1	4
24	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	3	1
25	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	3	2
26	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	3	3
27	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	3	4
28	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	0	1
29	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	0	2
30	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	0	3
31	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	0	4
32	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	1	1
33	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	1	2
34	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	1	3
35	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	1	4
36	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	2	1
37	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	2	2
38	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS	2	3

Entry index	Channel Access Type	The CP extension T_"ext" index defined in Clause 5.3.1 of [4, 38.211]	CAPC
	37.213 [14]		
39	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	2	4
40	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	3	1
41	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	3	2
42	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	3	3
43	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	3	4

Table 7.3.1.1.2-35A: Allowed entries for DCI format 0_1, DCI format 0_2 and DCI format 0_3, configured by higher layer parameter *ul-AccessConfigListDCI-0-1* in frequency range 2-2

Entry index	Channel Access Type
0	Type 1 channel access defined in clause 4.4.1 of TS 37.213 [14]
1	Type 2 channel access defined in clause 4.4.2 of TS 37.213 [14]
2	Type 3 channel access defined in clause 4.4.3 of TS 37.213 [14]

Table 7.3.1.1.2-36: SRS resource set indication

Bit field mapped to index	SRS resource set indication
0	SRS resource indicator field and Precoding information and number of layers field are associated with the first SRS resource set; Second SRS resource indicator field and Second Precoding information field are reserved; If there are two indicated joint/UL TCI states, the first indicated joint/UL TCI state is applied to the corresponding PUSCH transmission occasions.
1	SRS resource indicator field and Precoding information and number of layers field are associated with the second SRS resource set; Second SRS resource indicator field and Second Precoding information field are reserved; If there are two indicated joint/UL TCI states, the second indicated joint/UL TCI state is applied to the corresponding PUSCH transmission occasions.
2	SRS resource indicator field and Precoding information and number of layers field are associated with the first SRS resource set; Second SRS resource indicator field and Second Precoding information field are associated with the second SRS resource set; If there are two indicated joint/UL TCI states, the first indicated joint/UL TCI state is applied to the PUSCH transmission occasions/antenna ports associated with the first SRS resource set, and the second indicated joint/UL TCI state is applied to the PUSCH transmission occasions/antenna ports associated with the second SRS resource set.
3	SRS resource indicator field and Precoding information and number of layers field are associated with the first SRS resource set; Second SRS resource indicator field and Second Precoding information field are associated with the second SRS resource set; If there are two indicated joint/UL TCI states, the first indicated joint/UL TCI state is applied to the PUSCH transmission occasions associated with the first SRS resource set, and the second indicated joint/UL TCI state is applied to the PUSCH transmission occasions associated with the second SRS resource set. If *multipanelSchemeSDM* or *multipanelSchemeSFN* is configured, this row is reserved.
NOTE 1: The first and the second SRS resource sets are respectively the ones with lower and higher *srs-ResourceSetId* of the two SRS resources sets configured by higher layer parameter *srs-ResourceSetToAddModList* or *srs-ResourceSetToAddModListDCI-0-2*, and associated with the higher layer parameter *usage* of value '*nonCodeBook*' if *txConfig=nonCodebook* or '*codebook*' if *txConfig=codebook*. When only one SRS resource set is configured by higher layer parameter *srs-ResourceSetToAddModList* or *srs-ResourceSetToAddModListDCI-0-2*, and associated with the higher layer parameter usage of value '*codebook*' or '*nonCodeBook*' respectively, the first SRS resource set is the SRS resource set. The association of the first and second SRS resource sets to PUSCH repetitions for each bit field index value is as defined in Clause 6.1.2.1 of TS 38.214 [6]. NOTE 2: For DCI format 0_2, the first and second SRS resource sets configured by higher layer parameter *srs-ResourceSetToAddModListDCI-0-2* are composed of the first $N_{SRS,\,0_2}$ SRS resources together with other configurations in the first and second SRS resource sets configured by higher layer parameter *srs-ResourceSetToAddModList*, if any, and associated with the higher layer parameter *usage* of value '*codebook*' or '*nonCodeBook*', respectively, except for the higher layer parameters *'srs-ResourceSetId' and 'srs-ResourceIdList'.*	

Table 7.3.1.1.2-37: SRS offset indicator

Bit field mapped to index	Available slot offset, K=2	Bit field mapped to index	Available slot offset, K=3	Bit field mapped to index	Available slot offset, K=4
0	The 1st entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise	0	The 1st entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise	0	The 1st entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise
1	The 2nd entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise	1	The 2nd entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise	1	The 2nd entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise
		2	The 3rd entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise	2	The 3rd entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise
		3	Reserved	3	The 4th entry in *availableSlotOffsetList*, if configured for the aperiodic SRS resource set; 0, otherwise

Table 7.3.1.1.2-38: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	2	0
3	2	1
4	2	2
5	2	3
6	1	8
7	1	9
8	2	8
9	2	9
10	2	10
11	2	11
12-15	Reserved	Reserved

Table 7.3.1.1.2-39: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0,1
1	2	0,1
2	2	2,3
3	2	0,2
4	1	8,9
5	2	8,9
6	2	10,11
7	2	9,11
8-15	Reserved	Reserved

Table 7.3.1.1.2-40: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *multipanelSchemeSDM* is not configured, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	2	8-10
2	1	0,1,8
3	2	0,1,8
4	2	2,3,10
5-15	Reserved	Reserved

Table 7.3.1.1.2-40A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *multipanelSchemeSDM* is configured, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	2	8-10
2	1	0,1,8
3	2	0,1,8
4	2	2,3,10
5	2	0,2,3
6-15	Reserved	Reserved

Table 7.3.1.1.2-41: Antenna port(s), transform precoder is disabled, dmrs-Type=1, *dmrs-TypeEnh* is configured, maxLength=1, rank = 4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3
1	2	8-11
2	1	0,1,8,9
3	2	0,1,8,9
4	2	2,3,10,11
5-15	Reserved	Reserved

Table 7.3.1.1.2-42: Antenna port(s), transform precoder is disabled, dmrs-Type=1, *dmrs-TypeEnh* is configured, maxLength=1, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0,1,2,3,8
1-15	Reserved	Reserved

Table 7.3.1.1.2-43: Antenna port(s), transform precoder is disabled, dmrs-Type=1, *dmrs-TypeEnh* is configured, maxLength=1, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0,1,2,3,8,10
1-15	Reserved	Reserved

Table 7.3.1.1.2-44: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 7

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0,1,2,3,8,9,10
1-15	Reserved	Reserved

Table 7.3.1.1.2-45: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 8

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0,1,2,3,8,9,10,11
1-15	Reserved	Reserved

Table 7.3.1.1.2-46: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1
1	1	1	1
2	2	0	1
3	2	1	1
4	2	2	1
5	2	3	1
6	2	0	2
7	2	1	2
8	2	2	2
9	2	3	2
10	2	4	2
11	2	5	2
12	2	6	2
13	2	7	2
14	1	8	1
15	1	9	1
16	2	8	1
17	2	9	1
18	2	10	1
19	2	11	1
20	2	8	2
21	2	9	2
22	2	10	2
23	2	11	2
24	2	12	2
25	2	13	2
26	2	14	2
27	2	15	2
28	1	0	2
29	1	1	2
30	1	8	2
31	1	9	2

Table 7.3.1.1.2-47: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0,1	1
1	2	0,1	1
2	2	2,3	1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
3	2	0,2	1
4	2	0,1	2
5	2	2,3	2
6	2	4,5	2
7	2	6,7	2
8	2	0,4	2
9	2	2,6	2
10	1	8,9	1
11	2	8,9	1
12	2	10,11	1
13	2	8,9	2
14	2	10,11	2
15	2	12,13	2
16	2	14,15	2
17	2	9,11	1
18	2	1,3	1
19	2	0,2	2
20	2	1,3	2
21	2	4,6	2
22	2	5,7	2
23	2	8,10	2
24	2	9,11	2
25	2	12,14	2
26	2	13,15	2
27	1	0,1	2
28	1	8,9	2
29	1	4,5	2
30	1	12,13	2
31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-48: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *multipanelSchemeSDM* is not configured, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	2	0,1,4	2
2	2	2,3,6	2
3	2	9-11	1
4	2	8,9,12	2
5	2	10,11,14	2
6	1	0,1,8	1
7	2	0,1,8	1
8	2	2,3,10	1
9	2	0,1,8	2
10	2	4,5,12	2
11	2	2,3,10	2
12	2	6,7,14	2
13	2	5,8,9	2
14	2	7,10,11	2
15	2	7,12,13	2
16-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-48A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *multipanelSchemeSDM* is configured, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	2	0,1,4	2
2	2	2,3,6	2
3	2	9-11	1
4	2	8,9,12	2

5	2	10,11,14	2
6	1	0,1,8	1
7	2	0,1,8	1
8	2	2,3,10	1
9	2	0,1,8	2
10	2	4,5,12	2
11	2	2,3,10	2
12	2	6,7,14	2
13	2	5,8,9	2
14	2	7,10,11	2
15	2	7,12,13	2
16	2	0,2,3	1
17-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-49: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-3	1
1	2	0,1,4,5	2
2	2	2,3,6,7	2
3	2	0,2,4,6	2
4	2	8-11	1
5	2	8,9,12,13	2
6	2	10,11,14,15	2
7	2	1,3,5,7	2
8	1	0,1,8,9	1
9	2	0,1,8,9	1
10	2	2,3,10,11	1
11	1	0,1,8,9	2
12	1	4,5,12,13	2
13	2	0,1,8,9	2
14	2	4,5,12,13	2
15	2	2,3,10,11	2
16	2	6,7,14,15	2
17-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-50: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-4	2
1	2	0,1,2,3,8	1
2	1	0,1,4,5,8	2
3	2	0,1,4,5,8	2
4-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-51: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,4,6	2
1	2	0,1,2,3,8,10	1
2	1	0,1,4,5,8,12	2
3	2	0,1,4,5,8,12	2
4-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-52: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 7

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,4,5,6	2
1	2	0,1,2,3,8,9,10	1
2	1	0,1,4,5,8,9,12	2
3	2	0,1,4,5,8,9,12	2
4-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-53: Antenna port(s), transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 8

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,4,5,6,7	2
1	2	0,1,2,3,8,9,10,11	1
2	1	0,1,4,5,8,9,12,13	2
3	2	0,1,4,5,8,9,12,13	2
4-31	Reserved	Reserved	Reserved

Table 7.3.1.1.2-54: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0
1	1	1
2	2	0
3	2	1
4	2	2
5	2	3
6	3	0
7	3	1
8	3	2
9	3	3
10	3	4
11	3	5
12	1	12
13	1	13
14	2	12
15	2	13
16	2	14
17	2	15
18	3	12
19	3	13
20	3	14
21	3	15
22	3	16
23	3	17
24-31	Reserved	Reserved

Table 7.3.1.1.2-55: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0,1
1	2	0,1
2	2	2,3
3	3	0,1

4	3	2,3
5	3	4,5
6	2	0,2
7	1	12,13
8	2	12,13
9	2	14,15
10	3	12,13
11	3	14,15
12	3	16,17
13	3	13,15
14	2	13,15
15-31	Reserved	Reserved

Table 7.3.1.1.2-56: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *multipanelSchemeSDM* is not configured, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	3	0-2
2	3	3-5
3	1	0,1,12
4	2	0,1,12
5	2	2,3,14
6	3	0,1,12
7	3	2,3,14
8	3	4,5,16
9	3	13,15,17
10-31	Reserved	Reserved

Table 7.3.1.1.2-56A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *multipanelSchemeSDM* is configured, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-2
1	3	0-2
2	3	3-5
3	1	0,1,12
4	2	0,1,12
5	2	2,3,14
6	3	0,1,12
7	3	2,3,14
8	3	4,5,16
9	3	13,15,17
10	2	0,2,3
11-31	Reserved	Reserved

Table 7.3.1.1.2-57: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3
1	3	0-3
2	1	0,1,12,13
3	2	0,1,12,13
4	2	2,3,14,15
5	3	0,1,12,13
6	3	2,3,14,15
7	3	4,5,16,17
8-31	Reserved	Reserved

Table 7.3.1.1.2-58: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	3	0-4
1	2	0,1,2,3,12
2	3	0,1,2,3,12
3-31	Reserved	Reserved

Table 7.3.1.1.2-59: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	3	0-5
1	2	0,1,2,3,12,14
2	3	0,1,2,3,12,14
3-31	Reserved	Reserved

Table 7.3.1.1.2-60: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 7

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3,12-14
1	3	0-3,12-14
2-31	Reserved	Reserved

Table 7.3.1.1.2-61: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1, rank = 8

Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	2	0-3,12-15
1	3	0-3,12-15
2-31	Reserved	Reserved

Table 7.3.1.1.2-62: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 1

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1
1	1	1	1
2	2	0	1
3	2	1	1
4	2	2	1
5	2	3	1
6	3	0	1
7	3	1	1
8	3	2	1
9	3	3	1
10	3	4	1
11	3	5	1
12	3	0	2
13	3	1	2
14	3	2	2
15	3	3	2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
16	3	4	2
17	3	5	2
18	3	6	2
19	3	7	2
20	3	8	2
21	3	9	2
22	3	10	2
23	3	11	2
24	1	0	2
25	1	1	2
26	1	6	2
27	1	7	2
28	1	12	1
29	1	13	1
30	2	12	1
31	2	13	1
32	2	14	1
33	2	15	1
34	3	12	1
35	3	13	1
36	3	14	1
37	3	15	1
38	3	16	1
39	3	17	1
40	3	12	2
41	3	13	2
42	3	14	2
43	3	15	2
44	3	16	2
45	3	17	2
46	3	18	2
47	3	19	2
48	3	20	2
49	3	21	2
50	3	22	2
51	3	23	2
52	1	12	2
53	1	13	2
54	1	18	2
55	1	19	2
56-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-63: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 2

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0,1	1
1	2	0,1	1
2	2	2,3	1
3	3	0,1	1
4	3	2,3	1
5	3	4,5	1
6	2	0,2	1
7	3	0,1	2
8	3	2,3	2
9	3	4,5	2
10	3	6,7	2
11	3	8,9	2
12	3	10,11	2
13	1	0,1	2
14	1	6,7	2

15	2	0,1	2
16	2	2,3	2
17	2	6,7	2
18	2	8,9	2
19	1	12,13	1
20	2	12,13	1
21	2	14,15	1
22	3	12,13	1
23	3	14,15	1
24	3	16,17	1
25	3	12,13	2
26	3	14,15	2
27	3	16,17	2
28	3	18,19	2
29	3	20,21	2
30	3	22,23	2
31	1	12,13	2
32	1	18,19	2
33	2	12,13	2
34	2	14,15	2
35	2	18,19	2
36	2	20,21	2
37	3	13,15	1
38	2	13,15	1
39-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-64: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *multipanelSchemeSDM* is not configured, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	3	0-2	1
2	3	3-5	1
3	3	0,1,6	2
4	3	2,3,8	2
5	3	4,5,10	2
6	1	0,1,12	1
7	2	0,1,12	1
8	2	2,3,14	1
9	3	0,1,12	1
10	3	2,3,14	1
11	3	4,5,16	1
12	3	7,12,13	2
13	3	9,14,15	2
14	3	11,16,17	2
15	3	9,18,19	2
16	3	18,19,20	2
17	3	21,22,23	2
18	3	13,15,17	1
19-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-64A: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *multipanelSchemeSDM* is configured, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 3

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-2	1
1	3	0-2	1
2	3	3-5	1
3	3	0,1,6	2
4	3	2,3,8	2
5	3	4,5,10	2

6	1	0,1,12	1
7	2	0,1,12	1
8	2	2,3,14	1
9	3	0,1,12	1
10	3	2,3,14	1
11	3	4,5,16	1
12	3	7,12,13	2
13	3	9,14,15	2
14	3	11,16,17	2
15	3	9,18,19	2
16	3	18,19,20	2
17	3	21,22,23	2
18	3	13,15,17	1
19	2	0,2,3	1
20-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-65: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 4

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0-3	1
1	3	0-3	1
2	3	0,1,6,7	2
3	3	2,3,8,9	2
4	3	4,5,10,11	2
5	3	12,13,18,19	2
6	3	14,15,20,21	2
7	3	16,17,22,23	2
8	1	0,1,12,13	1
9	2	0,1,12,13	1
10	2	2,3,14,15	1
11	3	0,1,12,13	1
12	3	2,3,14,15	1
13	3	4,5,16,17	1
14-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-66: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 5

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	3	0-4	1
1	2	0,1,2,3,6	2
2	2	0,1,2,3,12	1
3	3	0,1,2,3,12	1
4	1	0,1,6,7,12	2
5	2	0,1,6,7,12	2
6	3	0,1,6,7,12	2
7-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-67: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 6

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	3	0-5	1
1	2	0,1,2,3,6,8	2
2	2	0-3,12,14	1
3	3	0-3,12,14	1
4	1	0,1,6,7,12,18	2
5	2	0,1,6,7,12,18	2

6	3	0,1,6,7,12,18	2
7-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-68: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 7

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,6,7,8	2
1	2	0-3,12-14	1
2	3	0-3,12-14	1
3	1	0,1,6,7,12,13,18	2
4	2	0,1,6,7,12,13,18	2
5	3	0,1,6,7,12,13,18	2
6-63	Reserved	Reserved	Reserved

Table 7.3.1.1.2-69: Antenna port(s), transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2, rank = 8

Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	2	0,1,2,3,6,7,8,9	2
1	2	0-3,12-15	1
2	3	0-3,12-15	1
3	1	0,1,6,7,12,13,18,19	2
4	2	0,1,6,7,12,13,18,19	2
5	3	0,1,6,7,12,13,18,19	2
6-63	Reserved	Reserved	Reserved

7.3.1.1.3 Format 0_2

DCI format 0_2 is used for the scheduling of PUSCH in one cell.

The following information is transmitted by means of the DCI format 0_2 with CRC scrambled by C-RNTI or CS-RNTI or SP-CSI-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bit
  - The value of this bit field is always set to 0, indicating an UL DCI format
- Carrier indicator - 0, 1, 2 or 3 bits determined by higher layer parameter *carrierIndicatorSizeDCI-0-2*, as defined in Clause 10.1 of [5, TS38.213]. This field is reserved when this format is carried by PDCCH on the primary cell and the UE is configured for scheduling on the primary cell from an SCell, with the same number of bits as that in this format carried by PDCCH on the SCell for scheduling on the primary cell.
- UL/SUL indicator - 0 bit for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell or UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell but only one carrier in the cell is configured for PUSCH transmission; otherwise, 1 bit as defined in Table 7.3.1.1.1-1.
- Bandwidth part indicator - 0, 1 or 2 bits as determined by the number of UL BWPs $n_{BWP,\,RRC}$ configured by higher layers, excluding the initial UL bandwidth part. The bitwidth for this field is determined as $\lceil \log_2(n_{BWP}) \rceil$ bits, where
  - $n_{BWP}=n_{BWP,\,RRC}+1$ if $n_{BWP,\,RRC}\leq 3$, in which case the bandwidth part indicator is equivalent to the ascending order of the higher layer parameter *BWP-Id*;
  - otherwise $n_{BWP}=n_{BWP,\,RRC}$, in which case the bandwidth part indicator is defined in Table 7.3.1.1.2-1;

  If a UE does not support active BWP change via DCI, the UE ignores this bit field.
- Frequency domain resource assignment - number of bits determined by the following:
  - N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 6.1.2.2.1 of [6, TS 38.214]

- $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil$ bits if only resource allocation type 1 is configured, or $max\left(\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil,\ N_{RBG}\right)+1$ bits if *resourceAllocationDCI-0-2-r16* is configured as '*dynamicSwitch*', where $N_{RBG,\,K1}=\lceil \left(N_{RB}^{UL,\,BWP}+\left(N_{UL,\,BWP}^{start} mod\, K1\right)\right)/K1 \rceil$, $N_{RB}^{UL,\,BWP}$ is the size of the active UL bandwidth part, $N_{UL,\,BWP}^{start}$ is defined as in clause 4.4.4.4 of [4, TS 38.211] and $K1$ is given by higher layer parameter *resourceAllocationType1GranularityDCI-0-2*. If the higher layer parameter *resourceAllocationType1GranularityDCI-0-2* is not configured, $K1$ is equal to 1.
- If *resourceAllocationDCI-0-2-r16* is configured as '*dynamicSwitch*', the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
- For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 6.1.2.2.1 of [6, TS 38.214].
- For resource allocation type 1, the $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil$ LSBs provide the resource allocation as follows:
  - For PUSCH hopping with resource allocation type 1:
    - N_{UL_hop} MSB bits are used to indicate the frequency offset according to Clause 6.3 of [6, TS 38.214], where $N_{UL_hop}=1$ if the higher layer parameter *frequencyHoppingOffsetListsDCI-0-2* contains two offset values and $N_{UL_hop}=2$ if the higher layer parameter *frequencyHoppingOffsetListsDCI-0-2* contains four offset values
    - $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil - N_{UL_hop}$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
  - For non-PUSCH hopping with resource allocation type 1:
    - $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and if *resourceAllocationDCI-0-2-r16* is configured as '*dynamicSwitch*' for the indicated bandwidth part, the UE assumes resource allocation type 0 for the indicated bandwidth part if the bitwidth of the "Frequency domain resource assignment" field of the active bandwidth part is smaller than the bitwidth of the "Frequency domain resource assignment" field of the indicated bandwidth part.
- Time domain resource assignment - 0, 1, 2, 3, 4, 5 or 6 bits as defined in Clause 6.1.2.1 of [6, TS38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *pusch-TimeDomainAllocationListDCI-0-2* if the higher layer parameter is configured, or I is the number of entries in the higher layer parameter *PUSCH-TimeDomainResourceAllocationList* if the higher layer parameter *PUSCH-TimeDomainResourceAllocationList* is configured and the higher layer parameter *pusch-TimeDomainAllocationListDCI-0-2* is not configured; otherwise I is the number of entries in the default table.
- Frequency hopping flag - 0 or 1 bit:
  - 0 bit if the higher layer parameter *frequencyHoppingDCI-0-2* is not configured;
  - 1 bit according to Table 7.3.1.1.1-3 otherwise, only applicable to resource allocation type 1, as defined in Clause 6.3 of [6, TS 38.214].
- Modulation and coding scheme -5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-0-2*
  - If 0 bit is configured, rv_{id} to be applied is 0;
  - 1 bit according to Table 7.3.1.2.3-1;
  - 2 bits according to Table 7.3.1.1.1-2.
- Transform precoder indicator - 0 or 1 bit
  - 1 bit if the higher layer parameter *dynamicTransformPrecoderFieldPresenceDCI-0-2* is configured to 'enabled' and if the UE is configured to monitor DCI format 0_2 with CRC scrambled by C-RNTI or CS-RNTI or MCS-C-RNTI, where the bit value of 0 indicates that transform precoder is enabled and the bit value of 1 indicates that transform precoder is disabled. For a DCI format 0_2 with CRC scrambled by

CS-RNTI and the value indicated by new data indicator field is 0, or for a DCI format 0_2 with CRC scrambled by SP-CSI-RNTI, the bit is reserved.
  - 0 bit otherwise.
- HARQ process number - number of bits determined by the following:
  - 5 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-0-2-v1700* if configured;
  - 0, 1, 2, 3, 4 or 5 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-0-2-Ext* if configured;
  - otherwise 0, 1, 2, 3 or 4 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-0-2*.
- Downlink assignment index - 0, 1, 2 or 4 bits
  - 0 bit if the higher layer parameter *downlinkAssignmentIndexDCI-0-2* is not configured;
  - 1, 2, 3, 4, 5 or 6 bits otherwise,
    - 1st downlink assignment index - 1 or 2 bits:
      - 1 bit for semi-static HARQ-ACK codebook for unicast and multicast if *pdsch-HARQ-ACK-Codebook = semiStatic* is configured for both unicast and multicast and the higher layer parameter *fdmed-ReceptionMulticast* is not configured; otherwise for semi-static HARQ-ACK codebook for unicast;
      - 2 bits for dynamic HARQ-ACK codebook for unicast.
    - 2nd downlink assignment index - 0 or 2 bits
      - 2 bits for dynamic HARQ-ACK codebook with two HARQ-ACK sub-codebooks for unicast;
      - 0 bit otherwise.
    - 3rd downlink assignment index - 0, 1 or 2 bits
      - 1 bit for semi-static HARQ-ACK codebook for multicast if the higher layer parameter *fdmed-ReceptionMulticast* is configured;
      - 2 bits for the dynamic HARQ-ACK codebook for multicast;
      - 0 bit otherwise.

  When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookList* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-2* is configured, if the bit width of the 1st or 2nd Downlink assignment index in DCI format 0_2 for one HARQ-ACK codebook is not equal to that of the 1st or 2nd Downlink assignment index in DCI format 0_2 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 1st or 2nd Downlink assignment index until the bit width of the 1st or 2nd Downlink assignment index in DCI format 0_2 for the two HARQ-ACK codebooks are the same.

  When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookListMulticast* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-2* is configured, if the bit width of the 3rd downlink assignment index in DCI format 0_2 for one HARQ-ACK codebook is not equal to that of the 3rd downlink assignment index in DCI format 0_2 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 3rd downlink assignment index until the bit width of the 3rd downlink assignment index in DCI format 0_2 for the two HARQ-ACK codebooks are the same.

- TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS38.213]
- Second TPC command for scheduled PUSCH - 2 bits as defined in Clause 7.1.1 of [5, TS38.213] if higher layer parameter *SecondTPCFieldDCI-0-2* is configured; 0 bit otherwise.
- SRS resource set indicator - 0 or 2 bits
  - 2 bits according to Table 7.3.1.1.2-36 if
    - *txConfig = nonCodeBook*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModListDCI-0-2* and associated with the *usage* of value '*nonCodeBook*', and is not configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided, or
    - *txConfig=codebook*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModListDCI-0-2* and associated with *usage* of value '*codebook*', and is not configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided;
  - 0 bit otherwise.
- SRS resource indicator - number of bits determined by the following: ′

- $\lceil \log_2 \left(\sum_{k=1}^{\min(L_{max},\, N_{SRS,0_2})} \binom{N_{SRS,0_2}}{k} \right) \rceil$ bits according to Tables 7.3.1.1.2-28/28A/29/29B/30/30B/31/31B if the higher layer parameter *txConfig = nonCodebook*, where
  - $N_{SRS,\,0_2}$ is the number of configured SRS resources in the SRS resource set indicated by SRS resource set indicator field if present,
  - N_{SRS} is the number of configured SRS resources in the SRS resource set associated with the *coresetPoolIndex* value for the CORESET used for the PDCCH carrying the DCI format 0_2, if the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*,
  - otherwise $N_{SRS,\,0_2}$ is the number of configured SRS resources in the SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModListDCI-0-2* and associated with the higher layer parameter *usage* of value '*nonCodeBook*', where the SRS resource set is composed of the first $N_{SRS,\,0_2}$ SRS resources together with other configurations in the SRS resource set, or in the SRS resource set with lower srs-ResourceSetId of two SRS resources sets, configured by higher layer parameter *srs-ResourceSetToAddModList*, if any, and associated with the higher layer parameter *usage* of value '*nonCodeBook*', except for the higher layer parameters *'srs-ResourceSetId' and 'srs-ResourceIdList'*,

  and
  - if UE supports operation with *maxMIMO-LayersDCI-0-2* and the higher layer parameter *maxMIMO-LayersDCI-0-2* of *PUSCH-ServingCellConfig* of the serving cell is configured,
    - L_{max} is given by max{*maxMIMO-LayersDCI-0-2*, *maxMIMO-LayersforSDM-DCI-0-2*} if *maxMIMO-LayersforSDM-DCI-0-2* is configured
    - L_{max} is given by max{*maxMIMO-LayersDCI-0-2*, *maxMIMO-LayersforSFN-DCI-0-2*} if *maxMIMO-LayersforSFN-DCI-0-2* is configured
    - L_{max} is given by *maxMIMO-LayersDCI-0-2* otherwise
  - otherwise, L_{max} is given by the maximum number of layers for PUSCH supported by the UE for the serving cell for non-codebook based operation.
- $\lceil \log_2 N_{SRS,\,0_2} \rceil$ bits according to Tables 7.3.1.1.2-32/32A/32B if the higher layer parameter *txConfig = codebook*, where
  - $N_{SRS,\,0_2}$ is the number of configured SRS resources in the SRS resource set indicated by SRS resource set indicator field if present,
  - N_{SRS} is the number of configured SRS resources in the SRS resource set associated with the *coresetPoolIndex* value for the CORESET used for the PDCCH carrying the DCI format 0_2, if the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*,
  - otherwise $N_{SRS,\,0_2}$ is the number of configured SRS resources in the SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModListDCI-0-2* and associated with the higher layer parameter *usage* of value '*codeBook*', where the SRS resource set is composed of the first $N_{SRS,\,0_2}$ SRS resources together with other configurations in the SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModList*, if any, and associated with the higher layer parameter *usage* of value '*codeBook*', except for the higher layer parameters *'srs-ResourceSetId' and 'srs-ResourceIdList'*.

  When the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModListDCI-0-2* and associated with *usage* of value '*codebook*' or '*nonCodeBook*', the first SRS resource set is associated with *coresetPoolIndex* value 0 and the second SRS resource set is associated with *coresetPoolIndex* value 1, where the first and the second SRS resource sets are respectively the ones with lower and higher *srs-ResourceSetId* of the two SRS resources sets, and the first and second SRS resource sets are composed of the first $N_{SRS,\,0_2}$ SRS resources together with other configurations in the first and second SRS resource sets configured by higher layer parameter *srs-ResourceSetToAddModList*, if any, and associated with the higher layer parameter *usage* of value '*codebook*' or '*nonCodeBook*', respectively, except for the higher layer parameters *'srs-ResourceSetId' and 'srs-ResourceIdList'*.

- Second SRS resource indicator - number of bits determined by the following:

- $\lceil \log_2 \left(\max_{k \in \{1,2,\ldots,\min\{L_{max}, N_{SRS,0_2}\}\}} \binom{N_{SRS,0_2}}{k} \right) \rceil$ bits according to Tables 7.3.1.1.2-28/29A/30A/31A with the same number of layers indicated by SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook*, the higher layer parameter *maxMIMO-LayersforSDM-DCI-0-2* is not configured, and SRS resource set indicator field is present, where $N_{SRS,\,0_2}$ is the number of configured SRS resources in the second SRS resource set, and
  - if UE supports operation with *maxMIMO-LayersDCI-0-2* and the higher layer parameter *maxMIMO-LayersDCI-0-2* of *PUSCH-ServingCellConfig* of the serving cell is configured,
    - L_{max} is given by *maxMIMO-LayersforSFN-DCI-0-2* if *maxMIMO-LayersforSFN-DCI-0-2* is configured
    - L_{max} is given by *maxMIMO-LayersDCI-0-2* otherwise
  - otherwise, L_{max} is given by the maximum number of layers for PUSCH supported by the UE for the serving cell for non-codebook based operation.
- $\lceil \log_2 \left(\sum_{k=1}^{\min\{L_{max},\ N_{SRS,0_2}\}} \binom{N_{SRS,0_2}}{k} \right) \rceil$ bits according to Tables 7.3.1.1.2-28/29 if the higher layer parameter *txConfig = nonCodebook*, the higher layer parameter *maxMIMO-LayersforSDM-DCI-0-2* is configured and SRS resource set indicator field is present, where $N_{SRS,\,0_2}$ is the number of configured SRS resources in the second SRS resource set, where the second SRS resource set is composed of the first $N_{SRS,\,0_2}$ SRS resources together with other configurations in the SRS resource set, or in the SRS resource set with higher srs-ResourceSetId of two SRS resources sets, configured by higher layer parameter *srs-ResourceSetToAddModList*, if any, and associated with the higher layer parameter *usage* of value '*nonCodeBook*', except for the higher layer parameters *'srs-ResourceSetId' and 'srs-ResourceIdList'*, and L_{max} is given by *maxMIMO-LayersforSDM-DCI-0-2*.
- $\lceil \log_2 N_{SRS,\,0_2} \rceil$ bits according to Tables 7.3.1.1.2-32/32A/32B if the higher layer parameter *txConfig = codebook* and SRS resource set indicator field is present, where $N_{SRS,\,0_2}$ is the number of configured SRS resources in the second SRS resource set.
- 0 bit otherwise.

- Precoding information and number of layers - number of bits determined by the following:
  - 0 bits if the higher layer parameter *txConfig = nonCodeBook*;
  - 0 bits for 1 antenna port and if the higher layer parameter *txConfig = codebook*;
  - 4, 5, or 6 bits according to Table 7.3.1.1.2-2 for 4 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRankDCI-0-2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*} if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*} if *multipanelSchemeSDM* is configured, and *codebookSubsetDCI-0-2*;
  - 4 or 5 bits according to Table 7.3.1.1.2-2A for 4 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission =fullpowerMode1*, the values of higher layer parameters *maxRankDCI-0-2=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*} = 2 if *multipanelSchemeSFN* is configured or max{*maxRanDCI-0-2k*, *maxRankSDM-DCI-0-2*} = 2 if *multipanelSchemeSDM* is configured, transform precoder is disabled, and according to the value of higher layer parameter *codebookSubsetDCI-0-2*;
  - 4 or 6 bits according to Table 7.3.1.1.2-2B for 4 antenna ports by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2* respectively, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1*, the values of higher layer parameters *maxRankDCI-0-2=3 or 4*, transform precoder is disabled, and according to the value of higher layer parameter *codebookSubsetDCI-0-2*;
  - 2, 4, or 5 bits according to Table 7.3.1.1.2-3 for 4 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and *maxRankDCI-0-2=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*,

maxRankSFN-DCI-0-2} = 1 if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*} = 1 if *multipanelSchemeSDM* is configured, and *codebookSubsetDCI-0-2*;

- 3 or 4 bits according to Table 7.3.1.1.2-3A for 4 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission =fullpowerMode1*, *maxRankDCI-0-2=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*} = 1 if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*} = 1 if *multipanelSchemeSDM* is configured, and according to whether transform precoder is enabled or disabled, and the value of higher layer parameter *codebookSubsetDCI-0-2*;
- 2 or 4 bits according to Table7.3.1.1.2-4 for 2 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRankDCI-0-2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*} if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*} if *multipanelSchemeSDM* is configured, and *codebookSubsetDCI-0-2*;
- 2 bits according to Table 7.3.1.1.2-4A for 2 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission =fullpowerMode1*, transform precoder is disabled, the *maxRankDCI-0-2=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*} = 2 if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*} = 2 if *multipanelSchemeSDM* is configured, and *codebookSubsetDCI-0-2=nonCoherent*;
- 1 or 3 bits according to Table7.3.1.1.2-5 for 2 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and *maxRankDCI-0-2=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*}*=1* if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*}*=1* if *multipanelSchemeSDM* is configured, and *codebookSubsetDCI-0-2*;
- 2 bits according to Table 7.3.1.1.2-5A for 2 antenna ports by replacing *maxRank*, *maxRankSFN*, *maxRankSDM* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*, *maxRankSDM-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission =fullpowerMode1*, *maxRankDCI-0-2=1* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSFN-DCI-0-2*} = 1 if *multipanelSchemeSFN* is configured or max{*maxRankDCI-0-2*, *maxRankSDM-DCI-0-2*} = 1 if *multipanelSchemeSDM* is configured, and according to whether transform precoder is enabled or disabled, and the value of higher layer parameter *codebookSubsetDCI-0-2*.
- 7 bits according to Table 7.3.1.1.2-5D for 8 antenna ports by replacing *maxRank-n8* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook1*, transform precoder is disabled, *maxRankDCI-0-2* =4, and according to *maxRankDCI-0-2*;
- 4, 6 or 7 bits according to Table 7.3.1.1.2-5E for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook1*, transform precoder is enabled or *maxRankDCI-0-2* =1, 2 or 3 if transform precoder is disabled, and according to transform precoder and *maxRankDCI-0-2*;
- 6 or 7 or 8 bits according to Table 7.3.1.1.2-5G for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook4*, transform precoder is disabled, *maxRankDCI-0-2*=2, 3 or 4, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to *maxRankDCI-0-2*;
- 3 bits according to Table 7.3.1.1.2-5H for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook4*, transform precoder is enabled or *maxRankDCI-0-2*=1 if transform precoder is disabled, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*.
- 5, 9 or 10 bits according to Table 7.3.1.1.2-5J for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook2*, transform precoder is enabled or *maxRankDCI-0-2* =1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to transform precoder and *maxRankDCI-0-2*;

- 4, 7, 9 or 10 bits according to Table 7.3.1.1.2-5L for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook3*, transform precoder is enabled or *maxRankDCI-0-2* =1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to transform precoder and *maxRankDCI-0-2*;
- 6 or 7 or 8 bits according to Table 7.3.1.1.2-5M for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook4*, transform precoder is disabled, *maxRankDCI-0-2*=2, 3 or 4, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to *maxRankDCI-0-2*;
- 4 bits according to Table 7.3.1.1.2-5N for 8 antenna ports, if *CodebookTypeUL=codebook4*, transform precoder is enabled or *maxRankDCI-0-2*=1 if transform precoder is disabled, *ul-FullPowerTransmission* is configured to *fullpowerMode1*.
- 6, 9 or 10 bits according to Table 7.3.1.1.2-5O for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook2*, transform precoder is enabled or *maxRankDCI-0-2* =1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to transform precoder and *maxRankDCI-0-2*;
- 5, 7, 9 or 10 bits according to Table 7.3.1.1.2-5P for 8 antenna ports by replacing *maxRank* with *maxRankDCI-0-2*, if *CodebookTypeUL=codebook3*, transform precoder is enabled or *maxRankDCI-0-2* =1, 2, 3 or 4 if transform precoder is disabled, *ul-FullPowerTransmission* is configured to *fullpowerMode1*, and according to transform precoder and *maxRankDCI-0-2*;
- 3 bits according to Table 7.3.1.1.2-5T for 3 antenna ports by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRankDCI-0-2*;
- 2 bits according to Table 7.3.1.1.2-5U for 3 antenna ports by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRankDCI-0-2*;

For the higher layer parameter *txConfig=codebook*, if *ul-FullPowerTransmission* is configured to *fullpowerMode2*, the values of higher layer parameters *maxRankDCI-0-2* is configured to be larger than 2, and at least one SRS resource with 4 antenna ports is configured in the SRS resource set indicated by SRS resource set indicator field if present, otherwise in an SRS resource set with usage set to 'codebook', and an SRS resource with 2 antenna ports is indicated via SRI in the same SRS resource set, then Table 7.3.1.1.2-4 is used by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2* respectively.

For the higher layer parameter *txConfig = codebook*, if different SRS resources with different number of antenna ports are configured, the bitwidth is determined according to the maximum number of ports in an SRS resource among the configured SRS resources in all SRS resource set(s) with usage set to 'codebook'. If the number of ports for a configured SRS resource in the set is less than the maximum number of ports in an SRS resource among the configured SRS resources, a number of most significant bits with value set to '0' are inserted to the field.

For the higher layer parameter *txConfig = codebook*, when the Transform precoder indicator field is present, if the bit width of the Precoding information and number of layers field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the Precoding information and number of layers field for the case with smaller bit width until the bit width of the Precoding information and number of layers field for the two cases are the same.

When the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs, and is provided *sTx-2Panel*, and there are two SRS resource sets configured by *srs-ResourceSetToAddModListDCI-0-2* and associated with *usage* of value '*codebook*' or '*nonCodeBook*', the Precoding information and number of layers field is associated with the SRS resource set that is associated with the *coresetPoolIndex* value for the CORESET used for the PDCCH carrying the DCI format 0_2.

- Second Precoding information - number of bits determined by the following:
  - 0 bits if SRS resource set indicator field is not present;
  - 0 bits if the higher layer parameter *txConfig = nonCodeBook*;
  - 0 bits for 1 antenna port and if the higher layer parameter *txConfig = codebook*;
  - 3, 4, or 5 bits according to Table 7.3.1.1.2-2C with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports by replacing *maxRank*, *maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2*, *maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*,

respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* transform precoder is disabled, and according to the values of higher layer parameters *maxRankDCI-0-2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2* if *multipanelSchemeSFN* is configured, and *codebookSubsetDCI-0-2*;

- 3 or 4 bits according to Table 7.3.1.1.2-2D with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1,* the values of higher layer parameters *maxRankDCI-0-2=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2=2* if *multipanelSchemeSFN* is configured, transform precoder is disabled, and according to the value of higher layer parameter *codebookSubsetDCI-0-2*;
- 3 or 4 bits according to Table 7.3.1.1.2-2E with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2* respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1, maxRankDCI-0-2=3 or 4,* transform precoder is disabled, and according to the value of higher layer parameter *codebookSubsetDCI-0-2*;
- 2, 4, or 5 bits according to Table 7.3.1.1.2-3 with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRankDCI-0-2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2* if *multipanelSchemeSFN* is configured, and *codebookSubsetDCI-0-2*;
- 3 or 4 bits according to Table 7.3.1.1.2-3A with the same number of layers indicated by Precoding information and number of layers field for 4 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1,* and according to whether transform precoder is enabled, or disabled and *maxRankDCI-0-2*=1 if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2*=1 if *multipanelSchemeSFN* is configured, and the value of higher layer parameter *codebookSubsetDCI-0-2*;
- 1 or 3 bits according to Table7.3.1.1.2-4B with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* transform precoder is disabled, and according to the values of higher layer parameters *maxRankDCI-0-2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2* if *multipanelSchemeSFN* is configured, and *codebookSubsetDCI-0-2*;
- 2 bits according to Table 7.3.1.1.2-4C with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1,* transform precoder is disabled, the *maxRankDCI-0-2=2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2=2* if *multipanelSchemeSFN* is configured, and *codebookSubsetDCI-0-2=nonCoherent*;
- 1 or 3 bits according to Table7.3.1.1.2-5 with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower,* and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRankDCI-0-2* if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2* if *multipanelSchemeSFN* is configured, and *codebookSubsetDCI-0-2*;
- 2 bits according to Table 7.3.1.1.2-5A with the same number of layers indicated by Precoding information and number of layers field for 2 antenna ports by replacing *maxRank, maxRankSFN* and *codebookSubset* with *maxRankDCI-0-2, maxRankSFN-DCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook, ul-*

FullPowerTransmission =fullpowerMode1, and according to whether transform precoder is enabled, or disabled and *maxRankDCI-0-2*=1 if neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured or *maxRankSFN-DCI-0-2*=1 if *multipanelSchemeSFN* is configured, and the value of higher layer parameter *codebookSubsetDCI-0-2*.

- 2 bits according to Table 7.3.1.1.2-5V with the same number of layers indicated by Precoding information and number of layers field for 3 antenna ports by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRankDCI-0-2*;
- 2 bits according to Table 7.3.1.1.2-5U with the same number of layers indicated by Precoding information and number of layers field for 3 antenna ports by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2*, respectively, if SRS resource set indicator field is present, *txConfig = codebook*, *ul-FullPowerTransmission* is not configured or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRankDCI-0-2*;

For the higher layer parameter *txConfig=codebook*, if *ul-FullPowerTransmission* is configured to *fullpowerMode2*, the values of higher layer parameters *maxRankDCI-0-2* is configured to be larger than 2, and at least one SRS resource with 4 antenna ports is configured in the SRS resource set indicated by SRS resource set indicator field, and an SRS resource with 2 antenna ports is indicated via Second SRS resource indicator field in the same SRS resource set, then Table 7.3.1.1.2-4B is used by replacing *maxRank* and *codebookSubset* with *maxRankDCI-0-2* and *codebookSubsetDCI-0-2* respectively.

For the higher layer parameter *txConfig = codebook*, if different SRS resources with different number of antenna ports are configured, the bitwidth is determined according to the maximum number of ports in an SRS resource among the configured SRS resources in the second SRS resource set with usage set to 'codebook' as defined in Table 7.3.1.1.2-36. If the number of ports for a configured SRS resource in the set is less than the maximum number of ports in an SRS resource among the configured SRS resources, a number of most significant bits with value set to '0' are inserted to the field.

For the higher layer parameter *txConfig = codebook*, when the Transform precoder indicator field is present, if the bit width of the Second Precoding information field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the Second Precoding information field for the case with smaller bit width until the bit width of the Second Precoding information field for the two cases are the same.

- Antenna ports - number of bits determined by the following:
  - 0 bit if higher layer parameter *antennaPortsFieldPresenceDCI-0-2* is not configured;
  - 2, 3, 4, 5 or 6 bits otherwise,
    - 2 bits as defined by Tables 7.3.1.1.2-6, if transform precoder is enabled, *dmrs-Type*=1, and *maxLength*=1, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2 BPSK modulation is used;
    - 2 bits as defined by 7.3.1.1.2-6A, if transform precoder is enabled, and *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2 BPSK modulation is used, *dmrs-Type*=1, and *maxLength*=1, where n_{SCID} is the scrambling identity for antenna ports defined in Clause 6.4.1.1.1.2, in [4, TS38.211];
    - 4 bits as defined by Tables 7.3.1.1.2-7, if transform precoder is enabled, *dmrs-Type*=1, and *maxLength*=2, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2 BPSK modulation is used;
    - 4 bits as defined by Tables 7.3.1.1.2-7A, if transform precoder is enabled, and *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2 BPSK modulation is used, *dmrs-Type*=1, and *maxLength*=2, where n_{SCID} is the scrambling identity for antenna ports defined in Clause 6.4.1.1.1.2, in [4, TS38.211];
    - 3 bits as defined by Tables 7.3.1.1.2-8/9/10/10A/11 according to the value of rank, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, and *maxLength*=1;
    - 4 bits as defined by Tables 7.3.1.1.2-12/13/14/14A/15 according to the value of rank, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, and *maxLength*=2;
    - 4 bits as defined by Tables 7.3.1.1.2-16/17/18/18A/19 according to the value of rank, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, and *maxLength*=1;
    - 5 bits as defined by Tables 7.3.1.1.2-20/21/22/22A/23 according to the value of rank, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, and *maxLength*=2.

- 4 bits as defined by Tables 7.3.1.1.2-38/39/40/40A/41, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, and *maxLength*=1;
- 5 bits as defined by Tables 7.3.1.1.2-46/47/48/48A/49, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, and *maxLength*=2;
- 5 bits as defined by Tables 7.3.1.1.2-54/55/56/56A/57, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, and *maxLength*=1;
- 6 bits as defined by Tables 7.3.1.1.2-62/63/64/64A/65, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, and *maxLength*=2.

where the number of CDM groups without data of values 1, 2, and 3 in Tables 7.3.1.1.2-6 to 7.3.1.1.2-23 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively, and the value of rank is

- the sum of the value determined according to the SRS resource indicator field and the value determined according to the second SRS resource indicator field, if *txConfig = nonCodebook* and *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "10"
- the sum of the value determined according to the Precoding information and number of layers field and the value determined according to the Second Precoding information, if *txConfig = codebook* and *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "10"
- determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and *multipanelSchemeSDM* is not configured, , or if the higher layer parameter *txConfig = nonCodebook*, *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "00" or “01”
- determined according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook* and *multipanelSchemeSDM* is not configured, or if the higher layer parameter *txConfig = codebook*, *multipanelSchemeSDM* is configured and SRS resource set indicator field equals "00" or "01".

If a UE is configured with both *dmrs-UplinkForPUSCH-MappingTypeA-DCI-0-2* and *dmrs-UplinkForPUSCH-MappingTypeB-DCI-0-2* and is configured with *antennaPortsFieldPresenceDCI-0-2*, the bitwidth of this field equals $max\left\{x_A, x_B\right\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-UplinkForPUSCH-MappingTypeA-DCI-0-2* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-UplinkForPUSCH-MappingTypeB-DCI-0-2*. A number of $\left|x_A - x_B\right|$ zeros are padded in the MSB of this field, if the mapping type of the PUSCH corresponds to the smaller value of x_A and x_B.

If a UE is not configured with higher layer parameter *antennaPortsFieldPresenceDCI-0-2*, antenna port(s) are defined assuming bit field index value 0 in Tables 7.3.1.1.2-6 to 7.3.1.1.2-23.

When the Transform precoder indicator field is present, if the bit width of the Antenna ports field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the Antenna ports field for the case with smaller bit width until the bit width of the Antenna ports field for the two cases are the same.

- SRS request - 0, 1, 2 or 3 bits
  - 0 bit if the higher layer parameter *srs-RequestDCI-0-2* is not configured;
  - 1 bit as defined by Table 7.3.1.1.3-1 if higher layer parameter *srs-RequestDCI-0-2 = 1* and for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell;
  - 2 bits if higher layer parameter *srs-RequestDCI-0-2 = 1* and for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell, where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second bit is defined by Table 7.3.1.1.3-1;
  - 2 bits as defined by Table 7.3.1.1.2-24 if higher layer parameter *srs-RequestDCI-0-2 = 2* and for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell;
  - 3 bits if higher layer parameter *srs-RequestDCI-0-2 = 2* and for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell, where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second and third bits are defined by Table 7.3.1.1.2-24;
- SRS offset indicator - 0, 1 or 2 bits.
  - 0 bit if higher layer parameter *AvailableSlotOffset* is not configured for any aperiodic SRS resource set in the scheduled cell, or if higher layer parameter *AvailableSlotOffset* is configured for at least one aperiodic SRS resource set in the scheduled cell and the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) is 1;

  - otherwise, $\lceil \log_2(K) \rceil$ bits are used to indicate available slot offset according to Table 7.3.1.1.2-37 and Clause 6.2.1 of [6, TS 38.214], where K is the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) in the scheduled cell;
- CSI request - 0, 1, 2, 3, 4, 5, or 6 bits determined by higher layer parameter *reportTriggerSizeDCI-0-2*.
- PTRS-DMRS association - number of bits determined as follows
  - 0 bit if *PTRS-UplinkConfi*g is not configured in either *dmrs-UplinkForPUSCH-MappingTypeA* or *dmrs-UplinkForPUSCH-MappingTypeB* and transform precoder is disabled, or if transform precoder is enabled, or if *maxRankDCI-0-2=1* and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, or if *maxMIMO-LayersDCI-0-2*=1 and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, or if *maxRankDCI-0-2=1* and *maxRankSFN-DCI-0-2=1*, or if *maxMIMO-LayersDCI-0-2*=1 and *maxMIMO-LayersforSFN-DCI-0-2*=1, or if *maxRankDCI-0-2=1* and *maxRankSDM-DCI-0-2=1* when two PTRS ports are configured by *maxNrofPorts-SDM*, or if *maxMIMO-LayersDCI-0-2*=1 and *maxMIMO-LayersforSDM-DCI-0-2*=1 when two PTRS ports are configured by *maxNrofPorts-SDM*;
  - 1 or 2 bits otherwise, where Table 7.3.1.1.2-25/7.3.1.1.2-25A/7.3.1.1.2-25B/7.3.1.1.2-26/7.3.1.1.2-26B are used to indicate the association between PTRS port(s) and DMRS port(s), and the DMRS ports are indicated by the Antenna ports field.
    - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 2, 4, or 8 antenna ports, SRS resource set indicator field is absent or SRS resource set indicator field is present and equals "00" or “01” and maxRank*DCI-0-2*<=4 or *maxMIMO-LayersDCI-0-2*<=4, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Table 7.3.1.1.2-25 and 7.3.1.1.2-26.
    - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 4 or 8 antenna ports, the SRS resource set indicator field is present and equals "10" or “11”, *maxRankDCI-0-2=3 or 4* or *maxMIMO-LayersDCI-0-2*=3 or 4 and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Table 7.3.1.1.2-25 and 7.3.1.1.2-26.
    - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 2, 4, or 8 antenna ports, the SRS resource set indicator field is present and equals "10" or "11" and *maxRankDCI-0-2=2* or *maxMIMO-LayersDCI-0-2*=2 and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, the MSB of this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field, and the LSB of this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to Second SRS resource indicator field and/or Second Precoding information field, according to Table 7.3.1.1.2-25A.
    - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, SRS resource set indicator field is absent or SRS resource set indicator field is present and equals "00" or "01", *maxRankDCI-0-2*<=3 or *maxMIMO-LayersDCI-0-2*<=3, this field indicates the association between PTRS port and DMRS port corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25.
    - 1 bit when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, SRS resource set indicator field is absent or SRS resource set indicator field is present and equals "00" or "01", *maxRankDCI-0-2*<=3 or *maxMIMO-LayersDCI-0-2*<=3, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-26B.
    - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRankDCI-0-2*=3 or *maxMIMO-LayersDCI-0-2*=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port and DMRS port corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-25.
    - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRankDCI-0-2*=2 or *maxMIMO-LayersDCI-0-2*=2, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, the MSB of this field indicates the association between PTRS port and DMRS port corresponding to SRS resource indicator and/or Precoding information and number of layers field, and the LSB of this field indicates the association between PTRS port and DMRS port corresponding to

Second SRS resource indicator field and/or Second Precoding information field, according to Table 7.3.1.1.2-25A.

- 1 bit when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, the SRS resource set indicator field is present and equals "10" or "11", *maxRankDCI-0-2*<=3 or *maxMIMO-LayersDCI-0-2*<=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Tables 7.3.1.1.2-26B.
- 2 bits when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig*, the SRS resource set indicator field is present and equals "10" and *multipanelSchemeSDM* is configured, the MSB of this field indicates the association between PTRS port 0 and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field, and the LSB of this field indicates the association between PTRS port 1 and DMRS port(s) corresponding to Second SRS resource indicator field and/or Second Precoding information field, according to Table 7.3.1.1.2-25A.
- 2 bits when one PTRS port is configured by *maxNrofPorts-SDM* in *PTRS-UplinkConfig*, SRS resource set indicator field is present and equals "10" and *multipanelSchemeSDM* is configured, this field indicates the association between PTRS port and DMRS ports corresponding to SRS resource indicator field and Second SRS resource indicator field and/or Precoding information and number of layers field and Second Precoding information field according to Table 7.3.1.1.2-25.
- 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig*, SRS resource set indicator field is present and equals "10", *multipanelSchemeSFN* is configured, this field indicates the association between PTRS port(s) and DMRS port(s) corresponding to SRS resource indicator field and/or Precoding information and number of layers field according to Table 7.3.1.1.2-25 and 7.3.1.1.2-26.

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and the "PTRS-DMRS association" field is present for the indicated bandwidth part but not present for the active bandwidth part, the UE assumes the "PTRS-DMRS association" field is not present for the indicated bandwidth part.

When the Transform precoder indicator field is present, if the bit width of PTRS-DMRS association field for the case with transform precoder enabled is not equal to that for the case with transform precoder disabled, a number of most significant bits with value set to '0' are inserted to the PTRS-DMRS association field for the case with smaller bit width until the bit width of the PTRS-DMRS association field for the two cases are the same.

- Second PTRS-DMRS association

  - 2 bits when one PTRS port or two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 4 or 8 antenna ports, PTRS-DMRS association field is present, SRS resource set indicator field is present and equals "10" or "11", *maxRankDCI-0-2*>2 or *maxMIMO-LayersDCI-0-2*>2, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured;
  - 2 bits when one PTRS port is configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, PTRS-DMRS association field is present, SRS resource set indicator field is present and equals "10" or "11", *maxRankDCI-0-2*=3 or *maxMIMO-LayersDCI-0-2*=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured;
  - 1 bit when two PTRS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* for 3 antenna ports, PTRS-DMRS association field is present, SRS resource set indicator field is present and equals "10" or "11", *maxRankDCI-0-2*<=3 or *maxMIMO-LayersDCI-0-2*<=3, and neither *multipanelSchemeSDM* nor *multipanelSchemeSFN* is configured;
  - 0 bit otherwise.

  Tables 7.3.1.1.2-25 and 7.3.1.1.2-26/7.3.1.1.2-26B are used to indicate the association between PTRS port(s) and DMRS port(s) corresponding to Second SRS resource indicator field and/or Second precoding information field when one PT-RS port and two PT-RS ports are configured by maxNrofPorts in PTRS-UplinkConfig respectively, and the DMRS ports are indicated by the Antenna ports field.

- beta_offset indicator - 0 bit if the higher layer parameter *betaOffsetsDCI-0-2 = semiStaticDCI-0-2*; otherwise 1 bit if 2 offset indexes are configured by higher layer parameter *dynamicDCI-0-2* as defined by Table 9.3-3A in [5, TS 38.213], and 2 bits if 4 offset indexes are configured by higher layer parameter *dynamicDCI-0-2* as defined by Table 9.3-3 in [5, TS 38.213].

  When two HARQ-ACK codebooks are configured by *pdsch-HARQ-ACK-CodebookList* or by *pdsch-HARQ-ACK-CodebookListMulticast* for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-2* is configured, if the bit width of the beta_offset indicator in DCI format 0_2 for one HARQ-ACK codebook

is not equal to that of the beta_offset indicator in DCI format 0_2 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller beta_offset indicator until the bit width of the beta_offset indicator in DCI format 0_2 for the two HARQ-ACK codebooks are the same.

- DMRS sequence initialization - 0 or 1 bit
  - 0 bit if the higher layer parameter *dmrs-SequenceInitializationDCI-0-2* is not configured, or if transform precoder is enabled by higher layers and the Transform precoder indicator field is not present;
  - 1 bit if transform precoder is disabled by higher layers and the higher layer parameter *dmrs-SequenceInitializationDCI-0-2* is configured, or if the Transform precoder indicator field is present and the higher layer parameter *dmrs-SequenceInitializationDCI-0-2* is configured. If the Transform precoder indicator field is present and set to '0', the bit is reserved.
- UL-SCH indicator - 1 bit. A value of "1" indicates UL-SCH shall be transmitted on the PUSCH and a value of "0" indicates UL-SCH shall not be transmitted on the PUSCH. If a UE does not support triggering SRS only in DCI, except for DCI format 0_2 with CRC scrambled by SP-CSI-RNTI, the UE is not expected to receive a DCI format 0_2 with UL-SCH indicator of "0" and CSI request of all zero(s). If a UE supports triggering SRS only in DCI, except for DCI format 0_2 with CRC scrambled by SP-CSI-RNTI, the UE is not expected to receive a DCI format 0_2 with UL-SCH indicator of "0", CSI request of all zero(s) and SRS request of all zero(s).
- ChannelAccess-CPext-CAPC - 0, 1, 2, 3, 4, 5 or 6 bits. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *ul-AccessConfigListDCI-0-2* or in Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1, or the number of entries in the high layer parameter *ul-AccessConfigListDCI-0-1* for operation in frequency range 2-2 if *ChannelAccessMode2-r17* is provided; otherwise 0 bit. One or more entries from Table 7.3.1.1.2-35 are configured by the higher layer parameter *ul-AccessConfigListDCI-0-2* in frequency range 1. One or more entries from Table 7.3.1.1.2-35A are configured by the higher layer parameter *ul-AccessConfigListDCI-0-1* in frequency range 2-2.
- Open-loop power control parameter set indication - 0 or 1 or 2 bits.
  - 0 bit if the higher layer parameter *p0-PUSCH-SetList* is not configured;
  - 1 or 2 bits otherwise,
    - 1 bit if SRS resource indicator is present in the DCI format 0_2;
    - 1 or 2 bits as determined by higher layer parameter *olpc-ParameterSetDCI-0-2* if SRS resource indicator is not present in the DCI format 0_2;
- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-0-2* is not configured; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213].
- Invalid symbol pattern indicator - 0 bit if higher layer parameter *invalidSymbolPatternIndicatorDCI-0-2* is not configured; otherwise 1 bit as defined in Clause 6.1.2.1 in [6, TS 38.214].
- PDCCH monitoring adaptation indication - 0, 1 or 2 bits
  - 1 or 2 bits, if *searchSpaceGroupIdList-r17* is not configured and if *pdcch-SkippingDurationList* is configured
    - 1 bit if the UE is configured with only one duration by *pdcch-SkippingDurationList;*
    - 2 bits if the UE is configured with more than one duration by *pdcch-SkippingDurationList*.
  - 1 or 2 bits, if *pdcch-SkippingDurationList* is not configured and if *searchSpaceGroupIdList-r17* is configured
    - 1 bit if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0 and search space set(s) with group index 1, and if the UE is not configured by *searchSpaceGroupIdList-r17* with any search space set with group index 2;
    - 2 bits if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0, search space set(s) with group index 1 and search space set(s) with group index 2;
  - 2 bits, if *pdcch-SkippingDurationList* is configured and if *searchSpaceGroupIdList-r17* is configured
  - 0 bit, otherwise
- Measurement gap cancellation – 0 bit if higher layer parameter *mg-CancellationDCI-0-2* is not configured; otherwise 1 bit as defined in Clause 10.6 in [5, TS 38.213].

A UE does not expect that the bit width of a field in DCI format 0_2 with CRC scrambled by CS-RNTI is larger than corresponding bit width of same field in DCI format 0_2 with CRC scrambled by C-RNTI for the same serving cell. If the bit width of a field in the DCI format 0_2 with CRC scrambled by CS-RNTI is not equal to that of the corresponding field in the DCI format 0_2 with CRC scrambled by C-RNTI for the same serving cell, a number of

most significant bits with value set to '0' are inserted to the field in DCI format 0_2 with CRC scrambled by CS-RNTI until the bit width equals that of the corresponding field in the DCI format 0_2 with CRC scrambled by C-RNTI for the same serving cell.

For a UE configured with scheduling on the primary cell from an SCell, if prior to padding the number of information bits in DCI format 0_2 carried by PDCCH on the primary cell is not equal to the number of information bits in DCI format 0_2 carried by PDCCH on the SCell for scheduling on the primary cell, zeros shall be appended to the DCI format 0_2 with smaller size until the payload size is the same.

- If application of step 4B in clause 7.3.1.0 results in additional zero padding for DCI format 0_2 for scheduling on the primary cell, corresponding zeros shall be appended to both DCI format 0_2 monitored on the primary cell and DCI format 0_2 monitored on the SCell for scheduling on the primary cell.
- If the SCell is deactivated and *firstActiveDownlinkBWP-Id* is not set to dormant BWP, the UE determines the number of information bits in DCI format 0_2 carried by PDCCH on the primary cell based on a DL BWP provided by *firstActiveDownlinkBWP-Id* for the SCell. If the active DL BWP of the SCell is a dormant DL BWP, or if the SCell is deactivated and *firstActiveDownlinkBWP-Id* is set to dormant BWP, the UE determines the number of information bits in DCI format 0_2 carried by PDCCH on the primary cell based on a DL BWP provided by *firstWithinActiveTimeBWP-Id* for the SCell if provided; otherwise, based on a DL BWP provided by *firstOutsideActiveTimeBWP-Id* for the SCell.

Table 7.3.1.1.3-1: 1 bit SRS request in DCI format 0_2 and DCI format 1_2

Value of SRS request field	Triggered aperiodic SRS resource set(s) for DCI format 0_2 and 1_2
0	No aperiodic SRS resource set triggered
1	SRS resource set(s) configured with higher layer parameter *aperiodicSRS-ResourceTrigger* set to 1 or an entry in *aperiodicSRS-ResourceTriggerList* set to 1

7.3.1.1.4 Format 0_3

DCI format 0_3 is used for the scheduling of one or multiple PUSCHs in one cell, or multiple PUSCHs in multiple cells with one or multiple PUSCHs per cell.

The following information is transmitted by means of the DCI format 0_3 with CRC scrambled by C-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bit
  - The value of this bit field is always set to 0, indicating an UL DCI format
- Scheduled cell set indicator - $\lceil \log_2 N_{set} \rceil$ bits, where N_{set} is the number of cell sets which are configured by higher layer parameter *mc-DCI-SetofCellsToAddModList* to be respectively scheduled by DCI format 0_3/1_3 from the cell on which this format is carried by PDCCH. If present, this field is used to indicate the scheduled cell set according to Table 7.3.1.1.4-1; otherwise, the scheduled cell set is the cell set configured to be scheduled by DCI format 0_3/1_3 from the cell by higher layer parameter *mc-DCI-SetofCellsToAddModList*.
- Scheduled cells indicator - number of bits determined by the following:
  - 0 bit if the higher layer parameter *scheduledCellComboListDCI-0-3* for the scheduled cell set is not configured;
  - otherwise $\lceil \log_2 I_{UL} \rceil$ bits indicating the scheduled cells in the scheduled cell set according to Table 7.3.1.1.4-2, where I_{UL} is the number of entries in the higher layer parameter *scheduledCellComboListDCI-0-3*. If only one entry is configured in the higher layer parameter *scheduledCellComboListDCI-0-3*, the scheduled cells are the cells configured by higher layer parameter *scheduledCellComboListDCI-0-3*.
- Bandwidth part indicator - 0, 1 or 2 bits determined as $\lceil \log_2 n_{BWP,\,max} \rceil$, where
  - $n_{BWP,\,max} = n_{BWP,RRC}^{max} + 1$ if $n_{BWP,RRC}^{max} \leq 3$, $n_{BWP,RRC}^{max}$ is the maximum number of UL BWPs configured by higher layers, excluding the initial UL bandwidth part, across all the cells configured by higher layer parameter *scheduledCellListDCI-0-3* in the scheduled cell set, in which case the bandwidth part indicator is equivalent to the ascending order of the higher layer parameter *BWP-Id*;

- otherwise $n_{BWP,\,max} = n_{BWP,RRC}^{max}$, in which case the bandwidth part indicator is defined in Table 7.3.1.1.2-1;

The field is only applicable to a scheduled cell with the number of configured UL BWPs larger than 1, including the initial UL bandwidth part, and is applied to the applicable scheduled cells in the scheduled cell set independently. If a UE does not support active BWP change via DCI, the UE ignores this bit field. If this field indicates a code point that does not correspond to a configured BWP of a scheduled cell, the UE ignores this bit field for the scheduled cell, and operates on the active BWP of the scheduled cell.

- Frequency domain resource assignment - number of bits determined by the following, where $N_{RB}^{UL,\,BWP}$ is the size of the active UL bandwidth part:

  - block number 1, block number 2,…, block number N_{cell}^{UL}

  If *scheduledCellComboListDCI-0-3* for the scheduled cell set is configured with more than one entry, N_{cell}^{UL} is the number of scheduled cells indicated by Scheduled cells indicator field; if *scheduledCellComboListDCI-0-3* for the scheduled cell set is configured with only one entry, N_{cell}^{UL} is the number of cells configured by higher layer parameter *scheduledCellComboListDCI-0-3*; otherwise, N_{cell}^{UL} is the number of cells configured by higher layer parameter *scheduledCellListDCI-0-3* in the scheduled cell set. Each block corresponds to the frequency domain resource assignment for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the frequency domain resource assignment for the cell with the smallest serving cell index. Each block is defined by the following fields:

  - If higher layer parameter *useInterlacePUCCH-PUSCH* in *BWP-UplinkDedicated* is not configured
    - N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 6.1.2.2.1 of [6, TS 38.214]
    - $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil$ bits if only resource allocation type 1 is configured, or $max\left(\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil,\ N_{RBG}\right)+1$ bits if *resourceAllocationDCI-0-3* is configured as '*dynamicSwitch*', where $N_{RBG,\,K1} = \lceil \left(N_{RB}^{UL,\,BWP} + \left(N_{UL,\,BWP}^{start} \bmod K1\right)\right)/K1 \rceil$, $N_{RB}^{UL,\,BWP}$ is the size of the active UL bandwidth part, $N_{UL,\,BWP}^{start}$ is defined as in clause 4.4.4.4 of [4, TS 38.211] and $K1$ is given by higher layer parameter *resourceAllocationType1GranularityDCI-0-3*. If the higher layer parameter *resourceAllocationType1GranularityDCI-0-3* is not configured, $K1$ is equal to 1.
    - If *resourceAllocationDCI-0-3* is configured as '*dynamicSwitch*', the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
    - For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 6.1.2.2.1 of [6, TS 38.214].
    - For resource allocation type 1, the $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil$ LSBs provide the resource allocation as follows:
      - For PUSCH hopping with resource allocation type 1:
        - N_{UL_hop} MSB bits are used to indicate the frequency offset according to Clause 6.3 of [6, TS 38.214], where $N_{UL_hop}=1$ if the higher layer parameter *frequencyHoppingOffsetLists* contains two offset values and $N_{UL_hop}=2$ if the higher layer parameter *frequencyHoppingOffsetLists* contains four offset values
        - $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil - N_{UL_hop}$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]
      - For non-PUSCH hopping with resource allocation type 1:
        - $\lceil \log_2\left(N_{RBG,\,K1}\left(N_{RBG,\,K1}+1\right)/2\right) \rceil$ bits provide the frequency domain resource allocation according to Clause 6.1.2.2.2 of [6, TS 38.214]

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and if *resourceAllocationDCI-0-3* is configured as '*dynamicSwitch*' for the indicated bandwidth part, the UE assumes resource allocation type 0 for the indicated bandwidth part if the bitwidth of the "Frequency domain resource assignment" field of the active bandwidth part is smaller than the bitwidth of the "Frequency domain resource assignment" field of the indicated bandwidth part.

- If the higher layer parameter *useInterlacePUCCH-PUSCH* in *BWP-UplinkDedicated* is configured
  - 5 + Y bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 30 kHz. The 5 MSBs provide the interlace allocation and the Y LSBs provide the RB set allocation.
  - 6 + Y bits provide the frequency domain resource allocation according to Clause 6.1.2.2.3 of [6, TS 38.214] if the subcarrier spacing for the active UL bandwidth part is 15 kHz. The 6 MSBs provide the interlace allocation and the Y LSBs provide the RB set allocation.

  The value of Y is determined by $\lceil \log_2\left(\frac{N_{RB\text{-}set,UL}^{BWP}\left(N_{RB\text{-}set,UL}^{BWP}+1\right)}{2}\right)\rceil$ where $N_{\text{RB-set,UL}}^{\text{BWP}}$ is the number of RB sets contained in the active UL BWP as defined in clause 7 of [6, TS38.214].

If the higher layer parameter *scheduledCellComboListDCI-0-3* for the scheduled cell set is not configured, each block is also used to indicate whether the corresponding cell is scheduled or not as follows:

- if all bits of a block are set to 0 for resource allocation type 0, or set to 1 for resource allocation type 1, or set to 0 or 1 for dynamic switch resource allocation type, or set to 0 for resource allocation type 2 with μ=1, or set to 1 for resource allocation type 2 with μ=0, the cell corresponding to the block is not scheduled;
- otherwise, the cell corresponding to the block is scheduled.

- Time domain resource assignment - $\lceil \log_2(I_{TDRA}) \rceil$ bits, where I_{TDRA} is the number of entries in the higher layer parameter *tdra-FieldIndexListDCI-0-3*. This field is used to indicate an entry in the higher layer parameter *tdra-FieldIndexListDCI-0-3* according to Table 7.3.1.1.4-3. Each entry in the higher layer parameter *tdra-FieldIndexListDCI-0-3* contains the 'Time domain resource assignment' index for each BWP of each cell in the scheduled cell set, where the 'Time domain resource assignment' indexes for all the cells are placed according to an ascending order of a serving cell index, and the 'Time domain resource assignment' indexes for all the BWPs of a cell are placed according to an ascending order of the higher layer parameter *BWP-Id*.

- Frequency hopping flag - 0 or 1 bit
  - 0 bit if the higher layer parameter *frequencyHopping* is not configured for any cell configured by higher layer parameter *scheduledCellListDCI-0-3* in the scheduled cell set;
  - 1 bit according to Table 7.3.1.1.1-3 otherwise, only applicable to resource allocation type 1, as defined in Clause 6.3 of [6, TS 38.214].

  The field is only applicable to a scheduled cell configured with *frequencyHopping*, and is applied to the applicable scheduled cells independently.

- Modulation and coding scheme - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the modulation and coding scheme for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the modulation and coding scheme for the cell with the smallest serving cell index. Each block is 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214].

- New data indicator - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the new data indicator for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the new data indicator for the cell with the smallest serving cell index. If *pusch-TimeDomainAllocationListForMultiPUSCH-DCI-0-3* is configured for a cell, the number of bits for the corresponding block is equal to the maximum number of schedulable PUSCHs among all entries in the higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH-DCI-0-3* for the cell, where each bit corresponds to one scheduled PUSCH as defined in clause 6.1.4 in [6, TS 38.214]; otherwise, the corresponding block is 1 bit.

- Redundancy version - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the redundancy version for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the redundancy version for the cell with the smallest serving cell index. The number of bits for each block is determined by following:

- if *pusch-TimeDomainAllocationListForMultiPUSCH-DCI-0-3* is configured for a cell, the number of bits for the corresponding block is determined by $m_A \times m_B$, where m_A is the maximum number of schedulable PUSCHs among all entries in the higher layer parameter *pusch-TimeDomainAllocationListForMultiPUSCH-DCI-0-3* for the cell, m_B is 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-0-3* for the cell, and each m_B bit(s) corresponds to one scheduled PUSCH as defined in clause 6.1.4 in [6, TS 38.214],
  - If 0 bit is configured, rv_{id} to be applied is 0;
  - 1 bit according to Table 7.3.1.2.3-1;
  - 2 bits according to Table 7.3.1.1.1-2.
- otherwise, the corresponding block is 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-0-3* configured for the cell,
  - If 0 bit is configured, rv_{id} to be applied is 0;
  - 1 bit according to Table 7.3.1.2.3-1;
  - 2 bits according to Table 7.3.1.1.1-2.

- HARQ process number - number of bits determined by the following:

  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the HARQ process number for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the HARQ process number for the cell with the smallest serving cell index. Each block is 0, 1, 2, 3, 4 or 5 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-0-3* or *harq-ProcessNumberSizeDCI-0-3-Ext* configured for the cell corresponding to the block.

- 1st downlink assignment index - 1 or 2 bits

  - 1 bit for semi-static HARQ-ACK codebook;
  - 2 bits for dynamic HARQ-ACK codebook.

  When two HARQ-ACK codebooks are configured for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-3* is configured, if the bit width of the 1st downlink assignment index in DCI format 0_3 for one HARQ-ACK codebook is not equal to that of the 1st downlink assignment index in DCI format 0_3 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 1st downlink assignment index until the bit width of the 1st downlink assignment index in DCI format 0_3 for the two HARQ-ACK codebooks are the same.

- 2nd downlink assignment index - 0 or 2 bits:

  - 2 bits for dynamic HARQ-ACK codebook with two HARQ-ACK sub-codebooks;
  - 0 bit otherwise.

  When two HARQ-ACK codebooks are configured for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-3* is configured, if the bit width of the 2nd downlink assignment index in DCI format 0_3 for one HARQ-ACK codebook is not equal to that of the 2nd downlink assignment index in DCI format 0_3 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller 2nd downlink assignment index until the bit width of the 2nd downlink assignment index in DCI format 0_3 for the two HARQ-ACK codebooks are the same.

- TPC command for scheduled PUSCH - number of bits determined by the following:

  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the TPC command for the scheduled PUSCH for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the TPC command for the scheduled PUSCH for the cell with the smallest serving cell index. Each block is 2 bits as defined in Clause 7.1.1 of [5, TS38.213].

- SRS resource indicator - number of bits determined by the following:

  - If *sri-DCI0-3= type1a* is configured by higher layer,
    - $\max_{r \in \{1,2,\ldots,N_{cell}^{UL,2}\}} M_s(r)$ bits applying to the scheduled cells with $M_s(r) > 0$ independently, where $N_{cell}^{UL,\,2}$ is the number of cells configured by higher layer parameter *scheduledCellListDCI-0-3* in the scheduled cell set, r is mapped to the cells according to an ascending order of a serving cell index with $r = 1$ corresponding to the cell with the smallest serving cell index, and $M_s(r)$ is defined below.

- If *sri-DCI0-3= type2* is configured by higher layer,
  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the SRS resource indicator for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the SRS resource indicator for the cell with the smallest serving cell index. Each block is defined below.

$M_s(r)$ above for the case of *sri-DCI0-3= type1a* or each block above for the case of *sri-DCI0-3= type2* is defined by the following:

- $\lceil \log_2 \left(\sum_{k=1}^{\min(L_{max},\, N_{SRS})} \binom{N_{SRS}}{k} \right) \rceil$ bits according to Tables 7.3.1.1.2-28/29/30/31 if the higher layer parameter *txConfig = nonCodebook*, where N_{SRS} is the number of configured SRS resources in the first SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModList*, and associated with the higher layer parameter *usage* of value '*nonCodeBook*' and
  - if UE supports operation with *maxMIMO-Layers* and the higher layer parameter *maxMIMO-Layers* of *PUSCH-ServingCellConfig* of the serving cell is configured, L_{max} is given by that parameter
  - otherwise, L_{max} is given by the maximum number of layers for PUSCH supported by the UE for the serving cell for non-codebook based operation.
- $\lceil \log_2(N_{SRS}) \rceil$ bits according to Tables 7.3.1.1.2-32, 7.3.1.1.2-32A and 7.3.1.1.2-32B if the higher layer parameter *txConfig = codebook*, where N_{SRS} is the number of configured SRS resources in the first SRS resource set configured by higher layer parameter *srs-ResourceSetToAddModList*, and associated with the higher layer parameter *usage* of value '*codeBook*'.

- Precoding information and number of layers - number of bits determined by the following:
  - If *tpmi-DCI0-3= type1a* is configured by higher layer,
    - $\max_{r \in \{1,2,\ldots,N_{cell}^{UL,2}\}} M_p(r)$ bits applying to the scheduled cells with $M_p(r) > 0$ independently, where r is mapped to the cells according to an ascending order of a serving cell index with $r = 1$ corresponding to the cell with the smallest serving cell index, and $M_p(r)$ is defined below.
  - If *tpmi-DCI0-3= type2* is configured by higher layer,
    - block number 1, block number 2,…, block number N_{cell}^{UL}

    Each block corresponds to the precoding information and number of layers for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the precoding information and number of layers for the cell with the smallest serving cell index. Each block is defined below.

$M_p(r)$ above for the case of *tpmi-DCI0-3= type1a* or each block above for the case of *tpmi-DCI0-3= type2* is defined by the following:

- 0 bits if the higher layer parameter *txConfig = nonCodeBook*;
- 0 bits for 1 antenna port and if the higher layer parameter *txConfig = codebook*;
- 4, 5, or 6 bits according to Table 7.3.1.1.2-2 for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRank*, and *codebookSubset*;
- 4 or 5 bits according to Table 7.3.1.1.2-2A for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1, maxRank=2*, transform precoder is disabled, and according to the value of higher layer parameter *codebookSubset*;
- 4 or 6 bits according to Table 7.3.1.1.2-2B for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1, maxRank=3 or 4*, transform precoder is disabled, and according to the value of higher layer parameter *codebookSubset*;
- 2, 4, or 5 bits according to Table 7.3.1.1.2-3 for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRank* and *codebookSubset*;
- 3 or 4 bits according to Table 7.3.1.1.2-3A for 4 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1*, and according to whether transform precoder is enabled, or disabled and *maxRank*=1, and the value of higher layer parameter *codebookSubset*;

- 2 or 4 bits according to Table7.3.1.1.2-4 for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, transform precoder is disabled, and according to the values of higher layer parameters *maxRank* and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-4A for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1*, transform precoder is disabled, *maxRank=2*, and *codebookSubset=nonCoherent*;
- 1 or 3 bits according to Table7.3.1.1.2-5 for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission* is not configured or configured to *fullpowerMode2* or configured to *fullpower*, and according to whether transform precoder is enabled or disabled, and the values of higher layer parameters *maxRank* and *codebookSubset*;
- 2 bits according to Table 7.3.1.1.2-5A for 2 antenna ports, if *txConfig = codebook, ul-FullPowerTransmission =fullpowerMode1*, and according to whether transform precoder is enabled, or disabled and *maxRank*=1, and the value of higher layer parameter *codebookSubset*.

For the higher layer parameter *txConfig=codebook*, if *ul-FullPowerTransmission* is configured to *fullpowerMode2*, *maxRank* is configured to be larger than 2, and at least one SRS resource with 4 antenna ports is configured in an SRS resource set with usage set to 'codebook', and an SRS resource with 2 antenna ports is indicated via SRI in the same SRS resource set, then Table 7.3.1.1.2-4 is used.

For the higher layer parameter *txConfig = codebook*, if different SRS resources with different number of antenna ports are configured, the bitwidth is determined according to the maximum number of ports in an SRS resource among the configured SRS resources in an SRS resource set with usage set to 'codebook'. If the number of ports for a configured SRS resource in the set is less than the maximum number of ports in an SRS resource among the configured SRS resources, a number of most significant bits with value set to '0' are inserted to the field.

- Antenna ports - number of bits determined by the following:
  - If *antennaPortsDCI0-3= type1a* is configured by higher layer,
    - $\max_{r \in \{1,2,\ldots,N_{cell}^{UL,2}\}} M_A(r)$ bits applying to the scheduled cells independently, where r is mapped to the cells according to an ascending order of a serving cell index with $r=1$ corresponding to the cell with the smallest serving cell index, and $M_A(r)$ is defined below.
  - If *antennaPortsDCI0-3= type2* is configured by higher layer,
    - block number 1, block number 2,…, block number N_{cell}^{UL}

    Each block corresponds to the Antenna ports information for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the Antenna ports information for the cell with the smallest serving cell index. Each block is defined below.

$M_A(r)$ above for the case of *antennaPortsDCI0-3= type1a* or each block above for the case of *antennaPortsDCI0-3= type2* is defined by the following:

- 2 bits as defined by Tables 7.3.1.1.2-6, if transform precoder is enabled, *dmrs-Type*=1, and *maxLength*=1, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2 BPSK modulation is used;
- 2 bits as defined by Tables 7.3.1.1.2-6A, if transform precoder is enabled and *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2 BPSK modulation is used, *dmrs-Type*=1, and *maxLength*=1, where n_{SCID} is the scrambling identity for antenna ports defined in clause 6.4.1.1.1.2, TS38.211];
- 4 bits as defined by Tables 7.3.1.1.2-7, if transform precoder is enabled, *dmrs-Type*=1, and *maxLength*=2, except that *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured and π/2 BPSK modulation is used;
- 4 bits as defined by Tables 7.3.1.1.2-7A, if transform precoder is enabled and *dmrs-UplinkTransformPrecoding* and *tp-pi2BPSK* are both configured, π/2 BPSK modulation is used, *dmrs-Type*=1, and *maxLength*=2, where n_{SCID} is the scrambling identity for antenna ports defined in clause 6.4.1.1.1.2, TS38.211];
- 3 bits as defined by Tables 7.3.1.1.2-8/9/10/11, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, and *maxLength*=1, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;

- 4 bits as defined by Tables 7.3.1.1.2-12/13/14/15, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, and *maxLength*=2, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;
- 4 bits as defined by Tables 7.3.1.1.2-16/17/18/19, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, and *maxLength*=1, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;
- 5 bits as defined by Tables 7.3.1.1.2-20/21/22/23, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, and *maxLength*=2, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;
- 4 bits as defined by Tables 7.3.1.1.2-38/39/40/40A/41, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, and *maxLength*=1, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;
- 5 bits as defined by Tables 7.3.1.1.2-46/47/48/48A/49, if transform precoder is disabled, *dmrs-Type*=1, *dmrs-TypeEnh* is configured, and *maxLength*=2, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;
- 5 bits as defined by Tables 7.3.1.1.2-54/55/56/56A/57, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, and *maxLength*=1, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*;
- 6 bits as defined by Tables 7.3.1.1.2-62/63/64/64A/65, if transform precoder is disabled, *dmrs-Type*=2, *dmrs-TypeEnh* is configured, and *maxLength*=2, and the value of rank is determined according to the SRS resource indicator field if the higher layer parameter *txConfig = nonCodebook* and according to the Precoding information and number of layers field if the higher layer parameter *txConfig = codebook*.

where the number of CDM groups without data of values 1, 2, and 3 in Tables 7.3.1.1.2-6 to 7.3.1.1.2-23 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively.

If a UE is configured with both *dmrs-UplinkForPUSCH-MappingTypeA* and *dmrs-UplinkForPUSCH-MappingTypeB*, the bitwidth of this field equals $max\left\{x_A, x_B\right\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-UplinkForPUSCH-MappingTypeA* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-UplinkForPUSCH-MappingTypeB*. A number of $\left|x_A - x_B\right|$ zeros are padded in the MSB of this field, if the mapping type of the PUSCH corresponds to the smaller value of x_A and x_B.

- SRS request - $\lceil \log_2(I_{SRS}) \rceil$ bits, where I_{SRS} is the number of entries in the higher layer parameter *srs-RequestListDCI-0-3*, or 0 bit if the higher layer parameter *srs-RequestListDCI-0-3* is not configured. This field is used to indicate an entry in the higher layer parameter *srs-RequestListDCI-0-3* according to Table 7.3.1.1.4-4. Each entry in the higher layer parameter *srs-RequestListDCI-0-3* contains the 'SRS request' index for each cell in the scheduled cell set, where the 'SRS request' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'SRS request' index is defined by the following:
  - 2 bits as defined by Table 7.3.1.1.2-24 for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell; 3 bits for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second and third bits are defined by Table 7.3.1.1.2-24. This bit field may also indicate the associated CSI-RS according to Clause 6.1.1.2 of [6, TS 38.214].
- SRS offset indicator - $\lceil \log_2(I_{offset}) \rceil$ bits, where I_{offset} is the number of entries in the higher layer parameter *srs-OffsetListDCI-0-3*, or 0 bit if the higher layer parameter *srs-OffsetListDCI-0-3* is not configured. This field is used to indicate an entry in the higher layer parameter *srs-OffsetListDCI-0-3* according

to Table 7.3.1.1.4-5. Each entry in the higher layer parameter *srs-OffsetListDCI-0-3* contains the 'SRS offset indicator' index for each cell in the scheduled cell set, where the 'SRS offset indicator' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'SRS offset indicator' index is defined by the following:

- 0 bit if higher layer parameter *AvailableSlotOffset* is not configured for any aperiodic SRS resource set in the scheduled cell, or if higher layer parameter *AvailableSlotOffset* is configured for at least one aperiodic SRS resource set in the scheduled cell and the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) is 1;
- otherwise, $\lceil \log_2(K) \rceil$ bits are used to indicate available slot offset according to Table 7.3.1.1.2-37 and Clause 6.2.1 of [6, TS 38.214], where K is the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) in the scheduled cell;

- CSI request - 0, 1, 2, 3, 4, 5, or 6 bits determined by higher layer parameter *reportTriggerSize*. This field is applied to the cell with the smallest serving cell index among the scheduled cells indicated by Scheduled cells indicator field or Frequency domain resource assignment field.

- PTRS-DMRS association - number of bits determined by the following:

  - block number 1, block number 2,…, block number N_{cell}^{UL}

  Each block corresponds to the PTRS-DMRS association information for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the PTRS-DMRS association information for the cell with the smallest serving cell index. Each block is defined by the following:

  - 0 bit if *PTRS-UplinkConfig* is not configured in either *dmrs-UplinkForPUSCH-MappingTypeA* or *dmrs-UplinkForPUSCH-MappingTypeB* and transform precoder is disabled, or if transform precoder is enabled, or if *maxRank=1*;
  - 2 bits otherwise, where Table 7.3.1.1.2-25 and 7.3.1.1.2-26 are used to indicate the association between PTRS port(s) and DMRS port(s) when one PT-RS port and two PT-RS ports are configured by *maxNrofPorts* in *PTRS-UplinkConfig* respectively, and the DMRS ports are indicated by the Antenna ports field.

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and the "PTRS-DMRS association" field is present for the indicated bandwidth part but not present for the active bandwidth part, the UE assumes the "PTRS-DMRS association" field is not present for the indicated bandwidth part.

- beta_offset indicator - 0 or 2 bits

  - 0 bit if *betaOffsets = semiStatic* is configured in *uci-OnPUSCH-ListDCI-0-3* for all the cells configured by higher layer parameter *scheduledCellListDCI-0-3* in the scheduled cell set;
  - otherwise 2 bits as defined by Table 9.3-3 in [5, TS 38.213].

  When two HARQ-ACK codebooks are configured for the same serving cell and if higher layer parameter *priorityIndicatorDCI-0-3* is configured, if the bit width of the beta_offset indicator in DCI format 0_3 for one HARQ-ACK codebook is not equal to that of the beta_offset indicator in DCI format 0_3 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller beta_offset indicator until the bit width of the beta_offset indicator in DCI format 0_3 for the two HARQ-ACK codebooks are the same.

  The field is only applicable to a scheduled cell configured with *betaOffsets = dynamic* in *uci-OnPUSCH-ListDCI-0-3*, and is applied to the applicable scheduled cells independently.

- DMRS sequence initialization - 1 bit if transform precoder is disabled at least for one cell configured by higher layer parameter *ScheduledCellListDCI-0-3* in the scheduled cell set; otherwise, 0 bit.

  This field is independently applied to all the scheduled cells with transform precoder disabled, and indicated by Scheduled cells indicator field or Frequency domain resource assignment field.

- UL-SCH indicator - 1 bit. A value of "1" indicates UL-SCH shall be transmitted on the PUSCH and a value of "0" indicates UL-SCH shall not be transmitted on the PUSCH. A UE is not expected to receive a DCI format 0_3 with UL-SCH indicator of "0" and CSI request of all zero(s). This field is applied to the PUSCH on the cell with the smallest serving cell index among the scheduled cells indicated by Scheduled cells indicator field or Frequency domain resource assignment field as defined in Clause 5.2.3 of [6, TS 38.214].

- ChannelAccess-CPext-CAPC - $\max_{r \in \{1,2,\ldots,N_{cell}^2\}} M_c(r)$ bits applying to the scheduled cells with $M_c(r)>0$ independently, where N_{cell}^2 is the number of cells configured by higher layer parameter *scheduledCellListDCI-0-3* in the scheduled cell set, r is mapped to the cells according to an ascending order of a serving cell index with $r=1$ corresponding to the cell with the smallest serving cell index, and $M_c(r)$ is defined by the following:
  - 0, 1, 2, 3, 4, 5 or 6 bits. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *ul-AccessConfigListDCI-0-1* or in Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1, or for operation in frequency range 2-2 if *ChannelAccessMode2-r17* is provided; otherwise 0 bit. One or more entries from Table 7.3.1.1.2-35 or Table 7.3.1.1.2-35A are configured by the higher layer parameter *ul-AccessConfigListDCI-0-1*.
- Open-loop power control parameter set indication - $\max_{r \in \{1,2,\ldots,N_{cell}^{UL,2}\}} M_o(r)$ bits applying to the scheduled cells with $M_o(r)>0$ independently, where r is mapped to the cells according to an ascending order of a serving cell index with $r=1$ corresponding to the cell with the smallest serving cell index, and $M_o(r)$ is defined by the following:
  - 0 bit if the higher layer parameter *p0-PUSCH-SetList* is not configured for a serving cell associated to index r;
  - 1 or 2 bits otherwise,
    - 1 bit if SRS resource indicator block number r is present in the DCI format 0_3 when *SRI-DCI0-3= type2* or if $M_s(r)>0$ when *SRI-DCI0-3= type1a*;
    - 1 or 2 bits as determined by higher layer parameter *olpc-ParameterSetDCI-0-1* if SRS resource indicator block number r is not present in the DCI format 0_3 when *SRI-DCI0-3= type2* or if $M_s(r)=0$ when *SRI-DCI0-3= type1a*.
- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-0-3* is not configured; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213]. This field is applied to all the scheduled cells indicated by Scheduled cells indicator field or Frequency domain resource assignment field.
- Minimum applicable scheduling offset indicator - 0 or 1 bit
  - 0 bit if higher layer parameter *minimumSchedulingOffsetK0DCI-0-3* is not configured;
  - 1 bit otherwise. The 1 bit indication is used to determine the minimum applicable K2 for the active UL BWP and the minimum applicable K0 for the active DL BWP, if configured respectively, according to Table 7.3.1.1.2-33. If the minimum applicable K0 is indicated, the minimum applicable value of the aperiodic CSI-RS triggering offset for an active DL BWP for each scheduled cell shall be the same as the minimum applicable K0.
- SCell dormancy indication - 0 bit if higher layer parameter *dormancyDCI-0-3* or *dormancyGroupWithinActiveTime* is not configured; otherwise 1, 2, 3, 4, or 5 bits bitmap determined according to the number of different *DormancyGroupID(s)* provided by higher layer parameter *dormancyGroupWithinActiveTime,* where each bit corresponds to one of the SCell group(s) configured by higher layers parameter *dormancyGroupWithinActiveTime,* with MSB to LSB of the bitmap corresponding to the first to last configured SCell group in ascending order of *DormancyGroupID*. The field is only present when this format is carried by PDCCH on the primary cell within DRX Active Time and the UE is configured with at least two DL BWPs for an SCell.
- PDCCH monitoring adaptation indication - 0, 1 or 2 bits
  - 0 bit if higher layer parameter *pdcchMonAdaptDCI-0-3* is not enabled;
  - otherwise,
    - 1 or 2 bits, if *searchSpaceGroupIdList-r17* is not configured and if *pdcch-SkippingDurationList* is configured
      - 1 bit if the UE is configured with only one duration by *pdcch-SkippingDurationList;*
      - 2 bits if the UE is configured with more than one duration by *pdcch-SkippingDurationList*.
    - 1 or 2 bits, if *pdcch-SkippingDurationList* is not configured and if *searchSpaceGroupIdList-r17* is configured

  - 1 bit if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0 and search space set(s) with group index 1, and if the UE is not configured by *searchSpaceGroupIdList-r17* with any search space set with group index 2;
  - 2 bits if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0, search space set(s) with group index 1 and search space set(s) with group index 2;
- 2 bits, if pdcch-SkippingDurationList is configured and if searchSpaceGroupIdList-r17 is configured
- Measurement gap cancellation – 0 bit if higher layer parameter *mg-CancellationDCI-0-3* is not configured; otherwise 1 bit as defined in Clause 10.6 in [5, TS 38.213].

If *scheduledCellComboListDCI-0-3* for the cell set is configured, zeros shall be appended to DCI format 0_3 if needed until the payload size equals the size of DCI format 0_3 that is determined by the configuration of the corresponding active bandwidth part(s) of the scheduled cells in the entry which results in the largest size among the entries in the higher layer parameter *scheduledCellComboListDCI-0-3*.

If an SCell within the scheduled cell set is deactivated, the UE determines the bitwidth of the fields in DCI format 0_3 based on a UL BWP provided by *firstActiveUplinkBWP-Id* for the SCell. If the active DL BWP of an SCell within the scheduled cell set is a dormant DL BWP, the UE determines the bitwidth of the fields in DCI format 0_3 based on the UL BWP provided by *BWP-id* equal to *firstWithinActiveTimeBWP-Id* for the SCell if provided; otherwise, based on a UL BWP provided by *BWP-id* equal to *firstOutsideActiveTimeBWP-Id* for the SCell.

Table 7.3.1.1.4-1: Scheduled cell set indicator in DCI format 0_3 and DCI format 1_3

Bit field mapped to index	Scheduled cell set
0	The cell set with the smallest set ID configured by *mc-DCI-SetofCellsToAddModList*
1	The cell set with the 2nd smallest set ID configured by *mc-DCI-SetofCellsToAddModList*
2	The cell set with the 3rd smallest set ID configured by *mc-DCI-SetofCellsToAddModList*, if any
3	The cell set with the 4th smallest set ID configured by *mc-DCI-SetofCellsToAddModList*, if any

Table 7.3.1.1.4-2: Scheduled cells indicator in DCI format 0_3

Bit field mapped to index	Scheduled cells
0	The cells configured by the 1st entry in *scheduledCellComboListDCI-0-3*
1	The cells configured by the 2nd entry in *scheduledCellComboListDCI-0-3*
2	The cells configured by the 3rd entry in *scheduledCellComboListDCI-0-3*, if any
3	The cells configured by the 4th entry in *scheduledCellComboListDCI-0-3*, if any
4	The cells configured by the 5th entry in *scheduledCellComboListDCI-0-3*, if any
5	The cells configured by the 6th entry in *scheduledCellComboListDCI-0-3*, if any
6	The cells configured by the 7th entry in *scheduledCellComboListDCI-0-3*, if any
7	The cells configured by the 8th entry in *scheduledCellComboListDCI-0-3*, if any
8	The cells configured by the 9th entry in *scheduledCellComboListDCI-0-3*, if any
9	The cells configured by the 10th entry in *scheduledCellComboListDCI-0-3*, if any
10	The cells configured by the 11th entry in *scheduledCellComboListDCI-0-3*, if any
11	The cells configured by the 12th entry in *scheduledCellComboListDCI-0-3*, if any
12	The cells configured by the 13th entry in *scheduledCellComboListDCI-0-3*, if any
13	The cells configured by the 14th entry in *scheduledCellComboListDCI-0-3*, if any
14	The cells configured by the 15th entry in *scheduledCellComboListDCI-0-3*, if any
15	The cells configured by the 16th entry in *scheduledCellComboListDCI-0-3*, if any

Table 7.3.1.1.4-3: Time domain resource assignment in DCI format 0_3

Bit field mapped to index	Indicated time domain resource allocation
0	The 1st entry in *tdra-FieldIndexListDCI-0-3*
1	The 2nd entry in *tdra-FieldIndexListDCI-0-3*
2	The 3rd entry in *tdra-FieldIndexListDCI-0-3*, if any
…	…
i	The $(i+1)^{th}$ entry in *tdra-FieldIndexListDCI-0-3*, if any
…	…

Table 7.3.1.1.4-4: SRS request in DCI format 0_3

Bit field mapped to index	Triggered aperiodic SRS resource set(s)
0	The 1st entry in *srs-RequestListDCI-0-3*
1	The 2nd entry in *srs-RequestListDCI-0-3*
2	The 3rd entry in *srs-RequestListDCI-0-3*, if any
…	…
15	The 16th entry in *srs-RequestListDCI-0-3*, if any

Table 7.3.1.1.4-5: SRS offset indicator in DCI format 0_3

Bit field mapped to index	Available slot offset
0	The 1st entry in *srs-OffsetListDCI-0-3*
1	The 2nd entry in *srs-OffsetListDCI-0-3*
2	The 3rd entry in *srs-OffsetListDCI-0-3*, if any
…	…
7	The 8th entry in *srs-OffsetListDCI-0-3*, if any

7.3.1.2 DCI formats for scheduling of PDSCH

7.3.1.2.1 Format 1_0

DCI format 1_0 is used for the scheduling of PDSCH in one DL cell.

The following information is transmitted by means of the DCI format 1_0 with CRC scrambled by C-RNTI or CS-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bits
  - The value of this bit field is always set to 1, indicating a DL DCI format
- Frequency domain resource assignment - $\lceil \log_2(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}+1)/2) \rceil$ bits where $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is given by Clause 7.3.1.0

If the CRC of the DCI format 1_0 is scrambled by C-RNTI and the "Frequency domain resource assignment" field are of all ones, the DCI format 1_0 is for random access procedure initiated by a PDCCH order, with all remaining fields set as follows:

- Random Access Preamble index - 6 bits according to *ra-PreambleIndex* in Clause 5.1.2 of [8, TS38.321]
- UL/SUL indicator - 1 bit.
  - If the Cell indicator field is absent or the Cell indicator field indicates serving cell, if the value of the "Random Access Preamble index" is not all zeros and if the UE is configured with *supplementaryUplink* in *ServingCellConfig* in the cell, this field indicates which UL carrier in the cell to transmit the PRACH according to Table 7.3.1.1.1-1;
  - If the Cell indicator field indicates a candidate cell, if the value of the "Random Access Preamble index" is not all zeros and if the UE is configured with *ltm-EarlyUL-SyncConfigSUL* in *LTM-Candidate* for the candidate cell, this field indicates which UL carrier in the candidate cell to transmit the PRACH according to Table 7.3.1.1.1-1;
  - Otherwise, this field is reserved.
- SS/PBCH index - 6 bits. If the value of the "Random Access Preamble index" is not all zeros, this field indicates the SS/PBCH that shall be used to determine the RACH occasion for the PRACH transmission; otherwise, this field is reserved.
- PRACH Mask index - 4 bits. If the value of the "Random Access Preamble index" is not all zeros, this field indicates the RACH occasion associated with the SS/PBCH indicated by "SS/PBCH index" for the PRACH transmission, according to Clause 5.1.1 of [8, TS38.321]; otherwise, this field is reserved
- Cell indicator - $\lceil \log_2(C+1) \rceil$ bits indicating the cell for the corresponding PRACH transmission if the UE is configured with higher layer parameter *EarlyUL-SyncConfig*, where C is the number of candidate cells configured with higher layer parameter *EarlyUL-SyncConfig*; 0 bit otherwise. The bit field index 0 of the cell

indicator field is mapped to the serving cell, and other bit field indexes are mapped to the candidate cells configured with higher layer parameter *EarlyUL-SyncConfig* according to an ascending order of a candidate identity configured by *ltm-CandidateId*, with the bit field index 1 mapped to the candidate cell with the smallest candidate identity.

- PRACH association indicator - 0 or 1 bit
  - 1bit if the UE is provided with *tag2-Id*, and the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for the first CORESETs, and is provided *coresetPoolIndex* with value 1 for the second CORESETs. This field is reserved if the cell indicated by Cell indicator field is a candidate cell.
    - This field indicates the PCI associated with the PRACH transmission if the UE is provided *SSB-MTC-AddtionalPCI*. The bit field index 0 of this field is mapped to the PCI of the serving cell, and the bit field index 1 of this field is mapped to the additional PCI associated with active TCI states.
    - This field indicates the PL-RS for the PRACH transmission if the UE is not provided *SSB-MTC-AddtionalPCI*. The bit field index 0 of this field is mapped to the DL RS that the DM-RS of the PDCCH order is quasi-collocated with, and the bit field index 1 of this field is mapped to the SS/PBCH indicated by the SS/PBCH index field in this DCI format.
  - 1bit if the UE is provided with *tag2-Id* and *SSB-MTC-AddtionalPCI*, and the UE is not configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided, and the UE is provided with *PrachAssociationIndicator_InDCI_format_1_0*, and the UE is not provided with *pl-Offset* for any *TCI-State* in *dl-OrJointTCI-StateList* or any *TCI-UL-State* in *ul-TCI-State-List*. This field is reserved if the cell indicated by Cell indicator field is a candidate cell.
    - This field indicates the PCI and the PL-RS for the PRACH transmission. The bit field index 0 of this field is mapped to the PCI of the serving cell and the DL RS that the DM-RS of the PDCCH order is quasi-collocated with, and the bit field index 1 of this field is mapped to the additional PCI associated with active TCI states and the SS/PBCH indicated by the SS/PBCH index field in this DCI format.
  - 0 bit otherwise.
- PRACH retransmission indicator - 0 or 1 bit
  - 1bit if the UE is configured with higher layer parameter *EarlyUL-SyncConfig*. This field indicates initial transmission or retransmission of PRACH according to Table 7.3.1.2.1-3 if the cell indicated by Cell indicator field is a candidate cell, and this field is reserved if the value of Cell indicator field is zero.
  - 0 bit otherwise.
- Pathloss offset indicator – 0 or 1 bit
  - 1 bit if the UE is configured with higher layer parameter *plOffsetInPrach_InDCI* and at least one configured TCI state for the serving cell is configured with *plOffset*.
    - If there is only one indicated joint/UL TCI state, the bit field index 0 of this field indicates that no pathloss offset is applied for the PRACH transmission, and the bit field index 1 of this field indicates that the pathloss offset configured in the indicated joint/UL TCI state is applied for the PRACH transmission.
    - If there are two indicated joint/UL TCI states, the bit field index 0 of this field indicates that the pathloss offset configured in the first indicated joint/UL TCI state is applied for the PRACH transmission, and the bit field index 1 of this field indicates that the pathloss offset configured in the second indicated joint/UL TCI state is applied for the PRACH transmission.
  - 0 bit otherwise.
- PRACH resource indicator - 0 or 1 bit
  - 1 bit if the UE is configured with higher layer parameter *addl-RACH-Config-Adaptation*. This field indicates the availability of the PRACH resource configured by *addl-RACH-Config-Adaptation* according to Table 7.3.1.2.1-5.
  - 0 bit otherwise.
- RACH occasion indicator - 0 or 1 bit
  - 1 bit if the UE is configured with higher layer parameter *sbfd-RACHSingleConfig* or *sbfd-RACHDualConfig*. If the value of the "Random Access Preamble index" is not all zeros, this field indicates the RACH occasion for PRACH transmission according to Table 7.3.1.2.1-6; otherwise, this field is reserved.
  - 0 bit otherwise.
- Reserved bits - a number of bits as determined by the following:

- $(12 - Y_1 - Y_2 - Y_3 - Y_4 - Y_5)$ bits for operation in a cell with shared spectrum channel access in frequency range 1 or when the DCI format is monitored in common search space for operation in a cell in frequency range 2-2;
- $(10 - Y_1 - Y_2 - Y_3 - Y_4 - Y_5)$ bits otherwise;

where,

- $Y_1 = 0$ if the UE is not configured with higher layer parameter *EarlyUL-SyncConfig*;
  $Y_1 = \lceil \log_2(C+1) \rceil + 1$ otherwise.
- $Y_2 = 0$ if the "PRACH association indicator" field is not present in this DCI format; $Y_2 = 1$ otherwise.
- $Y_3 = 0$ if the "Pathloss offset indicator" field is not present in this DCI format; $Y_3 = 1$ otherwise.
- $Y_4 = 0$ if the "PRACH resource indicator" field is not present in this DCI format; $Y_4 = 1$ otherwise.
- $Y_5 = 0$ if the "RACH occasion indicator " field is not present in this DCI format; $Y_5 = 1$ otherwise.

Otherwise, all remaining fields are set as follows:

- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS 38.214]
- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- HARQ process number - 4 bits
- Downlink assignment index - 2 bits as defined in Clause 9.1.3 of [5, TS 38.213], as counter DAI
- TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS 38.213]
- PUCCH resource indicator - 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]
- PDSCH-to-HARQ_feedback timing indicator - 3 bits as defined in Clause 9.2.3 of [5, TS38.213]
- ChannelAccess-CPext - 2 bits indicating combinations of channel access type and CP extension as defined in Table 7.3.1.1.1-4, or Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1; 2 bits indicating channel access type as defined in Table 7.3.1.1.1-4B if *ChannelAccessMode2-r17* is provided for operation in a cell in frequency range 2-2; 0 bits otherwise
- Reserved bits - 2 bits when the DCI format is monitored in common search space for operation in a cell in frequency range 2-2 and the number of bits for the field of 'ChannelAccess-CPext' is 0; 0 bits otherwise

The following information is transmitted by means of the DCI format 1_0 with CRC scrambled by P-RNTI:

- Short Messages Indicator - 2 bits according to Table 7.3.1.2.1-1. If *addl-RACH-Config-Adaptation* is configured and this field is set to "00", all the remaining fields are reserved except the "Short Messages" field.
- Short Messages - 8 bits, according to Clause 6.5 of [9, TS38.331]. If only the scheduling information for Paging, and TRS availability indication if *trs-ResourceSetConfig* or *trs-ResourceSetConfig-r18* is configured, are carried, all the bits in this bit field are reserved. If *addl-RACH-Config-Adaptation* is configured and the "Short Messages Indicator" field is set to "00" or "01", all the bits in this bit field are reserved, except the bit indicating the availability of the PRACH resource configured by *addl-RACH-Config-Adaptation* according to Clause 6.5 of [9, TS38.331].
- Frequency domain resource assignment - $\lceil \log_2(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}+1)/2) \rceil$ bits. If only the short message, and TRS availability indication if *trs-ResourceSetConfig* or *trs-ResourceSetConfig-r18* is configured, are carried, this bit field is reserved. If *addl-RACH-Config-Adaptation* is configured and the "Short Messages Indicator" field is set to "10", this bit field is reserved.
  - $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is the size of CORESET 0
- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS38.214]. If only the short message, and TRS availability indication if *trs-ResourceSetConfig* or *trs-ResourceSetConfig-r18* is

configured, are carried, this bit field is reserved. If *addl-RACH-Config-Adaptation* is configured and the "Short Messages Indicator" field is set to "10", this bit field is reserved.

- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5. If only the short message, and TRS availability indication if *trs-ResourceSetConfig* or *trs-ResourceSetConfig-r18* is configured, are carried, this bit field is reserved. If *addl-RACH-Config-Adaptation* is configured and the "Short Messages Indicator" field is set to "10", this bit field is reserved.
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3 of [6, TS38.214], using Table 5.1.3.1-1. If only the short message, and TRS availability indication if *trs-ResourceSetConfig* or *trs-ResourceSetConfig-r18* is configured, are carried, this bit field is reserved. If *addl-RACH-Config-Adaptation* is configured and the "Short Messages Indicator" field is set to "10", this bit field is reserved.
- TB scaling - 2 bits as defined in Clause 5.1.3.2 of [6, TS38.214]. If only the short message, and TRS availability indication if *trs-ResourceSetConfig* or *trs-ResourceSetConfig-r18* is configured, are carried, this bit field is reserved. If *addl-RACH-Config-Adaptation* is configured and the "Short Messages Indicator" field is set to "10", this bit field is reserved.
- TRS availability indication - 1, 2, 3, 4, 5, or 6 bits, where the number of bits is equal to one plus the highest value of all the *indBitID*(s) provided by the *trs-ResourceSetConfig* if configured or the number of bits is equal to one plus the highest value of all the *indBitID-r18*(s) provided by the *trs-ResourceSetConfig-r18* if configured; 0 bits otherwise.
- Reserved bits - (8 - *M*) bits for operation in a cell with shared spectrum channel access in frequency range 1 or for operation in a cell in frequency range 2-2; (6 - *M*) bits for operation in a cell without shared spectrum channel access, where the value of *M* is the number of bits for the field of 'TRS availability indication' as defined above.

The following information is transmitted by means of the DCI format 1_0 with CRC scrambled by SI-RNTI:

- Frequency domain resource assignment - $\lceil \log_2\left(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}\left(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}+1\right)/2\right)\rceil$ bits
  - $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is the size of CORESET 0
- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS38.214]
- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3 of [6, TS38.214], using Table 5.1.3.1-1
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- System information indicator - 1 bit as defined in Table 7.3.1.2.1-2
- Reserved bits - 17 bits for operation in a cell with shared spectrum channel access in frequency range 1 or for operation in a cell in frequency range 2-2; otherwise 15 bits

The following information is transmitted by means of the DCI format 1_0 with CRC scrambled by RA-RNTI or MsgB-RNTI:

- Frequency domain resource assignment - $\lceil \log_2\left(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}\left(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}+1\right)/2\right)\rceil$ bits
  - $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is the size of CORESET 0 if CORESET 0 is configured for the cell and $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is the size of initial DL bandwidth part if CORESET 0 is not configured for the cell
- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS38.214]
- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3 of [6, TS38.214], using Table 5.1.3.1-1
- TB scaling - 2 bits as defined in Clause 5.1.3.2 of [6, TS38.214]
- LSBs of SFN - 2 bits for the DCI format 1_0 with CRC scrambled by MsgB-RNTI as defined in Clause 8.2A of [5, TS 38.213] if *msgB-responseWindow* is configured to be larger than 10 ms; or 2 bits for the DCI format 1_0 with CRC scrambled by RA-RNTI as defined in Clause 8.2 of [5, TS 38.213] for operation in a cell with

shared spectrum channel access if *ra-ResponseWindow or ra-ResponseWindow-v1610* is configured to be larger than 10 ms; 0 bit otherwise

- Reserved bits - (16 - *A*) bits for operation in a cell without shared spectrum access in frequency range 1 and frequency range 2-1, (18 - *A*) for operation in a cell with shared spectrum access in frequency range 1 or for operation in a cell in frequency range 2-2, where the value of *A* is the number of bits for the field of 'LSBs of SFN' as defined above

The following information is transmitted by means of the DCI format 1_0 with CRC scrambled by TC-RNTI:

- Identifier for DCI formats - 1 bit
  - The value of this bit field is always set to 1, indicating a DL DCI format
- Frequency domain resource assignment - $\lceil \log_2(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}(N_{\mathrm{RB}}^{\mathrm{DL,BWP}}+1)/2) \rceil$ bits
  - $N_{\mathrm{RB}}^{\mathrm{DL,BWP}}$ is the size of CORESET 0
- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS38.214]
- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5
- Modulation and coding scheme - 5 bits
  - If the UE indicates the support of repetition of PDSCH scheduled by DCI format 1_0 with CRC scrambled by TC-RNTI, 5 bits as defined in Clause 5.1.2.1 and Clause 5.1.3.1 of [6, TS38.214];
  - otherwise 5 bits as defined in Clause 5.1.3 of [6, TS38.214], using Table 5.1.3.1-1.
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- HARQ process number - 4 bits
- Downlink assignment index - 2 bits
  - 2 bits indicating the number of repetitions for PUCCH as defined in clause 9.2.6 of [5, TS38.213] according to Table 7.3.1.2.1-4, if the higher layer parameter *numberOfMsg4HARQ-ACK-Repetitions* is configured with at least two repetition factors and the UE has indicated capability of PUCCH repetition on common PUCCH resource [8, TS38.321];
  - otherwise, reserved.
- TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS38.213]
- PUCCH resource indicator - 3 bits as defined in Clause 9.2.3 of [5, TS38.213]
- PDSCH-to-HARQ_feedback timing indicator - 3 bits as defined in Clause 9.2.3 of [5, TS38.213]
- ChannelAccess-CPext - 2 bits indicating combinations of channel access type and CP extension as defined in Table 7.3.1.1.1-4, or Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1; 2 bits indicating channel access type as defined in Table 7.3.1.1.1-4B if *ChannelAccessMode2-r17* is provided for operation in a cell in frequency range 2-2; otherwise 0 bit
- Reserved bits - 2 bits when the DCI format is monitored in common search space for operation in a cell in frequency range 2-2 and the number of bits for the field of 'ChannelAccess-CPext' is 0; 0 bits otherwise

Table 7.3.1.2.1-1: Short Message indicator

Bit field	Short Message indicator
00	Only indication of availability of the PRACH resource configured by *addl-RACH-Config-Adaptation* is present in the DCI, if *addl-RACH-Config-Adaptation* is configured; reserved otherwise
01	Only scheduling information for Paging, TRS availability indication if *trs-ResourceSetConfig* is configured, and indication of availability of the PRACH resource configured by *addl-RACH-Config-Adaptation* if *addl-RACH-Config-Adaptation* is configured, are present in the DCI
10	Only short message, TRS availability indication if *trs-ResourceSetConfig* is configured, and indication of availability of the PRACH resource configured by *addl-RACH-Config-Adaptation* if *addl-RACH-Config-Adaptation* is configured, are present in the DCI
11	Scheduling information for Paging, TRS availability indication if *trs-ResourceSetConfig* is configured, short message, and indication of availability of the PRACH resource configured by *addl-RACH-Config-Adaptation* if *addl-RACH-Config-Adaptation* is configured, are present in the DCI

Table 7.3.1.2.1-2: System information indicator

Bit field	System information indicator
0	SIB1 [9, TS38.331, Clause 5.2.1]
1	SI message [9, TS38.331, Clause 5.2.1]

Table 7.3.1.2.1-3: PRACH retransmission indicator

Bit field	PRACH retransmission indicator
0	Initial transmission of PRACH
1	Retransmission of PRACH

Table 7.3.1.2.1-4: Number of repetitions N_{PUCCH}^{repeat} as a function of 2 bits of Downlink assignment index field

Bit field	N_{PUCCH}^{repeat}
00	First repetition factor configured by *numberOfMsg4HARQ-ACK-Repetitions*
01	Second repetition factor configured by *numberOfMsg4HARQ-ACK-Repetitions*
10	Third repetition factor configured by *numberOfMsg4HARQ-ACK-Repetitions* if provided, otherwise reserved
11	Fourth repetition factor configured by *numberOfMsg4HARQ-ACK-Repetitions* if provided, otherwise reserved

Table 7.3.1.2.1-5: PRACH resource indicator

Bit field	PRACH resource indicator
0	The PRACH resource configured by *addl-RACH-Config-Adaptation* is not available for the PRACH transmission initiated by the PDCCH order
1	The PRACH resource configured by *addl-RACH-Config-Adaptation* is available for the PRACH transmission initiated by the PDCCH order

Table 7.3.1.2.1-6: RACH occasion indicator

Bit field	RACH occasion indicator
0	The RACH occasion for the PRACH transmission is from the first PRACH occasions as defined in Clause 8 of [5, TR38.213]
1	The RACH occasion for the PRACH transmission is from the second PRACH occasions as defined in Clause 8 of [5, TR38.213]

7.3.1.2.2 Format 1_1

DCI format 1_1 is used for the scheduling of one or multiple PDSCH in one cell.

The following information is transmitted by means of the DCI format 1_1 with CRC scrambled by C-RNTI or CS-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bits
  - The value of this bit field is always set to 1, indicating a DL DCI format
- Carrier indicator - 0 or 3 bits as defined in Clause 10.1 of [5, TS 38.213]. This field is reserved when this format is carried by PDCCH on the primary cell and the UE is configured for scheduling on the primary cell from an SCell, with the same number of bits as that in this format carried by PDCCH on the SCell for scheduling on the primary cell.
- Bandwidth part indicator - 0, 1 or 2 bits as determined by the number of DL BWPs $n_{\text{BWP,RRC}}$ configured by higher layers, excluding the initial DL bandwidth part. The bitwidth for this field is determined as $\lceil \log_2(n_{\text{BWP}}) \rceil$ bits, where
  - $n_{\text{BWP}} = n_{\text{BWP,RRC}} + 1$ if $n_{\text{BWP,RRC}} \leq 3$, in which case the bandwidth part indicator is equivalent to the ascending order of the higher layer parameter *BWP-Id*;
  - otherwise $n_{\text{BWP}} = n_{\text{BWP,RRC}}$, in which case the bandwidth part indicator is defined in Table 7.3.1.1.2-1;

  If a UE does not support active BWP change via DCI, the UE ignores this bit field.
- Frequency domain resource assignment - number of bits determined by the following, where $N_{\text{RB}}^{\text{DL,BWP}}$ is the size of the active DL bandwidth part:
  - N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 5.1.2.2.1 of [6, TS38.214],
  - $\lceil \log_2(N_{\text{RB}}^{\text{DL,BWP}}(N_{\text{RB}}^{\text{DL,BWP}}+1)/2) \rceil$ bits if only resource allocation type 1 is configured, or
  - $\max(\lceil \log_2(N_{\text{RB}}^{\text{DL,BWP}}(N_{\text{RB}}^{\text{DL,BWP}}+1)/2) \rceil, N_{\text{RBG}}) + 1$ bits if *resourceAllocation* is configured as '*dynamicSwitch*'.
  - If *resourceAllocation* is configured as '*dynamicSwitch*', the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
  - For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 5.1.2.2.1 of [6, TS 38.214].
  - For resource allocation type 1, the $\lceil \log_2(N_{\text{RB}}^{\text{DL,BWP}}(N_{\text{RB}}^{\text{DL,BWP}}+1)/2) \rceil$ LSBs provide the resource allocation as defined in Clause 5.1.2.2.2 of [6, TS 38.214]

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and if *resourceAllocation* is configured as '*dynamicSwitch*' for the indicated bandwidth part, the UE assumes resource allocation type 0 for the indicated bandwidth part if the bitwidth of the "Frequency domain resource assignment" field of the active bandwidth part is smaller than the bitwidth of the "Frequency domain resource assignment" field of the indicated bandwidth part.
- Time domain resource assignment - 0, 1, 2, 3, 4, 5 or 6 bits
  - If the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH* is not configured and if the higher layer parameter *pdsch-TimeDomainAllocationList* is configured, 0, 1, 2, 3 or 4 bits as defined in Clause 5.1.2.1 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I

is the number of entries in the higher layer parameter *pdsch-TimeDomainAllocationList* if the higher layer parameter is configured;
- if the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH* is configured, 0, 1, 2, 3, 4, 5 or 6 bits as defined in Clause 5.1.2.1 of [6, TS38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH*;
- otherwise I is the number of entries in the default table.

- VRB-to-PRB mapping - 0 or 1 bit:
  - 0 bit if only resource allocation type 0 is configured or if interleaved VRB-to-PRB mapping is not configured by high layers;
  - 1 bit according to Table 7.3.1.2.2-5 otherwise, only applicable to resource allocation type 1, as defined in Clause 7.3.1.6 of [4, TS 38.211].
- PRB bundling size indicator - 0 bit if the higher layer parameter *prb-BundlingType* is not configured or is set to 'staticBundling', or 1 bit if the higher layer parameter *prb-BundlingType* is set to 'dynamicBundling' according to Clause 5.1.2.3 of [6, TS 38.214].
- Rate matching indicator - 0, 1, or 2 bits according to higher layer parameters *rateMatchPatternGroup1* and *rateMatchPatternGroup2*, where the MSB is used to indicate *rateMatchPatternGroup1* and the LSB is used to indicate *rateMatchPatternGroup2* when there are two groups.
- ZP CSI-RS trigger - 0, 1, or 2 bits as defined in Clause 5.1.4.2 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(n_{ZP}+1) \rceil$ bits, where n_{ZP} is the number of aperiodic ZP CSI-RS resource sets configured by higher layer.

For transport block 1:

- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3.1 of [6, TS 38.214]
- New data indicator - 1 bit if the number of scheduled PDSCH indicated by the Time domain resource assignment field is 1; otherwise 2, 3, 4, 5, 6, 7 or 8 bits determined based on the maximum number of schedulable PDSCH among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH*, where each bit corresponds to one scheduled PDSCH as defined in clause 5.1.3 in [6, TS 38.214].
- Redundancy version - number of bits determined by the following:
  - 2 bits as defined in Table 7.3.1.1.1-2 if the number of scheduled PDSCH indicated by the Time domain resource assignment field is 1;
  - otherwise 2, 3, 4, 5, 6, 7 or 8 bits determined by the maximum number of schedulable PDSCHs among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH*, where each bit corresponds to one scheduled PDSCH as defined in clause 5.1.3 in [6, TS 38.214] and redundancy version is determined according to Table 7.3.1.1.2-34.

For transport block 2 (only present if *maxNrofCodeWordsScheduledByDCI* equals 2):

- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3.1 of [6, TS 38.214]
- New data indicator - 1 bit if the number of scheduled PDSCH indicated by the Time domain resource assignment field is 1; otherwise 2, 3, 4, 5, 6, 7 or 8 bits determined based on the maximum number of schedulable PDSCH among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH*, where each bit corresponds to one scheduled PDSCH as defined in clause 5.1.3 in [6, TS 38.214].
- Redundancy version - number of bits determined by the following:
  - 2 bits as defined in Table 7.3.1.1.1-2 if the number of scheduled PDSCH indicated by the Time domain resource assignment field is 1;
  - otherwise 2, 3, 4, 5, 6, 7 or 8 bits determined by the maximum number of schedulable PDSCHs among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH*, where each bit corresponds to one scheduled PDSCH as defined in clause 5.1.3 in [6, TS 38.214] and redundancy version is determined according to Table 7.3.1.1.2-34.

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and the value of *maxNrofCodeWordsScheduledByDCI* for the indicated bandwidth part equals 2 and the value of *maxNrofCodeWordsScheduledByDCI* for the active bandwidth part equals 1, the UE assumes zeros are padded when interpreting the "Modulation and coding scheme", "New data indicator", and "Redundancy version" fields of transport block 2 according to Clause 12 of [5, TS38.213], and the UE ignores the "Modulation and coding scheme", "New data indicator", and "Redundancy version" fields of transport block 2 for the indicated bandwidth part.

- HARQ process number - 5 bits if higher layer parameter *harq-ProcessNumberSizeDCI-1-1* or *harq-ProcessNumberSizeDCI-1-1-Ext* is configured; otherwise 4 bits
- Downlink assignment index - number of bits as defined in the following
  - 6 bits if more than one serving cell are configured in the DL and the higher layer parameter *nfi-TotalDAI-Included* is configured. The 4 MSB bits are the counter DAI and the total DAI for the scheduled PDSCH group, and the 2 LSB bits are the total DAI for the non-scheduled PDSCH group.
  - 4 bits if only one serving cell is configured in the DL and the higher layer parameter *nfi-TotalDAI-Included* is configured. The 2 MSB bits are the counter DAI for the scheduled PDSCH group, and the 2 LSB bits are the total DAI for the non-scheduled PDSCH group;
  - 4 bits if more than one serving cell are configured in the DL, the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic* or *pdsch-HARQ-ACK-Codebook-r16= enhancedDynamic*, and *nfi-TotalDAI-Included* is not configured, where the 2 MSB bits are the counter DAI and the 2 LSB bits are the total DAI;
  - 4 bits if one serving cell is configured in the DL, and the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic*, and the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for one or more first CORESETs and is provided *coresetPoolIndex* with value 1 for one or more second CORESETs, and is provided *ackNackFeedbackMode = joint*, where the 2 MSB bits are the counter DAI and the 2 LSB bits are the total DAI;
  - 2 bits if only one serving cell is configured in the DL, the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic* or *pdsch-HARQ-ACK-Codebook-r16=enhancedDynamic*, and *nfi-TotalDAI-Included* is not configured, when the UE is not configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided or the UE is not configured with *ackNackFeedbackMode = joint*, where the 2 bits are the counter DAI;
  - 0 bits otherwise.

  If the UE is configured with a PUCCH-SCell, the number of serving cells is determined within a PUCCH group.

  If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-Codebook* is replaced by *pdsch-HARQ-ACK-Codebook-secondaryPUCCHgroup-r16* if present for the secondary PUCCH group.

  If higher layer parameter *priorityIndicatorDCI-1-1* is configured, if the bit width of the Downlink assignment index in DCI format 1_1 for one HARQ-ACK codebook is not equal to that of the Downlink assignment index in DCI format 1_1 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller Downlink assignment index until the bit width of the Downlink assignment index in DCI format 1_1 for the two HARQ-ACK codebooks are the same.
- TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS 38.213]
- Second TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS 38.213] if higher layer parameter *SecondTPCFieldDCI-1-1* is configured; 0 bit otherwise.
- PUCCH resource indicator - 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]
- PDSCH-to-HARQ_feedback timing indicator - 0, 1, 2, or 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *dl-DataToUL-ACK*.

  If higher layer parameter *priorityIndicatorDCI-1-1* is configured, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_1 for one HARQ-ACK codebook is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_1 for the other HARQ-ACK codebook on the same cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_1 for the two HARQ-ACK codebooks are the same.

  If higher layer parameter *pucch-sSCellDyn* is configured, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_1 associated with one cell for PUCCH transmission is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_1 associated with the other cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_1 associated with the two cells are the same.

  If the UE is configured with a PUCCH-SCell, *pucch-sSCellDyn* is replaced by *pucch-sSCellDynSecondaryPUCCHgroup* for the secondary PUCCH group.
- One-shot HARQ-ACK request - 0 or 1 bit.

- 1 bit if higher layer parameter *pdsch-HARQ-ACK-OneShotFeedback-r16* or *pdsch-HARQ-ACK-EnhType3ToAddModList* is configured;
- 0 bit otherwise.

If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-EnhType3ToAddModList* is replaced by *pdsch-HARQ-ACK-EnhType3SecondaryToAddModList* for the secondary PUCCH group.

- Enhanced Type 3 codebook indicator - 0, 1, 2, or 3 bits.
  - 0 bit if *pdsch-HARQ-ACK-EnhType3DCI-Field* is not configured;
  - $\lceil \log_2(n_{CB}) \rceil$ bits otherwise, where n_{CB} is the number of entries in the higher layer parameter *pdsch-HARQ-ACK-EnhType3ToAddModList.*

  If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-EnhType3DCI-Field* is replaced by *pdsch-HARQ-ACK-EnhType3DCI-FieldSecondaryPUCCHgroup* for the secondary PUCCH group, and *pdsch-HARQ-ACK-EnhType3ToAddModList* is replaced by *pdsch-HARQ-ACK-EnhType3SecondaryList* for the secondary PUCCH group.
- PDSCH group index - 0 or 1 bit.
  - 1 bit if the higher layer parameter *pdsch-HARQ-ACK-Codebook-r16= enhancedDynamic*;
  - 0 bit otherwise.
- New feedback indicator - 0, 1 or 2 bits.
  - 1 bit if the higher layer parameter *pdsch-HARQ-ACK-Codebook-r16= enhancedDynamic* and the higher layer parameter *nfi-TotalDAI-Included* is not configured;
  - 2 bits if the higher layer parameter *pdsch-HARQ-ACK-Codebook-r16= enhancedDynamic* and the higher layer parameter *nfi-TotalDAI-Included=true*; the MSB corresponds to the scheduled PDSCH group, and the LSB corresponds to the non-scheduled PDSCH group, as defined in [TS38.213] clause 9.1.3.3
  - 0 bit otherwise.
- Number of requested PDSCH group(s) - 0 or 1 bit.
  - 1 bit if the higher layer parameter *pdsch-HARQ-ACK-Codebook-r16= enhancedDynamic*;
  - 0 bit otherwise.
- HARQ-ACK retransmission indicator - 0 or 1 bit.
  - 1 bit if higher layer parameter *pdsch-HARQ-ACK-Retx* is configured.
  - 0 bit otherwise.

  If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-Retx* is replaced by *pdsch-HARQ-ACK-RetxSecondaryPUCCHgroup* for the secondary PUCCH group.
- Antenna port(s) - 4, 5, 6, 7 or 8 bits as defined by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10 and Tables 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A, where the number of CDM groups without data of values 1, 2, and 3 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively. The antenna ports $\{p_{0,\ldots,}p_{\upsilon-1}\}$ shall be determined according to the ordering of DMRS port(s) given by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10 or Tables 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A. When a UE not configured with *dl-OrJointTCI-StateList* receives an activation command that maps at least one codepoint of DCI field '*Transmission Configuration Indication*' to two TCI states, or when a UE configured with *dl-OrJointTCI-StateList* is having two indicated TCI states, the UE shall use Table 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A; otherwise, it shall use Tables 7.3.1.2.2-1/2/3/4/7/8/9/10. The UE can receive an entry with DMRS ports equals to 1000, 1002, 1003 when two the UE is not configured with *dl-OrJointTCI-StateList* and TCI states are indicated in a codepoint of DCI field '*Transmission Configuration Indication*', or when the UE configured with *dl-OrJointTCI-StateList* is having two indicated TCI states to be applied to PDSCH.

  If a UE is configured with both *dmrs-DownlinkForPDSCH-MappingTypeA* and *dmrs-DownlinkForPDSCH-MappingTypeB*, the bitwidth of this field equals $\max\{x_A, x_B\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeA* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeB*. A number of $|x_A - x_B|$ zeros are padded in the MSB of this field, if the mapping type of the PDSCH corresponds to the smaller value of x_A and x_B.
- Transmission configuration indication - 0 bit if higher layer parameter *tci-PresentInDCI* is not enabled; otherwise 3 bits as defined in Clause 5.1.5 of [6, TS38.214].

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part,

- if the higher layer parameter *tci-PresentInDCI* is not enabled for the CORESET used for the PDCCH carrying the DCI format 1_1,
  - the UE assumes *tci-PresentInDCI* is not enabled for all CORESETs in the indicated bandwidth part;
- otherwise,
  - the UE assumes *tci-PresentInDCI* is enabled for all CORESETs in the indicated bandwidth part.

- TCI selection - 0 bit if higher layer parameter *tci-SelectionPresentInDCI* is not configured; otherwise 2 bits according to Table 7.3.1.2.2-11.

- SRS request - 2 bits as defined by Table 7.3.1.1.2-24 for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell; 3 bits for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second and third bits are defined by Table 7.3.1.1.2-24. This bit field may also indicate the associated CSI-RS according to Clause 6.1.1.2 of [6, TS 38.214].

- SRS offset indicator - 0, 1 or 2 bits.

  - 0 bit if higher layer parameter *AvailableSlotOffset* is not configured for any aperiodic SRS resource set in the scheduled cell, or if higher layer parameter *AvailableSlotOffset* is configured for at least one aperiodic SRS resource set in the scheduled cell and the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) is 1;
  - otherwise, $\lceil \log_2(K) \rceil$ bits are used to indicate available slot offset according to Table 7.3.1.1.2-37 and Clause 6.2.1 of [6, TS 38.214], where K is the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) in the scheduled cell;

- CBG transmission information (CBGTI) - 0 bit if higher layer parameter *PDSCH-CodeBlockGroupTransmission* for PDSCH is not configured, otherwise, 2, 4, 6, or 8 bits as defined in Clause 5.1.7 of [6, TS38.214], determined by the higher layer parameters *maxCodeBlockGroupsPerTransportBlock* and *maxNrofCodeWordsScheduledByDCI* for the PDSCH.

  If higher layer parameter *priorityIndicatorDCI-1-1* is configured, if the bit width of the CBG transmission information in DCI format 1_1 for one HARQ-ACK codebook is not equal to that of the CBG transmission information in DCI format 1_1 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller CBG transmission information until the bit width of the CBG transmission information in DCI format 1_1 for the two HARQ-ACK codebooks are the same.

- CBG flushing out information (CBGFI) - 1 bit if higher layer parameter *codeBlockGroupFlushIndicator* is configured as "TRUE", 0 bit otherwise.

  If higher layer parameter *priorityIndicatorDCI-1-1* is configured, if the bit width of the CBG flushing out information in DCI format 1_1 for one HARQ-ACK codebook is not equal to that of the CBG flushing out information in DCI format 1_1 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller CBG flushing out information until the bit width of the CBG flushing out information in DCI format 1_1 for the two HARQ-ACK codebooks are the same.

- DMRS sequence initialization - 1 bit.

- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-1-1* is not configured; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213].

- ChannelAccess-CPext - 0, 1, 2, 3 or 4 bits. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *ul-AccessConfigListDCI-1-1* or in Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1, or for operation in frequency range 2-2 if *ChannelAccessMode2-r17* is provided; otherwise 0 bit. One or more entries from Table 7.3.1.2.2-6 or Table 7.3.1.2.2-6A are configured by the higher layer parameter *ul-AccessConfigListDCI-1-1*.

- Minimum applicable scheduling offset indicator - 0 or 1 bit

  - 0 bit if higher layer parameter *minimumSchedulingOffsetK0* is not configured;
  - 1 bit if higher layer parameter *minimumSchedulingOffsetK0* is configured. The 1 bit indication is used to determine the minimum applicable K0 for the active DL BWP and the minimum applicable K2 value for the active UL BWP, if configured respectively, according to Table 7.3.1.1.2-33. If the minimum applicable K0 is

indicated, the minimum applicable value of the aperiodic CSI-RS triggering offset for an active DL BWP shall be the same as the minimum applicable K0 value.

- SCell dormancy indication - 0 bit if higher layer parameter *dormancyGroupWithinActiveTime* is not configured; otherwise 1, 2, 3, 4 or 5 bits bitmap determined according to the number of different *DormancyGroupID(s)* provided by higher layer parameter *dormancyGroupWithinActiveTime*, where each bit corresponds to one of the SCell group(s) configured by higher layers parameter *dormancyGroupWithinActiveTime*, with MSB to LSB of the bitmap corresponding to the first to last configured SCell group in ascending order of *DormancyGroupID*. The field is only present when this format is carried by PDCCH on the primary cell within DRX Active Time and the UE is configured with at least two DL BWPs for an SCell.

  If one-shot HARQ-ACK request is not present or set to '0', and all bits of frequency domain resource assignment are set to 0 for resource allocation type 0 or set to 1 for resource allocation type 1 or set to 0 or 1 for dynamic switch resource allocation type, this field is reserved and the following fields among the fields above are used for SCell dormancy indication, where each bit corresponds to one of the configured SCell(s), with MSB to LSB of the following fields concatenated in the order below corresponding to the SCell with lowest to highest SCell index

  - Modulation and coding scheme of transport block 1
  - New data indicator of transport block 1
  - Redundancy version of transport block 1
  - HARQ process number
  - Antenna port(s)
  - DMRS sequence initialization
- PDCCH monitoring adaptation indication - 0, 1 or 2 bits

  - 1 or 2 bits, if *searchSpaceGroupIdList-r17* is not configured and if *pdcch-SkippingDurationList* is configured
    - 1 bit if the UE is configured with only one duration by *pdcch-SkippingDurationList;*
    - 2 bits if the UE is configured with more than one duration by *pdcch-SkippingDurationList.*
  - 1 or 2 bits, if *pdcch-SkippingDurationList* is not configured and if *searchSpaceGroupIdList-r17* is configured
    - 1 bit if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0 and search space set(s) with group index 1, and if the UE is not configured by *searchSpaceGroupIdList-r17* with any search space set with group index 2;
    - 2 bits if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0, search space set(s) with group index 1 and search space set(s) with group index 2;
  - 2 bits, if *pdcch-SkippingDurationList* is configured and if *searchSpaceGroupIdList-r17* is configured
  - 0 bit, otherwise
- PUCCH Cell indicator - 0 or 1 bit.

  - 1 bit if higher layer parameter *pucch-sSCellDyn* is configured.
  - 0 bit otherwise.

  If the UE is configured with a PUCCH-SCell, *pucch-sSCellDyn* is replaced by *pucch-sSCellDynSecondaryPUCCHgroup* for the secondary PUCCH group.
- Co-scheduled UE information – 0 or 3 bits

  - 3 bits as defined in Table 7.3.1.2.2-12 if higher layer parameter *advReceiver-MU-MIMO-DCI-1-1* is configured. This field is reserved if two codewords are scheduled by this DCI format 1_1.
  - 0 bit otherwise.

- TPC command for SRS – 2 bits as defined in Clause 7.3.1 of [5, TS 38.213] if higher layer parameter *tpcOfSrsClosedLoopIndex_InDCI_format_1_1* is configured; 0 bit otherwise.

  - Closed loop indicator for SRS – 1 bit if higher layer parameter *srsClosedLoopIndexIndicator_InDCI_format_1_1* and higher layer parameter *enableTwoSeparatePowerControlAdjustmentStatesForSRS* are both configured; 0 bit otherwise.
- Measurement gap cancellation – 0 bit if higher layer parameter *mg-CancellationDCI-1-1* is not configured; otherwise 1 bit as defined in Clause 10.6 in [5, TS 38.213].

If DCI formats 1_1 are monitored in multiple search spaces associated with multiple CORESETs in a BWP for scheduling the same serving cell, zeros shall be appended until the payload size of the DCI formats 1_1 monitored in the multiple search spaces equal to the maximum payload size of the DCI format 1_1 monitored in the multiple search spaces.

If the number of information bits in DCI format 1_1 scheduling a single PDSCH prior to padding is not equal to the number of information bits in DCI format 1_1 scheduling multiple PDSCHs for the same serving cell, zeros shall be appended to the DCI format 1_1 with smaller size until the payload size is the same for scheduling a single PDSCH and multiple PDSCHs.

For a UE configured with scheduling on the primary cell from an SCell, if prior to padding the number of information bits in DCI format 1_1 carried by PDCCH on the primary cell is not equal to the number of information bits in DCI format 1_1 carried by PDCCH on the SCell for scheduling on the primary cell, zeros shall be appended to the DCI format 1_1 with smaller size until the payload size is the same:

- If application of step 4C in clause 7.3.1.0 results in additional zero padding for DCI format 1_1 for scheduling on the primary cell, corresponding zeros shall be appended to both DCI format 1_1 monitored on the primary cell and DCI format 1_1 monitored on the SCell for scheduling on the primary cell.
- If the SCell is deactivated and *firstActiveDownlinkBWP-Id* is not set to dormant BWP, the UE determines the number of information bits in DCI format 1_1 carried by PDCCH on the primary cell based on a DL BWP provided by *firstActiveDownlinkBWP-Id* for the SCell. If the active DL BWP of the SCell is a dormant DL BWP, or if the SCell is deactivated and *firstActiveDownlinkBWP-Id* is set to dormant BWP, the UE determines the number of information bits in DCI format 1_1 carried by PDCCH on the primary cell based on a DL BWP provided by *firstWithinActiveTimeBWP-Id* for the SCell if provided; otherwise, based on a DL BWP provided by *firstOutsideActiveTimeBWP-Id* for the SCell.

Table 7.3.1.2.2-1: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1

One Codeword: Codeword 0 enabled, Codeword 1 disabled		
Value	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**
0	1	0
1	1	1
2	1	0,1
3	2	0
4	2	1
5	2	2
6	2	3
7	2	0,1
8	2	2,3
9	2	0-2
10	2	0-3
11	2	0,2
12-15	Reserved	Reserved

Table 7.3.1.2.2-1A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=1

One Codeword: Codeword 0 enabled, Codeword 1 disabled		
Value	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**
0	1	0
1	1	1
2	1	0,1
3	2	0
4	2	1
5	2	2
6	2	3
7	2	0,1
8	2	2,3
9	2	0-2
10	2	0-3
11	2	0,2
12	2	0,2,3
13-15	Reserved	Reserved

Table 7.3.1.2.2-2: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2

One Codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	2	0-4	2
1	1	1	1	1	2	0,1,2,3,4,6	2
2	1	0,1	1	2	2	0,1,2,3,4,5,6	2
3	2	0	1	3	2	0,1,2,3,4,5,6,7	2
4	2	1	1	4-31	reserved	reserved	reserved
5	2	2	1				
6	2	3	1				
7	2	0,1	1				
8	2	2,3	1				
9	2	0-2	1				
10	2	0-3	1				
11	2	0,2	1				
12	2	0	2				
13	2	1	2				
14	2	2	2				
15	2	3	2				
16	2	4	2				
17	2	5	2				
18	2	6	2				
19	2	7	2				
20	2	0,1	2				
21	2	2,3	2				
22	2	4,5	2				
23	2	6,7	2				
24	2	0,4	2				
25	2	2,6	2				
26	2	0,1,4	2				
27	2	2,3,6	2				
28	2	0,1,4,5	2				
29	2	2,3,6,7	2				
30	2	0,2,4,6	2				
31	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-2A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is not configured, *maxLength*=2

One Codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	2	0-4	2
1	1	1	1	1	2	0,1,2,3,4,6	2
2	1	0,1	1	2	2	0,1,2,3,4,5,6	2
3	2	0	1	3	2	0,1,2,3,4,5,6,7	2
4	2	1	1	4-31	reserved	reserved	reserved
5	2	2	1				
6	2	3	1				
7	2	0,1	1				
8	2	2,3	1				
9	2	0-2	1				
10	2	0-3	1				
11	2	0,2	1				
12	2	0	2				
13	2	1	2				
14	2	2	2				
15	2	3	2				
16	2	4	2				
17	2	5	2				
18	2	6	2				
19	2	7	2				
20	2	0,1	2				
21	2	2,3	2				
22	2	4,5	2				
23	2	6,7	2				
24	2	0,4	2				
25	2	2,6	2				
26	2	0,1,4	2				
27	2	2,3,6	2				
28	2	0,1,4,5	2				
29	2	2,3,6,7	2				
30	2	0,2,4,6	2				
31	2	0,2,3	1				

Table 7.3.1.2.2-3: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1

One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**	**Value**	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**
0	1	0	0	3	0-4
1	1	1	1	3	0-5
2	1	0,1	2-31	reserved	reserved
3	2	0			
4	2	1			
5	2	2			
6	2	3			
7	2	0,1			
8	2	2,3			
9	2	0-2			
10	2	0-3			
11	3	0			
12	3	1			
13	3	2			
14	3	3			
15	3	4			
16	3	5			
17	3	0,1			
18	3	2,3			
19	3	4,5			
20	3	0-2			
21	3	3-5			
22	3	0-3			
23	2	0,2			
24-31	Reserved	Reserved			

Table 7.3.1.2.2-3A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=1

One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0	0	3	0-4
1	1	1	1	3	0-5
2	1	0,1	2-31	reserved	reserved
3	2	0			
4	2	1			
5	2	2			
6	2	3			
7	2	0,1			
8	2	2,3			
9	2	0-2			
10	2	0-3			
11	3	0			
12	3	1			
13	3	2			
14	3	3			
15	3	4			
16	3	5			
17	3	0,1			
18	3	2,3			
19	3	4,5			
20	3	0-2			
21	3	3-5			
22	3	0-3			
23	2	0,2			
24	2	0,2,3			
25-31	Reserved	Reserved			

Table 7.3.1.2.2-4: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	3	0-4	1
1	1	1	1	1	3	0-5	1
2	1	0,1	1	2	2	0,1,2,3,6	2
3	2	0	1	3	2	0,1,2,3,6,8	2
4	2	1	1	4	2	0,1,2,3,6,7,8	2
5	2	2	1	5	2	0,1,2,3,6,7,8,9	2
6	2	3	1	6-63	Reserved	Reserved	Reserved
7	2	0,1	1				
8	2	2,3	1				
9	2	0-2	1				
10	2	0-3	1				
11	3	0	1				
12	3	1	1				
13	3	2	1				
14	3	3	1				
15	3	4	1				

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
16	3	5	1				
17	3	0,1	1				
18	3	2,3	1				
19	3	4,5	1				
20	3	0-2	1				
21	3	3-5	1				
22	3	0-3	1				
23	2	0,2	1				
24	3	0	2				
25	3	1	2				
26	3	2	2				
27	3	3	2				
28	3	4	2				
29	3	5	2				
30	3	6	2				
31	3	7	2				
32	3	8	2				
33	3	9	2				
34	3	10	2				
35	3	11	2				
36	3	0,1	2				
37	3	2,3	2				
38	3	4,5	2				
39	3	6,7	2				
40	3	8,9	2				
41	3	10,11	2				
42	3	0,1,6	2				
43	3	2,3,8	2				
44	3	4,5,10	2				
45	3	0,1,6,7	2				
46	3	2,3,8,9	2				
47	3	4,5,10,11	2				
48	1	0	2				
49	1	1	2				
50	1	6	2				
51	1	7	2				
52	1	0,1	2				
53	1	6,7	2				
54	2	0,1	2				
55	2	2,3	2				
56	2	6,7	2				
57	2	8,9	2				
58-63	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-4A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is not configured, *maxLength*=2

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	3	0-4	1
1	1	1	1	1	3	0-5	1

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
2	1	0,1	1	2	2	0,1,2,3,6	2
3	2	0	1	3	2	0,1,2,3,6,8	2
4	2	1	1	4	2	0,1,2,3,6,7,8	2
5	2	2	1	5	2	0,1,2,3,6,7,8,9	2
6	2	3	1	6-63	Reserved	Reserved	Reserved
7	2	0,1	1				
8	2	2,3	1				
9	2	0-2	1				
10	2	0-3	1				
11	3	0	1				
12	3	1	1				
13	3	2	1				
14	3	3	1				
15	3	4	1				
16	3	5	1				
17	3	0,1	1				
18	3	2,3	1				
19	3	4,5	1				
20	3	0-2	1				
21	3	3-5	1				
22	3	0-3	1				
23	2	0,2	1				
24	3	0	2				
25	3	1	2				
26	3	2	2				
27	3	3	2				
28	3	4	2				
29	3	5	2				
30	3	6	2				
31	3	7	2				
32	3	8	2				
33	3	9	2				
34	3	10	2				
35	3	11	2				
36	3	0,1	2				
37	3	2,3	2				
38	3	4,5	2				
39	3	6,7	2				
40	3	8,9	2				
41	3	10,11	2				
42	3	0,1,6	2				
43	3	2,3,8	2				
44	3	4,5,10	2				
45	3	0,1,6,7	2				
46	3	2,3,8,9	2				
47	3	4,5,10,11	2				
48	1	0	2				
49	1	1	2				
50	1	6	2				
51	1	7	2				
52	1	0,1	2				
53	1	6,7	2				
54	2	0,1	2				
55	2	2,3	2				
56	2	6,7	2				
57	2	8,9	2				
58	2	0,2,3	1				
59-63	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-5: VRB-to-PRB mapping

Bit field mapped to index	VRB-to-PRB mapping
0	Non-interleaved
1	Interleaved

Table 7.3.1.2.2-6: Allowed entries for DCI format 1_1/1_3 and DCI format 1_2, configured by higher layer parameter *ul-AccessConfigListDCI-1-1* and *ul-AccessConfigListDCI-1-2*, respectively, in frequency range 1

Entry index	Channel Access Type	The CP extension Text index defined in Clause 5.3.1 of [4,TS 38.211]
0	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	0
1	Type2C-ULChannelAccess defined in clause 4.2.1.2.3 in TS 37.213 [14]	2
2	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	0
3	Type2B-ULChannelAccess defined in clause 4.2.1.2.2 in TS 37.213 [14]	2
4	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	0
5	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	1
6	Type2A-ULChannelAccess defined in clause 4.2.1.2.1 in TS 37.213 [14]	3
7	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	0
8	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	1
9	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	2
10	Type1-ULChannelAccess defined in clause 4.2.1.1 in TS 37.213 [14]	3

Table 7.3.1.2.2-6A: Allowed entries for DCI format 1_1, DCI format 1_2 and DCI format 1_3, configured by higher layer parameter *ul-AccessConfigListDCI-1-1* in frequency range 2-2

Entry index	Channel Access Type
0	Type 1 channel access defined in clause 4.4.1 of TS 37.213 [14]
1	Type 2 channel access defined in clause 4.4.2 of TS 37.213 [14]
2	Type 3 channel access defined in clause 4.4.3 of TS 37.213 [14]

Table 7.3.1.2.2-7: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=1

One Codeword: Codeword 0 enabled, Codeword 1 disabled			Two Codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0	0	2	0,1,2,3,8
1	1	1	1	2	0,1,2,3,8,10
2	1	0,1	2	2	0,1,2,3,8,9,10
3	2	0	3	2	0,1,2,3,8,9,10,11
4	2	1	4-31	Reserved	Reserved
5	2	2			
6	2	3			
7	2	0,1			
8	2	2,3			
9	2	0-2			
10	2	0-3			
11	2	0,2			
12	1	8			
13	1	9			
14	1	8,9			

One Codeword: Codeword 0 enabled, Codeword 1 disabled			Two Codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
15	2	8			
16	2	9			
17	2	10			
18	2	11			
19	2	8,9			
20	2	10,11			
21	1	0,1,8			
22	1	0,1,8,9			
23	2	0,1,8			
24	2	0,1,8,9			
25	2	2,3,10			
26	2	2,3,10,11			
27-31	Reserved	Reserved			

Table 7.3.1.2.2-7A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=1

One Codeword: Codeword 0 enabled, Codeword 1 disabled			Two Codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0	0	2	0,1,2,3,8
1	1	1	1	2	0,1,2,3,8,10
2	1	0,1	2	2	0,1,2,3,8,9,10
3	2	0	3	2	0,1,2,3,8,9,10,11
4	2	1	4-31	Reserved	Reserved
5	2	2			
6	2	3			
7	2	0,1			
8	2	2,3			
9	2	0-2			
10	2	0-3			
11	2	0,2			
12	1	8			
13	1	9			
14	1	8,9			
15	2	8			
16	2	9			
17	2	10			
18	2	11			
19	2	8,9			
20	2	10,11			
21	1	0,1,8			
22	1	0,1,8,9			
23	2	0,1,8			
24	2	0,1,8,9			
25	2	2,3,10			
26	2	2,3,10,11			
27	2	0,2,3			
28-31	Reserved	Reserved			

Table 7.3.1.2.2-8: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2

One Codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	2	0,1,2,3,8	1
1	1	1	1	1	2	0,1,2,3,8,10	1
2	1	0,1	1	2	2	0,1,2,3,8,9,10	1
3	2	0	1	3	2	0,1,2,3,8,9,10,11	1
4	2	1	1	4	2	0-4	2
5	2	2	1	5	2	0,1,2,3,4,6	2
6	2	3	1	6	2	0,1,2,3,4,5,6	2
7	2	0,1	1	7	2	0,1,2,3,4,5,6,7	2
8	2	2,3	1	8	1	0,1,4,5,8	2
9	2	0-2	1	9	1	0,1,4,5,8,12	2
10	2	0-3	1	10	1	0,1,4,5,8,9,12	2
11	2	0,2	1	11	1	0,1,4,5,8,9,12,13	2
12	1	8	1	12	2	0,1,4,5,8	2
13	1	9	1	13	2	0,1,4,5,8,12	2
14	1	8,9	1	14	2	0,1,4,5,8,9,12	2
15	2	8	1	15	2	0,1,4,5,8,9,12,13	2
16	2	9	1	16~127	Reserved	Reserved	Reserved
17	2	10	1				
18	2	11	1				
19	2	8,9	1				
20	2	10,11	1				
21	1	0,1,8	1				
22	1	0,1,8,9	1				
23	2	0,1,8	1				
24	2	0,1,8,9	1				
25	2	2,3,10	1				
26	2	2,3,10,11	1				
27	2	0	2				
28	2	1	2				
29	2	2	2				
30	2	3	2				
31	2	4	2				
32	2	5	2				
33	2	6	2				
34	2	7	2				
35	2	0,1	2				
36	2	2,3	2				
37	2	4,5	2				
38	2	6,7	2				
39	2	0,4	2				
40	2	2,6	2				
41	2	0,1,4	2				
42	2	2,3,6	2				
43	2	0,1,4,5	2				
44	2	2,3,6,7	2				
45	2	0,2,4,6	2				
46	2	8	2				
47	2	9	2				
48	2	10	2				
49	2	11	2				
50	2	12	2				
51	2	13	2				
52	2	14	2				
53	2	15	2				
54	2	8,9	2				

One Codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
55	2	10,11	2				
56	2	12,13	2				
57	2	14,15	2				
58	2	0,1,8	2				
59	2	0,1,8,9	2				
60	2	4,5,12	2				
61	2	4,5,12,13	2				
62	2	2,3,10	2				
63	2	2,3,10,11	2				
64	2	6,7,14	2				
65	2	6,7,14,15	2				
66-127	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-8A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=1, *dmrs-TypeEnh* is configured, *maxLength*=2

One Codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	2	0,1,2,3,8	1
1	1	1	1	1	2	0,1,2,3,8,10	1
2	1	0,1	1	2	2	0,1,2,3,8,9,10	1
3	2	0	1	3	2	0,1,2,3,8,9,10,11	1
4	2	1	1	4	2	0-4	2
5	2	2	1	5	2	0,1,2,3,4,6	2
6	2	3	1	6	2	0,1,2,3,4,5,6	2
7	2	0,1	1	7	2	0,1,2,3,4,5,6,7	2
8	2	2,3	1	8	1	0,1,4,5,8	2
9	2	0-2	1	9	1	0,1,4,5,8,12	2
10	2	0-3	1	10	1	0,1,4,5,8,9,12	2
11	2	0,2	1	11	1	0,1,4,5,8,9,12,13	2
12	1	8	1	12	2	0,1,4,5,8	2
13	1	9	1	13	2	0,1,4,5,8,12	2
14	1	8,9	1	14	2	0,1,4,5,8,9,12	2
15	2	8	1	15	2	0,1,4,5,8,9,12,13	2
16	2	9	1	16~127	Reserved	Reserved	Reserved
17	2	10	1				
18	2	11	1				
19	2	8,9	1				
20	2	10,11	1				
21	1	0,1,8	1				
22	1	0,1,8,9	1				
23	2	0,1,8	1				
24	2	0,1,8,9	1				
25	2	2,3,10	1				
26	2	2,3,10,11	1				
27	2	0	2				
28	2	1	2				
29	2	2	2				
30	2	3	2				
31	2	4	2				
32	2	5	2				

One Codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
33	2	6	2				
34	2	7	2				
35	2	0,1	2				
36	2	2,3	2				
37	2	4,5	2				
38	2	6,7	2				
39	2	0,4	2				
40	2	2,6	2				
41	2	0,1,4	2				
42	2	2,3,6	2				
43	2	0,1,4,5	2				
44	2	2,3,6,7	2				
45	2	0,2,4,6	2				
46	2	8	2				
47	2	9	2				
48	2	10	2				
49	2	11	2				
50	2	12	2				
51	2	13	2				
52	2	14	2				
53	2	15	2				
54	2	8,9	2				
55	2	10,11	2				
56	2	12,13	2				
57	2	14,15	2				
58	2	0,1,8	2				
59	2	0,1,8,9	2				
60	2	4,5,12	2				
61	2	4,5,12,13	2				
62	2	2,3,10	2				
63	2	2,3,10,11	2				
64	2	6,7,14	2				
65	2	6,7,14,15	2				
66	2	0,2,3	1				
67-127	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-9: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1

One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Value	Number of DMRS CDM group(s) without data	DMRS port(s)
0	1	0	0	3	0-4
1	1	1	1	3	0-5
2	1	0,1	2	2	0,1,2,3,12
3	2	0	3	2	0,1,2,3,12,14
4	2	1	4	2	0-3,12-14
5	2	2	5	2	0-3,12-15
6	2	3	6	3	0,1,2,3,12
7	2	0,1	7	3	0,1,2,3,12,14
8	2	2,3	8	3	0-3,12-14
9	2	0-2	9	3	0-3,12-15
10	2	0-3	10~63	Reserved	Reserved
11	3	0			
12	3	1			

One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**	**Value**	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**
13	3	2			
14	3	3			
15	3	4			
16	3	5			
17	3	0,1			
18	3	2,3			
19	3	4,5			
20	3	0-2			
21	3	3-5			
22	3	0-3			
23	2	0,2			
24	1	12			
25	1	13			
26	1	12,13			
27	2	12			
28	2	13			
29	2	14			
30	2	15			
31	2	12,13			
32	2	14,15			
33	3	12			
34	3	13			
35	3	14			
36	3	15			
37	3	16			
38	3	17			
39	3	12,13			
40	3	14,15			
41	3	16,17			
42	1	0,1,12			
43	1	0,1,12,13			
44	2	0,1,12			
45	2	0,1,12,13			
46	2	2,3,14			
47	2	2,3,14,15			
48	3	0,1,12			
49	3	0,1,12,13			
50	3	2,3,14			
51	3	2,3,14,15			
52	3	4,5,16			
53	3	4,5,16,17			
54-63	Reserved	Reserved			

Table 7.3.1.2.2-9A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=1

One codeword: Codeword 0 enabled, Codeword 1 disabled			Two codewords: Codeword 0 enabled, Codeword 1 enabled		
Value	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**	**Value**	**Number of DMRS CDM group(s) without data**	**DMRS port(s)**
0	1	0	0	3	0-4
1	1	1	1	3	0-5
2	1	0,1	2	2	0,1,2,3,12
3	2	0	3	2	0,1,2,3,12,14
4	2	1	4	2	0-3,12-14
5	2	2	5	2	0-3,12-15
6	2	3	6	3	0,1,2,3,12
7	2	0,1	7	3	0,1,2,3,12,14
8	2	2,3	8	3	0-3,12-14
9	2	0-2	9	3	0-3,12-15
10	2	0-3	10~63	Reserved	Reserved
11	3	0			
12	3	1			
13	3	2			
14	3	3			
15	3	4			
16	3	5			
17	3	0,1			
18	3	2,3			
19	3	4,5			
20	3	0-2			
21	3	3-5			
22	3	0-3			
23	2	0,2			
24	1	12			
25	1	13			
26	1	12,13			
27	2	12			
28	2	13			
29	2	14			
30	2	15			
31	2	12,13			
32	2	14,15			
33	3	12			
34	3	13			
35	3	14			
36	3	15			
37	3	16			
38	3	17			
39	3	12,13			
40	3	14,15			
41	3	16,17			
42	1	0,1,12			
43	1	0,1,12,13			
44	2	0,1,12			
45	2	0,1,12,13			
46	2	2,3,14			
47	2	2,3,14,15			
48	3	0,1,12			
49	3	0,1,12,13			
50	3	2,3,14			
51	3	2,3,14,15			
52	3	4,5,16			
53	3	4,5,16,17			
54	2	0,2,3			
55-63	Reserved	Reserved			

Table 7.3.1.2.2-10: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	3	0-4	1
1	1	1	1	1	3	0-5	1
2	1	0,1	1	2	2	0,1,2,3,12	1
3	2	0	1	3	2	0-3,12,14	1
4	2	1	1	4	2	0-3,12-14	1
5	2	2	1	5	2	0-3,12-15	1
6	2	3	1	6	3	0,1,2,3,12	1
7	2	0,1	1	7	3	0-3,12,14	1
8	2	2,3	1	8	3	0-3,12-14	1
9	2	0-2	1	9	3	0-3,12-15	1
10	2	0-3	1	10	2	0,1,2,3,6	2
11	3	0	1	11	2	0,1,2,3,6,8	2
12	3	1	1	12	2	0,1,2,3,6,7,8	2
13	3	2	1	13	2	0,1,2,3,6,7,8,9	2
14	3	3	1	14	1	0,1,6,7,12	2
15	3	4	1	15	1	0,1,6,7,12,18	2
16	3	5	1	16	1	0,1,6,7,12,13,18	2
17	3	0,1	1	17	1	0,1,6,7,12,13,18,19	2
18	3	2,3	1	18	2	0,1,6,7,12	2
19	3	4,5	1	19	2	0,1,6,7,12,18	2
20	3	0-2	1	20	2	0,1,6,7,12,13,18	2
21	3	3-5	1	21	2	0,1,6,7,12,13,18,19	2
22	3	0-3	1	22	3	0,1,6,7,12	2
23	2	0,2	1	23	3	0,1,6,7,12,18	2
24	1	12	1	24	3	0,1,6,7,12,13,18	2
25	1	13	1	25	3	0,1,6,7,12,13,18,19	2
26	1	12,13	1	26~255	Reserved	Reserved	Reserved
27	2	12	1				
28	2	13	1				
29	2	14	1				
30	2	15	1				
31	2	12,13	1				
32	2	14,15	1				
33	3	12	1				
34	3	13	1				
35	3	14	1				
36	3	15	1				
37	3	16	1				
38	3	17	1				
39	3	12,13	1				
40	3	14,15	1				
41	3	16,17	1				
42	1	0,1,12	1				
43	1	0,1,12,13	1				
44	2	0,1,12	1				
45	2	0,1,12,13	1				
46	2	2,3,14	1				
47	2	2,3,14,15	1				
48	3	0,1,12	1				
49	3	0,1,12,13	1				
50	3	2,3,14	1				
51	3	2,3,14,15	1				
52	3	4,5,16	1				
53	3	4,5,16,17	1				
54	3	0	2				
55	3	1	2				

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
56	3	2	2				
57	3	3	2				
58	3	4	2				
59	3	5	2				
60	3	6	2				
61	3	7	2				
62	3	8	2				
63	3	9	2				
64	3	10	2				
65	3	11	2				
66	3	0,1	2				
67	3	2,3	2				
68	3	4,5	2				
69	3	6,7	2				
70	3	8,9	2				
71	3	10,11	2				
72	3	0,1,6	2				
73	3	2,3,8	2				
74	3	4,5,10	2				
75	3	0,1,6,7	2				
76	3	2,3,8,9	2				
77	3	4,5,10,11	2				
78	1	0	2				
79	1	1	2				
80	1	6	2				
81	1	7	2				
82	1	0,1	2				
83	1	6,7	2				
84	2	0,1	2				
85	2	2,3	2				
86	2	6,7	2				
87	2	8,9	2				
88	3	12	2				
89	3	13	2				
90	3	14	2				
91	3	15	2				
92	3	16	2				
93	3	17	2				
94	3	18	2				
95	3	19	2				
96	3	20	2				
97	3	21	2				
98	3	22	2				
99	3	23	2				
100	3	12,13	2				
101	3	14,15	2				
102	3	16,17	2				
103	3	18,19	2				
104	3	20,21	2				
105	3	22,23	2				
106	1	12	2				
107	1	13	2				
108	1	18	2				
109	1	19	2				
110	1	12,13	2				
111	1	18,19	2				
112	2	12,13	2				
113	2	14,15	2				
114	2	18,19	2				

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
115	2	20,21	2				
116	2	0,1,12	2				
117	2	0,1,12,13	2				
118	2	6,7,18	2				
119	2	6,7,18,19	2				
120	2	2,3,14	2				
121	2	2,3,14,15	2				
122	2	8,9,20	2				
123	2	8,9,20,21	2				
124	3	0,1,12	2				
125	3	0,1,12,13	2				
126	3	6,7,18	2				
127	3	6,7,18,19	2				
128	3	2,3,14	2				
129	3	2,3,14,15	2				
130	3	8,9,20	2				
131	3	8,9,20,21	2				
132	3	4,5,16	2				
133	3	4,5,16,17	2				
134	3	10,11,22	2				
135	3	10,11,22,23	2				
136-255	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-10A: Antenna port(s) (1000 + DMRS port), *dmrs-Type*=2, *dmrs-TypeEnh* is configured, *maxLength*=2

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
0	1	0	1	0	3	0-4	1
1	1	1	1	1	3	0-5	1
2	1	0,1	1	2	2	0,1,2,3,12	1
3	2	0	1	3	2	0-3,12,14	1
4	2	1	1	4	2	0-3,12-14	1
5	2	2	1	5	2	0-3,12-15	1
6	2	3	1	6	3	0,1,2,3,12	1
7	2	0,1	1	7	3	0-3,12,14	1
8	2	2,3	1	8	3	0-3,12-14	1
9	2	0-2	1	9	3	0-3,12-15	1
10	2	0-3	1	10	2	0,1,2,3,6	2
11	3	0	1	11	2	0,1,2,3,6,8	2
12	3	1	1	12	2	0,1,2,3,6,7,8	2
13	3	2	1	13	2	0,1,2,3,6,7,8,9	2
14	3	3	1	14	1	0,1,6,7,12	2
15	3	4	1	15	1	0,1,6,7,12,18	2
16	3	5	1	16	1	0,1,6,7,12,13,18	2
17	3	0,1	1	17	1	0,1,6,7,12,13,18,19	2
18	3	2,3	1	18	2	0,1,6,7,12	2
19	3	4,5	1	19	2	0,1,6,7,12,18	2
20	3	0-2	1	20	2	0,1,6,7,12,13,18	2
21	3	3-5	1	21	2	0,1,6,7,12,13,18,19	2
22	3	0-3	1	22	3	0,1,6,7,12	2
23	2	0,2	1	23	3	0,1,6,7,12,18	2
24	1	12	1	24	3	0,1,6,7,12,13,18	2

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
25	1	13	1	25	3	0,1,6,7,12,13,18,19	2
26	1	12,13	1	26~255	Reserved	Reserved	Reserved
27	2	12	1				
28	2	13	1				
29	2	14	1				
30	2	15	1				
31	2	12,13	1				
32	2	14,15	1				
33	3	12	1				
34	3	13	1				
35	3	14	1				
36	3	15	1				
37	3	16	1				
38	3	17	1				
39	3	12,13	1				
40	3	14,15	1				
41	3	16,17	1				
42	1	0,1,12	1				
43	1	0,1,12,13	1				
44	2	0,1,12	1				
45	2	0,1,12,13	1				
46	2	2,3,14	1				
47	2	2,3,14,15	1				
48	3	0,1,12	1				
49	3	0,1,12,13	1				
50	3	2,3,14	1				
51	3	2,3,14,15	1				
52	3	4,5,16	1				
53	3	4,5,16,17	1				
54	3	0	2				
55	3	1	2				
56	3	2	2				
57	3	3	2				
58	3	4	2				
59	3	5	2				
60	3	6	2				
61	3	7	2				
62	3	8	2				
63	3	9	2				
64	3	10	2				
65	3	11	2				
66	3	0,1	2				
67	3	2,3	2				
68	3	4,5	2				
69	3	6,7	2				
70	3	8,9	2				
71	3	10,11	2				
72	3	0,1,6	2				
73	3	2,3,8	2				
74	3	4,5,10	2				
75	3	0,1,6,7	2				
76	3	2,3,8,9	2				
77	3	4,5,10,11	2				
78	1	0	2				
79	1	1	2				
80	1	6	2				
81	1	7	2				
82	1	0,1	2				
83	1	6,7	2				

One codeword: Codeword 0 enabled, Codeword 1 disabled				Two Codewords: Codeword 0 enabled, Codeword 1 enabled			
Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols	Value	Number of DMRS CDM group(s) without data	DMRS port(s)	Number of front-load symbols
84	2	0,1	2				
85	2	2,3	2				
86	2	6,7	2				
87	2	8,9	2				
88	3	12	2				
89	3	13	2				
90	3	14	2				
91	3	15	2				
92	3	16	2				
93	3	17	2				
94	3	18	2				
95	3	19	2				
96	3	20	2				
97	3	21	2				
98	3	22	2				
99	3	23	2				
100	3	12,13	2				
101	3	14,15	2				
102	3	16,17	2				
103	3	18,19	2				
104	3	20,21	2				
105	3	22,23	2				
106	1	12	2				
107	1	13	2				
108	1	18	2				
109	1	19	2				
110	1	12,13	2				
111	1	18,19	2				
112	2	12,13	2				
113	2	14,15	2				
114	2	18,19	2				
115	2	20,21	2				
116	2	0,1,12	2				
117	2	0,1,12,13	2				
118	2	6,7,18	2				
119	2	6,7,18,19	2				
120	2	2,3,14	2				
121	2	2,3,14,15	2				
122	2	8,9,20	2				
123	2	8,9,20,21	2				
124	3	0,1,12	2				
125	3	0,1,12,13	2				
126	3	6,7,18	2				
127	3	6,7,18,19	2				
128	3	2,3,14	2				
129	3	2,3,14,15	2				
130	3	8,9,20	2				
131	3	8,9,20,21	2				
132	3	4,5,16	2				
133	3	4,5,16,17	2				
134	3	10,11,22	2				
135	3	10,11,22,23	2				
136	2	0,2,3	1				
137-255	Reserved	Reserved	Reserved				

Table 7.3.1.2.2-11: TCI selection

Bit field mapped to index	TCI selection
0	The first indicated joint/DL TCI state is applied to the scheduled PDSCH
1	The second indicated joint/DL TCI state is applied to the scheduled PDSCH
2	Both indicated joint/DL TCI states are applied to the scheduled PDSCH
3	Reserved

Table 7.3.1.2.2-12: Co-scheduled UE information

Bit field mapped to index	Co-scheduled UE information
0	In all the PRBs allocated to the UE, there is no co-scheduled UE or there is co-scheduled UE but with a different root DMRS sequence
1	In all the PRBs allocated to the UE, all the co-scheduled UE(s), if any, which have the same root DMRS sequence as the UE, are scheduled with modulation scheme QPSK
2	In all the PRBs allocated to the UE, all the co-scheduled UE(s), if any, which have the same root DMRS sequence as the UE, are scheduled with modulation scheme 16QAM
3	In all the PRBs allocated to the UE, all the co-scheduled UE(s), if any, which have the same root DMRS sequence as the UE, are scheduled with modulation scheme 64QAM
4	In all the PRBs allocated to the UE, all the co-scheduled UE(s), if any, which have the same root DMRS sequence as the UE, are scheduled with modulation scheme 256QAM
5	In all the PRBs allocated to the UE, all the co-scheduled UE(s), if any, which have the same root DMRS sequence as the UE, are scheduled with modulation scheme 1024QAM
6	In each individual PRB allocated to the UE, all the co-scheduled UE(s), which have the same root DMRS sequence as the UE, are scheduled with the same modulation scheme, except the cases corresponding to index 0~5
7	All cases not covered above
Note: Root DMRS sequence is $r(n)$ as defined in clause 7.4.1.1.1 of [4, TS 38.211]	

7.3.1.2.3 Format 1_2

DCI format 1_2 is used for the scheduling of PDSCH in one cell.

The following information is transmitted by means of the DCI format 1_2 with CRC scrambled by C-RNTI or CS-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bits
  - The value of this bit field is always set to 1, indicating a DL DCI format.
- Carrier indicator - 0, 1, 2 or 3 bits determined by higher layer parameter *carrierIndicatorSizeDCI-1-2*, as defined in Clause 10.1 of [5, TS38.213]. This field is reserved when this format is carried by PDCCH on the primary cell and the UE is configured for scheduling on the primary cell from an SCell, with the same number of bits as that in this format carried by PDCCH on the SCell for scheduling on the primary cell.
- Bandwidth part indicator - 0, 1 or 2 bits as determined by the number of DL BWPs $n_{BWP,\,RRC}$ configured by higher layers, excluding the initial DL bandwidth part. The bitwidth for this field is determined as $\lceil \log_2(n_{BWP}) \rceil$ bits, where
  - $n_{BWP}=n_{BWP,\,RRC}+1$ if $n_{BWP,\,RRC}\leq 3$, in which case the bandwidth part indicator is equivalent to the ascending order of the higher layer parameter *BWP-Id*;
  - otherwise $n_{BWP}=n_{BWP,\,RRC}$, in which case the bandwidth part indicator is defined in Table 7.3.1.1.2-1;

  If a UE does not support active BWP change via DCI, the UE ignores this bit field.
- Frequency domain resource assignment - number of bits determined by the following:
  - N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 5.1.2.2.1 of [6, TS 38.214];

- $\lceil \log_2\left(N_{RBG,\,K2}\left(N_{RBG,\,K2}+1\right)/2\right) \rceil$ bits if only resource allocation type 1 is configured, or $max\left(\lceil \log_2\left(N_{RBG,\,K2}\left(N_{RBG,\,K2}+1\right)/2\right) \rceil,\ N_{RBG}\right)+1$ bits if resourceAllocationDCI-1-2-r16 is configured as 'dynamicSwitch', where $N_{RBG,\,K2}=\lceil \left(N_{RB}^{DL,\,BWP}+\left(N_{DL,\,BWP}^{start} mod\, K2\right)\right)/K2 \rceil$, $N_{RB}^{DL,\,BWP}$ is the size of the active DL bandwidth part, $N_{DL,\,BWP}^{start}$ is defined as in clause 4.4.4.4 of [4, TS 38.211] and $K2$ is determined by higher layer parameter resourceAllocationType1GranularityDCI-1-2. If the higher layer parameter resourceAllocationType1GranularityDCI-1-2 is not configured, $K2$ is equal to 1.
  - If *resourceAllocationDCI-1-2-r16* is configured as '*dynamicSwitch'*, the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
  - For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 5.1.2.2.1 of [6, TS 38.214].
  - For resource allocation type 1, the $\lceil \log_2\left(N_{RBG,\,K2}\left(N_{RBG,\,K2}+1\right)/2\right) \rceil$ LSBs provide the resource allocation as defined in Clause 5.1.2.2.2 of [6, TS 38.214]

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and if *resourceAllocationDCI-1-2-r16* is configured as '*dynamicSwitch'* for the indicated bandwidth part, the UE assumes resource allocation type 0 for the indicated bandwidth part if the bitwidth of the "Frequency domain resource assignment" field of the active bandwidth part is smaller than the bitwidth of the "Frequency domain resource assignment" field of the indicated bandwidth part.
- Time domain resource assignment - 0, 1, 2, 3, or 4 bits as defined in Clause 5.1.2.1 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *pdsch-TimeDomainAllocationListDCI-1-2* if the higher layer parameter is configured, or I is the number of entries in the higher layer parameter *pdsch-TimeDomainAllocationList* if the higher layer parameter *pdsch-TimeDomainAllocationList* is configured when the higher layer parameter *pdsch-TimeDomainAllocationListDCI-1-2* is not configured; otherwise I is the number of entries in the default table.
- VRB-to-PRB mapping - 0 or 1 bit:
  - 0 bit if the higher layer parameter *vrb-ToPRB-InterleaverDCI-1-2* is not configured;
  - 1 bit according to Table 7.3.1.2.2-5 otherwise, only applicable to resource allocation type 1, as defined in Clause 7.3.1.6 of [4, TS 38.211].
- PRB bundling size indicator - 0 bit if the higher layer parameter *prb-BundlingTypeDCI-1-2* is not configured or is set to 'static', or 1 bit if the higher layer parameter *prb-BundlingTypeDCI-1-2* is set to 'dynamic' according to Clause 5.1.2.3 of [6, TS 38.214].
- Rate matching indicator - 0, 1, or 2 bits according to higher layer parameters *rateMatchPatternGroup1DCI-1-2* and *rateMatchPatternGroup2DCI-1-2*, where the MSB is used to indicate *rateMatchPatternGroup1DCI-1-2* and the LSB is used to indicate *rateMatchPatternGroup2DCI-1-2* when there are two groups.
- ZP CSI-RS trigger - 0, 1, or 2 bits as defined in Clause 5.1.4.2 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(n_{ZP}+1) \rceil$ bits, where n_{ZP} is the number of aperiodic ZP CSI-RS resource sets configured by higher layer parameter *aperiodicZP-CSI-RS-ResourceSetsToAddModListDCI-1-2*.
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3.1 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-1-2*
  - If 0 bit is configured, rv_{id} to be applied is 0;
  - 1 bit according to Table 7.3.1.2.3-1;
  - 2 bits according to Table 7.3.1.1.1-2.
- HARQ process number - number of bits determined by the following:
  - 0, 1, 2, 3, 4 or 5 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-1-2-v1700* or *harq-ProcessNumberSizeDCI-1-2-Ext* if configured;
  - otherwise 0, 1, 2, 3 or 4 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-1-2*.
- Downlink assignment index - 0, 1, 2 or 4 bits

- 0 bit if the higher layer parameter *downlinkAssignmentIndexDCI-1-2* is not configured;
- 1, 2 or 4 bits determined by higher layer parameter *downlinkAssignmentIndexDCI-1-2* otherwise,
  - 4 bits if more than one serving cell are configured in the DL and the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic*, where the 2 MSB bits are the counter DAI and the 2 LSB bits are the total DAI
  - 4 bits if only one serving cell is configured in the DL and the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic*, and the UE is not provided *coresetPoolIndex* or is provided *coresetPoolIndex* with value 0 for one or more first CORESETs and is provided *coresetPoolIndex* with value 1 for one or more second CORESETs, and is provided *ackNackFeedbackMode = joint*, where the 2 MSB bits are the counter DAI and the 2 LSB bits are the total DAI.
  - 1 or 2 bits if only one serving cell is configured in the DL and the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic*, when the UE is not configured with *coresetPoolIndex* or the value of *coresetPoolIndex* is the same for all CORESETs if *coresetPoolIndex* is provided or the UE is not configured with *ackNackFeedbackMode = joint*, where the 1 bit or 2 bits are the counter DAI.

If the UE is configured with a PUCCH-SCell, the number of serving cells is determined within a PUCCH group.

If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-Codebook* is replaced by *pdsch-HARQ-ACK-Codebook-secondaryPUCCHgroup-r16* if present for the secondary PUCCH group.

If higher layer parameter *priorityIndicatorDCI-1-2* is configured, if the bit width of the Downlink assignment index in DCI format 1_2 for one HARQ-ACK codebook is not equal to that of the Downlink assignment index in DCI format 1_2 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller Downlink assignment index until the bit width of the Downlink assignment index in DCI format 1_2 for the two HARQ-ACK codebooks are the same.

- TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS 38.213]

- Second TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS 38.213] if higher layer parameter *SecondTPCFieldDCI-1-2* is configured; 0 bit otherwise.

- PUCCH resource indicator - 0 or 1 or 2 or 3 bits determined by higher layer parameter *numberOfBitsForPUCCH-ResourceIndicatorDCI-1-2*

  If higher layer parameter *pucch-sSCellPattern* or *pucch-sSCellDynDCI-1-2* is configured, if the bit width of the PUCCH resource indicator in DCI format 1_2 associated with one cell for PUCCH transmission is not equal to that of the PUCCH resource indicator in DCI format 1_2 associated with the other cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PUCCH resource indicator until the bit width of the PUCCH resource indicator in DCI format 1_2 associated with the two cells for PUCCH transmissions are the same.

  If the UE is configured with a PUCCH-SCell, *pucch-sSCellPattern* is replaced by *pucch-sSCellPatternSecondaryPUCCHgroup* for the secondary PUCCH group.

- PDSCH-to-HARQ_feedback timing indicator - 0, 1, 2, or 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *DL-DataToUL-ACK-DCI-1-2*.

  If higher layer parameter *priorityIndicatorDCI-1-2* is configured, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_2 for one HARQ-ACK codebook is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_2 for the other HARQ-ACK codebook on the same cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_2 for the two HARQ-ACK codebooks are the same.

  If higher layer parameter *pucch-sSCellDynDCI-1-2* is configured, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_2 associated with one cell for PUCCH transmission is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_2 associated with the other cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_2 associated with the two cells are the same.

- One-shot HARQ-ACK request -0 or 1 bit.
  - 1 bit if higher layer parameter *pdsch-HARQ-ACK-OneShotFeedbackDCI-1-2* or *pdsch-HARQ-ACK-EnhType3DCI-1-2* is configured;
  - 0 bit otherwise.

- Enhanced Type 3 codebook indicator - 0, 1, 2, or 3 bits.
  - 0 bit if *pdsch-HARQ-ACK-EnhType3DCI-Field-1-2* is not configured;
  - $\lceil \log_2(n_{CB}) \rceil$ bits otherwise, where n_{CB} is the number of entries in the higher layer parameter *pdsch-HARQ-ACK-EnhType3ToAddModList.*

  If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-EnhType3ToAddModList* is replaced by *pdsch-HARQ-ACK-EnhType3SecondaryToAddModList* for the secondary PUCCH group.
- HARQ-ACK retransmission indicator - 0 or 1 bit.
  - 1 bit if higher layer parameter *pdsch-HARQ-ACK-retxDCI-1-2* is configured.
  - 0 bit otherwise.
- Antenna port(s) - 0, 4, 5, 6, 7 or 8 bits
  - 0 bit if higher layer parameter *antennaPortsFieldPresenceDCI-1-2* is not configured;
  - Otherwise 4, 5, 6, 7 or 8 bits as defined by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10 and Tables 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A, where the number of CDM groups without data of values 1, 2, and 3 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively. The antenna ports $\{p_0, \ldots, p_{v-1}\}$ shall be determined according to the ordering of DMRS port(s) given by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10 or Tables 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A. When a UE not configured with *dl-OrJointTCI-StateList* receives an activation command that maps at least one codepoint of DCI field '*Transmission Configuration Indication*' to two TCI states, or when a UE configured with *dl-OrJointTCI-StateList* is having two indicated TCI states, the UE shall use Table 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A; otherwise, it shall use Tables 7.3.1.2.2-1/2/3/4/7/8/9/10.
  - If a UE is configured with both *dmrs-DownlinkForPDSCH-MappingTypeA-DCI-1-2* and *dmrs-DownlinkForPDSCH-MappingTypeB-DCI-1-2* and is configured with higher layer parameter *antennaPortsFieldPresenceDCI-1-2*, the bitwidth of this field equals $max\{x_A, x_B\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeA-DCI-1-2* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeB-DCI-1-2*. A number of $|x_A - x_B|$ zeros are padded in the MSB of this field, if the mapping type of the PDSCH corresponds to the smaller value of x_A and x_B.

  If a UE is not configured with higher layer parameter *antennaPortsFieldPresenceDCI-1-2*, antenna port(s) are defined assuming bit field index value 0 in Tables 7.3.1.2.2-1/2/3/4/7/8/9/10.

- Transmission configuration indication - 0 bit if higher layer parameter *tci-PresentDCI-1-2* is not configured; otherwise 1 or 2 or 3 bits determined by higher layer parameter *tci-PresentDCI-1-2* as defined in Clause 5.1.5 of [6, TS38.214].

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part,
  - if the higher layer parameter *tci-PresentDCI-1-2* is not configured for the CORESET used for the PDCCH carrying the DCI format 1_2,
    - the UE assumes *tci-PresentDCI-1-2* is not configured for all CORESETs in the indicated bandwidth part;
  - otherwise,
    - the UE assumes *tci-PresentDCI-1-2* is configured for all CORESETs in the indicated bandwidth part with the same value configured for the CORESET used for the PDCCH carrying the DCI format 1_2.
- TCI selection - 0 bit if higher layer parameter *tci-SelectionPresentInDCI* is not configured; otherwise 2 bits according to Table 7.3.1.2.2-11.
- SRS request - 0, 1, 2 or 3 bits
  - 0 bit if the higher layer parameter *srs-RequestDCI-1-2* is not configured;
  - 1 bit as defined by Table 7.3.1.1.3-1 if the higher layer parameter *srs-RequestDCI-1-2 = 1* and for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell;
  - 2 bits if the higher layer parameter *srs-RequestDCI-1-2 = 1* and for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell, where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second bit is defined by Table 7.3.1.1.3-1;
  - 2 bits as defined by Table 7.3.1.1.2-24 if the higher layer parameter *srs-RequestDCI-1-2 = 2* and for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell;

  - 3 bits if the higher layer parameter *srs-RequestDCI-1-2 = 2* and for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell, where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second and third bits are defined by Table 7.3.1.1.2-24;
- SRS offset indicator - 0, 1 or 2 bits.
  - 0 bit if higher layer parameter *AvailableSlotOffset* is not configured for any aperiodic SRS resource set in the scheduled cell, or if higher layer parameter *AvailableSlotOffset* is configured for at least one aperiodic SRS resource set in the scheduled cell and the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) is 1;
  - otherwise, $\lceil \log_2(K) \rceil$ bits are used to indicate available slot offset according to Table 7.3.1.1.2-37 and Clause 6.2.1 of [6, TS 38.214], where K is the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) in the scheduled cell;
- DMRS sequence initialization - 0 or 1 bit
  - 0 bit if the higher layer parameter *dmrs-SequenceInitializationDCI-1-2* is not configured;
  - 1 bit otherwise.
- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-1-2* is not configured; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213].
- PDCCH monitoring adaptation indication - 0, 1 or 2 bits
  - 1 or 2 bits, if *searchSpaceGroupIdList-r17* is not configured and if *pdcch-SkippingDurationList* is configured
    - 1 bit if the UE is configured with only one duration by *pdcch-SkippingDurationList;*
    - 2 bits if the UE is configured with more than one duration by *pdcch-SkippingDurationList*.
  - 1 or 2 bits, if *pdcch-SkippingDurationList* is not configured and if *searchSpaceGroupIdList-r17* is configured
    - 1 bit if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0 and search space set(s) with group index 1, and if the UE is not configured by *searchSpaceGroupIdList-r17* with any search space set with group index 2;
    - 2 bits if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0, search space set(s) with group index 1 and search space set(s) with group index 2;
  - 2 bits, if *pdcch-SkippingDurationList* is configured and if *searchSpaceGroupIdList-r17* is configured
  - 0 bit, otherwise
- ChannelAccess-CPext - 0, 1, 2, 3 or 4 bits. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *ul-AccessConfigListDCI-1-2* or in Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1, or the number of entries in the high layer parameter *ul-AccessConfigListDCI-1-1* for operation in frequency range 2-2 if *ChannelAccessMode2-r17* is provided; otherwise 0 bit. One or more entries from Table 7.3.1.2.2-6 are configured by the higher layer parameter *ul-AccessConfigListDCI-1-2* in frequency range 1. One or more entries from Table 7.3.1.1.2-6A are configured by the higher layer parameter *ul-AccessConfigListDCI-1-1* in frequency range 2-2.
- PUCCH Cell indicator - 0 or 1 bit.
  - 1 bit if higher layer parameter *pucch-sSCellDynDCI-1-2* is configured.
  - 0 bit otherwise.
- Measurement gap cancellation – 0 bit if higher layer parameter *mg-CancellationDCI-1-2* is not configured; otherwise 1 bit as defined in Clause 10.6 in [5, TS 38.213].

If DCI formats 1_2 are monitored in multiple search spaces associated with multiple CORESETs in a BWP for scheduling the same serving cell, zeros shall be appended until the payload size of the DCI formats 1_2 monitored in the multiple search spaces equal to the maximum payload size of the DCI format 1_2 monitored in the multiple search spaces.

For a UE configured with scheduling on the primary cell from an SCell, if prior to padding the number of information bits in DCI format 1_2 carried by PDCCH on the primary cell is not equal to the number of information bits in DCI format 1_2 carried by PDCCH on the SCell for scheduling on the primary cell, zeros shall be appended to the DCI format 1_2 with smaller size until the payload size is the same.

- If application of step 4B in clause 7.3.1.0 results in additional zero padding for DCI format 1_2 for scheduling on the primary cell, corresponding zeros shall be appended to both DCI format 1_2 monitored on the primary cell and DCI format 1_2 monitored on the SCell for scheduling on the primary cell.

- If the SCell is deactivated and *firstActiveDownlinkBWP-Id* is not set to dormant BWP, the UE determines the number of information bits in DCI format 1_2 carried by PDCCH on the primary cell based on a DL BWP provided by *firstActiveDownlinkBWP-Id* for the SCell. If the active DL BWP of the SCell is a dormant DL BWP, or if the SCell is deactivated and *firstActiveDownlinkBWP-Id* is set to dormant BWP, the UE determines the number of information bits in DCI format 1_2 carried by PDCCH on the primary cell based on a DL BWP provided by *firstWithinActiveTimeBWP-Id* for the SCell if provided; otherwise, based on a DL BWP provided by *firstOutsideActiveTimeBWP-Id* for the SCell.

Table 7.3.1.2.3-1: Redundancy version

Value of the Redundancy version field	Value of rv_{id} to be applied
0	0
1	3

7.3.1.2.4 Format 1_3

DCI format 1_3 is used for the scheduling of one or multiple PDSCHs in one cell, or multiple PDSCHs in multiple cells with one or multiple PDSCHs per cell.

The following information is transmitted by means of the DCI format 1_3 with CRC scrambled by C-RNTI or MCS-C-RNTI:

- Identifier for DCI formats - 1 bits
  - The value of this bit field is always set to 1, indicating a DL DCI format
- Scheduled cell set indicator - $\lceil \log_2 N_{set} \rceil$ bits, where N_{set} is the number of cell sets which are configured by higher layer parameter *mc-DCI-SetofCellsToAddModList* to be respectively scheduled by DCI format 0_3/1_3 from the cell on which this format is carried by PDCCH. If present, this field is used to indicate the scheduled cell set according to Table 7.3.1.1.4-1; otherwise, the scheduled cell set is the cell set configured to be scheduled by DCI format 0_3/1_3 from the cell by higher layer parameter *mc-DCI-SetofCellsToAddModList*.
- Scheduled cells indicator - number of bits determined by the following:
  - 0 bit if the higher layer parameter *scheduledCellComboListDCI-1-3* for the scheduled cell set is not configured;
  - otherwise $\lceil \log_2 I_{DL} \rceil$ bits indicating the scheduled cells in the scheduled cell set according to Table 7.3.1.2.4-1, where I_{DL} is the number of entries in the higher layer parameter *scheduledCellComboListDCI-1-3*. If only one entry is configured in the higher layer parameter *scheduledCellComboListDCI-1-3*, the scheduled cells are the cells configured by higher layer parameter *scheduledCellComboListDCI-1-3*.
- Bandwidth part indicator - 0, 1 or 2 bits determined as $\lceil \log_2 n_{BWP,\,max} \rceil$, where
  - $n_{BWP,\,max} = n_{BWP,RRC}^{max} + 1$ if $n_{BWP,RRC}^{max} \leq 3$, $n_{BWP,RRC}^{max}$ is the maximum number of DL BWPs configured by higher layers, excluding the initial DL bandwidth part, across all the cells configured by higher layer parameter *scheduledCellListDCI-1-3* in the scheduled cell set, in which case the bandwidth part indicator is equivalent to the ascending order of the higher layer parameter *BWP-Id*;
  - otherwise $n_{BWP,\,max} = n_{BWP,RRC}^{max}$, in which case the bandwidth part indicator is defined in Table 7.3.1.1.2-1;

  The field is only applicable to a scheduled cell with the number of configured DL BWPs larger than 1, including the initial DL bandwidth part, and is applied to the applicable scheduled cells in the scheduled cell set independently. If a UE does not support active BWP change via DCI, the UE ignores this bit field. If this field indicates a code point that does not correspond to a configured BWP of a scheduled cell, the UE ignores this bit field for the scheduled cell, and operates on the active BWP of the scheduled cell.
- Frequency domain resource assignment - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{DL}

If *scheduledCellComboListDCI-1-3* for the scheduled cell set is configured with more than one entry, N_{cell}^{DL} is the number of scheduled cells indicated by Scheduled cells indicator field; if *scheduledCellComboListDCI-1-3* for the scheduled cell set is configured with only one entry, N_{cell}^{DL} is the number of cells configured by higher layer parameter *scheduledCellComboListDCI-1-3*; otherwise, N_{cell}^{DL} is the number of cells in the scheduled cell set configured by higher layer parameter *scheduledCellListDCI-1-3*. Each block corresponds to the frequency domain resource assignment for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the frequency domain resource assignment for the cell with the smallest serving cell index. Each block is defined by the following fields:

- N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 5.1.2.2.1 of [6, TS 38.214]
- $\lceil \log_2\left(N_{RBG,\,K2}\left(N_{RBG,\,K2}+1\right)/2\right) \rceil$ bits if only resource allocation type 1 is configured, or $max\left(\lceil \log_2\left(N_{RBG,\,K2}\left(N_{RBG,\,K2}+1\right)/2\right) \rceil,\ N_{RBG}\right)+1$ bits if *resourceAllocationDCI-1-3* is configured as '*dynamicSwitch*', where $N_{RBG,\,K2}=\lceil \left(N_{RB}^{DL,\,BWP}+\left(N_{DL,\,BWP}^{start} mod\, K2\right)\right)/K2 \rceil$, $N_{RB}^{DL,\,BWP}$ is the size of the active DL bandwidth part, $N_{DL,\,BWP}^{start}$ is defined as in clause 4.4.4.4 of [4, TS 38.211] and $K2$ is given by higher layer parameter *resourceAllocationType1GranularityDCI-1-3*. If the higher layer parameter *resourceAllocationType1GranularityDCI-1-3* is not configured, $K2$ is equal to 1.
- If *resourceAllocationDCI-1-3* is configured as '*dynamicSwitch*', the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
- For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 5.1.2.2.1 of [6, TS 38.214].
- For resource allocation type 1, the $\lceil \log_2\left(N_{RBG,\,K2}\left(N_{RBG,\,K2}+1\right)/2\right) \rceil$ LSBs provide the resource allocation as defined in Clause 5.1.2.2.2 of [6, TS 38.214].

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and if *resourceAllocationDCI-1-3* is configured as '*dynamicSwitch*' for the indicated bandwidth part, the UE assumes resource allocation type 0 for the indicated bandwidth part if the bitwidth of the "Frequency domain resource assignment" field of the active bandwidth part is smaller than the bitwidth of the "Frequency domain resource assignment" field of the indicated bandwidth part.

If the higher layer parameter *scheduledCellComboListDCI-1-3* for the scheduled cell set is not configured, each block is also used to indicate whether the corresponding cell is scheduled or not as follows:

- if all bits of a block are set to 0 for resource allocation type 0 or set to 1 for resource allocation type 1 or set to 0 or 1 for dynamic switch resource allocation type, the cell corresponding to the block is not scheduled;
- otherwise, the cell corresponding to the block is scheduled.

- Time domain resource assignment - $\lceil \log_2(I_{TDRA}) \rceil$ bits, where I_{TDRA} is the number of entries in the higher layer parameter *tdra-FieldIndexListDCI-1-3*. This field is used to indicate an entry in the higher layer parameter *tdra-FieldIndexListDCI-1-3* according to Table 7.3.1.2.4-2. Each entry in the higher layer parameter *tdra-FieldIndexListDCI-1-3* contains the 'Time domain resource assignment' index for each BWP of each cell in the scheduled cell set, where the 'Time domain resource assignment' indexes for all the cells are placed according to an ascending order of a serving cell index, and the 'Time domain resource assignment' indexes for all the BWPs of a cell are placed according to an ascending order of the higher layer parameter *BWP-Id*.

- VRB-to-PRB mapping - 0 or 1 bit
  - 0 bit if the higher layer parameter *vrb-ToPRB-Interleaver* is not configured for any cell configured by higher layer parameter *scheduledCellListDCI-1-3* in the scheduled cell set;
  - 1 bit according to Table 7.3.1.2.2-5 otherwise, only applicable to resource allocation type 1, as defined in Clause 7.3.1.6 of [4, TS 38.211].

  The field is only applicable to a scheduled cell configured with *vrb-ToPRB-Interleaver*, and is applied to the applicable scheduled cells independently.

- PRB bundling size indicator - 0 or 1 bit
  - 0 bit if the higher layer parameter *prb-BundlingType* is not configured or is set to 'staticBundling' for any cell configured by higher layer parameter *scheduledCellListDCI-1-3* in the scheduled cell set;

- 1 bit according to Clause 5.1.2.3 of [6, TS 38.214] otherwise.

The field is only applicable to a scheduled cell configured with *prb-BundlingType* set to 'dynamicBundling', and is applied to the applicable scheduled cells independently.

- Rate matching indicator - $\lceil \log_2(I_{RM}) \rceil$ bits, where I_{RM} is the number of entries in the higher layer parameter *rateMatchListDCI-1-3*, or 0 bit if the higher layer parameter *rateMatchListDCI-1-3* is not configured. This field is used to indicate an entry in the higher layer parameter *rateMatchListDCI-1-3* according to Table 7.3.1.2.4-3. Each entry in the higher layer parameter *rateMatchListDCI-1-3* contains the 'Rate matching indicator' index for each cell configured with *rateMatchPatternGroup1* or *rateMatchPatternGroup2* on at least one DL BWP in the scheduled cell set, where the 'Rate matching indicator' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'Rate matching indicator' index is defined by the following:
  - 0, 1, or 2 bits according to higher layer parameters *rateMatchPatternGroup1* and *rateMatchPatternGroup2*, where the MSB is used to indicate *rateMatchPatternGroup1* and the LSB is used to indicate *rateMatchPatternGroup2* when there are two groups.
- ZP CSI-RS trigger - $\lceil \log_2(I_{CSIRS}) \rceil$ bits, where I_{CSIRS} is the number of entries in the higher layer parameter *zp-CSI-RSListDCI-1-3*, or 0 bit if the higher layer parameter *zp-CSI-RSListDCI-1-3* is not configured. This field is used to indicate an entry in the higher layer parameter *zp-CSI-RSListDCI-1-3* according to Table 7.3.1.2.4-4. Each entry in the higher layer parameter *zp-CSI-RSListDCI-1-3* contains the 'ZP CSI-RS trigger' index for each cell configured with *aperiodicZP-CSI-RS-ResourceSetsToAddModList* on at least one DL BWP in the scheduled cell set, where the 'ZP CSI-RS trigger' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'ZP CSI-RS trigger' index is defined by the following:
  - 0, 1, or 2 bits as defined in Clause 5.1.4.2 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(n_{ZP}+1) \rceil$ bits, where n_{ZP} is the number of aperiodic ZP CSI-RS resource sets configured by higher layer parameter *aperiodicZP-CSI-RS-ResourceSetsToAddModList*.

For transport block 1:

- Modulation and coding scheme - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{DL}

  Each block corresponds to the modulation and coding scheme for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the modulation and coding scheme for the cell with the smallest serving cell index. Each block is 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214].

- New data indicator - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{DL}

  Each block corresponds to the new data indicator for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the new data indicator for the cell with the smallest serving cell index. If *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* is configured for a cell, the number of bits for the corresponding block is equal to the maximum number of schedulable PDSCHs among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* for the cell, where each bit corresponds to one scheduled PDSCH as defined in clause 6.1.4 in [6, TS 38.214]; otherwise, the corresponding block is 1 bit.

- Redundancy version - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{DL}

  Each block corresponds to the redundancy version for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the redundancy version for the cell with the smallest serving cell index. The number of bits for each block is determined by following:
  - if *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* is configured for a cell, the number of bits for the corresponding block is determined by $m_A \times m_B$, where m_A is the maximum number of schedulable PDSCHs among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* for the cell, m_B is 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-1-3* for the cell, and each m_B bit(s) corresponds to one scheduled PDSCH as defined in clause 6.1.4 in [6, TS 38.214],
    - If 0 bit is configured, rv_{id} to be applied is 0;

- 1 bit according to Table 7.3.1.2.3-1;
- 2 bits according to Table 7.3.1.1.1-2.

- otherwise, the corresponding block is 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-1-3* configured for the cell,
  - If 0 bit is configured, rv_{id} to be applied is 0;
  - 1 bit according to Table 7.3.1.2.3-1;
  - 2 bits according to Table 7.3.1.1.1-2.

For transport block 2:

- Modulation and coding scheme - number of bits determined by the following:
  - block number 1, block number 2,…, block number $N_{cell}^{DL,3}$

  If *scheduledCellComboListDCI-1-3* for the scheduled cell set is configured with more than one entry, $N_{cell}^{DL,3}$ is the number of scheduled cells indicated by Scheduled cells indicator field and configured with *maxNrofCodeWordsScheduledByDCI = 2*; if *scheduledCellComboListDCI-1-3* for the scheduled cell set is configured with only one entry, $N_{cell}^{DL,3}$ is the number of cells configured by higher layer parameter *scheduledCellComboListDCI-1-3* and configured with *maxNrofCodeWordsScheduledByDCI = 2*; otherwise, $N_{cell}^{DL,3}$ is the number of cells configured by higher layer parameter *scheduledCellListDCI-1-3* in the scheduled cell set and configured with *maxNrofCodeWordsScheduledByDCI = 2*. Each block corresponds to the modulation and coding scheme for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the modulation and coding scheme for the cell with the smallest serving cell index. Each block is 5 bits as defined in Clause 6.1.4.1 of [6, TS 38.214].
- New data indicator - number of bits determined by the following:
  - block number 1, block number 2,…, block number $N_{cell}^{DL,3}$

  Each block corresponds to the new data indicator for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the new data indicator for the cell with the smallest serving cell index. If *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* is configured for a cell, the number of bits for the corresponding block is equal to the maximum number of schedulable PDSCHs among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* for the cell, where each bit corresponds to one scheduled PDSCH as defined in clause 6.1.4 in [6, TS 38.214]; otherwise, the corresponding block is 1 bit.

- Redundancy version - number of bits determined by the following:
  - block number 1, block number 2,…, block number $N_{cell}^{DL,3}$

  Each block corresponds to the redundancy version for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the redundancy version for the cell with the smallest serving cell index. The number of bits for each block is determined by following:

  - if *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* is configured for a cell, the number of bits for the corresponding block is determined by $m_A \times m_B$, where m_A is the maximum number of schedulable PDSCHs among all entries in the higher layer parameter *pdsch-TimeDomainAllocationListForMultiPDSCH-DCI-1-3* for the cell, m_B is 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-1-3* for the cell, and each m_B bit(s) corresponds to one scheduled PDSCH as defined in clause 6.1.4 in [6, TS 38.214],
    - If 0 bit is configured, rv_{id} to be applied is 0;
    - 1 bit according to Table 7.3.1.2.3-1;
    - 2 bits according to Table 7.3.1.1.1-2.
  - otherwise, the corresponding block is 0, 1 or 2 bits determined by higher layer parameter *numberOfBitsForRV-DCI-1-3* configured for the cell,
    - If 0 bit is configured, rv_{id} to be applied is 0;
    - 1 bit according to Table 7.3.1.2.3-1;
    - 2 bits according to Table 7.3.1.1.1-2.

If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part and the value of *maxNrofCodeWordsScheduledByDCI* for the indicated bandwidth part equals 2 and the value of *maxNrofCodeWordsScheduledByDCI* for the active bandwidth part equals 1, the UE assumes zeros are padded when interpreting the "Modulation and coding scheme", "New data indicator", and "Redundancy version" fields of transport block 2 according to Clause 12 of [5, TS38.213], and the UE

ignores the "Modulation and coding scheme", "New data indicator", and "Redundancy version" fields of transport block 2 for the indicated bandwidth part.

- HARQ process number - number of bits determined by the following:
  - block number 1, block number 2,…, block number N_{cell}^{DL}

  Each block corresponds to the HARQ process number for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the HARQ process number for the cell with the smallest serving cell index. Each block is 0, 1, 2, 3, 4 or 5 bits determined by higher layer parameter *harq-ProcessNumberSizeDCI-1-3* or *harq-ProcessNumberSizeDCI-1-3-Ext* configured for the cell corresponding to the block.

- Downlink assignment index - number of bits as defined in the following
  - 4 bits if the higher layer parameter *pdsch-HARQ-ACK-Codebook=dynamic*, where the 2 MSB bits are the counter DAI and the 2 LSB bits are the total DAI;
  - 0 bits otherwise.

  If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-Codebook* is replaced by *pdsch-HARQ-ACK-Codebook-secondaryPUCCHgroup-r16* if present for the secondary PUCCH group.

  If higher layer parameter *priorityIndicatorDCI-1-3* is configured, if the bit width of the Downlink assignment index in DCI format 1_3 for one HARQ-ACK codebook is not equal to that of the Downlink assignment index in DCI format 1_3 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller Downlink assignment index until the bit width of the Downlink assignment index in DCI format 1_3 for the two HARQ-ACK codebooks are the same.

- TPC command for scheduled PUCCH - 2 bits as defined in Clause 7.2.1 of [5, TS 38.213]
- PUCCH resource indicator - 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]
- PDSCH-to-HARQ_feedback timing indicator - 0, 1, 2, or 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *dL-DataToUL-ACK*.

  If higher layer parameter *priorityIndicatorDCI-1-3* is configured, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_3 for one HARQ-ACK codebook is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_3 for the other HARQ-ACK codebook on the same cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_3 for the two HARQ-ACK codebooks are the same.

  If higher layer parameter *pucch-sSCellDynDCI-1-3* is configured, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_3 associated with one cell for PUCCH transmission is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_3 associated with the other cell for PUCCH transmission, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 1_3 associated with the two cells are the same.

- One-shot HARQ-ACK request - 0 or 1 bit.
  - 1 bit if higher layer parameter pdsch-HARQ-ACK-OneShotFeedbackDCI-1-3 or pdsch-HARQ-ACK-enhType3DCI-1-3 is configured;
  - 0 bit otherwise.
- Enhanced Type 3 codebook indicator - 0, 1, 2, or 3 bits.
  - 0 bit if pdsch-HARQ-ACK-enhType3DCIfieldDCI-1-3 is not configured;
  - $\lceil \log_2(n_{CB}) \rceil$ bits otherwise, where n_{CB} is the number of entries in the higher layer parameter pdsch-HARQ-ACK-EnhType3ToAddModList.

  If the UE is configured with a PUCCH-SCell, *pdsch-HARQ-ACK-EnhType3ToAddModList* is replaced by *pdsch-HARQ-ACK-EnhType3SecondaryList* for the secondary PUCCH group.

- HARQ-ACK retransmission indicator - 0 or 1 bit.
  - 1 bit if higher layer parameter *pdsch-HARQ-ACK-retxDCI-1-3* is configured.
  - 0 bit otherwise.

- Antenna ports - number of bits determined by the following:
  - If *antennaPortsDCI-1-3= type1a* is configured by higher layer,
    - $\max_{r \in \{1,2,\ldots,N_{cell}^{DL,2}\}} M_A(r)$ bits applying to the scheduled cells independently, where $N_{cell}^{DL,2}$ is the number of cells configured by higher layer parameter *scheduledCellListDCI-1-3* in the scheduled cell set, r is mapped to the cells according to an ascending order of a serving cell index with $r=1$ corresponding to the cell with the smallest serving cell index, and $M_A(r)$ is defined below.
  - If *antennaPortsDCI-1-3= type2* is configured by higher layer,
    - block number 1, block number 2,…, block number N_{cell}^{DL}

    Each block corresponds to the Antenna ports information for a cell, and the blocks are placed according to an ascending order of a serving cell index, with block number 1 corresponding to the Antenna ports information for the cell with the smallest serving cell index. Each block is defined below.

  $M_A(r)$ above for the case of *antennaPortsDCI-1-3= type1A* or each block above for the case of *antennaPortsDCI-1-3= type2* is defined by the following:

  - 4, 5, 6, 7 or 8 bits as defined by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10 and Tables 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A, where the number of CDM groups without data of values 1, 2, and 3 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively. The antenna ports $\{p_0,\ldots,p_{v-1}\}$ shall be determined according to the ordering of DMRS port(s) given by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10 or Tables 7.3.1.2.2-1A/2A/3A/4A/7A/8A/9A/10A.

  If a UE is configured with both *dmrs-DownlinkForPDSCH-MappingTypeA* and *dmrs-DownlinkForPDSCH-MappingTypeB*, the bitwidth of this field equals $\max\{x_A, x_B\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeA* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeB*. A number of $|x_A - x_B|$ zeros are padded in the MSB of this field, if the mapping type of the PDSCH corresponds to the smaller value of x_A and x_B.

- Transmission configuration indication - number of bits determined by the following:
  - 0 bit if higher layer parameter *tci-PresentInDCI* is not enabled or if higher layer parameter *tci-ListDCI-1-3* is not configured;
  - otherwise $\lceil \log_2(I_{TCI}) \rceil$ bits, where I_{TCI} is the number of entries in the higher layer parameter *tci-ListDCI-1-3*. This field is used to indicate an entry in the higher layer parameter *tci-ListDCI-1-3* according to Table 7.3.1.2.4-5. Each entry in the higher layer parameter *tci-ListDCI-1-3* contains the 'Transmission configuration indication' index for each cell in the scheduled cell set, where the 'Transmission configuration indication' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'Transmission configuration indication' index is 3 bits as defined in Clause 5.1.5 of [6, TS38.214].

  If "Bandwidth part indicator" field indicates a bandwidth part other than the active bandwidth part,

  - if the higher layer parameter *tci-PresentInDCI* is not enabled for the CORESET used for the PDCCH carrying the DCI format 1_3,
    - the UE assumes *tci-PresentInDCI* is not enabled for all CORESETs in the indicated bandwidth part;
  - otherwise,
    - the UE assumes *tci-PresentInDCI* is enabled for all CORESETs in the indicated bandwidth part.
- SRS request - $\lceil \log_2(I_{SRS}) \rceil$ bits, where I_{SRS} is the number of entries in the higher layer parameter *srs-RequestListDCI-1-3*, or 0 bit if the higher layer parameter *srs-RequestListDCI-1-3* is not configured. This field is used to indicate an entry in the higher layer parameter *srs-RequestListDCI-1-3* according to Table 7.3.1.2.4-6. Each entry in the higher layer parameter *srs-RequestListDCI-1-3* contains the 'SRS request' index for each cell in the scheduled cell set, where the 'SRS request' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'SRS request' index is defined by the following:

  - 2 bits as defined by Table 7.3.1.1.2-24 for UEs not configured with *supplementaryUplink* in *ServingCellConfig* in the cell; 3 bits for UEs configured with *supplementaryUplink* in *ServingCellConfig* in the cell where the first bit is the non-SUL/SUL indicator as defined in Table 7.3.1.1.1-1 and the second and third bits are defined by Table 7.3.1.1.2-24. This bit field may also indicate the associated CSI-RS according to Clause 6.1.1.2 of [6, TS 38.214].

- SRS offset indicator - $\lceil \log_2(I_{offset}) \rceil$ bits, where I_{offset} is the number of entries in the higher layer parameter *srs-OffsetListDCI-1-3*, or 0 bit if the higher layer parameter *srs-OffsetListDCI-1-3* is not configured. This field is used to indicate an entry in the higher layer parameter *srs-OffsetListDCI-1-3* according to Table 7.3.1.2.4-7. Each entry in the higher layer parameter *srs-OffsetListDCI-1-3* contains the 'SRS offset indicator' index for each cell in the scheduled cell set, where the 'SRS offset indicator' indexes for all the cells are placed according to an ascending order of a serving cell index. Each 'SRS offset indicator' index is defined by the following:
  - 0 bit if higher layer parameter *AvailableSlotOffset* is not configured for any aperiodic SRS resource set in the scheduled cell, or if higher layer parameter *AvailableSlotOffset* is configured for at least one aperiodic SRS resource set in the scheduled cell and the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) is 1;
  - otherwise, $\lceil \log_2(K) \rceil$ bits are used to indicate available slot offset according to Table 7.3.1.1.2-37 and Clause 6.2.1 of [6, TS 38.214], where K is the maximum number of entries of *availableSlotOffsetList* configured for all aperiodic SRS resource set(s) in the scheduled cell;
- DMRS sequence initialization - 1 bit. This field is applied to all the scheduled cells indicated by Scheduled cells indicator field or Frequency domain resource assignment field independently.
- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-1-3* is not configured; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213].
- ChannelAccess-CPext - 0, 1, 2, 3 or 4 bits. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *ul-AccessConfigListDCI-1-1* or in Table 7.3.1.1.1-4A if *channelAccessMode-r16* = "*semiStatic*" is provided, for operation in a cell with shared spectrum channel access in frequency range 1, or for operation in frequency range 2-2 if *ChannelAccessMode2-r17* is provided; otherwise 0 bit. One or more entries from Table 7.3.1.2.2-6 or Table 7.3.1.2.2-6A are configured by the higher layer parameter *ul-AccessConfigListDCI-1-1*.
- Minimum applicable scheduling offset indicator - 0 or 1 bit
  - 0 bit if higher layer parameter *minimumSchedulingOffsetK0DCI-1-3* is not configured;
  - 1 bit otherwise. The 1 bit indication is used to determine the minimum applicable K0 for the active DL BWP and the minimum applicable K2 for the active UL BWP, if configured respectively, according to Table 7.3.1.1.2-33. If the minimum applicable K0 is indicated, the minimum applicable value of the aperiodic CSI-RS triggering offset for an active DL BWP for each scheduled cell shall be the same as the minimum applicable K0.
- SCell dormancy indication - 0 bit if higher layer parameter *dormancyDCI-1-3* or *dormancyGroupWithinActiveTime* is not configured; otherwise 1, 2, 3, 4, or 5 bits bitmap determined according to the number of different *DormancyGroupID(s)* provided by higher layer parameter *dormancyGroupWithinActiveTime*, where each bit corresponds to one of the SCell group(s) configured by higher layers parameter *dormancyGroupWithinActiveTime*, with MSB to LSB of the bitmap corresponding to the first to the last configured SCell group in ascending order of *DormancyGroupID*. The field is only present when this format is carried by PDCCH on the primary cell within DRX Active Time and the UE is configured with at least two DL BWPs for an SCell.

  If the "One-shot HARQ-ACK request" field is not present or set to '0', and if the "HARQ-ACK retransmission indicator" field is not present or set to '0', and if all bits of the corresponding block(s) of the frequency domain resource assignment field are set to 0 for resource allocation type 0 or set to 1 for resource allocation type 1 or set to 0 or 1 for dynamic switch resource allocation type for one or more cells in the scheduled cell set, this field is reserved and the following fields, corresponding to the cell with smallest serving cell index among the one or more cells, among the fields above are used for SCell dormancy indication, where each bit corresponds to one of the configured SCell(s), with MSB to LSB of the following fields concatenated in the order below corresponding to the SCell with lowest to highest SCell index

  - Modulation and coding scheme of transport block 1
  - New data indicator of transport block 1
  - Redundancy version of transport block 1
  - HARQ process number
  - Antenna port(s) if *antennaPortsDCI1-3= type2* is configured by higher layer.
- PDCCH monitoring adaptation indication - 0, 1 or 2 bits
  - 0 bit if higher layer parameter *pdcchMonAdaptDCI-1-3* is not enabled;
  - otherwise,

- 1 or 2 bits, if *searchSpaceGroupIdList-r17* is not configured and if *pdcch-SkippingDurationList* is configured
  - 1 bit if the UE is configured with only one duration by *pdcch-SkippingDurationList;*
  - 2 bits if the UE is configured with more than one duration by *pdcch-SkippingDurationList*.
- 1 or 2 bits, if *pdcch-SkippingDurationList* is not configured and if *searchSpaceGroupIdList-r17* is configured
  - 1 bit if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0 and search space set(s) with group index 1, and if the UE is not configured by *searchSpaceGroupIdList-r17* with any search space set with group index 2;
  - 2 bits if the UE is configured by *searchSpaceGroupIdList-r17* with search space set(s) with group index 0, search space set(s) with group index 1 and search space set(s) with group index 2;
- 2 bits, if pdcch-SkippingDurationList is configured and if searchSpaceGroupIdList-r17 is configured

- PUCCH Cell indicator - 0 or 1 bit.
  - 1 bit if higher layer parameter *pucch-sSCellDynDCI-1-3* is configured.
  - 0 bit otherwise.
- Measurement gap cancellation – 0 bit if higher layer parameter *mg-CancellationDCI-1-3* is not configured; otherwise 1 bit as defined in Clause 10.6 in [5, TS 38.213].

If *scheduledCellComboListDCI-1-3* for the cell set is configured, zeros shall be appended to DCI format 1_3 if needed until the payload size equals the size of DCI format 1_3 that is determined by the configuration of the corresponding active bandwidth part(s) of the scheduled cells in the entry which results in the largest size among the entries in the higher layer parameter *scheduledCellComboListDCI-1-3*.

If an SCell within the scheduled cell set is deactivated and the *firstActiveDownlinkBWP-Id* corresponding to the SCell is not set to dormant BWP, the UE determines the bitwidth of the fields in DCI format 1_3 based on a DL BWP provided by *firstActiveDownlinkBWP-Id* for the SCell. If the active DL BWP of an SCell within the scheduled cell set is a dormant DL BWP, or if an SCell within the scheduled cell set is deactivated and the *firstActiveDownlinkBWP-Id* corresponding to the SCell is set to dormant BWP, the UE determines the bitwidth of the fields in DCI format 1_3 based on a DL BWP provided by *firstWithinActiveTimeBWP-Id* for the SCell if provided; otherwise, based on a DL BWP provided by *firstOutsideActiveTimeBWP-Id* for the SCell.

Table 7.3.1.2.4-1: Scheduled cells indicator in DCI format 1_3

Bit field mapped to index	Scheduled cells
0	The cells configured by the 1st entry in *scheduledCellComboListDCI-1-3*
1	The cells configured by the 2nd entry in *scheduledCellComboListDCI-1-3*
2	The cells configured by the 3rd entry in *scheduledCellComboListDCI-1-3*, if any
3	The cells configured by the 4th entry in *scheduledCellComboListDCI-1-3*, if any
4	The cells configured by the 5th entry in *scheduledCellComboListDCI-1-3*, if any
5	The cells configured by the 6th entry in *scheduledCellComboListDCI-1-3*, if any
6	The cells configured by the 7th entry in *scheduledCellComboListDCI-1-3*, if any
7	The cells configured by the 8th entry in *scheduledCellComboListDCI-1-3*, if any
8	The cells configured by the 9th entry in *scheduledCellComboListDCI-1-3*, if any
9	The cells configured by the 10th entry in *scheduledCellComboListDCI-1-3*, if any
10	The cells configured by the 11th entry in *scheduledCellComboListDCI-1-3*, if any
11	The cells configured by the 12th entry in *scheduledCellComboListDCI-1-3* if any
12	The cells configured by the 13th entry in *scheduledCellComboListDCI-1-3*, if any
13	The cells configured by the 14th entry in *scheduledCellComboListDCI-1-3*, if any
14	The cells configured by the 15th entry in *scheduledCellComboListDCI-1-3*, if any
15	The cells configured by the 16th entry in *scheduledCellComboListDCI-1-3*, if any

Table 7.3.1.2.4-2: Time domain resource assignment in DCI format 1_3

Bit field mapped to index	Indicated time domain resource allocation
0	The 1st entry in *tdra-FieldIndexListDCI-1-3*
1	The 2nd entry in *tdra-FieldIndexListDCI-1-3*
2	The 3rd entry in *tdra-FieldIndexListDCI-1-3*, if any
…	…
i	The $(i+1)^{th}$ entry in *tdra-FieldIndexListDCI-1-3*, if any
…	…

Table 7.3.1.2.4-3: Rate matching indicator

Bit field mapped to index	Indicated rate matching pattern
0	The 1st entry in *rateMatchListDCI-1-3*
1	The 2nd entry in *rateMatchListDCI-1-3*
2	The 3rd entry in *rateMatchListDCI-1-3*, if any
…	…
15	The 16th entry in *rateMatchListDCI-1-3*, if any

Table 7.3.1.2.4-4: ZP CSI-RS trigger

Bit field mapped to index	Triggered aperiodic ZP CSI-RS
0	The 1st entry in *zp-CSI-RSListDCI-1-3*
1	The 2nd entry in *zp-CSI-RSListDCI-1-3*
2	The 3rd entry in *zp-CSI-RSListDCI-1-3*, if any
…	…
7	The 8th entry in *zp-CSI-RSListDCI-1-3*, if any

Table 7.3.1.2.4-5: Transmission configuration indication

Bit field mapped to index	Indicated transmission configuration indication
0	The 1st entry in *tci-ListDCI-1-3*
1	The 2nd entry in *tci-ListDCI-1-3*
2	The 3rd entry in *tci-ListDCI-1-3*, if any
…	…
15	The 16th entry in *tci-ListDCI-1-3*, if any

Table 7.3.1.2.4-6: SRS request in DCI format 1_3

Bit field mapped to index	Triggered aperiodic SRS resource set(s)
0	The 1st entry in *srs-RequestListDCI-1-3*
1	The 2nd entry in *srs-RequestListDCI-1-3*
2	The 3rd entry in *srs-RequestListDCI-1-3*, if any
…	…
15	The 16th entry in *srs-RequestListDCI-1-3*, if any

Table 7.3.1.2.4-7: SRS offset indicator in DCI format 1_3

Bit field mapped to index	Available slot offset
0	The 1st entry in *srs-OffsetListDCI-1-3*
1	The 2nd entry in *srs-OffsetListDCI-1-3*
2	The 3rd entry in *srs-OffsetListDCI-1-3*, if any
…	…
7	The 8th entry in *srs-OffsetListDCI-1-3*, if any

7.3.1.3 DCI formats for other purposes

7.3.1.3.1 Format 2_0

DCI format 2_0 is used for notifying the slot format, COT duration, available RB set, and search space set group switching.

The following information is transmitted by means of the DCI format 2_0 with CRC scrambled by SFI-RNTI:

- If the higher layer parameter *slotFormatCombToAddModList* is configured,
  - Slot format indicator 1, Slot format indicator 2, …, Slot format indicator *N*,
- If the higher layer parameter *availableRB-SetsToAddModList* is configured,

  - Available RB set Indicator 1, Available RB set Indicator 2, …, Available RB set Indicator *N1*,
- If the higher layer parameter *co-DurationsPerCellToAddModList* is configured
  - COT duration indicator 1, COT duration indicator 2, …, COT duration indicator *N2*.
- If the higher layer parameter *switchTriggerToAddModList* is configured
  - Search space set group switching flag 1, Search space set group switching flag 2, …, Search space set group switching flag *M*.

The size of DCI format 2_0 is configurable by higher layers up to 128 bits, according to Clause 11.1.1 of [5, TS 38.213].

7.3.1.3.2 Format 2_1

DCI format 2_1 is used for notifying the PRB(s) and OFDM symbol(s) where UE may assume no transmission is intended for the UE.

The following information is transmitted by means of the DCI format 2_1 with CRC scrambled by INT-RNTI:

- Pre-emption indication 1, Pre-emption indication 2, ⋯, Pre-emption indication *N*.

The size of DCI format 2_1 is configurable by higher layers up to 126 bits, according to Clause 11.2 of [5, TS 38.213]. Each pre-emption indication is 14 bits.

7.3.1.3.3 Format 2_2

DCI format 2_2 is used for the transmission of TPC commands for PUCCH and PUSCH.

The following information is transmitted by means of the DCI format 2_2 with CRC scrambled by TPC-PUSCH-RNTI or TPC-PUCCH-RNTI:

- block number 1, block number 2,…, block number *N*

The parameter *tpc-PUSCH* or *tpc-PUCCH* provided by higher layers determines the index to the block number for an UL of a cell, with the following fields defined for each block:

- Closed loop indicator - 0 or 1 bit.
  - For DCI format 2_2 with TPC-PUSCH-RNTI, 0 bit if the UE is not configured with high layer parameter *twoPUSCH-PC-AdjustmentStates*, in which case UE assumes each block in the DCI format 2_2 is of 2 bits; 1 bit otherwise, in which case UE assumes each block in the DCI format 2_2 is of 3 bits;
  - For DCI format 2_2 with TPC-PUCCH-RNTI, 0 bit if the UE is not configured with high layer parameter *twoPUCCH-PC-AdjustmentStates*, in which case UE assumes each block in the DCI format 2_2 is of 2 bits; 1 bit otherwise, in which case UE assumes each block in the DCI format 2_2 is of 3 bits;
- TPC command -2 bits

The number of information bits in format 2_2 shall be equal to or less than the payload size of format 1_0 monitored in common search space in the same serving cell. If the number of information bits in format 2_2 is less than the payload size of format 1_0 monitored in common search space in the same serving cell, zeros shall be appended to format 2_2 until the payload size equals that of format 1_0 monitored in common search space in the same serving cell.

7.3.1.3.4 Format 2_3

DCI format 2_3 is used for the transmission of a group of TPC commands for SRS transmissions by one or more UEs. Along with a TPC command, a SRS request may also be transmitted.

The following information is transmitted by means of the DCI format 2_3 with CRC scrambled by TPC-SRS-RNTI:

- block number 1, block number 2, …, block number B

  where the starting position of a block is determined by the parameter *startingBitOfFormat2-3* or *startingBitOfFormat2-3SUL-v1530* provided by higher layers for the UE configured with the block.

If the UE is configured with higher layer parameter *srs-TPC-PDCCH-Group = typeA* for an UL without PUCCH and PUSCH or an UL on which the SRS power control is not tied with PUSCH power control, one or two blocks are configured for the UE by higher layers where one block applies to non-SUL carriers and another block applies to SUL carriers, with the following fields defined for each block:

- SRS request - 0 or 2 bits. The presence of this field is according to the definition in Clause 11.4 of [5, TS38.213]. If present, this field is interpreted as defined by Table 7.3.1.1.2-24.
- TPC command number 1, TPC command number 2, ..., TPC command number *N*, where each TPC command applies to a respective UL carrier provided by higher layer parameter *cc-IndexInOneCC-Set*
- Closed loop indicator 1, Closed loop indicator 2, ..., Closed loop indicator N_1, where each Closed loop indicator applies to a respective UL carrier provided by higher layer parameter *cc-IndexInOneCC-Set*. A Closed loop indicator is present only if the UE is configured with higher layer parameter *enableTwoSeparatePowerControlAdjustmentStatesForSRS* for the UL carrier.

If the UE is configured with higher layer parameter *srs-TPC-PDCCH-Group = typeB* for an UL without PUCCH and PUSCH or an UL on which the SRS power control is not tied with PUSCH power control, one block or more blocks is configured for the UE by higher layers where each block applies to an UL carrier, with the following fields defined for each block:

- SRS request - 0 or 2 bits. The presence of this field is according to the definition in Clause 11.4 of [5, TS38.213]. If present, this field is interpreted as defined by Table 7.3.1.1.2-24.
- TPC command -2 bits
- Closed loop indicator - 1 bit if the UE is configured with higher layer parameter *enableTwoSeparatePowerControlAdjustmentStatesForSRS* for the UL carrier; 0 bit otherwise.

The number of information bits in format 2_3 shall be equal to or less than the payload size of format 1_0 monitored in common search space in the same serving cell. If the number of information bits in format 2_3 is less than the payload size of format 1_0 monitored in common search space in the same serving cell, zeros shall be appended to format 2_3 until the payload size equals that of format 1_0 monitored in common search space in the same serving cell.

7.3.1.3.5 Format 2_4

DCI format 2_4 is used for notifying the PRB(s) and OFDM symbol(s) where UE cancels the corresponding UL transmission from the UE according to Clause 11.2A of [5, TS 38.213].

The following information is transmitted by means of the DCI format 2_4 with CRC scrambled by CI-RNTI:

- Cancellation indication 1, Cancellation indication 2, …, Cancellation indication indication *N*.

The size of DCI format 2_4 is configurable by higher layers parameter *dci-PayloadSizeForCI* up to 126 bits, according to Clause 11.2A of [5, TS 38.213]. The number of bits for each cancellation indication is configurable by higher layer parameter *ci-PayloadSize*. For a UE, there is at most one cancellation indication for an UL carrier.

7.3.1.3.6 Format 2_5

DCI format 2_5 is used for notifying the availability of soft resources as defined in Clause 9.3.1 of [10, TS 38.473]

The following information is transmitted by means of the DCI format 2_5 with CRC scrambled by AI-RNTI:

- Availability indicator 1, Availability indicator 2, ⋯, Availability indicator *N*.

The size of DCI format 2_5 is configurable by higher layers up to 128 bits, according to Clause 14 of [5, TS 38.213].

7.3.1.3.7 Format 2_6

DCI format 2_6 is used for notifying the power saving information outside DRX Active Time for one or more UEs.

The following information is transmitted by means of the DCI format 2_6 with CRC scrambled by PS-RNTI:

- block number 1, block number 2,…, block number *N*

  where the starting position of a block is determined by the parameter *ps-PositionDCI-2-6* provided by higher layers for the UE configured with the block.

If the UE is configured with higher layer parameter *ps-RNTI* and *dci-Format2-6*, one block is configured for the UE by higher layers, with the following fields defined for the block:

- Wake-up indication - 1 bit
- SCell dormancy indication - 0 bit if higher layer parameter *dormancyGroupOutsideActiveTime* is not configured; otherwise 1, 2, 3, 4 or 5 bits bitmap determined according to the number of different *DormancyGroupID(s)* provided by higher layer parameter *dormancyGroupOutsideActiveTime*, where each bit corresponds to one of the SCell group(s) configured by higher layers parameter *dormancyGroupOutsideActiveTime*, with MSB to LSB of the bitmap corresponding to the first to last configured SCell group in ascending order of *DormancyGroupID*.

The size of DCI format 2_6 is indicated by the higher layer parameter *sizeDCI-2-6*, according to Clause 10.3 of [5, TS 38.213].

7.3.1.3.8 Format 2_7

DCI format 2_7 is used for notifying the paging early indication and TRS availability indication for one or more UEs.

The following information is transmitted by means of the DCI format 2_7 with CRC scrambled by PEI-RNTI:

- Paging indication field - $N_{PO}^{PEI} N_{SG}^{PO}$ bit(s), where
  - N_{PO}^{PEI} is the number of paging occasions configured by higher layer parameter *po-NumPerPEI* as defined in Clause 10.4A in [5, TS 38.213];
  - N_{SG}^{PO} is the number of sub-groups of a paging occasion configured by higher layer parameter *subgroupsNumPerPO*.
  - Each bit in the field indicates one UE subgroup of a paging occasion.
- TRS availability indication - 1, 2, 3, 4, 5, or 6 bits, where the number of bits is equal to one plus the highest value of all the *indBitID*(s) provided by the *trs-ResourceSetConfig* or the number of bits is equal to one plus the highest value of all the *indBitID-r18*(s) provided by the *trs-ResourceSetConfig-r18* if configured if configured; 0 bits otherwise.

The size of DCI format 2_7 is indicated by the higher layer parameter *payloadSizeDCI-2-7*, according to Clause 10.4A of [5, TS 38.213]. The number of information bits in format 2_7 shall be equal to or less than the payload size of format 2_7. If the number of information bits in format 2_7 is less than the size of format 2_7, the remaining bits are reserved.

7.3.1.3.9 Format 2_8

DCI format 2_8 is used for notifying the aperiodic beam indication and associated time resources

The following information is transmitted by means of the DCI format 2_8 with CRC scrambled by NCR-RNTI:

- Beam index 1, Beam index 2, …, Beam index N

  The bitwidth of each beam index field is determined by the higher layer parameter *aperiodicBeamFieldWidth*.

- Time resource indication 1, Time resource indication 2, …, Time resource indication N

  The bitwidth of each time resource indication field is determined by max $\left\{ \lceil \log_2(I) \rceil, 1 \right\}$, where I is the number of time domain resources configured by *aperiodicFwdConfig*. The bit field indexes of a time resource indication field are mapped to the time domain resources configured by *aperiodicFwdConfig* according to an ascending order of a resource identity configured by *aperiodicFwdTimeRsrcId*, with the bit field index 0 mapped to the time resource with the smallest resource identity.

The N beam indexes are sequentially associated with the N time resource indications with one to one mapping. N is configured by the higher layer parameter *numberOfFields*. The size of DCI format 2_8 is up to 128 bits.

7.3.1.3.10 Format 2_9

DCI format 2_9 with CRC scrambled by cellDTRX-RNTI is used for activating or de-activating the cell DTX and/or DRX configuration of one or multiple serving cells for one or more UEs, and/or for providing NES-mode indication of the primary cell for one or more UEs. DCI format 2_9 with CRC scrambled by ssbPeriodicityIndication-RNTI is used for adapting SSB periodicity of one or multiple serving cells for one or more UEs.

The following information is transmitted by means of the DCI format 2_9 with CRC scrambled by cellDTRX-RNTI:

- block number 1, block number 2,…, block number *N*

  where the starting position of a block associated with a serving cell is determined by the parameter *positionInDCI-cellDTRX* provided by higher layers for the UE.

If the UE is configured to monitor DCI 2_9 with CRC scrambled by cellDTRX-RNTI and for a DCI format 2_9 with CRC scrambled by cellDTRX-RNTI, one or more blocks are configured for the UE by higher layers, with the following fields defined for each block:

- Cell DTX/DRX indication - number of bits determined by the following:
  - If higher layer parameter *cellDTX-DRX-L1activation* is configured
    - 2 bits as defined in Clause 11.5 of [5, TS38.213] if *cellDTX-DRX-ConfigType* is configured to *dtxdrx* for the associated serving cell of the block, with the MSB corresponding to cell DTX configuration and the LSB corresponding to cell DRX configuration;
    - 1 bit as defined in Clause 11.5 of [5, TS38.213] if *cellDTX-DRX-ConfigType* is configured to either *dtx* or *drx* for the associated serving cell of the block;
  - 0 bit otherwise.
- NES-mode indication – 1 bit indicating NES-specific CHO execution condition as defined in Clause 11.5 of [5, TS38.213], if the higher layer parameter *nesEvent* is configured and the associated serving cell of the block is primary cell; 0 bit otherwise.

The following information is transmitted by means of the DCI format 2_9 with CRC scrambled by ssbPeriodicityIndication-RNTI:

- block number 1, block number 2,…, block number *N1*

  where the starting position of a block associated with a serving cell is determined by the parameter *positionInDCI-ssbPeriodicityIndicationForScell* provided by higher layers for the UE.

If the UE is configured to monitor DCI 2_9 with CRC scrambled by ssbPeriodicityIndication-RNTI and for a DCI format 2_9 with CRC scrambled by ssbPeriodicityIndication-RNTI, one or more blocks are configured for the UE by higher layers, with the following fields defined for each block:

- SSB periodicity indication - number of bits determined by the following:
  - If higher layer parameter *addl-ssb-Periodicity* is configured
    - 1 bit as defined in Clause 11.6 of [5, TS38.213], if one additional SSB periodicity is configured by higher layer parameter *addl-ssb-Periodicity*;
    - 2 bits as defined in Clause 11.6 of [5, TS38.213], if two additional SSB periodicities are configured by higher layer parameter *addl-ssb-Periodicity*;
  - 0 bit otherwise.

The size of DCI format 2_9 is indicated by the higher layer parameter *sizeDCI-2-9*. If the number of information bits in format 2_9 is less than the size of format 2_9, the remaining bits are reserved.

7.3.1.4 DCI formats for scheduling of sidelink

7.3.1.4.1 Format 3_0

DCI format 3_0 is used for scheduling of NR PSCCH and NR PSSCH in one cell, or scheduling of NR PSCCH, NR PSSCH and NR SL PRS for a shared SL PRS resource pool in one cell.

The following information is transmitted by means of the DCI format 3_0 with CRC scrambled by SL-RNTI or SL-CS-RNTI:

- Resource pool index - $\lceil \log_2 I \rceil$ bits, where I is the total number of resource pools for transmission configured by the higher layer parameter *sl-TxPoolScheduling*, if configured, and *sl-DiscTxPoolScheduling*, if configured.
- Time gap - 3 bits determined by higher layer parameter *sl-DCI-ToSL-Trans*, as defined in clause 8.1.2.1 of [6, TS 38.214]
- HARQ process number - 4 bits.
- New data indicator - 1 bit.
- Lowest index of the subchannel allocation to the initial transmission - $\lceil \log_2(N_{\text{subChannel}}^{\text{SL}}) \rceil$ bits as defined in Clause 8.1.2.2 of [6, TS 38.214].
- Lowest index of the RB set allocation to the initial transmission - $\lceil \log_2(N_{RBset}) \rceil$ bits as defined in Clause 8.1.2.2 of [6, TS 38.214] if the higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured to 'interlaceRB'; 0 bit otherwise.
- SCI format 1-A fields according to clause 8.3.1.1:
  - Frequency resource assignment.
  - Time resource assignment.
- PSFCH-to-HARQ feedback timing indicator - $\lceil \log_2 N_{fb_timing} \rceil$ bits, where N_{fb_timing} is the number of entries in the higher layer parameter *sl-PSFCH-ToPUCCH*, as defined in clause 16.5 of [5, TS 38.213]
- PUCCH resource indicator - 3 bits as defined in clause 16.5 of [5, TS 38.213].
- Configuration index - 0 bit if the UE is not configured to monitor DCI format 3_0 with CRC scrambled by SL-CS-RNTI; otherwise 3 bits as defined in clause 8.1.2 of [6, TS 38.214]. If the UE is configured to monitor DCI format 3_0 with CRC scrambled by SL-CS-RNTI, this field is reserved for DCI format 3_0 with CRC scrambled by SL-RNTI.
- Counter sidelink assignment index - 2 bits
  - 2 bits as defined in clause 16.5.2 of [5, TS 38.213] if the UE is configured with *pdsch-HARQ-ACK-Codebook = dynamic*
  - 2 bits as defined in clause 16.5.1 of [5, TS 38.213] if the UE is configured with *pdsch-HARQ-ACK-Codebook = semi-static*
- Padding bits, if required

If the total number of transmit resource pools provided in *sl-TxPoolScheduling*, if configured, and *sl-DiscTxPoolScheduling*, if configured, is larger than one, zeros shall be appended to the DCI format 3_0 until the payload size is equal to the size of a DCI format 3_0 given by a configuration of the transmit resource pool resulting in the largest number of information bits for DCI format 3_0.

If the UE is configured to monitor DCI format 3_1 and/or DCI format 3_2 and the number of information bits in DCI format 3_0 is less than the larger payload size of DCI format 3_1 if configured and DCI format 3_2 if configured, zeros shall be appended to DCI format 3_0 until the payload size equals the larger payload size of DCI format 3_1 if configured and DCI format 3_2 if configured.

7.3.1.4.2 Format 3_1

DCI format 3_1 is used for scheduling of LTE PSCCH and LTE PSSCH in one cell.

The following information is transmitted by means of the DCI format 3_1 with CRC scrambled by SL Semi-Persistent Scheduling V-RNTI:

- Timing offset - 3 bits determined by higher layer parameter *sl-TimeOffsetEUTRA-List*, as defined in clause 16.6 of [5, TS 38.213]
- Carrier indicator -3 bits as defined in 5.3.3.1.9A of [11, TS 36.212].

- Lowest index of the subchannel allocation to the initial transmission - $\lceil \log_2(N_{\text{subchannel}}^{\text{SL}}) \rceil$ bits as defined in 5.3.3.1.9A of [11, TS 36.212].
- Frequency resource location of initial transmission and retransmission, as defined in 5.3.3.1.9A of [11, TS 36.212]
- Time gap between initial transmission and retransmission, as defined in 5.3.3.1.9A of [11, TS 36.212]
- SL index - 2 bits as defined in 5.3.3.1.9A of [11, TS 36.212]
- SL SPS configuration index - 3 bits as defined in clause 5.3.3.1.9A of [11, TS 36.212].
- Activation/release indication - 1 bit as defined in clause 5.3.3.1.9A of [11, TS 36.212].

If the UE is configured to monitor DCI format 3_0 and/or DCI format 3_2 and the number of information bits in DCI format 3_1 is less than the larger payload size of DCI format 3_0 if configured and DCI format 3_2 if configured, zeros shall be appended to DCI format 3_1 until the payload size equals the larger payload size of DCI format 3_0 if configured and DCI format 3_2 if configured.

7.3.1.4.3 Format 3_2

DCI format 3_2 is used for scheduling of NR PSCCH and NR SL PRS for a dedicated SL PRS resource pool in one cell.

The following information is transmitted by means of the DCI format 3_2 with CRC scrambled by SL-PRS-RNTI or SL-PRS-CS-RNTI:

- Resource pool index - $\lceil \log_2 I \rceil$ bits, where I is the total number of dedicated SL PRS resource pools for transmission configured by the higher layer parameter *sl-PRS-TxPoolScheduling*, if configured.
- Time gap - 3 bits determined by higher layer parameter *sl-DCI-ToSL-Trans*, as defined in clause 8.2.4.1.1 of [6, TS 38.214]
- First SL PRS indicator - $\lceil \log_2 N_{SL-PRS} \rceil$ bits indicating the SL PRS resource ID for the first SL PRS transmission, where the value N_{SL-PRS} is the total number of SL PRS resources within a slot in a dedicated SL PRS resource pool and provided by the higher layer parameter *sl-PRS-ResourcesDedicatedSL-PRS-RP*.
- SCI format 1-B fields according to clause 8.3.1.2:
  - Time resource assignment
  - Resource ID indication
- Configuration index – 0 bit if the UE is not configured to monitor DCI format 3_2 with CRC scrambled by SL-PRS-CS-RNTI; otherwise 3 bits as defined in clause 8.2.4.1 of [6, TS 38.214]. If the UE is configured to monitor DCI format 3_2 with CRC scrambled by SL-PRS-CS-RNTI, this field is reserved for DCI format 3_2 with CRC scrambled by SL-PRS-RNTI.
- Activation/release indication – 0 bit if the UE is not configured to monitor DCI format 3_2 with CRC scrambled with SL-PRS-CS-RNTI; otherwise 1 bit, where value 0 indicates release and value 1 indicates activation. If the UE is configured to monitor DCI format 3_2 with CRC scrambled with SL-PRS-CS-RNTI, this field is reserved for DCI format 3_2 with CRC scrambled by SL-PRS-RNTI.
- Padding bits, if required.

If the total number of transmit resource pools provided in *sl-PRS-TxPoolScheduling*, if configured, is larger than one, zeros shall be appended to the DCI format 3_2 until the payload size is equal to the size of a DCI format 3_2 given by a configuration of the transmit resource pool resulting in the largest number of information bits for DCI format 3_2.

If the UE is configured to monitor DCI format 3_0 and/or DCI format 3_1 and the number of information bits in DCI format 3_2 is less than the larger payload size of DCI format 3_0 if configured and DCI format 3_1 if configured, zeros shall be appended to DCI format 3_2 until the payload size equals the larger payload size of DCI format 3_0 if configured and DCI format 3_1 if configured.

7.3.1.5 DCI formats for scheduling of MBS

7.3.1.5.1 Format 4_0

DCI format 4_0 is used for the scheduling of PDSCH for broadcast or for multicast in RRC_INACTIVE state in DL cell.

The following information is transmitted by means of the DCI format 4_0 with CRC scrambled by MCCH-RNTI or G-RNTI for broadcast configured by *MBS-SessionInfo*, or by Multicast MCCH-RNTI:

- Frequency domain resource assignment - $\lceil \log_2(N_{RB}^{DL,CFR}(N_{RB}^{DL,CFR}+1)/2) \rceil$ bits where $N_{RB}^{DL,CFR}$ equals to
  - the size of CORESET 0 if CORESET 0 is configured for the cell; and
  - the size of initial DL bandwidth part if CORESET 0 is not configured for the cell.
- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS38.214]
- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3 of [6, TS38.214]
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- MCCH change notification - 2 bits as defined in Clause 5.9.1.3 and Clause 5.10.1.3 of [9, TS38.331] if the CRC of the DCI format 4_0 is scrambled by MCCH-RNTI and Multicast MCCH-RNTI respectively. Otherwise, this bit field is reserved.
- Reserved bits - 14bits

7.3.1.5.2 Format 4_1

DCI format 4_1 is used for the scheduling of PDSCH for multicast in DL cell.

The following information is transmitted by means of the DCI format 4_1 with CRC scrambled by G-RNTI for multicast or G-CS-RNTI configured by *MBS-RNTI-SpecificConfig*, or by G-RNTI for multicast configured by *MBS-SessionInfoListMulticast*:

- Frequency domain resource assignment - $\lceil \log_2(N_{RB}^{DL,CFR}(N_{RB}^{DL,CFR}+1)/2) \rceil$ bits where $N_{RB}^{DL,CFR}$ equals to
  - the size of CORESET 0 if CORESET 0 is configured for the cell; and
  - the size of initial DL bandwidth part if CORESET 0 is not configured for the cell.
- Time domain resource assignment - 4 bits as defined in Clause 5.1.2.1 of [6, TS38.214]
- VRB-to-PRB mapping - 1 bit according to Table 7.3.1.2.2-5
- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3 of [6, TS38.214]
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- HARQ process number - 4 bits
- Downlink assignment index - 2 bits as defined in Clause 9.1.3 of [5, TS 38.213], as counter DAI
- PUCCH resource indicator - 3 bits as defined in Clause 9.2.3 of [5, TS38.213]
- PDSCH-to-HARQ_feedback timing indicator - 3 bits as defined in Clause 9.2.3 of [5, TS38.213]
- Reserved bits - 3 bits

7.3.1.5.3 Format 4_2

DCI format 4_2 is used for the scheduling of PDSCH for multicast in DL cell.

The following information is transmitted by means of the DCI format 4_2 with CRC scrambled by G-RNTI for multicast or G-CS-RNTI configured by *MBS-RNTI-SpecificConfig*:

- Frequency domain resource assignment - number of bits determined by the following, where $N_{RB}^{DL,CFR}$ is the size of the common frequency resource as defined in Clause 18 of [5, TS38.213].
  - N_{RBG} bits if only resource allocation type 0 is configured, where N_{RBG} is defined in Clause 5.1.2.2.1 of [6, TS38.214],
  - $\lceil \log_2(N_{RB}^{DL,CFR}(N_{RB}^{DL,CFR}+1)/2) \rceil$ bits if only resource allocation type 1 is configured, or
  - $max\left(\lceil \log_2(N_{RB}^{DL,CFR}(N_{RB}^{DL,CFR}+1)/2) \rceil, N_{RBG}\right)+1$ bits if *resourceAllocation* in *pdsch-ConfigMulticast* is configured as '*dynamicSwitch*'.
  - If *resourceAllocation* in *pdsch-ConfigMulticast* is configured as '*dynamicSwitch*', the MSB bit is used to indicate resource allocation type 0 or resource allocation type 1, where the bit value of 0 indicates resource allocation type 0 and the bit value of 1 indicates resource allocation type 1.
  - For resource allocation type 0, the N_{RBG} LSBs provide the resource allocation as defined in Clause 5.1.2.2.1 of [6, TS 38.214].
  - For resource allocation type 1, the $\lceil \log_2(N_{RB}^{DL,CFR}(N_{RB}^{DL,CFR}+1)/2) \rceil$ LSBs provide the resource allocation as defined in Clause 5.1.2.2.2 of [6, TS 38.214]
- Time domain resource assignment - 0, 1, 2, 3, or 4 bits as defined in Clause 5.1.2.1 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *pdsch-TimeDomainAllocationList* if the higher layer parameter is configured; otherwise I is the number of entries in the default table.
- VRB-to-PRB mapping - 0 or 1 bit:
  - 0 bit if only resource allocation type 0 is configured or if *vrb-ToPRB-Interleaver* in *pdsch-ConfigMulticast* is not configured;
  - 1 bit according to Table 7.3.1.2.2-5 otherwise, only applicable to resource allocation type 1, as defined in Clause 7.3.1.6 of [4, TS 38.211].
- PRB bundling size indicator - 0 bit if the higher layer parameter *prb-BundlingType* is not configured in *pdsch-ConfigMulticast* or is set to 'staticBundling', or 1 bit if the higher layer parameter *prb-BundlingType* in *pdsch-ConfigMulticast* is set to 'dynamicBundling' according to Clause 5.1.2.3 of [6, TS 38.214].
- Rate matching indicator - 0, 1, or 2 bits according to higher layer parameters *rateMatchPatternGroup1* and *rateMatchPatternGroup2* in *pdsch-ConfigMulticast*, where the MSB is used to indicate *rateMatchPatternGroup1* and the LSB is used to indicate *rateMatchPatternGroup2* when there are two groups.
- ZP CSI-RS trigger - 0, 1, or 2 bits as defined in Clause 5.1.4.2 of [6, TS 38.214]. The bitwidth for this field is determined as $\lceil \log_2(n_{ZP}+1) \rceil$ bits, where n_{ZP} is the number of aperiodic ZP CSI-RS resource sets configured in *pdsch-ConfigMulticast*.

For transport block 1:

- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3.1 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2

For transport block 2 (only present if *maxNrofCodeWordsScheduledByDCI* configured in *pdsch-ConfigMulticast* equals 2):

- Modulation and coding scheme - 5 bits as defined in Clause 5.1.3.1 of [6, TS 38.214]
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2

- HARQ process number - 4 bits
- Downlink assignment index - number of bits as defined in the following
  - 2 bits if the higher layer parameter *pdsch-HARQ-ACK-Codebook =dynamic* is configured for multicast, where the 2 bits are the counter DAI;
  - 0 bits otherwise.

  If higher layer parameter *priorityIndicatorDCI-4-2* is configured in *pdsch-ConfigMulticast*, if the bit width of the Downlink assignment index in DCI format 4_2 for one HARQ-ACK codebook is not equal to that of the Downlink assignment index in DCI format 4_2 for the other HARQ-ACK codebook, a number of most

significant bits with value set to '0' are inserted to smaller Downlink assignment index until the bit width of the Downlink assignment index in DCI format 4_2 for the two HARQ-ACK codebooks are the same.

- PUCCH resource indicator - 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]
- PDSCH-to-HARQ_feedback timing indicator - 0, 1, 2, or 3 bits as defined in Clause 9.2.3 of [5, TS 38.213]. The bitwidth for this field is determined as $\lceil \log_2(I) \rceil$ bits, where I is the number of entries in the higher layer parameter *dl-DataToUL-ACK* in *pucch-ConfigMulticast1* if configured or *pucch-ConfigMulticast2* if configured; otherwise, I is the number of entries in the higher layer parameter *dl-DataToUL-ACK* in *PUCCH-Config*.

  If higher layer parameter *priorityIndicatorDCI-4-2* is configured in *pdsch-ConfigMulticast*, if the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 4_2 for one HARQ-ACK codebook is not equal to that of the PDSCH-to-HARQ_feedback timing indicator in DCI format 4_2 for the other HARQ-ACK codebook, a number of most significant bits with value set to '0' are inserted to smaller PDSCH-to-HARQ_feedback timing indicator until the bit width of the PDSCH-to-HARQ_feedback timing indicator in DCI format 4_2 for the two HARQ-ACK codebooks are the same.
- Antenna port(s) - 4, 5, 6, 7 or 8 bits as defined by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10, where the number of CDM groups without data of values 1, 2, and 3 refers to CDM groups {0}, {0,1}, and {0, 1,2} respectively. The antenna ports $\{p_0, \ldots, p_{v-1}\}$ shall be determined according to the ordering of DMRS port(s) given by Tables 7.3.1.2.2-1/2/3/4/7/8/9/10.

  If a UE is configured with both *dmrs-DownlinkForPDSCH-MappingTypeA* and *dmrs-DownlinkForPDSCH-MappingTypeB*, the bitwidth of this field equals $max\{x_A, x_B\}$, where x_A is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeA* and x_B is the "Antenna ports" bitwidth derived according to *dmrs-DownlinkForPDSCH-MappingTypeB*. A number of $|x_A - x_B|$ zeros are padded in the MSB of this field, if the mapping type of the PDSCH corresponds to the smaller value of x_A and x_B.
- Transmission configuration indication - 0 bit if higher layer parameter *tci-PresentInDCI* in *pdcch-ConfigMulticast* is not enabled; otherwise 3 bits as defined in Clause 5.1.5 of [6, TS38.214].
- DMRS sequence initialization - 1 bit.
- Priority indicator - 0 bit if higher layer parameter *priorityIndicatorDCI-4-2* is not configured in *pdsch-ConfigMulticast*; otherwise 1 bit as defined in Clause 9 in [5, TS 38.213].
- Enabling/disabling HARQ-ACK feedback indication -1 bit if higher layer parameter *harq-FeedbackEnablerMulticast* indicates *dci-enabler*, where value 1 indicates enabling HARQ-ACK feedback and value 0 indicates disabling HARQ-ACK feedback; 0 bit, otherwise.

The size of DCI format 4_2 is configurable by higher layer parameter *sizeDCI-4-2* from 20 bits and up to 140 bits. If the number of information bits in DCI format 4_2 is less than the size of DCI format 4_2, the remaining bits are reserved.

7.3.2 CRC attachment

Error detection is provided on DCI transmissions through a Cyclic Redundancy Check (CRC).

The entire payload is used to calculate the CRC parity bits. Denote the bits of the payload by $a_0, a_1, a_2, a_3, \ldots, a_{A-1}$, and the parity bits by $p_0, p_1, p_2, p_3, \ldots, p_{L-1}$, where A is the payload size and L is the number of parity bits. Let $a'_0, a'_1, a'_2, a'_3, \ldots, a'_{A+L-1}$ be a bit sequence such that $a'_i = 1$ for $i = 0,1,\ldots,L-1$ and $a'_i = a_{i-L}$ for $i = L, L+1, \ldots, A+L-1$. The parity bits are computed with input bit sequence $a'_0, a'_1, a'_2, a'_3, \ldots, a'_{A+L-1}$ and attached according to Clause 5.1 by setting L to 24 bits and using the generator polynomial $g_{CRC24C}(D)$. The output bit $b_0, b_1, b_2, b_3, \ldots, b_{K-1}$ is

$$b_k = a_k \text{ for } k = 0,1,2,\ldots,A-1$$

$b_k = p_{k-A}$ for $k = A, A+1, A+2, \ldots, A+L-1$,

where $K = A + L$.

After attachment, the CRC parity bits are scrambled with the corresponding RNTI $x_{rnti,0}, x_{rnti,1}, \ldots, x_{rnti,15}$, where $x_{rnti,0}$ corresponds to the MSB of the RNTI, to form the sequence of bits $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$. The relation between c_k and b_k is:

$c_k = b_k$ for k = 0, 1, 2, …, $A+7$

$c_k = (b_k + x_{rnti,k-A-8}) \bmod 2$ for k = $A+8$, $A+9$, $A+10$, ..., $A+23$.

7.3.3 Channel coding

Information bits are delivered to the channel coding block. They are denoted by $c_0, c_1, c_2, c_3, \ldots, c_{K-1}$, where K is the number of bits, and they are encoded via Polar coding according to Clause 5.3.1, by setting $n_{\max} = 9$, $I_{IL} = 1$, $n_{PC} = 0$, and $n_{PC}^{wm} = 0$.

After encoding the bits are denoted by $d_0, d_1, d_2, d_3, \ldots, d_{N-1}$, where N is the number of coded bits.

7.3.4 Rate matching

The input bit sequence to rate matching is $d_0, d_1, d_2, \ldots, d_{N-1}$.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL} = 0$.

The output bit sequence after rate matching is denoted as $f_0, f_1, f_2, \ldots, f_{E-1}$.

7.4 Wake-up information

The wake-up information is carried by a wake-up signal as defined in clause 7.4.4 of [4, TS 38.211].

- For a UE configured with higher layer parameter *LP-WUS_LP-SS_startRB_IDLE_INACTIVE* and operating in the RRC_IDLE or RRC_INACTIVE state, the wake-up information bit sequence $c_0, c_1, \ldots, c_{K-1}$ is the binary sequence of the codepoint as defined by Clause 10.4C of [5, TS38.213], where c_0 is the most significant bit and K is
  - $max(\log_2 N_{PO}, 1)$ if $N_{subgroup}^{PO} = 1$, where N_{PO} is configured by higher layer parameter *PO-to-LO association* and $N_{subgroup}^{PO}$ is configured by higher layer parameter *subgroupNumber_PO_LPWUS*;
  - $\lceil \log_2 (N_{PO}(N_{subgroup}^{PO} + 1)) \rceil$ if $N_{subgroup}^{PO} > 1$, where $N_{subgroup}^{PO}$ is configured by higher layer parameter *subgroupNumber_PO_LPWUS*;
- For a UE configured with higher layer parameter *LP-WUS_startRB_CONNECTED* and operating in the RRC_CONNECTED state, the wake-up information bit sequence $c_0, c_1, \ldots, c_{K-1}$ is the binary sequence of the codepoint as defined by Clause 10.4D of [5, TS38.213], where c_0 is the most significant bit, K is provided by the higher layer parameter *Num_info_bits_WUS_CONNECTED*.

The following coding steps can be identified for the wake-up information:

- For the first bit block in case of OOK modulation
  - Channel coding
  - Rate matching

  - Line coding
- For the second bit block in case of sequence modulation
  - Rate matching

The first bit block and the second bit block, if both are present, are generated based on the same information bits.

7.4.1 Channel coding

Information bits are delivered to the channel coding block. They are denoted by $c_0, c_1, \ldots, c_{K-1}$, where K is the number of bits and $K \leq 5$.

The information bits are encoded according to Clause 5.3.3, where Q_m is set to 1 when Table 5.3.3.1-1 or Table 5.3.3.2-1 is applied.

After encoding the bits are denoted by $d_{00}, d_{01}, \ldots, d_{0(N_0-1)}$, where N_0 is the number of coded bits.

7.4.2 Rate matching

7.4.2.1 Rate matching for OOK modulation

The input bit sequence to rate matching is $d_{00}, d_{01}, \ldots, d_{0(N_0-1)}$.

Rate matching is performed according to Clause 5.4.3 by setting the rate matching output sequence length $E = E_0$, where $E_0 = E_{WUS}$, $E_{WUS} = N_{OS} \times M_{LP} / 2$, where

- for a UE configured with higher layer parameter *LP-WUS_LP-SS_startRB_IDLE_INACTIVE* and operating in the RRC_IDLE or RRC_INACTIVE state, N_{OS} is configured by higher layer parameter *LP-WUS_ActualMO_duration_IDLE/INACTIVE* and M_{LP} is configured by higher layer parameter *LP-WUS_Mvalue_IDLE/INACTIVE;*
- for a UE configured with higher layer parameter *LP-WUS_startRB_CONNECTED* and operating in the RRC_CONNECTED state, N_{OS} is configured by higher layer parameter *LP-WUS_ActualMO_duration_CONNECTED* and M_{LP} is configured by higher layer parameter *LP-WUS_Mvalue_CONNECTED*.

The output bit sequence after rate matching is denoted as $f_{00}, f_{01}, f_{02}, \ldots, f_{0(E_0-1)}$.

7.4.2.2 Rate matching for sequence modulation

Information bits for the second bit block are delivered to the rate matching block. They are denoted by $c_0, c_1, \ldots, c_{K-1}$, where K is the number of bits and $K \leq 5$.

If the number of sequences configured by higher layer parameter *LP-WUS_num_overlaidSeq_CONNECTED* or *LP-WUS_num_overlaidSeq_IDLE/INACTIVE*, denoted as L_1, is larger than one, padding is performed and the bits after padding are denoted by $d_{10}, d_{11}, \ldots, d_{1(N_1-1)}$, where $N_1 = K + L$, $L = (-K) \bmod \log_2 L_1$. The relation between c_k and d_{1k} is:

$$d_{1k} = 0 \text{ for } k = 0, 1, \ldots, L-1$$

$$d_{1k} = c_{k-L} \text{ for } k = L, L+1, \ldots, N_1 - 1.$$

Rate matching is performed according to Clause 5.4.3 by setting the rate matching output sequence length $E = E_1$, where $E_1 = E_{WUS} \times \log_2 L_1$, E_{WUS} is as defined in Clause 7.4.2.1, and the output bit sequence after rate matching is denoted as $f_{10}, f_{11}, f_{12}, \ldots, f_{1(E_1-1)}$.

7.4.3 Line coding

The input bit sequence to line coding block are the sequences $f_{00}, f_{01}, f_{02}, \ldots, f_{0(E_0-1)}$.

The bits after line coding are denoted by $g_{00}, g_{01}, \ldots, g_{0(G_0-1)}$, where $G_0 = 2 \times E_0$.

Line coding is performed according to the following by setting $i = 0, 1, \ldots, E_0 - 1$:

$$g_{0(2i)} = 1 - f_{0i};$$

$$g_{0(2i+1)} = f_{0i};$$

8 Sidelink transport channels and control information

8.1 Sidelink broadcast channel

The processing for SL-BCH transport channel follows the BCH according to clause 7.1, with the following changes:

- In Clause 7.1, 'maximum of one transport block every 80ms' is replaced with 'maximum of one transport block'.
- Clause 7.1.1 for PBCH payload generation is not performed.
- Clause 7.1.2 for scrambling is not performed.
- In clause 7.1.5, the rate matching output sequence length E = 1386 when higher layer parameter *cyclicPrefix* is configured, otherwise, E = 1782.

8.1.1 Void

8.2 Sidelink shared channel

The processing for SL-SCH transport channel follows the UL-SCH according to clause 6.2, with the following changes:

- Rate matching of SL-SCH follows the rate matching according to clause 6.2.5 by setting $I_{LBRM} = 0$
- Clause 6.2.7 is replaced by clause 8.2.1

8.2.1 Data and control multiplexing

Denote the coded bits for SL-SCH as $g_0^{SL\text{-}SCH}$, $g_1^{SL\text{-}SCH}$, $g_2^{SL\text{-}SCH}$, $g_3^{SL\text{-}SCH}$, ⋯, $g_{G^{SL\text{-}SCH}-1}^{SL\text{-}SCH}$.

Denote the coded bits for the 2nd-stage SCI, as $g_0^{SCI\,2}$, $g_1^{SCI\,2}$, $g_2^{SCI\,2}$, $g_3^{SCI\,2}$, ⋯, $g_{G^{SCI\,2}-1}^{SCI\,2}$.

Denote the multiplexed data and control coded bit sequence as $g_0, g_1, \cdots, g_{G-1}$, where G is the total number of coded bits for transmission.

Assuming that N_L is the number of layers onto which the SL-SCH transport block is mapped, the multiplexed data and control coded bit sequence $g_0, g_1, \cdots, g_{G-1}$ is obtained as follows:

Denote $Q_m^{SCI\,2}$ is modulation order of the 2nd-stage SCI.

if $N_L = 1$,

for $i = 0$ to $G^{SCI\,2} + G^{SL\text{-}SCH} - 1$

if $0 \le i < G^{SCI\,2}$

$g_i = g_i^{SCI\,2}$

end if

if $G^{SCI\,2} \le i \le G^{SCI\,2} + G^{SL\text{-}SCH} - 1$

$g_i = g_{i-G^{SCI\,2}}^{SL\text{-}SCH}$

end if

end for

end if

if $N_L = 2,$

let $M_{count,SCI\,2}^{\mathrm{RE}} = G^{SCI\,2} / Q_m^{SCI\,2}$

set $m_{count}^{\mathrm{RE}} = 0$

for $i = 0$ to $M_{count,SCI\,2}^{\mathrm{RE}} - 1$

for $v = 0$ to $N_L - 1$

for $q = 0$ to $Q_m^{SCI\,2} - 1$

if $v = 0$

$g_{m_{count}^{\mathrm{RE}}} = g_{i \cdot Q_m^{SCI\,2} + q}^{SCI\,2}$

else

$g_{m_{count}^{\mathrm{RE}}} = x$ // placeholder bit

end if

$m_{count}^{\mathrm{RE}} = m_{count}^{\mathrm{RE}} + 1$

end for

end for

end for

for $i = 0$ to $G^{SL\text{-}SCH} - 1$

$g_{m_{count}^{\mathrm{RE}}} = g_i^{SL\text{-}SCH}$

$m_{count}^{\mathrm{RE}} = m_{count}^{\mathrm{RE}} + 1$

end for

end if

8.3 Sidelink control information on PSCCH

SCI carried on PSCCH is a 1st-stage SCI, which transports sidelink scheduling information.

8.3.1 1st-stage SCI formats

The fields defined in each of the 1st-stage SCI formats below are mapped to the information bits a_0 to a_{A-1} as follows:

Each field is mapped in the order in which it appears in the description, with the first field mapped to the lowest order information bit a_0 and each successive field mapped to higher order information bits. The most significant bit of each field is mapped to the lowest order information bit for that field, e.g. the most significant bit of the first field is mapped to a_0.

8.3.1.1 SCI format 1-A

SCI format 1-A is used for the scheduling of PSSCH and 2nd-stage-SCI on PSSCH

The following information is transmitted by means of the SCI format 1-A:

- Priority - 3 bits as specified in clause 5.4.3.3 of [12, TS 23.287] and clause 5.22.1.3.1 of [8, TS 38.321]. Value '000' of Priority field corresponds to priority value '1', value '001' of Priority field corresponds to priority value '2', and so on.
- Frequency resource assignment - number of bits determined by the following:
  - If higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is not configured or configured to 'contiguousRB'
    - $\lceil \log_2(\frac{N_{subChannel}^{SL}(N_{subChannel}^{SL}+1)}{2}) \rceil$ bits when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 2; otherwise $\lceil \log_2(\frac{N_{subChannel}^{SL}(N_{subChannel}^{SL}+1)(2N_{subChannel}^{SL}+1)}{6}) \rceil$ bits when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 3, as defined in clause 8.1.5 of [6, TS 38.214].
  - If the higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured to 'interlaceRB'
    - X + Y bits provide the frequency domain resource allocation according to Clause 8.1.5 of [6, TS 38.214], where the X MSBs provide the RB set allocation and the Y LSBs provide the sub-channel allocation,
      - the value of X is determined by $\lceil \log_2(\frac{N_{RBset}(N_{RBset}+1)}{2}) \rceil$ when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 2, or determined by $\lceil \log_2(\frac{N_{RBset}(N_{RBset}+1)(2N_{RBset}+1)}{6}) \rceil$ when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 3, where N_{RBset} is the number of RB sets in a resource pool
      - the value of Y is determined by $\lceil \log_2(\frac{N_{subChannel}^{SL}(N_{subChannel}^{SL}+1)}{2}) \rceil$ when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 2, or determined by $\lceil \log_2(\frac{N_{subChannel}^{SL}(N_{subChannel}^{SL}+1)(2N_{subChannel}^{SL}+1)}{6}) \rceil$ when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 3, as defined in clause 8.1.5 of [6, TS 38.214].
- Time resource assignment - 5 bits when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 2; otherwise 9 bits when the value of the higher layer parameter *sl-MaxNumPerReserve* is configured to 3, as defined in clause 8.1.5 of [6, TS 38.214].
- Resource reservation period - $\lceil \log_2 N_{rsv_period} \rceil$ bits as defined in clause 16.4 of [5, TS 38.213], where N_{rsv_period} is the number of entries in the higher layer parameter *sl-ResourceReservePeriodList*, if higher layer parameter *sl-MultiReserveResource* is configured; 0 bit otherwise.
- DMRS pattern - $\lceil \log_2 N_{pattern} \rceil$ bits as defined in clause 8.4.1.1.2 of [4, TS 38.211], where $N_{pattern}$ is the number of DMRS patterns configured by higher layer parameter *sl-PSSCH-DMRS-TimePatternList*.
- 2nd-stage SCI format - 2 bits as defined in Table 8.3.1.1-1.
- Beta_offset indicator - 2 bits as provided by higher layer parameter *sl-BetaOffsets2ndSCI* and Table 8.3.1.1-2.
- Number of DMRS port - 1 bit as defined in Table 8.3.1.1-3.
- Modulation and coding scheme - 5 bits as defined in clause 8.1.3 of [6, TS 38.214].
- Additional MCS table indicator - as defined in clause 8.1.3.1 of [6, TS 38.214]: 1 bit if one MCS table is configured by higher layer parameter *sl-Additional-MCS-Table*; 2 bits if two MCS tables are configured by higher layer parameter *sl-Additional-MCS-Table*; 0 bit otherwise.
- PSFCH overhead indication - 1 bit as defined in clause 8.1.3.2 of [6, TS 38.214] if higher layer parameter *sl-PSFCH-Period* = 2 or 4; 0 bit otherwise.

- Reserved - a number of bits as determined by the following:
  - $N_{reserved}$ bits as configured by higher layer parameter *sl-NumReservedBits*, with value set to zero, if higher layer parameter *sl-IndicationUE-B* is not configured, or if higher layer parameter *sl-IndicationUE-B* is configured to 'disabled', and if higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is not configured;
  - $(N_{reserved}-2)$ bits if higher layer parameter *sl-IndicationUE-B* is configured to 'enabled', and if higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured, with value set to zero.
  - $(N_{reserved}-1)$ bits otherwise, with value set to zero.
- COT sharing flag – 0 or 1 bit
  - 1 bit if the higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured;
  - 0 bit otherwise.
- Conflict information receiver flag - 0 or 1 bit
  - 1 bit if higher layer parameter *sl-IndicationUE-B* is configured to 'enabled', where the bit value of 0 indicates that the UE cannot be a UE to receive conflict information and the bit value of 1 indicates that the UE can be a UE to receive conflict information as defined in Clause 16.3.0 of [5, TS 38.213];
  - 0 bit otherwise.

Table 8.3.1.1-1: 2nd-stage SCI formats

Value of 2nd-stage SCI format field	2nd-stage SCI format
00	SCI format 2-A
01	SCI format 2-B; or reserved if higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured
10	SCI format 2-C; or reserved if higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured and the COT sharing flag field is set to '1'
11	SCI format 2-D; or reserved if higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured

Table 8.3.1.1-2: Mapping of Beta_offset indicator values to indexes in Table 9.3-2 of [5, TS38.213]

Value of Beta_offset indicator	Beta_offset index in Table 9.3-2 of TS 38.213 [5]
00	1st index provided by higher layer parameter *sl-BetaOffsets2ndSCI*
01	2nd index provided by higher layer parameter *sl-BetaOffsets2ndSCI*
10	3rd index provided by higher layer parameter *sl-BetaOffsets2ndSCI*
11	4th index provided by higher layer parameter *sl-BetaOffsets2ndSCI*

Table 8.3.1.1-3: Number of DMRS port(s)

Value of the Number of DMRS port field	Antenna ports
0	1000
1	1000 and 1001

8.3.1.2 SCI format 1-B

SCI format 1-B is used for the scheduling of SL PRS for a dedicated SL PRS resource pool.

The following information is transmitted by means of the SCI format 1-B:

- Priority - 3 bits as specified in clause 5.7 of [12, TS 23.586] and clause 5.22 of [8, TS 38.321]. Value '000' of Priority field corresponds to priority value '1', value '001' of Priority field corresponds to priority value '2', and so on.
- Source ID – 12 or 24 bits determined by higher layer parameter *sl-SRC-ID-LenDedicatedSL-PRS-RP*, as defined in clause 16.4A of [5, TS 38.213].
- Destination ID – 24 bits as defined in clause 16.4A of [5, TS 38.213].
- Cast type indicator – 2 bits as defined in Table 8.3.1.2-1 and in clause 16.4A of [5, TS 38.213].
- Resource reservation period – $\lceil \log_2 N_{rsv_period} \rceil$ bits as defined in clause 16.4A of [5, TS 38.213], where N_{rsv_period} is the number of entries in the higher layer parameter *sl-PRS-ResourceReservePeriodList*, if higher layer parameter *sl-PRS-ResourceReservePeriodList* is configured; 0 bit otherwise.
- Time resource assignment – 5 bits when the value of the higher layer parameter *sl-MaxNumPerReserveDedicatedSL-PRS-RP* is configured to 2; otherwise 9 bits when the value of the higher layer parameter *sl-MaxNumPerReserveDedicatedSL-PRS-RP* is configured to 3, as defined in clause 8.2.4.2A of [6, TS 38.214].
- Resource ID indication – $\lceil \log_2 N_{SL-PRS} \rceil$ bits when the value of the higher layer parameter *sl-MaxNumPerReserveDedicatedSL-PRS-RP* is configured to 2; otherwise $\lceil 2\log_2 N_{SL-PRS} \rceil$ bits when the value of the higher layer parameter *sl-MaxNumPerReserveDedicatedSL-PRS-RP* is configured to 3. The value N_{SL-PRS} is the total number of SL PRS resources within a slot in a dedicated SL PRS resource pool and provided by the higher layer parameter *sl-PRS-ResourcesDedicatedSL-PRS-RP*.
- SL PRS request – 1 bit as defined in clause 8.4.4 of [6, TS 38.214] when the higher layer parameter *sl-SCI-basedSL-PRS-TxTriggerSCI1-B* is provided; 0 bit otherwise.
- Reserved – $N_{reserved}$ bits as configured by higher layer parameter *sl-NumReservedBitsSCI1B-DedicatedSL-PRS-RP*, with value set to zero.

Table 8.3.1.2-1: Cast type indicator

Value of Cast type indicator	Cast type
00	Broadcast
01	Groupcast
10	Unicast
11	Reserved

8.3.2 CRC attachment

CRC attachment is performed according to clause 7.3.2 except that scrambling is not performed.

8.3.3 Channel coding

Channel coding is performed according to clause 7.3.3.

8.3.4 Rate Matching

Rate matching is performed according to clause 7.3.4.

8.4 Sidelink control information on PSSCH

SCI carried on PSSCH is a 2nd-stage SCI, which transports sidelink scheduling information, and/or inter-UE coordination related information.

8.4.1 2nd-stage SCI formats

The fields defined in each of the 2nd-stage SCI formats below are mapped to the information bits a_0 to a_{A-1} as follows:

Each field is mapped in the order in which it appears in the description, with the first field mapped to the lowest order information bit a_0 and each successive field mapped to higher order information bits. The most significant bit of each field is mapped to the lowest order information bit for that field, e.g. the most significant bit of the first field is mapped to a_0.

8.4.1.1 SCI format 2-A

SCI format 2-A is used for the decoding of PSSCH, with HARQ operation when HARQ-ACK information includes ACK or NACK, when HARQ-ACK information includes only NACK, or when there is no feedback of HARQ-ACK information.

The following information is transmitted by means of the SCI format 2-A:

- HARQ process number - 4 bits.
- New data indicator - 1 bit.
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2.
- Source ID - 8 bits as defined in clause 8.1 of [6, TS 38.214].
- Destination ID - 16 bits as defined in clause 8.1 of [6, TS 38.214].
- HARQ feedback enabled/disabled indicator - 1 bit as defined in clause 16.3 of [5, TS 38.213].
- Cast type indicator - 2 bits as defined in Table 8.4.1.1-1 and in clause 8.1 of [6, TS 38.214].
- CSI request - 1 bit as defined in clause 8.2.1 of [6, TS 38.214] and in clause 8.1 of [6, TS 38.214].

If the 'COT sharing flag' field in SCI format 1-A is present and set to '1', all the remaining fields are present and set as follows:

- CAPC – 2 bits. Values '00', '01', '10' and '11' correspond to CAPC values '1', '2', '3' and '4' as defined in Table 4.5-1 of [14, TS 37.213], respectively.
- COT sharing cast type – 2 bits as defined in Table 8.4.1.1-1.
- COT sharing additional ID – 24 bits. The 16 LSBs provide layer 1 destination ID and the 8 MSBs provide layer 1 source ID, as defined in [6, TS 38.214]. The 8 MSBs are reserved when the COT sharing cast type field is set to '00' or '01'.
- Remaining COT duration – $\lceil \log_2(10 \cdot 2^{\mu}) \rceil$ bits as defined in clause 4.5.3 of [14, TS 37.213], where μ is defined in Table 4.2-1 of Clause 4.2 of [4, TS 38.211].

Table 8.4.1.1-1: Cast type indicator or COT sharing cast type

Value of Cast type indicator or COT sharing cast type	Cast type
00	Broadcast
01	Groupcast when HARQ-ACK information includes ACK or NACK
10	Unicast
11	Groupcast when HARQ-ACK information includes only NACK; or reserved, if higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured

8.4.1.2 SCI format 2-B

SCI format 2-B is used for the decoding of PSSCH, with HARQ operation when HARQ-ACK information includes only NACK, or when there is no feedback of HARQ-ACK information.

The following information is transmitted by means of the SCI format 2-B:

- HARQ process number - 4 bits.
- New data indicator - 1 bit.
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2.
- Source ID - 8 bits as defined in clause 8.1 of [6, TS 38.214].
- Destination ID - 16 bits as defined in clause 8.1 of [6, TS 38.214].
- HARQ feedback enabled/disabled indicator - 1 bit as defined in clause 16.3 of [5, TS 38.213].
- Zone ID - 12 bits as defined in clause 5.8.11 of [9, TS 38.331].
- Communication range requirement - 4 bits determined by higher layer parameter *sl-ZoneConfigMCR-Index*.

8.4.1.3 SCI format 2-C

SCI format 2-C is used for the decoding of PSSCH, and providing inter-UE coordination information or requesting inter-UE coordination information. SCI format 2-C can be used only for unicast.

The following information is transmitted by means of the SCI format 2-C:

- HARQ process number - 4 bits
- New data indicator - 1 bit
- Redundancy version - 2 bits as defined in Table 7.3.1.1.1-2
- Source ID - 8 bits as defined in clause 8.1 of [6, TS 38.214]
- Destination ID - 16 bits as defined in clause 8.1 of [6, TS 38.214]
- HARQ feedback enabled/disabled indicator - 1 bit as defined in clause 16.3 of [5, TS 38.213]
- CSI request - 1 bit as defined in clause 8.2.1 of [6, TS 38.214] and in clause 8.1 of [6, TS 38.214]
- Providing/Requesting indicator - 1 bit, where value 0 indicates SCI format 2-C is used for providing inter-UE coordination information and value 1 indicates SCI format 2-C is used for requesting inter-UE coordination information

If the 'Providing/Requesting indicator' field is set to 0, all the remaining fields are set as follows:

- Resource combinations - number of bits determined by the following:
  - If higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is not configured or configured to 'contiguousRB'
    - $2\cdot\left(\lceil \log_2(\frac{N_{subChannel}^{SL}\left(N_{subChannel}^{SL}+1\right)\left(2N_{subChannel}^{SL}+1\right)}{6})\rceil+9+Y\right)$ bits as defined in Clause 8.1.5A of [6, TS 38.214];
  - If the higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured to 'interlaceRB'
    - $2\cdot\left(\lceil \log_2(\frac{N_{subChannel}^{SL}\left(N_{subChannel}^{SL}+1\right)\left(2N_{subChannel}^{SL}+1\right)}{6})\rceil+\lceil \log_2(\frac{N_{RBset}\left(N_{RBset}+1\right)\left(2N_{RBset}+1\right)}{6})\rceil$ bits as defined in Clause 8.1.5A of [6, TS 38.214];

  where

  - $Y = \lceil \log_2 N_{rsv_period} \rceil$ and N_{rsv_period} is the number of entries in the higher layer parameter *sl-ResourceReservePeriodList*, if higher layer parameter *sl-MultiReserveResource* is configured; $Y = 0$ otherwise.
  - $N_{\text{subChannel}}^{\text{SL}}$ is provided by the higher layer parameter *sl-NumSubchannel* as defined in Clause 8.1.5 of [6, TS 38.214].
  - N_{RBset} is the number of RB sets in a resource pool.
- First resource location - 8 bits as defined in Clause 8.1.5A of [6, TS 38.214].
- Reference slot location - $(10 + \lceil \log_2(10 \cdot 2^{\mu}) \rceil)$ bits as defined in Clause 8.1.5A of [6, TS 38.214], where μ is defined in Table 4.2-1 of Clause 4.2 of [4, TS 38.211].
- Resource set type - 1 bit, where value 0 indicates preferred resource set and value 1 indicates non-preferred resource set.
- Lowest subChannel indices - $2 \cdot \lceil \log_2 N_{\text{subChannel}}^{\text{SL}} \rceil$ bits as defined in Clause 8.1.5A of [6, TS 38.214].
- Lowest RB set indices - $2 \cdot \lceil \log_2 N_{RBset} \rceil$ bits as defined in Clause 8.1.5A of [6, TS 38.214] if the higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured to 'interlaceRB'; 0 bit otherwise.

If the 'Providing/Requesting indicator' field is set to 1, all the remaining fields are set as follows:

- Priority - 3 bits as specified in clause 5.4.3.3 of [12, TS 23.287] and clause 5.22.1.3.1 of [8, TS 38.321]. Value '000' of Priority field corresponds to priority value '1', value '001' of Priority field corresponds to priority value '2', and so on.
- Number of subchannels - $\lceil \log_2 N_{\text{subChannel}}^{\text{SL}} \rceil$ bits as defined in Clause 8.1.4A of [6, TS 38.214].
- Number of RB sets - $\lceil \log_2 N_{RBset} \rceil$ bits as defined in Clause 8.1.4A of [6, TS 38.214] if the higher layer parameter *sl-TransmissionStructureForPSCCHandPSSCH* in *SL-BWP-Config* is configured to 'interlaceRB'; 0 bit otherwise.
- Resource reservation period - $\lceil \log_2 N_{rsv_period} \rceil$ bits as defined in Clause 8.1.4A of [6, TS 38.214], where N_{rsv_period} is the number of entries in the higher layer parameter *sl-ResourceReservePeriodList*, if higher layer parameter *sl-MultiReserveResource* is configured; 0 bit otherwise.
- Resource selection window location - $2 \cdot \left(10 + \lceil \log_2(10 \cdot 2^{\mu}) \rceil\right)$ bits as defined in Clause 8.1.4A of [6, TS 38.214], where μ is defined in Table 4.2-1 of Clause 4.2 of [4, TS 38.211].
- Resource set type - 1 bit, where value 0 indicates a request for inter-UE coordination information providing preferred resource set and value 1 indicates a request for inter-UE coordination information providing non-preferred resource set, if higher layer parameter *sl-DetermineResourceType* is configured to 'ueb'; otherwise, 0 bit.
- Padding bits.

For operation in a same resource pool, zeros shall be appended to SCI format 2-C of which 'Providing/Requesting indicator' field is set to 1 until the payload size equals that of SCI format 2-C of which 'Providing/Requesting indicator' field is set to 0.

8.4.1.4 SCI format 2-D

SCI format 2-D is used for the decoding of PSSCH and the scheduling of SL PRS for a shared SL PRS resource pool.

The following information is transmitted by means of the SCI format 2-D:

- SL PRS resource ID - $\lceil \log_2 N_{SL\text{-}PRS} \rceil$ bits, where the value $N_{SL\text{-}PRS}$ is the total number of SL PRS resource IDs within a slot in a shared SL PRS resource pool and provided by the higher layer parameter *sl-PRS-ResourcesSharedSL-PRS-RP*.
- SL PRS request – 1 bit as defined in clause 8.4.4 of [6, TS 38.214] when the higher layer parameter *sl-SCI-basedSL-PRS-TxTriggerSCI2-D* is provided; 0 bit otherwise.

- Embedded SCI format - 2 bits. This field indicates the embedded SCI format as defined in Table 8.4.1.4-1.
- Embedded SCI format payload - number of bits determined according to Table 8.4.1.4-1. This field is set to the associated payload of the embedded SCI format indicated by the 'Embedded SCI format' field as defined in Table 8.4.1.4-1.

Table 8.4.1.4-1: Embedded SCI format and payload

Value of the Embedded SCI format field	Embedded SCI format	Embedded SCI format payload
00	SCI format 2-A	Set to all fields included in SCI format 2-A. Padding bits, if necessary, are appended to the 'Embedded SCI format payload' field untill the bitwidth equals the larger payload size of SCI format 2-A and SCI format 2-B.
01	SCI format 2-B	Set to all fields included in SCI format 2-B. Padding bits, if necessary, are appended to the 'Embedded SCI format payload' field untill the bitwidth equals the larger payload size of SCI format 2-A and SCI format 2-B.
10	Reserved	Reserved
11	Reserved	Reserved

8.4.2 CRC attachment

CRC attachment is performed according to clause 7.3.2 except that scrambling is not performed.

8.4.3 Channel coding

Channel coding is performed according to clause 7.3.3.

8.4.4 Rate Matching

For 2nd-stage SCI transmission on PSSCH with SL-SCH, the number of coded modulation symbols generated for 2nd-stage SCI transmission prior to duplication for the 2nd layer if present, denoted as $Q'_{SCI\,2}$, is determined as follows:

$$Q'_{SCI\,2} = \min\left\{ \left\lceil \frac{\left(O_{SCI\,2} + L_{SCI\,2}\right)\cdot \beta_{offset}^{SCI\,2}}{Q_m^{SCI\,2}\cdot R} \right\rceil,\ \left\lceil \alpha \sum_{l=0}^{N_{symbol}^{PSSCH}-1} M_{sc}^{SCI\,2}(l) \right\rceil \right\} + \gamma$$

where

- $O_{SCI\,2}$ is the number of the 2nd-stage SCI bits
- $L_{SCI\,2}$ is the number of CRC bits for the 2nd-stage SCI, which is 24 bits.
- $\beta_{offset}^{SCI\,2}$ is indicated in the corresponding 1st-stage SCI.
- $M_{sc}^{PSSCH}(l)$ is the number of allocated PRBs of PSSCH transmission n_{PRB} according to clause 8.1.3.2 in [6, TS 38.214], expressed as a number of subcarriers.
- $M_{sc}^{PSCCH}(l)$ is the number of subcarriers in OFDM symbol l that carry PSCCH and PSCCH DMRS associated with the PSSCH transmission.
- $M_{sc}^{SCI\,2}(l)$ is the number of resource elements that can be used for transmission of the 2nd-stage SCI in OFDM symbol l, for $l = 0,1,2\cdots, N_{symbol}^{PSSCH} - 1$ and for $N_{symbol}^{PSSCH} = N_{symb}^{sh} - N_{symb}^{PSFCH} - N_{symb}^{SL\ PRS}$, in PSSCH transmission, where N_{symb}^{sh} = *sl-lengthSymbols* - 2, where *sl-lengthSymbols* is the number of sidelink symbols within the slot provided by higher layers as defined in [6, TS 38.214]. $N_{symb}^{SL\ PRS}$ is the number of symbols for SL PRS provided by the higher layer parameter *numSym-SL-PRS-2ndStageSCI* if the 2nd-stage SCI

is SCI format 2-D, and $N_{symb}^{SL\ PRS}$ = 0 otherwise. If *sl-StartingSymbolFirst* and *sl-StartingSymbolSecond* are provided for the SL-BWP, N_{symb}^{sh} = *sl-NumRefSymbolLength* - 2, where *sl-NumRefSymbolLength* is provided by higher layers. If higher layer parameter *sl-PSFCH-Period* = 2 or 4, N_{symb}^{PSFCH} = 3 if "PSFCH overhead indication" field of SCI format 1-A indicates "1", and N_{symb}^{PSFCH} = 0 otherwise. If higher layer parameter *sl-PSFCH-Period* = 0, $N_{symb}^{PSFCH}=0$. If higher layer parameter *sl-PSFCH-Period* is 1, $N_{symb}^{PSFCH}=3$.

- $M_{sc}^{SCI\,2}(l) = M_{sc}^{PSSCH}(l) - M_{sc}^{PSCCH}(l)$

- γ is the number of vacant resource elements in the resource block to which the last coded symbol of the 2nd-stage SCI belongs.
- R is the coding rate as indicated by "Modulation and coding scheme" field in SCI format 1-A.
- α is configured by higher layer parameter *sl-Scaling*.

The input bit sequence to rate matching is $d_0, d_1, d_2, d_3, \cdots, d_{N-1}$, where N is the number of coded bits.

Rate matching is performed according to Clause 5.4.1 by setting $I_{BIL}=1$.

The output bit sequence after rate matching is denoted as $g_0^{SCI\,2}, g_1^{SCI\,2}, g_2^{SCI\,2}, g_3^{SCI\,2}, \cdots, g_{G^{SCI\,2}-1}^{SCI\,2}$, where $G^{SCI\,2}=Q_{SCI\,2}^{'}\cdot Q_m^{SCI\,2}$ and $Q_m^{SCI\,2}$ is modulation order of the 2nd-stage SCI. A UE is not expected to have $G^{SCI\,2}>4096$.

8.4.5 Multiplexing of coded 2nd-stage SCI bits to PSSCH

The coded 2nd-stage SCI bits are multiplexed onto PSSCH according to the procedures in Clause 8.2.1.

Annex A (informative): Change history

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2017-05	RAN1#89	R1-1707082				Draft skeleton	0.0.0
2017-07	AH_NR2	R1-1712014				Inclusion of LDPC related agreements	0.0.1
2017-08	RAN1#90	R1-1714564				Inclusion of Polar coding related agreements	0.0.2
2017-08	RAN1#90	R1-1714659				Endorsed version by RAN1#90 as basis for further updates	0.1.0
2017-09	RAN1#90	R1-1715322				Capturing additional agreements on LDPC and Polar code from RAN1 #90	0.1.1
2017-09	RAN#77	RP-171991				For information to plenary	1.0.0
2017-09	RAN1#90b	R1-1716928				Capturing additional agreements on LDPC and Polar code from RAN1 NR AH#3	1.0.1
2017-10	RAN1#90b	R1-1719106				Endorsed as v1.1.0	1.1.0
2017-11	RAN1#91	R1-1719225				Capturing additional agreements on channel coding, etc.	1.1.1
2017-11	RAN1#91	R1-1719245				Capturing additional agreements on DCI format, channel coding, etc.	1.1.2
2017-11	RAN1#91	R1-1721049				Endorsed as v1.2.0	1.2.0
2017-12	RAN1#91	R1-1721342				Capturing additional agreements on UCI, DCI, channel coding, etc.	1.2.1
2017-12	RAN#78	RP-172668				Endorsed version for approval by plenary.	2.0.0
2017-12	RAN#78					Approved by plenary - Rel-15 spec under change control	15.0.0
2018-03	RAN#79	RP-180200	0001	-	F	CR capturing the Jan18 ad-hoc and RAN1#92 meeting agreements	15.1.0
2018-04	RAN#79					MCC: correction of typo in DCI format 0_1 (time domain resource assignment) - higher layer parameter should be *pusch-AllocationList*	15.1.1
2018-06	RAN#80	RP-181172	0002	1	F	CR to 38.212 capturing the RAN1#92bis and RAN1#93 meeting agreements	15.2.0
2018-06	RAN#80	RP-181257	0003	-	B	CR to 38.212 capturing the RAN1#92bis and RAN1#93 meeting agreements related to URLLC	15.2.0
2018-09	RAN#81	RP-181789	0004	-	F	CR to 38.212 capturing the RAN1#94 meeting agreements	15.3.0
2018-12	RAN#82	RP-182523	0005	3	F	Combined CR of all essential corrections to 38.212 from RAN1#94bis and RAN1#95	15.4.0
2019-03	RAN#83	RP-190448	0006	-	F	Correction of wrong implementation on frequency domain resource assignment bitwidth	15.5.0
2019-03	RAN#83	RP-190448	0008	-	F	Correction to UCI multiplexing	15.5.0
2019-03	RAN#83	RP-190448	0009	-	F	Correction on DCI format 2_3 for SUL cell in TS 38.212	15.5.0
2019-03	RAN#83	RP-190448	0010	-	F	Corrections to TS38.212	15.5.0
2019-03	RAN#83	RP-190448	0011	-	F	On bitwidth calculation for DCI fields using RRC parameter indicating maximum number of MIMO layers per serving cell	15.5.0
2019-03	RAN#83	RP-190448	0012	-	F	CR on zero-padding of DCI 1_1 in cross-carrier scheduling case	15.5.0
2019-03	RAN#83	RP-190448	0013	-	F	Clarification on UL_SUL indicator field and SRS request field	15.5.0
2019-06	RAN#84	RP-191282	0014	-	F	CR on correction to bitwidth of NNZC indicator	15.6.0
2019-06	RAN#84	RP-191282	0015	-	F	Correction on DCI size alignment in TS 38.212	15.6.0
2019-06	RAN#84	RP-191282	0016	-	F	Correction on UL/SUL indicator in DCI format 0_0	15.6.0
2019-06	RAN#84	RP-191282	0017	-	F	Corrections to 38.212 including alignment of terminology across specifications	15.6.0
2019-06	RAN#84	RP-191282	0018	-	F	CR on maximum modulation order configured for serving cell	15.6.0
2019-06	RAN#84	RP-191282	0019	1	F	Corrections to 38.212 including alignment of terminology across specifications from RAN1#97	15.6.0
2019-09	RAN#85	RP-191941	0020	-	F	Corrections to 38.212 including alignment of terminology across specifications in RAN1#98	15.7.0
2019-12	RAN#86	RP-192625	0021	-	F	CR on UL/SUL indicator in DCI format 0_1	15.8.0
2019-12	RAN#86	RP-192625	0022	-	F	Corrections to 38.212 including alignment of terminology across specifications in RAN1#98bis and RAN1#99	15.8.0
2019-12	RAN#86	RP-192636	0023	-	B	Introduction of NR based access to unlicensed spectrum into 38.212	16.0.0
2019-12	RAN#86	RP-192637	0024	-	B	Introduction of IAB into 38.212	16.0.0
2019-12	RAN#86	RP-192638	0025	-	B	Introduction of 5G V2X sidelink features into TS 38.212	16.0.0

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2019-12	RAN#86	RP-192639	0026	-	B	Introduction of Physical Layer Enhancements for NR URLLC	16.0.0
2019-12	RAN#86	RP-192641	0027	-	B	Introduction of Enhancements on NR MIMO	16.0.0
2019-12	RAN#86	RP-192642	0028	-	B	Introduction of power saving in 38.212	16.0.0
2019-12	RAN#86	RP-192645	0029	-	B	Introduction of MR DC/CA	16.0.0
2019-12	RAN#86	RP-192643	0030	-	B	Introduction of NR positioning support	16.0.0
2019-12	RAN#86	RP-192635	0031	-	B	Introduction of two-step RACH	16.0.0
2020-03	RAN#87-e	RP-200185	0032	-	F	Corrections for Rel-16 NR-U after RAN1#100-e	16.1.0
2020-03	RAN#87-e	RP-200190	0033	-	F	Corrections for NR MIMO after RAN1#100-e	16.1.0
2020-03	RAN#87-e	RP-200188	0034	-	F	Corrections for URLLC after RAN1#100-e	16.1.0
2020-03	RAN#87-e	RP-200191	0035	-	F	Corrections for power saving after RAN1#100-e	16.1.0
2020-03	RAN#87-e	RP-200187	0036	-	F	Corrections on 5G V2X sidelink features after RAN1#100-e	16.1.0
2020-06	RAN#88-e	RP-200683	0038	-	A	CR on L1-RSRP report on PUSCH	16.2.0
2020-06	RAN#88-e	RP-200693	0039	1	F	Corrections for power saving	16.2.0
2020-06	RAN#88-e	RP-200689	0040	1	F	Corrections on 5G V2X sidelink features after RAN1#100bis-e and RAN1#101-e	16.2.0
2020-06	RAN#88-e	RP-200694	0041	1	F	Corrections in TS 38.212 for NR positioning	16.2.0
2020-06	RAN#88-e	RP-200692	0042	1	F	Corrections in TS 38.212 for NR MIMO	16.2.0
2020-06	RAN#88-e	RP-200696	0043	-	F	Corrections for Rel-16 MR-DC/CA after RAN1#100bis-e	16.2.0
2020-06	RAN#88-e	RP-200690	0044	1	F	Corrections on NR eURLLC	16.2.0
2020-06	RAN#88-e	RP-200687	0045	1	F	Corrections for Rel-16 NR-U	16.2.0
2020-06	RAN#88-e	RP-200688	0046	-	F	Corrections for NR IAB	16.2.0
2020-09	RAN#89-e	RP-201814	0047	-	F	Correction on UCI bit sequence generation	16.3.0
2020-09	RAN#89-e	RP-201803	0049	-	A	CR on PTRS for TS 38.212	16.3.0
2020-09	RAN#89-e	RP-201810	0050	-	F	Alignment of RRC parameter *ps-RNTI*	16.3.0
2020-09	RAN#89-e	RP-201813	0051	-	F	CR to 38.212 on RRC parameter alignment for SCell dormancy	16.3.0
2020-09	RAN#89-e	RP-201807	0052	-	F	Corrections on 5G V2X sidelink features	16.3.0
2020-09	RAN#89-e	RP-201809	0053	-	F	Corrections to MIMO enhancements	16.3.0
2020-09	RAN#89-e	RP-201805	0054	-	F	Corrections to MIMO enhancements	16.3.0
2020-09	RAN#89-e	RP-201808	0055	-	F	Corrections on NR eURLLC	16.3.0
2020-12	RAN#90-e	RP-202390	0056	-	F	RRC IE name fix to dynamic frequency domain resource allocation type selection (Rel-15 origin)	16.4.0
2020-12	RAN#90-e	RP-202384	0057	-	F	Correction on Transmission configuration indication in DCI format 1_2	16.4.0
2020-12	RAN#90-e	RP-202398	0058	-	F	Alignment CR for TS 38.212	16.4.0
2021-03	RAN#91-e	RP-210052	0059	-	F	CR on DMRS	16.5.0
2021-03	RAN#91-e	RP-210049	0060	-	F	Correction to description of FDRA field size in DCI 0_0	16.5.0
2021-03	RAN#91-e	RP-210049	0061	-	F	Correction to description of FDRA field interpretation in DCI 0_1	16.5.0

Change history							
Date	Meeting	TDoc	CR	Rev	Cat	Subject/Comment	New version
2021-03	RAN#91-e	RP-210050	0062	-	F	Correction on Sidelink Broadcast channel	16.5.0
2021-03	RAN#91-e	RP-210049	0063	-	F	Correction on LBT Type and CP Extension Indication for Semi-Static Channel Occupancy	16.5.0
2021-03	RAN#91-e	RP-210059	0064	-	F	Alignment CR for TS 38.212	16.5.0
2021-06	RAN#92-e	RP-211252	0066	-	F	38.212 CR on DAI size determination for DCI format 1_1/1-2 in CA	16.6.0
2021-06	RAN#92-e	RP-211236	0067	-	F	Corrections on parameter of MCS table set to qam256	16.6.0
2021-06	RAN#92-e	RP-211234	0068	-	D	Alignment CR for TS 38.212 (post RAN1#104bis-e)	16.6.0
2021-06	RAN#92-e	RP-211234	0069	-	F	Correction on HARQ-ACK codebook RRC parameter	16.6.0
2021-06	RAN#92-e	RP-211236	0070	-	F	Correction on SRS resource set configuration in TS 38.212	16.6.0
2021-06	RAN#92-e	RP-211243	0071	-	F	Alignment CR for TS 38.212 (post RAN1#105-e)	16.6.0
2021-09	RAN#93-e	RP-211843	0072	-	F	Correction on SRS resource set configuration for DCI format 0_2 in TS 38.212	16.7.0
2021-09	RAN#93-e	RP-211841	0074	-	A	Rel-15 editorial corrections for TS 38.212 (mirrored to Rel-16)	16.7.0
2021-09	RAN#93-e	RP-211850	0075	-	F	Alignment CR for TS 38.212	16.7.0
2021-12	RAN#94-e	RP-212959	0076	-	F	Correction on mapping between priority field value and priority value in SCI format 1-A	16.8.0
2021-12	RAN#94-e	RP-212961	0077	-	F	Changes of channel access types tables in TS 38.212	16.8.0
2021-12	RAN#94-e	RP-212961	0078	-	F	Corrections on CG-UCI multiplexing in TS38.212	16.8.0
2021-12	RAN#94-e	RP-212958	0080	-	A	Clarify UCI bitwidth and UCI mapping order for non-PMI based CSI feedback	16.8.0
2021-12	RAN#94-e	RP-213238	0081	-	F	Clarification on KNZ to codepoint mapping for eType II CSI	16.8.0
2021-12	RAN#94-e	RP-212958	0083	-	A	Rel-15 editorial corrections for TS 38.212 (mirrored to Rel-16)	16.8.0
2021-12	RAN#94-e	RP-212964	0084	-	F	Alignment CR for TS 38.212	16.8.0
2021-12	RAN#94-e	RP-212967	0085	-	B	Introduction of features to extend current NR operation to 71 GHz	17.0.0
2021-12	RAN#94-e	RP-212982	0086	-	B	Introduction of NR DL 1024QAM for FR1	17.0.0
2021-12	RAN#94-e	RP-212973	0087	-	B	Introduction of Coverage Enhancements	17.0.0
2021-12	RAN#94-e	RP-212979	0088	-	B	Introduction of NR Multicast and Broadcast Services	17.0.0
2021-12	RAN#94-e	RP-212966	0089	-	B	Introduction of Further enhancements on MIMO for NR	17.0.0
2021-12	RAN#94-e	RP-212969	0090	-	B	Introduction of NR non-terrestrial networks (NTN)	17.0.0
2021-12	RAN#94-e	RP-212972	0091	-	B	Introduction of Rel-17 UE power saving enhancements	17.0.0
2021-12	RAN#94-e	RP-212968	0092	-	B	Introduction of Rel-17 enhanced IIoT and URLLC	17.0.0
2021-12	RAN#94-e	RP-212980	0093	-	B	Introduction of NR dynamic spectrum sharing enhancements	17.0.0
2021-12	RAN#94-e	RP-212978	0094	-	B	Introduction of NR sidelink enhancement	17.0.0
2022-03	RAN#95-e	RP-220269	0096	-	A	Correction of NZC partitioning in eType II CSI	17.1.0
2022-03	RAN#95-e	RP-220248	0098	-	A	Correction on Rel-16 UE dormancy adaptation	17.1.0
2022-03	RAN#95-e	RP-220252	0099	-	F	Corrections on enhanced IIoT and URLLC in 38.212	17.1.0
2022-03	RAN#95-e	RP-220262	0100	-	F	Corrections on NR sidelink enhancement in 38.212	17.1.0
2022-03	RAN#95-e	RP-220257	0101	-	F	Corrections on coverage enhancements in 38.212	17.1.0

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2022-03	RAN#95-e	RP-220263	0102	-	F	Corrections on NR Multicast and Broadcast Services in 38.212	17.1.0
2022-03	RAN#95-e	RP-220256	0103	-	F	Corrections on UE power saving enhancements in 38.212	17.1.0
2022-03	RAN#95-e	RP-220251	0104	-	F	Correction on extension of current NR operation to 71 GHz in 38.212	17.1.0
2022-03	RAN#95-e	RP-220264	0105	-	F	Corrections on NR dynamic spectrum sharing enhancements in 38.212	17.1.0
2022-03	RAN#95-e	RP-220250	0106	-	F	Corrections on Further enhancements on MIMO for NR in TS 38.212	17.1.0
2022-06	RAN#96	RP-221617	0108	-	A	Clarification of TPMI indication for UL full power transmission	17.2.0
2022-06	RAN#96	RP-221602	0109	-	F	Corrections on enhanced IIoT and URLLC in 38.212	17.2.0
2022-06	RAN#96	RP-221612	0110	-	F	Corrections on NR Multicast and Broadcast Services in 38.212	17.2.0
2022-06	RAN#96	RP-221606	0111	-	F	Corrections on UE power saving enhancements in 38.212	17.2.0
2022-06	RAN#96	RP-221601	0112	-	F	Correction on extension of current NR operation to 71 GHz in 38.212	17.2.0
2022-06	RAN#96	RP-221600	0113	-	F	Corrections on Further enhancements on MIMO for NR in TS 38.212	17.2.0
2022-06	RAN#96	RP-221599	0115	-	A	Rel-16 editorial corrections for TS 38.212 (mirrored to Rel-17)	17.2.0
2022-09	RAN#97-e	RP-222403	0116	1	F	CR on DCI size for Rel-17 NTN HARQ in 38.212	17.3.0
2022-09	RAN#97-e	RP-222400	0117	-	F	CR on the description of the SRS resource set indication for PUSCH repetition	17.3.0
2022-09	RAN#97-e	RP-222401	0118	-	F	CR on ChannelAccess-Cpext in Fallback DCI	17.3.0
2022-09	RAN#97-e	RP-222413	0119	-	F	CR on DCI size alignment for Cross-carrier scheduling from SCell to PCell	17.3.0
2022-09	RAN#97-e	RP-222406	0120	-	F	Corrections on UE Power Saving Enhancements for NR in TS 38.212	17.3.0
2022-09	RAN#97-e	RP-222412	0121	-	F	Corrections on NR Multicast and Broadcast Services in 38.212	17.3.0
2022-09	RAN#97-e	RP-222411	0122	-	F	Correction on NR sidelink enhancement	17.3.0
2022-09	RAN#97-e	RP-222422	0123	-	F	Rel-17 editorial corrections for TS 38.212	17.3.0
2022-12	RAN#98-e	RP-222863	0124	-	F	Corrections on resource pool index	17.4.0
2022-12	RAN#98-e	RP-222853	0125	-	F	CR on channel access type indication in non-fallback DCI	17.4.0
2022-12	RAN#98-e	RP-222853	0126	-	F	Correction to support up to 32 HARQ process numbers for FR2-2	17.4.0
2022-12	RAN#98-e	RP-222853	0127	-	F	Correction on TDRA for multiple PUSCH scheduling in TS 38.212	17.4.0
2022-12	RAN#98-e	RP-222854	0128	-	F	CR on priority of CG-UCI	17.4.0
2022-12	RAN#98-e	RP-222864	0129	-	F	CR on number of HARQ-ACK codebooks configurable for multicast	17.4.0
2022-12	RAN#98-e	RP-222865	0130	-	F	CR on DCI size alignment for Cross-carrier schduling from SCell to PCell	17.4.0
2022-12	RAN#98-e	RP-222868	0131	1	F	Rel-17 editorial corrections for TS 38.212	17.4.0
2022-12	RAN#98-e	RP-222858	0132	-	F	Correction on the short message indicator when TRS availability indication is present	17.4.0
2022-12	RAN#98-e	RP-222864	0133	-	F	CR on format 4_0 DCI size alignment in SCell	17.4.0
2022-12	RAN#98-e	RP-222870	0134	-	F	CR on CSI reporting	17.4.0

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2023-03	RAN#99	RP-230451	0135	-	F	CR on aligning DCI sizes when configuring two HARQ-ACK codebooks for multicast	17.5.0
2023-03	RAN#99	RP-230443	0136	-	F	Corrections on intra-UE multiplexing and semi-static channel occupancy	17.5.0
2023-03	RAN#99	RP-230442	0137	-	F	CR on DCI field sizes for multiple PDSCHs scheduled by single DCI	17.5.0
2023-03	RAN#99	RP-230442	0138	-	F	Corrections to ChanneAccess-CPext field in DCI formats x_2 in TS38.212	17.5.0
2023-03	RAN#99	RP-230440	0140	-	A	Rel-16 editorial corrections for TS 38.212 (mirrored to Rel-17)	17.5.0
2023-03	RAN#99	RP-230453	0141	-	F	Rel-17 editorial corrections for TS 38.212	17.5.0
2023-09	RAN#101	RP-232445	0142	-	F	Correction for the mapping of rank combination value for Rel-17 NCJT CSI	17.6.0
2023-09	RAN#101	RP-232531	0143	-	F	Rel-17 editorial corrections for TS 38.212	17.6.0
2023-09	RAN#101	RP-232471	0144	-	B	Introduction of Rel-18 Multi-carrier enhancements	18.0.0
2023-09	RAN#101	RP-232458	0145	-	B	Introduction of Rel-18 MIMO Evolution for Downlink and Uplink	18.0.0
2023-09	RAN#101	RP-232473	0146	-	B	Introduction of Rel-18 Further NR mobility enhancements	18.0.0
2023-09	RAN#101	RP-232474	0147	-	B	Introduction of Rel-18 NR NTN enhancements	18.0.0
2023-09	RAN#101	RP-232480	0148	-	B	Introduction of NR positioning enhancement in Rel-18	18.0.0
2023-09	RAN#101	RP-232469	0149	-	B	Introduction of Rel-18 NR sidelink evolution	18.0.0
2023-09	RAN#101	RP-232479	0150	-	B	Introduction of Rel-18 network controlled repeaters	18.0.0
2023-09	RAN#101	RP-232481	0151	-	B	Introduction of Rel-18 network energy saving for NR	18.0.0
2023-09	RAN#101	RP-232472	0152	-	B	Introduction of Rel-18 further NR Coverage enhancement	18.0.0
2023-09	RAN#101	RP-232477	0154	-	B	Introduction of Rel-18 NR support for dedicated spectrum less than 5MHz for FR1	18.0.0
2023-09	RAN#101	RP-232482	0155	-	B	Introduction of Rel-18 XR enhancements for NR	18.0.0
2023-12	RAN#102	RP-233703	0158	1	A	Correction on the rate matching when HARQ-ACK multiplexed with CG-PUSCH	18.1.0
2023-12	RAN#102	RP-233727	0160	-	A	Correction on CSI reporting for 1 CSI-RS port	18.1.0
2023-12	RAN#102	RP-233728	0162	-	A	Rel-17 editorial corrections for TS 38.212 (mirrored to Rel-18)	18.1.0
2023-12	RAN#102	RP-233719	0163	-	F	Corrections on NR positioning enhancement in 38.212	18.1.0
2023-12	RAN#102	RP-233706	0164	-	F	Corrections on Rel-18 NR sidelink evolution in 38.212	18.1.0
2023-12	RAN#102	RP-233709	0165	-	F	Corrections on Rel-18 further NR Coverage enhancement in 38.212	18.1.0
2023-12	RAN#102	RP-233710	0166	-	F	Corrections on Rel-18 Further NR mobility enhancements in 38.212	18.1.0
2023-12	RAN#102	RP-233705	0167	-	F	Corrections on Rel-18 MIMO Evolution for Downlink and Uplink in 38.212	18.1.0
2023-12	RAN#102	RP-233708	0168	-	F	Corrections on Rel-18 Multi-carrier enhancements in 38.212	18.1.0
2023-12	RAN#102	RP-233720	0169	-	F	Corrections on Rel-18 network energy saving for NR in 38.212	18.1.0
2023-12	RAN#102	RP-233714	0170	-	F	Corrections on Rel-18 NR NTN enhancements in 38.212	18.1.0
2023-12	RAN#102	RP-233721	0171	-	F	Corrections on Rel-18 XR enhancements for NR in 38.212	18.1.0
2023-12	RAN#102	RP-233729	0172	-	B	Introduction of Rel-18 NR demodulation performance evoluation	18.1.0

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2023-12	RAN#102	RP-233733	0173	-	B	Introduction of Rel-18 enhancements of NR Multicast and Broadcast Services	18.1.0
2023-12	RAN#102	RP-233718	0174	-	F	Corrections on Rel-18 network controlled repeaters in 38.212	18.1.0
2024-03	RAN#103	RP-240534	0176	-	A	Clarification on typeA SRS TPC commands for SUL	18.2.0
2024-03	RAN#103	RP-240515	0178	-	A	CR on reportQuantity for RSRP/SINR in TS38.212	18.2.0
2024-03	RAN#103	RP-240530	0179	-	F	Correction of RRC parameter names for UTO-UCI indication	18.2.0
2024-03	RAN#103	RP-240536	0181	-	A	Rel-16 editorial corrections for TS 38.212 (mirrored to Rel-18)	18.2.0
2024-03	RAN#103	RP-240537	0183	-	F	Rel-18 editorial corrections for TS 38.212	18.2.0
2024-03	RAN#103	RP-240520	0184	-	F	Corrections on Rel-18 Multi-carrier enhancements in 38.212	18.2.0
2024-03	RAN#103	RP-240518	0185	-	F	Corrections on Rel-18 MIMO Evolution for Downlink and Uplink in 38.212	18.2.0
2024-03	RAN#103	RP-240528	0186	-	F	Corrections on NR positioning enhancement in 38.212	18.2.0
2024-03	RAN#103	RP-240519	0187	-	F	Corrections on Rel-18 NR sidelink evolution in 38.212	18.2.0
2024-06	RAN#104	RP-241068	0188	-	F	Correction on UL/SUL field in DCI format 1_0 in LTM	18.3.0
2024-06	RAN#104	RP-241064	0191	-	A	Correction to maxRank configuration restriction with fullpowerMode1 and transform precoding ‘enabled’	18.3.0
2024-06	RAN#104	RP-241067	0192	-	F	Corrections on PRACH association indicator in PDCCH order in 38.212	18.3.0
2024-06	RAN#104	RP-241067	0193	-	F	Corrections for Transmission with more than 4 Layers for 8TX UE	18.3.0
2024-06	RAN#104	RP-241075	0194	-	F	CR for 38.212 on TRS occasions for idle/inactive UEs	18.3.0
2024-06	RAN#104	RP-241067	0195	-	F	Correction on PTRS-DMRS association field in DCI format 0_1 and DCI format 0_2	18.3.0
2024-06	RAN#104	RP-241066	0196	-	F	Corrections on Rel-18 Multi-carrier enhancements in 38.212	18.3.0
2024-06	RAN#104	RP-241060	0199	-	F	Rel-18 editorial corrections for TS 38.212	18.3.0
2024-06	RAN#104	RP-241067	0200	-	F	Corrections on Rel-18 MIMO Evolution for Downlink and Uplink in 38.212	18.3.0
2024-06	RAN#104	RP-241058	0202	-	A	Rel-16 editorial corrections for TS 38.212 (mirrored to Rel-18)	18.3.0
2024-06	RAN#104	RP-241059	0203	-	A	Rel-17 editorial corrections for TS 38.212 (mirrored to Rel-18)	18.3.0
2024-09	RAN#105	RP-242209	0204	-	F	CR on PTRS-DMRS Association for 8 Tx UL MIMO	18.4.0
2024-09	RAN#105	RP-242213	0205	-	F	Correction on SL-U COT sharing flag indication	18.4.0
2024-09	RAN#105	RP-242204	0206	-	F	Rel-18 editorial corrections for TS 38.212	18.4.0
2024-12	RAN#106	RP-242932	0208	-	F	Correction on PTRS-DMRS Association	18.5.0
2024-12	RAN#106	RP-242925	0209	1	F	Rel-18 editorial corrections for TS 38.212	18.5.0
2025-03	RAN#107	RP-250234	0210	-	F	Corrections on determination of DCI format 0_3 in case of dormancy	18.6.0
2025-03	RAN#107	RP-250231	0212	-	A	Correction on LBRM TBS determination for UE indicating 1024QAM rank restricted capability	18.6.0
2025-03	RAN#107	RP-250227	0213	-	F	CR on the PRACH retransmission indicator field included in the PDCCH order	18.6.0
2025-03	RAN#107	RP-250226	0214	-	F	Rel-18 editorial corrections for TS 38.212	18.6.0

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2025-06	RAN#108	RP-251562	0224	-	A	Rel-17 editorial corrections for TS 38.212 (mirrored to Rel-18)	18.7.0
2025-06	RAN#108	RP-251563	0225	-	F	Rel-18 editorial corrections for TS 38.212	18.7.0
2025-06	RAN#108	RP-251571	0215	-	B	Introduction of Rel-19 enhancements of network energy savings for NR	19.0.0
2025-06	RAN#108	RP-251572	0216	-	B	Introduction of Rel-19 AI/ML for NR Air Interface	19.0.0
2025-06	RAN#108	RP-251577	0217	-	B	Introduction of Rel-19 low-power Wake-up Signal for NR	19.0.0
2025-06	RAN#108	RP-251836	0218	-	B	Introduction of Rel-19 evolution of NR duplex operation	19.0.0
2025-06	RAN#108	RP-251578	0219	-	B	Introduction of Rel-19 Multi-carrier enhancements	19.0.0
2025-06	RAN#108	RP-251580	0220	-	B	Introduction of Rel-19 NR MIMO Phase 5	19.0.0
2025-06	RAN#108	RP-251582	0221	-	B	Introduction of Rel-19 NR mobility enhancements Phase 4 in 38.212	19.0.0
2025-06	RAN#108	RP-251583	0226	-	B	Introduction of Rel-19 NR NTN Phase 3 in 38.212	19.0.0
2025-06	RAN#108	RP-251588	0227	-	B	Introduction of Rel-19 XR enhancements for NR	19.0.0
2025-09	RAN#109	RP-252634	0222	1	B	Introduction of 32 HARQ process numbers in Rel-19 [TN32HARQ]	19.1.0
2025-09	RAN#109	RP-252629	0228	-	F	Corrections on Rel-19 low-power Wake-up Signal for NR	19.1.0
2025-09	RAN#109	RP-252627	0229	-	F	Corrections on Rel-19 AI/ML for NR Air Interface	19.1.0
2025-09	RAN#109	RP-252630	0230	-	F	Corrections on Rel-19 Multi-carrier enhancements	19.1.0
2025-09	RAN#109	RP-252631	0231	-	F	Corrections on Rel-19 NR MIMO Phase 5	19.1.0
2025-12	RAN#110	RP-253034	0233	-	A	Rel-18 editorial corrections for TS 38.212 (mirrored to Rel-19)	19.2.0
2025-12	RAN#110	RP-253020	0234	-	F	Correction on Rel-19 evolution of NR duplex operation	19.2.0
2025-12	RAN#110	RP-253027	0235	-	B	Introduction of Rel-19 less than 5MHz for NTN	19.2.0
2025-12	RAN#110	RP-253016	0236	-	B	Introduction of Rel-19 early CSI acquisition for L3 handover to TS 38.212 [EarlyCSI_L3HO]	19.2.0
2025-12	RAN#110	RP-253019	0237	-	F	Corrections on Rel-19 NR MIMO Phase 5	19.2.0
2025-12	RAN#110	RP-253022	0238	-	F	Correction on Rel-19 enhancements of network energy savings for NR	19.2.0
2025-12	RAN#110	RP-253036	0242	-	A	Rel-16 editorial corrections for TS 38.212 (mirrored to Rel-19)	19.2.0